

Report: Core vs Divergent Metabolism + Clinical + Survival

Core patients: 1127 | Divergent: 39

Metabolic variables analyzed: 248 | Significant ($p_{\text{adj}} < 0.05$): 93

FBA solutions: FBA, pFBA, L1w, L2, L2w

Structure: Clinical → TNBC → Metabolism → Integration → Survival

Effect size: Cliff's δ | ≥ 0.147 small · ≥ 0.330 medium · ≥ 0.474 large

SECTION I

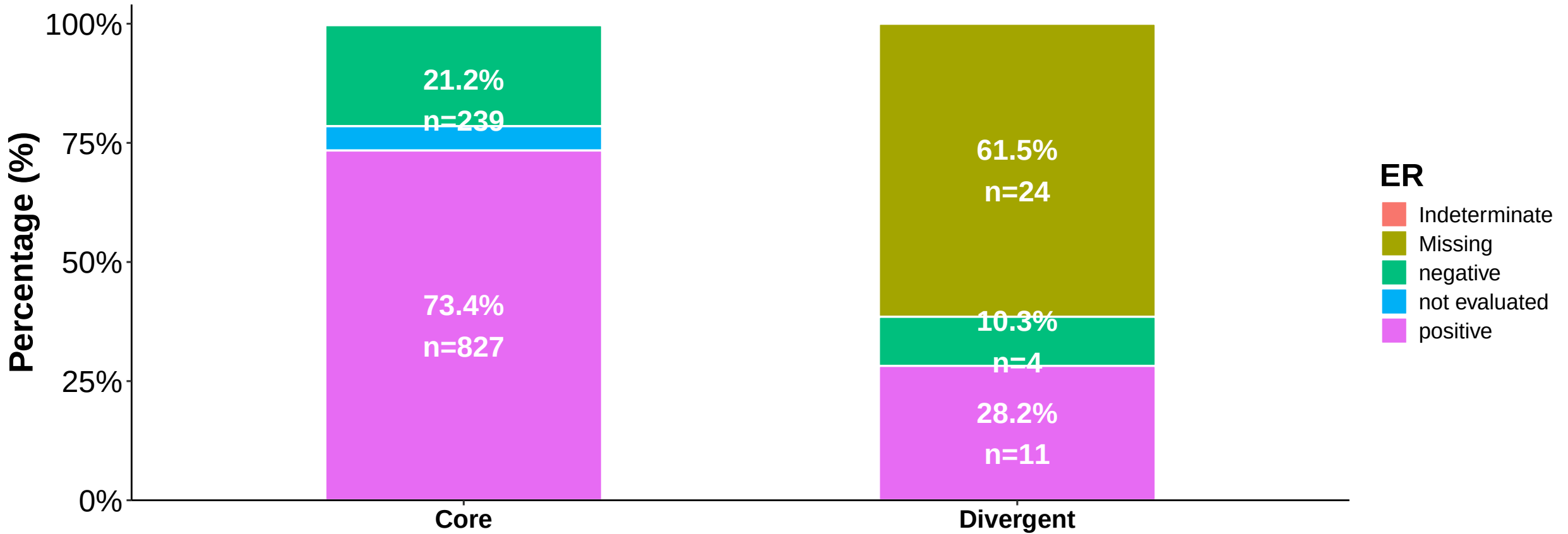
Clinical Differences: Core vs Divergent

Macro-categories: Tumor Biology | Stage | Treatment History | Patient Factors

Categorical Clinical – . A. Tumor Biology

ER

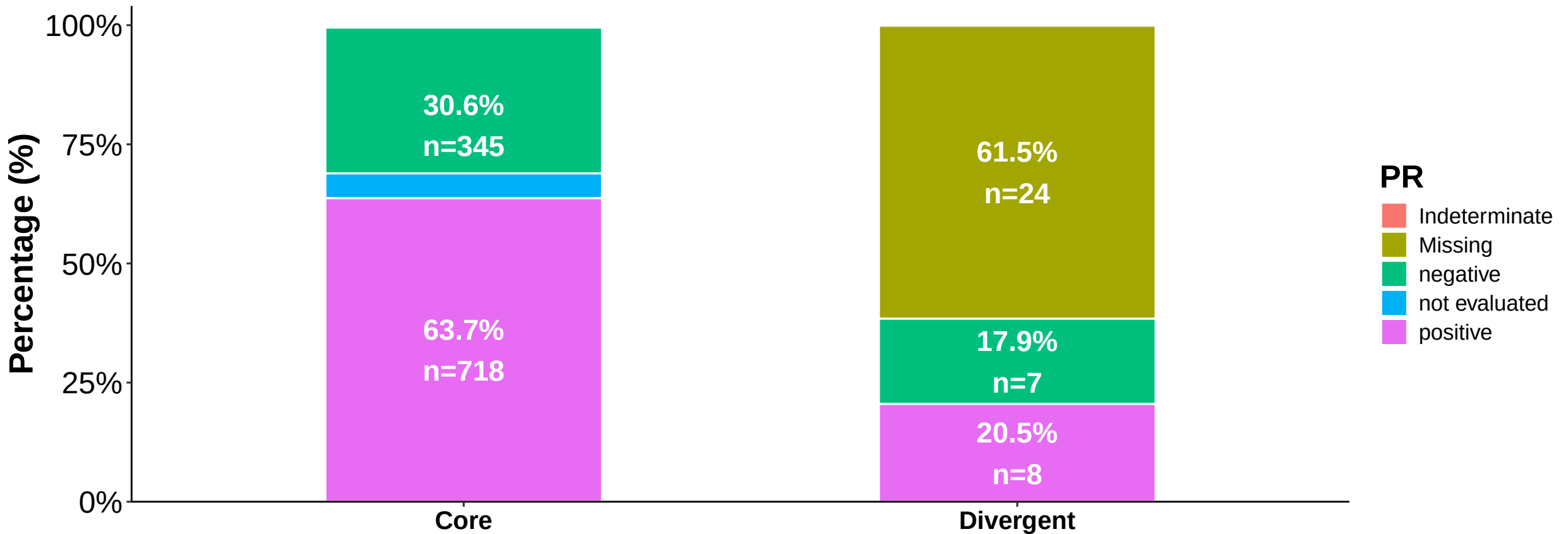
Chi² p: 4.98e-152



Group

PR

Chi² p: 3.82e-152

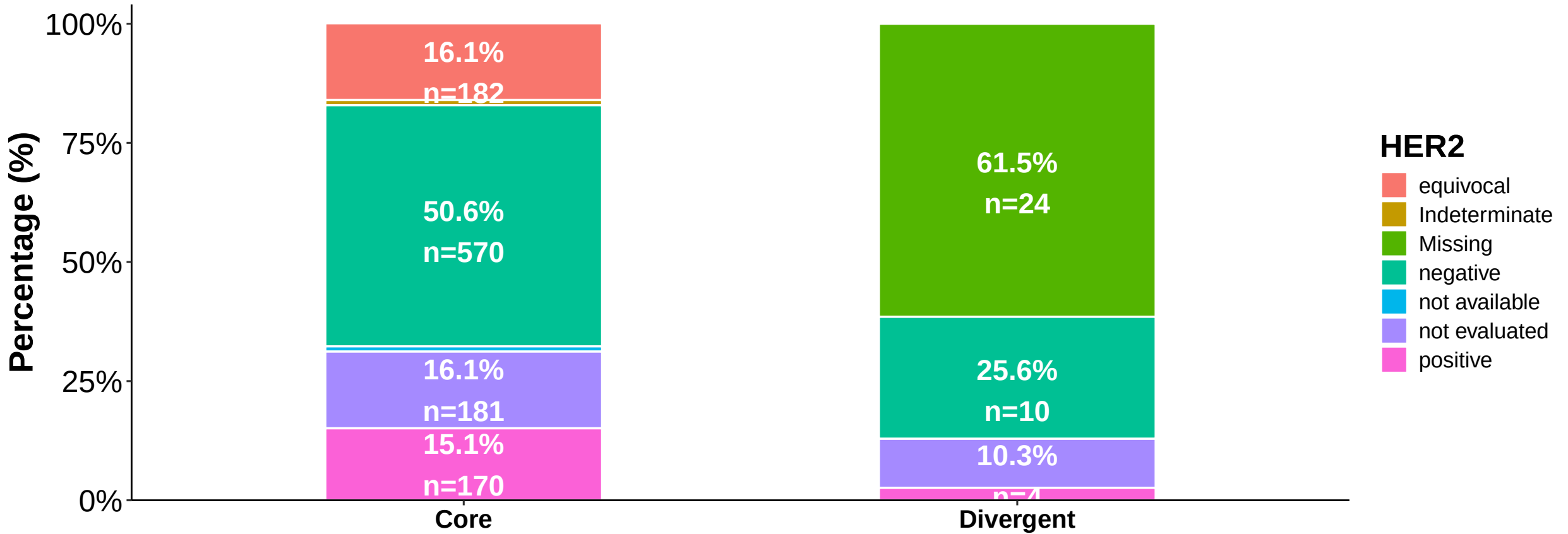


Group

Categorical Clinical – . A. Tumor Biology

HER2

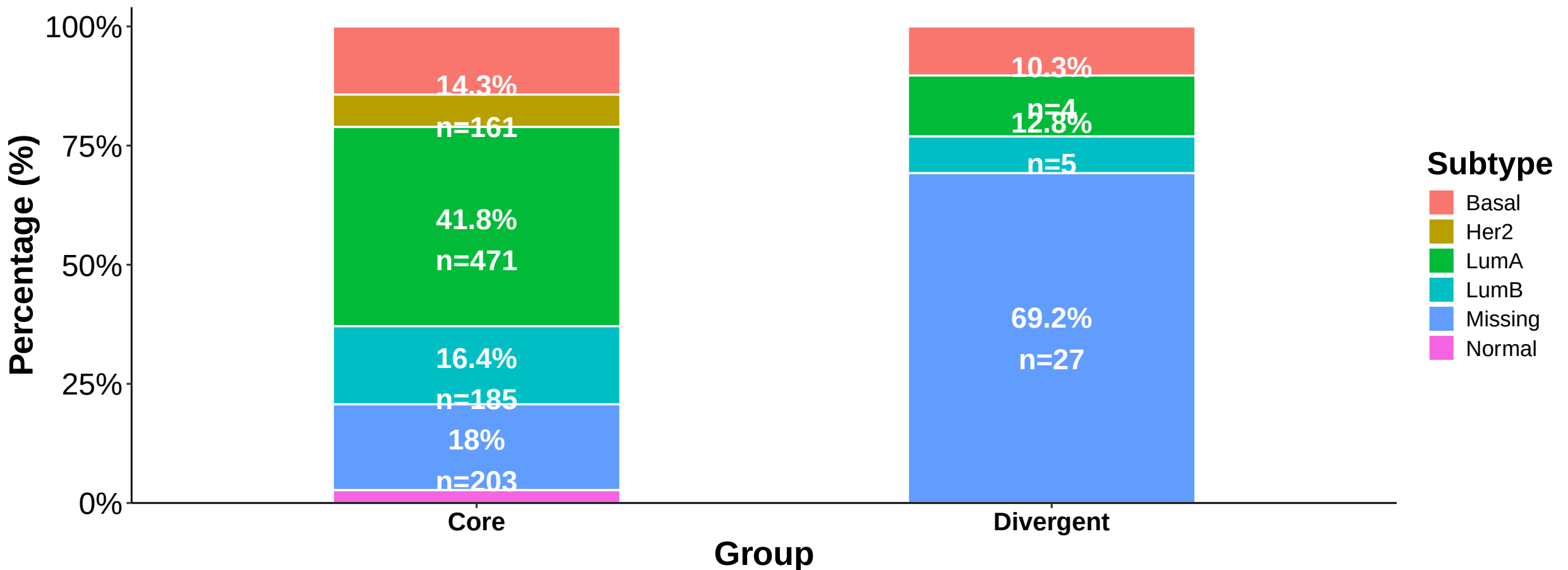
Chi² p: 3.82e-150



Subtype

Subtype

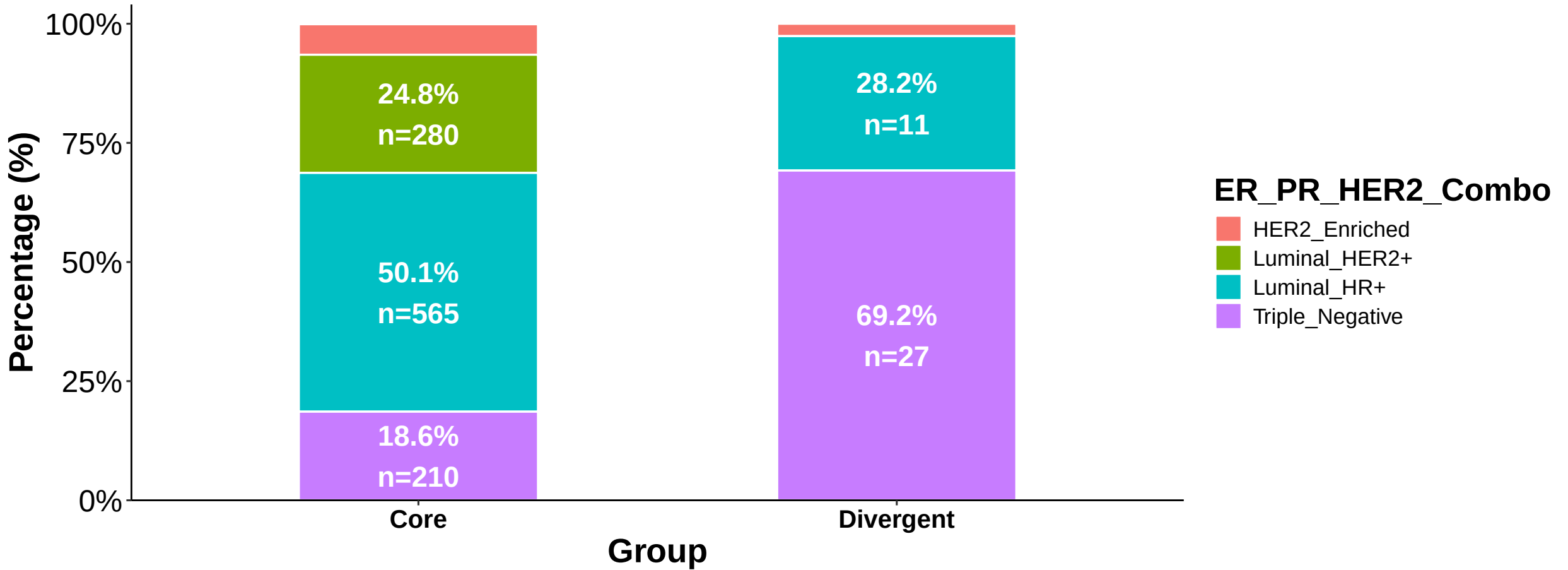
Chi² p: 1.99e-12



Categorical Clinical – A. Tumor Biology

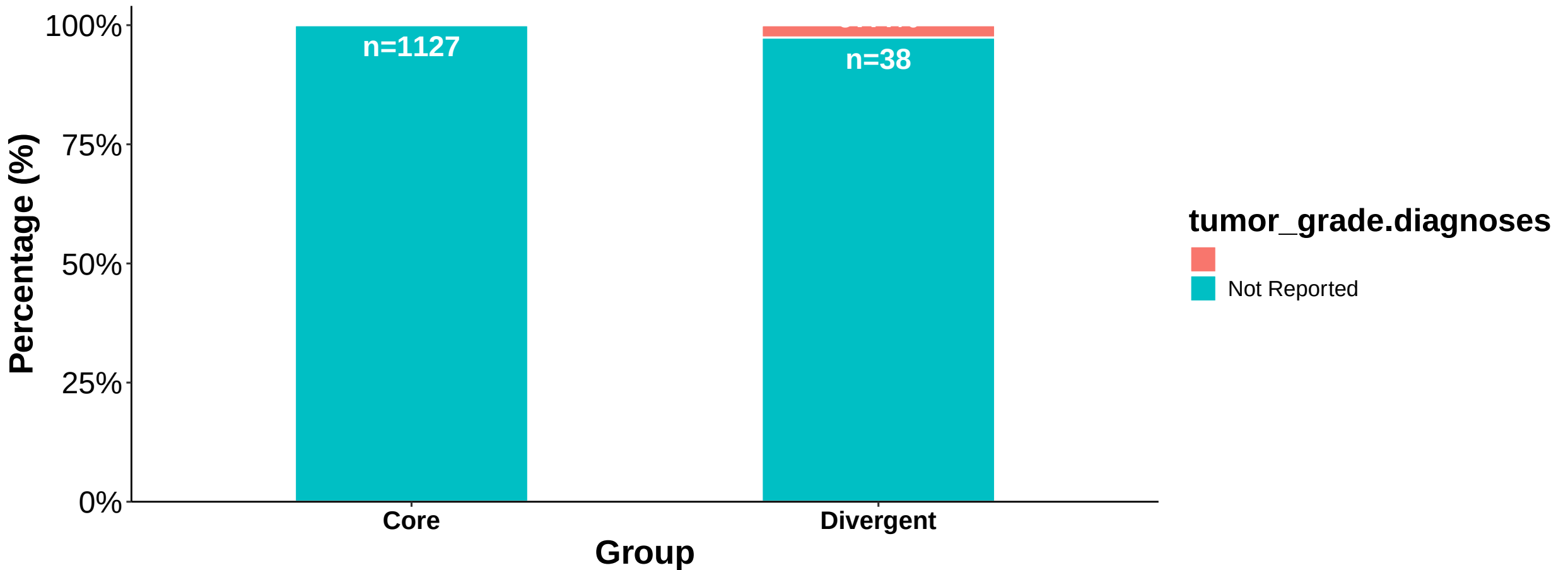
ER_PR_HER2_Combo

Chi² p: 2.52e-13



tumor_grade.diagnoses

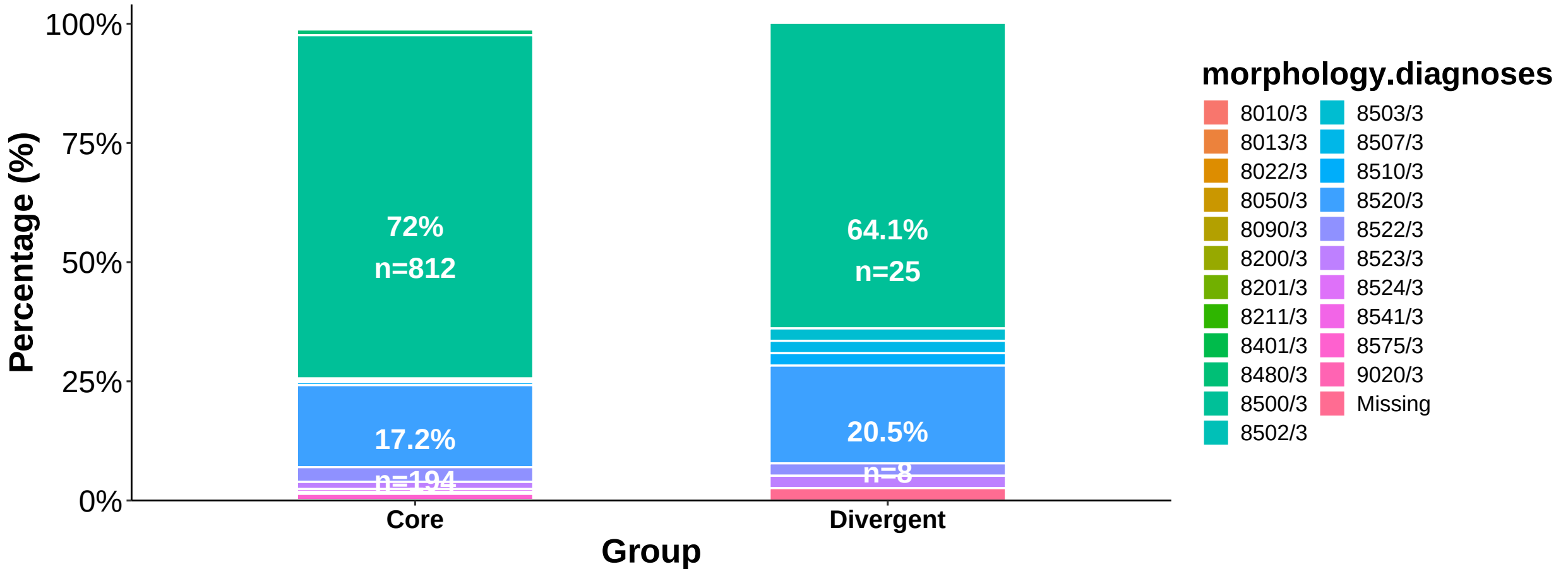
Chi² p: 0.00943



Categorical Clinical – A. Tumor Biology

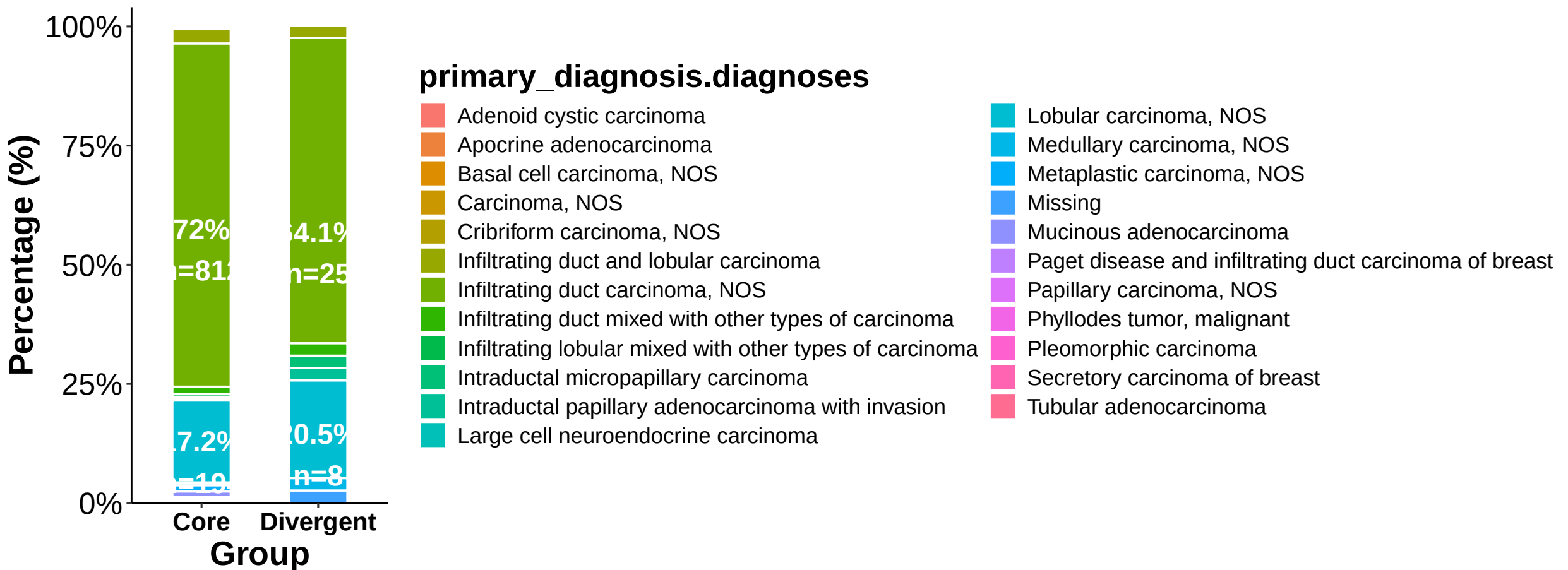
morphology.diagnoses

Chi² p: 0.00498



primary_diagnosis.diagnoses

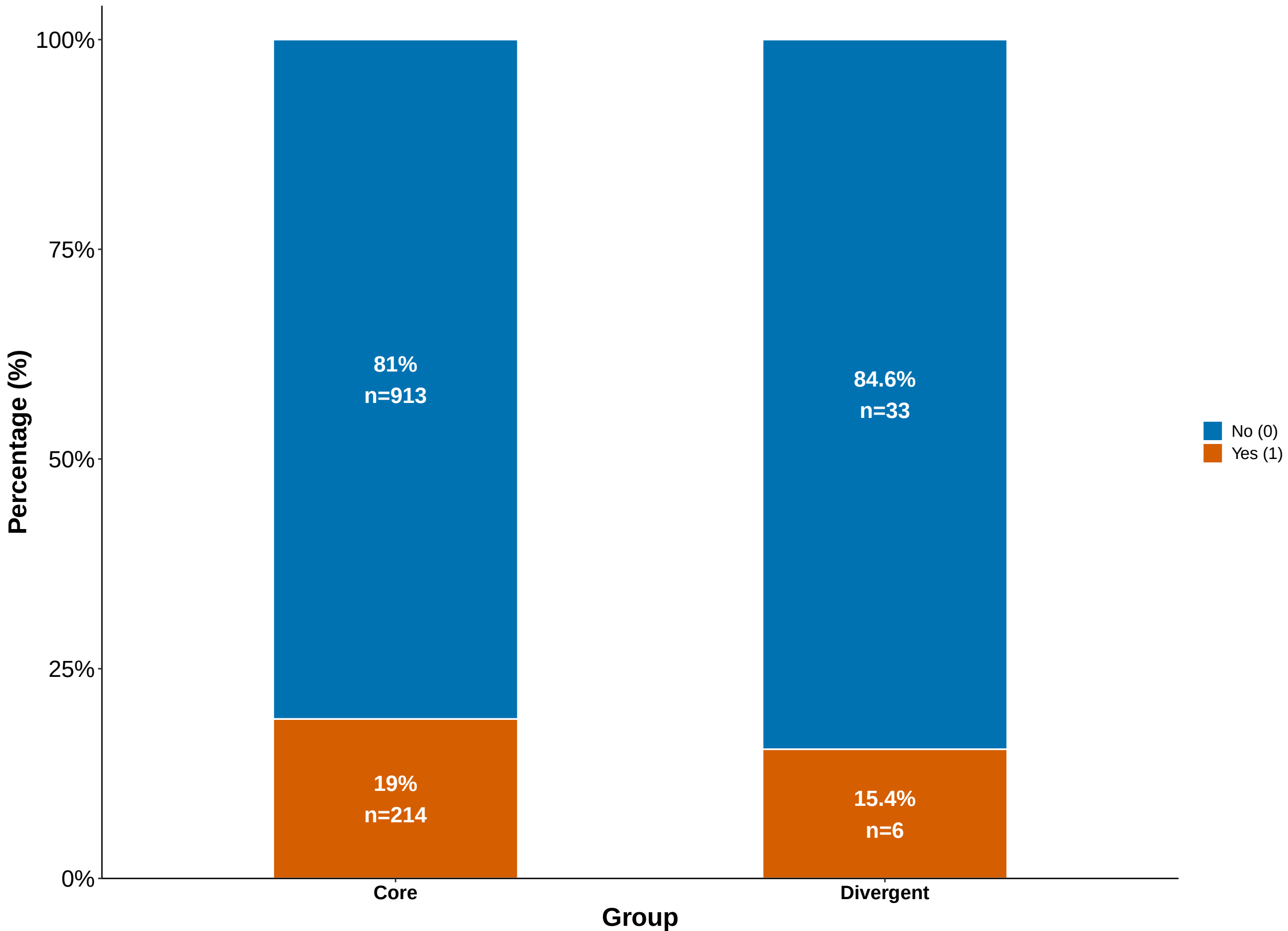
Chi² p: 0.00498



Numeric Clinical – . B. Stage and Tumor Burden

Metastasis_Flag

p-adj: 0.572 | .=-0.04 (Negligible) |



Cliff's .: Clinical: . B. Stage and Tumor Burden

Blue = Core > Divergent | Orange = Divergent > Core | * p_adj<0.05

Cliff's .
Div vs Core1.00.5.00.51.0



Metastasis_Flag

ns .=-0.04

Cliff's . – Clinical: .B. Stage and Tumor Burden

Cliff's . + .% median | . higher in Core · . higher in Divergent

Metastasis_Flag

ns .=-0.04 (NA%)



-0.6

Large

Medium

Small

0

Small

Medium

Large

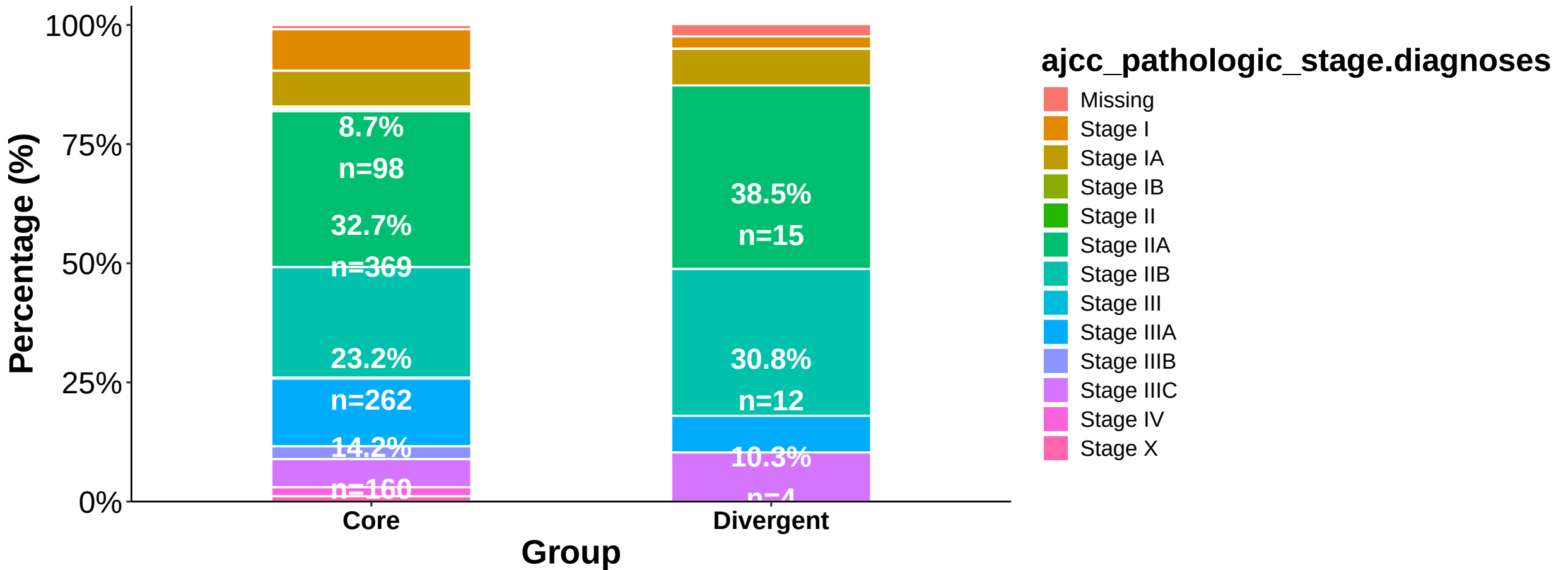
0.6

Cliff's delta [Core vs Divergent]

Categorical Clinical – . B. Stage and Tumor Burden

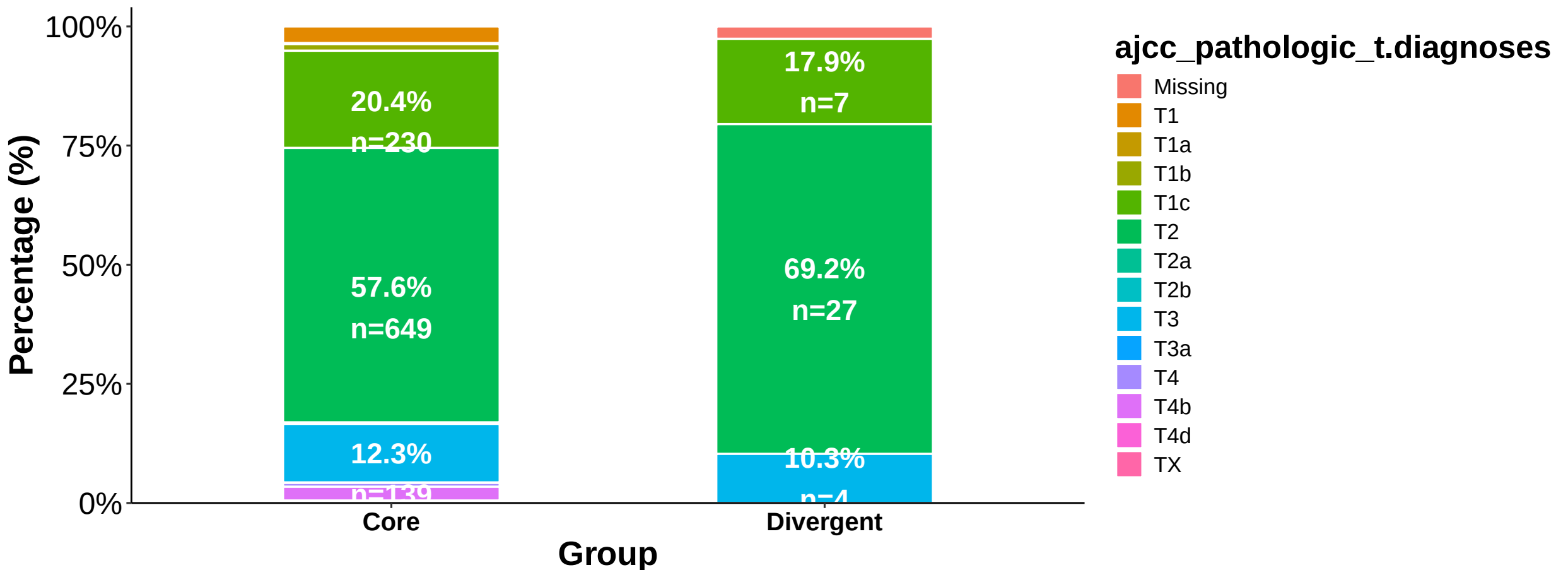
ajcc_pathologic_stage.diagnoses

Chi² p: 0.7



ajcc_pathologic_t.diagnoses

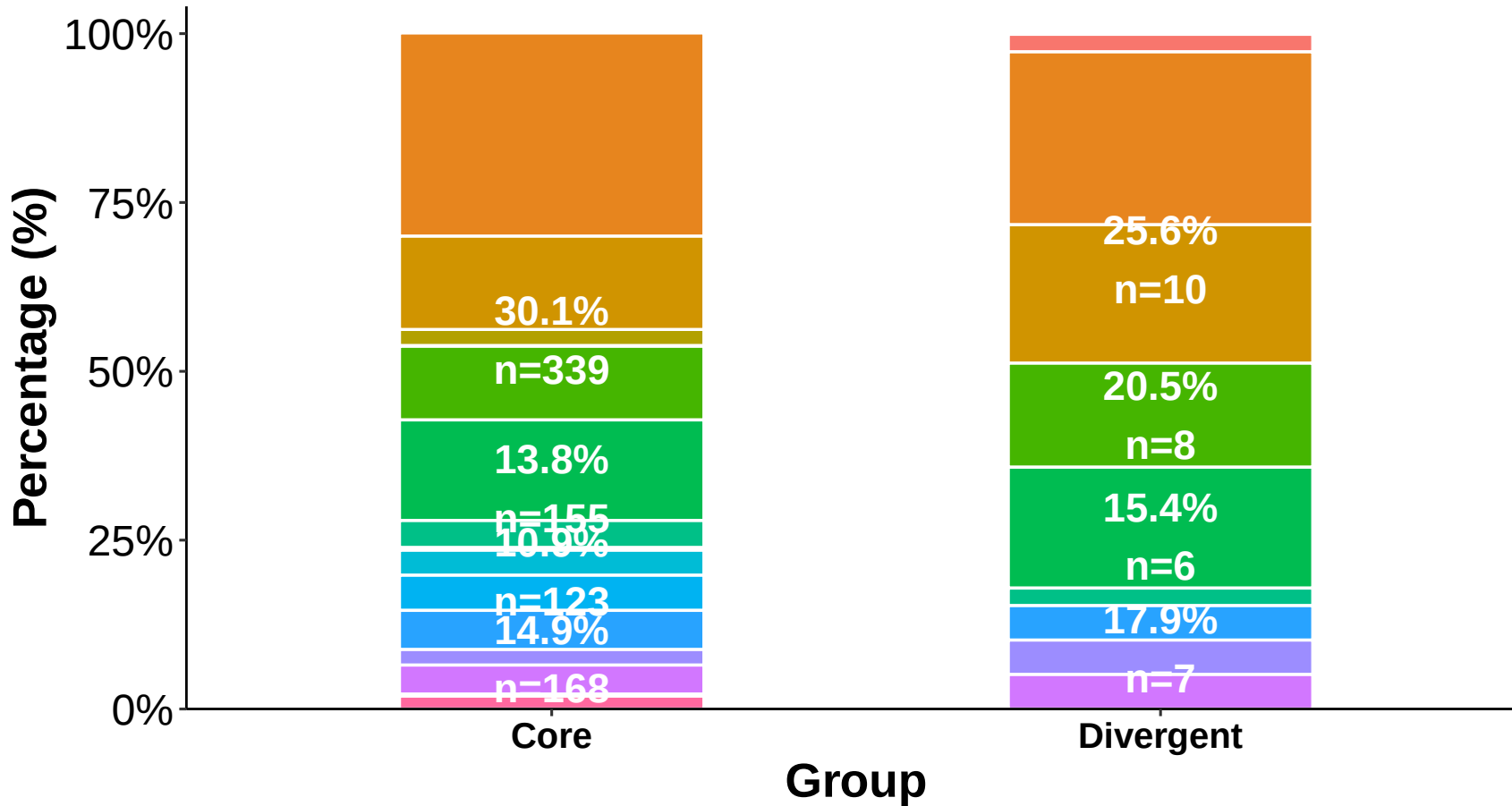
Chi² p: 0.00129



Categorical Clinical – . B. Stage and Tumor Burden

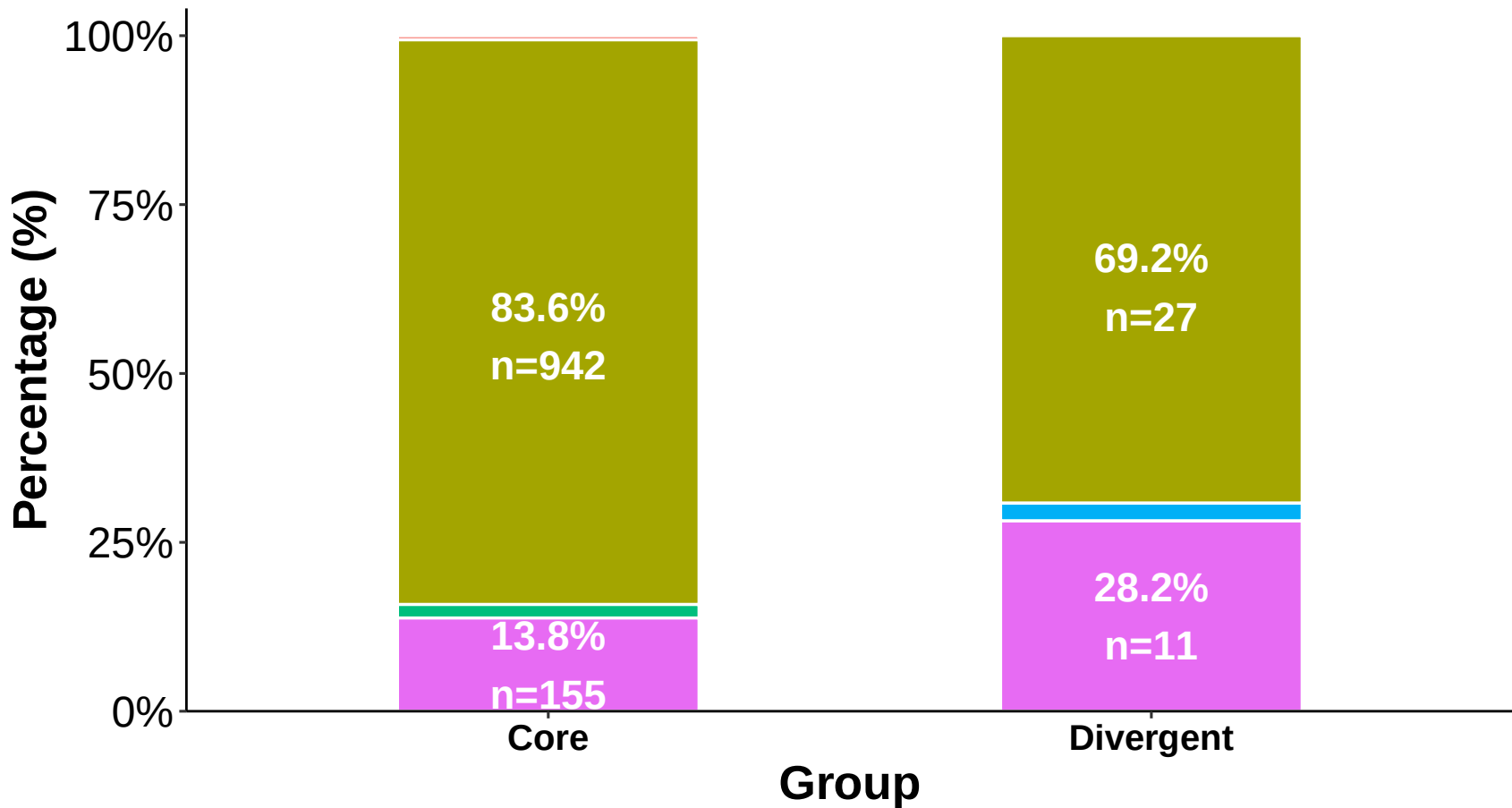
ajcc_pathologic_n.diagnoses

Chi² p: 0.00139



ajcc_pathologic_m.diagnoses

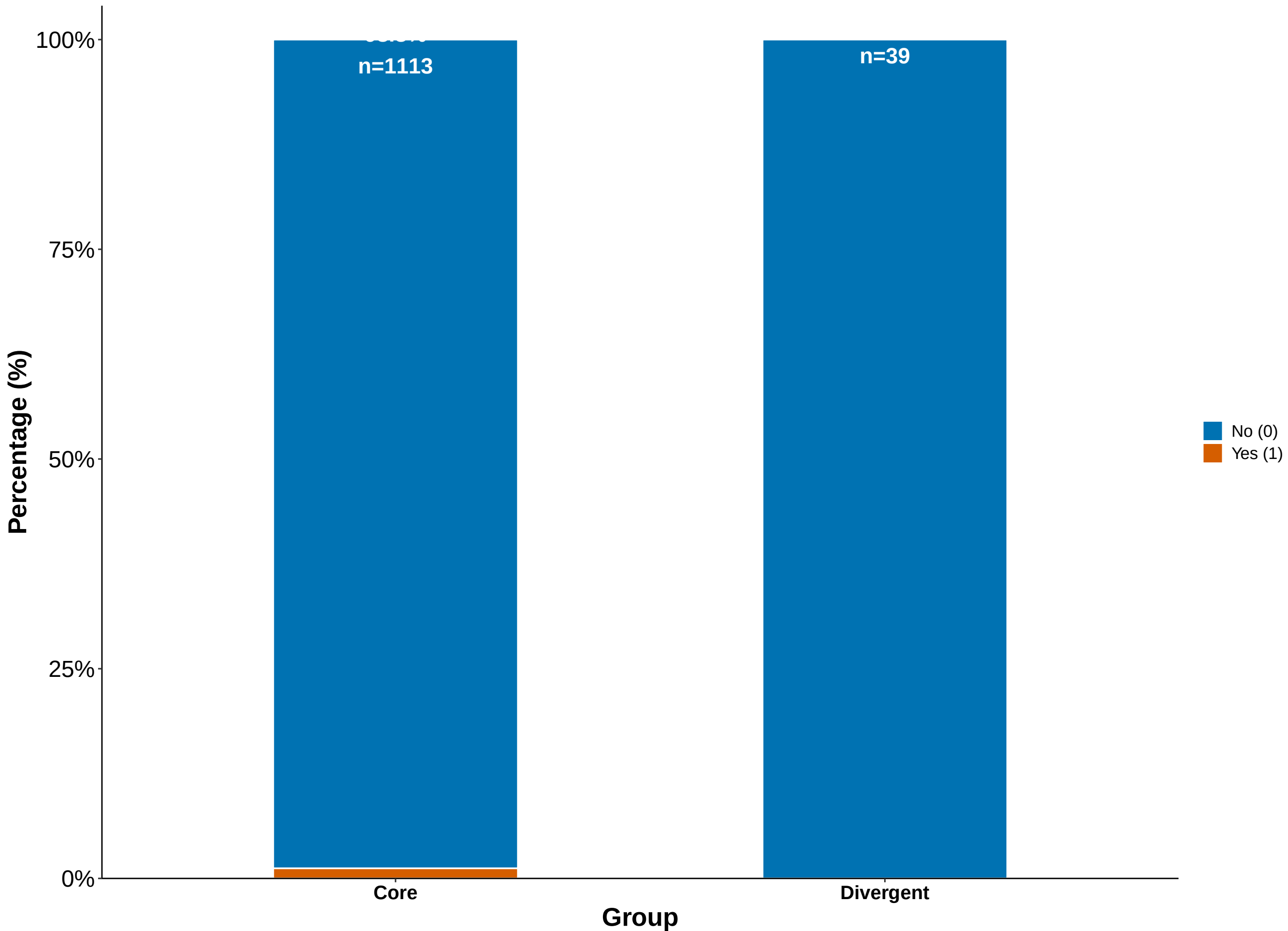
Chi² p: 2.39e-07



Numeric Clinical – . C. Treatment History

Prior_Treatment_Flag

p-adj: 0.572 | .=-0.01 (Negligible) |



Cliff's .: Clinical: . C. Treatment History

Blue = Core > Divergent | Orange = Divergent > Core | * p_adj<0.05

Cliff's .
Div vs Core



-1.0 -0.5 0.0 0.5 1.0

Prior_Treatment_Flag

ns .=-0.01

Cliff's . – Clinical: . C. Treatment History

Cliff's . + .% median | . higher in Core · . higher in Divergent

Prior_Treatment_Flag

ns .=-0.01 (NA%)

-0.6

Large

Medium

Small

0

Small

Medium

Large

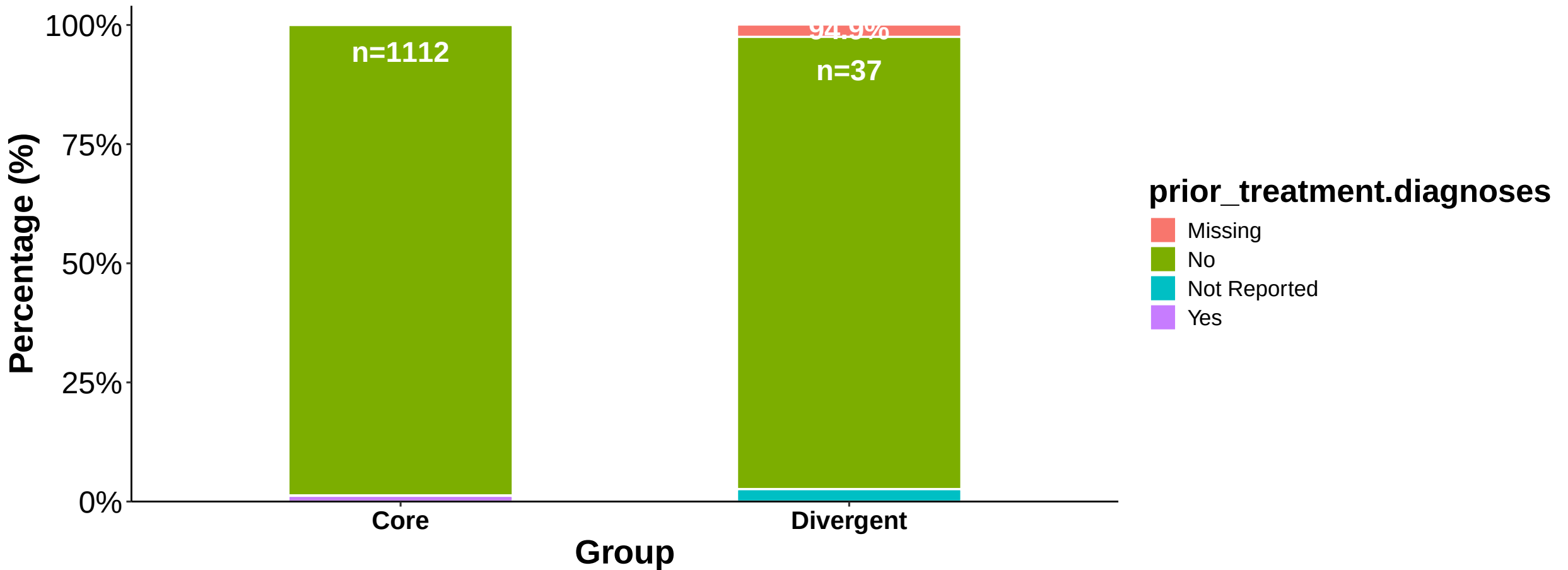
0.6

Cliff's delta [Core vs Divergent]

Categorical Clinical – .C. Treatment History

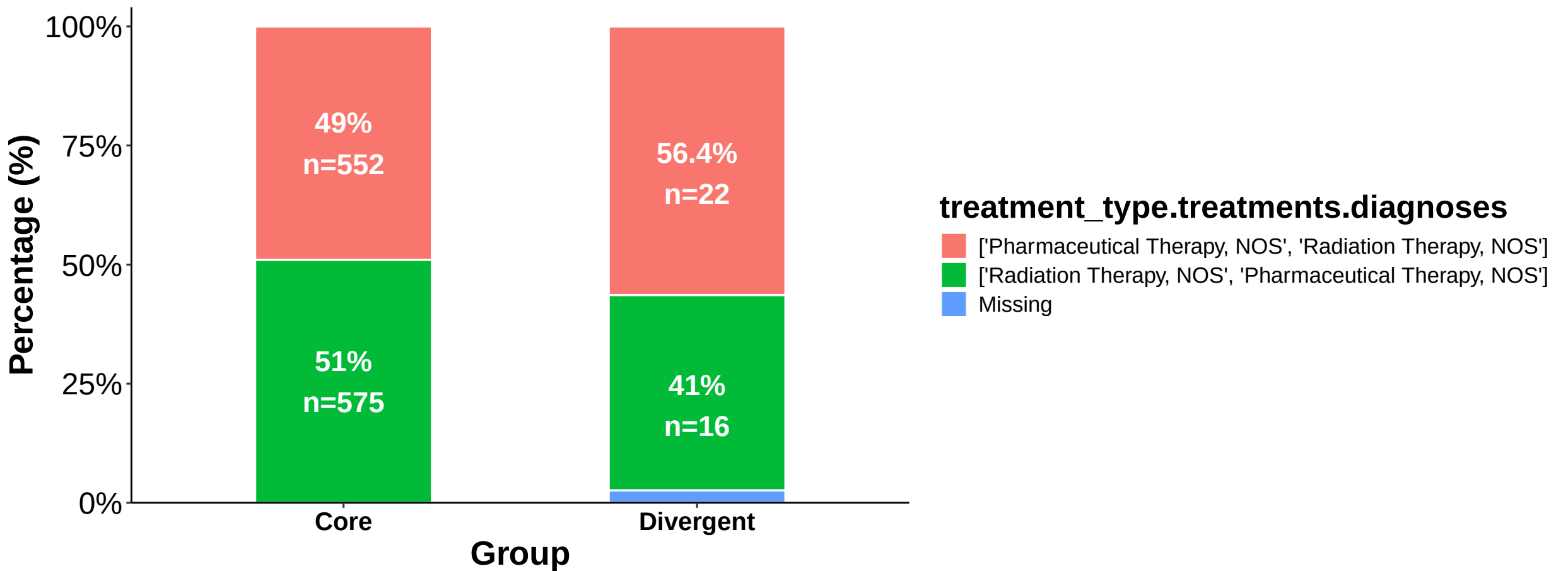
prior_treatment.diagnoses

Chi² p: 2.58e-09



treatment_type.treatments.diagnoses

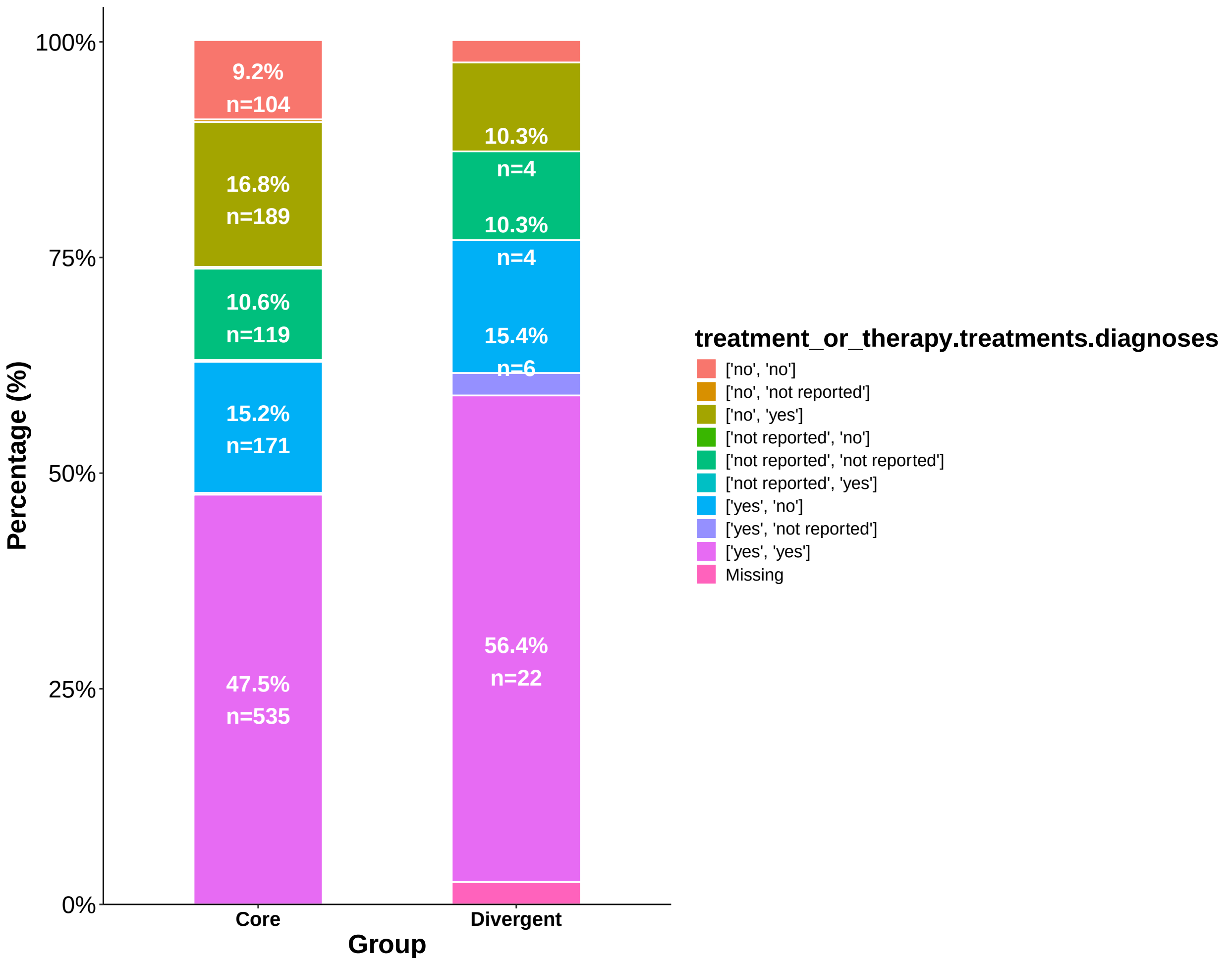
Chi² p: 2.96e-07



Categorical Clinical – .C. Treatment History

treatment_or_therapy.treatments.diagnoses

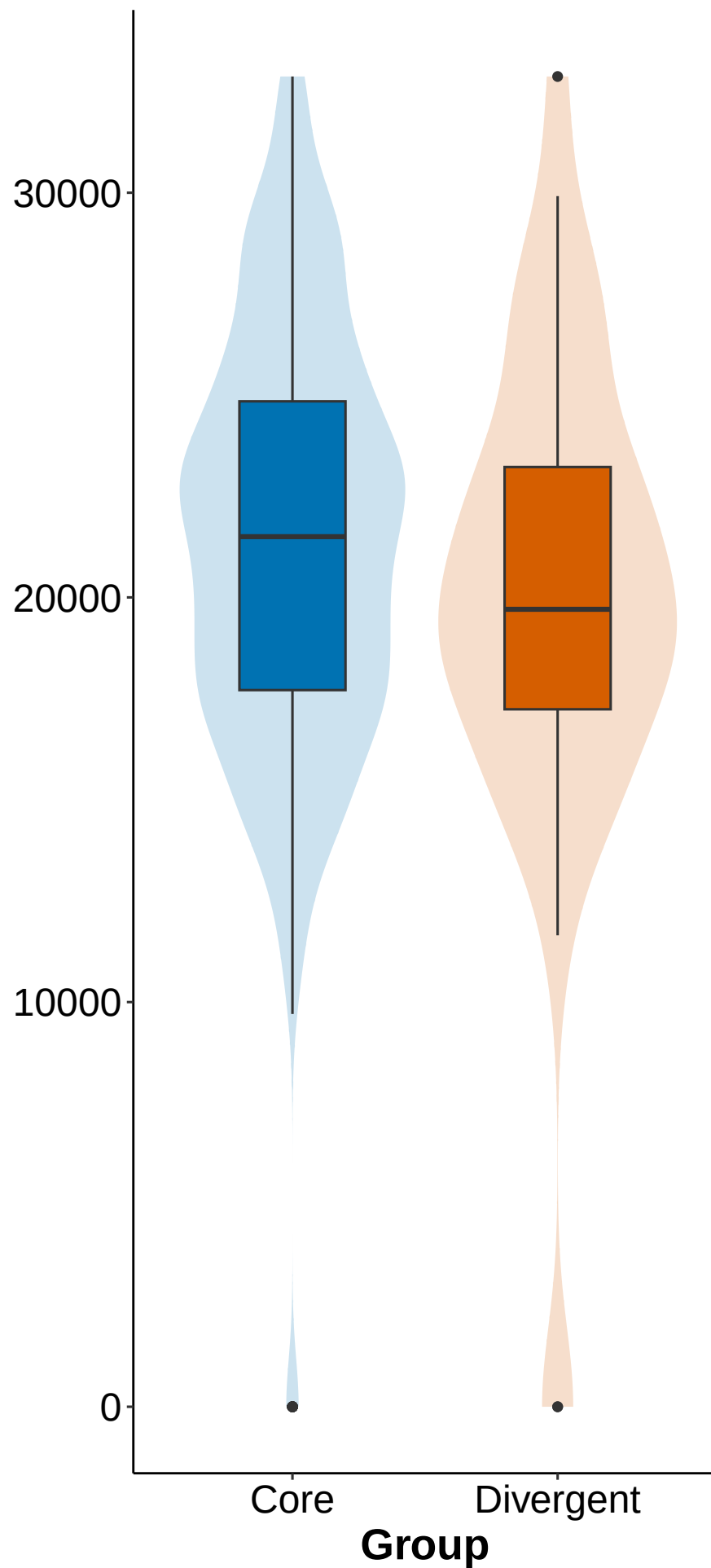
Chi² p: 5.12e-06



Numeric Clinical – .D. Patient Factors

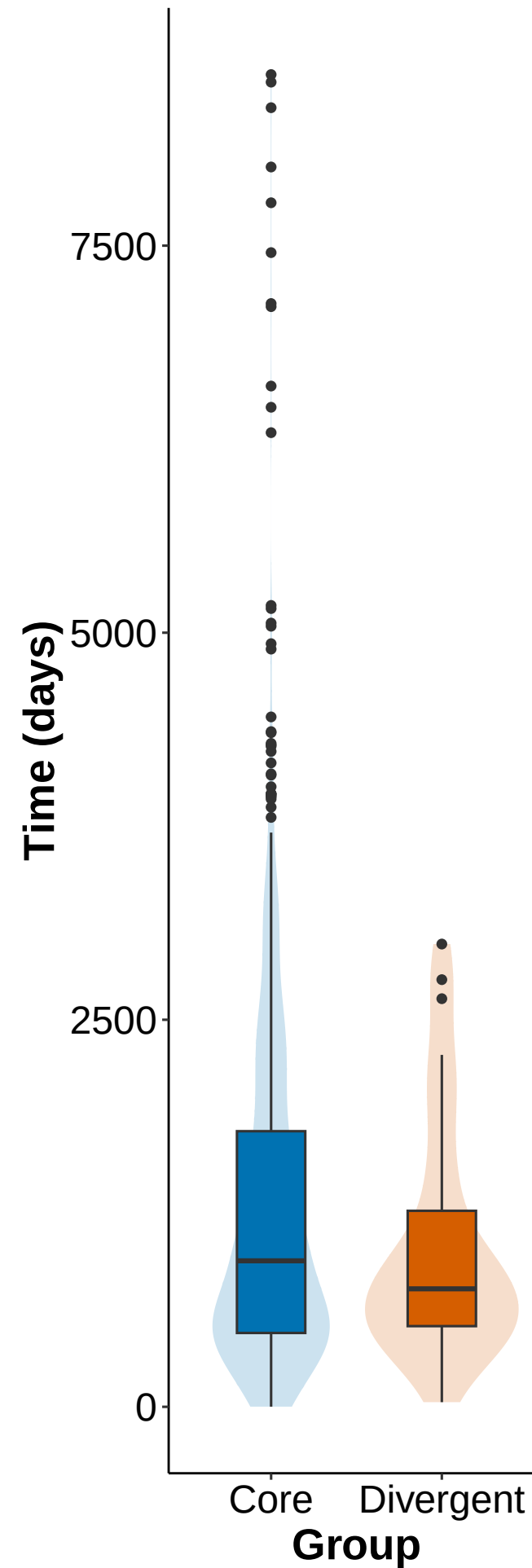
age_at_diagnosis.diagnoses

p-adj: 0.414 | .=-0.14 (Negligible) | .%: -8.3

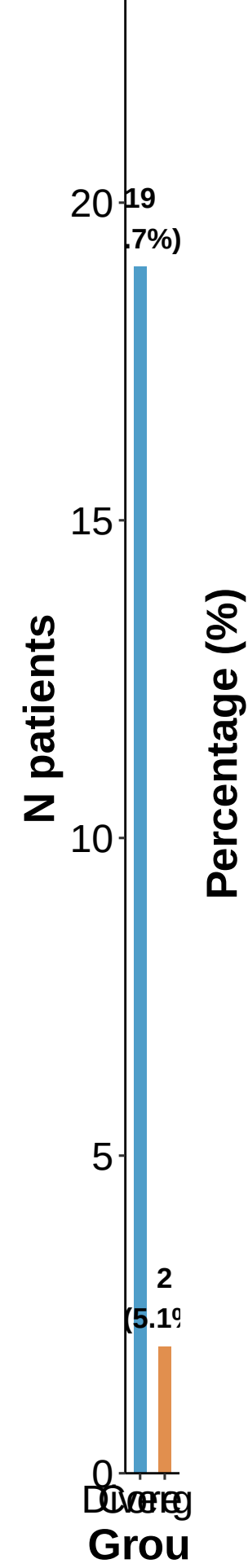


OS.time

p-adj: 0.572 | .=-0.08 (Negligible)

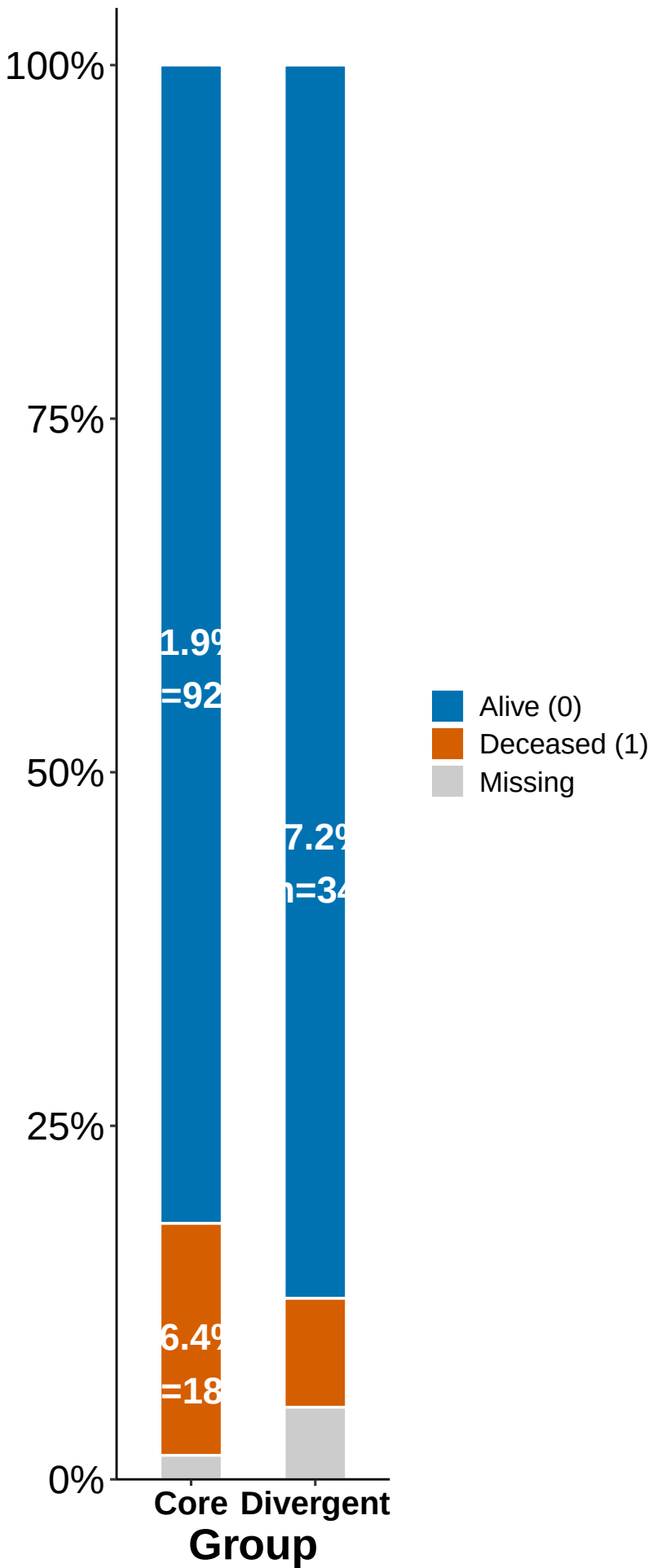


Miss



OS

p-adj: 0.414 | .=-0.09 (Negligible) |



Cliff's η^2 : Clinical: . D. Patient Factors

Blue = Core > Divergent | Orange = Divergent > Core | * $p_{adj} < 0.05$

Cliff's η^2
Div vs Core



1.0 0.5 0.5 1.0

OS.time

ns $\eta^2 = -0.08$

OS

ns $\eta^2 = -0.09$

age_at_diagnosis.diagnoses

ns $\eta^2 = -0.14$

Cliff's Δ – Clinical: Δ D. Patient Factors

Cliff's Δ + Δ.% median | Δ higher in Core · Δ higher in Divergent

OS.time

ns Δ=-0.08 (-19.2%)

OS

ns Δ=-0.09 (NA%)

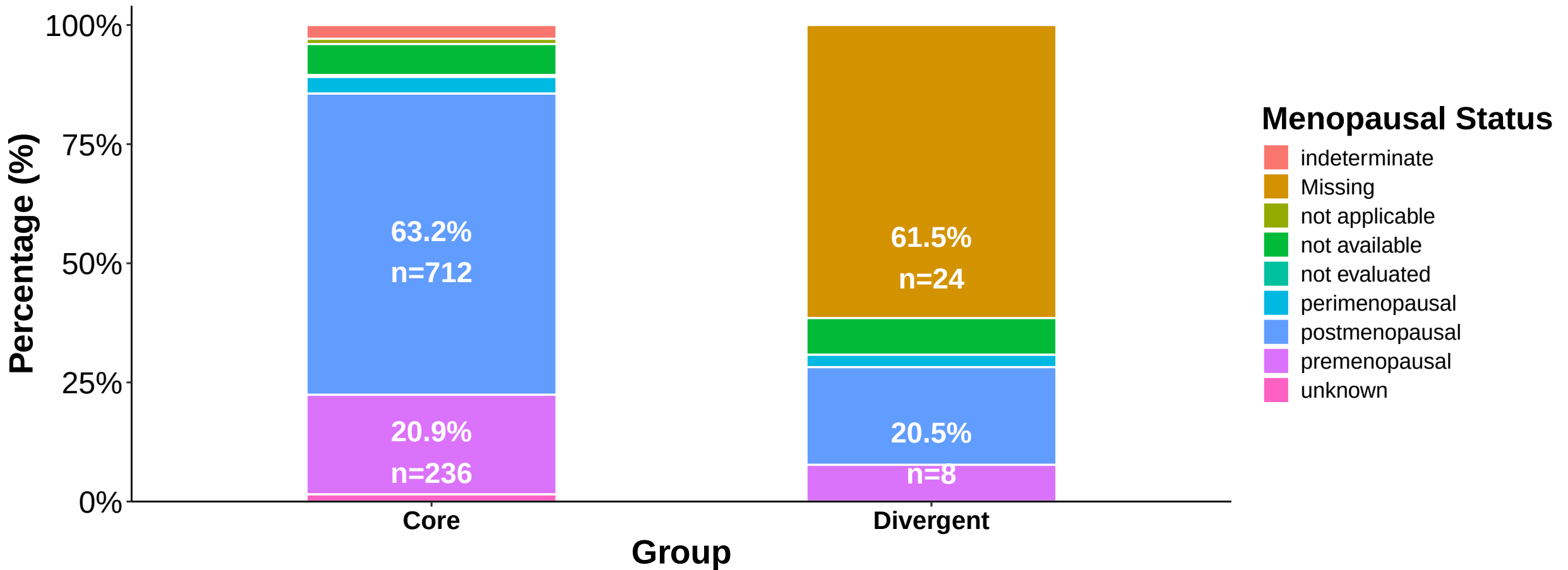
age_at_diagnosis.diagnoses

ns Δ=-0.14 (-8.3%)

Cliff's delta [Core vs Divergent]

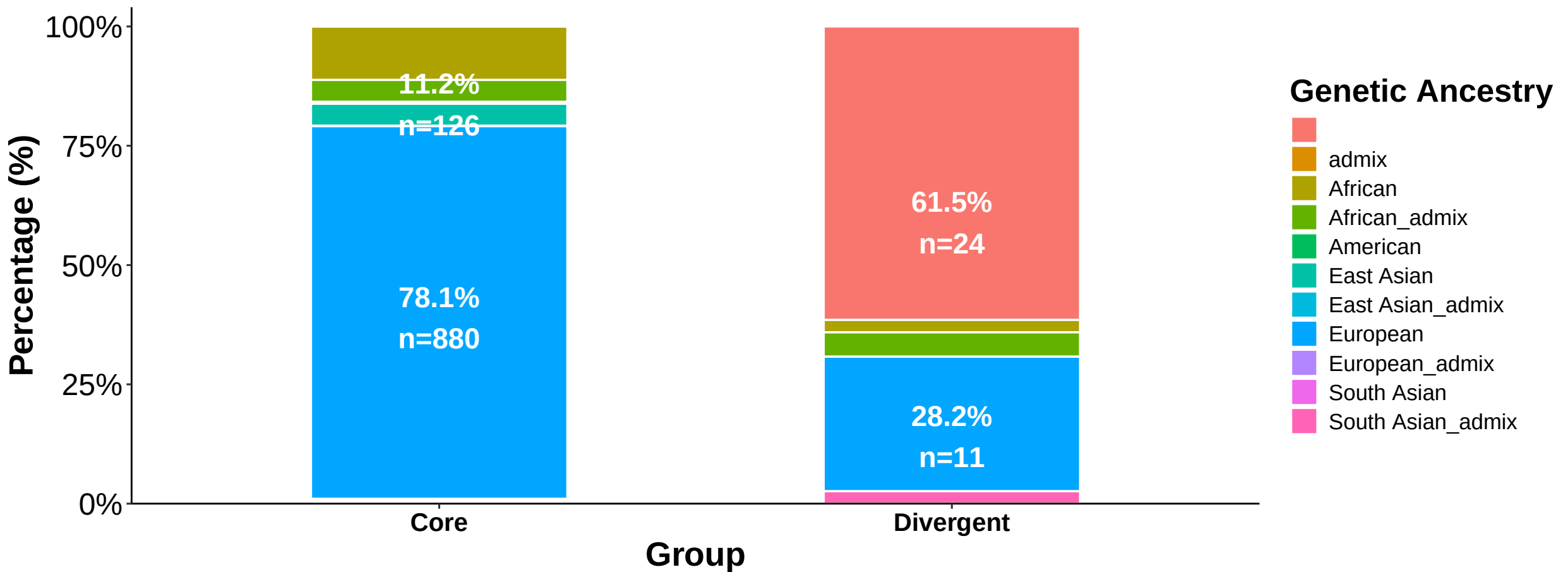
Categorical Clinical – . D. Patient Factors

Menopausal Status

Chi² p: 4.23e-148

Genetic Ancestry

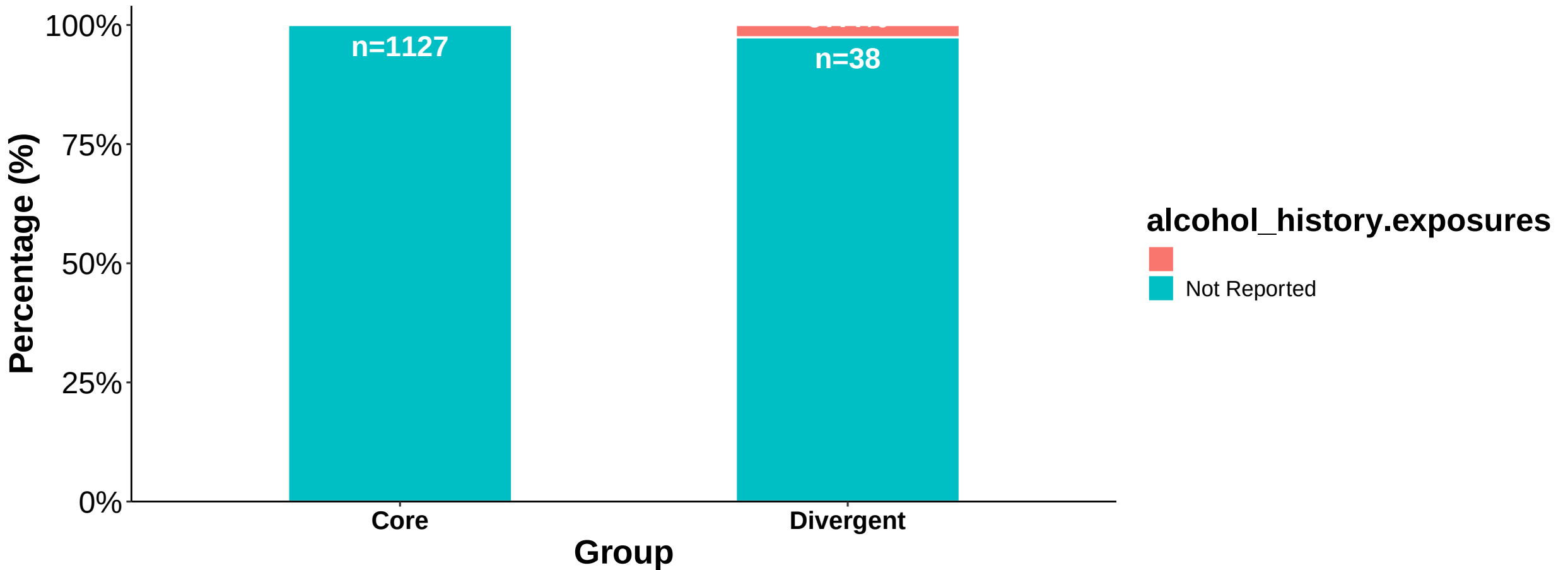
Chi² p: 1.81e-147



Categorical Clinical – . D. Patient Factors

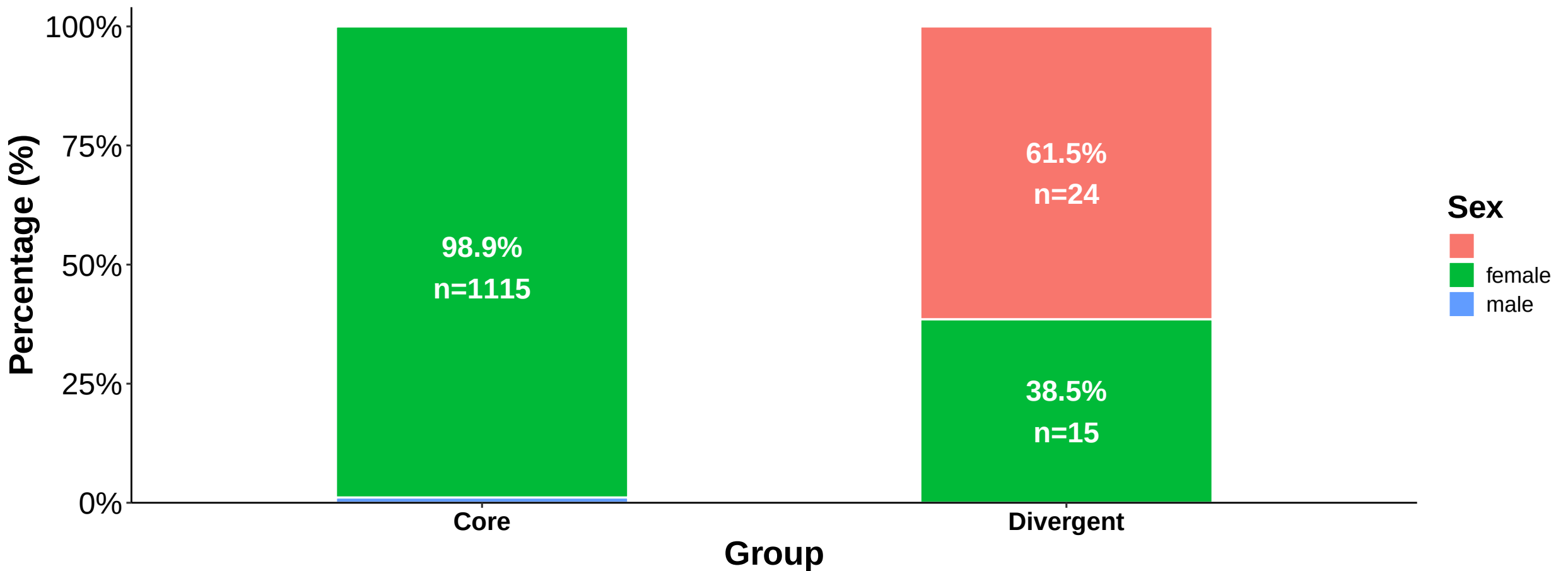
alcohol_history.exposures

Chi² p: 0.00943



Sex

Chi² p: 1.66e-154



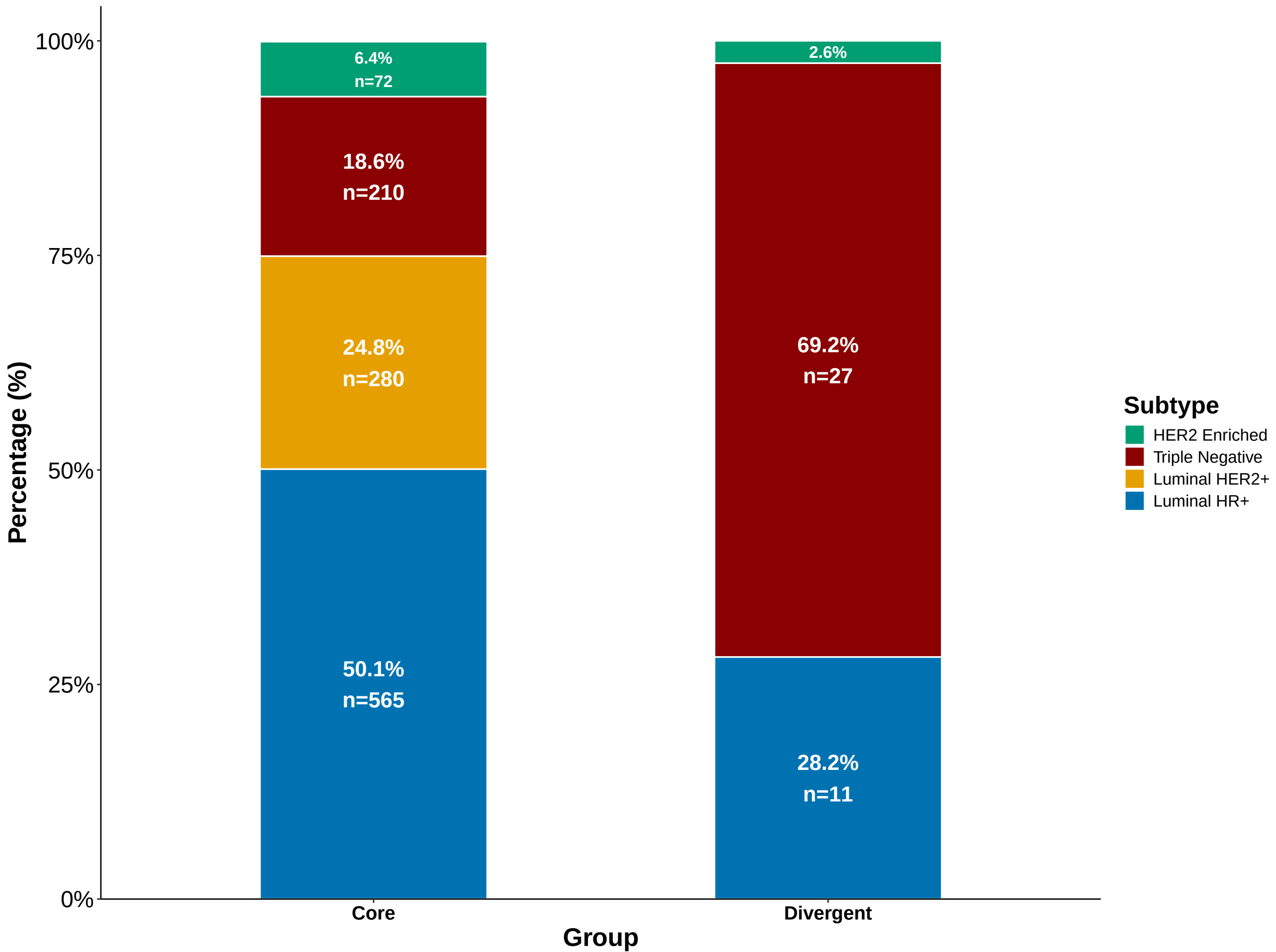
SECTION I–B

Molecular Subtype Analysis: Core vs Divergent

Source: ER_PR_HER2_Combo | Triple Negative · Luminal HR+ · Luminal HER2+ · HER2 Enriched · Missing

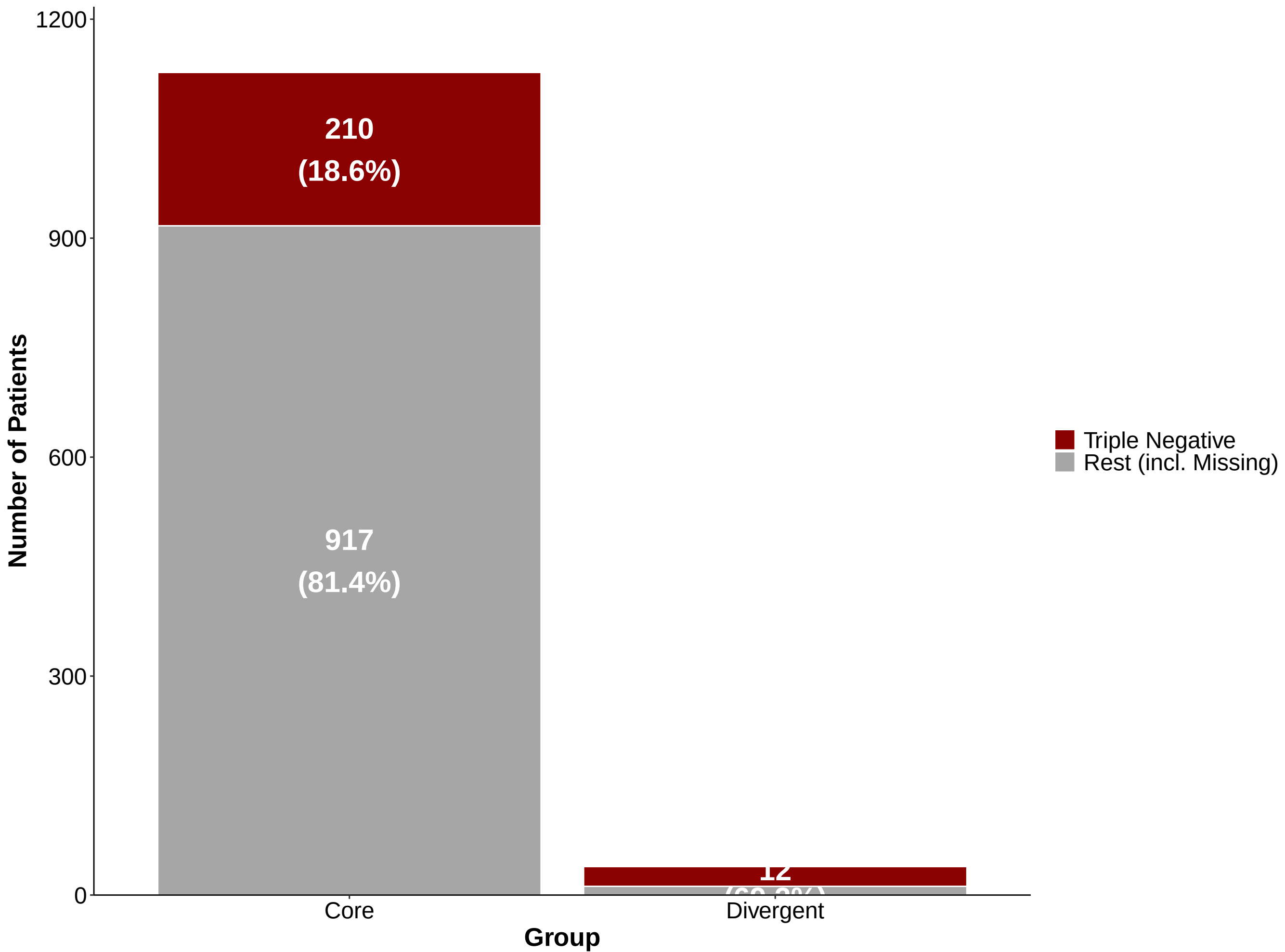
Molecular Subtypes: Proportion (%)

Chi² p = 2.52e-13



Triple Negative vs Rest – Core vs Divergent

Chi² p = 2.52e-13



Triple Negative Summary (source: ER_PR_HER2_Combo):

Core: 210 TNBC out of 1127 total (18.6%)

Divergent: 27 TNBC out of 39 total (69.2%)

Chi² test subtypes vs Group: Chi² p = 2.52e−13

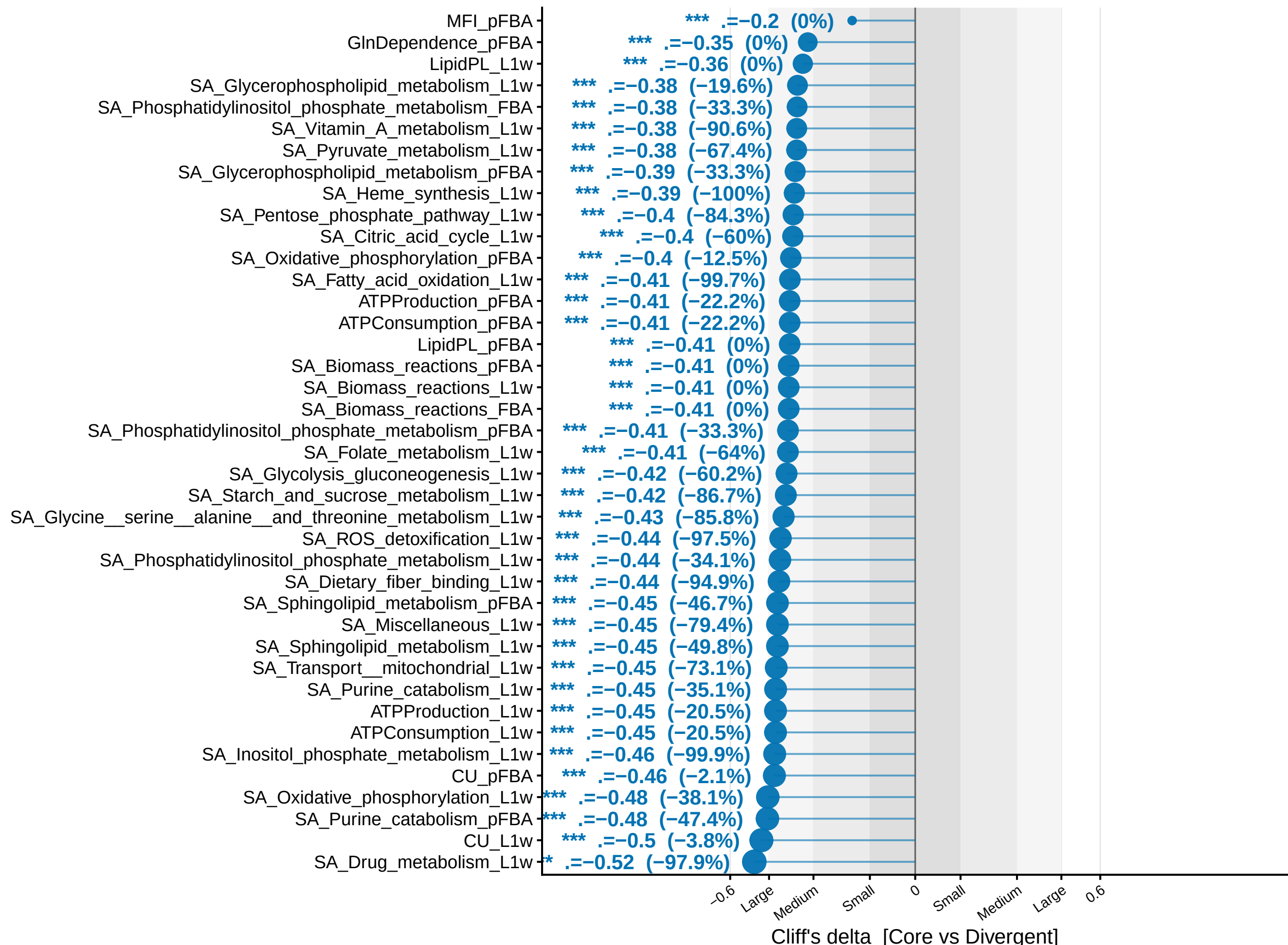
SECTION II

Metabolic Differences: Core vs Divergent

5 functional blocks: Energy | Redox | Lipids | Dependencies | Systemic Reprogramming

Top 40 Metabolic Variables – Cliff's .

Cliff's . + .% median | . higher in Core · . higher in Divergent



Cliff's .: . 1. Energy and Bioenergetics

Blue = Core > Divergent | Orange = Divergent > Core | * p_adj<0.05

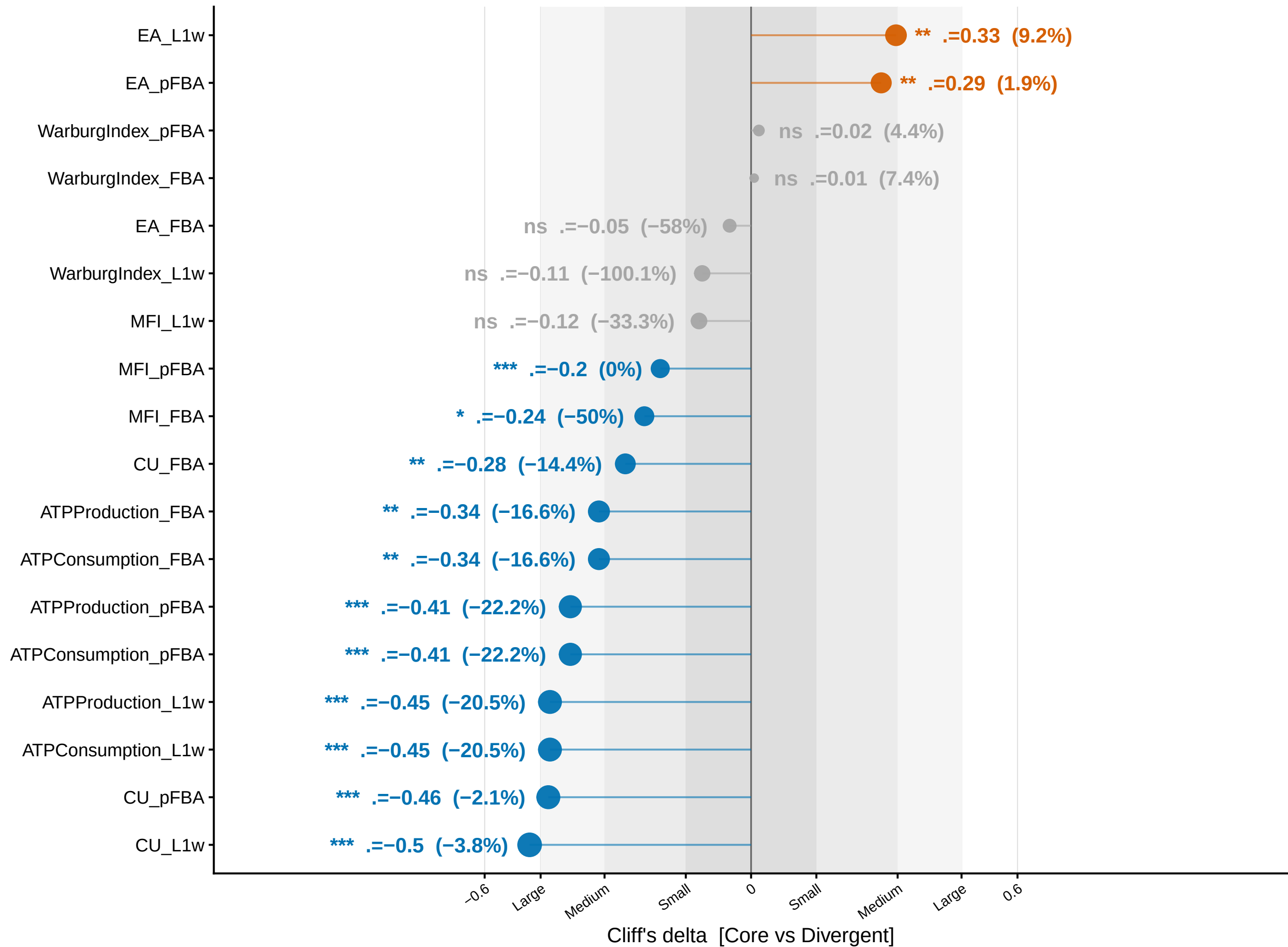
Cliff's .
Div vs Core



EA_L1w	** .=0.33
EA_pFBA	** .=0.29
WarburgIndex_pFBA	ns .=0.02
WarburgIndex_FBA	ns .=0.01
EA_FBA	ns .=-0.05
WarburgIndex_L1w	ns .=-0.11
MFI_L1w	ns .=-0.12
MFI_pFBA	*** .=-0.2
MFI_FBA	* .=-0.24
CU_FBA	** .=-0.28
ATPProduction_FBA	** .=-0.34
ATPConsumption_FBA	** .=-0.34
ATPProduction_pFBA	*** .=-0.41
ATPConsumption_pFBA	*** .=-0.41
ATPProduction_L1w	*** .=-0.45
ATPConsumption_L1w	*** .=-0.45
CU_pFBA	*** .=-0.46
CU_L1w	*** .=-0.5

Cliff's . – . 1. Energy and Bioenergetics

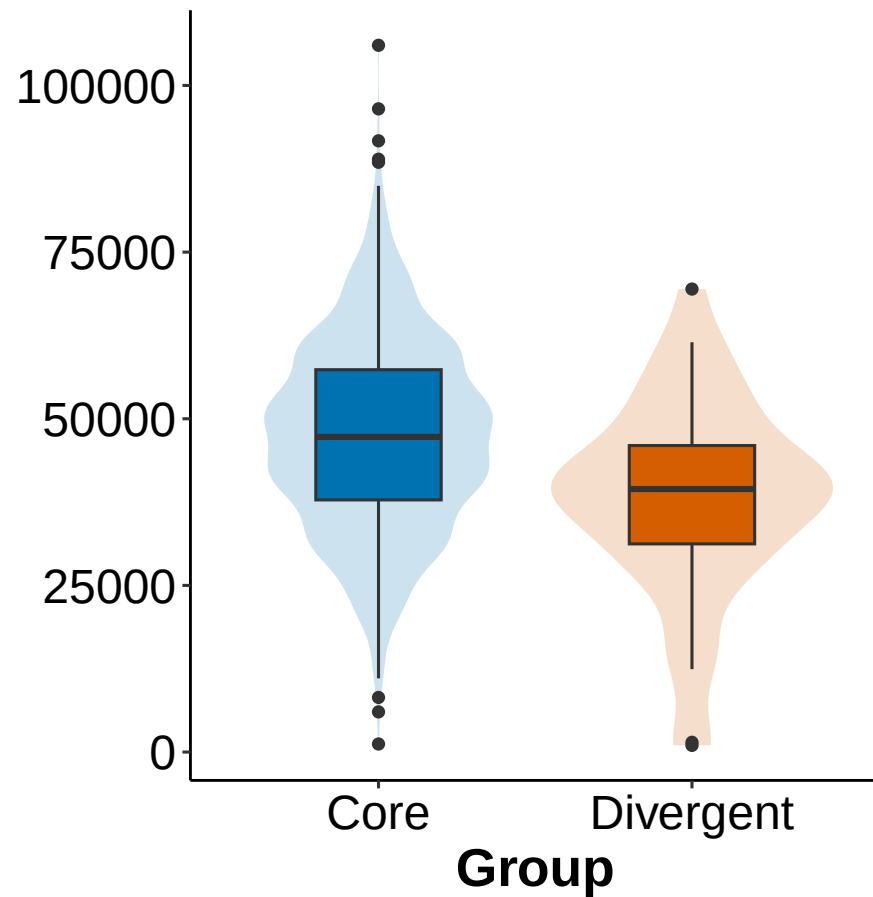
Cliff's . + .% median | . higher in Core · . higher in Divergent



Metabolism – . 1. Energy and Bioenergetics

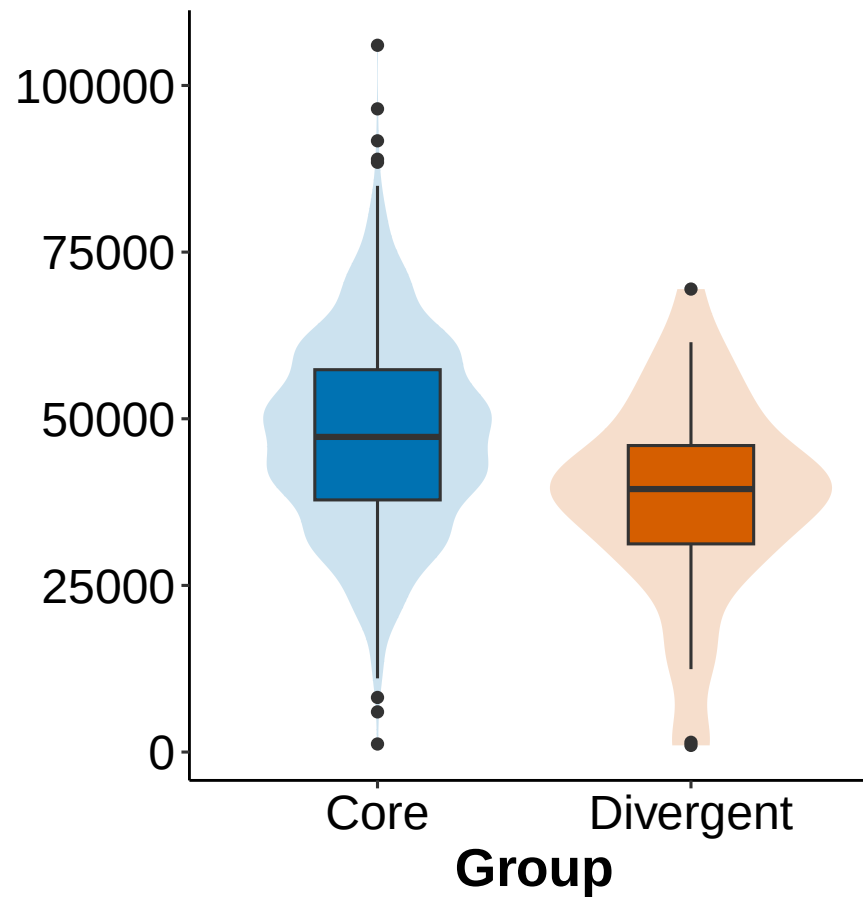
ATPConsumption_FBA

p-adj: 0.00119 | .=-0.34 (Medium) | .%: -16.4



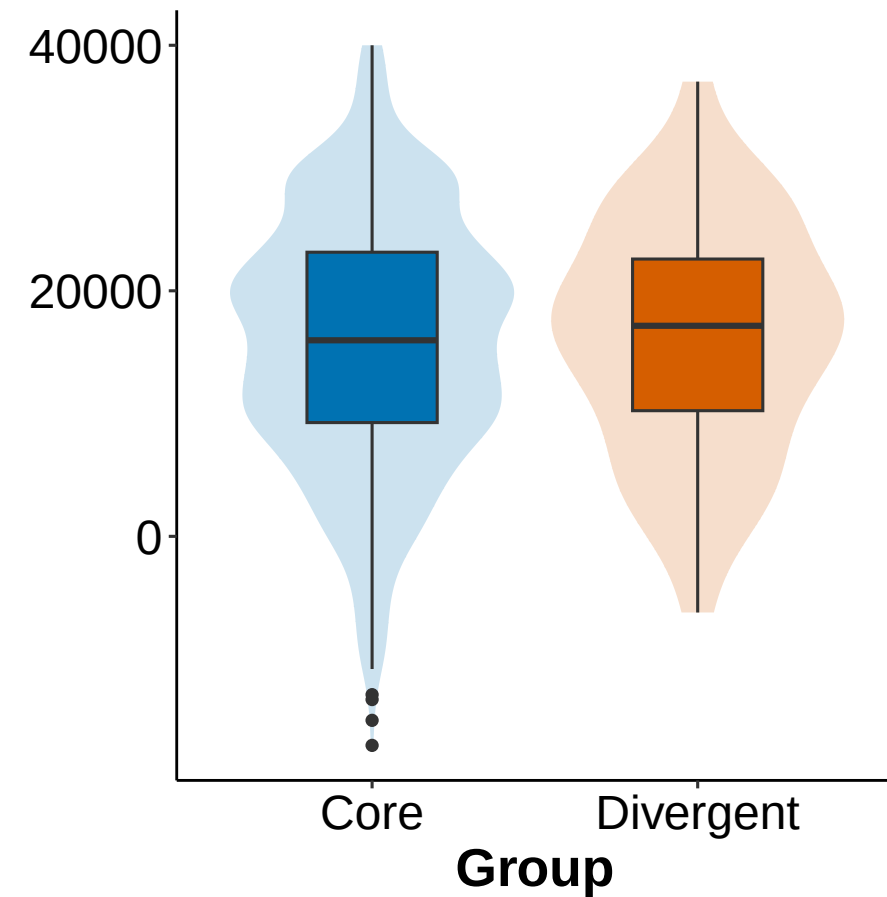
ATPProduction_FBA

p-adj: 0.00119 | .=-0.34 (Medium) | .%: -16.4



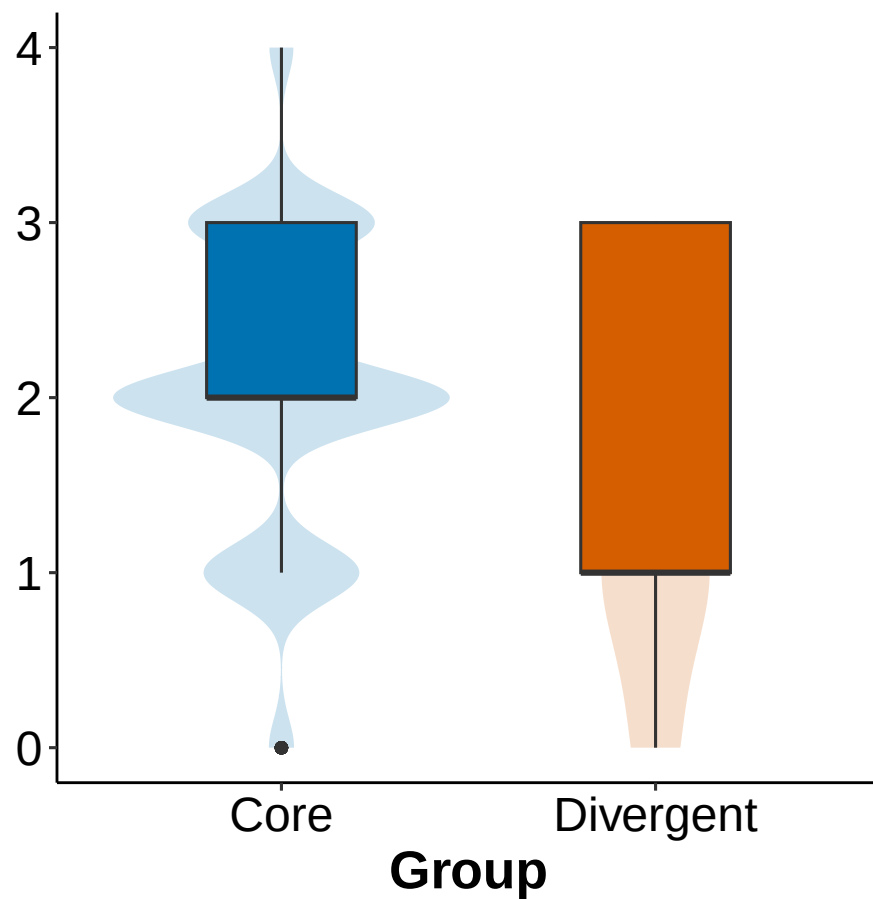
WarburgIndex_FBA

p-adj: 0.952 | .>=0.01 (Negligible) | .%: 7.4



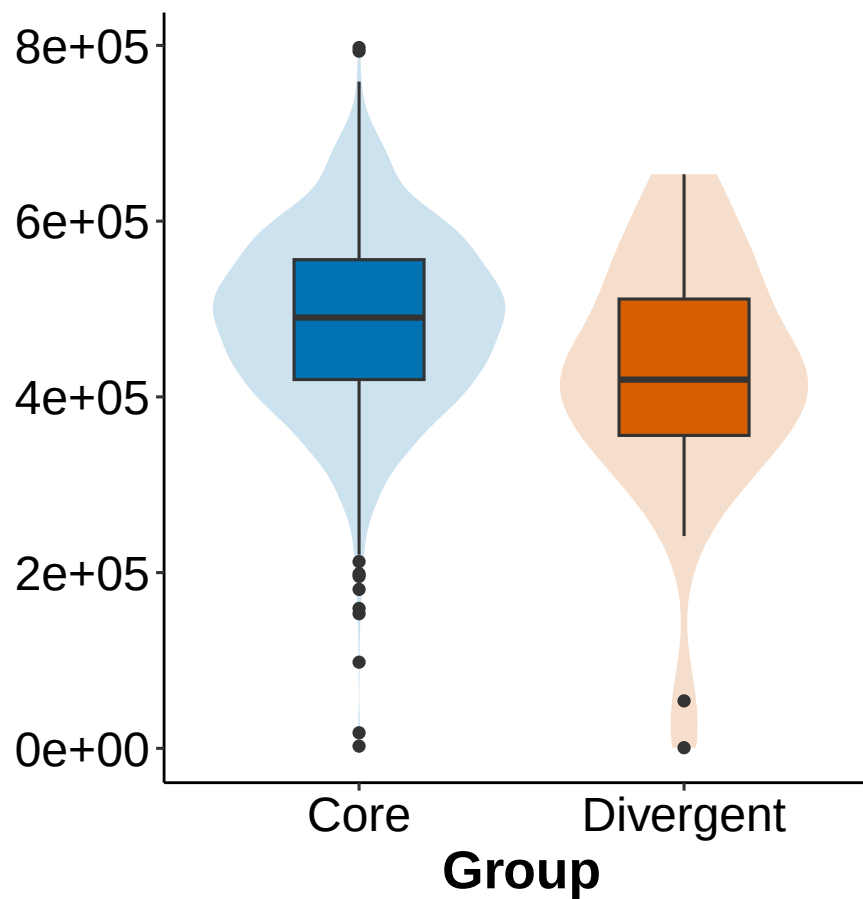
MFI_FBA

p-adj: 0.0192 | .=-0.24 (Small) | .%: -50



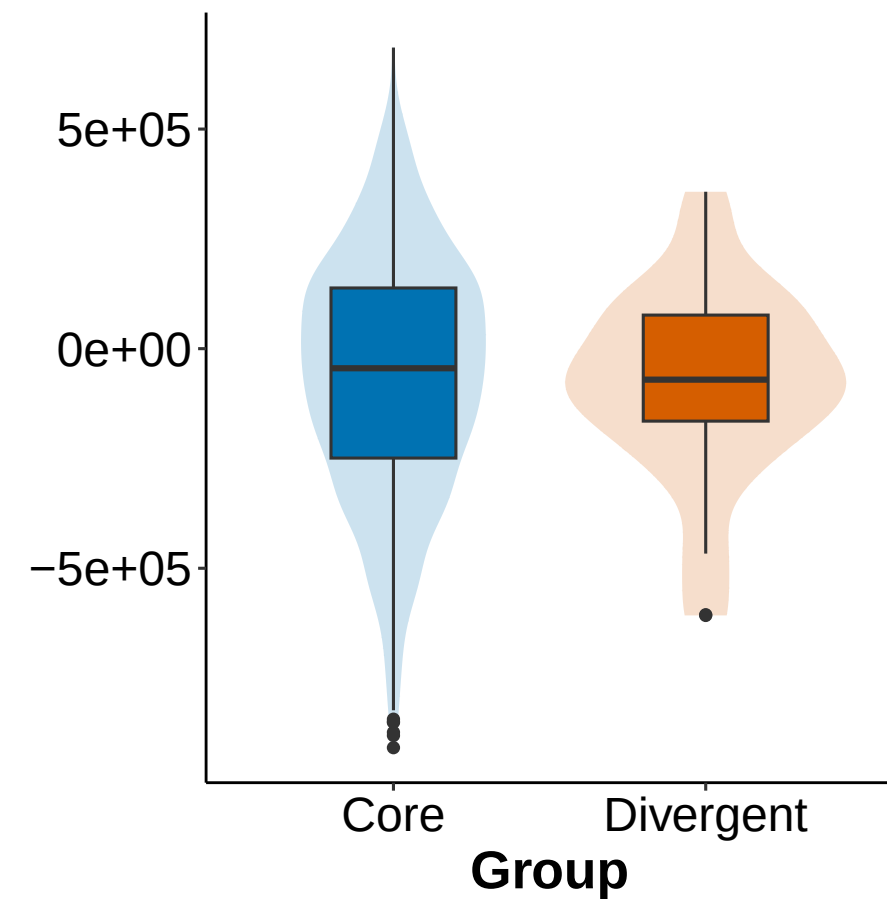
CU_FBA

p-adj: 0.00851 | .=-0.28 (Small) | .%: -14.4



EA_FBA

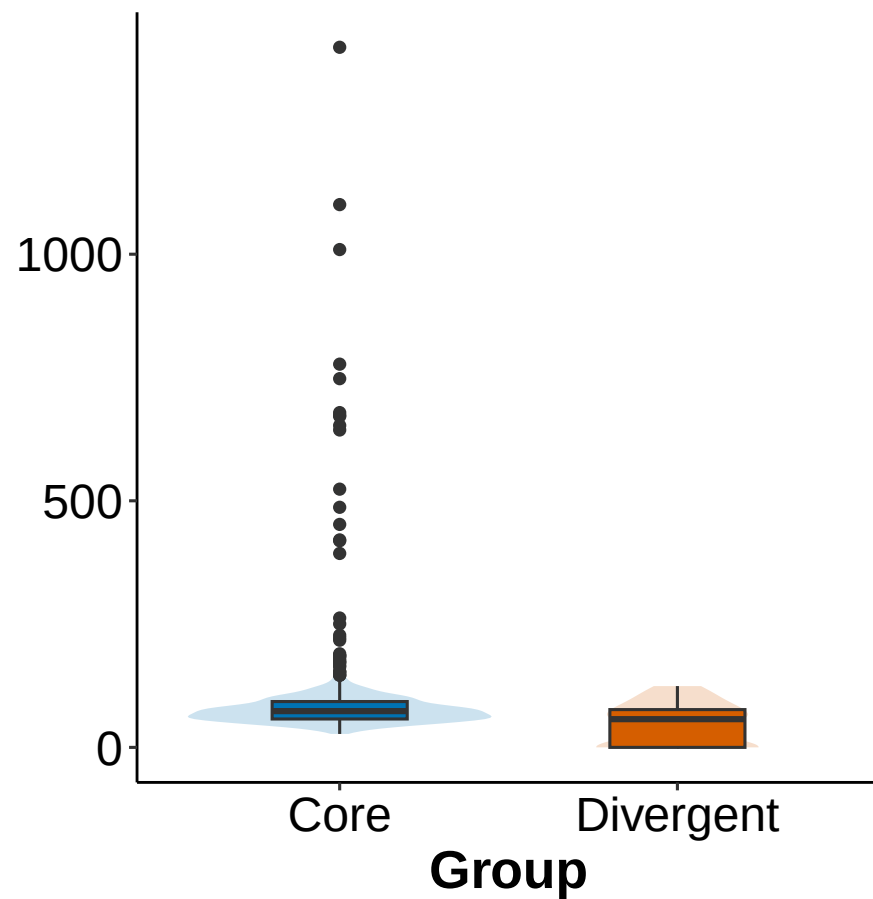
p-adj: 0.733 | .=-0.05 (Negligible) | .%: -58



Metabolism – . 1. Energy and Bioenergetics

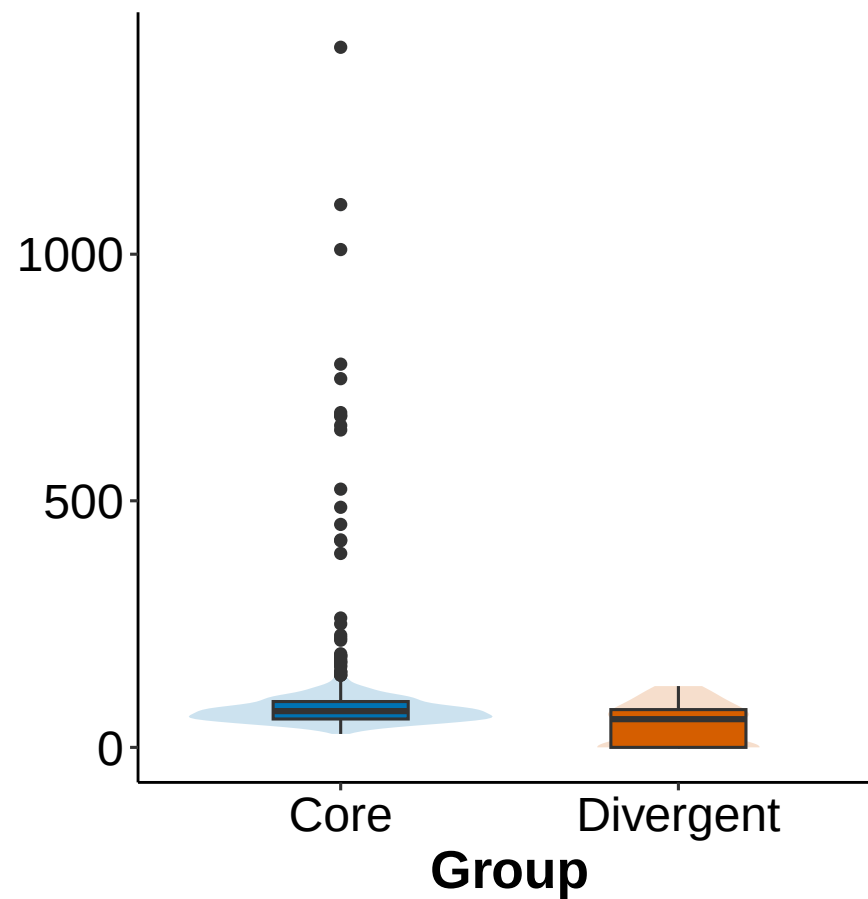
ATPConsumption_pFBA

p-adj: 0.000134 | .=-0.41 (Medium) | .%: -22.2



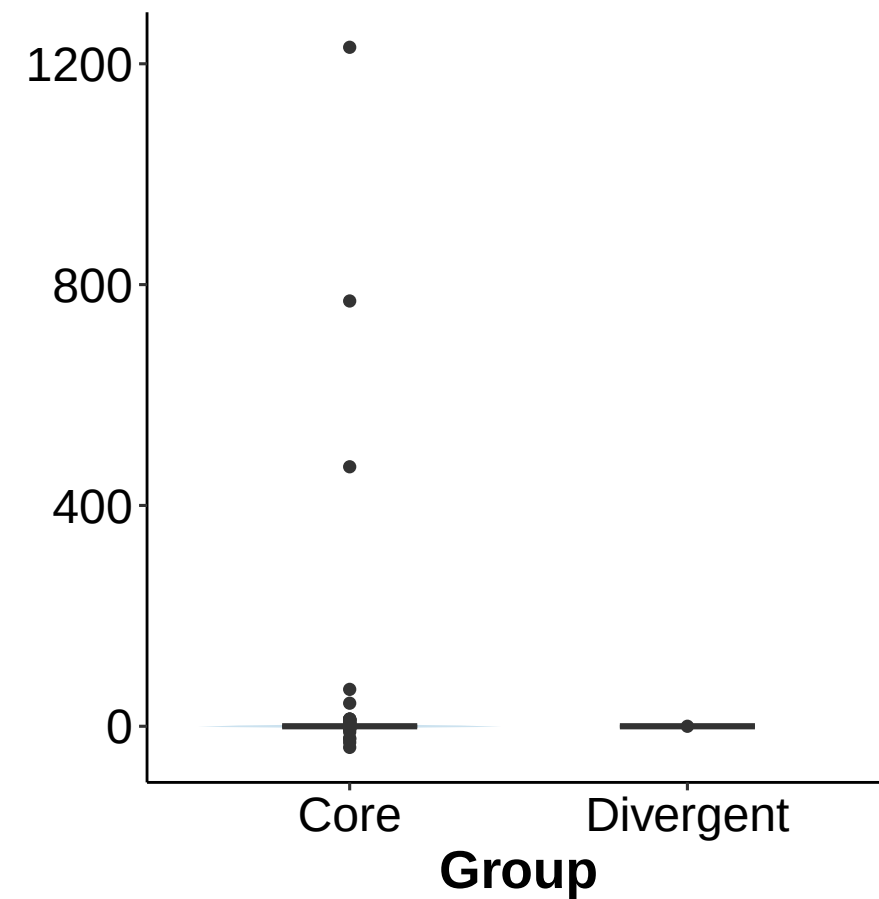
ATPProduction_pFBA

p-adj: 0.000134 | .=-0.41 (Medium) | .%: -22.2



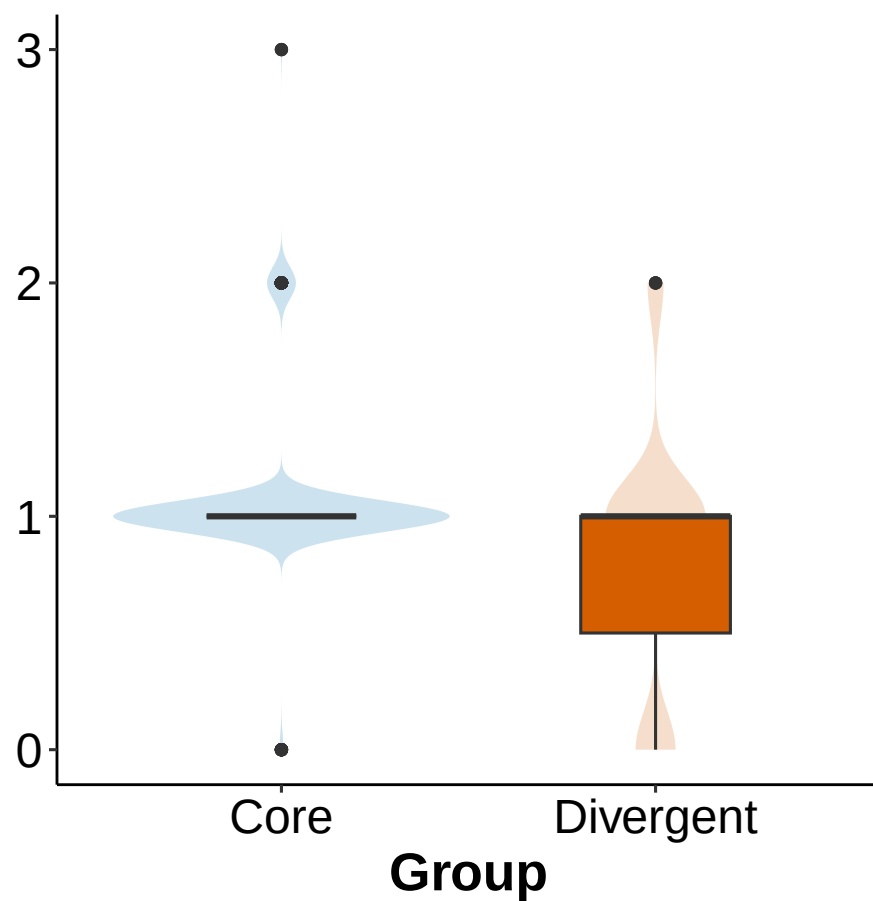
WarburgIndex_pFBA

p-adj: 0.902 | .=0.02 (Negligible) | .%: 4.4



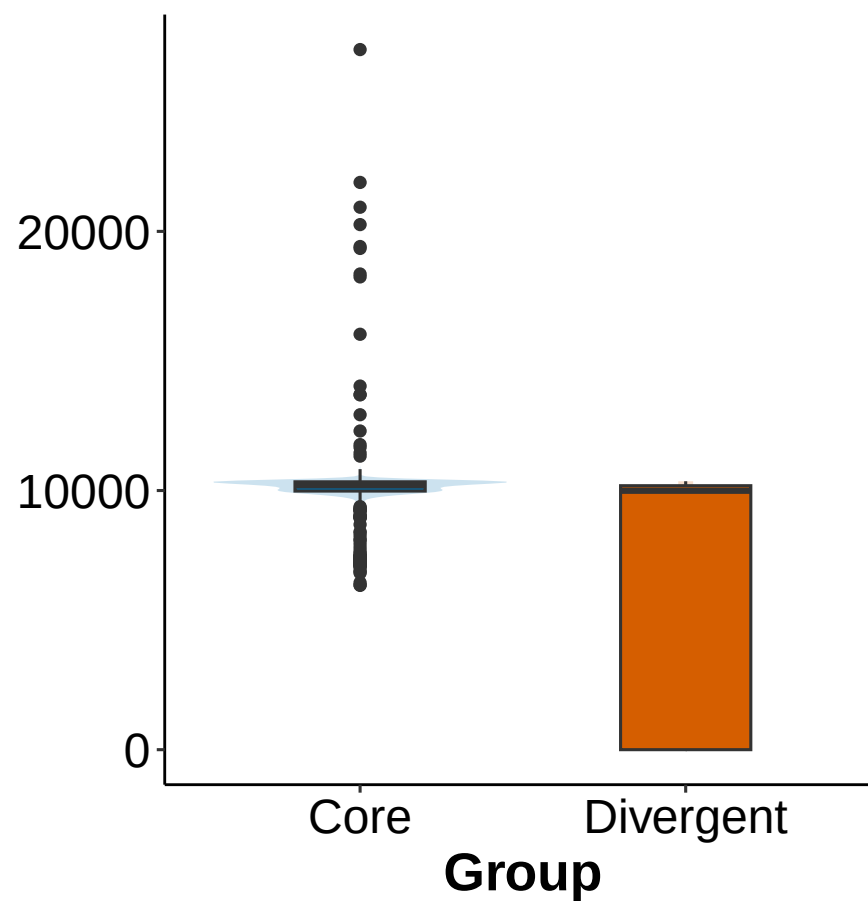
MFI_pFBA

p-adj: 0.000219 | .=-0.2 (Small) | .%: 0



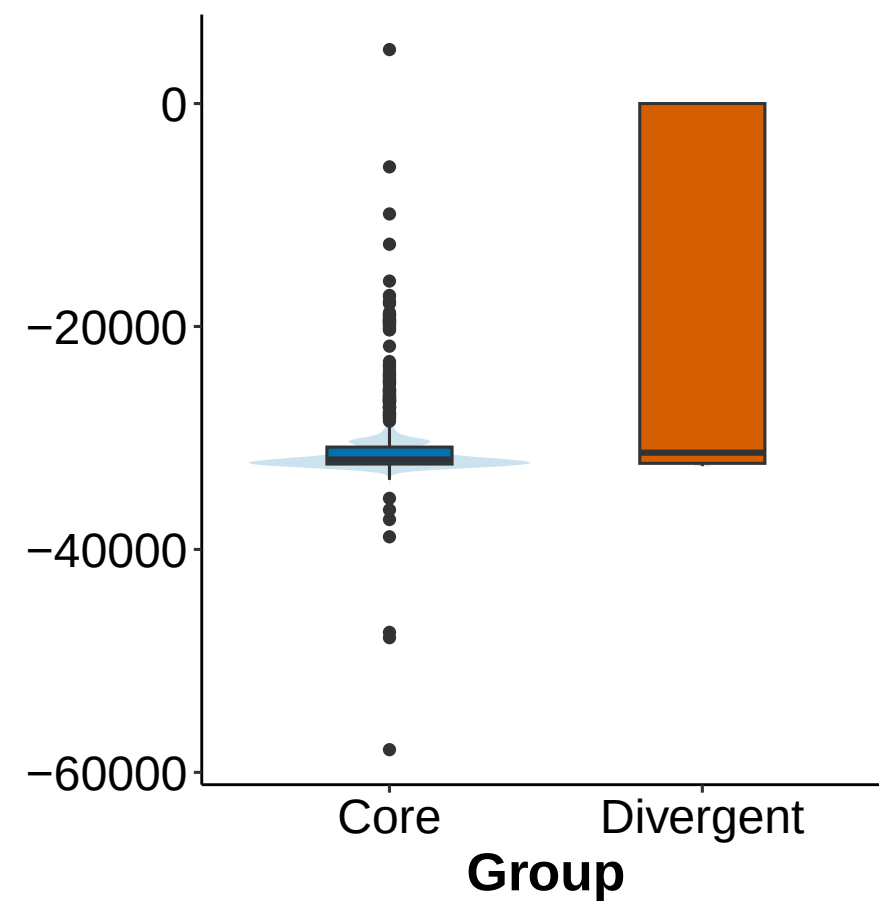
CU_pFBA

p-adj: 2.66e-05 | .=-0.46 (Medium) | .%: -2.1



EA_pFBA

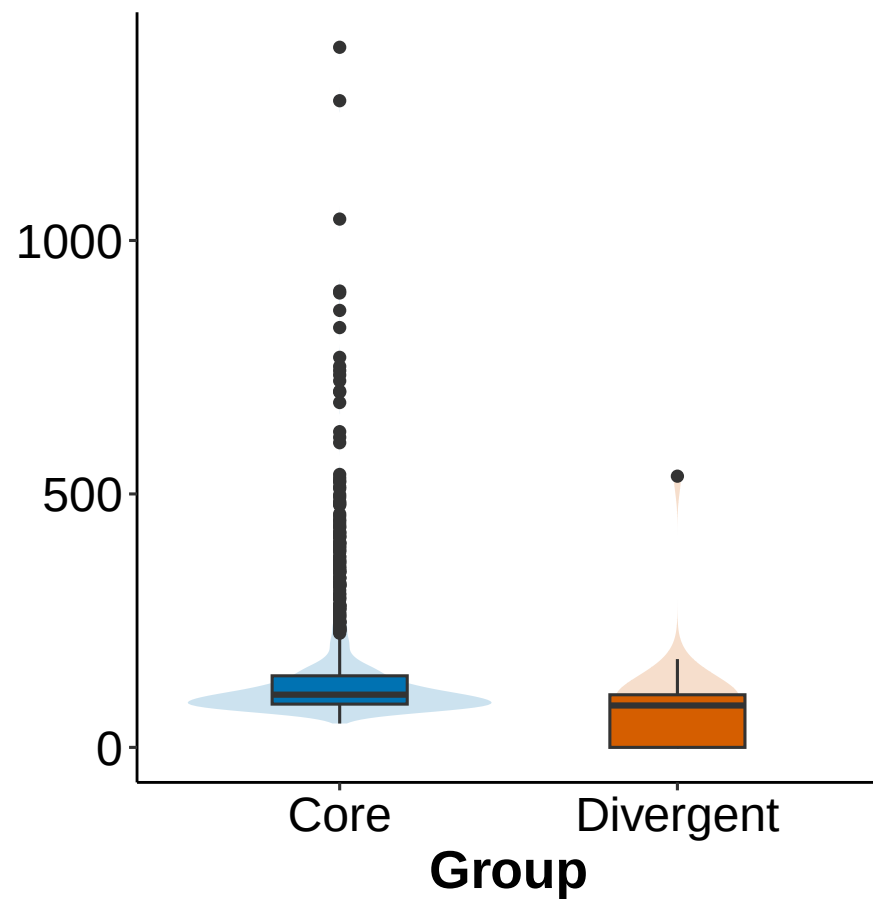
p-adj: 0.00621 | .=0.29 (Small) | .%: 1.9



Metabolism – . 1. Energy and Bioenergetics

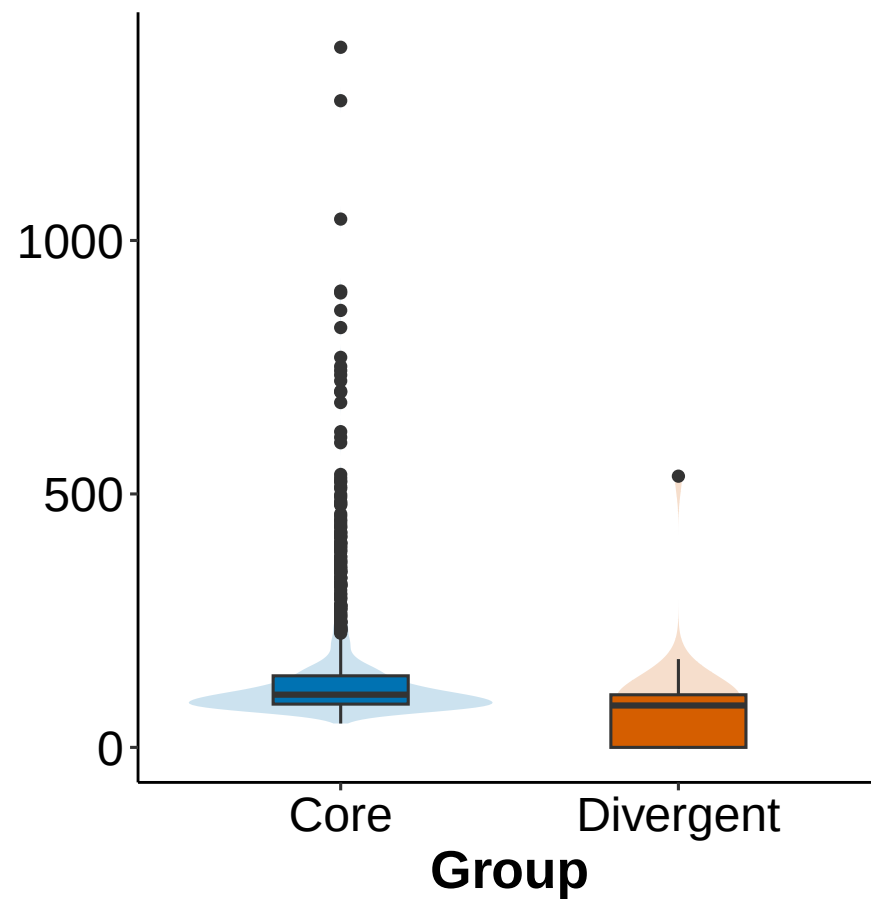
ATPConsumption_L1w

p-adj: 2.66e-05 | .=-0.45 (Medium) | .%: -20.5



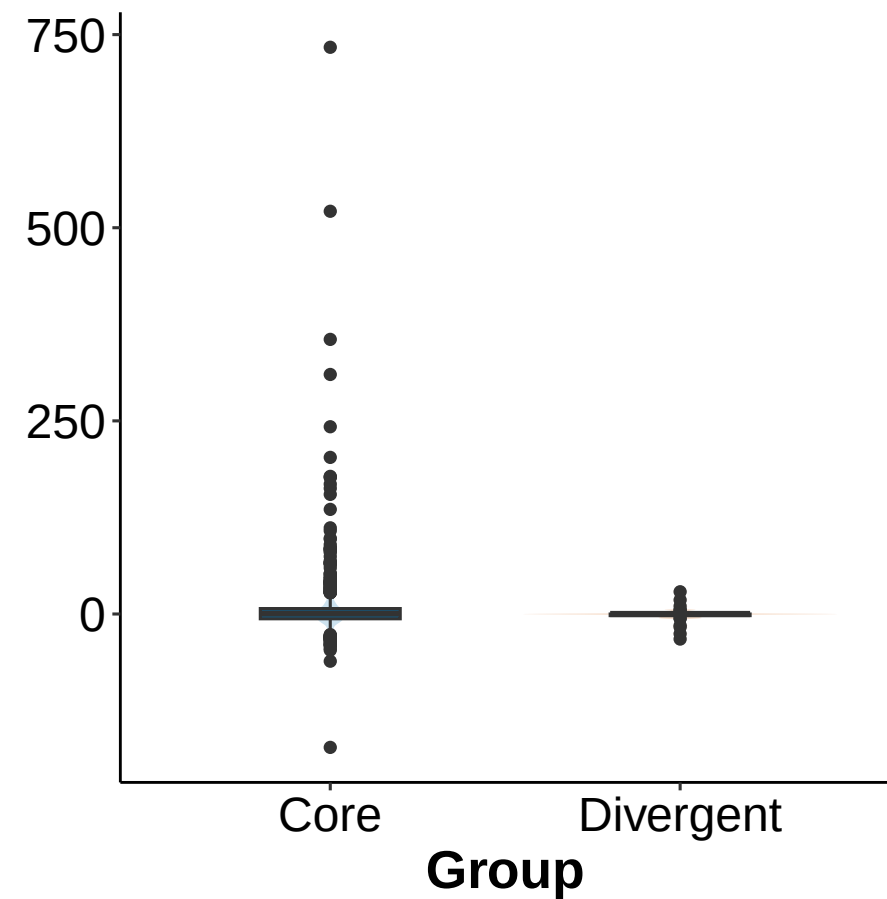
ATPProduction_L1w

p-adj: 2.66e-05 | .=-0.45 (Medium) | .%: -20.5



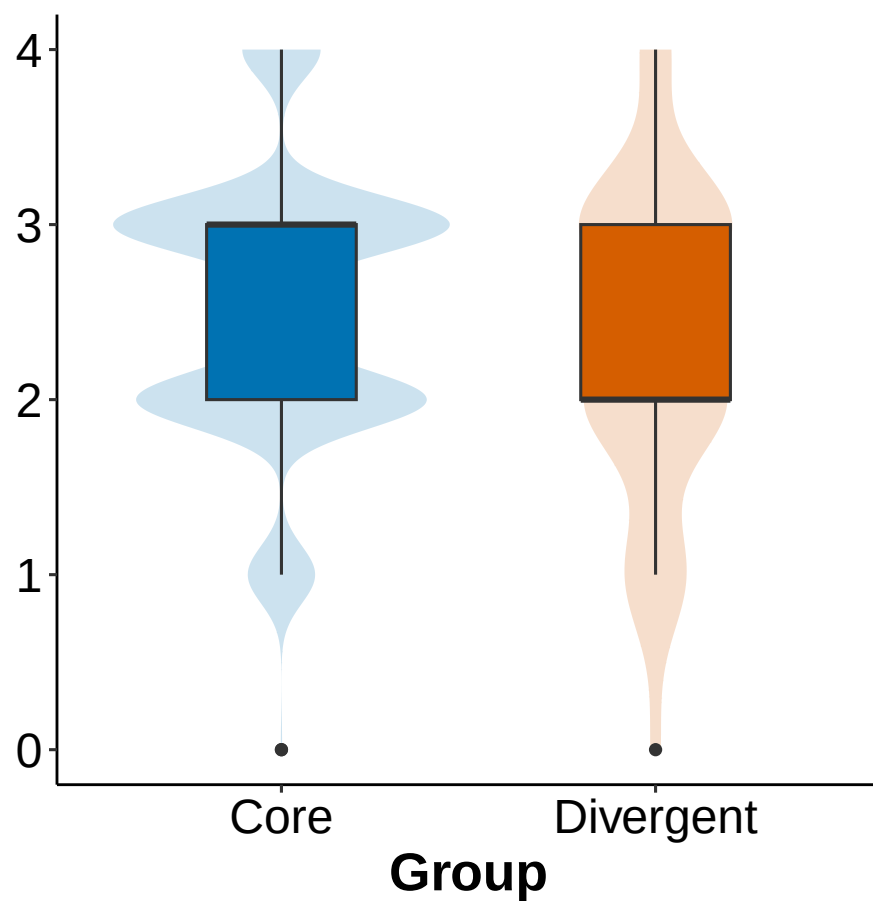
WarburgIndex_L1w

p-adj: 0.411 | .=-0.11 (Negligible) | .%: -100.1



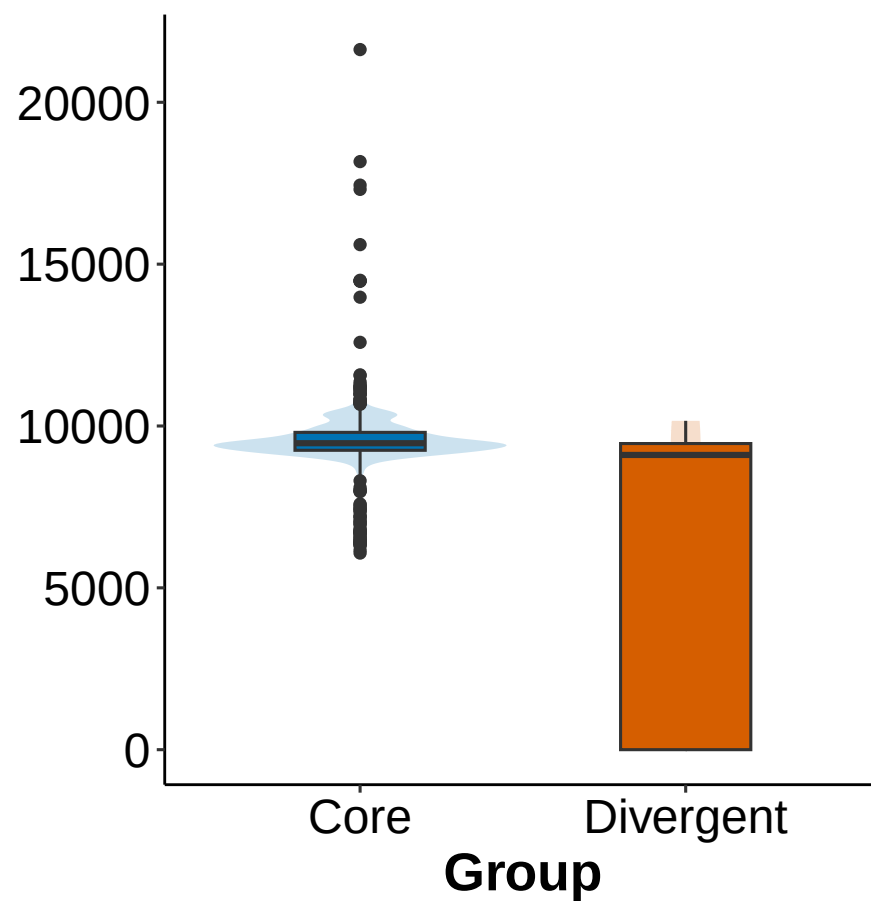
MFI_L1w

p-adj: 0.333 | .=-0.12 (Negligible) | .%: -33.3



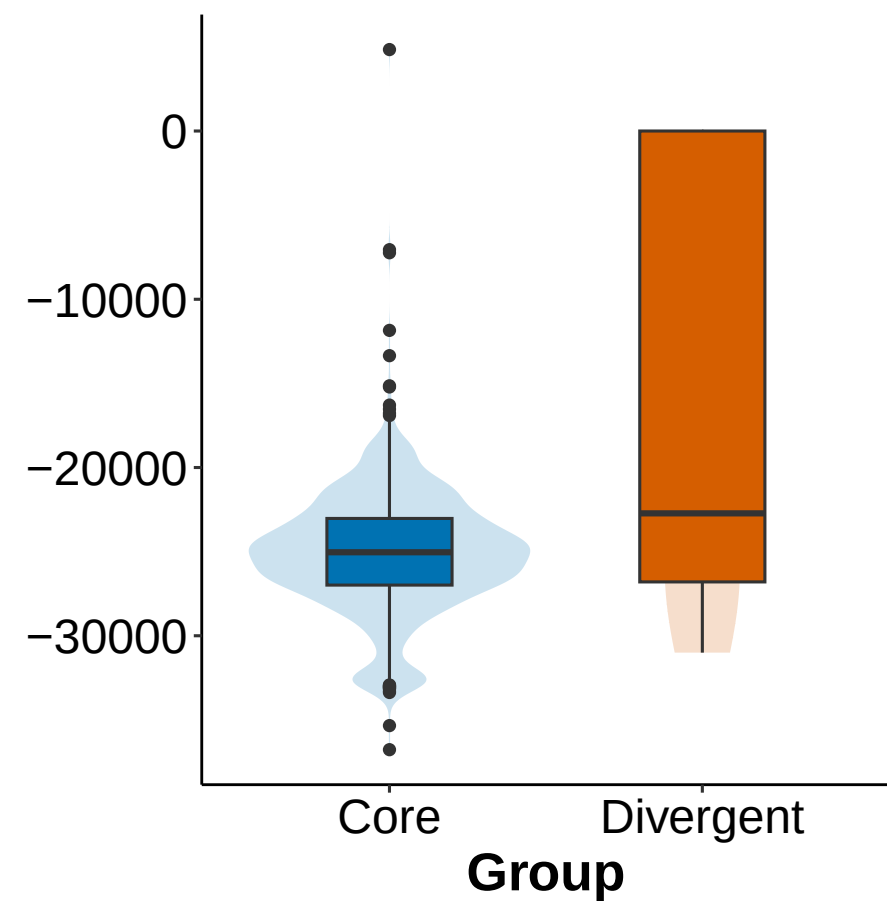
CU_L1w

p-adj: 4.07e-06 | .=-0.5 (Large) | .%: -3.8



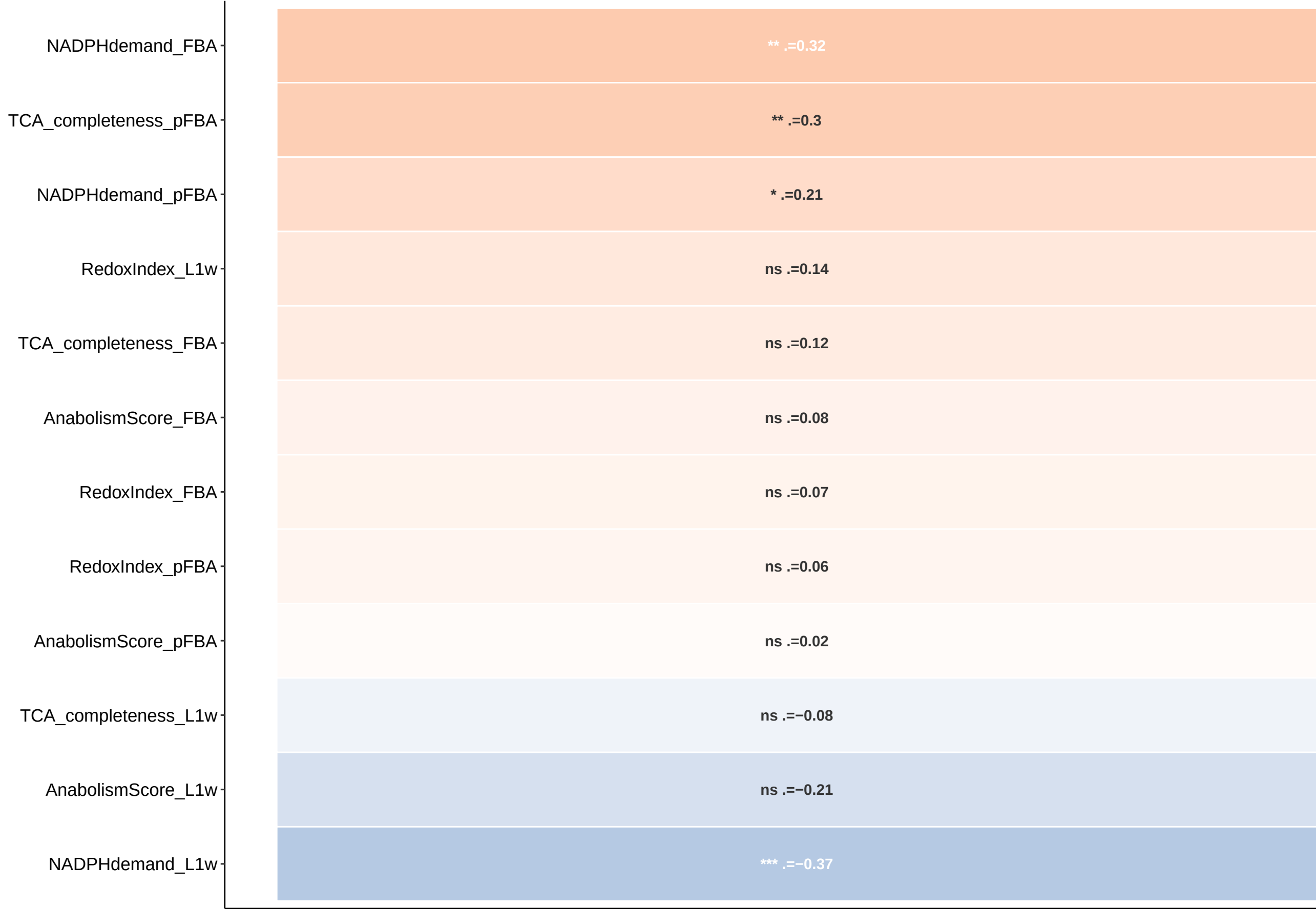
EA_L1w

p-adj: 0.00213 | .=0.33 (Small) | .%: 9.2



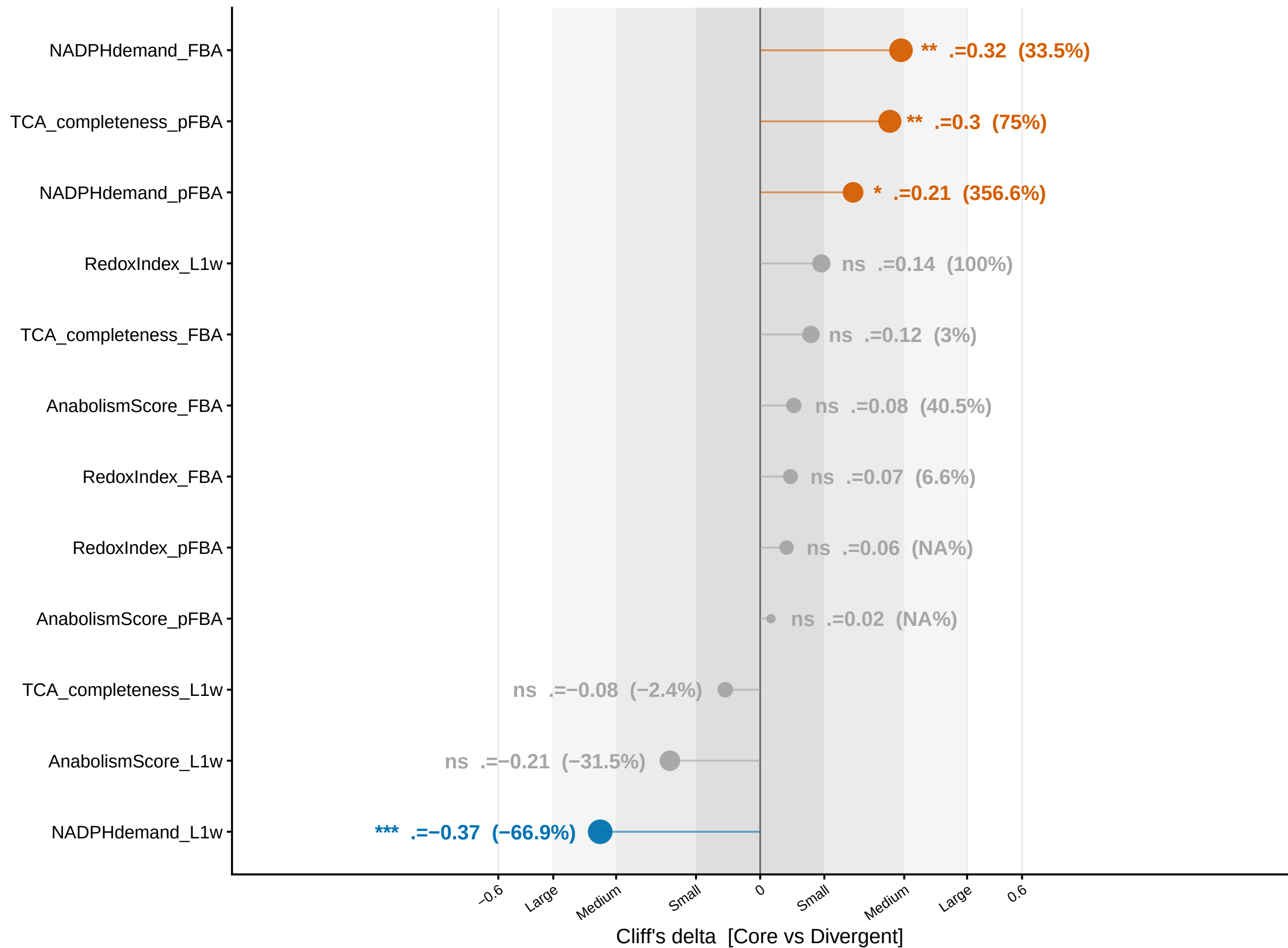
Cliff's .: . 2. Redox State and Anabolism

Blue = Core > Divergent | Orange = Divergent > Core | * p_adj<0.05



Cliff's . – . 2. Redox State and Anabolism

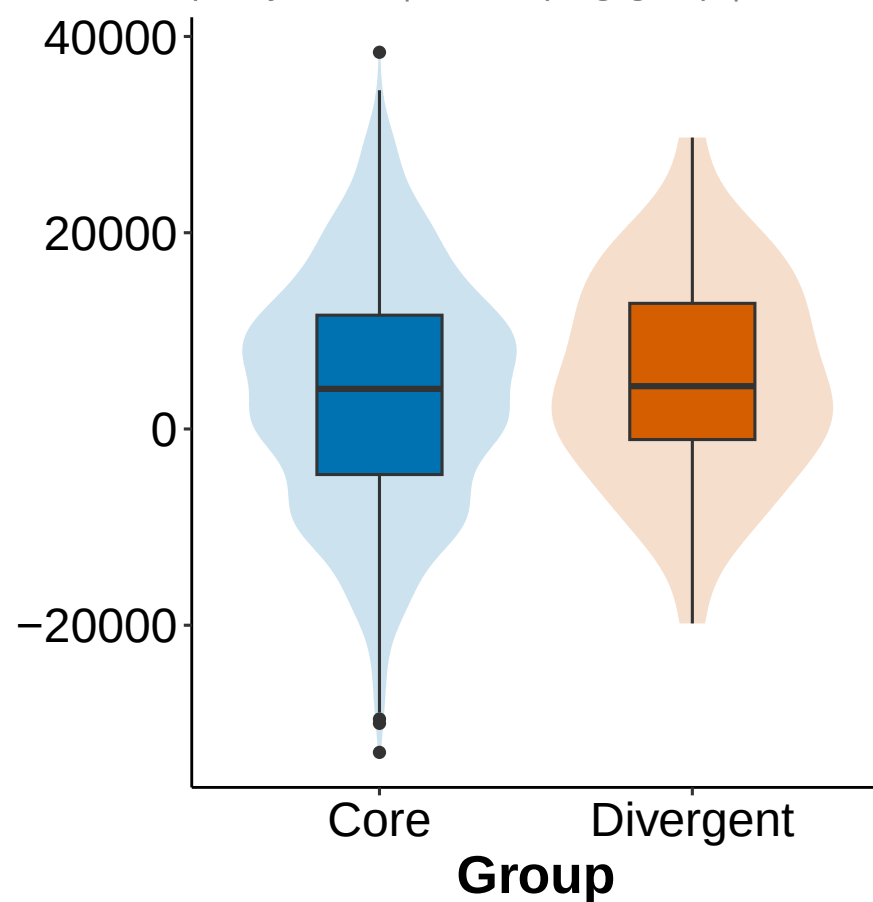
Cliff's . + .% median | . higher in Core · . higher in Divergent



Metabolism – . 2. RedoxState and Anabolism

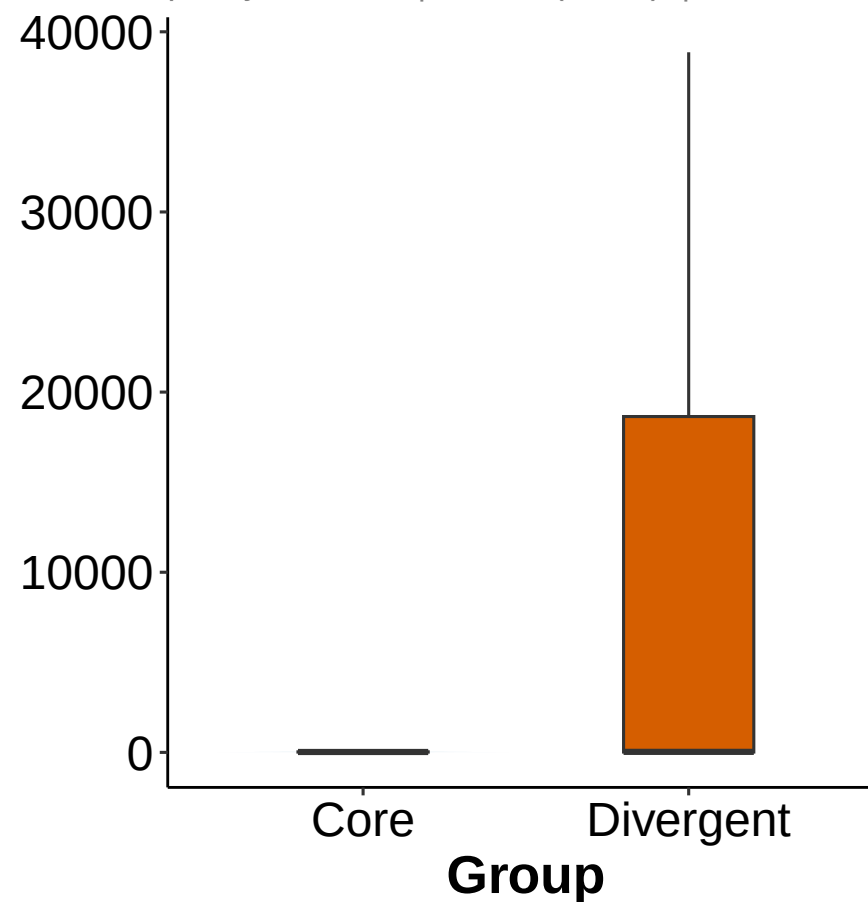
RedoxIndex_FBA

p-adj: 0.624 | .:=0.07 (Negligible) | .%: 6.6



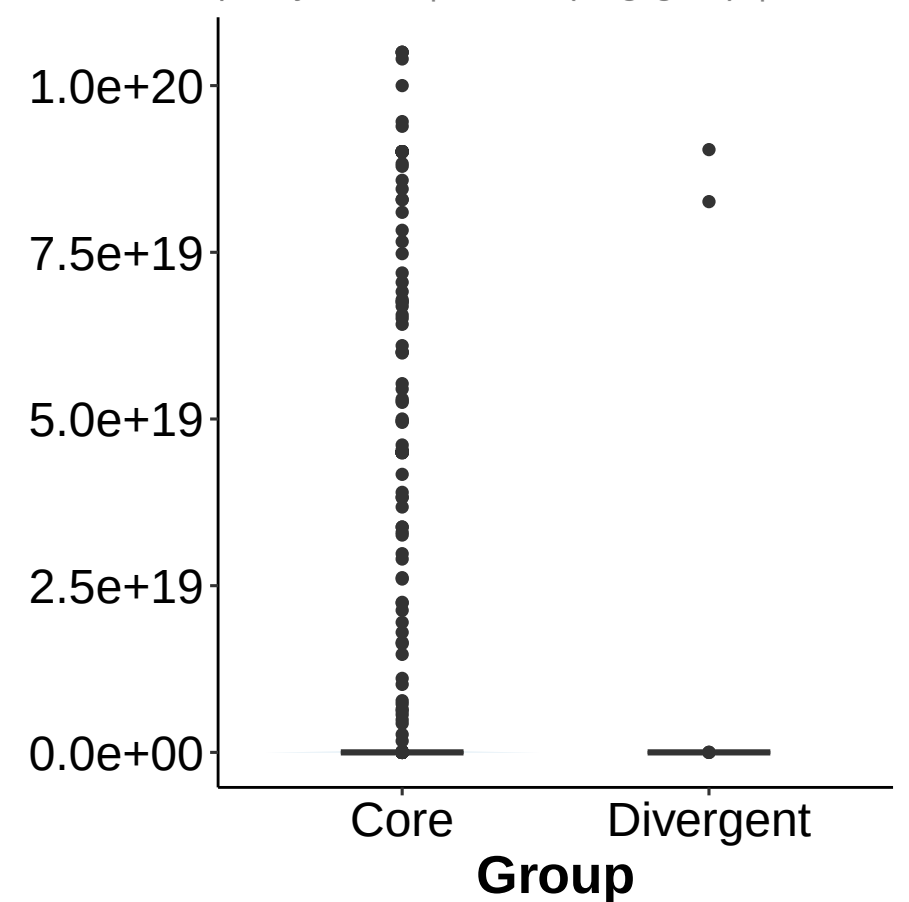
NADPHdemand_FBA

p-adj: 0.00241 | .:=0.32 (Small) | .%: 33.5



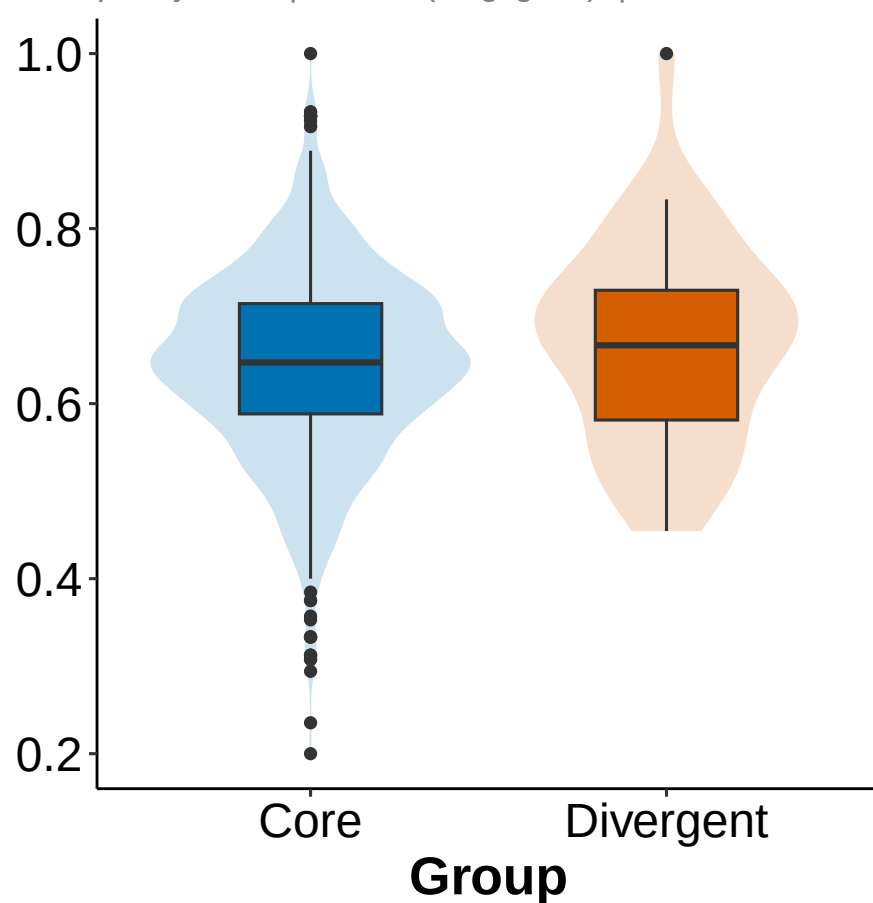
AnabolismScore_FBA

p-adj: 0.668 | .:=0.08 (Negligible) | .%: 40.5



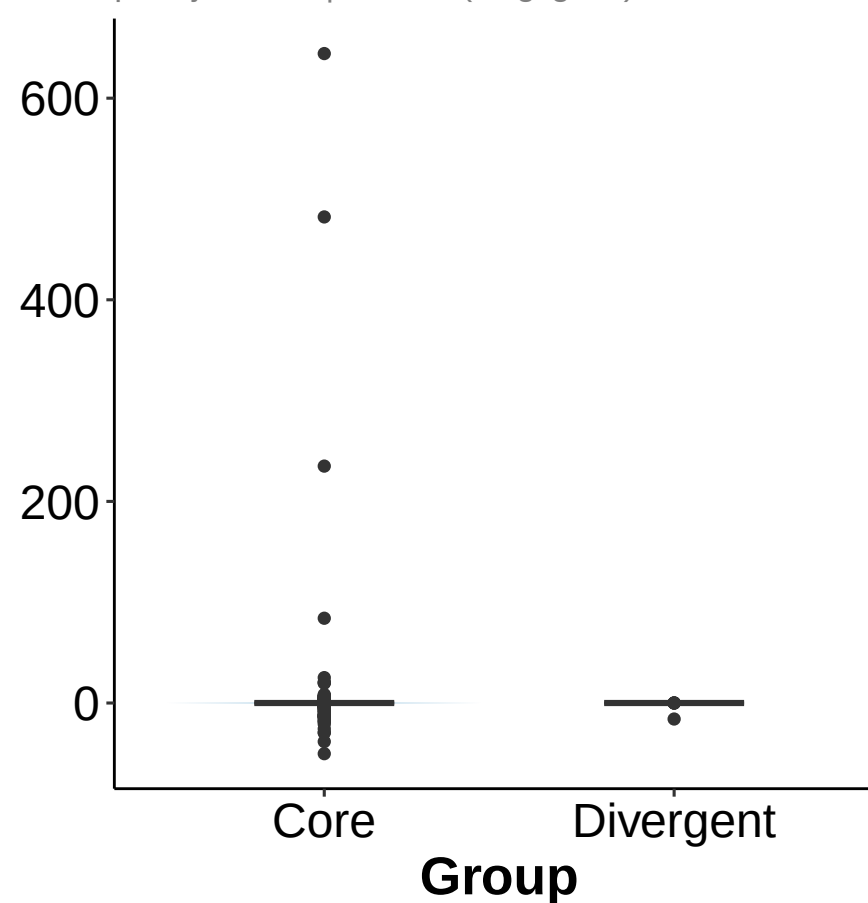
TCA_completeness_FBA

p-adj: 0.38 | .:=0.12 (Negligible) | .%: 3



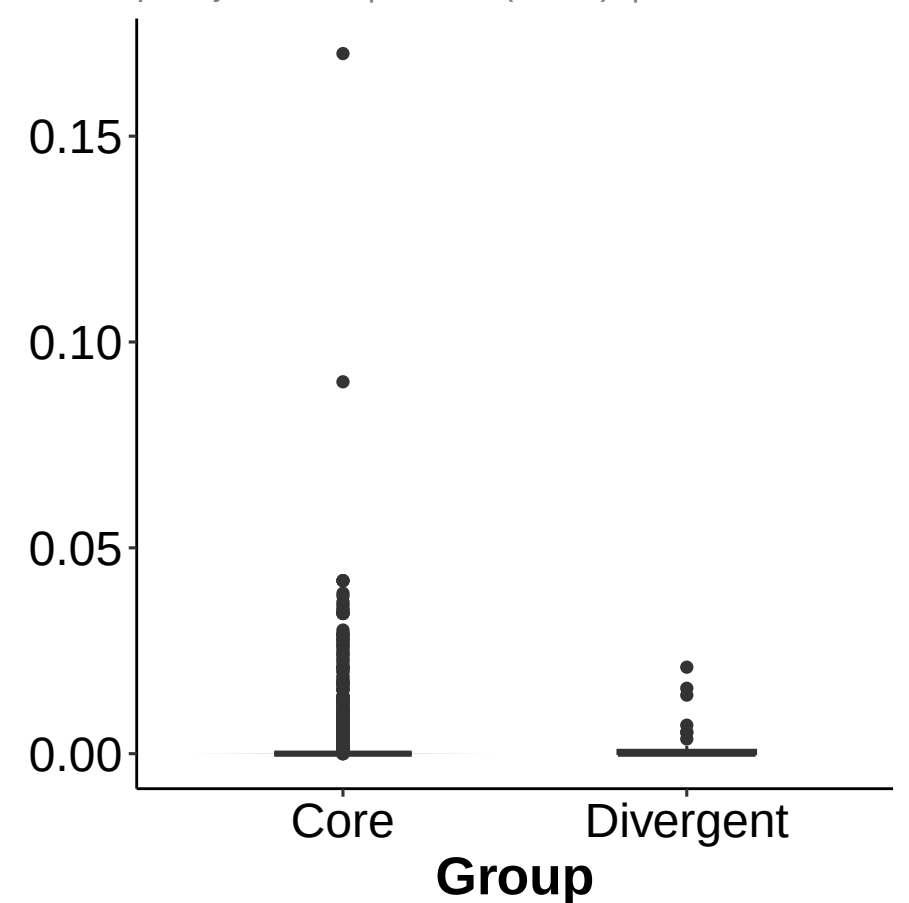
RedoxIndex_pFBA

p-adj: 0.671 | .:=0.06 (Negligible)

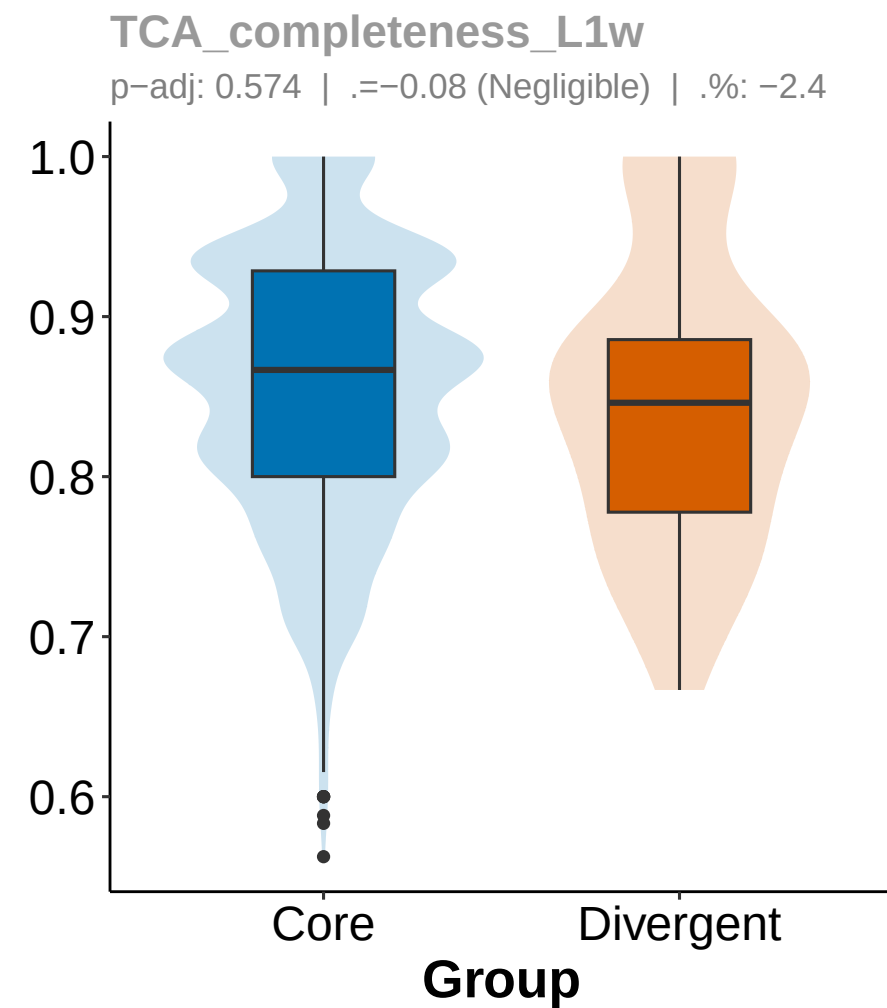
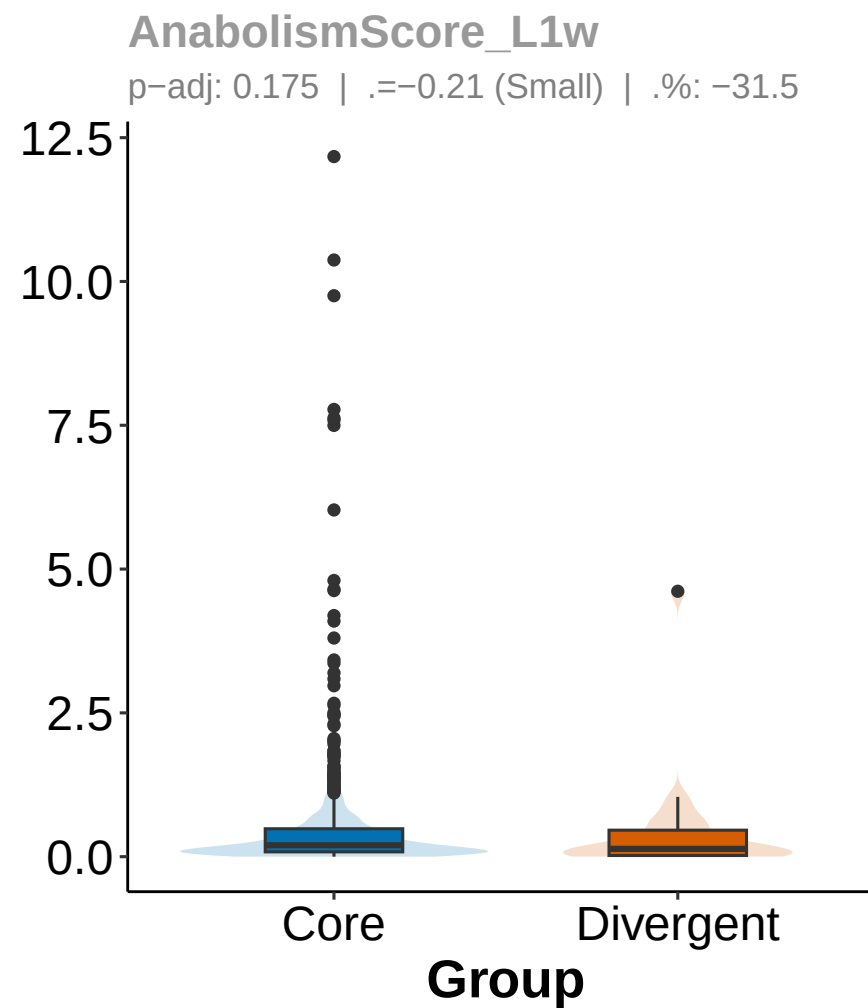
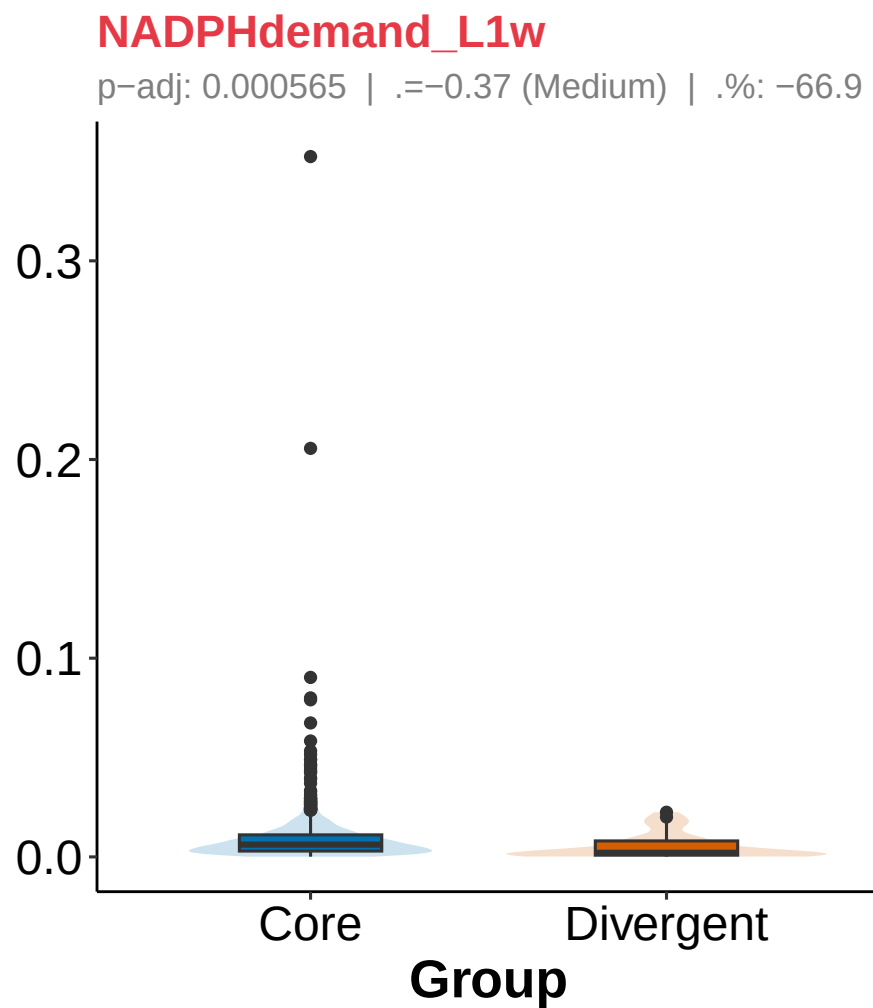
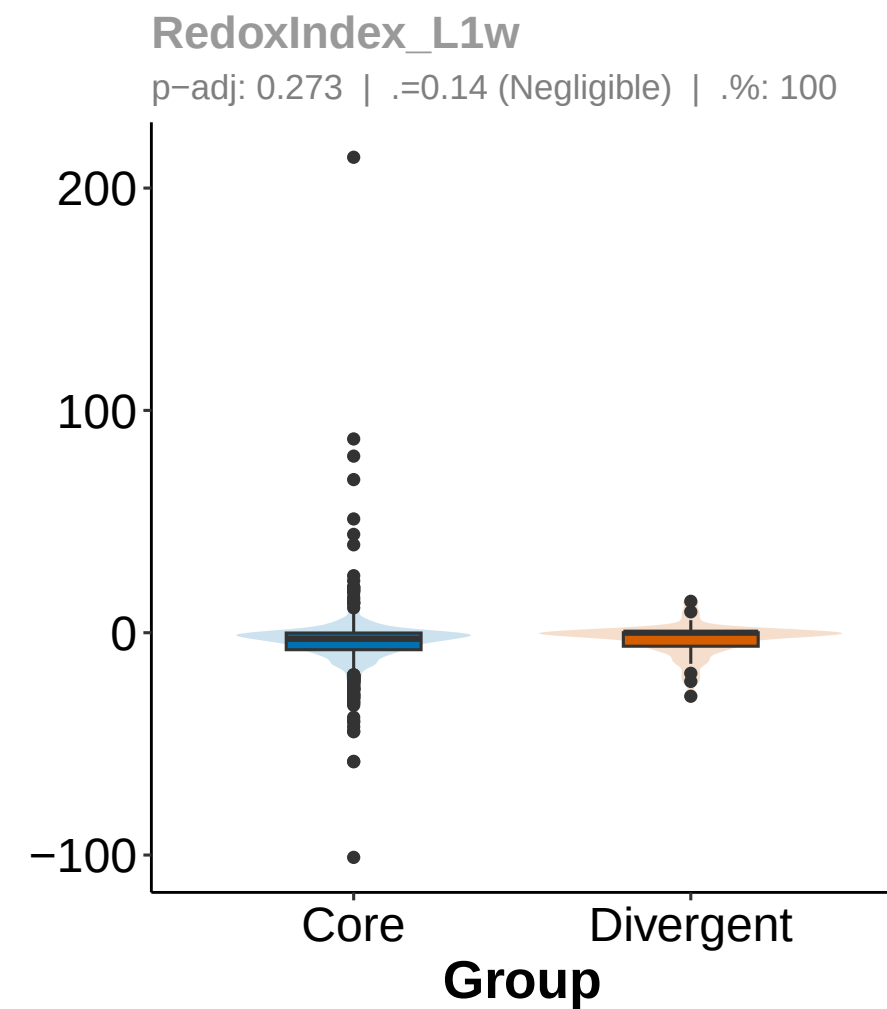
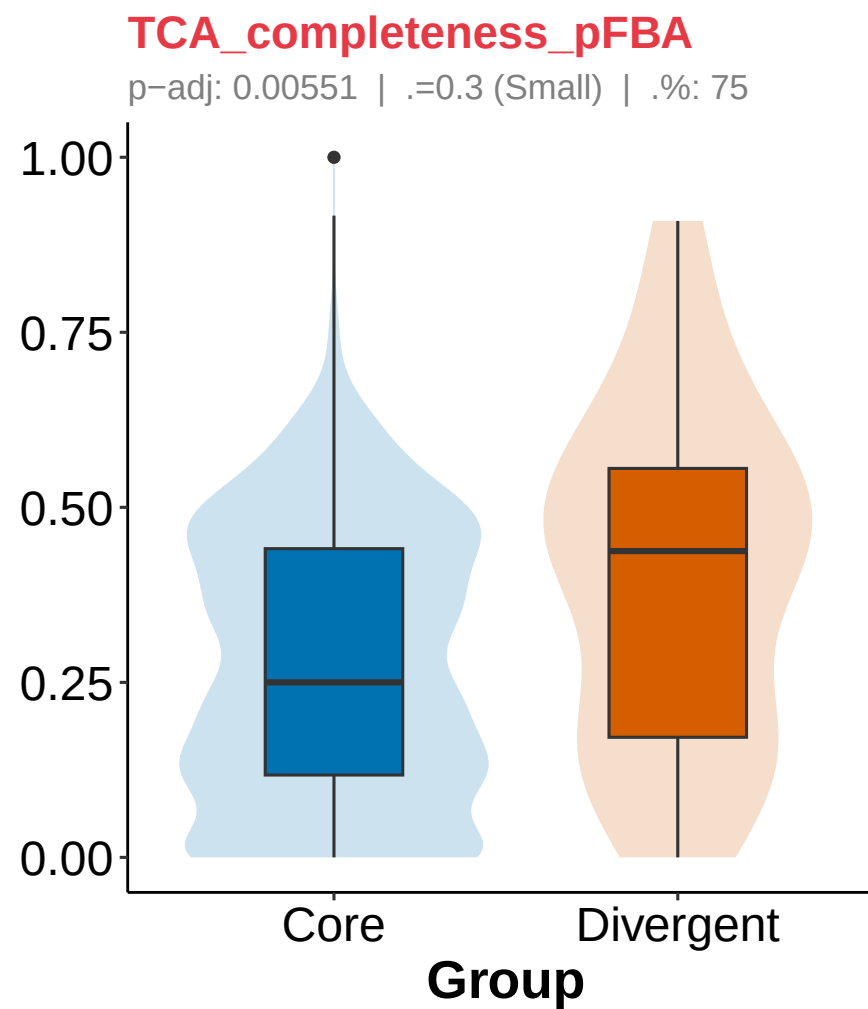
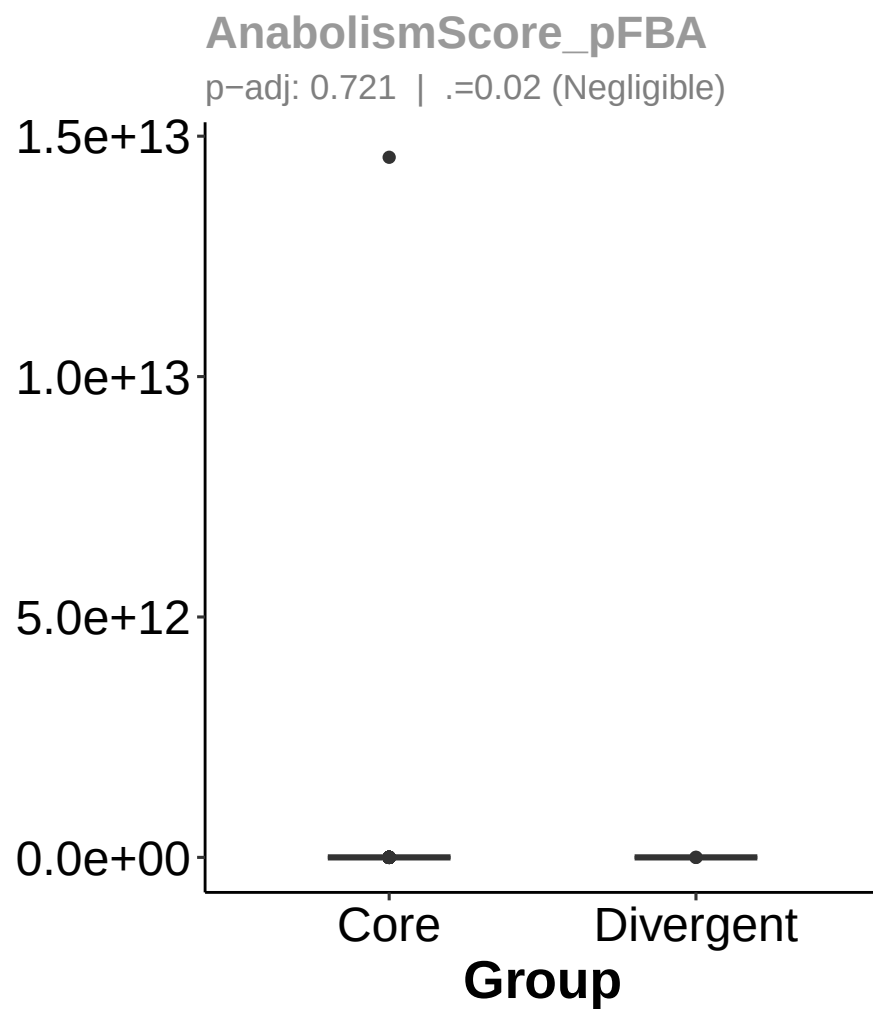


NADPHdemand_pFBA

p-adj: 0.0478 | .:=0.21 (Small) | .%: 356.6



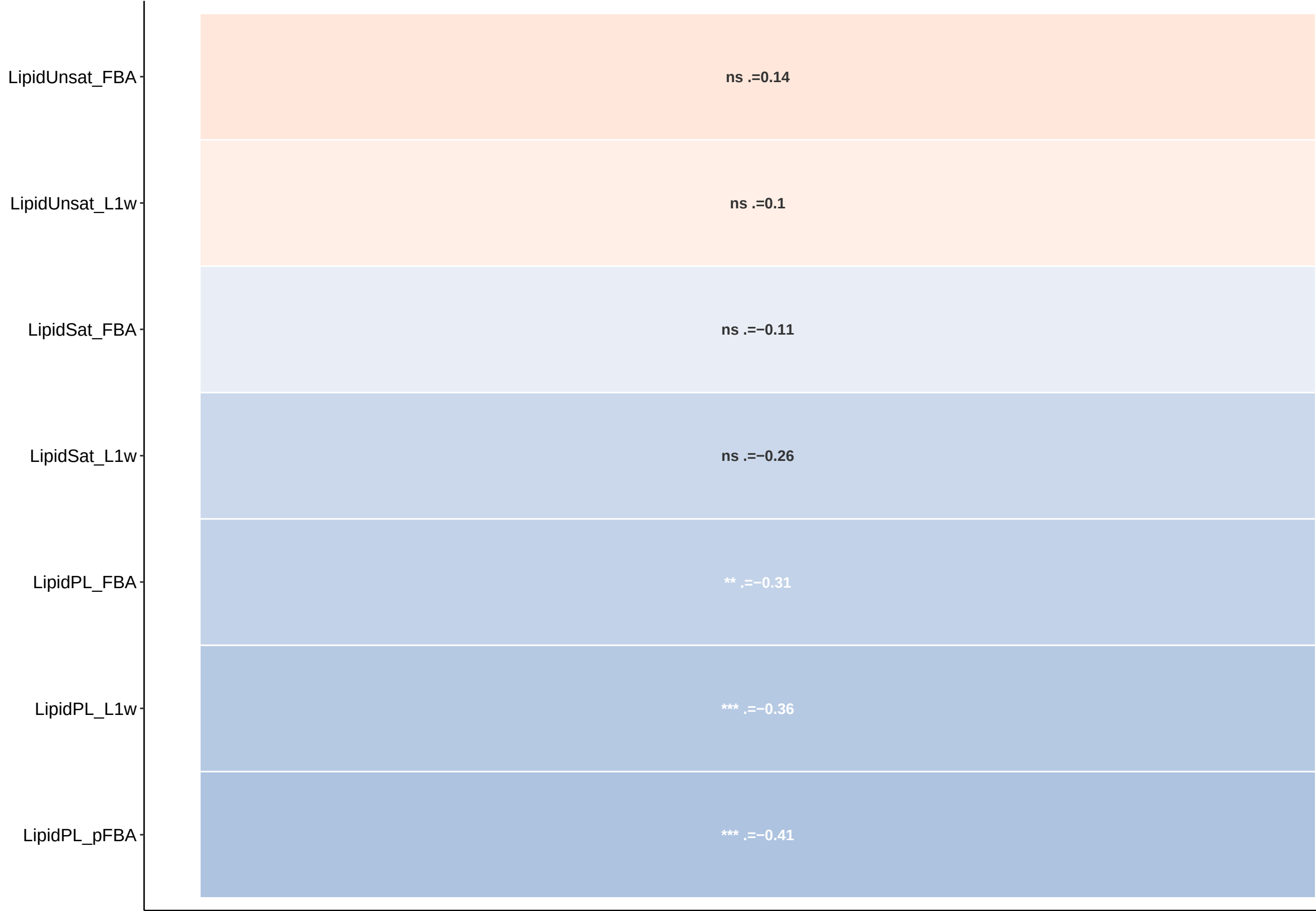
Metabolism – . 2. RedoxState and Anabolism



Cliff's .: . 3. Lipid Metabolism

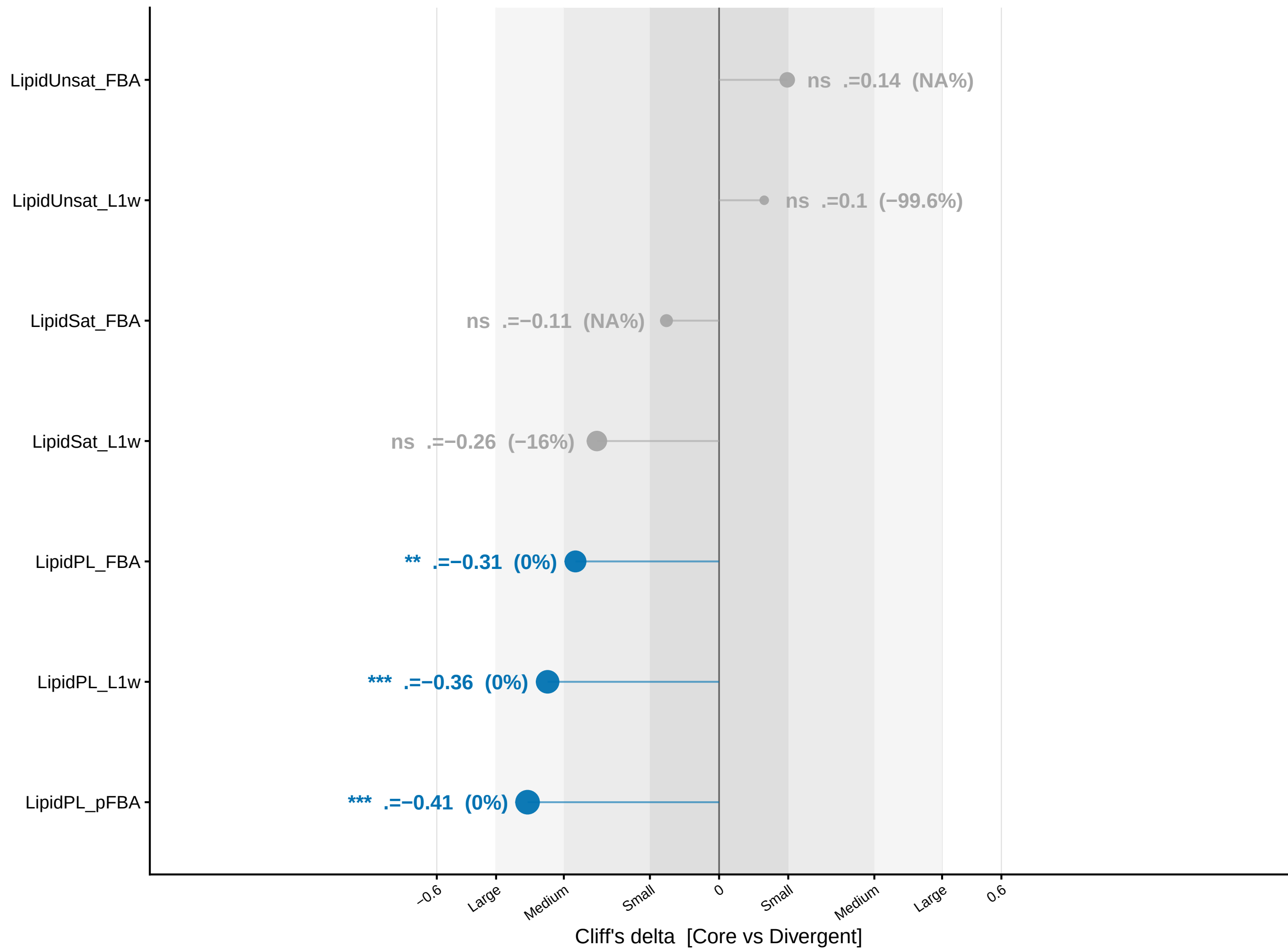
Blue = Core > Divergent | Orange = Divergent > Core | * p_adj<0.05

Cliff's .
Div vs Core1.00.5.00.51.0

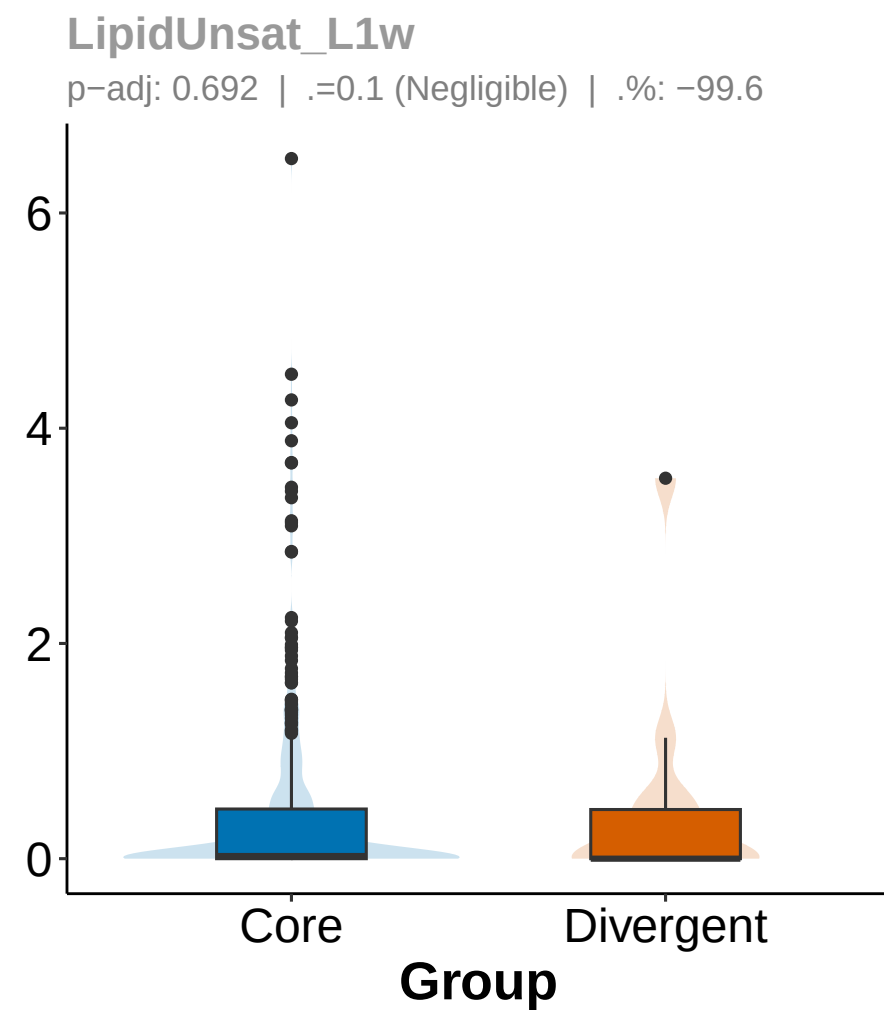
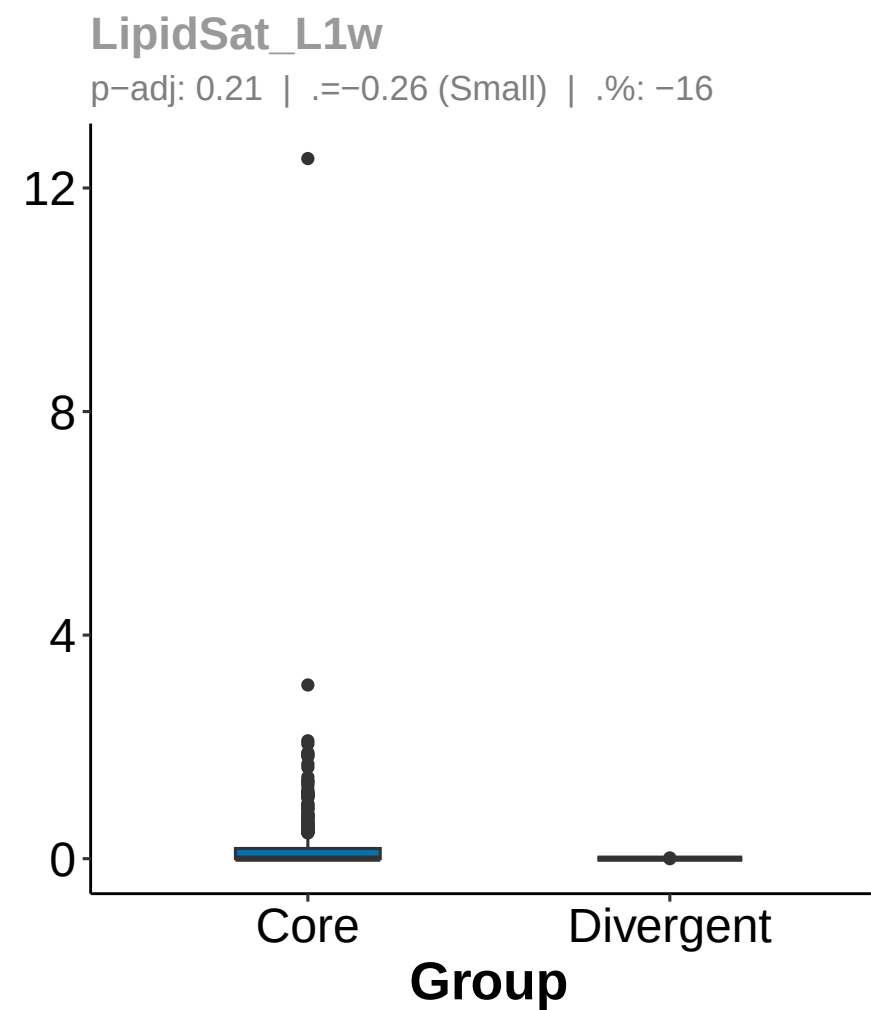
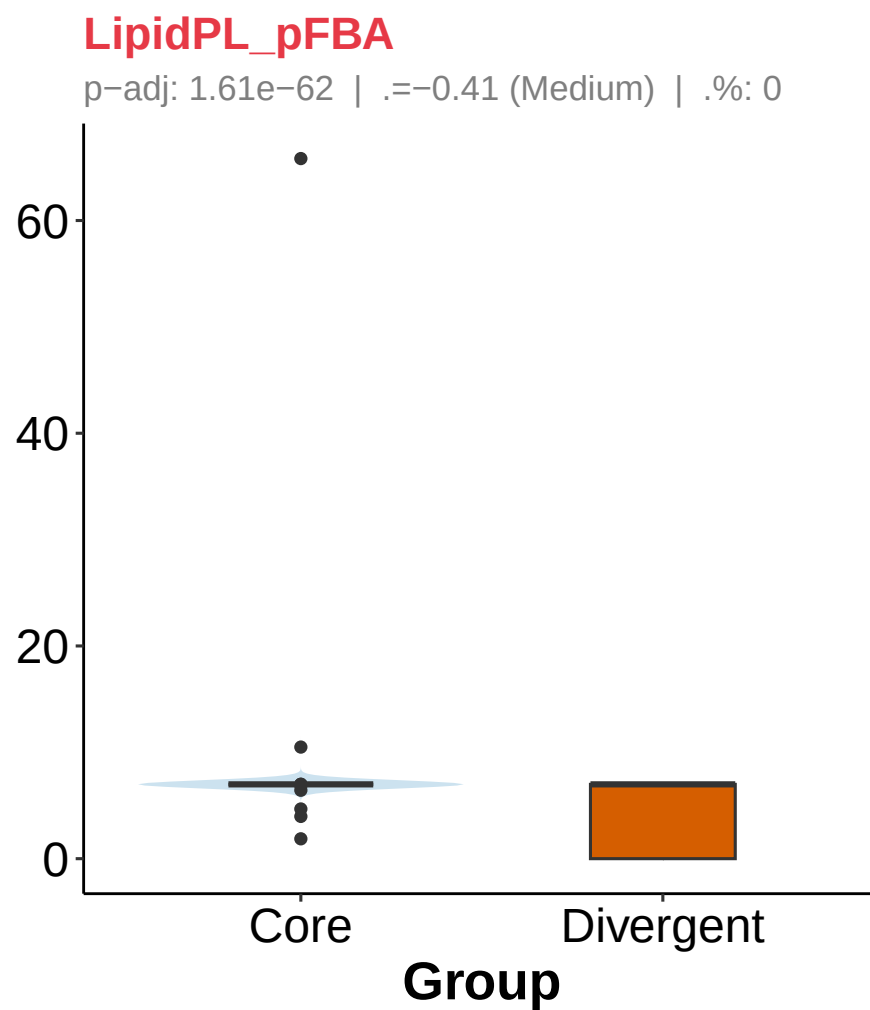
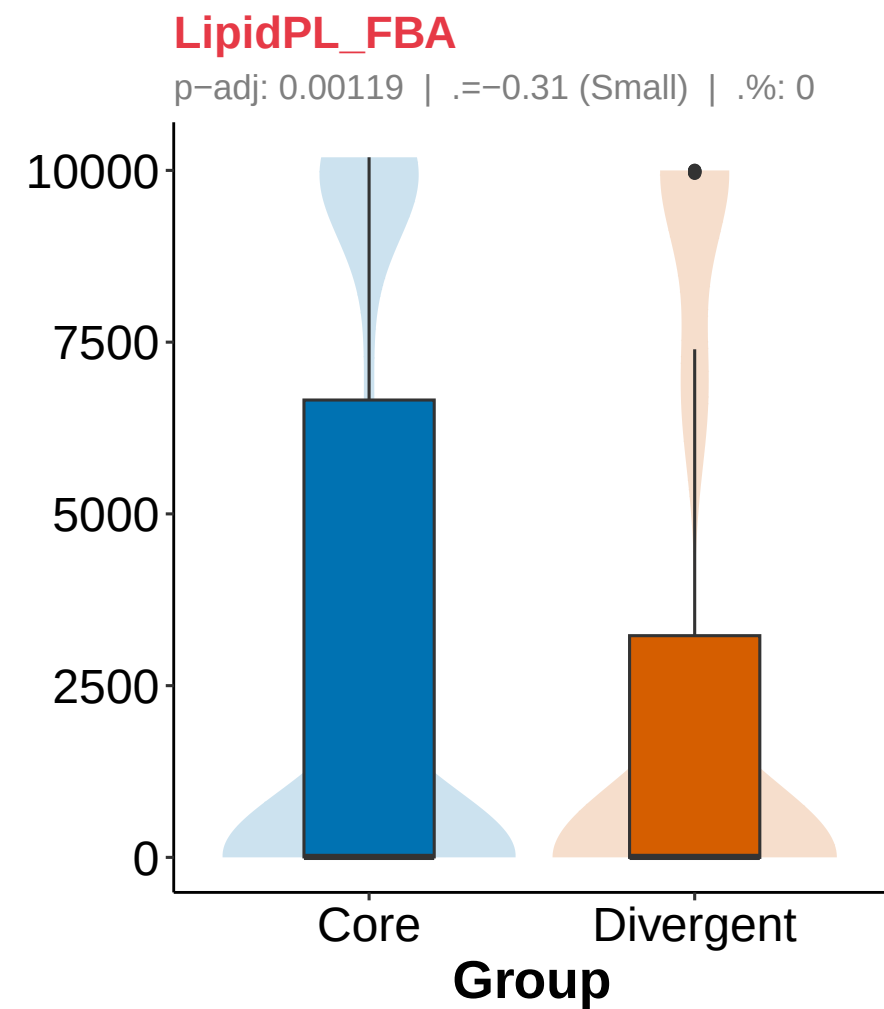
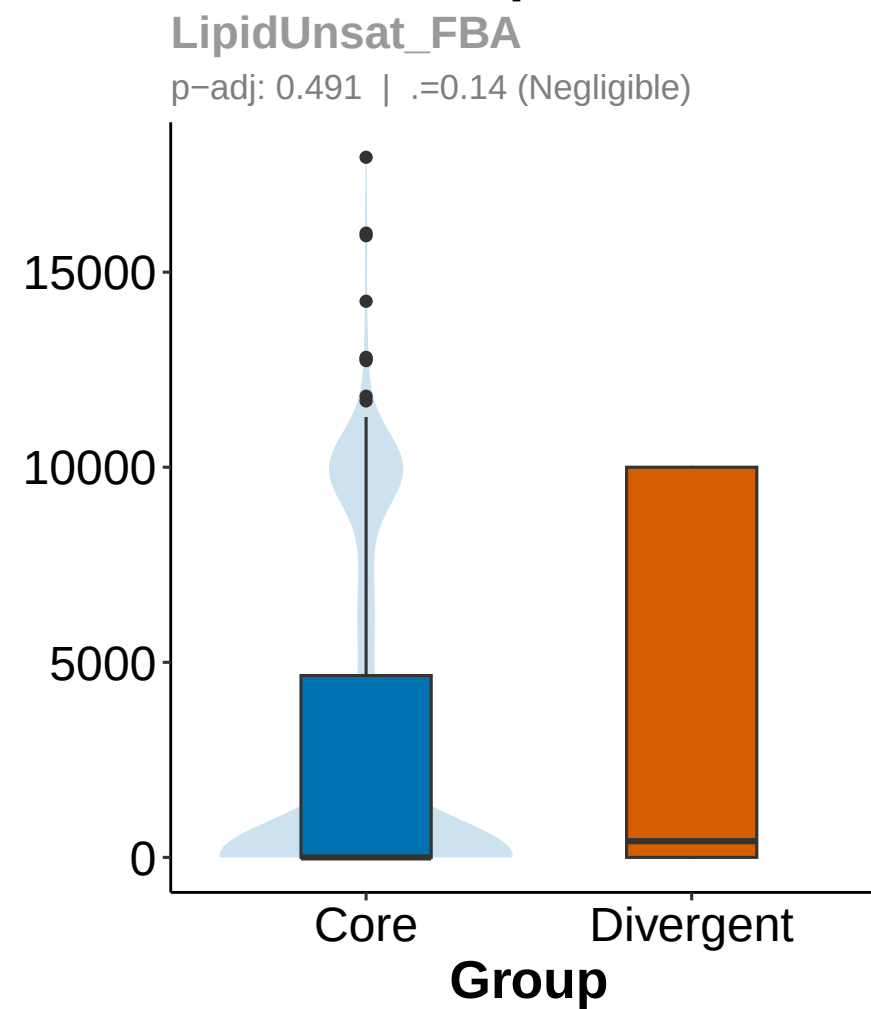
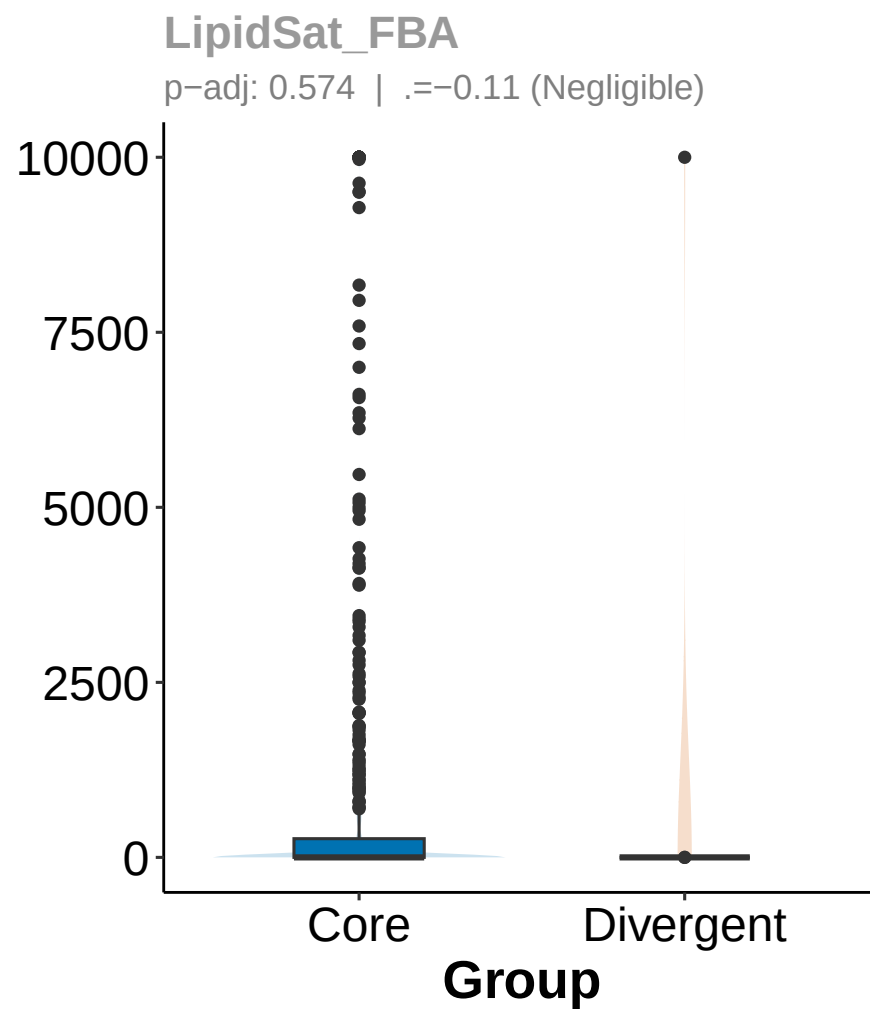


Cliff's . – . 3. Lipid Metabolism

Cliff's . + .% median | . higher in Core · . higher in Divergent



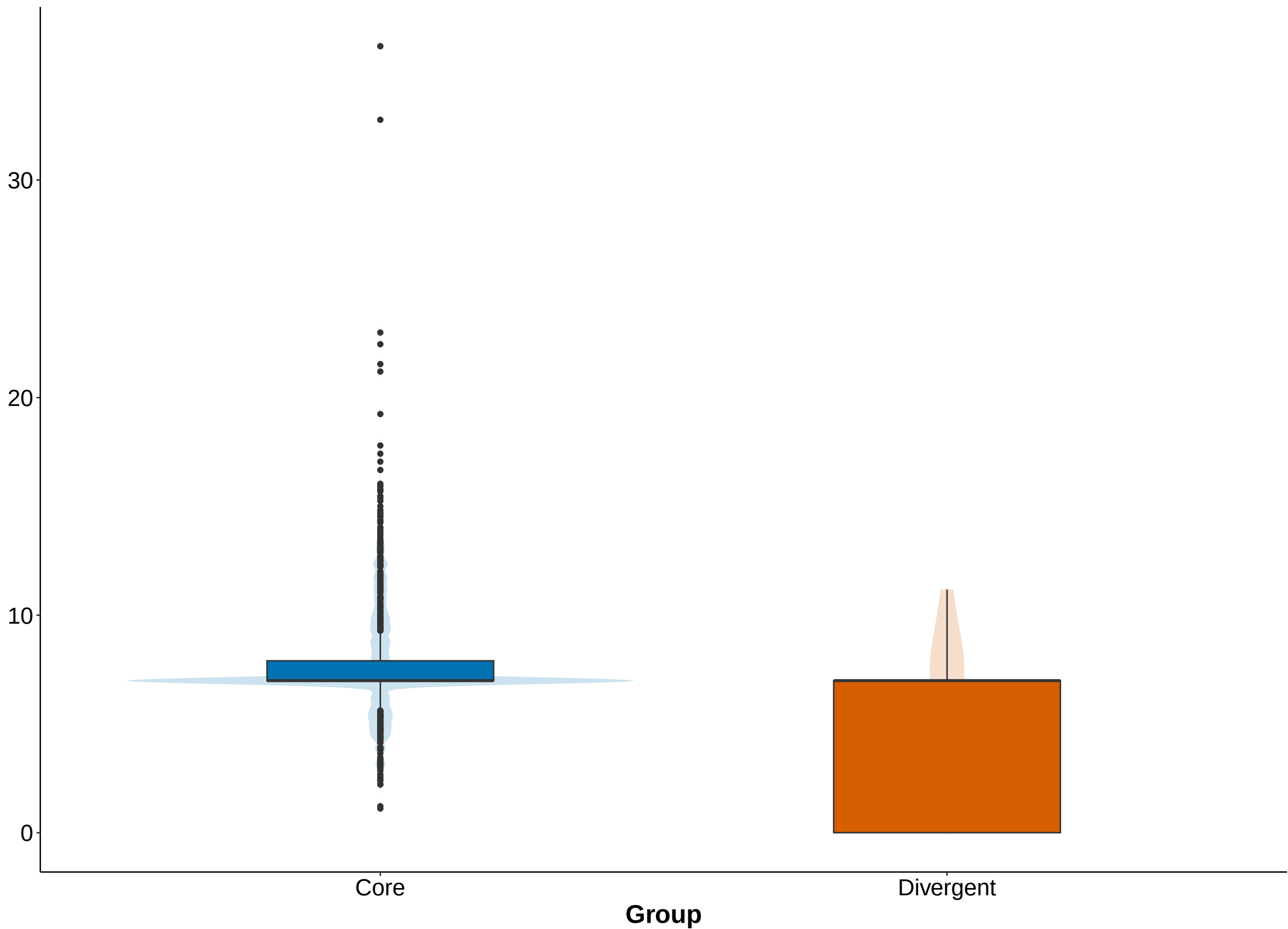
Metabolism – . 3. Lipid Metabolism



Metabolism - . 3. Lipid Metabolism

LipidPL_L1w

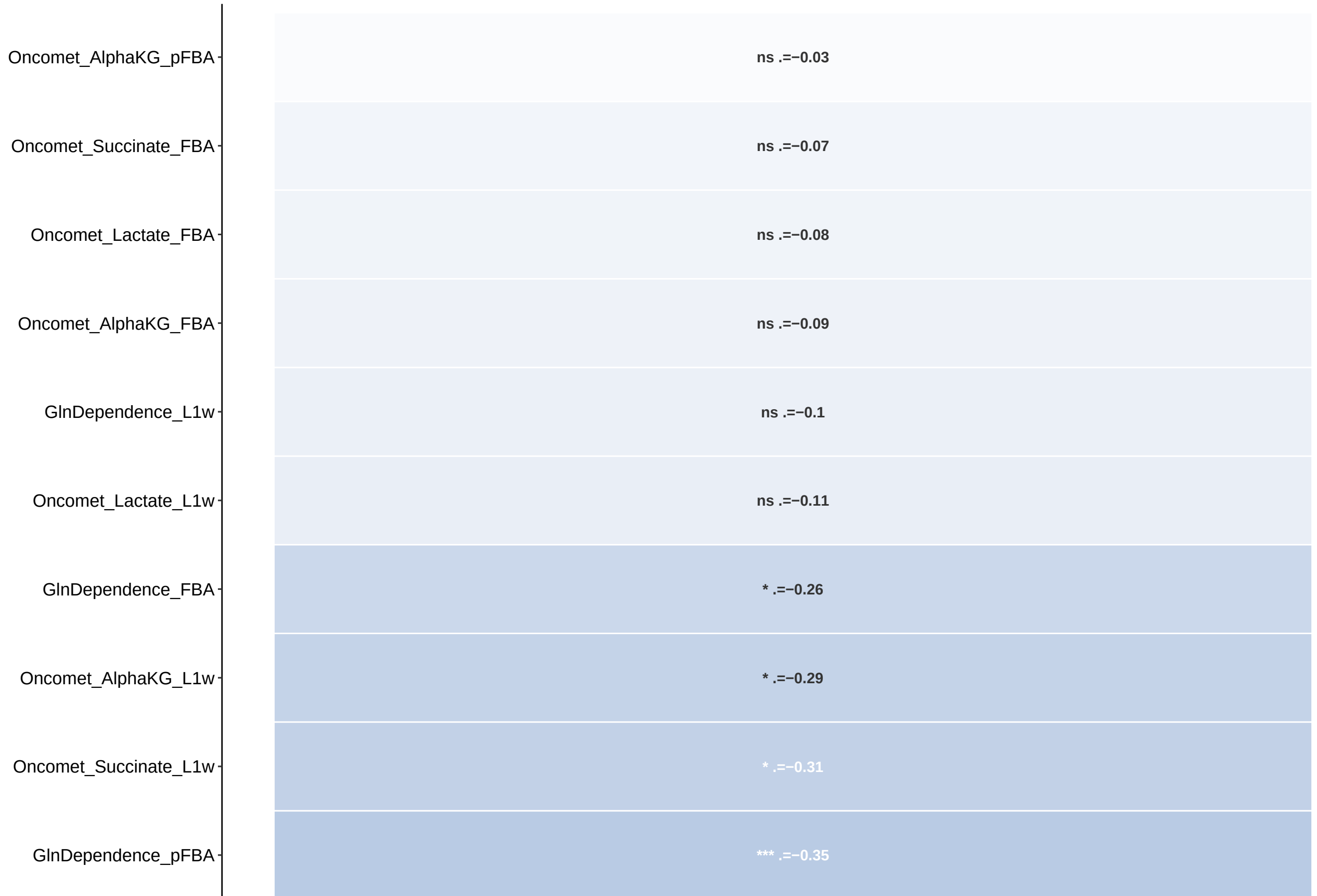
p-adj: 0.000219 | .=-0.36 (Medium) | .%: 0



Cliff's .: . 4. Specific Metabolic Dependencies

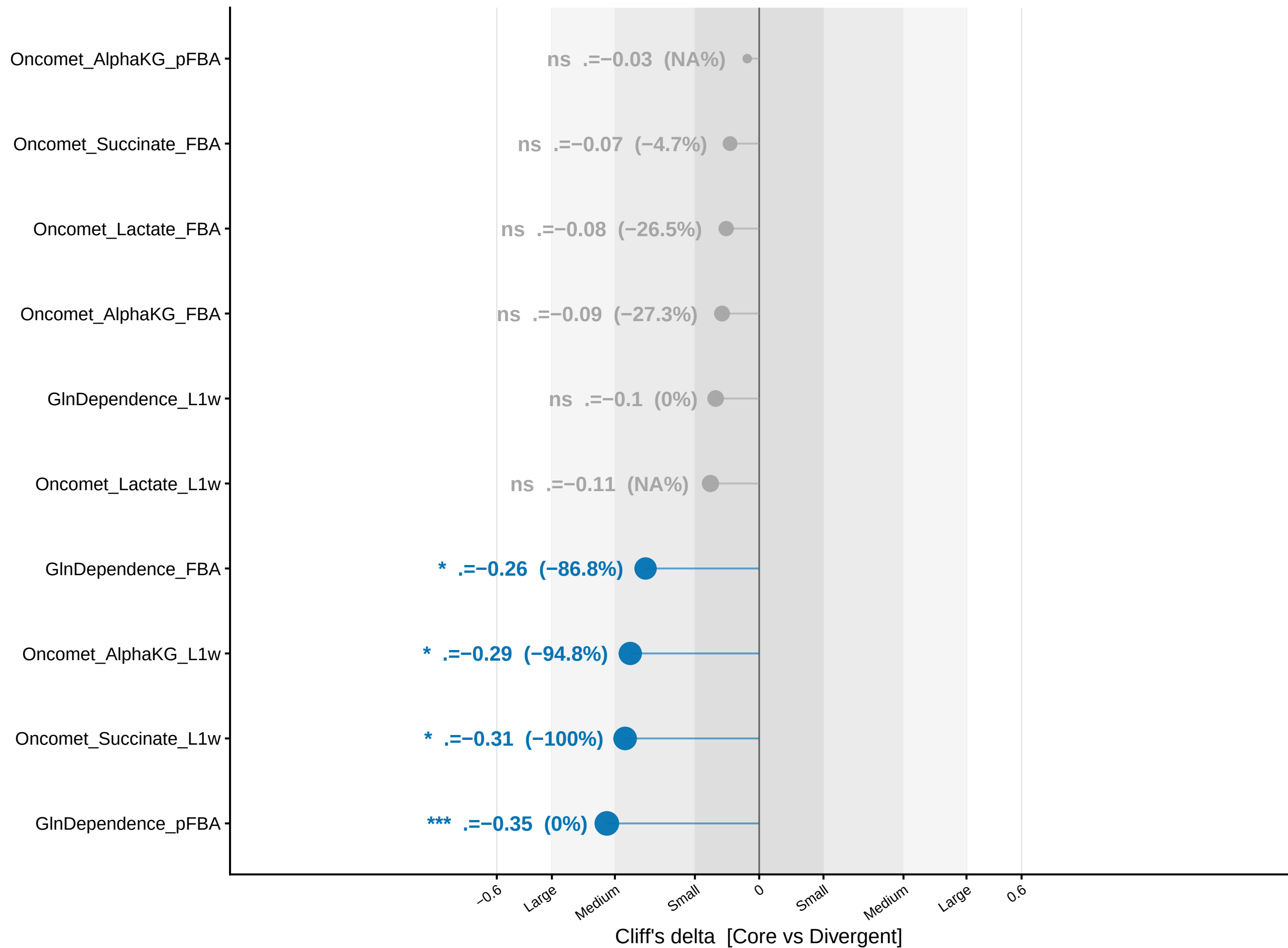
Blue = Core > Divergent | Orange = Divergent > Core | * p_adj<0.05

Cliff's .
Div vs Core



Cliff's . – . 4. Specific Metabolic Dependencies

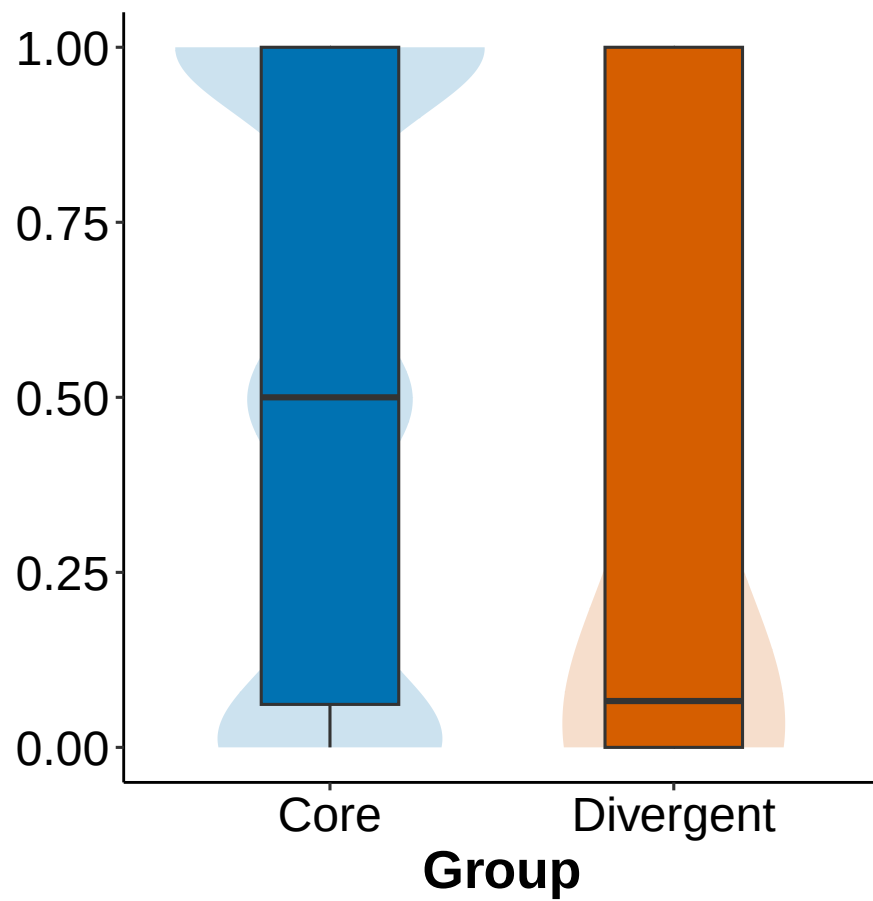
Cliff's . + .% median | . higher in Core · . higher in Divergent



Metabolism – . 4. Specific Metabolic Dependencies

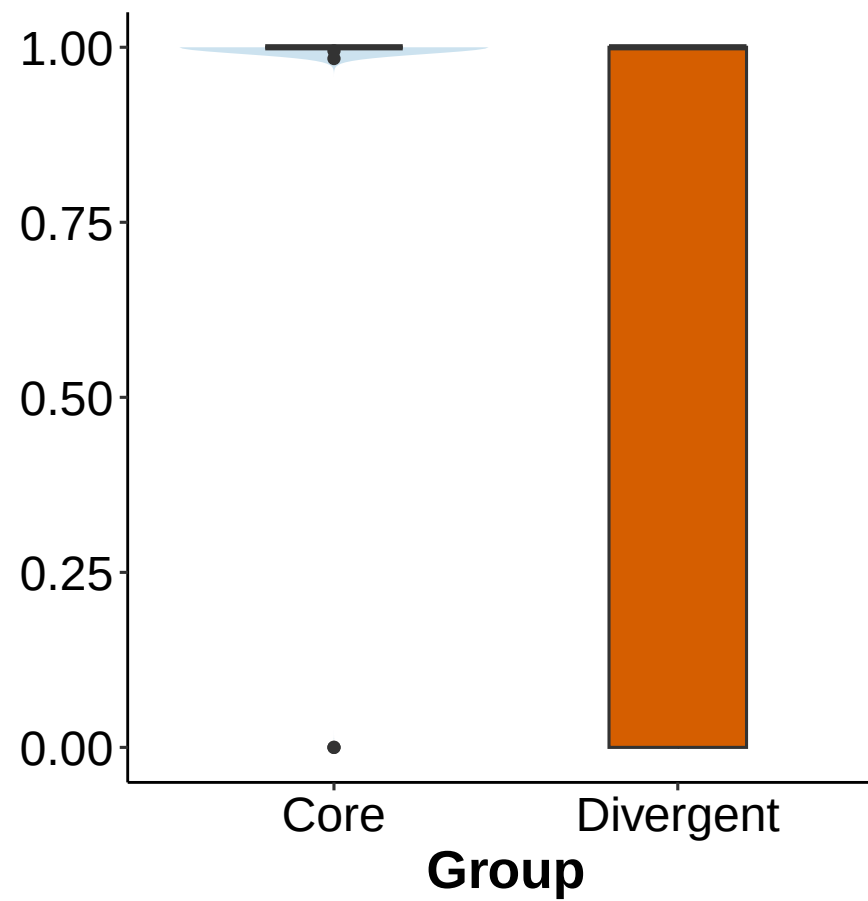
GlnDependence_FBA

p-adj: 0.0151 | .=-0.26 (Small) | .%: -86.8



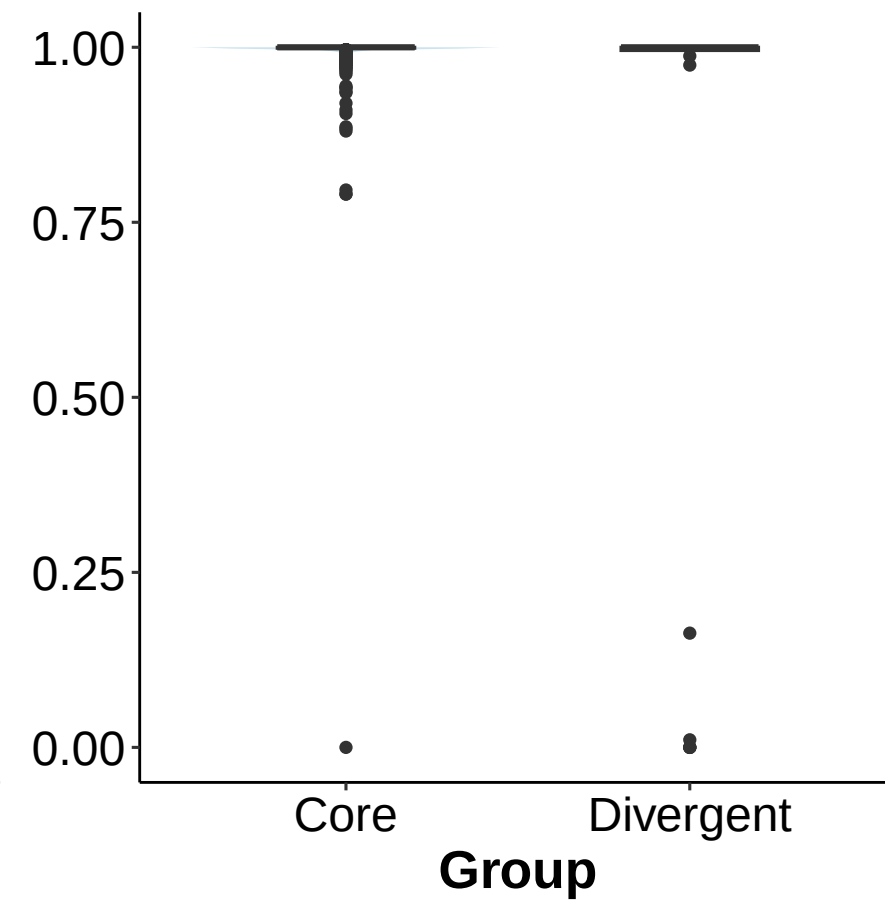
GlnDependence_pFBA

p-adj: 4e-65 | .=-0.35 (Medium) | .%: 0



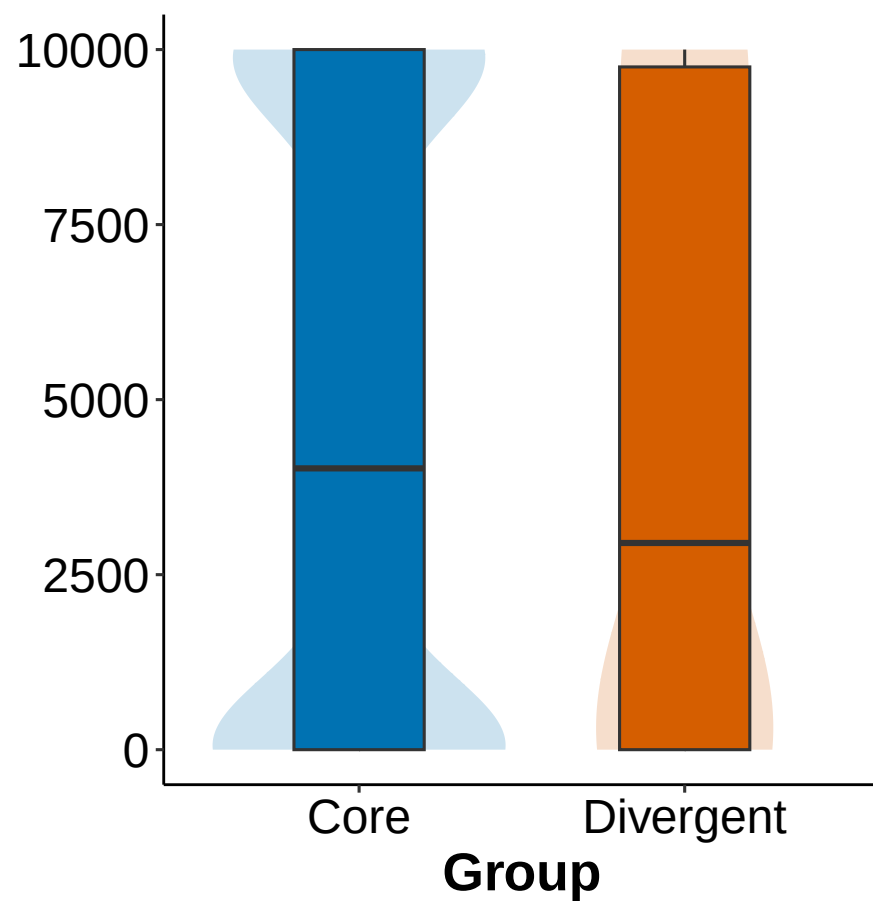
GlnDependence_L1w

p-adj: 0.451 | .=-0.1 (Negligible) | .%: 0



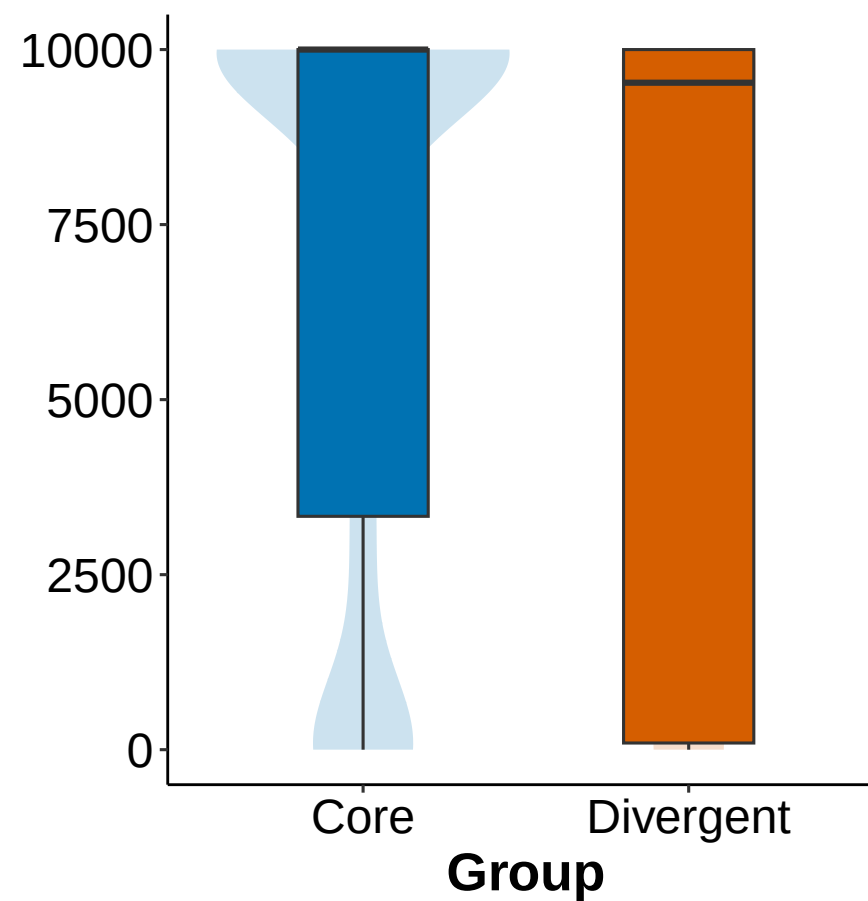
Oncomet_Lactate_FBA

p-adj: 0.575 | .=-0.08 (Negligible) | .%: -26.5



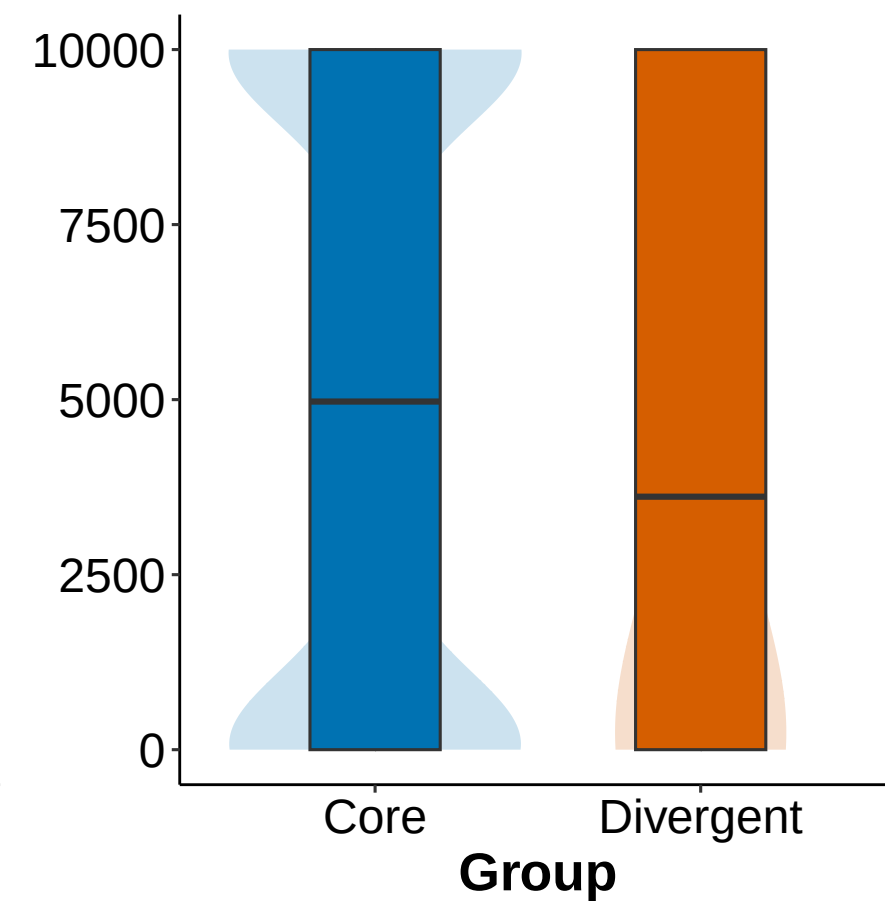
Oncomet_Succinate_FBA

p-adj: 0.689 | .=-0.07 (Negligible) | .%: -4.7



Oncomet_AlphaKG_FBA

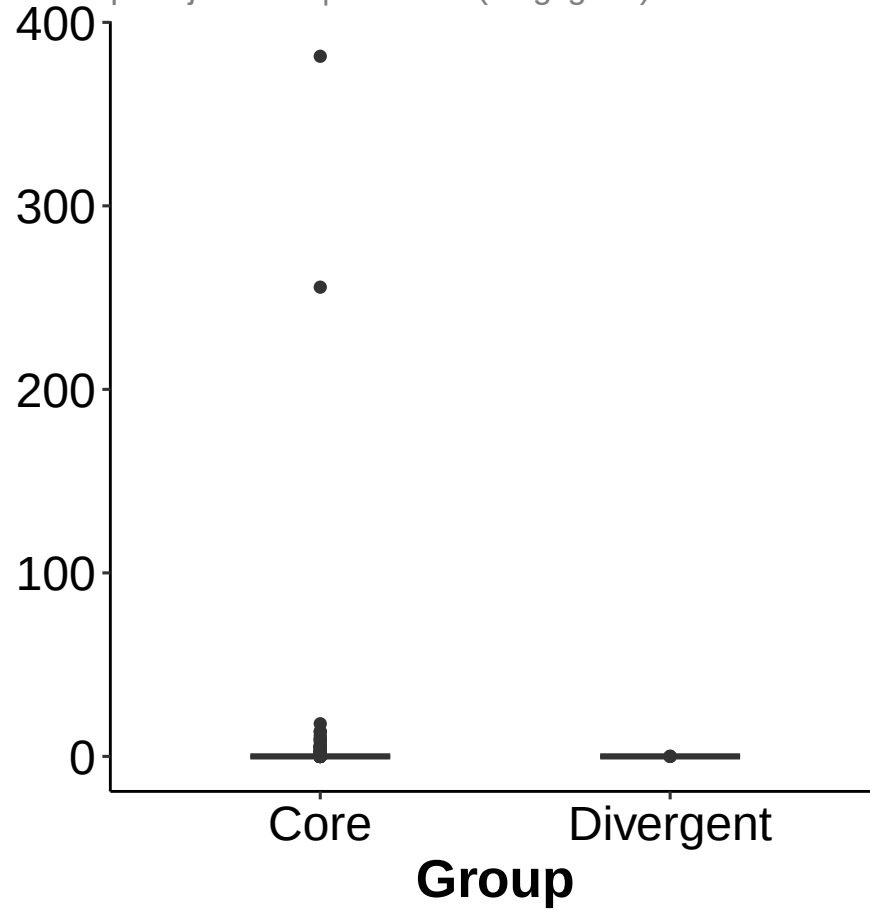
p-adj: 0.574 | .=-0.09 (Negligible) | .%: -27.3



Metabolism – . 4. Specific Metabolic Dependencies

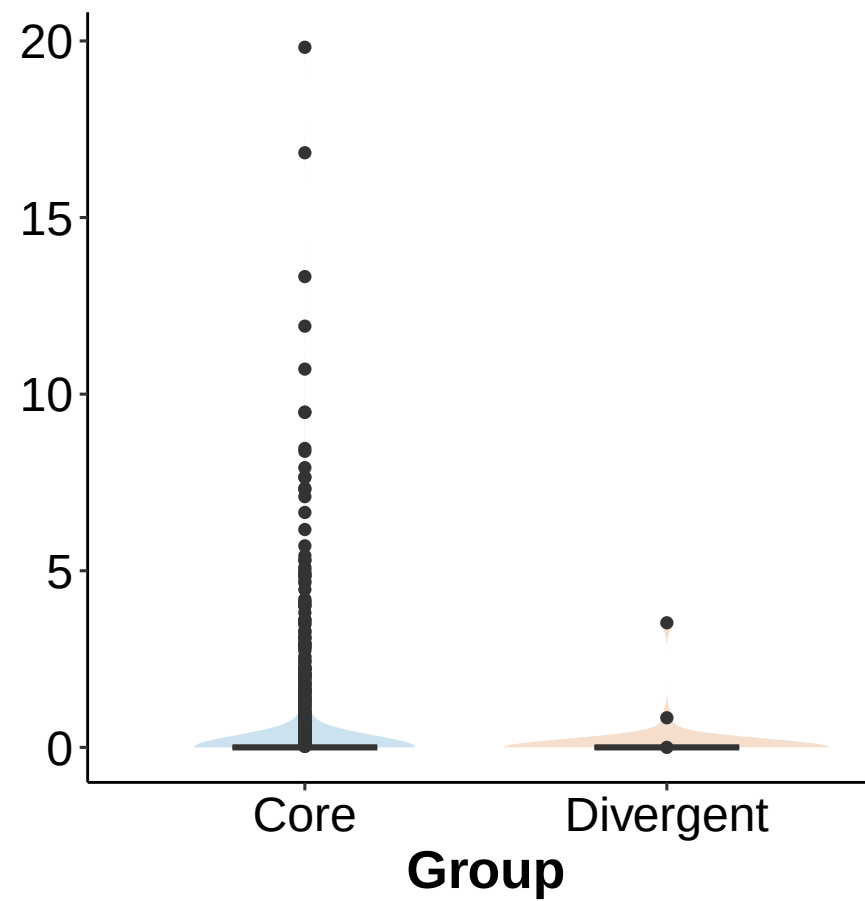
Oncomet_AlphaKG_pFBA

p-adj: 0.762 | .=-0.03 (Negligible)



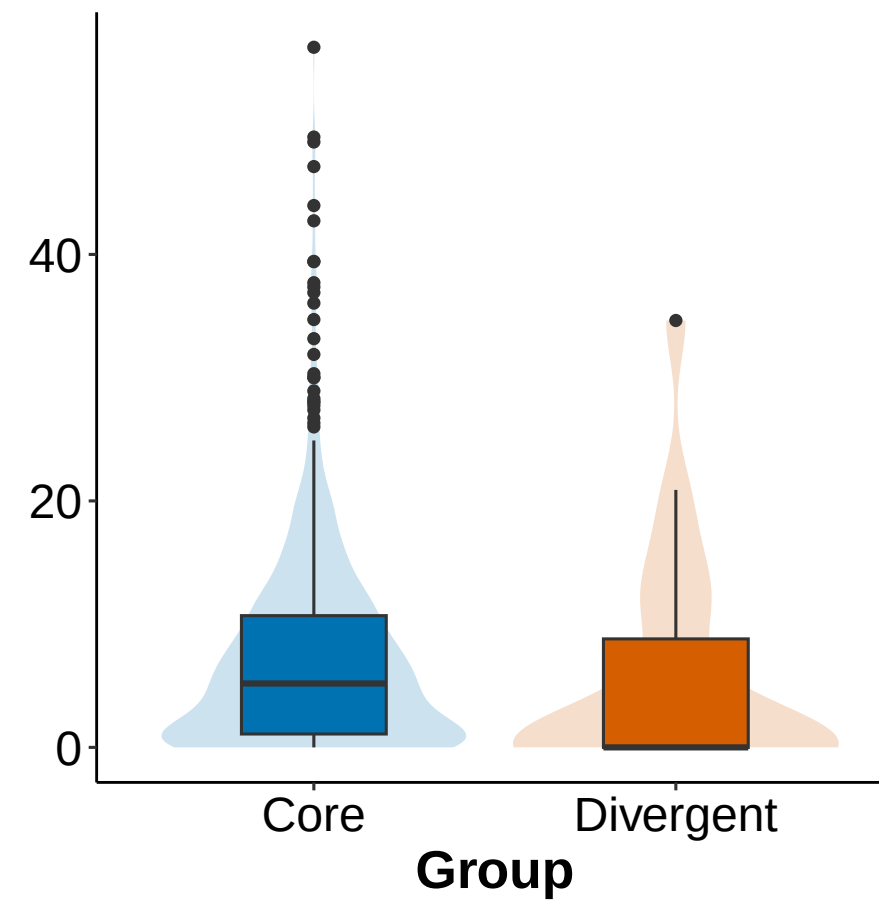
Oncomet_Lactate_L1w

p-adj: 0.175 | .=-0.11 (Negligible)



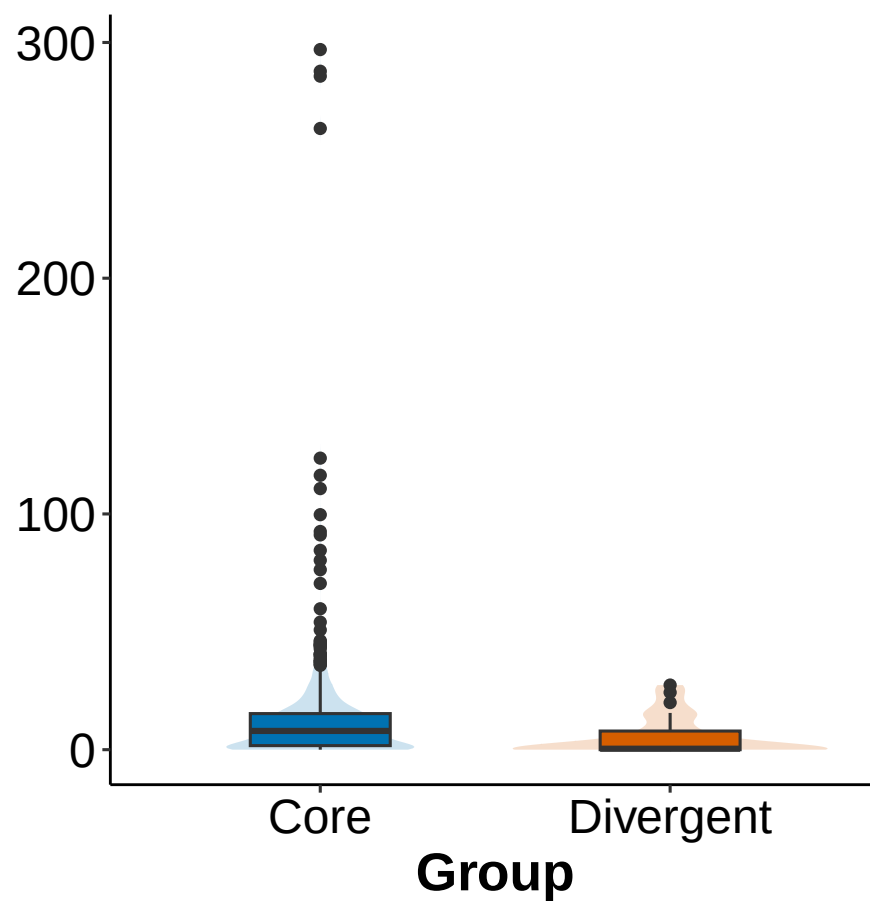
Oncomet_Succinate_L1w

p-adj: 0.0249 | .=-0.31 (Small) | .%: -100



Oncomet_AlphaKG_L1w

p-adj: 0.0174 | .=-0.29 (Small) | .%: -94.8



Cliff's .: . 5. Systemic Reprogramming (SA_*)

Blue = Core > Divergent | Orange = Divergent > Core | * p_adj<0.05

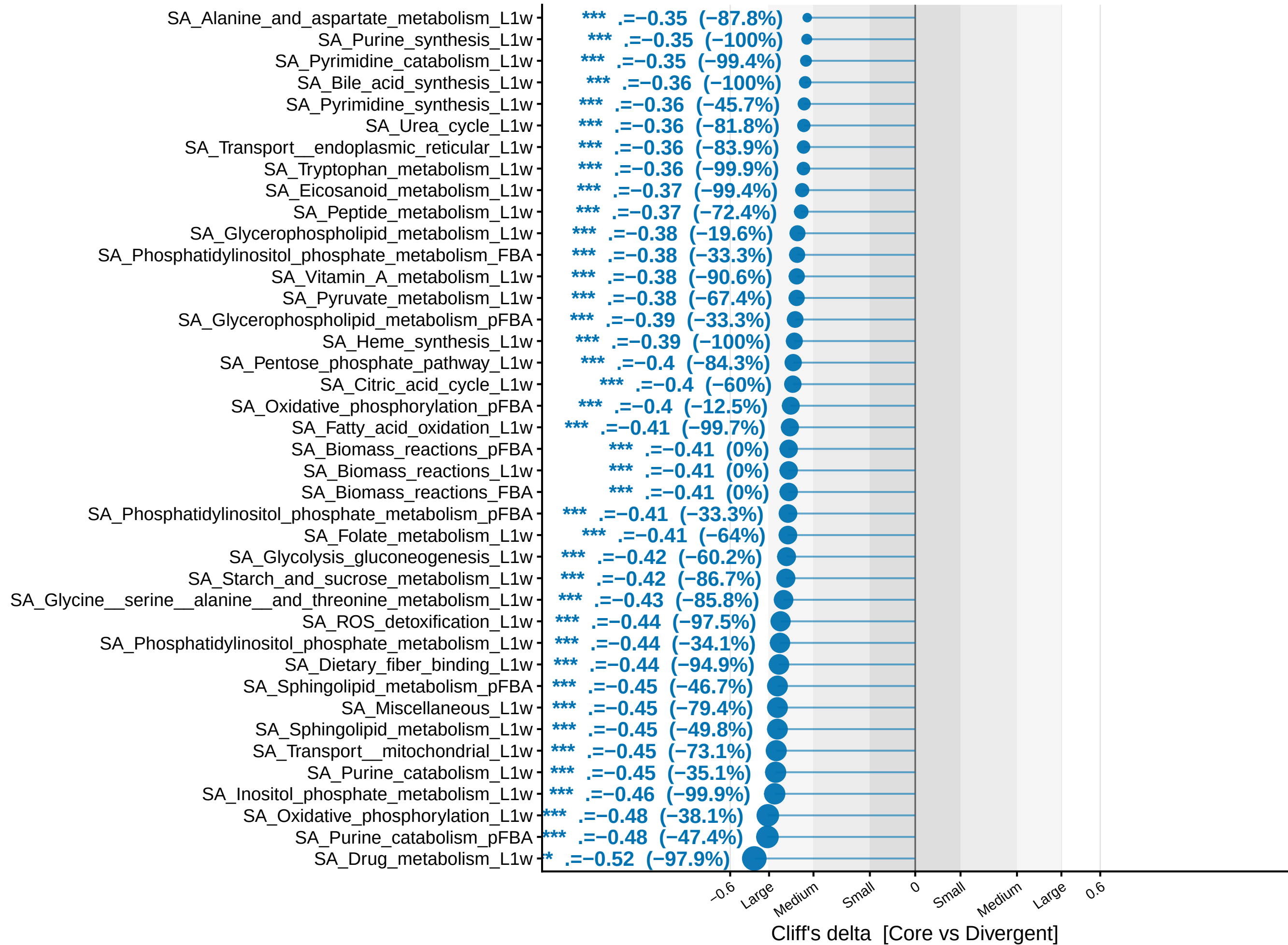
Cliff's .
Div vs Core1.00.0.00.51.0



SA_Alanine_and_aspartate_metabolism_L1w	***	.-0.35
SA_Purine_synthesis_L1w	***	.-0.35
SA_Pyrimidine_catabolism_L1w	***	.-0.35
SA_Bile_acid_synthesis_L1w	***	.-0.36
SA_Pyrimidine_synthesis_L1w	***	.-0.36
SA_Urea_cycle_L1w	***	.-0.36
SA_Transport__endoplasmic_reticular_L1w	***	.-0.36
SA_Tryptophan_metabolism_L1w	***	.-0.36
SA_Eicosanoid_metabolism_L1w	***	.-0.37
SA_Peptide_metabolism_L1w	***	.-0.37
SA_Glycerophospholipid_metabolism_L1w	***	.-0.38
SA_Phosphatidylinositol_phosphate_metabolism_FBA	***	.-0.38
SA_Vitamin_A_metabolism_L1w	***	.-0.38
SA_Pyruvate_metabolism_L1w	***	.-0.38
SA_Glycerophospholipid_metabolism_pFBA	***	.-0.39
SA_Heme_synthesis_L1w	***	.-0.39
SA_Pentose_phosphate_pathway_L1w	***	.-0.4
SA_Citric_acid_cycle_L1w	***	.-0.4
SA_Oxidative_phosphorylation_pFBA	***	.-0.4
SA_Fatty_acid_oxidation_L1w	***	.-0.41
SA_Biomass_reactions_pFBA	***	.-0.41
SA_Biomass_reactions_L1w	***	.-0.41
SA_Biomass_reactions_FBA	***	.-0.41
SA_Phosphatidylinositol_phosphate_metabolism_pFBA	***	.-0.41
SA_Folate_metabolism_L1w	***	.-0.41
SA_Glycolysis_gluconeogenesis_L1w	***	.-0.42
SA_Starch_and_sucrose_metabolism_L1w	***	.-0.42
SA_Glycine__serine__alanine__and_threonine_metabolism_L1w	***	.-0.43
SA_ROS_detoxification_L1w	***	.-0.44
SA_Phosphatidylinositol_phosphate_metabolism_L1w	***	.-0.44
SA_Dietary_fiber_binding_L1w	***	.-0.44
SA_Sphingolipid_metabolism_pFBA	***	.-0.45
SA_Miscellaneous_L1w	***	.-0.45
SA_Sphingolipid_metabolism_L1w	***	.-0.45
SA_Transport__mitochondrial_L1w	***	.-0.45
SA_Purine_catabolism_L1w	***	.-0.45
SA_Inositol_phosphate_metabolism_L1w	***	.-0.46
SA_Oxidative_phosphorylation_L1w	***	.-0.48
SA_Purine_catabolism_pFBA	***	.-0.48
SA_Drug_metabolism_L1w	***	.-0.52

Cliff's . – . 5. Systemic Reprogramming (SA_*)

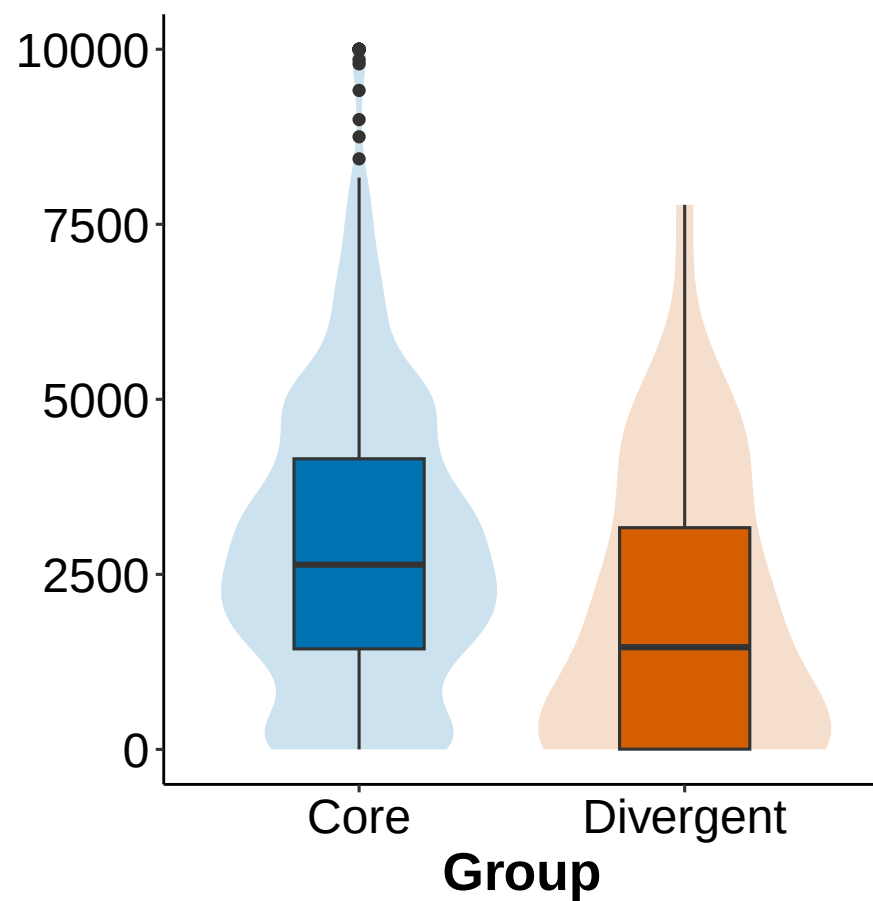
Cliff's . + .% median | . higher in Core . higher in Divergent



Metabolism – . 5. Systemic Reprogramming (SA_*)

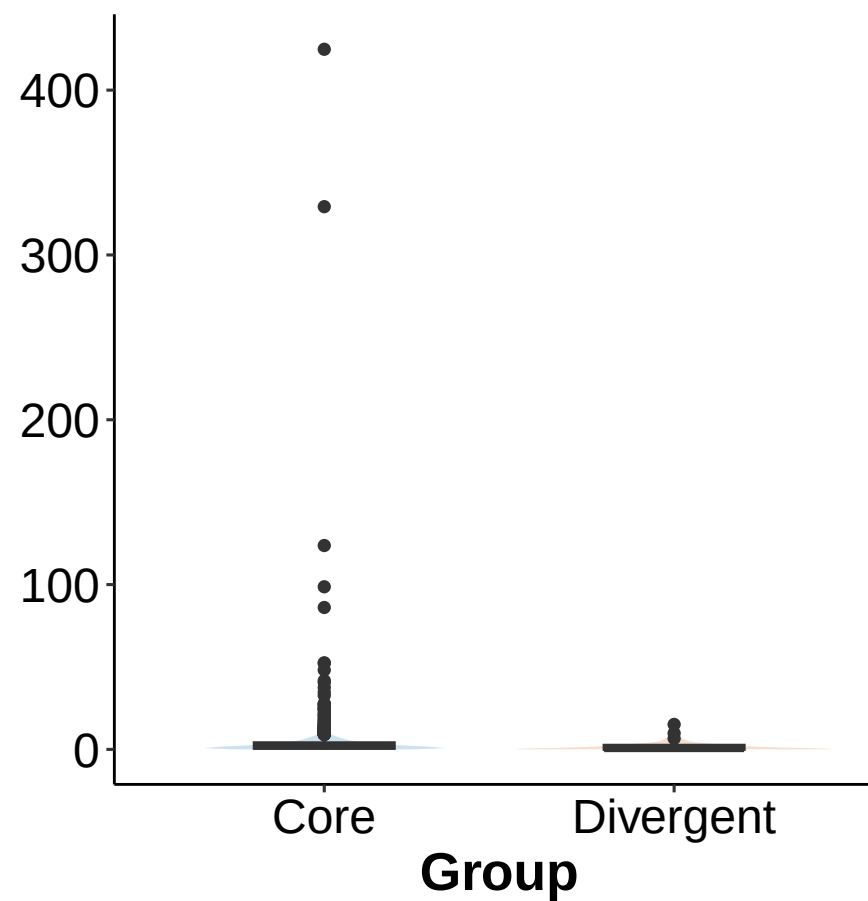
SA_Alanine_and_aspartate_metabolism

p-adj: 0.00414 | .=-0.31 (Small) | .%: -44.7



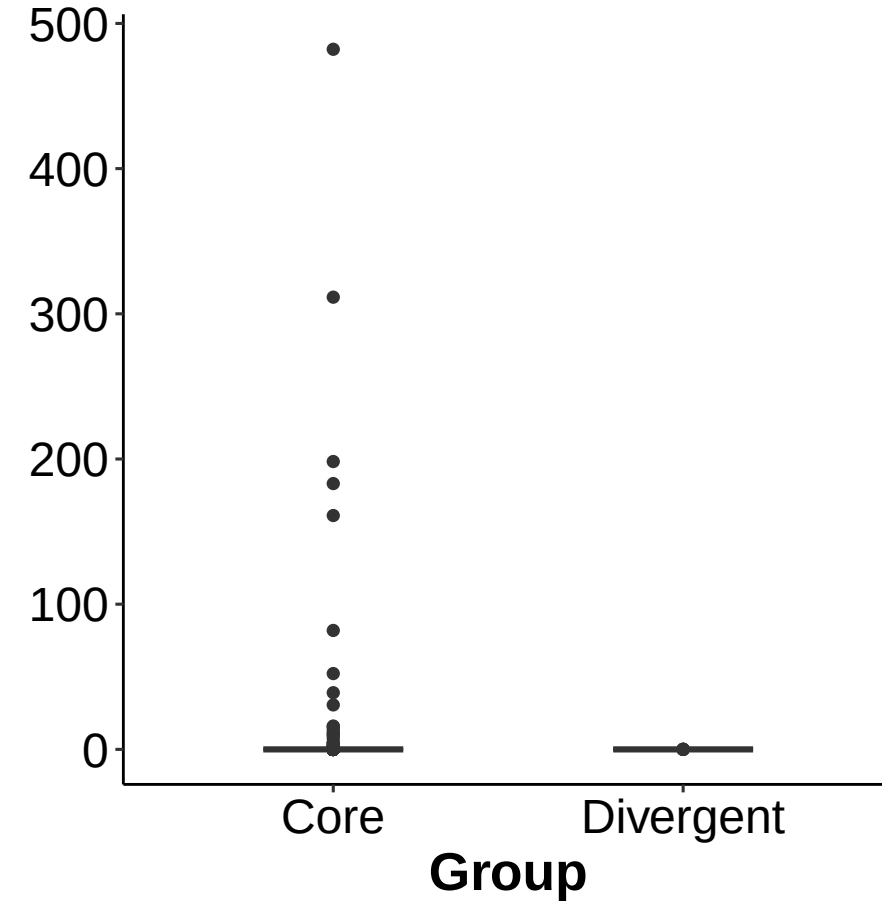
SA_Alanine_and_aspartate_metabolism

p-adj: 0.000961 | .=-0.35 (Medium) | .%: -87.8



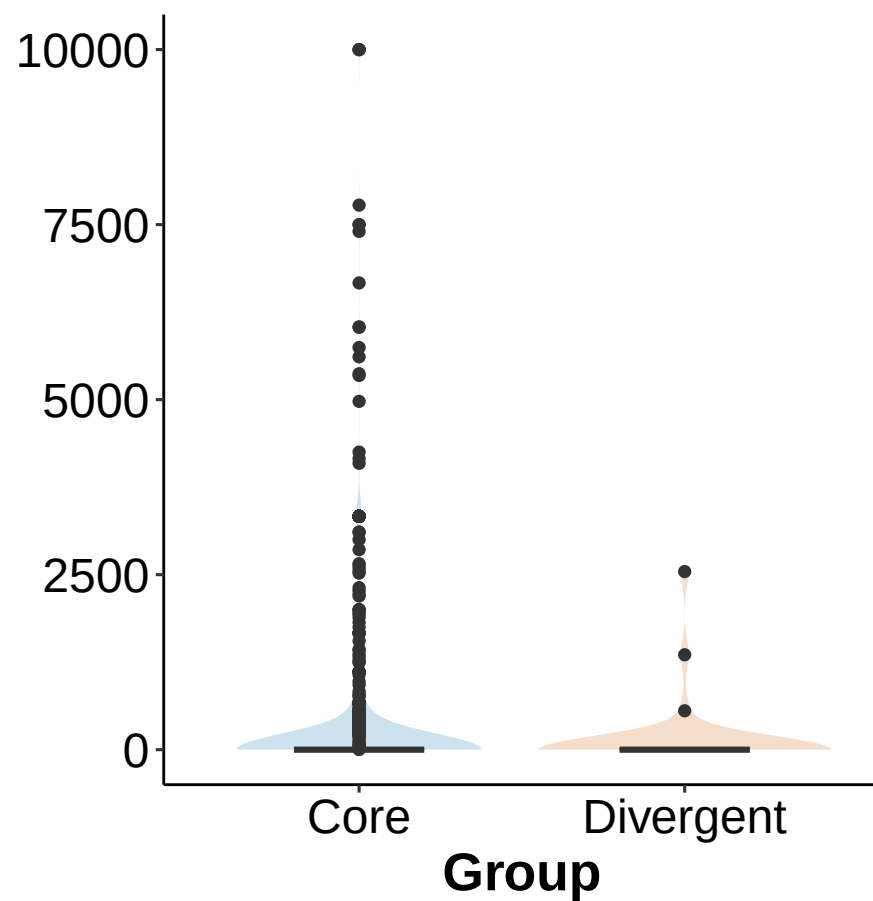
SA_Alanine_and_aspartate_metabolism

p-adj: 0.823 | .=-0.02 (Negligible)



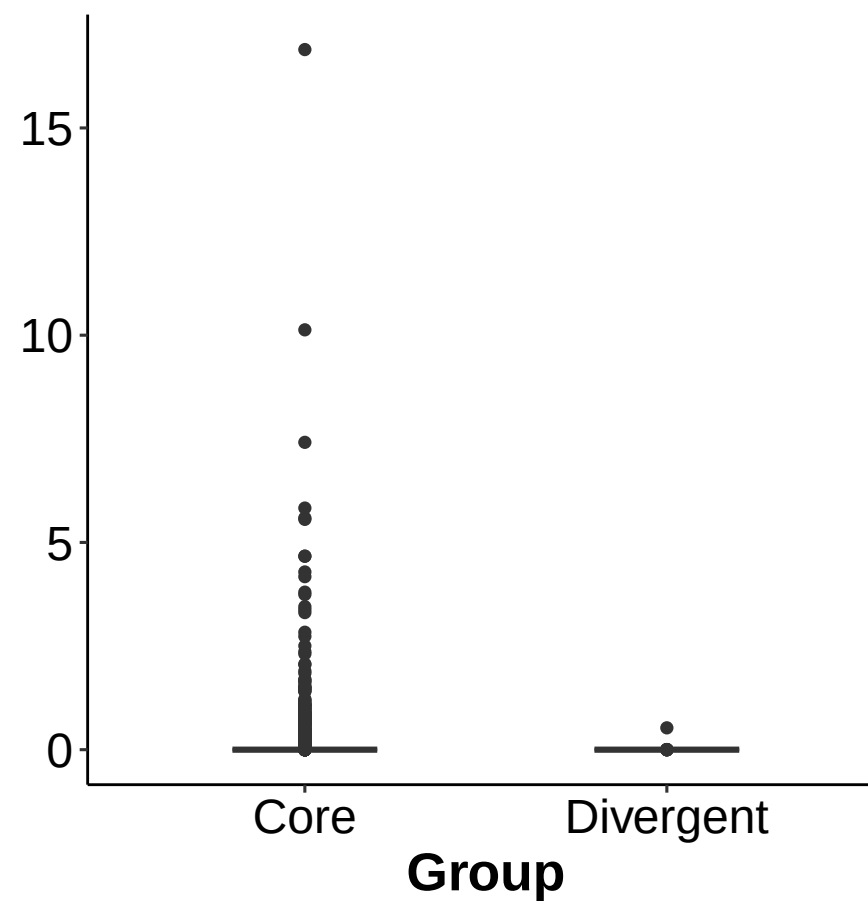
SA_Aminosugar_metabolism_FBA

p-adj: 0.352 | .=-0.08 (Negligible)



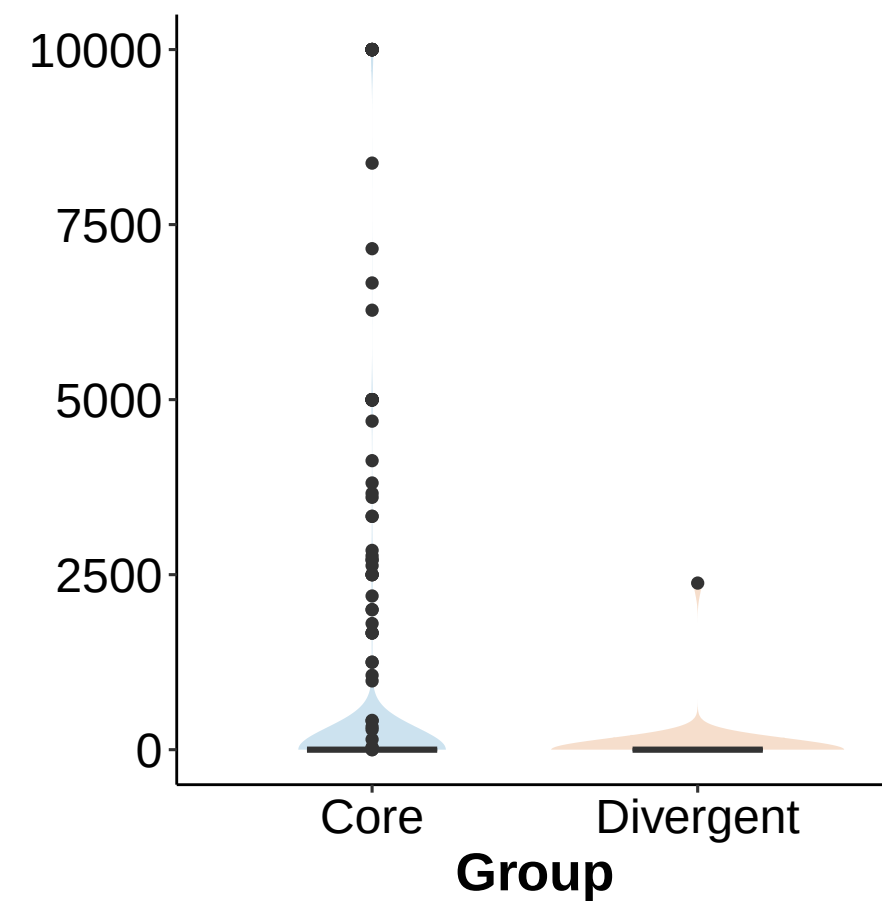
SA_Aminosugar_metabolism_L1w

p-adj: 0.0207 | .=-0.25 (Small) | .%: -100



SA_Androgen_and_estrogen_synthesis

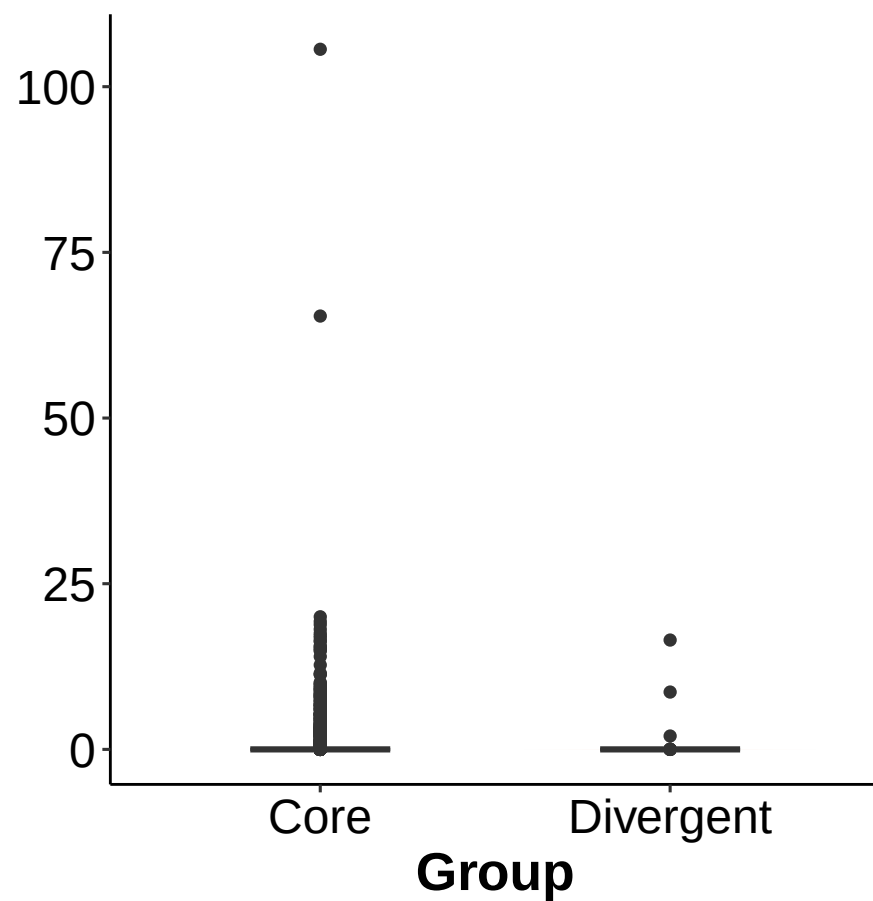
p-adj: 0.563 | .=-0.04 (Negligible)



Metabolism – . 5. Systemic Reprogramming (SA_*)

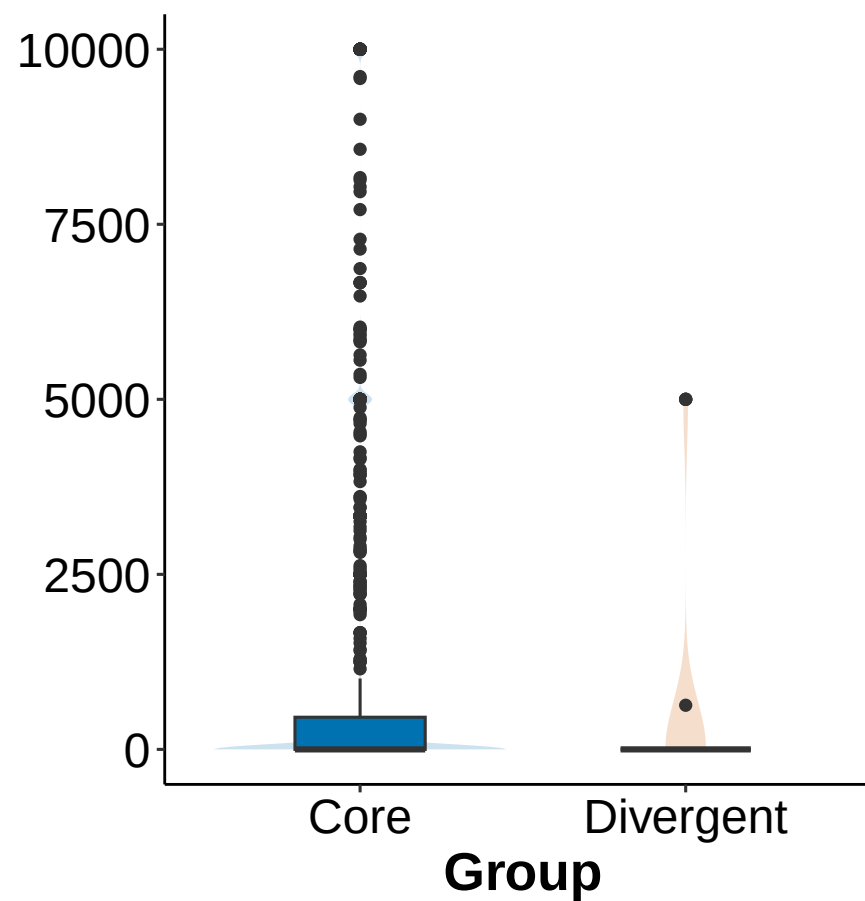
SA_Androgen_and_estrogen_synthe

p-adj: 0.721 | .=-0.04 (Negligible)



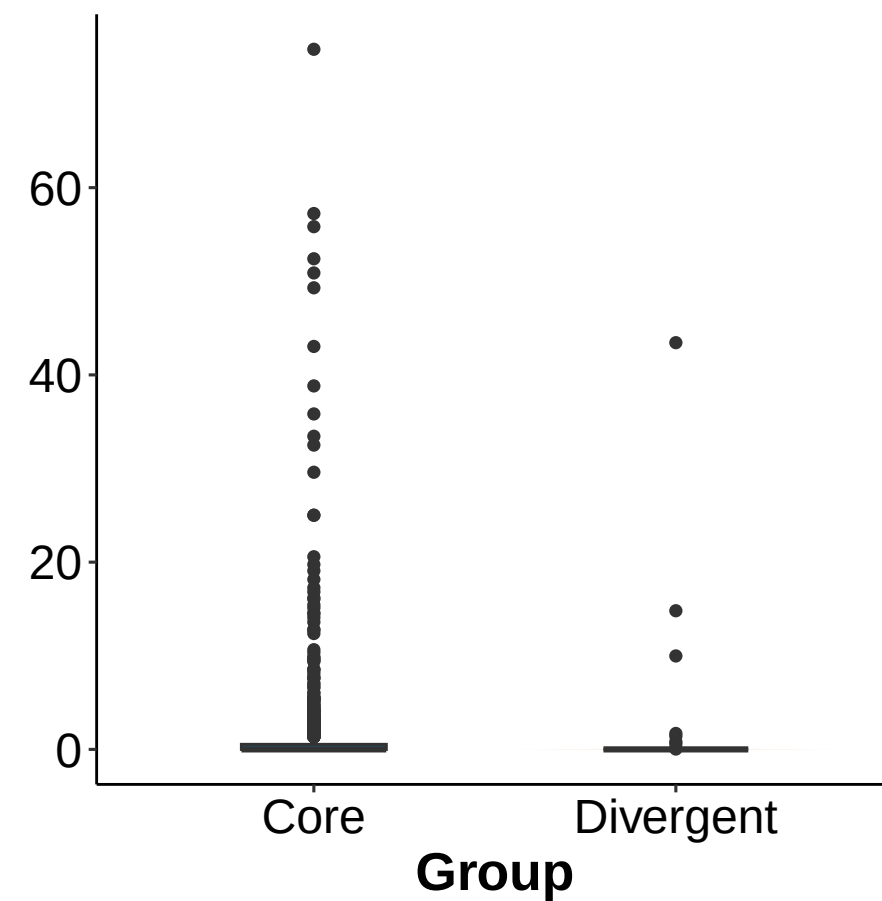
SA_Arachidonic_acid_metabolism

p-adj: 0.162 | .=-0.13 (Negligible)



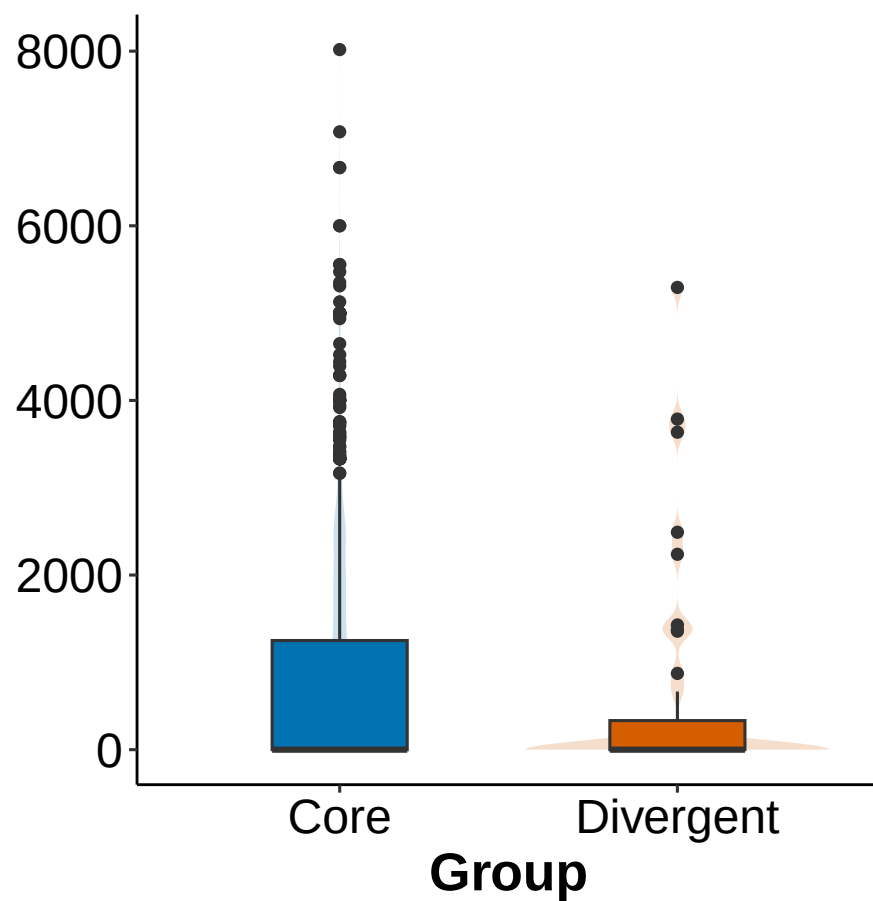
SA_Arachidonic_acid_metabolism_L1

p-adj: 0.078 | .=-0.2 (Small) | .%: -99.8



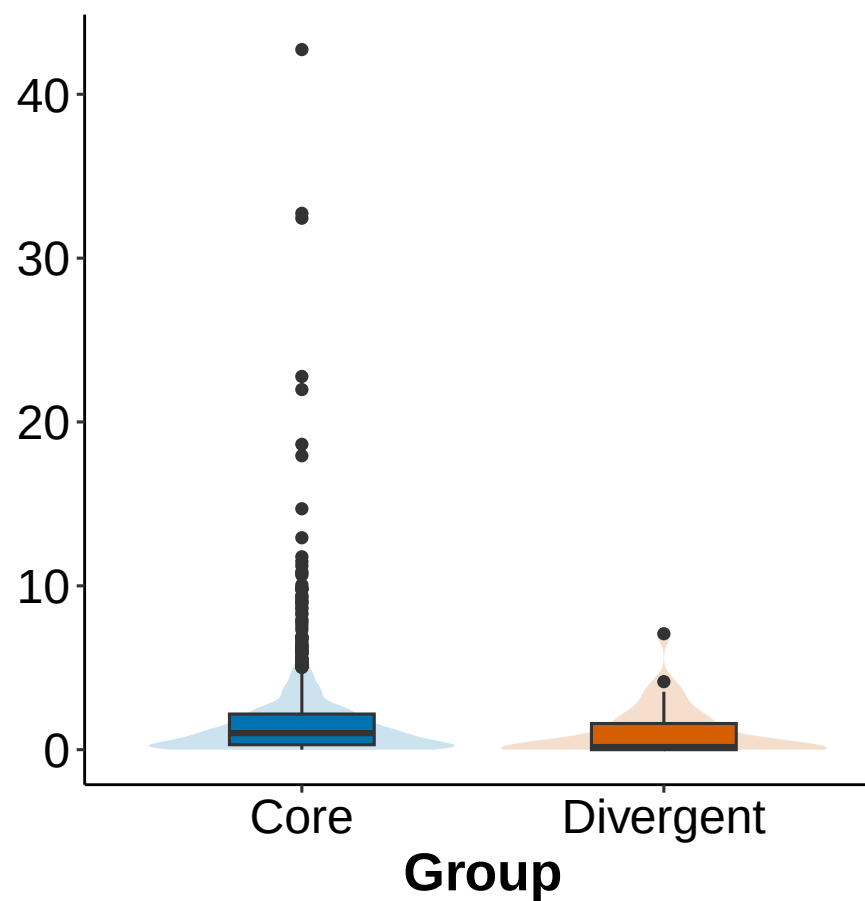
SA_Arginine_and_proline_metaboli

p-adj: 0.251 | .=-0.13 (Negligible)



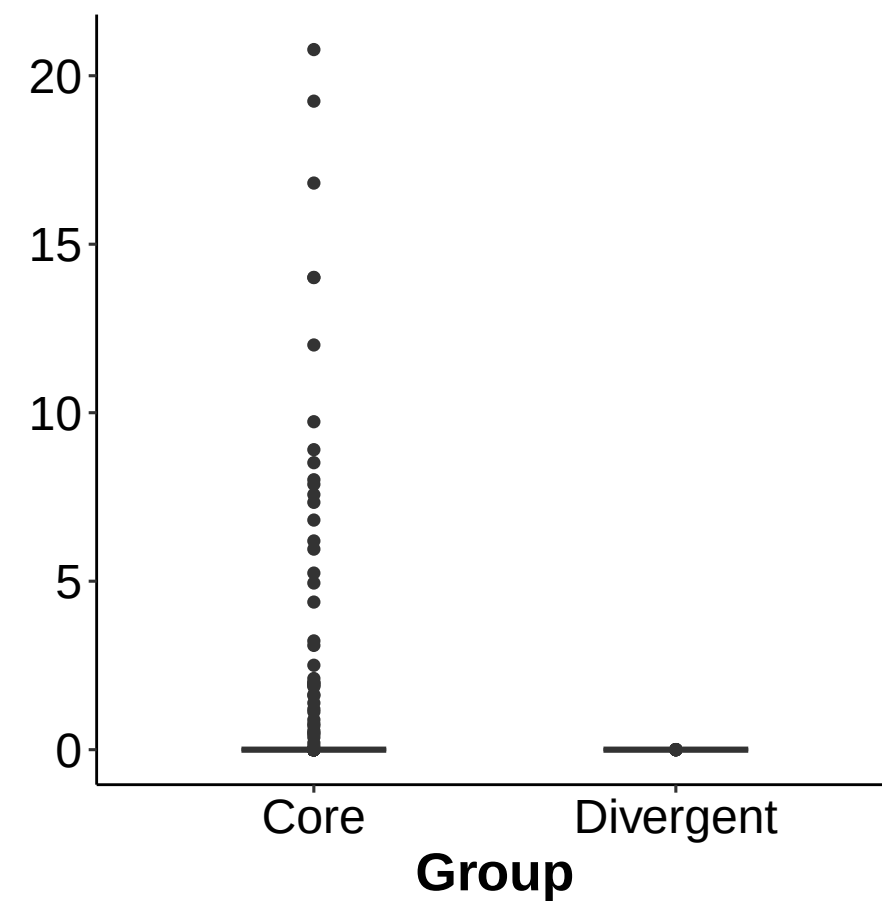
SA_Arginine_and_proline_metabolism

p-adj: 0.0151 | .=-0.26 (Small) | .%: -84.2



SA_Arginine_and_proline_metabolism

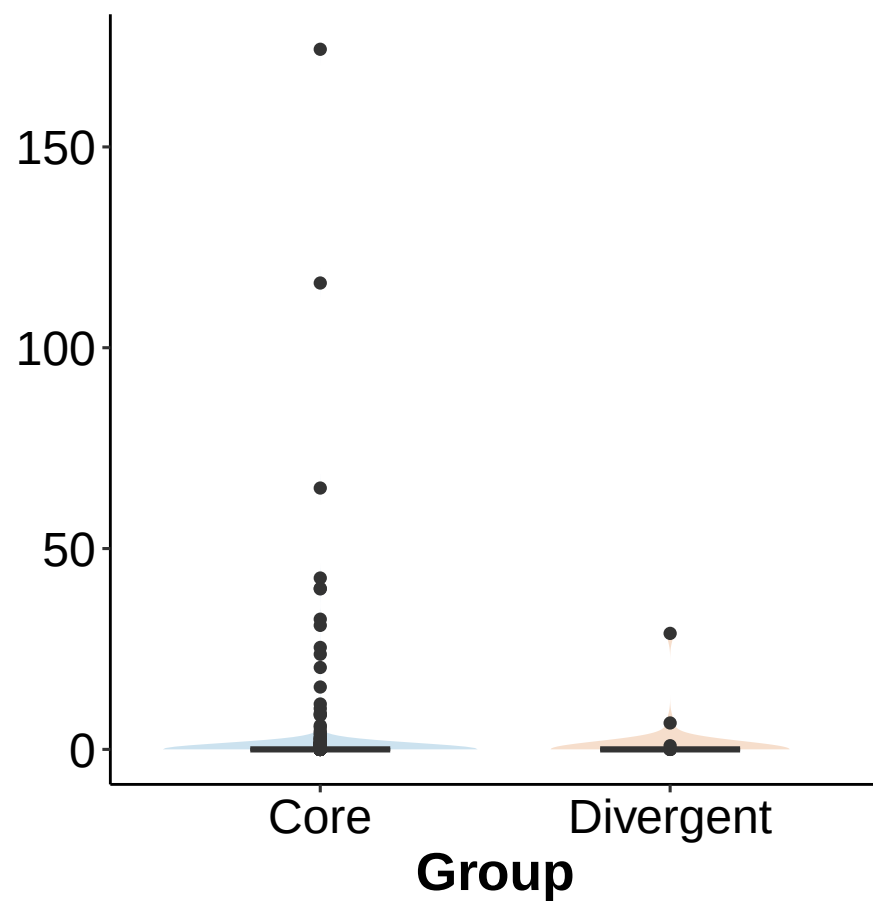
p-adj: 0.625 | .=0.05 (Negligible)



Metabolism – . 5. Systemic Reprogramming (SA_*)

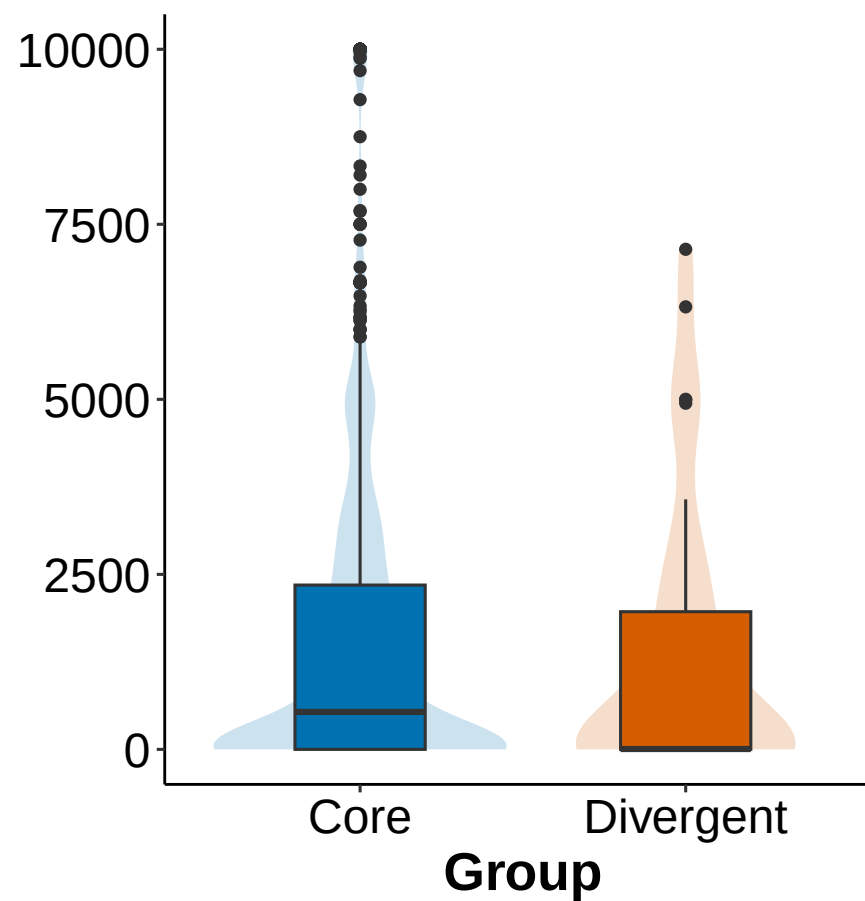
SA_Beta_Alanine_metabolism_L1w

p-adj: 0.566 | .:=0.05 (Negligible)



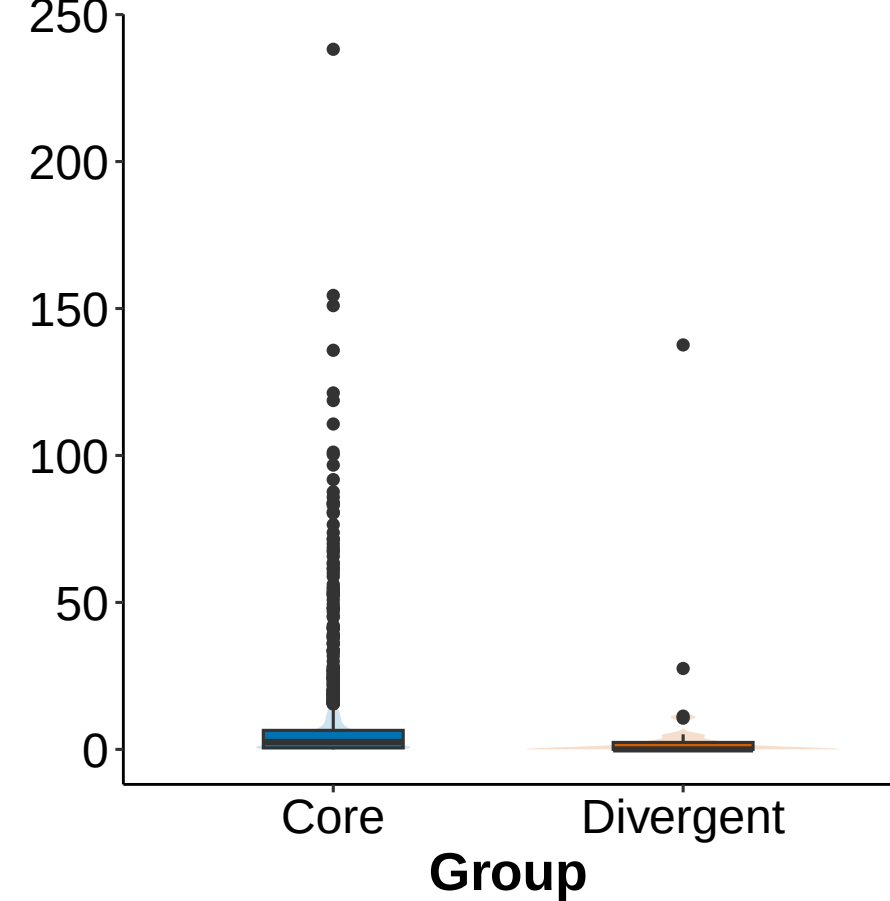
SA_Bile_acid_synthesis_FBA

p-adj: 0.574 | .:=−0.08 (Negligible) | .%: −99



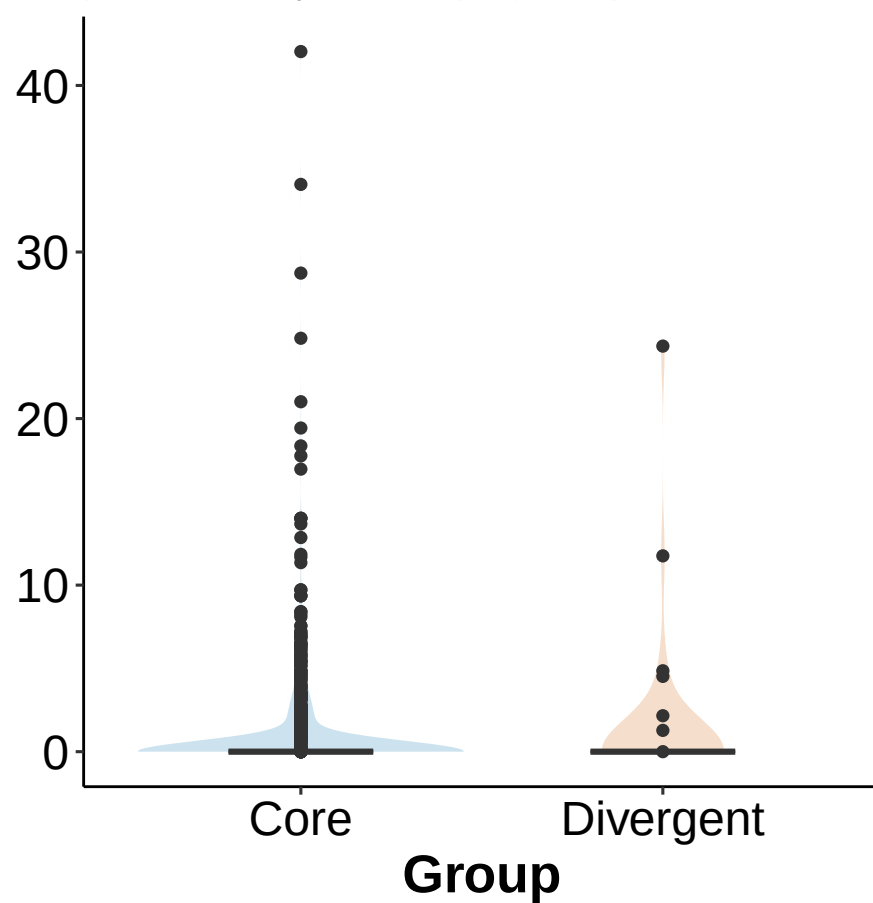
SA_Bile_acid_synthesis_L1w

p-adj: 0.000769 | .:=−0.36 (Medium) | .%: −100



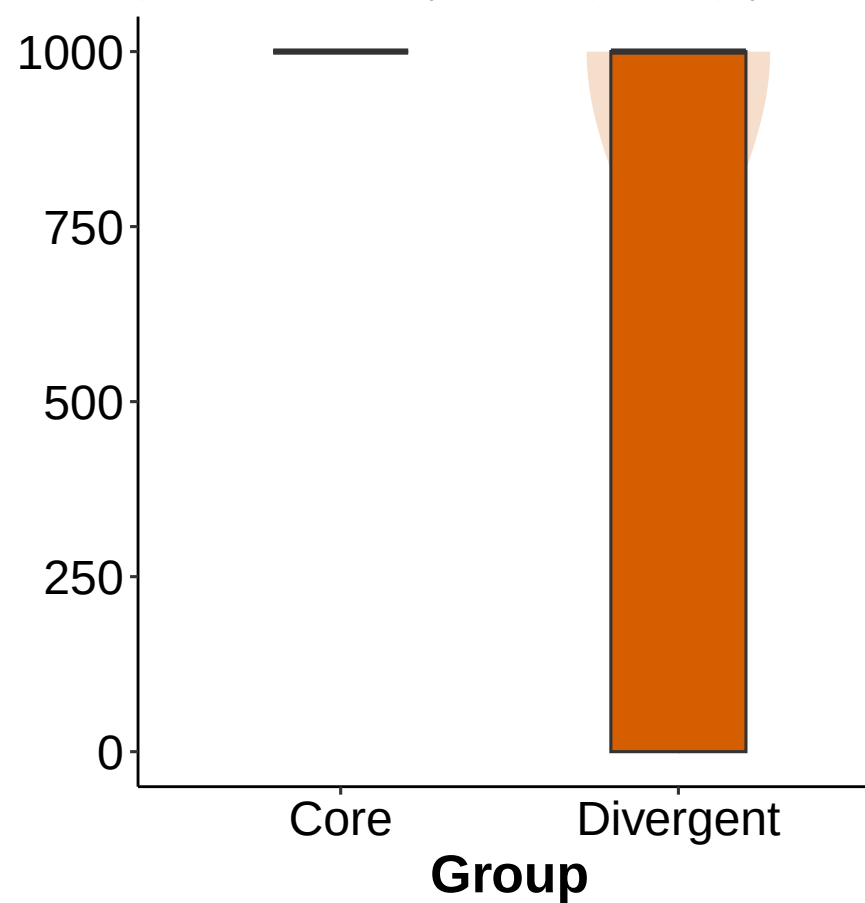
SA_Bile_acid_synthesis_pFBA

p-adj: 0.715 | .:=−0.04 (Negligible)



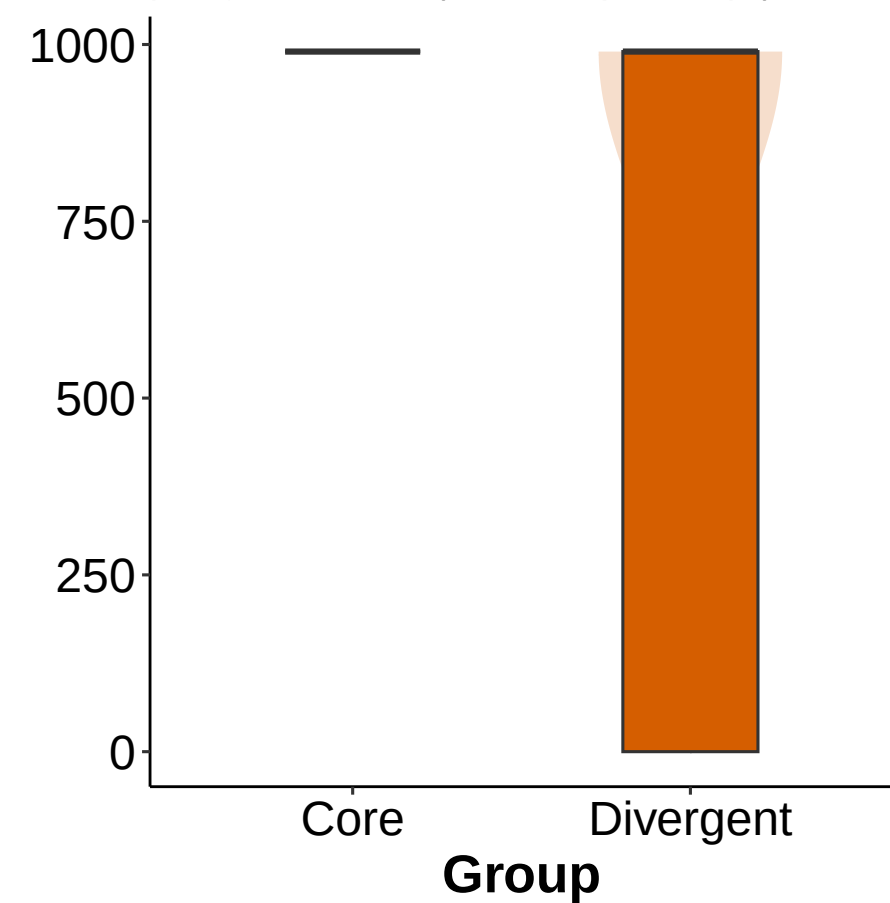
SA_Biomass_reactions_FBA

p-adj: 6.09e−102 | .:=−0.41 (Medium) | .%: 0



SA_Biomass_reactions_L1w

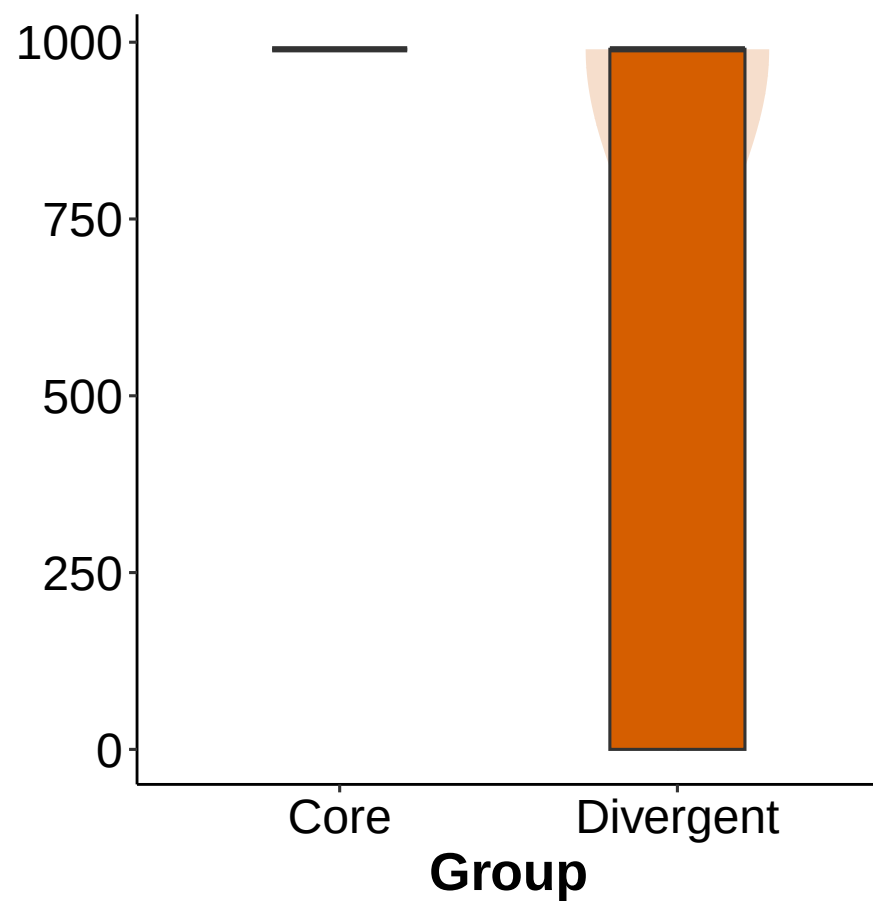
p-adj: 6.09e−102 | .:=−0.41 (Medium) | .%: 0



Metabolism – . 5. Systemic Reprogramming (SA_*)

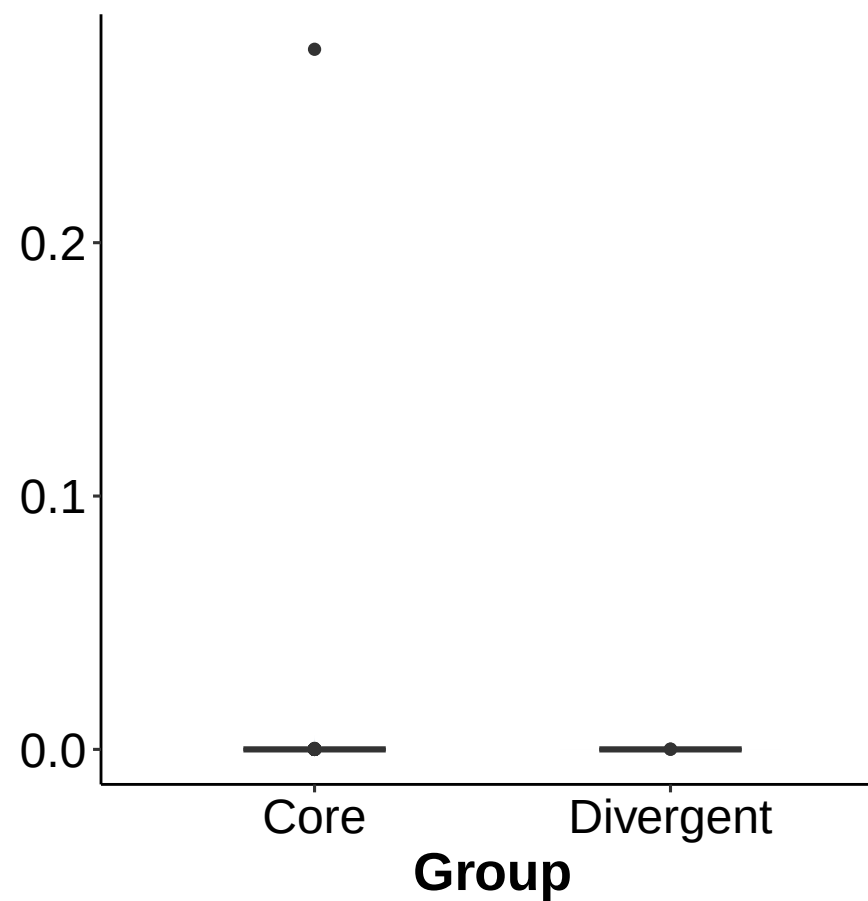
SA_Biomass_reactions_pFBA

p-adj: 6.09e-102 | .=-0.41 (Medium) | .%: 0



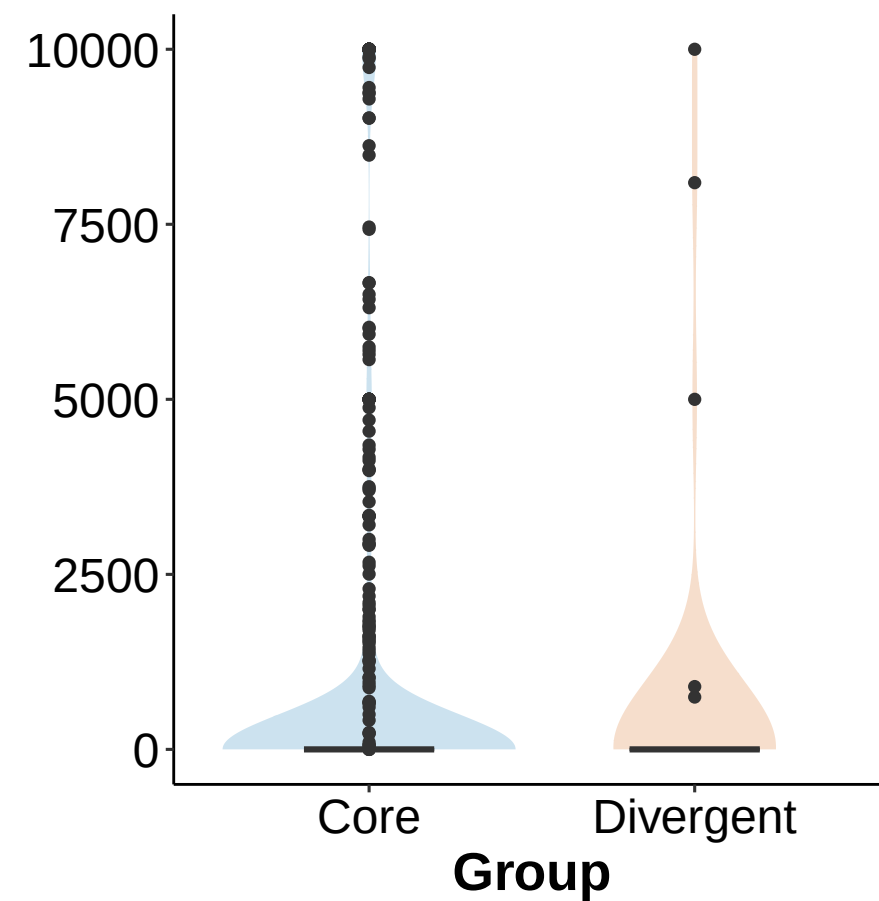
SA_Biotin_metabolism_L1w

p-adj: 0.507 | .=-0.05 (Negligible)



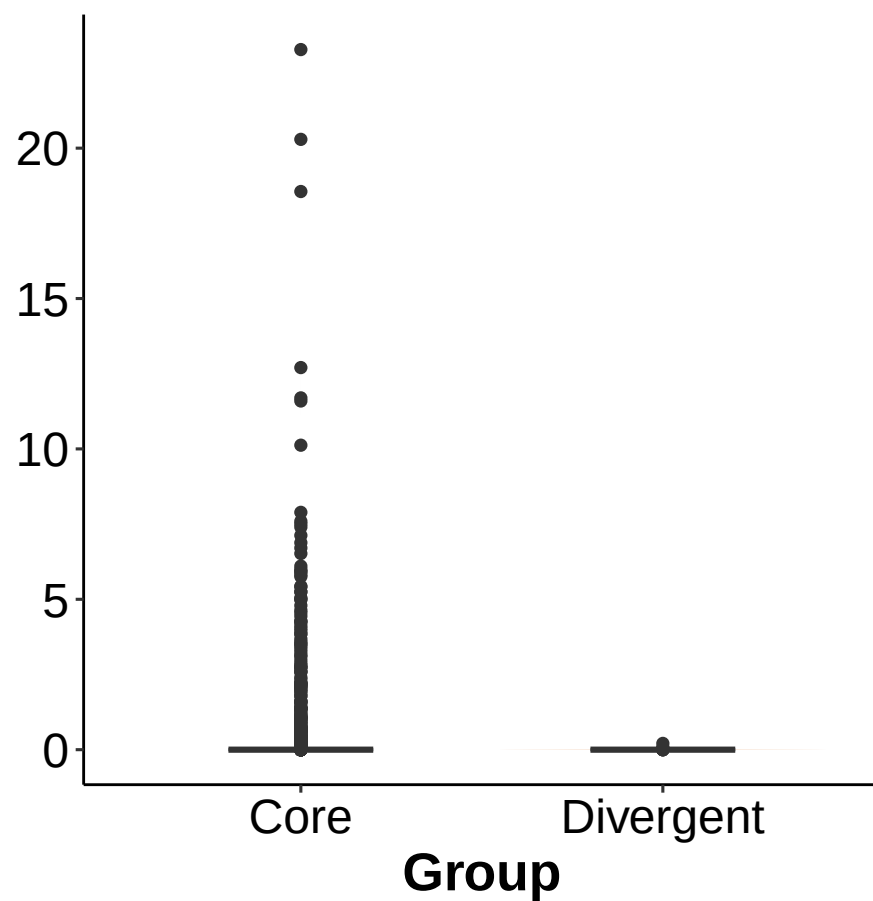
SA_Butanoate_metabolism_FBA

p-adj: 0.908 | .=-0.01 (Negligible)



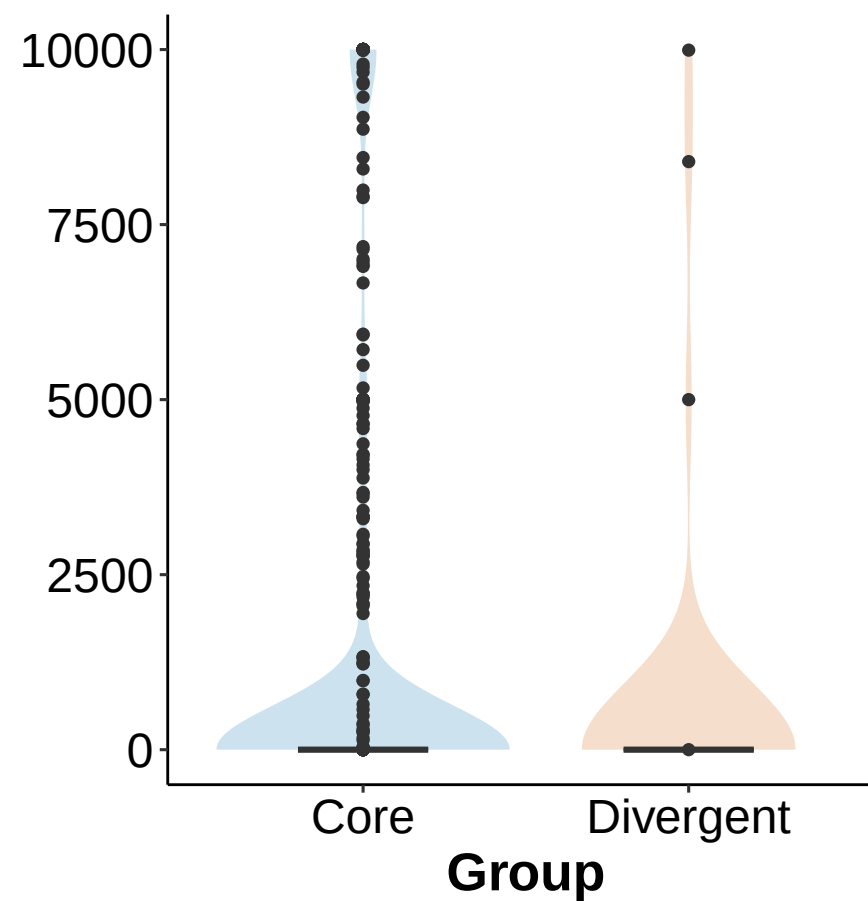
SA_Butanoate_metabolism_L1w

p-adj: 0.599 | .=-0.06 (Negligible)



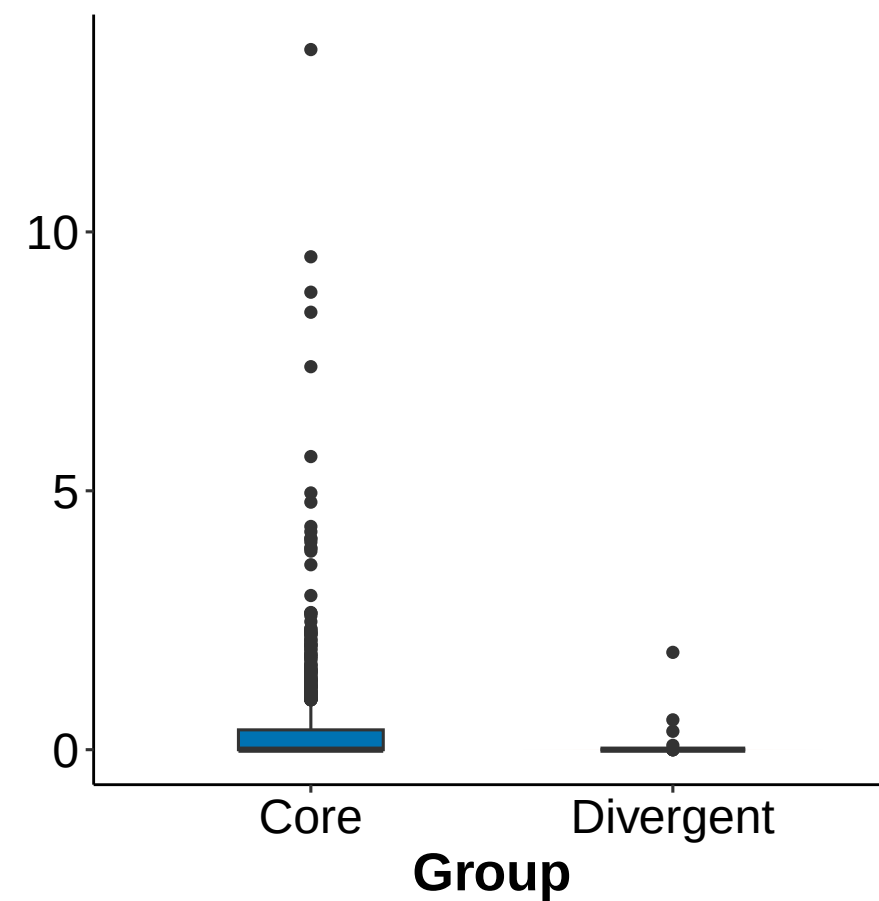
SA_C5_branched_dibasic_acid_m

p-adj: 0.323 | .=-0.09 (Negligible)



SA_C5_branched_dibasic_acid_metal

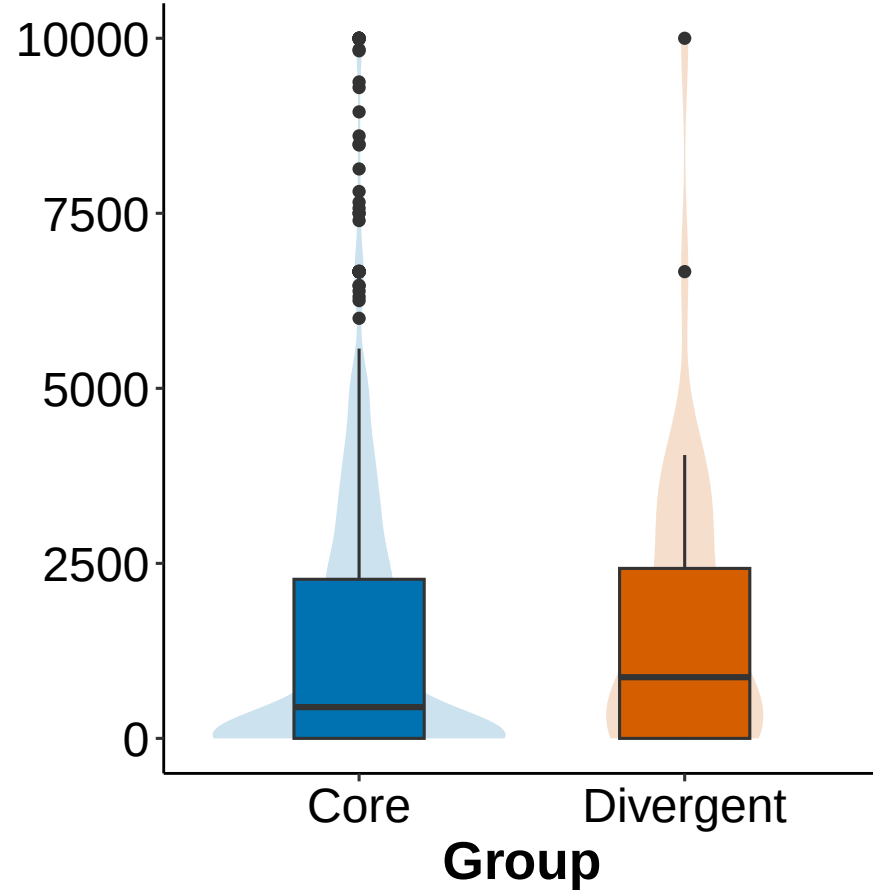
p-adj: 0.078 | .=-0.18 (Small)



Metabolism – . 5. Systemic Reprogramming (SA_*)

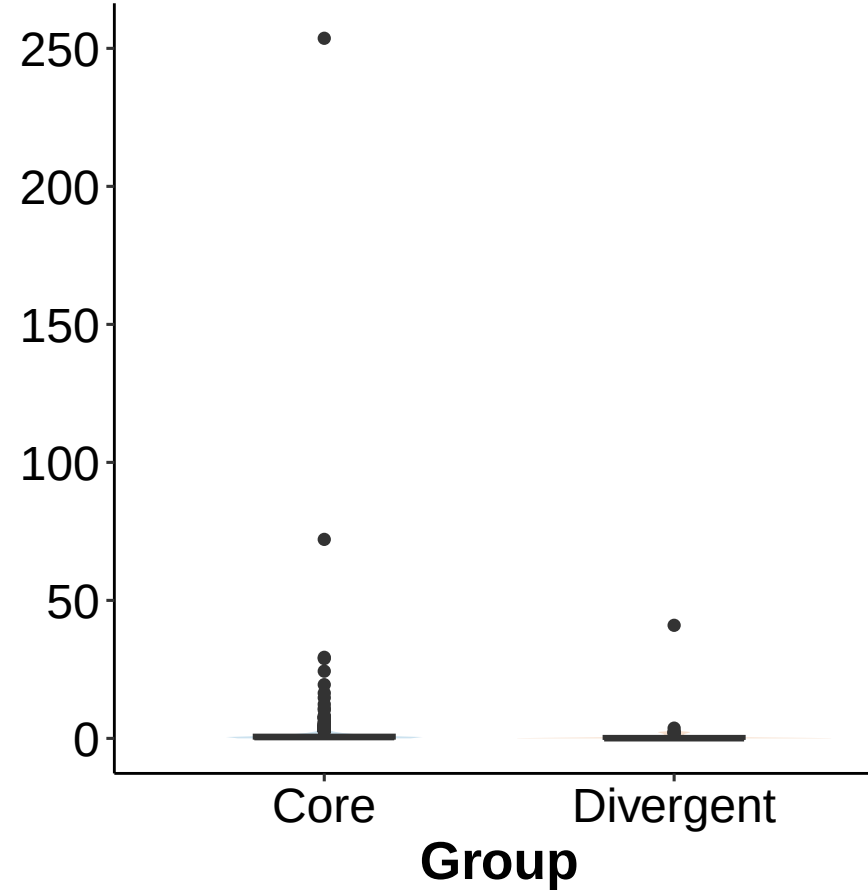
SA_Cholesterol_metabolism_FBA

p-adj: 0.721 | . = 0.05 (Negligible) | .%: 95.9



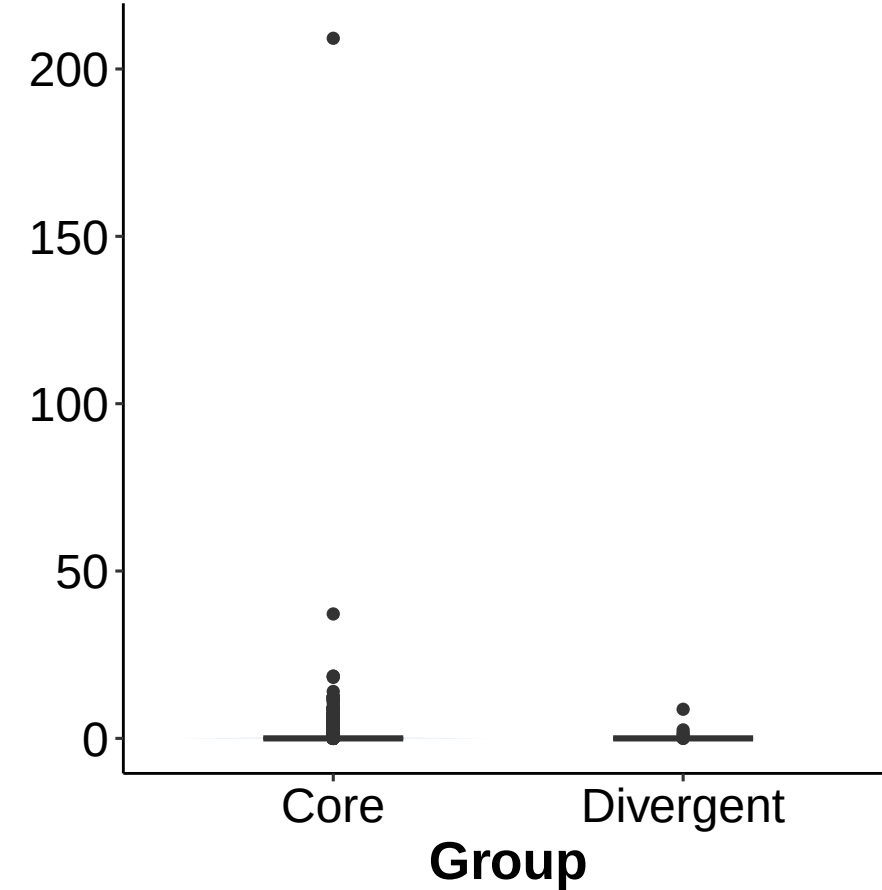
SA_Cholesterol_metabolism_L1w

p-adj: 0.0207 | . = -0.25 (Small) | .%: -99.4



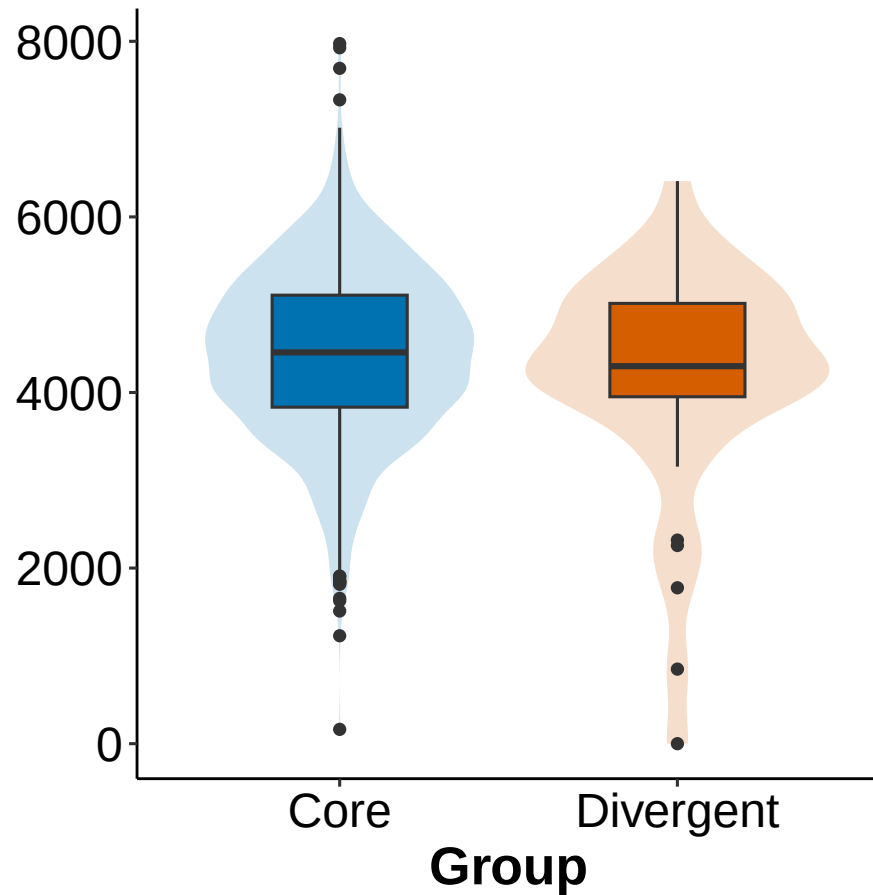
SA_Cholesterol_metabolism_pFBA

p-adj: 0.771 | . = 0.03 (Negligible)



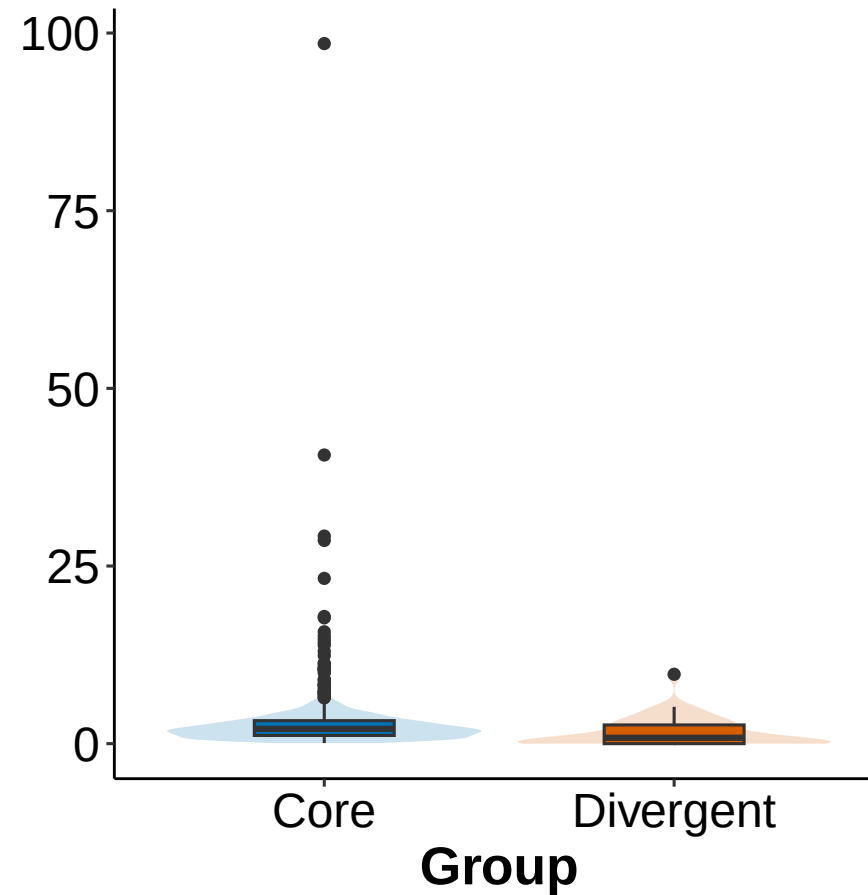
SA_Citric_acid_cycle_FBA

p-adj: 0.614 | . = -0.07 (Negligible) | .%: -3.5



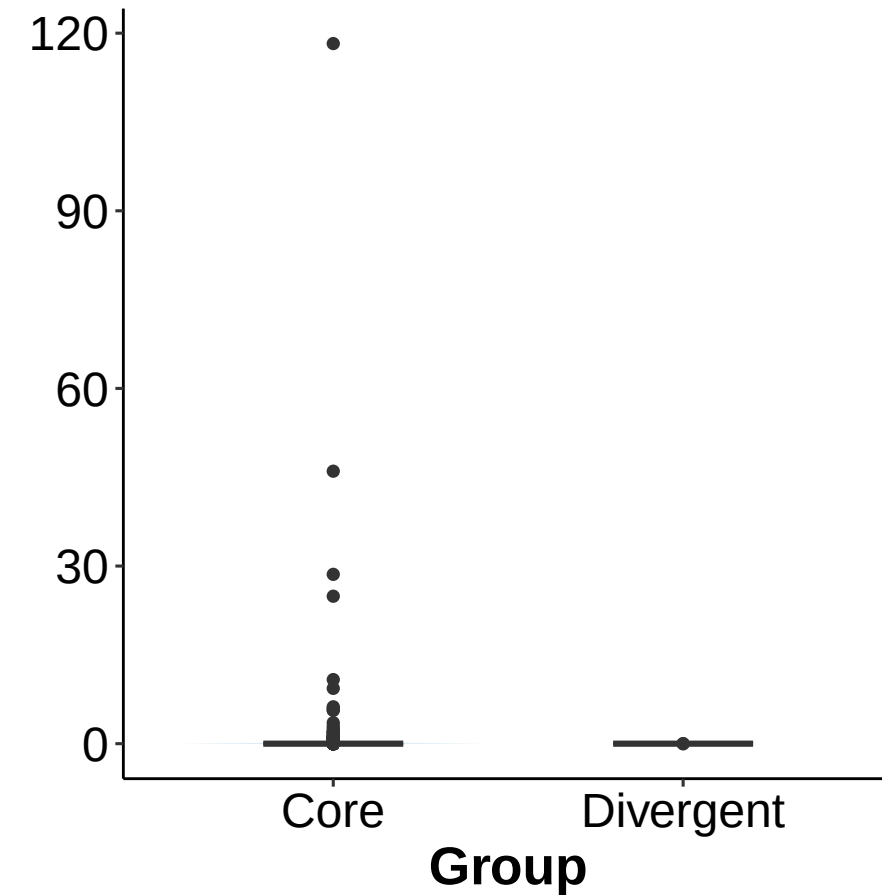
SA_Citric_acid_cycle_L1w

p-adj: 0.000197 | . = -0.4 (Medium) | .%: -60



SA_Citric_acid_cycle_pFBA

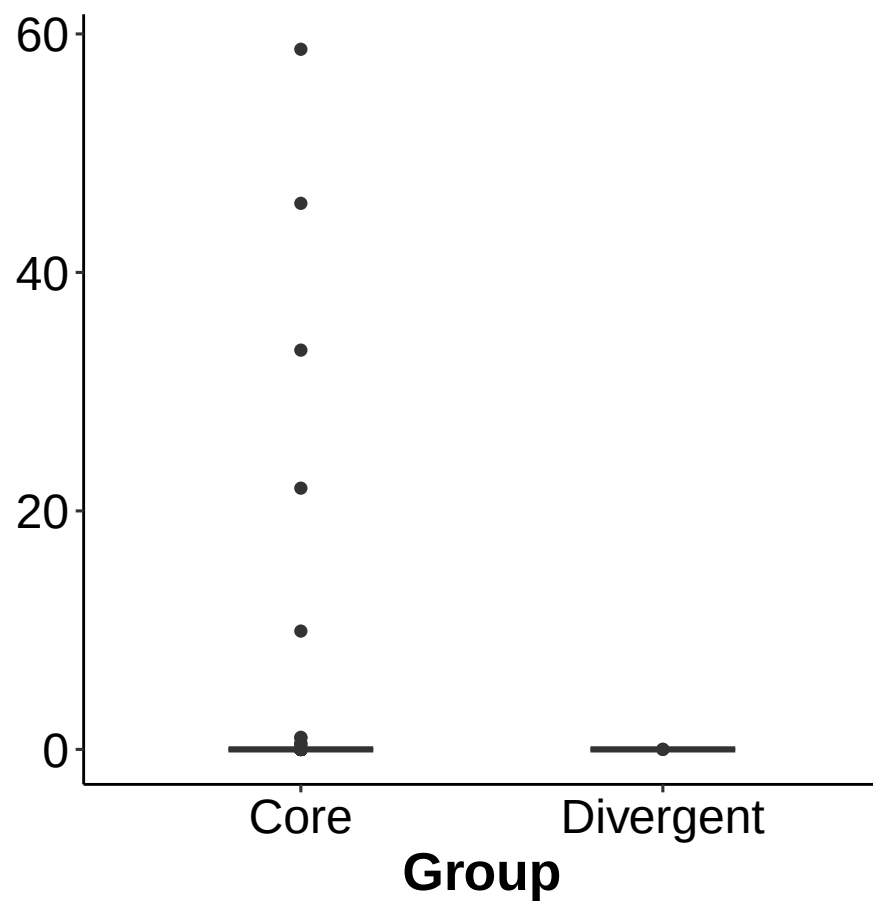
p-adj: 0.103 | . = 0.19 (Small) | .%: 31.7



Metabolism – 5. Systemic Reprogramming (SA_*)

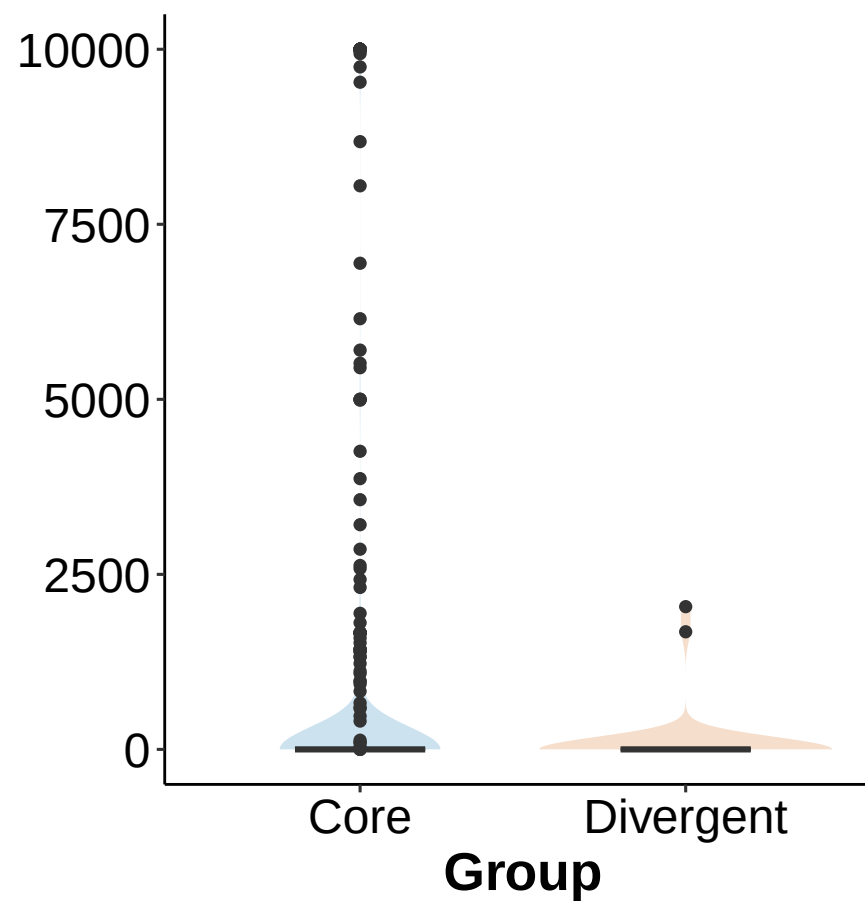
SA_CoA_catabolism_L1w

p-adj: 0.761 | .=-0.02 (Negligible)



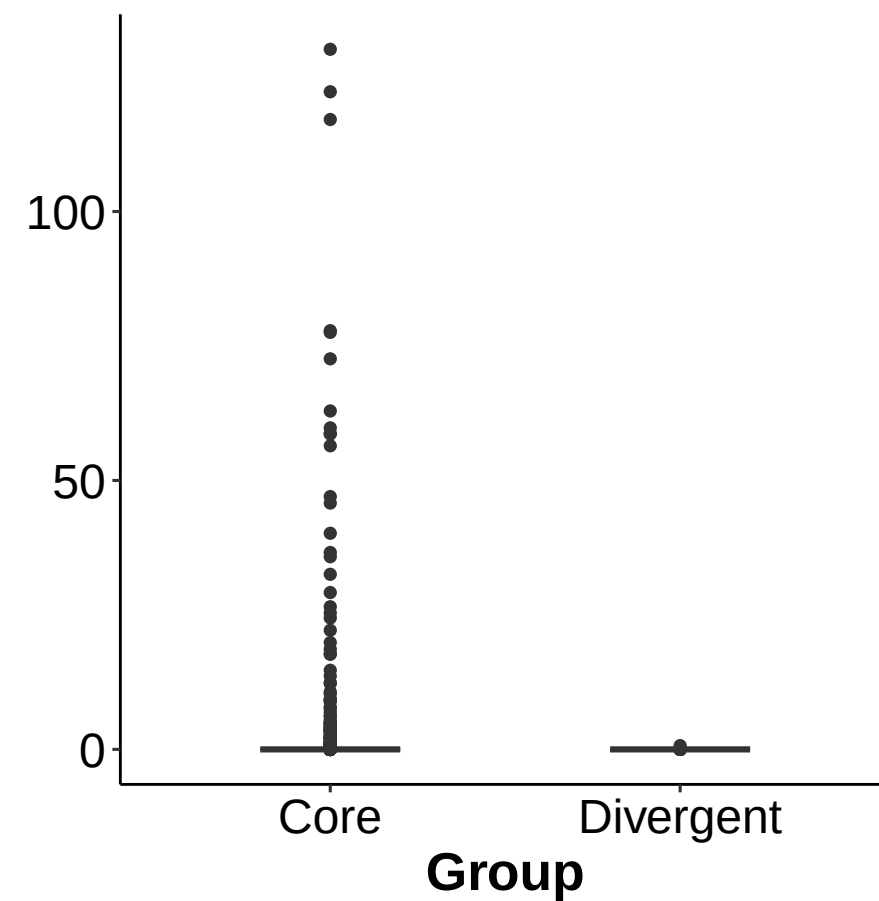
SA_CoA_synthesis_FBA

p-adj: 0.762 | .=-0.02 (Negligible)



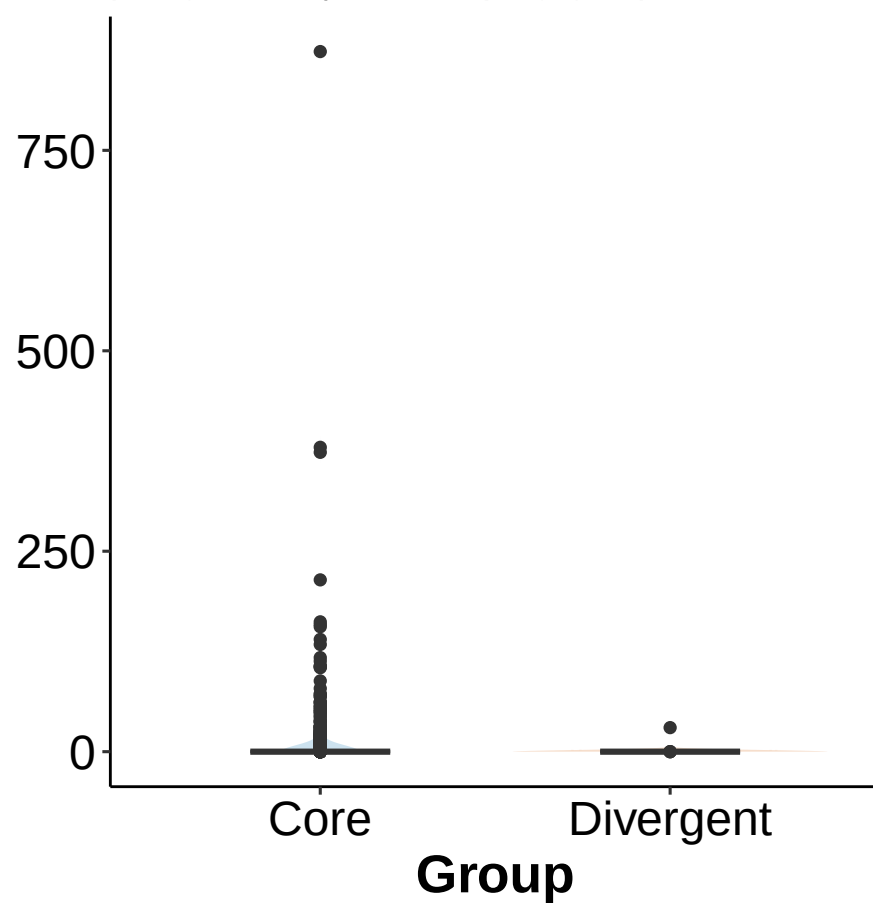
SA_CoA_synthesis_L1w

p-adj: 0.563 | .=-0.08 (Negligible)



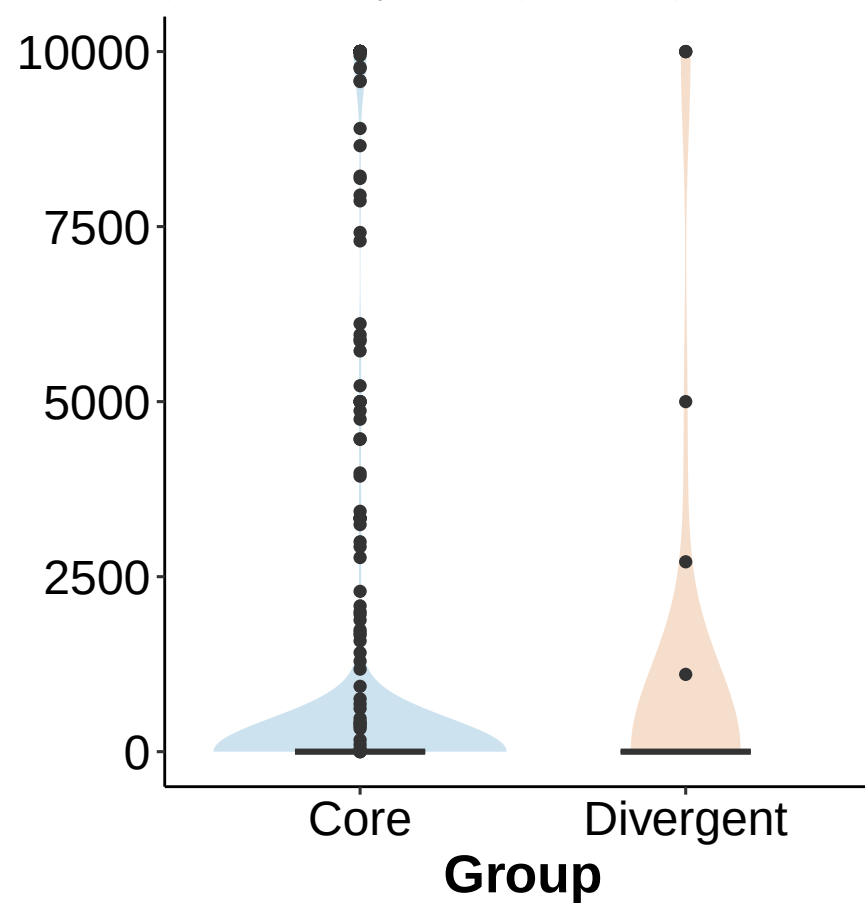
SA_Cytochrome_metabolism_L1w

p-adj: 0.352 | .=-0.08 (Negligible)



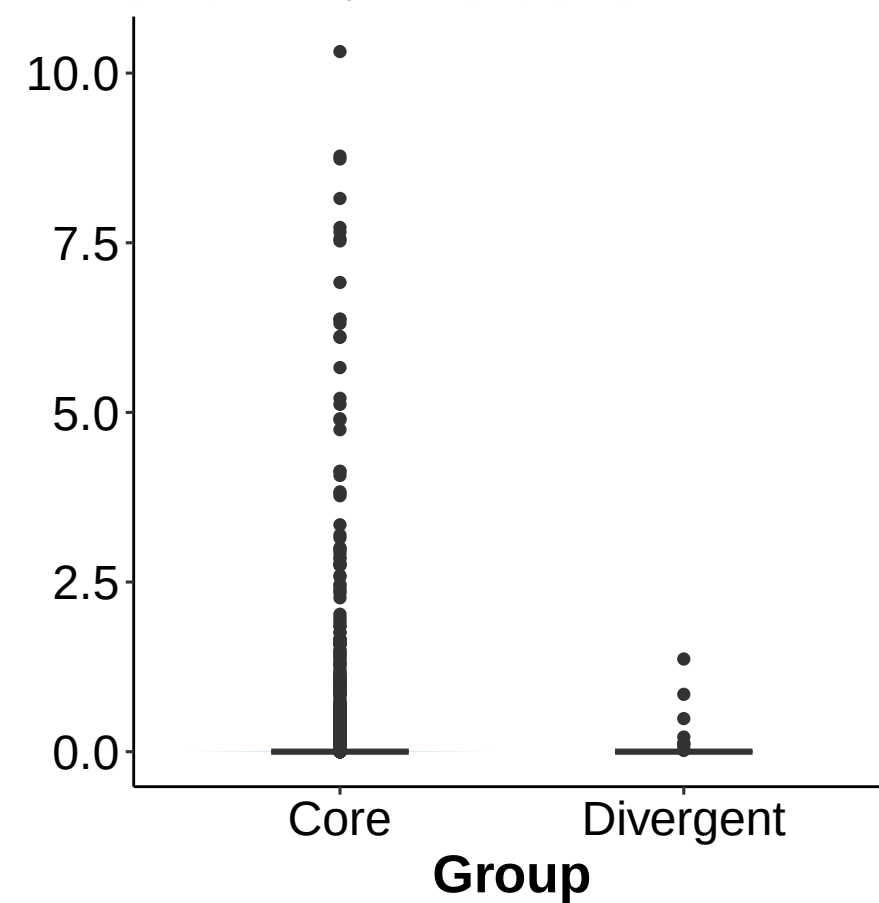
SA_D_alanine_metabolism_FBA

p-adj: 0.404 | .=0.06 (Negligible)



SA_D_alanine_metabolism_L1w

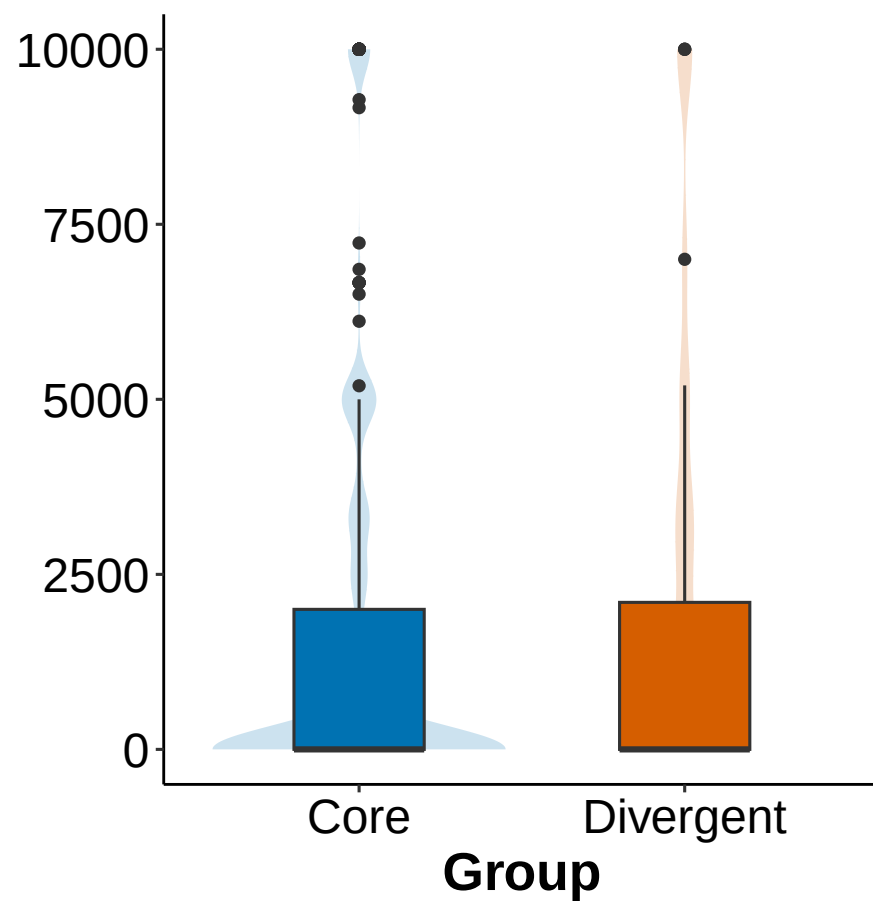
p-adj: 0.388 | .=0.1 (Negligible)



Metabolism – . 5. Systemic Reprogramming (SA_*)

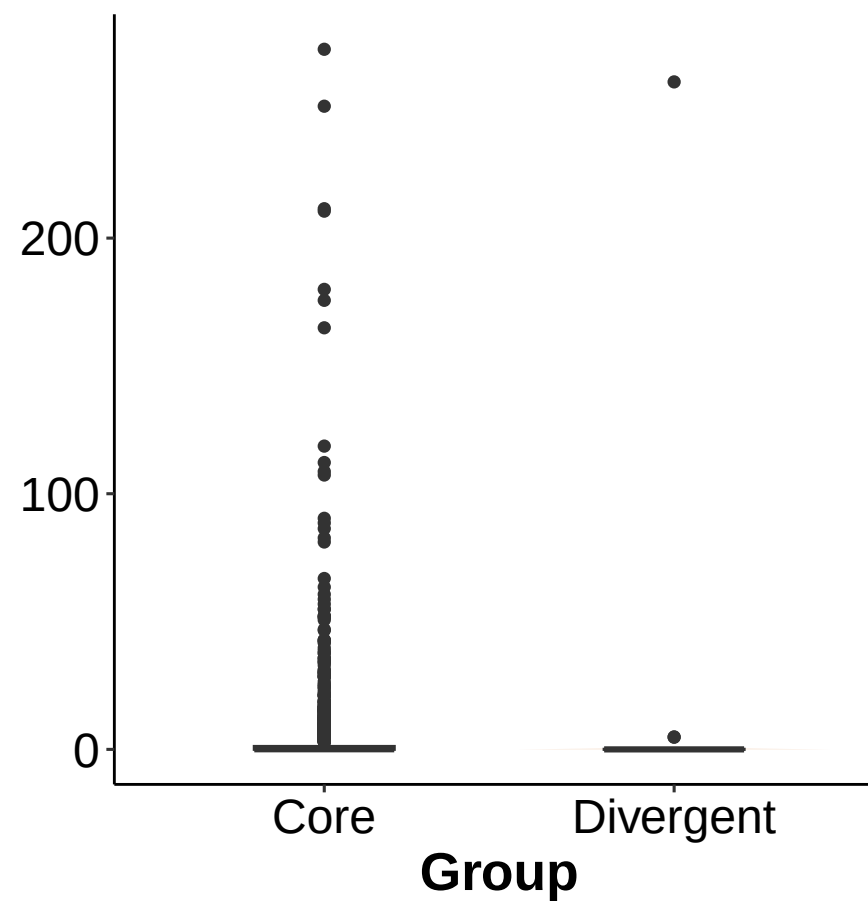
SA_Dietary_fiber_binding_FBA

p-adj: 0.663 | .:=0.05 (Negligible)



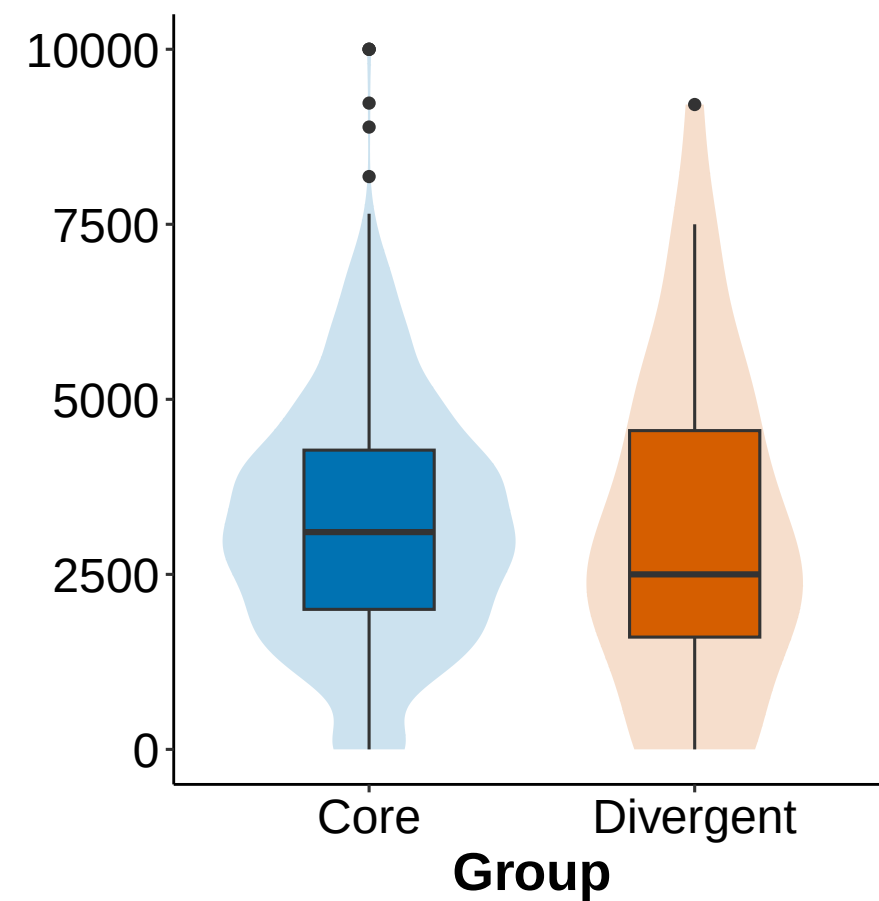
SA_Dietary_fiber_binding_L1w

p-adj: 3.35e-05 | .:= -0.44 (Medium) | .%: -94.9



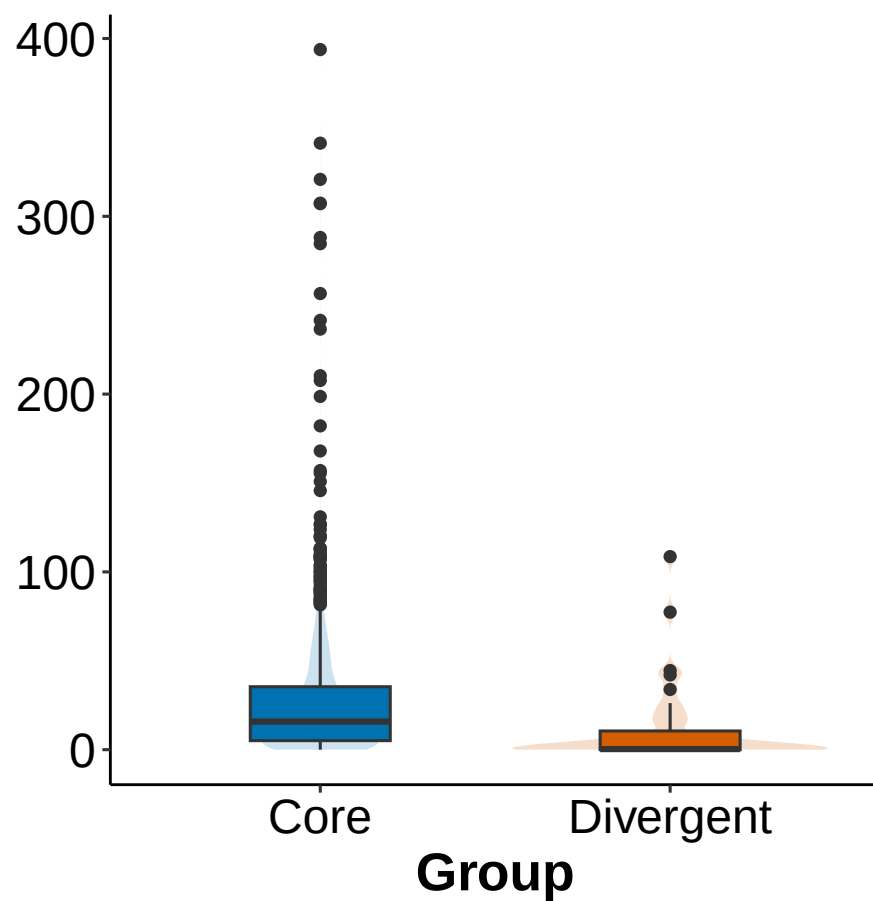
SA_Drug_metabolism_FBA

p-adj: 0.58 | .:= -0.08 (Negligible) | .%: -19.4



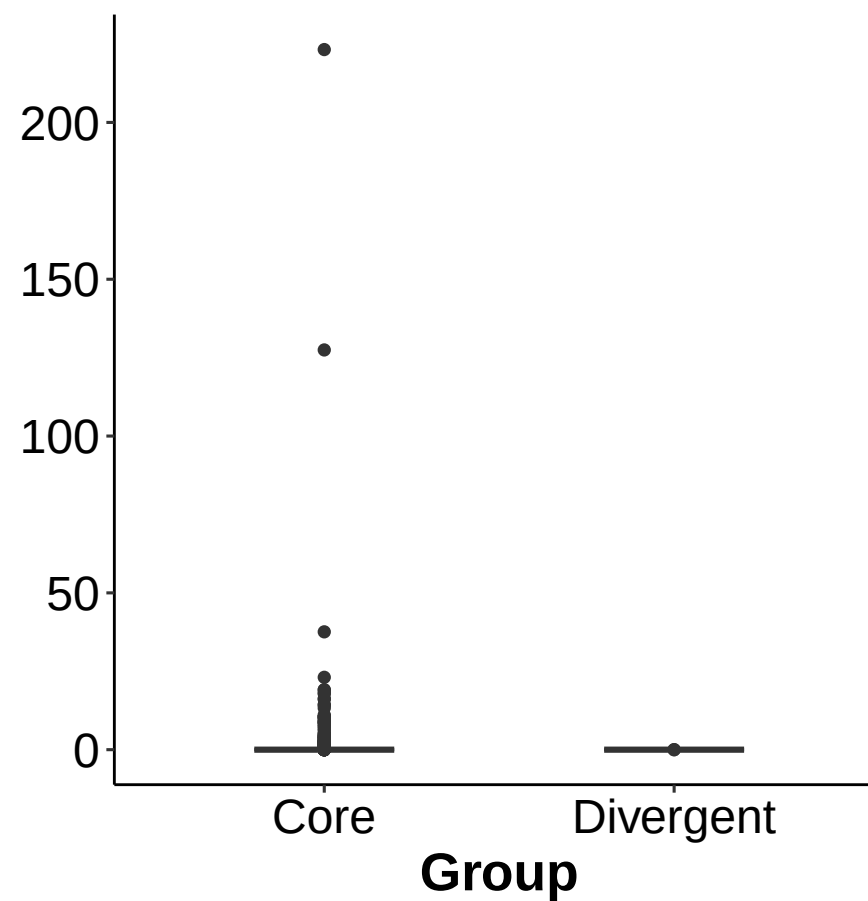
SA_Drug_metabolism_L1w

p-adj: 1.2e-06 | .:= -0.52 (Large) | .%: -97.9



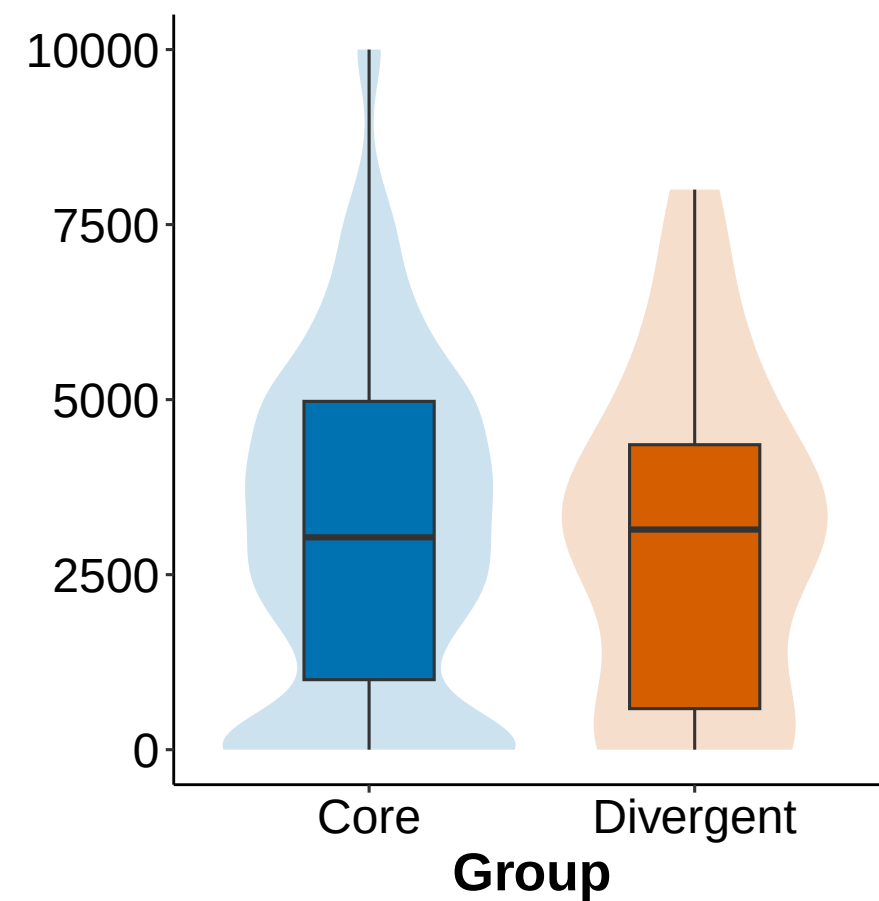
SA_Drug_metabolism_pFBA

p-adj: 0.28 | .:= -0.08 (Negligible)



SA_Eicosanoid_metabolism_FBA

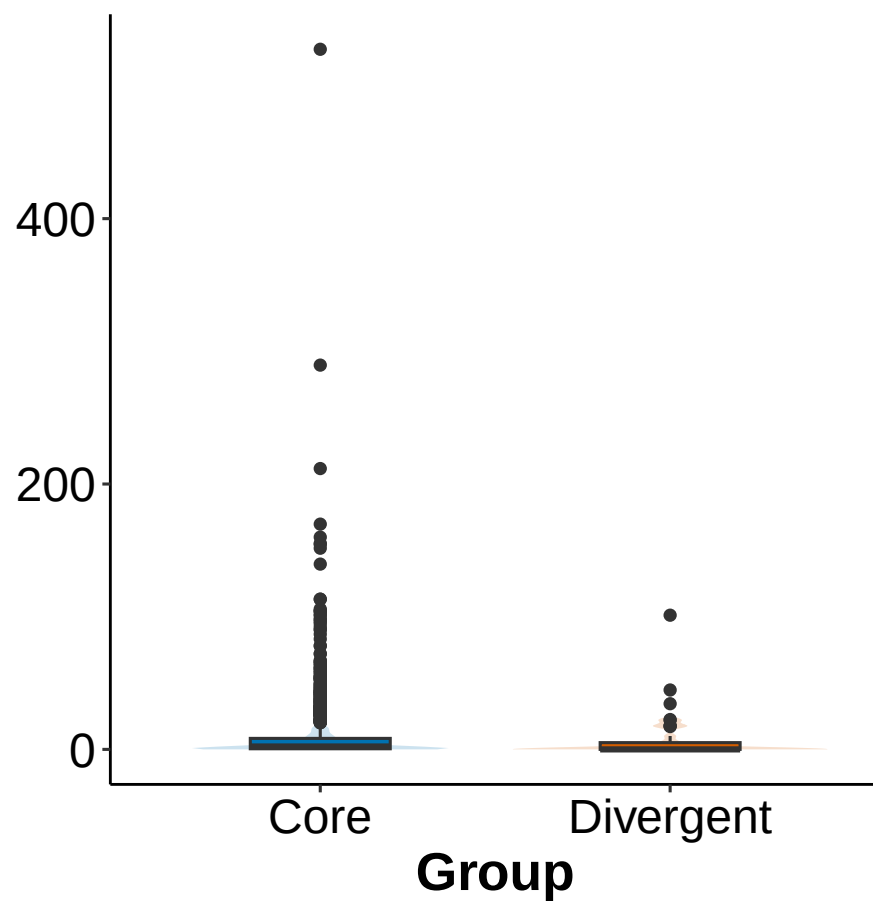
p-adj: 0.762 | .:= -0.04 (Negligible) | .%: 3.7



Metabolism – . 5. Systemic Reprogramming (SA_*)

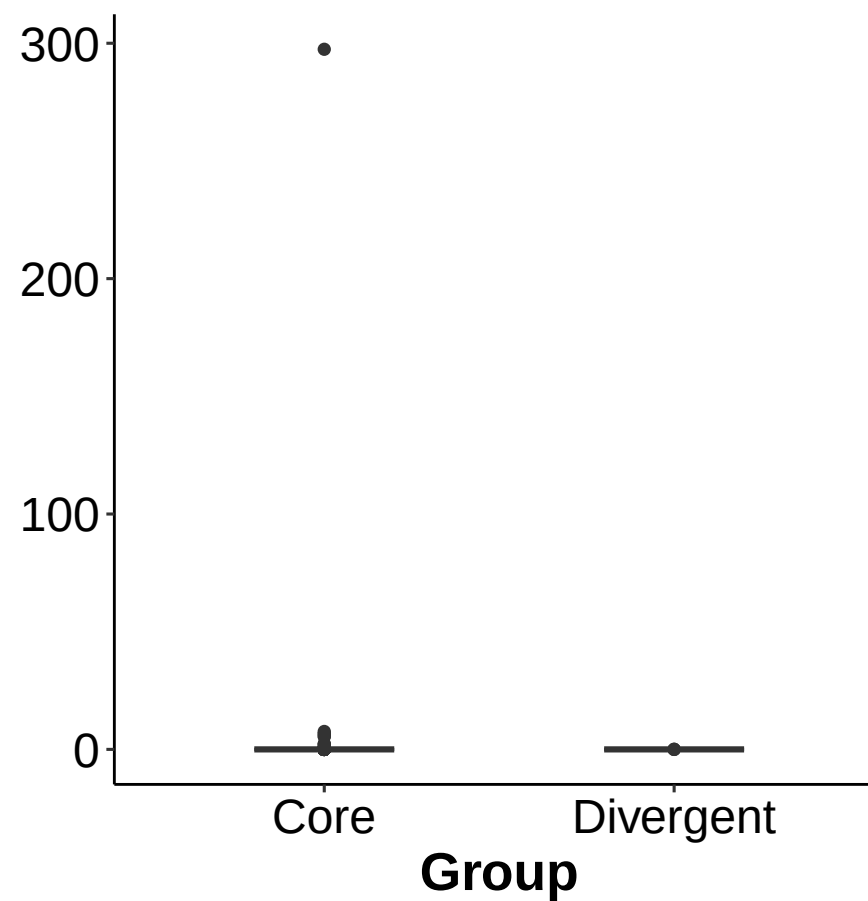
SA_Eicosanoid_metabolism_L1w

p-adj: 0.000565 | .=-0.37 (Medium) | .%: -99.4



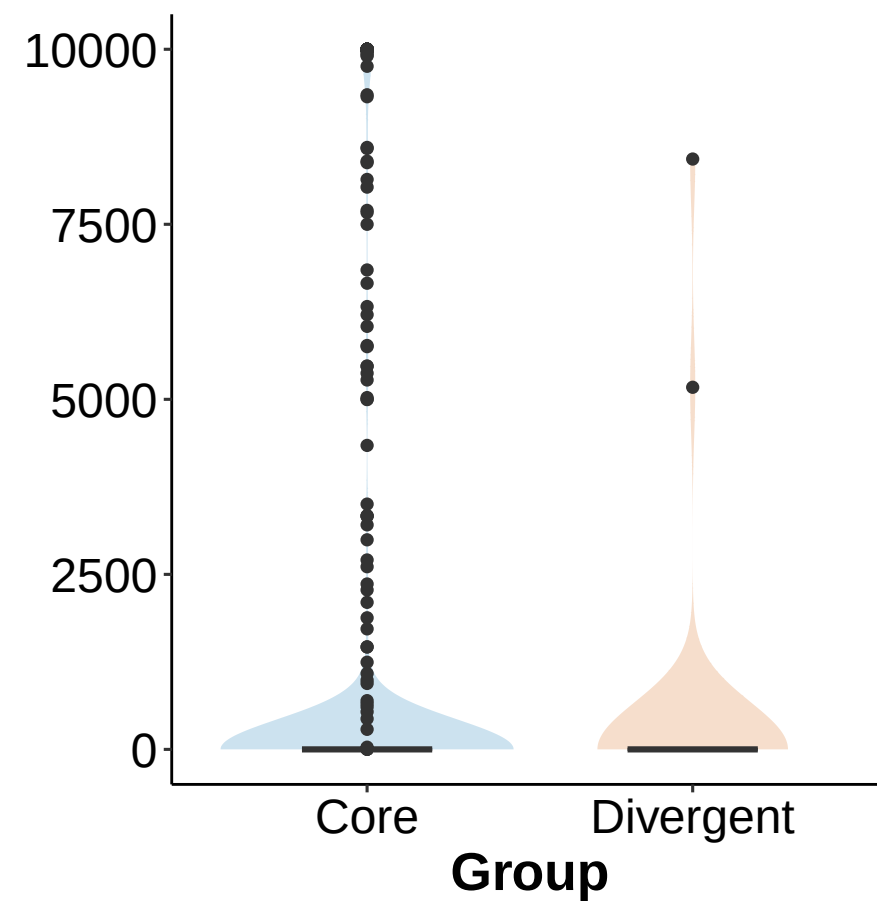
SA_Eicosanoid_metabolism_pFBA

p-adj: 0.82 | .=-0.02 (Negligible)



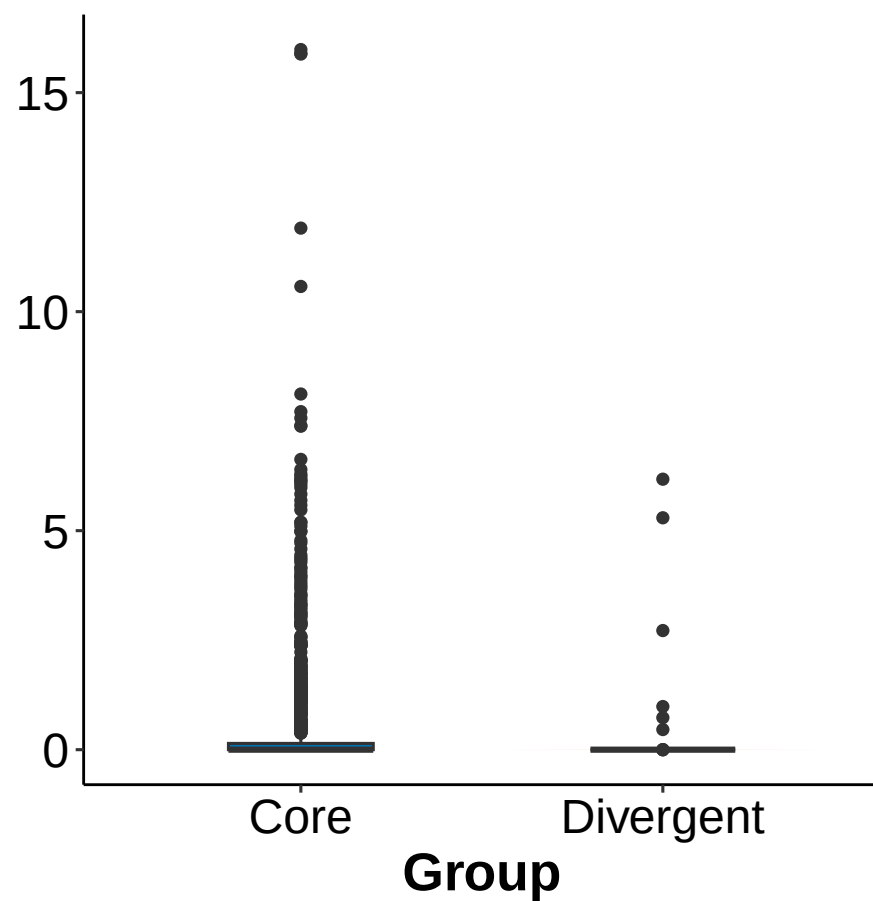
SA_Exchange_demand_reaction_L1w

p-adj: 0.58 | .=-0.04 (Negligible)



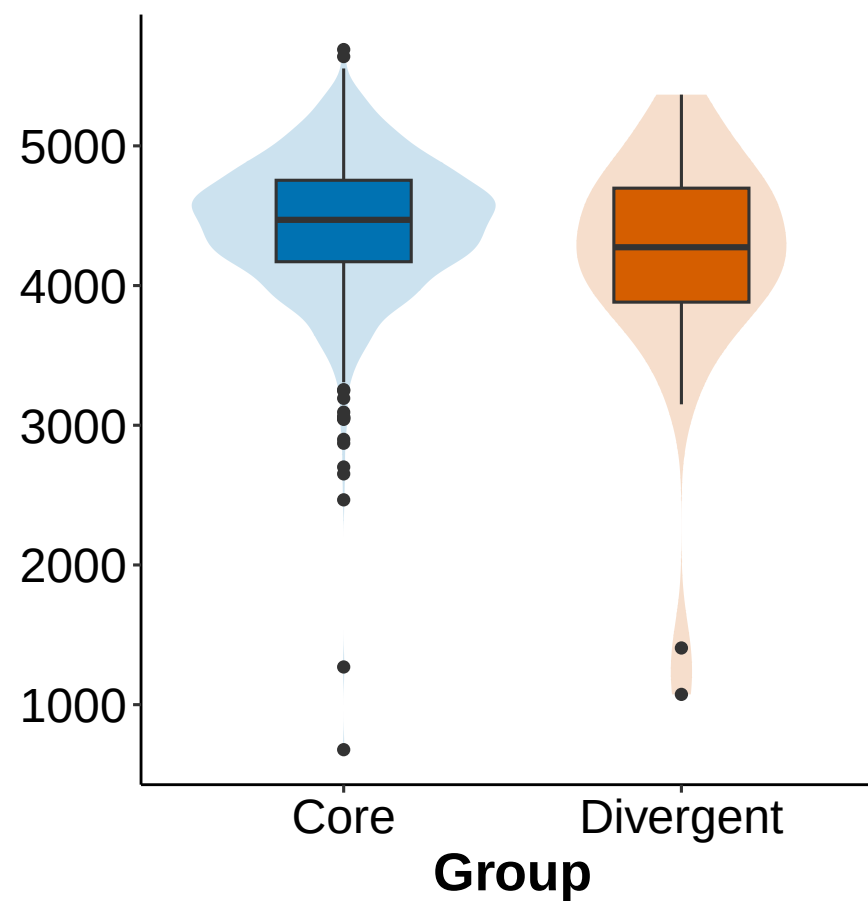
SA_Exchange_demand_reaction_L1w

p-adj: 0.949 | .=-0.01 (Negligible)



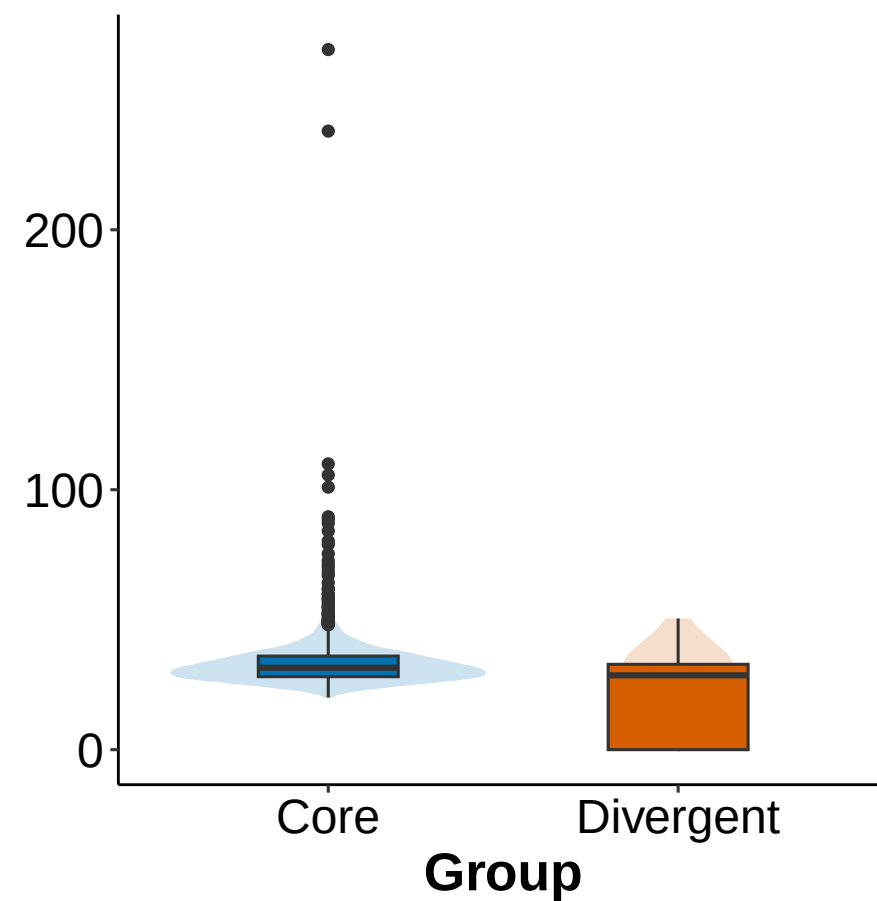
SA_Extracellular_exchange_FBA

p-adj: 0.111 | .=-0.19 (Small) | .%: -4.4



SA_Extracellular_exchange_L1w

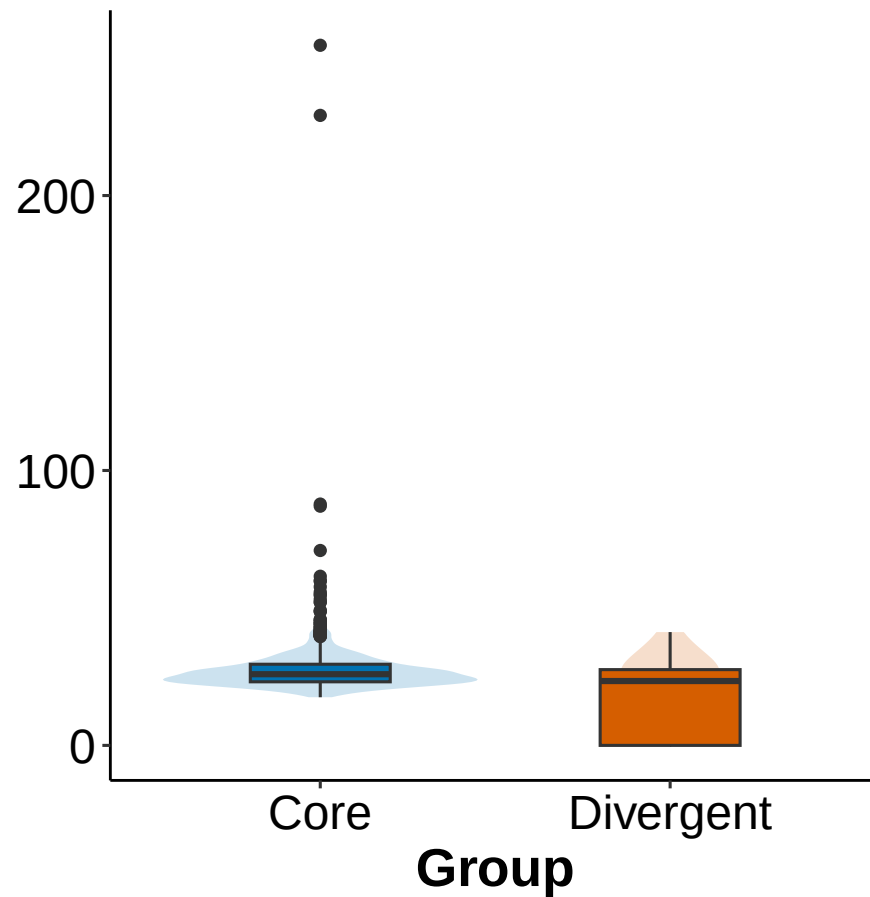
p-adj: 0.00158 | .=-0.34 (Medium) | .%: -9.3



Metabolism – 5. Systemic Reprogramming (SA_*)

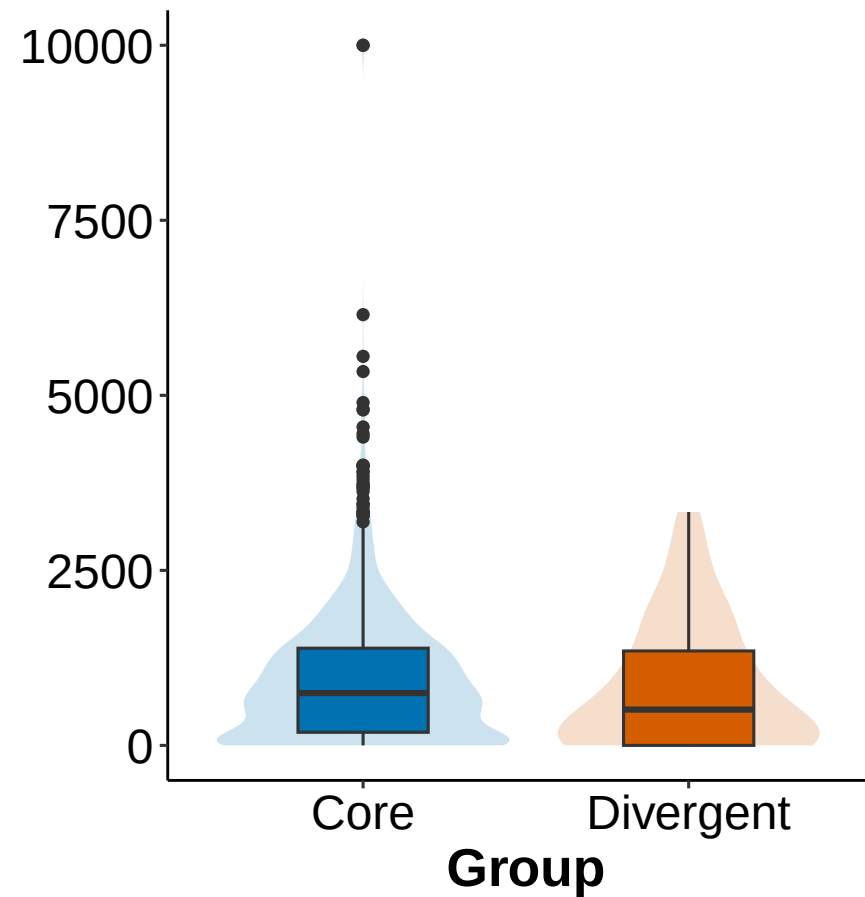
SA_Extracellular_exchange_pFBA

p-adj: 0.00241 | .=-0.32 (Small) | .%: -9.6



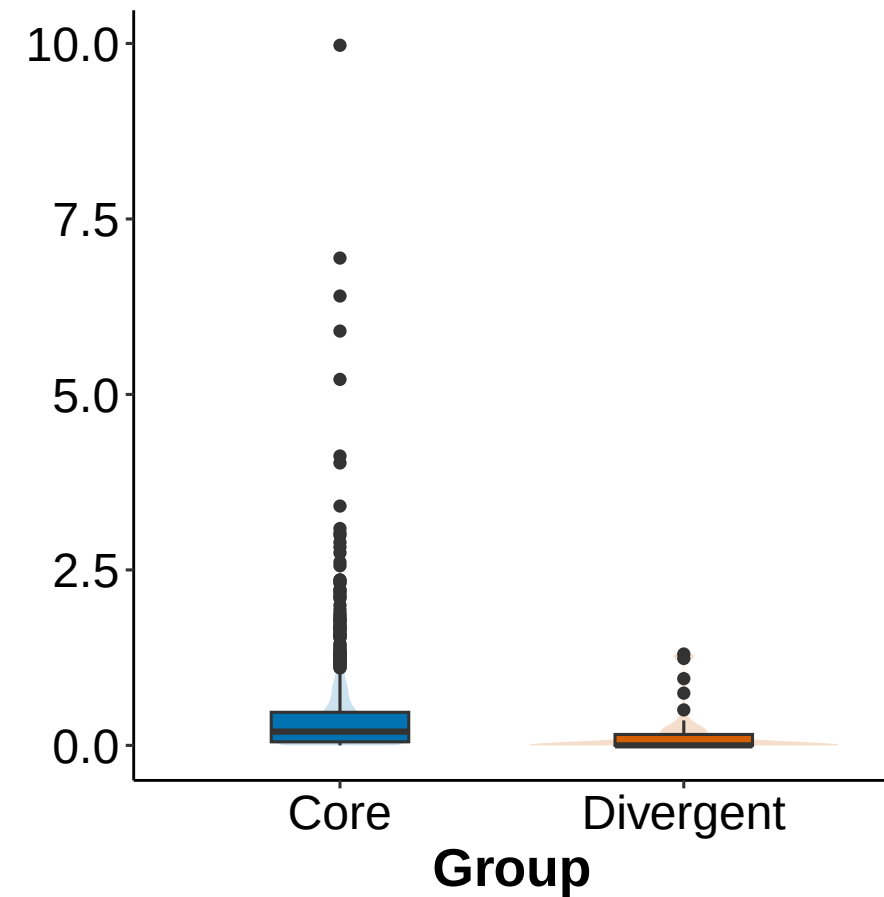
SA_Fatty_acid_oxidation_FBA

p-adj: 0.425 | .=-0.11 (Negligible) | .%: -31.5



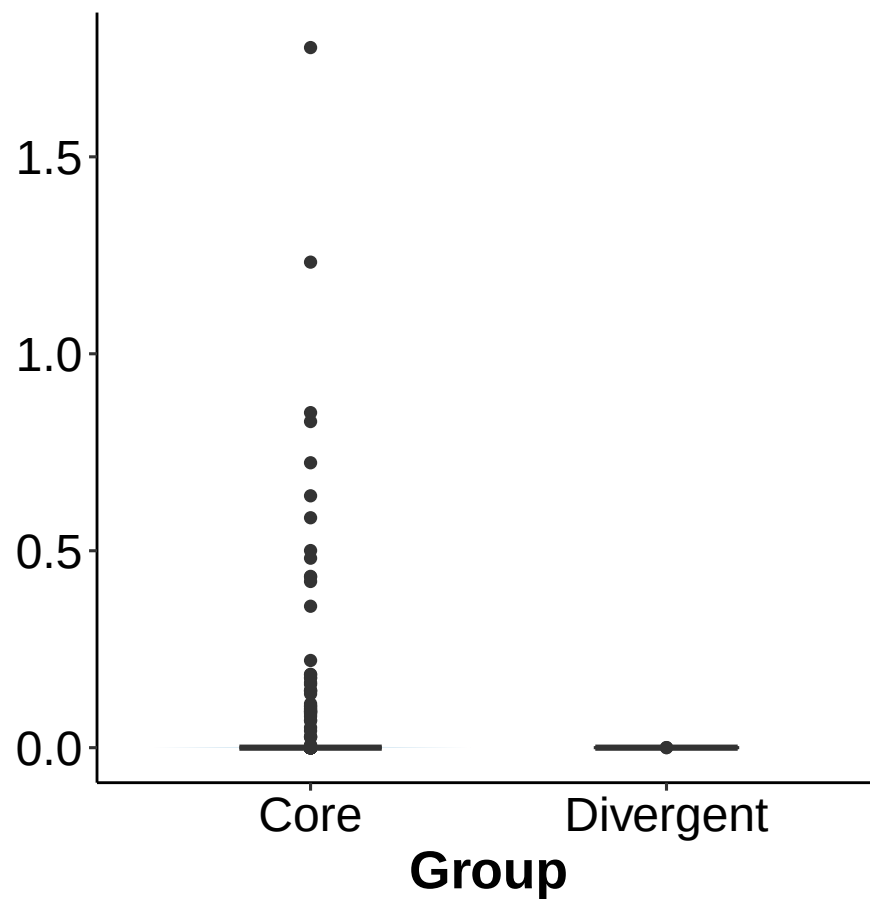
SA_Fatty_acid_oxidation_L1w

p-adj: 0.000134 | .=-0.41 (Medium) | .%: -99.7



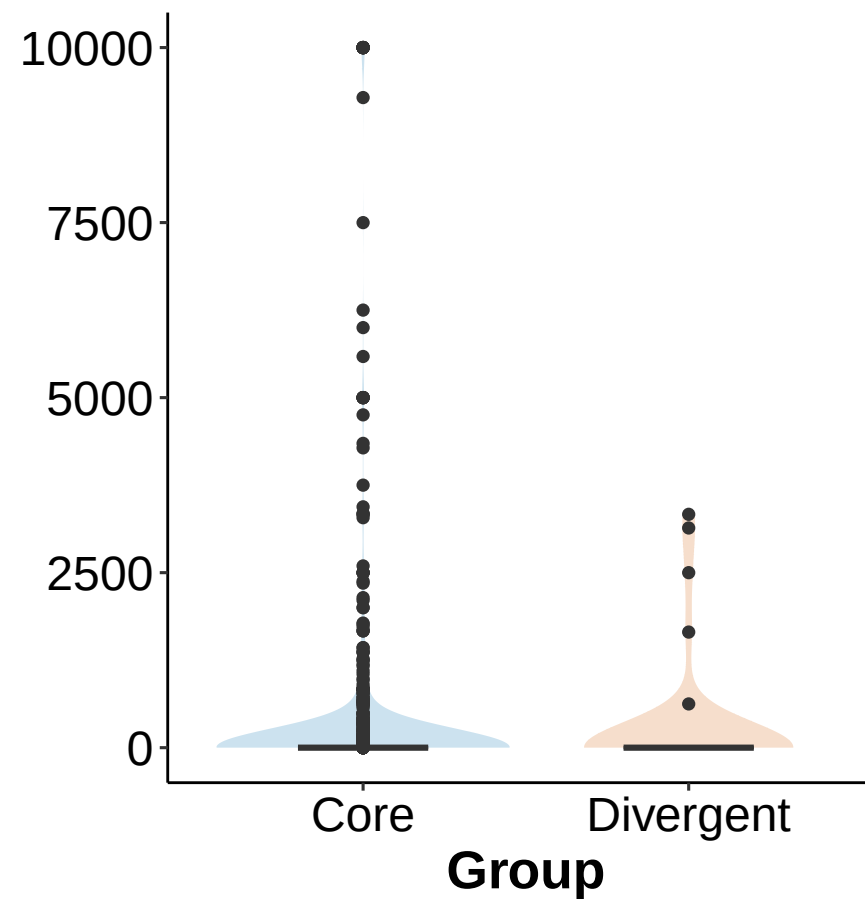
SA_Fatty_acid_oxidation_pFBA

p-adj: 0.762 | .=-0.04 (Negligible) | .%: 21.2



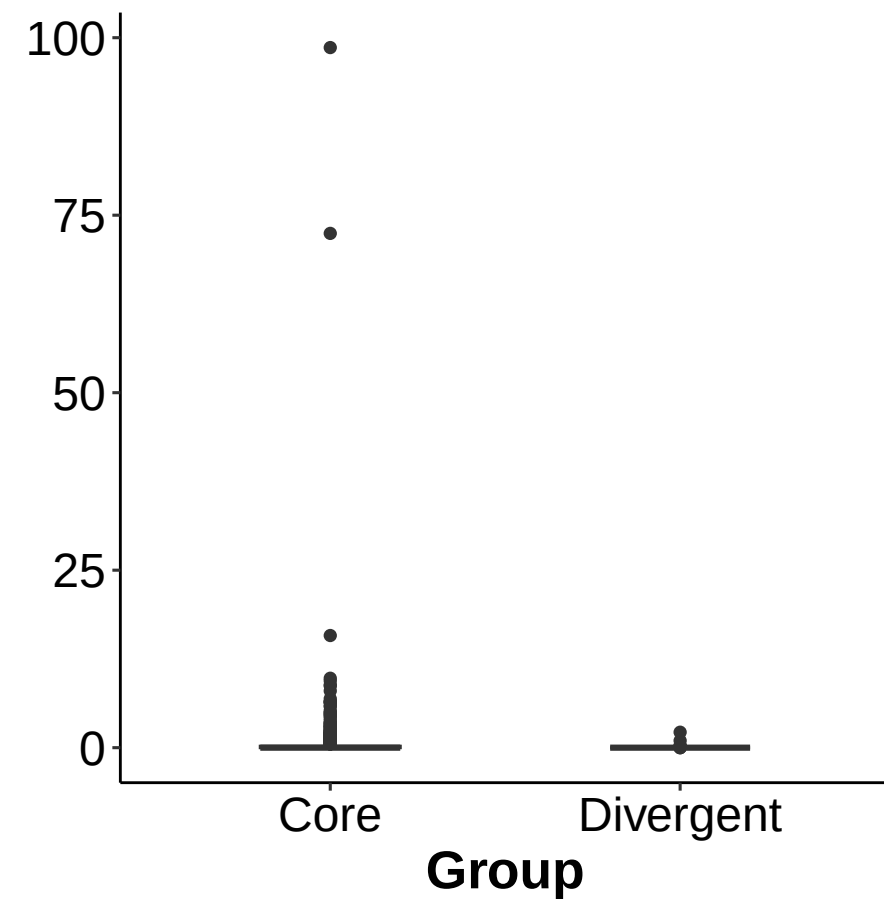
SA_Fatty_acid_synthesis_FBA

p-adj: 0.826 | .=-0.02 (Negligible)



SA_Fatty_acid_synthesis_L1w

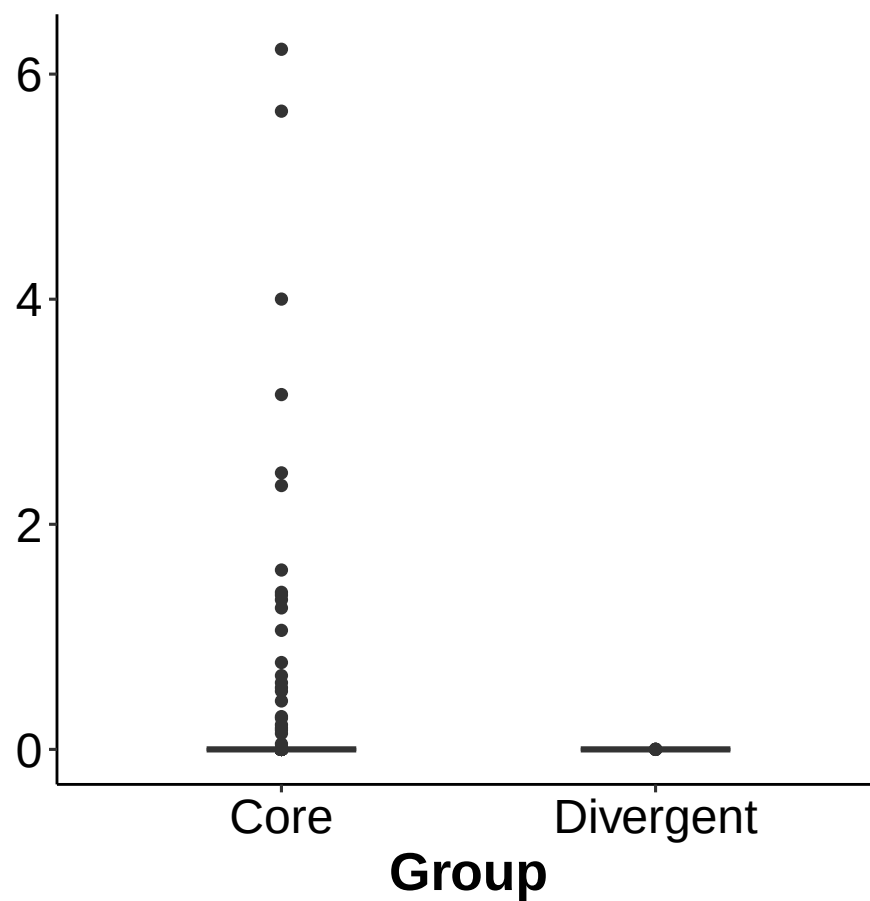
p-adj: 0.0531 | .=-0.22 (Small) | .%: -76.7



Metabolism – 5. Systemic Reprogramming (SA_*)

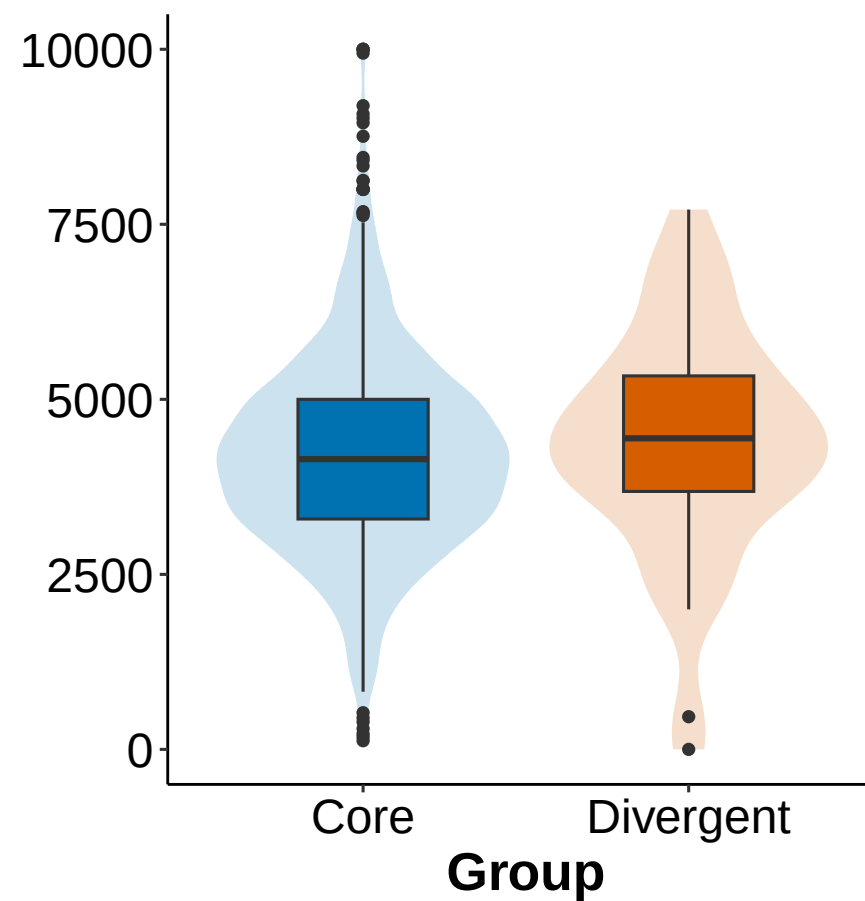
SA_Fatty_acid_synthesis_pFBA

p-adj: 0.671 | .=-0.04 (Negligible)



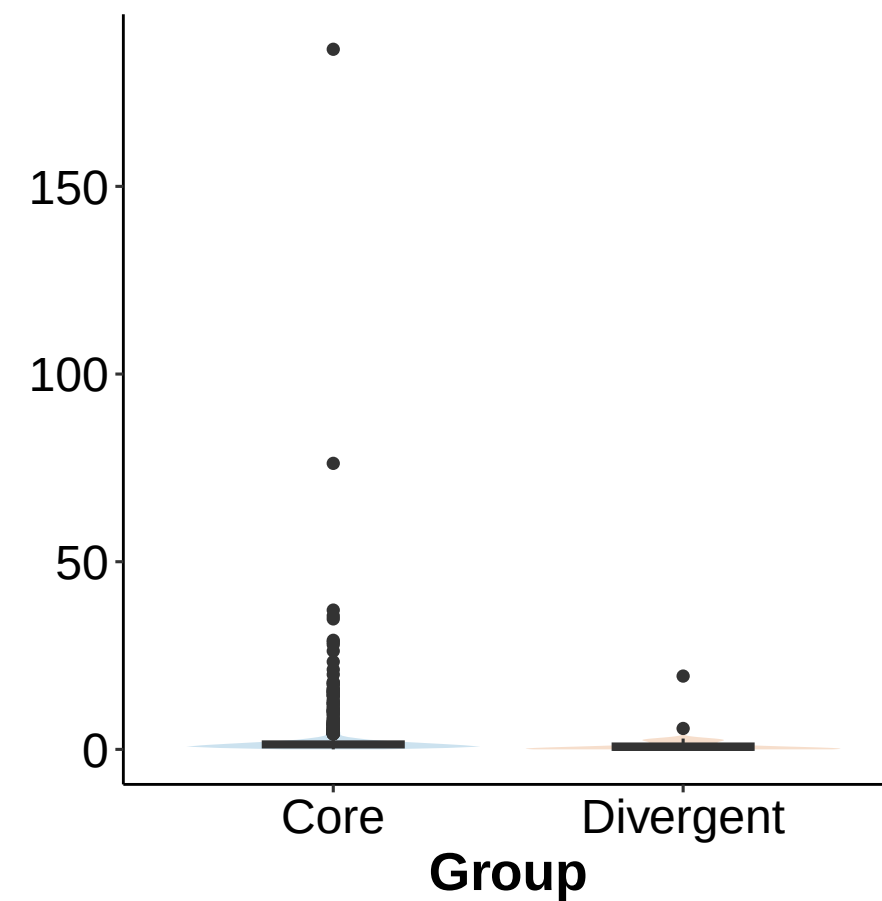
SA_Folate_metabolism_FBA

p-adj: 0.452 | . = 0.1 (Negligible) | .%: 7.2



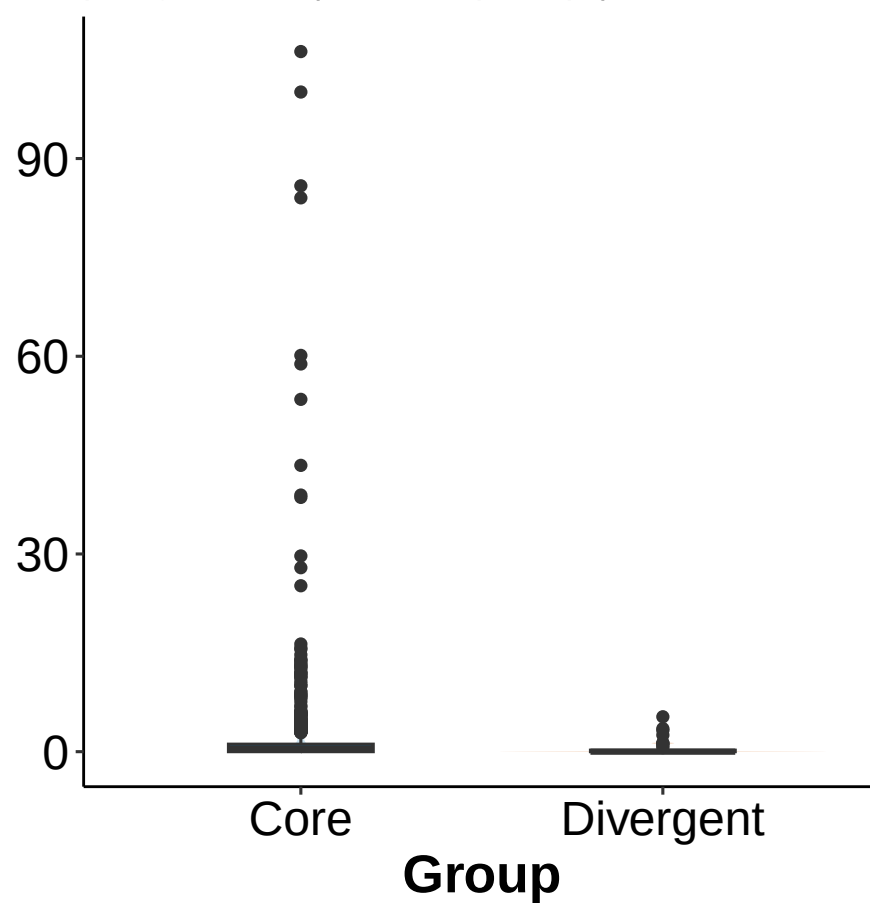
SA_Folate_metabolism_L1w

p-adj: 0.000109 | .=-0.41 (Medium) | .%: -64



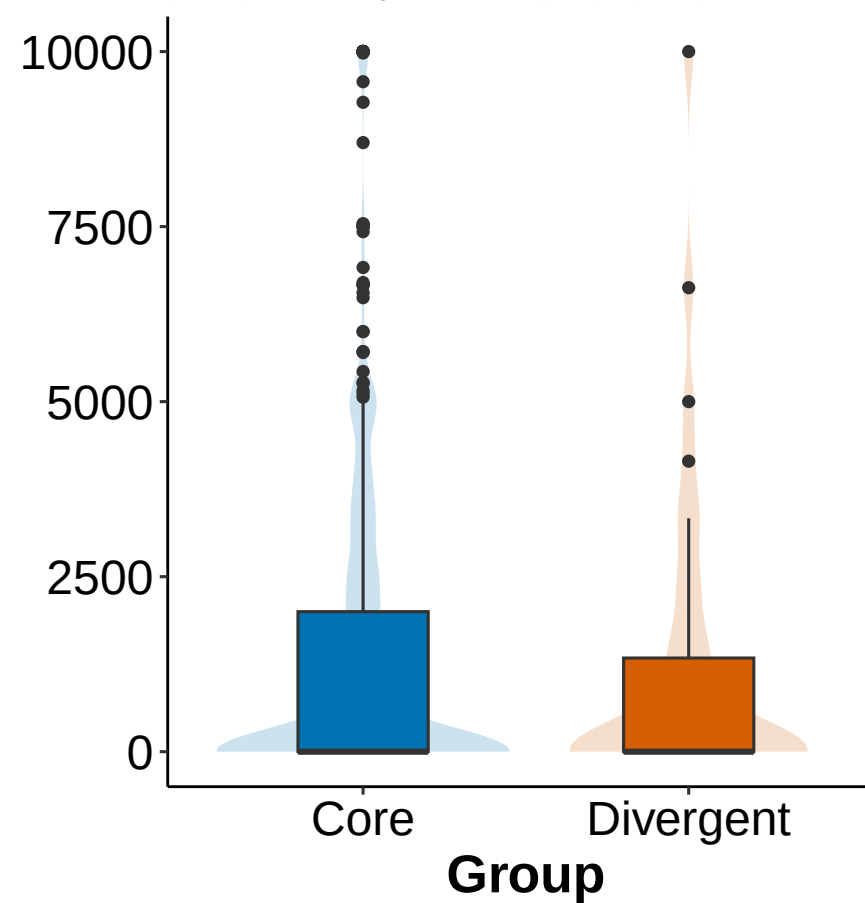
SA_Folate_metabolism_pFBA

p-adj: 0.0289 | .=-0.24 (Small) | .%: -100



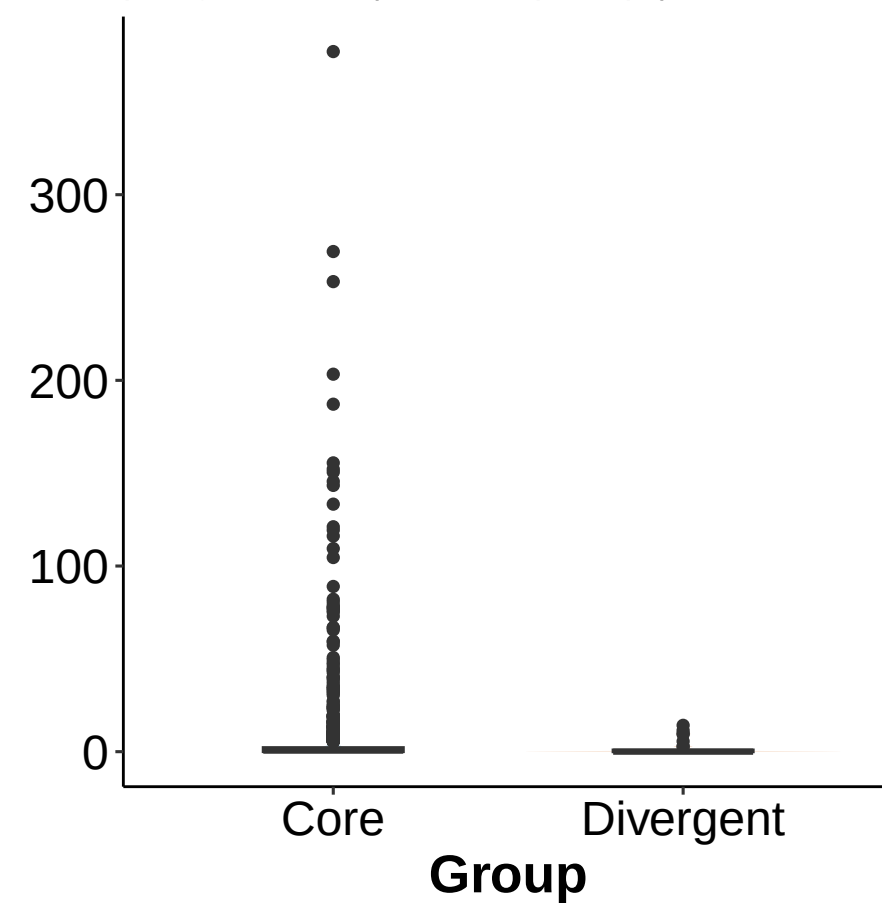
SA_Fructose_and_mannose_metabo

p-adj: 0.696 | .=-0.05 (Negligible)



SA_Fructose_and_mannose_metabo

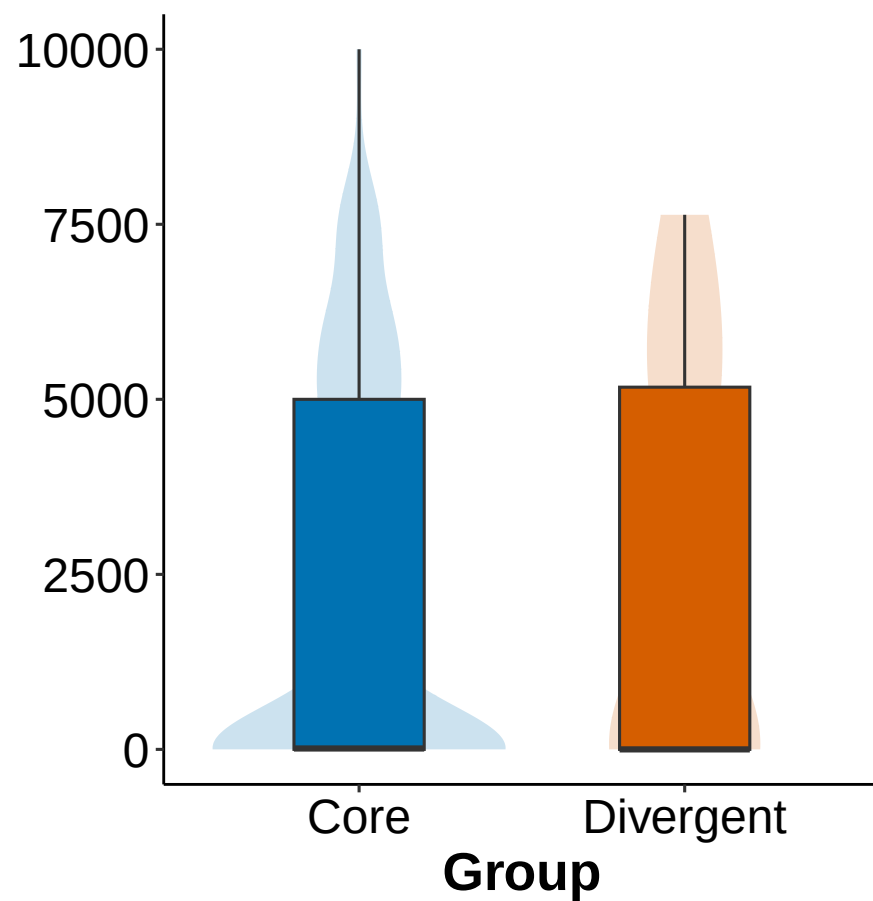
p-adj: 0.00427 | .=-0.31 (Small) | .%: -100



Metabolism – . 5. Systemic Reprogramming (SA_*)

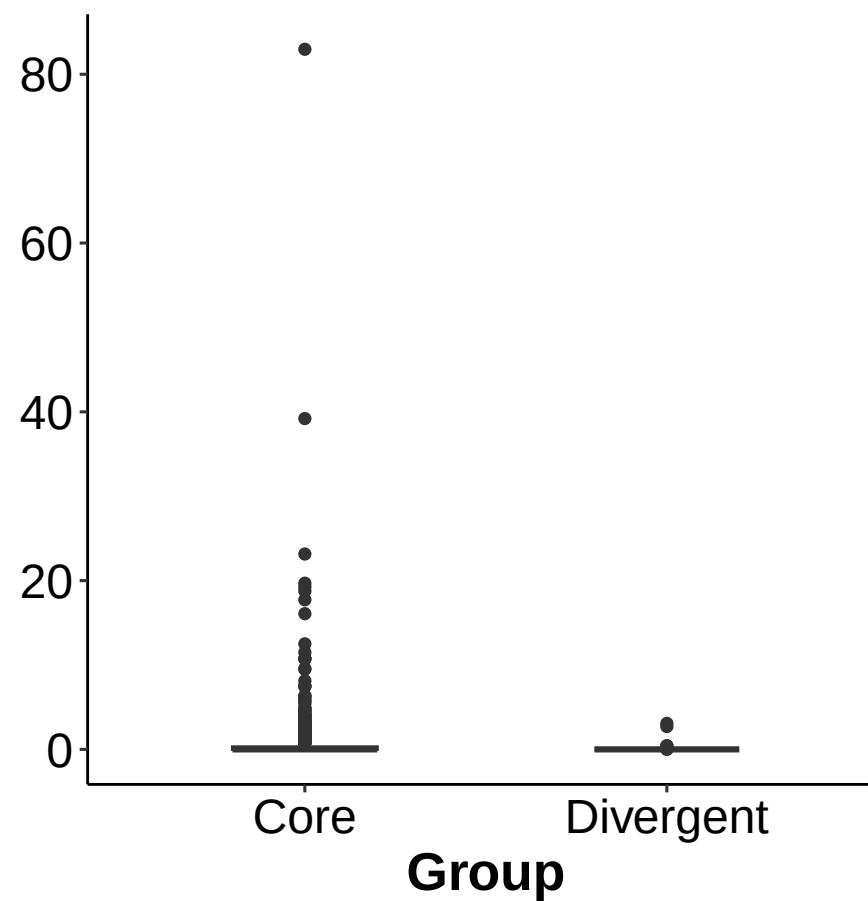
SA_Galactose_metabolism_FBA

p-adj: 0.911 | .:=0.01 (Negligible) | .%: -100



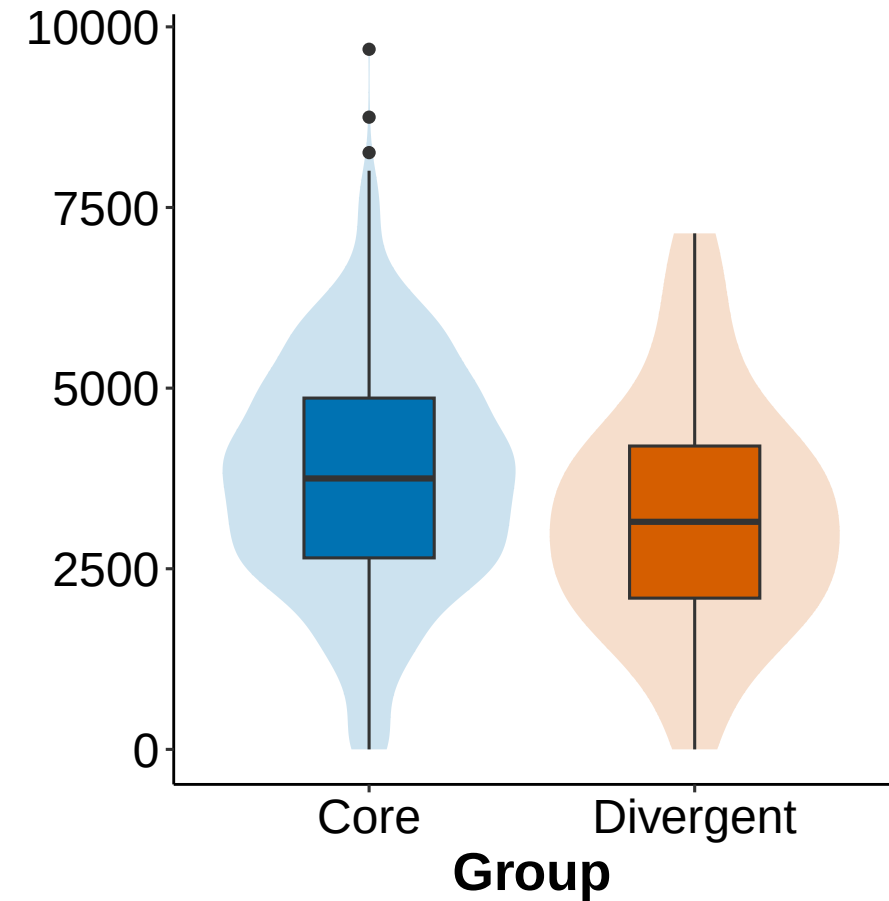
SA_Galactose_metabolism_L1w

p-adj: 0.223 | .:= -0.15 (Small) | .%: -76.5



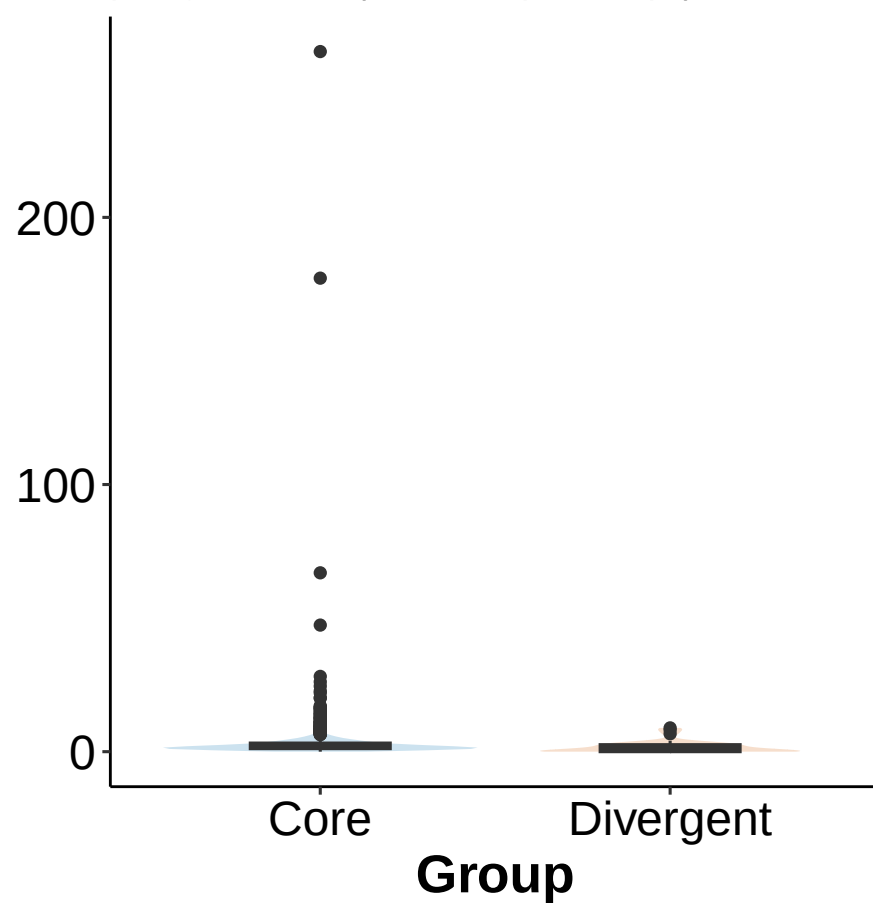
SA_Glutamate_metabolism_FBA

p-adj: 0.0894 | .:= -0.2 (Small) | .%: -16



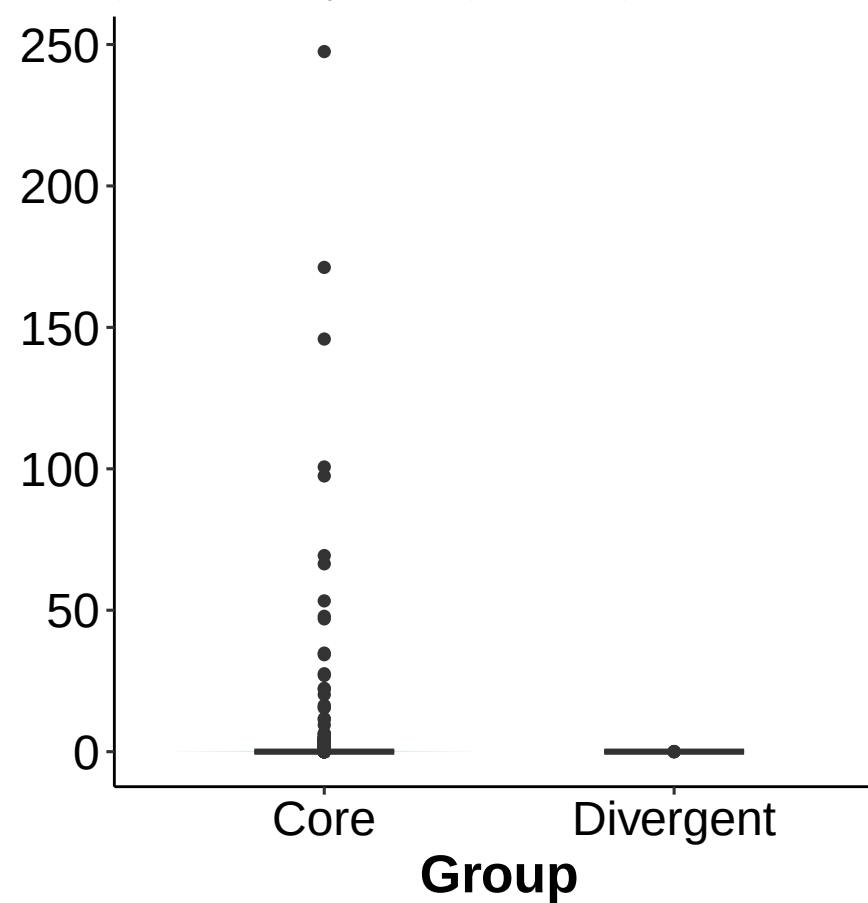
SA_Glutamate_metabolism_L1w

p-adj: 0.00177 | .:= -0.33 (Medium) | .%: -40.2



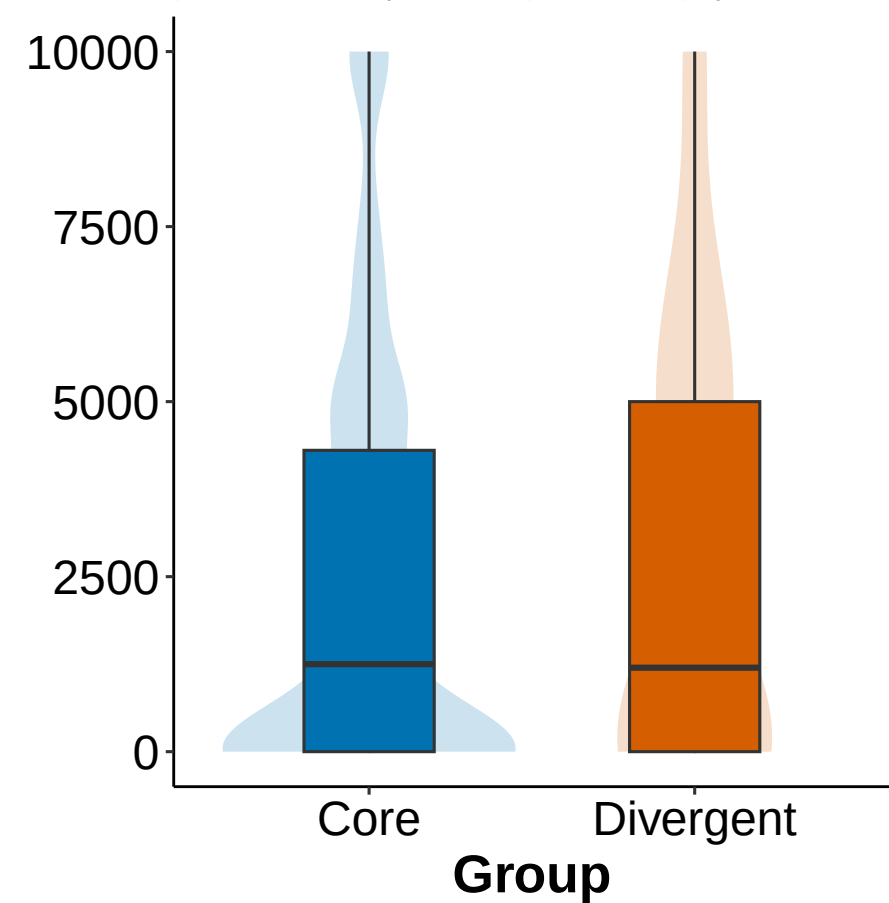
SA_Glutamate_metabolism_pFBA

p-adj: 0.733 | .:=0.04 (Negligible)

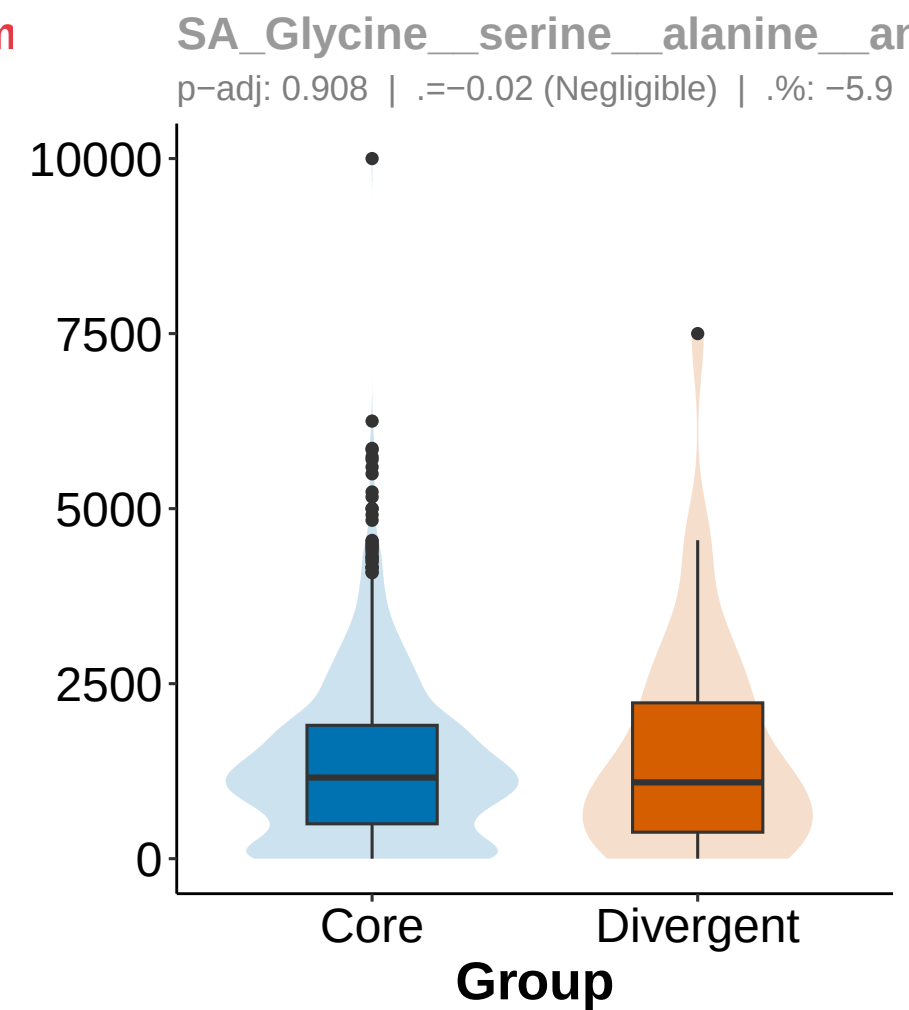
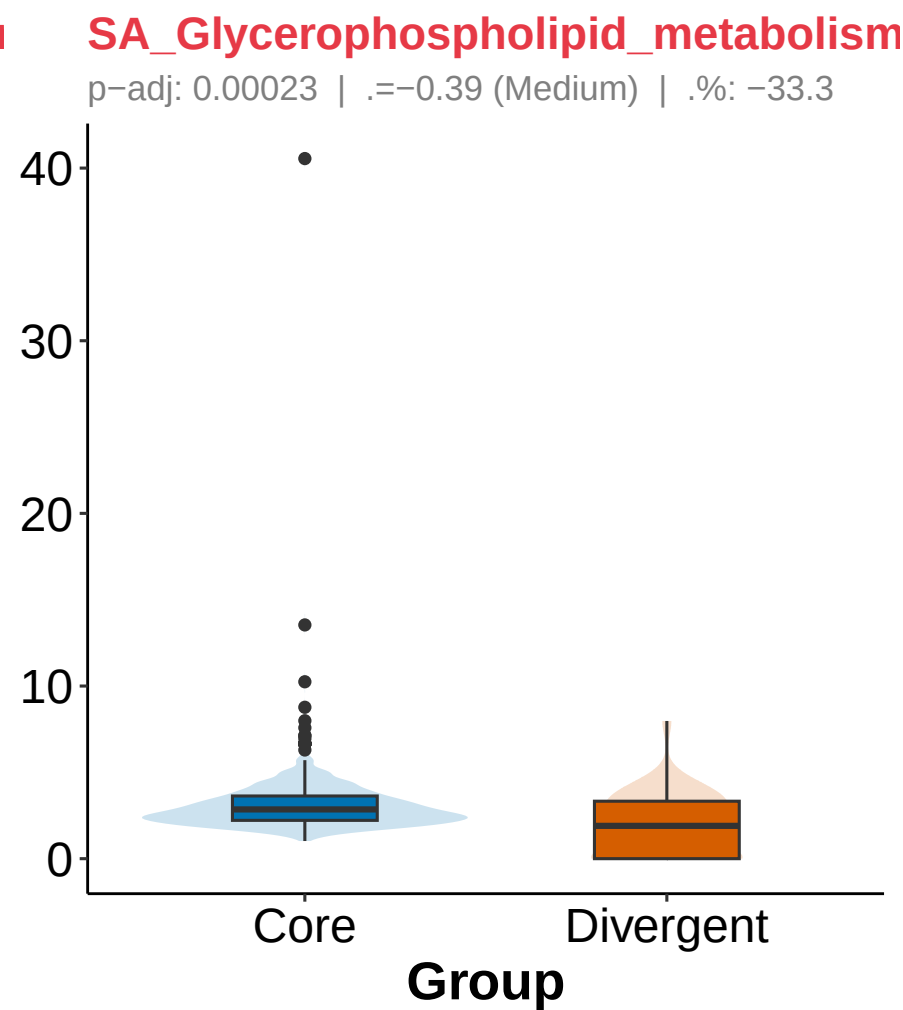
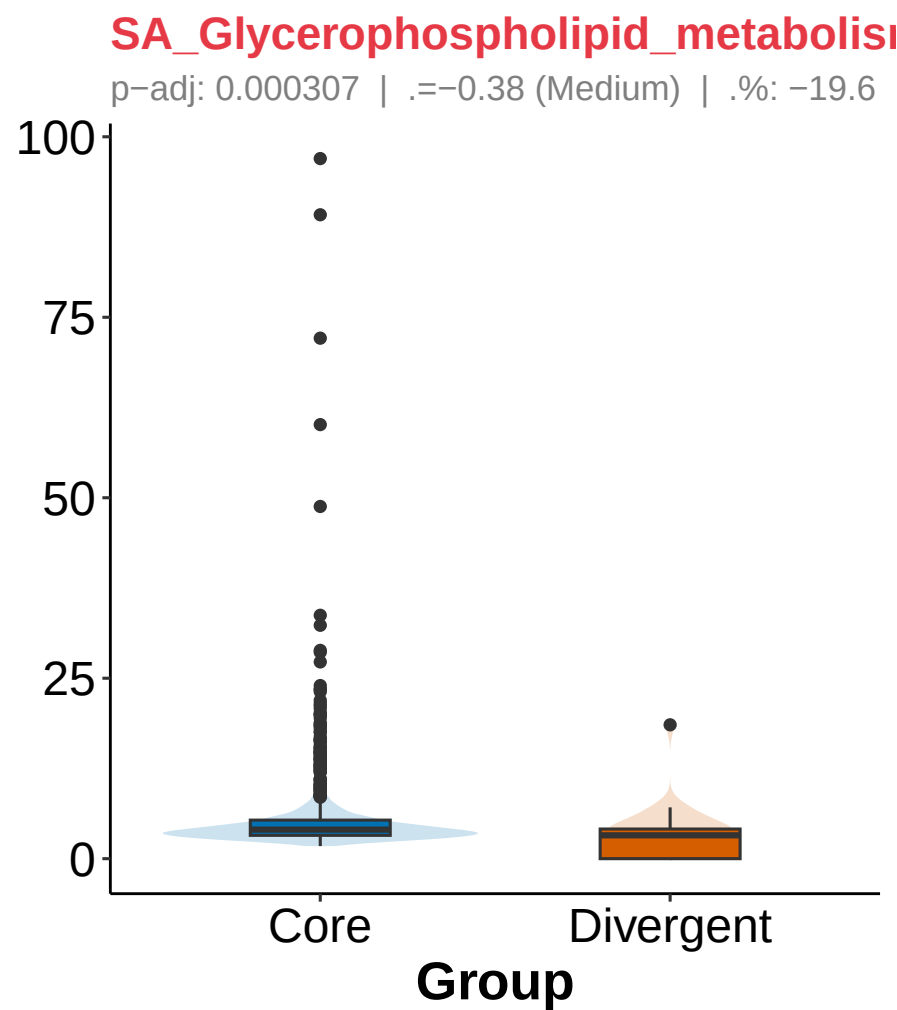
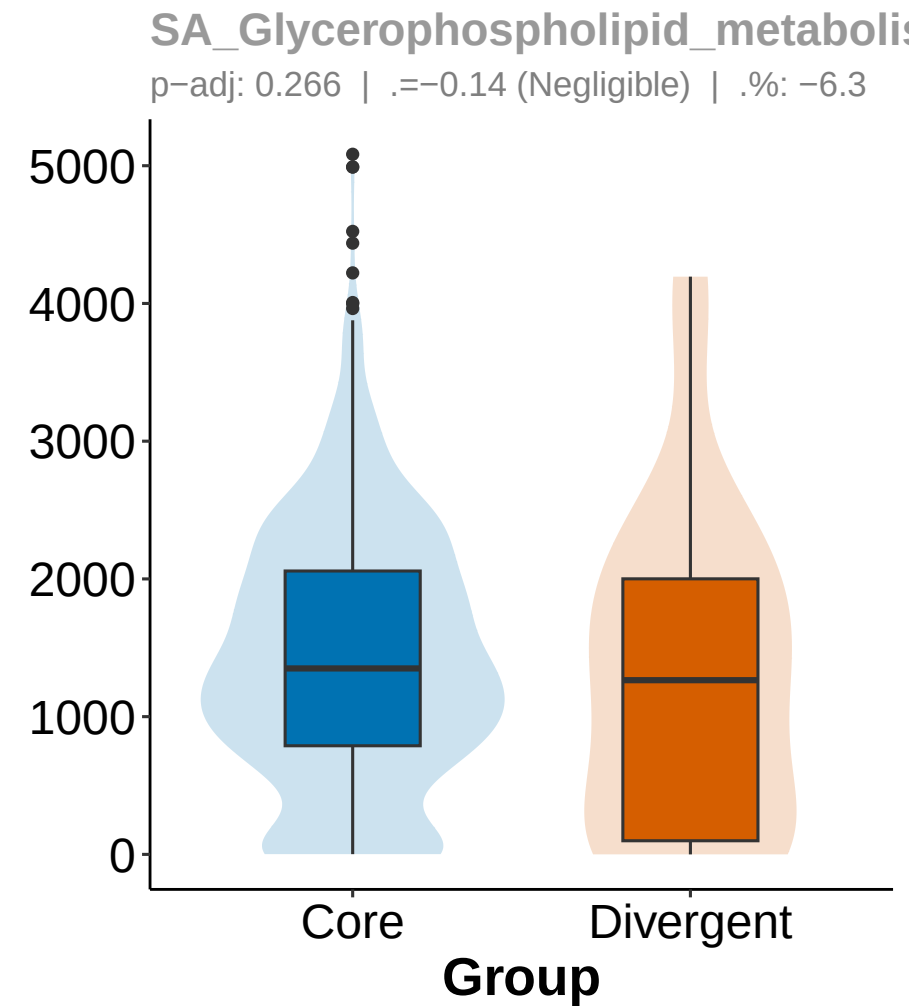
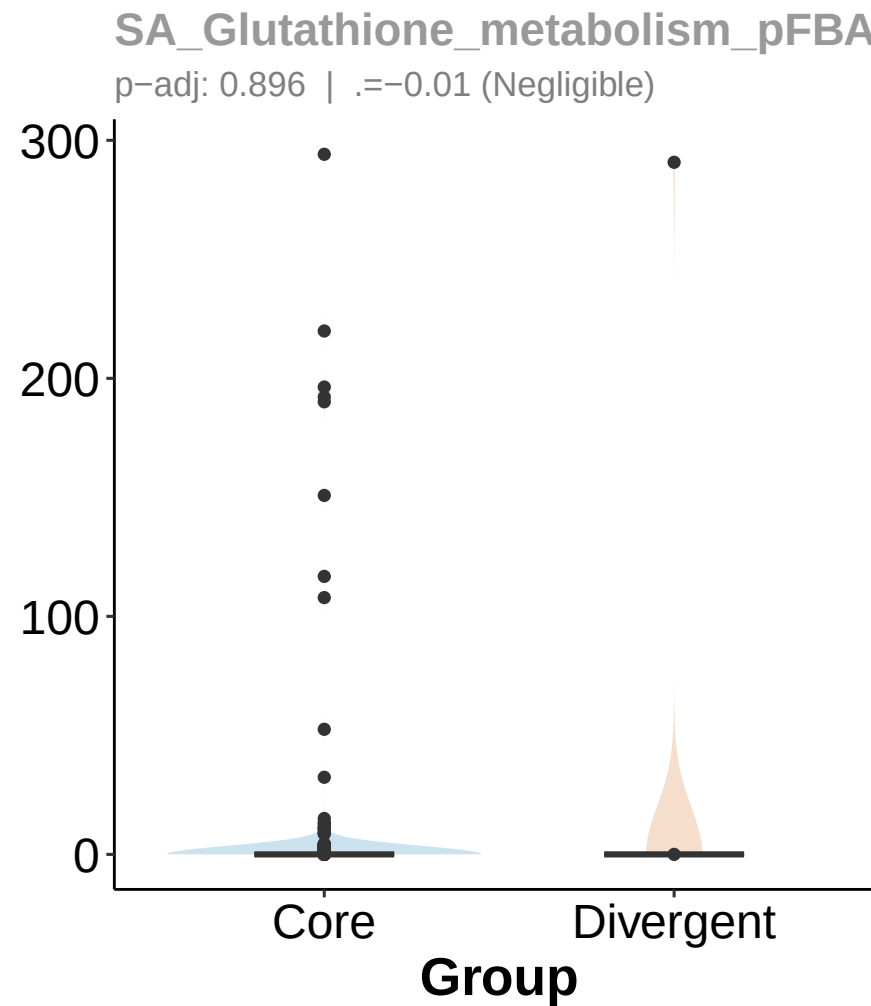
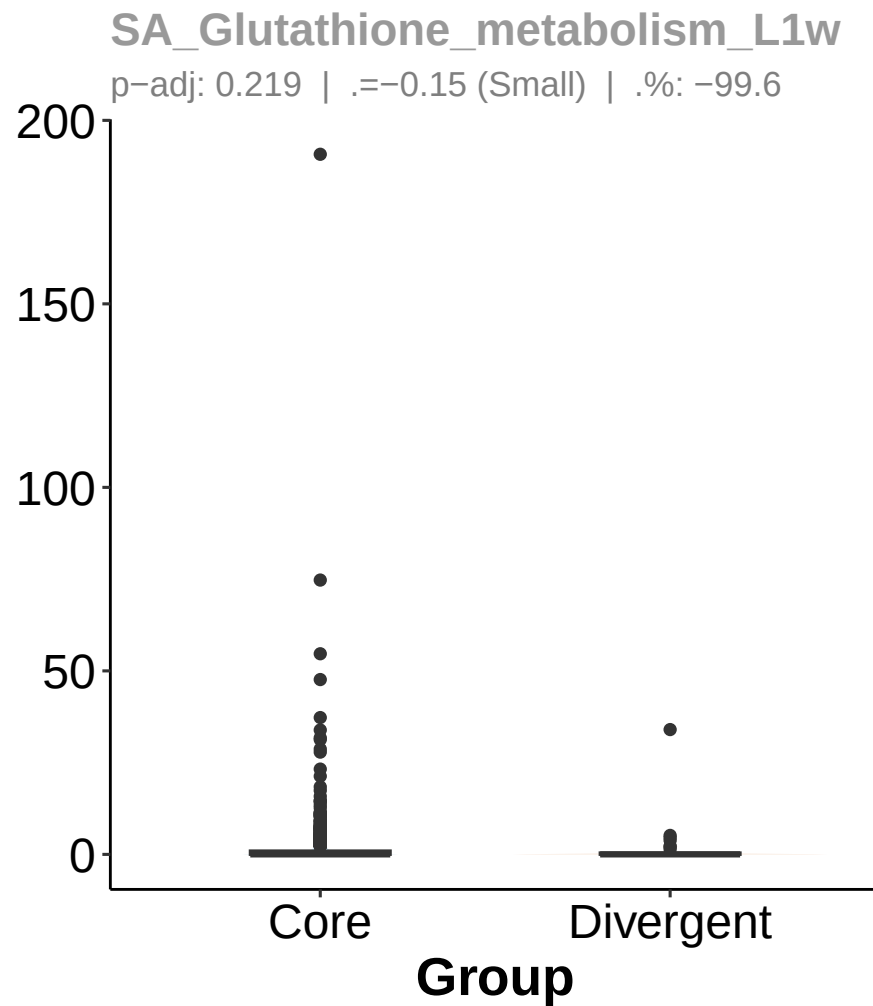


SA_Glutathione_metabolism_FBA

p-adj: 0.711 | .:=0.05 (Negligible) | .%: -4



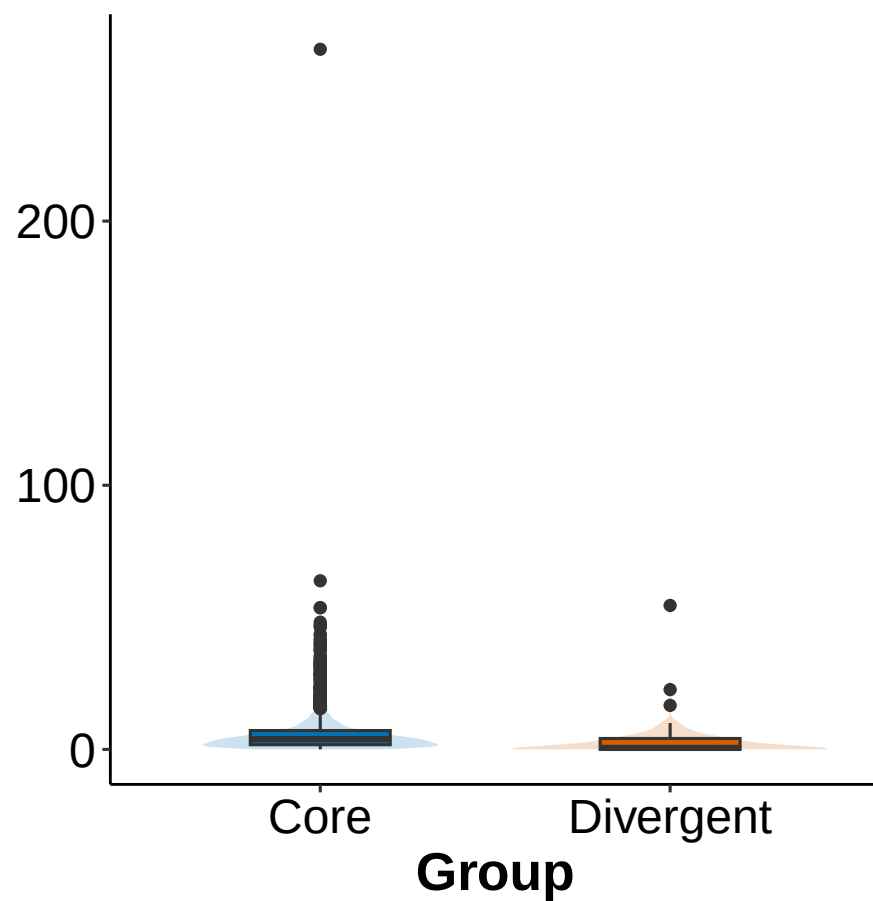
Metabolism – . 5. Systemic Reprogramming (SA_*)



Metabolism – . 5. Systemic Reprogramming (SA_*)

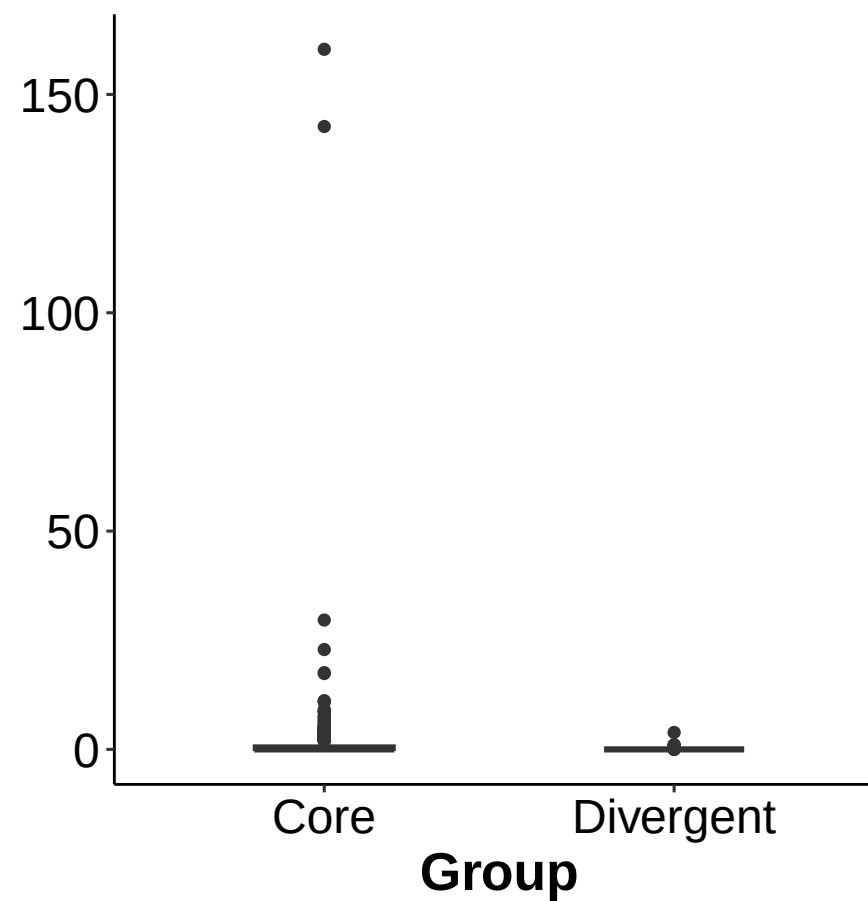
SA_Glycine_serine_alanine_and_

p-adj: 6.43e-05 | .=-0.43 (Medium) | .%: -85.8



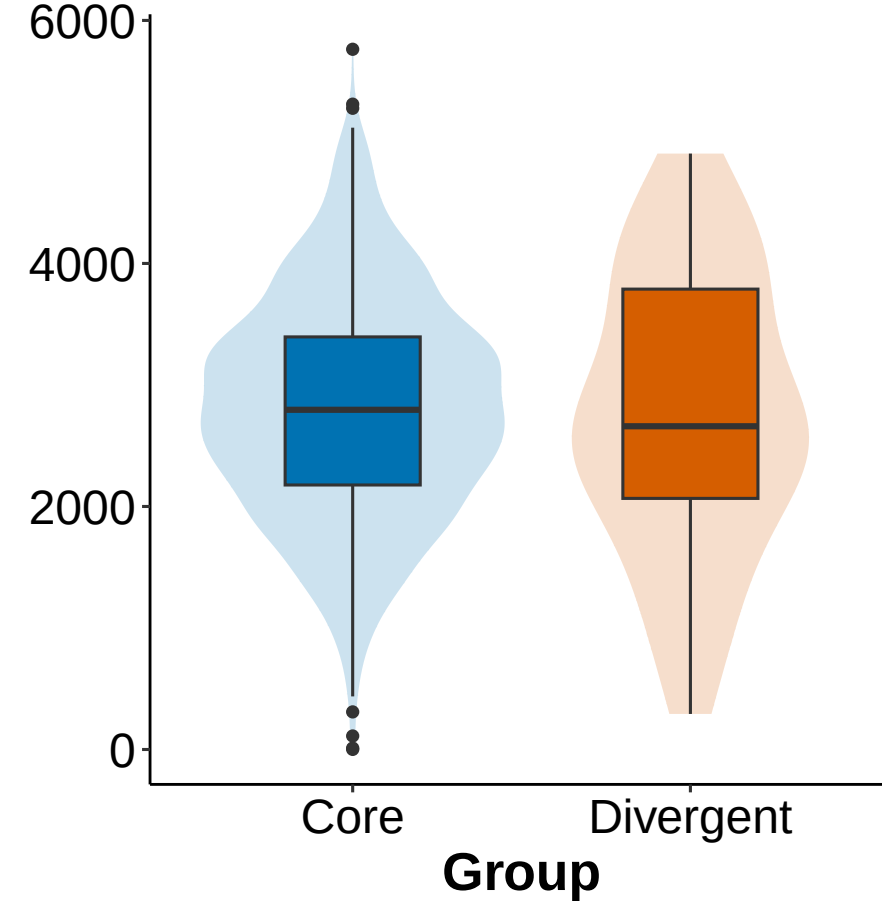
SA_Glycine_serine_alanine_and_

p-adj: 0.00475 | .=-0.29 (Small) | .%: -100



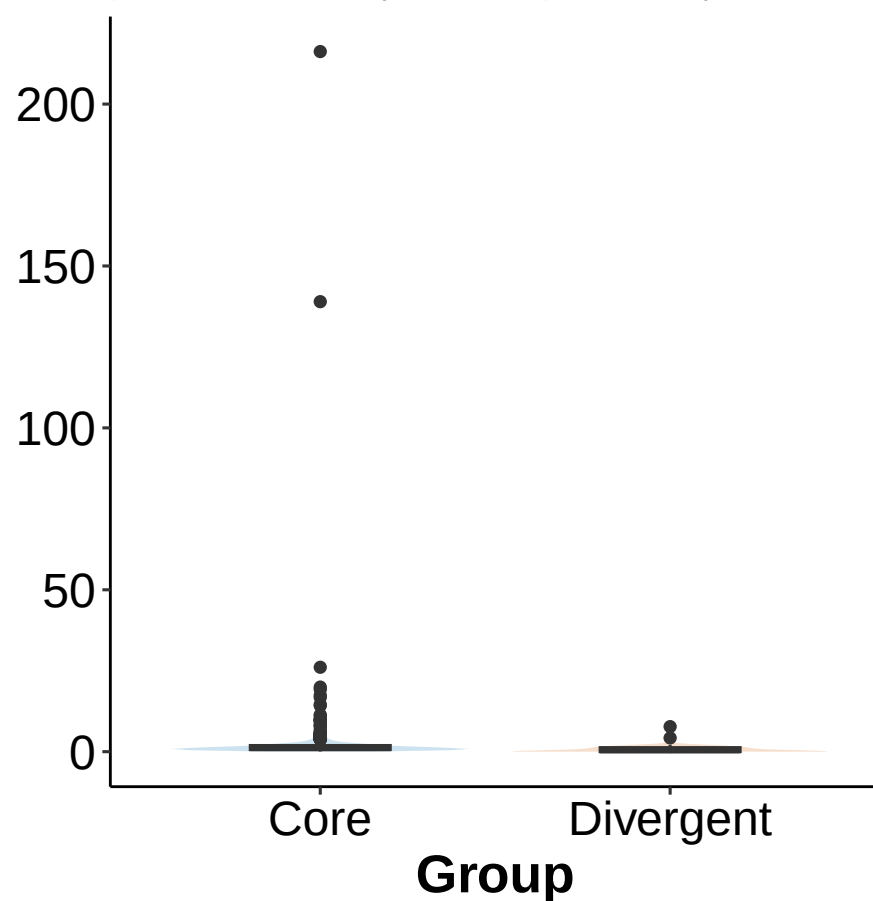
SA_Glycolysis_gluconeogenesis_F

p-adj: 0.845 | .=-0.03 (Negligible) | .%: -4.9



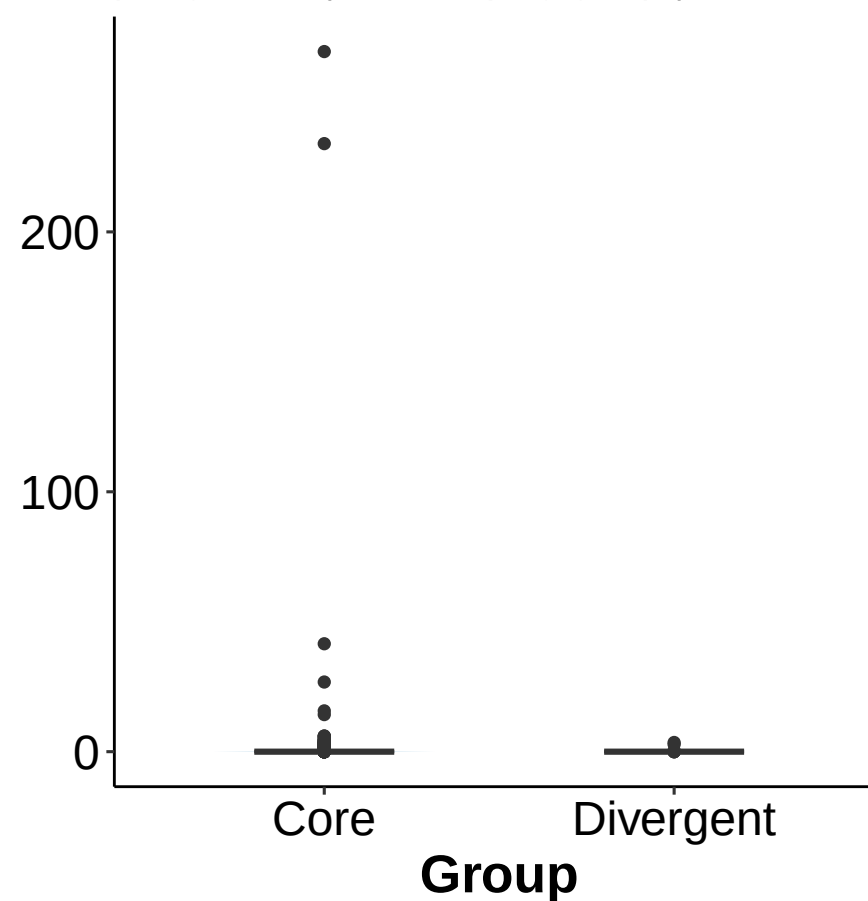
SA_Glycolysis_gluconeogenesis_L1

p-adj: 9.14e-05 | .=-0.42 (Medium) | .%: -60.2



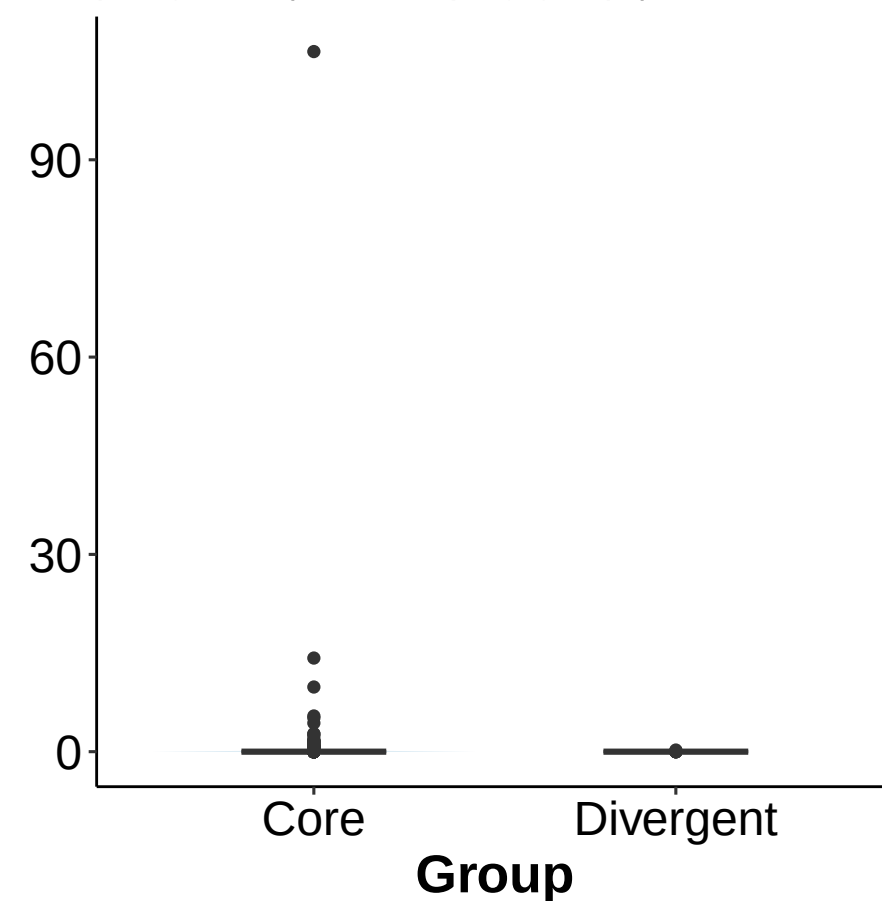
SA_Glycolysis_gluconeogenesis_pF

p-adj: 0.574 | .=-0.08 (Negligible) | .%: -50.5



SA_Glycosphingolipid_metabolism_L

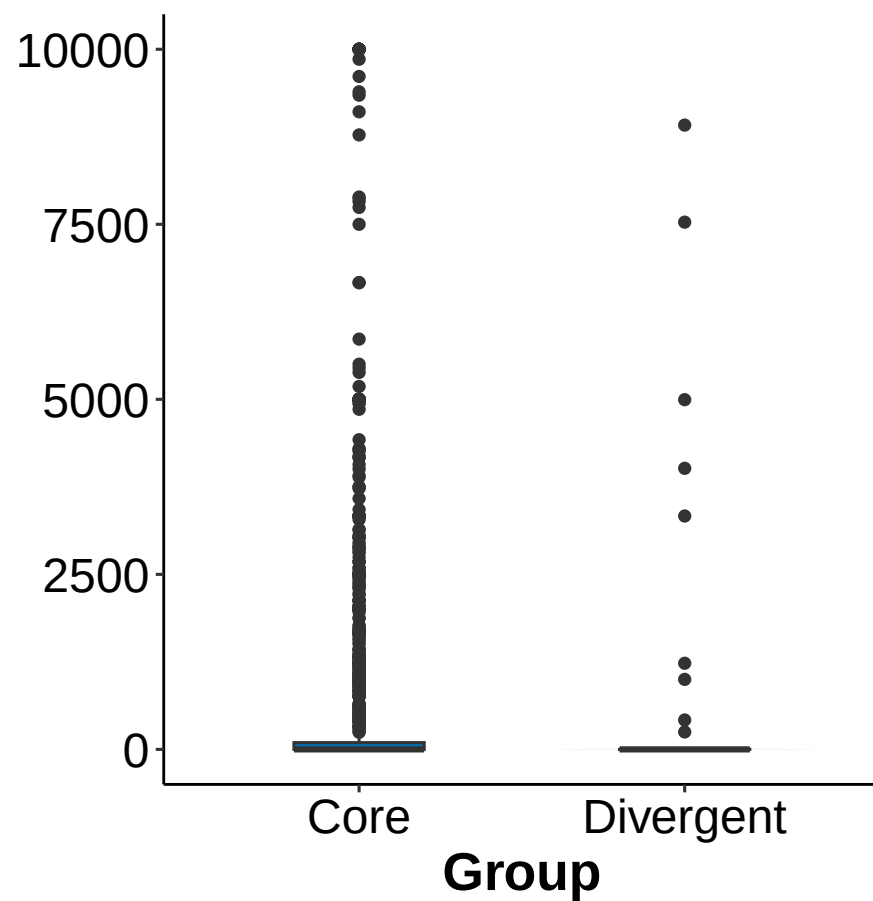
p-adj: 0.59 | .=-0.07 (Negligible) | .%: -100



Metabolism – . 5. Systemic Reprogramming (SA_*)

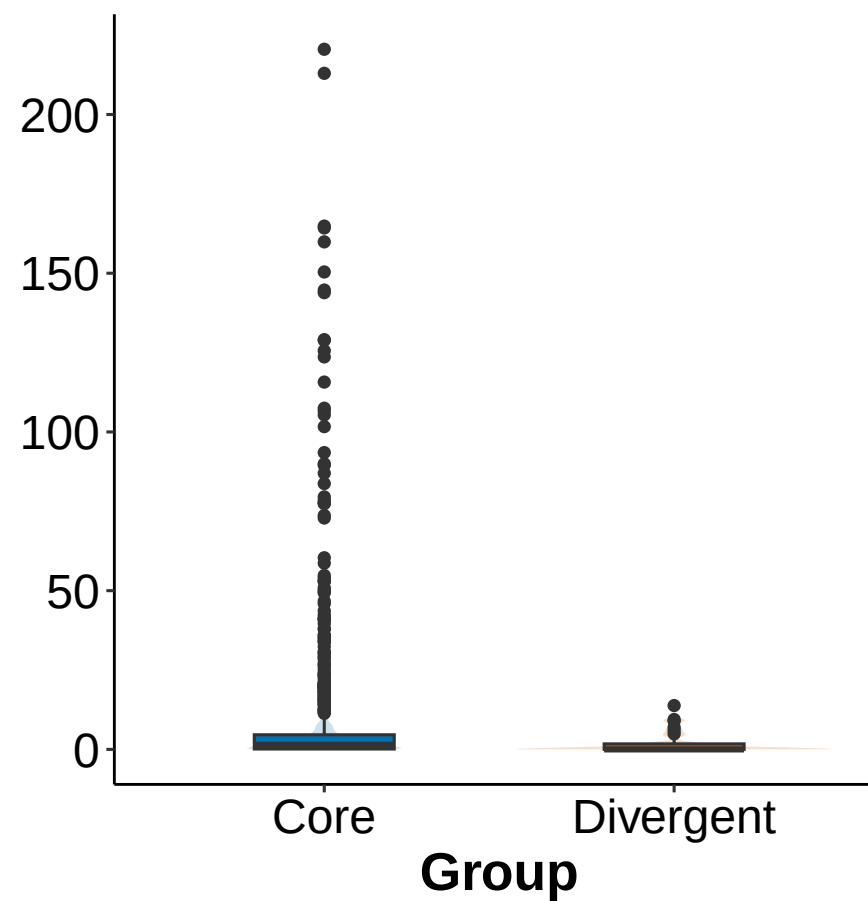
SA_Glyoxylate_and_dicarboxylate

p-adj: 0.949 | .=-0.01 (Negligible)



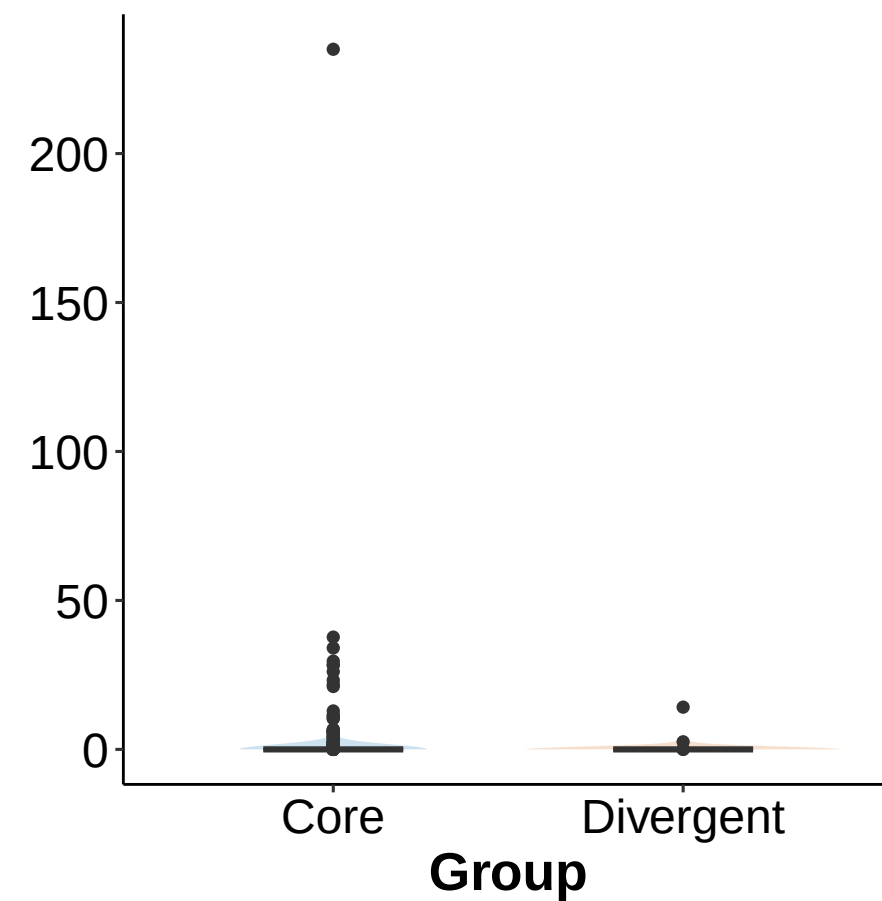
SA_Glyoxylate_and_dicarboxylate_n

p-adj: 0.00344 | .=-0.31 (Small) | .%: -100



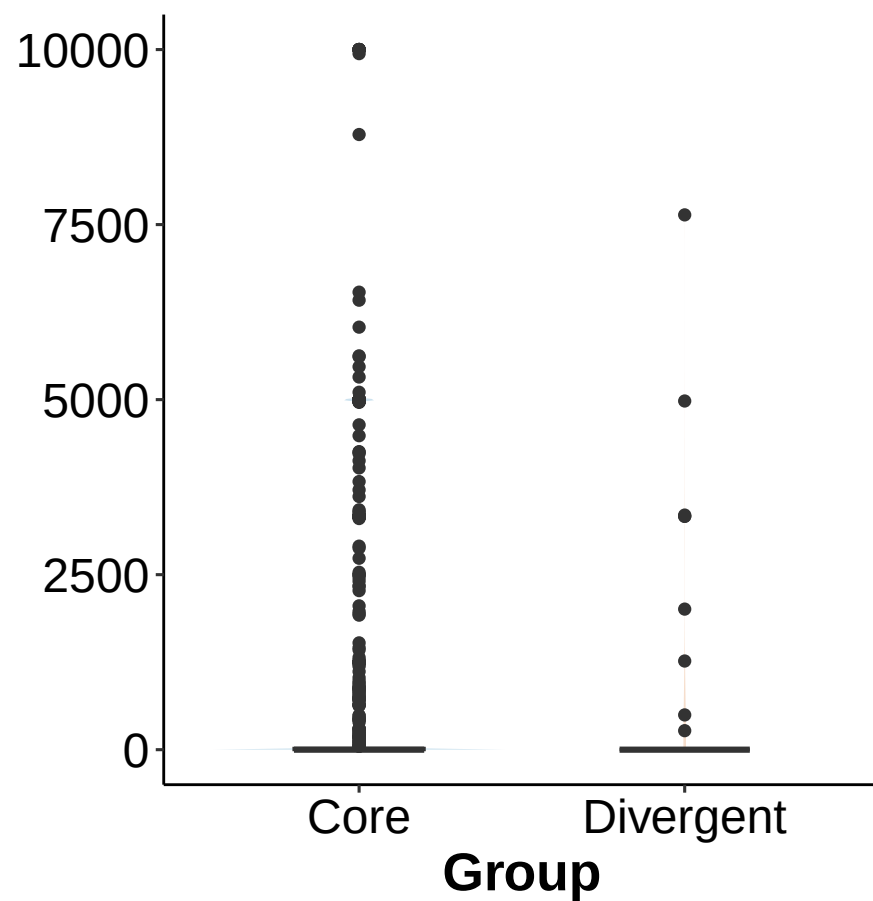
SA_Glyoxylate_and_dicarboxylate_n

p-adj: 0.574 | .=-0.04 (Negligible)



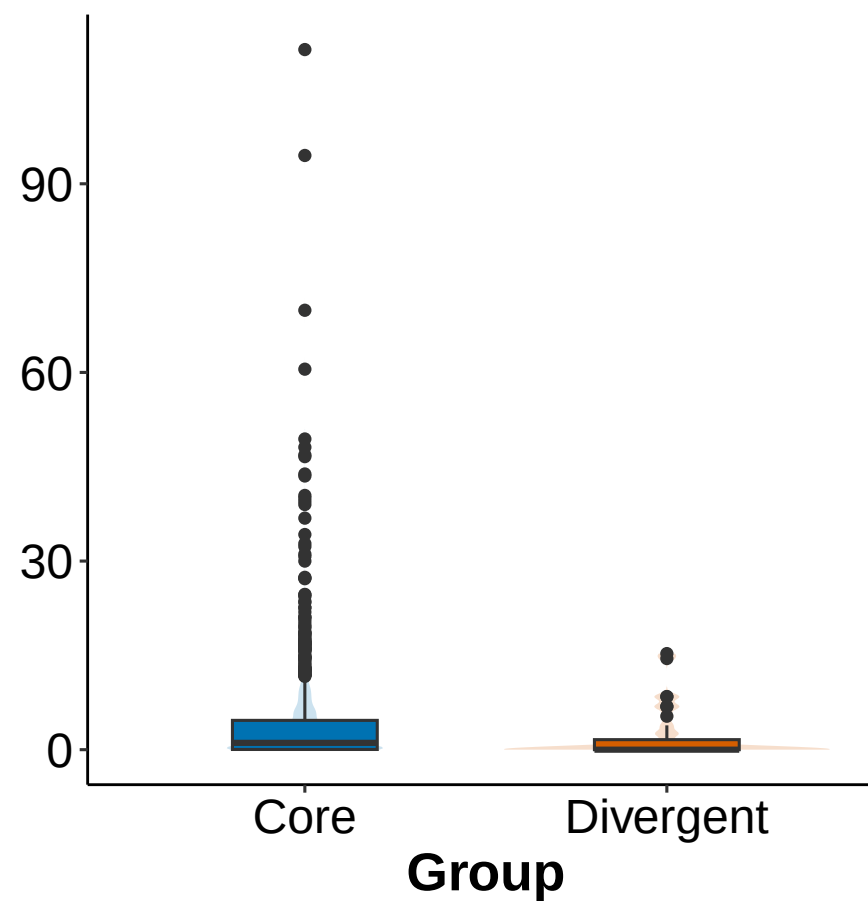
SA_Heme_degradation_FBA

p-adj: 0.743 | .=-0.04 (Negligible)



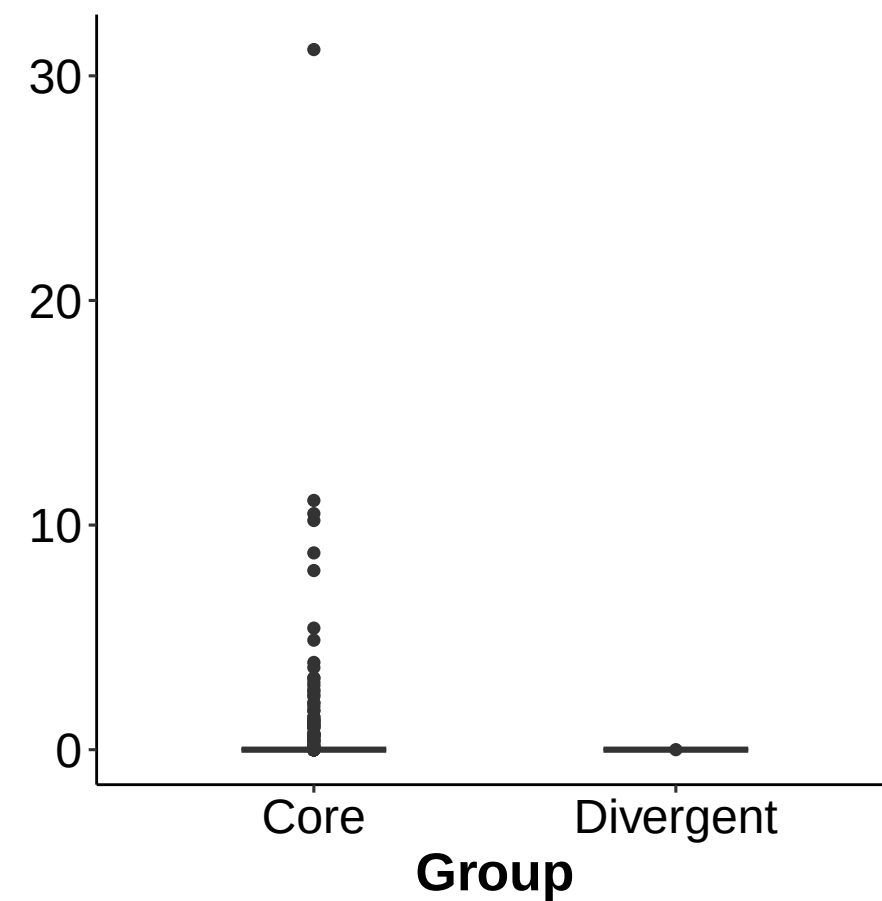
SA_Heme_degradation_L1w

p-adj: 0.0151 | .=-0.26 (Small) | .%: -100

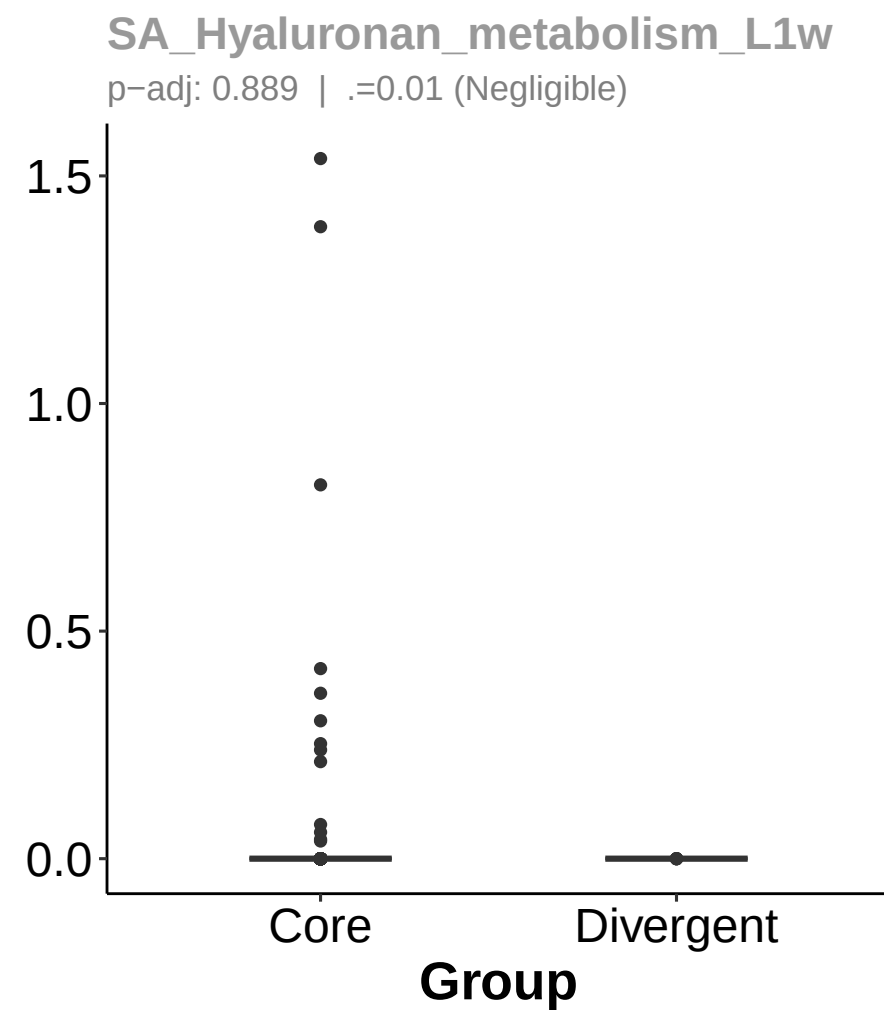
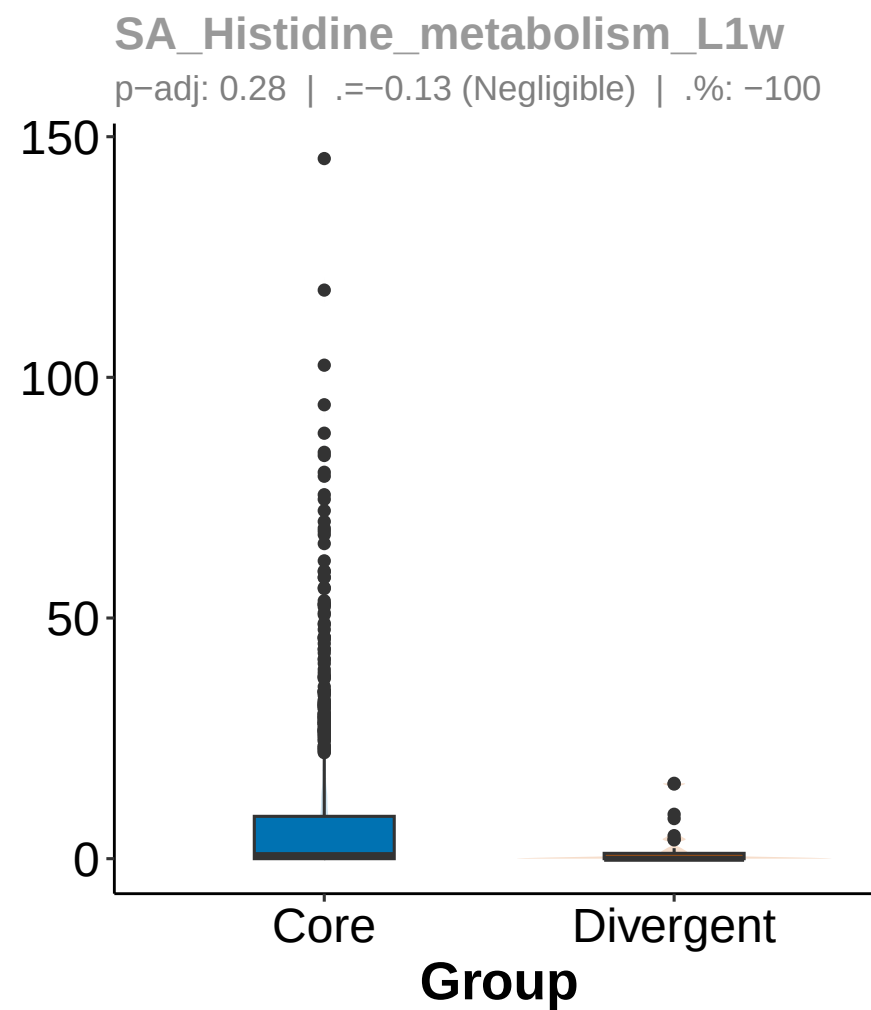
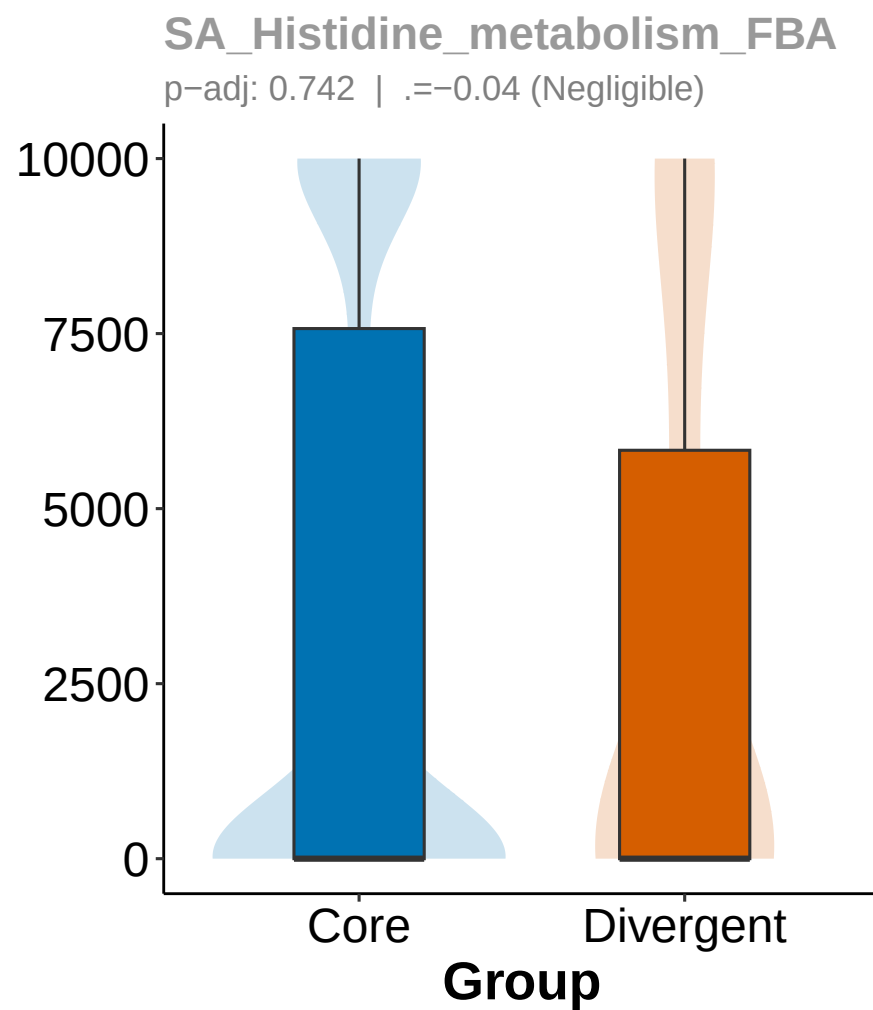
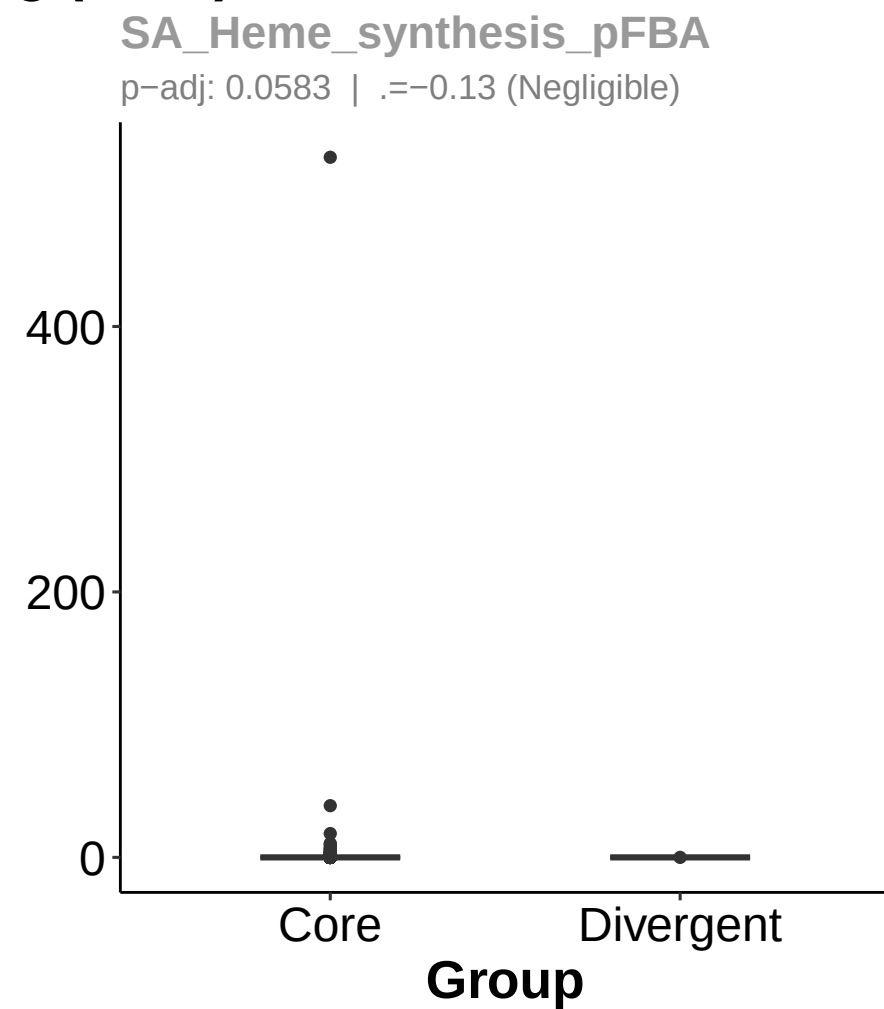
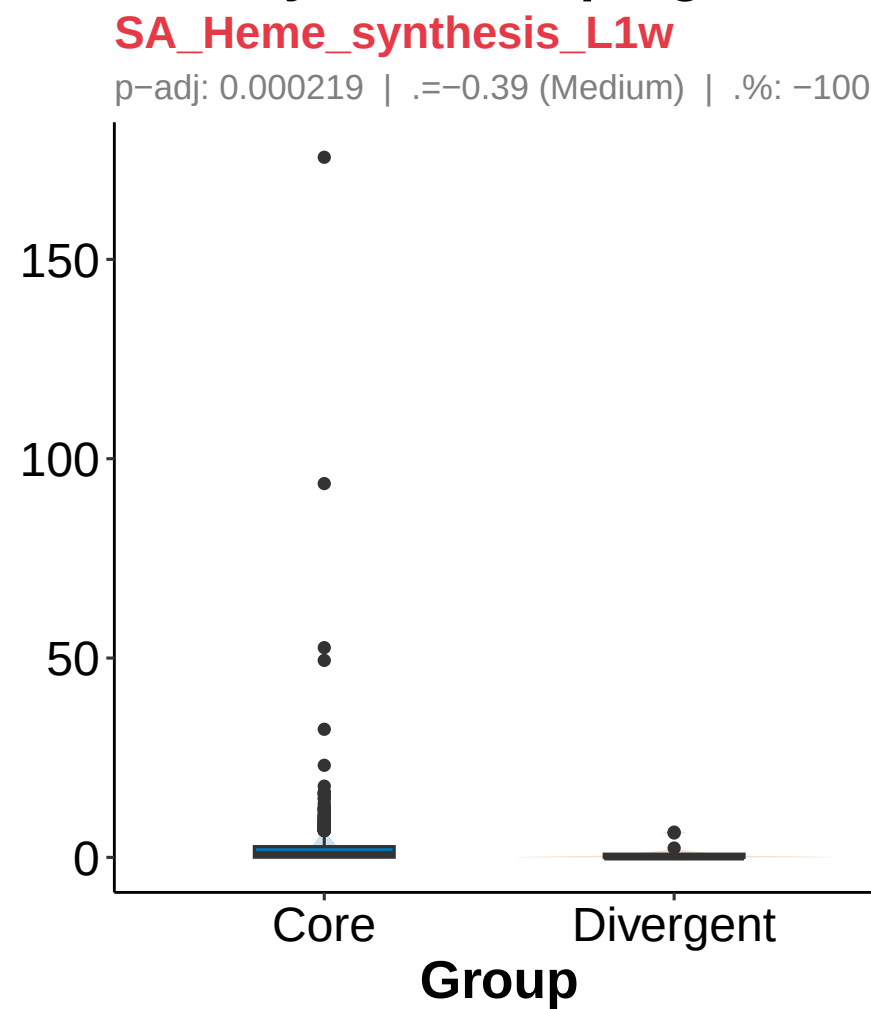
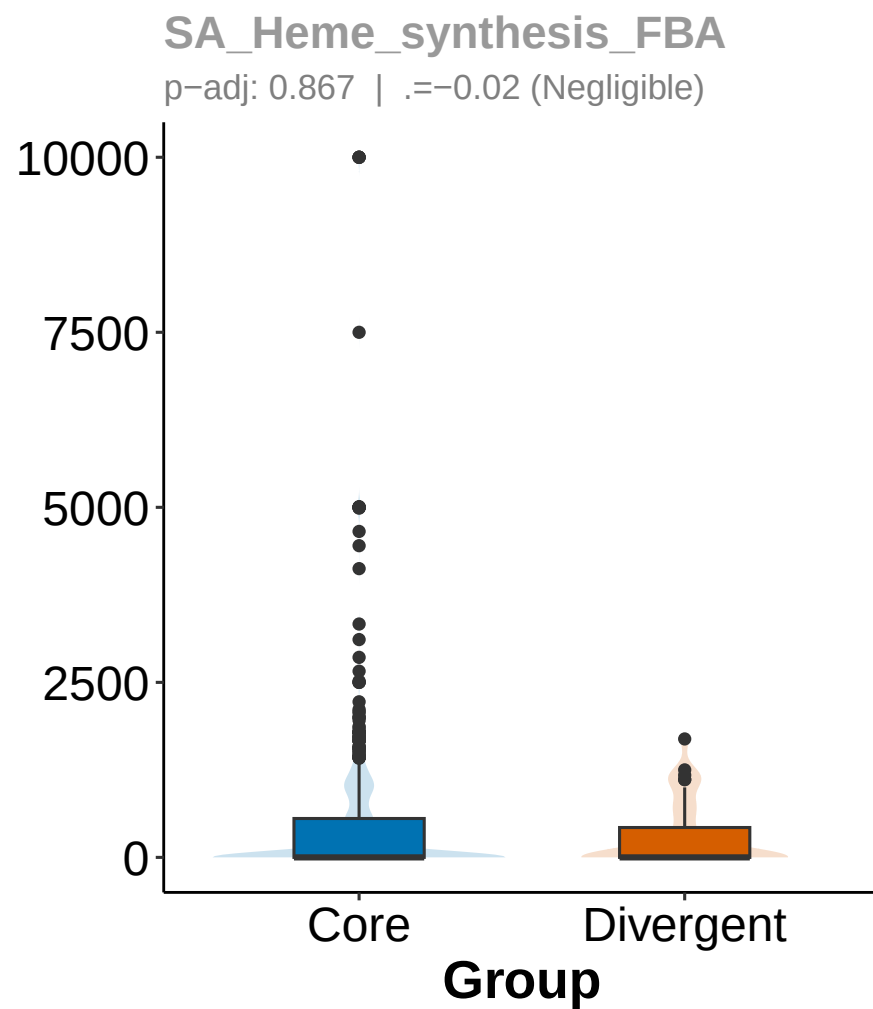


SA_Heme_degradation_pFBA

p-adj: 0.068 | .=-0.13 (Negligible)



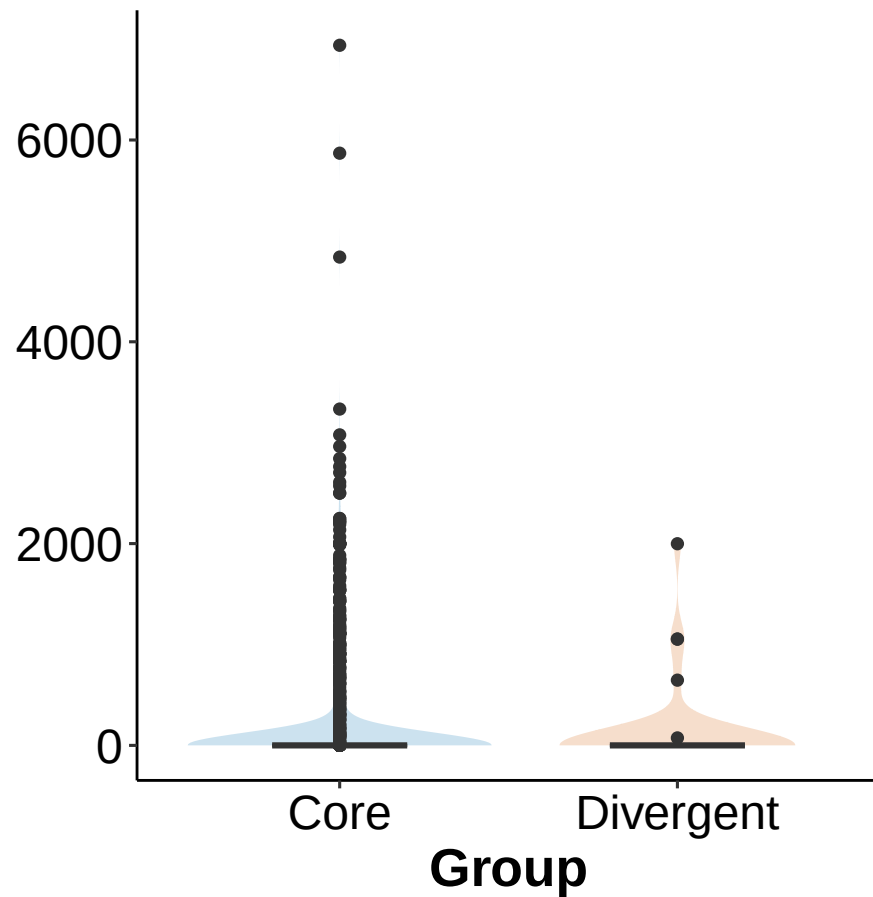
Metabolism – . 5. Systemic Reprogramming (SA_*)



Metabolism – . 5. Systemic Reprogramming (SA_*)

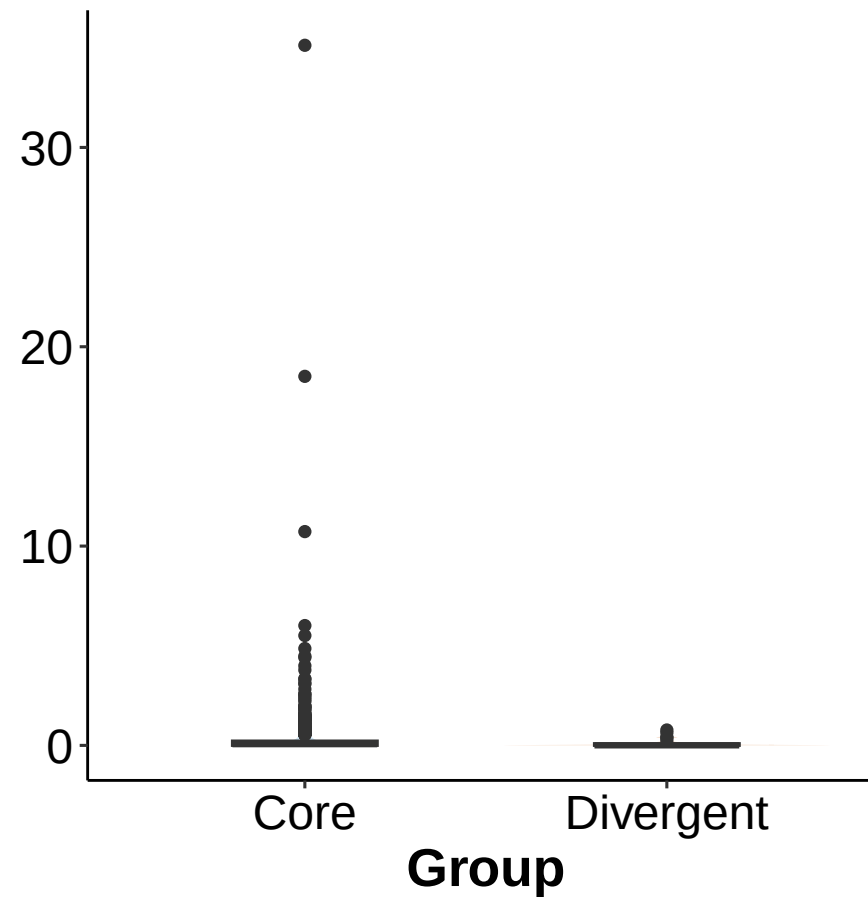
SA_Inositol_phosphate_metabolism

p-adj: 0.367 | .=-0.09 (Negligible)



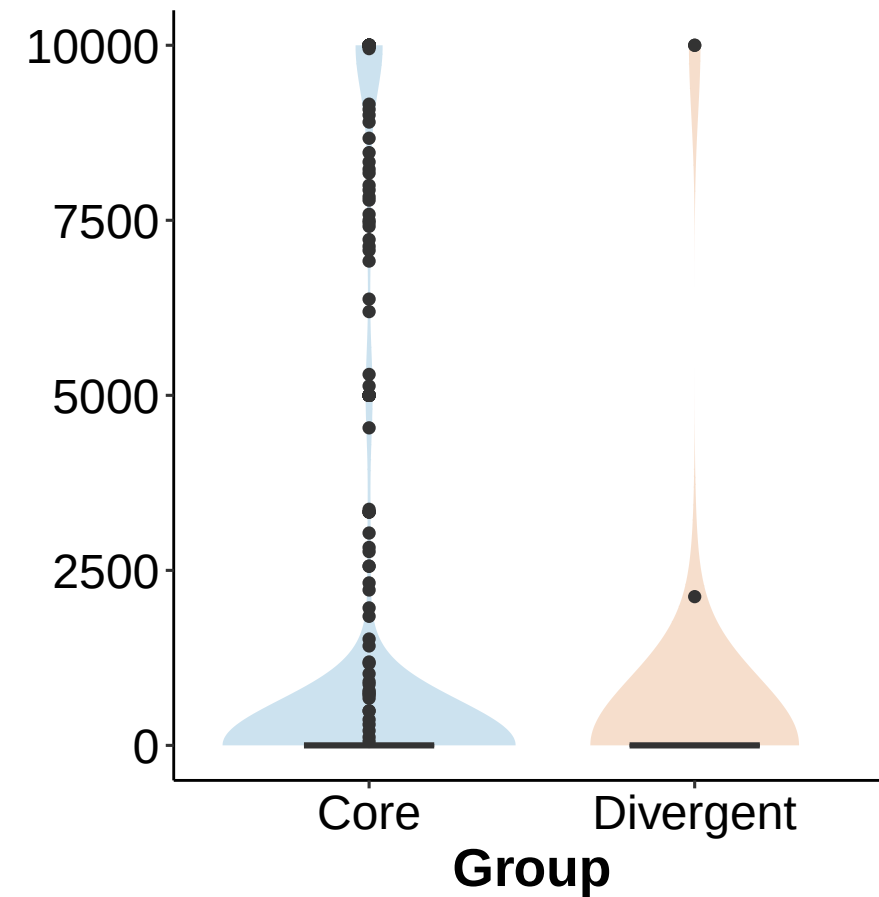
SA_Inositol_phosphate_metabolism_I

p-adj: 2.66e-05 | .=-0.46 (Medium) | .%: -99.9



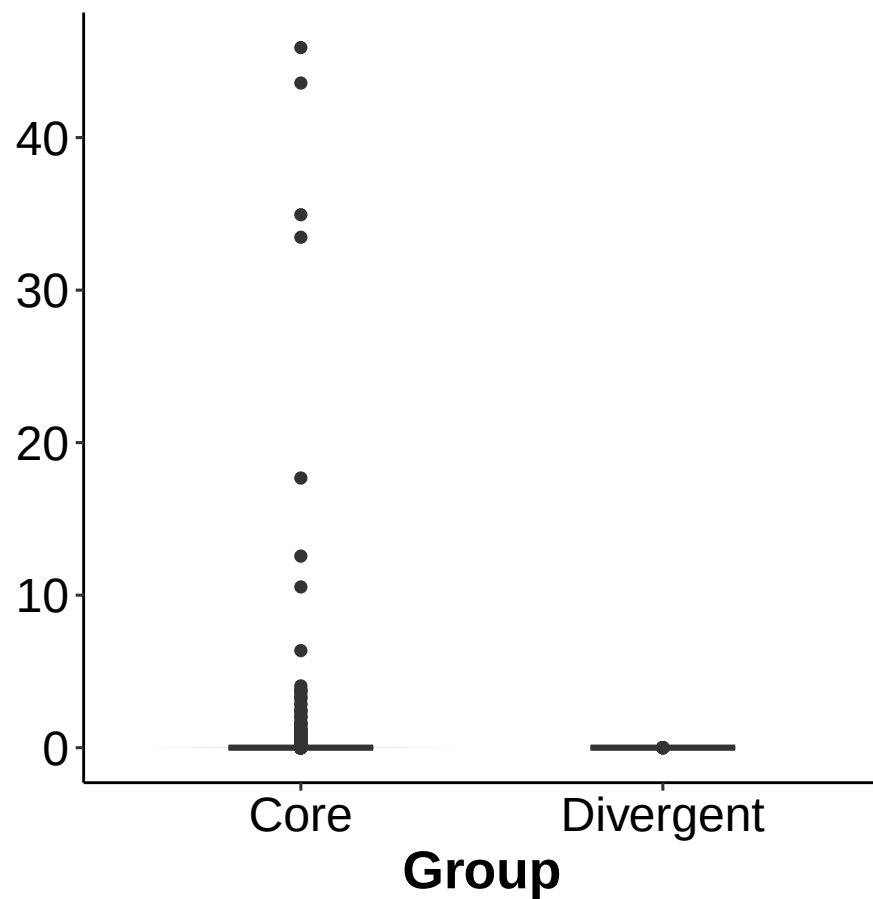
SA_Leukotriene_metabolism_FBA

p-adj: 0.388 | .=-0.07 (Negligible)



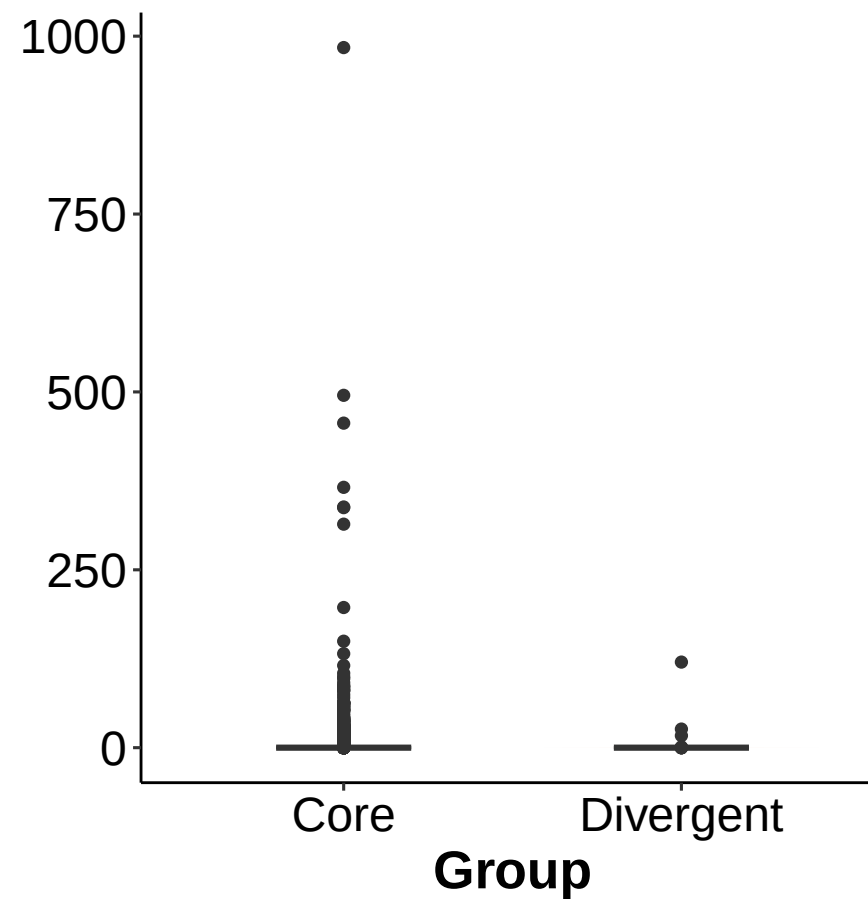
SA_Leukotriene_metabolism_L1w

p-adj: 0.964 | .=0 (Negligible)



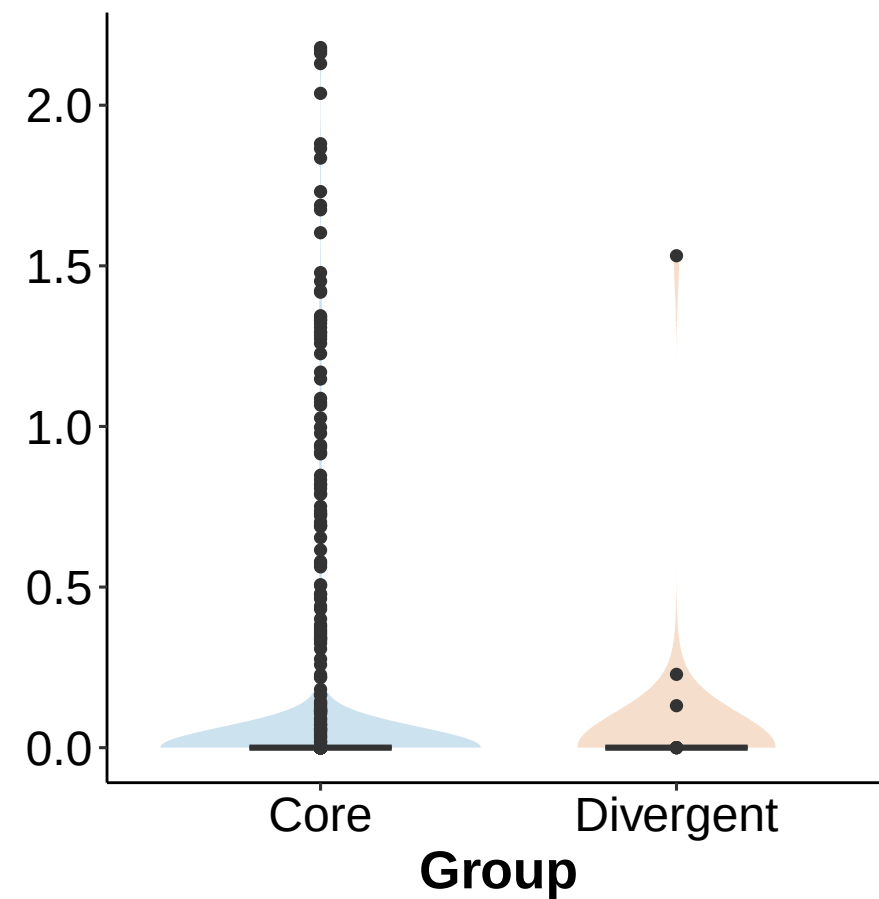
SA_Limonene_and_pinene_degrad

p-adj: 0.914 | .=-0.01 (Negligible)



SA_Linoleate_metabolism_L1w

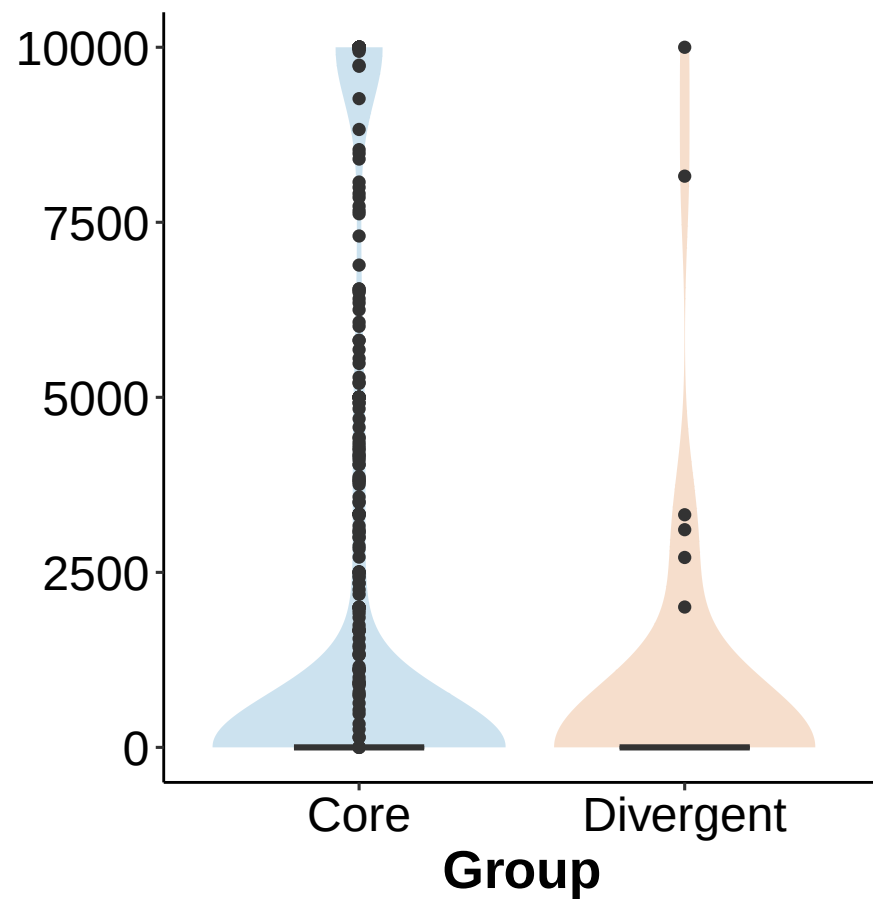
p-adj: 0.574 | .=-0.06 (Negligible)



Metabolism – . 5. Systemic Reprogramming (SA_*)

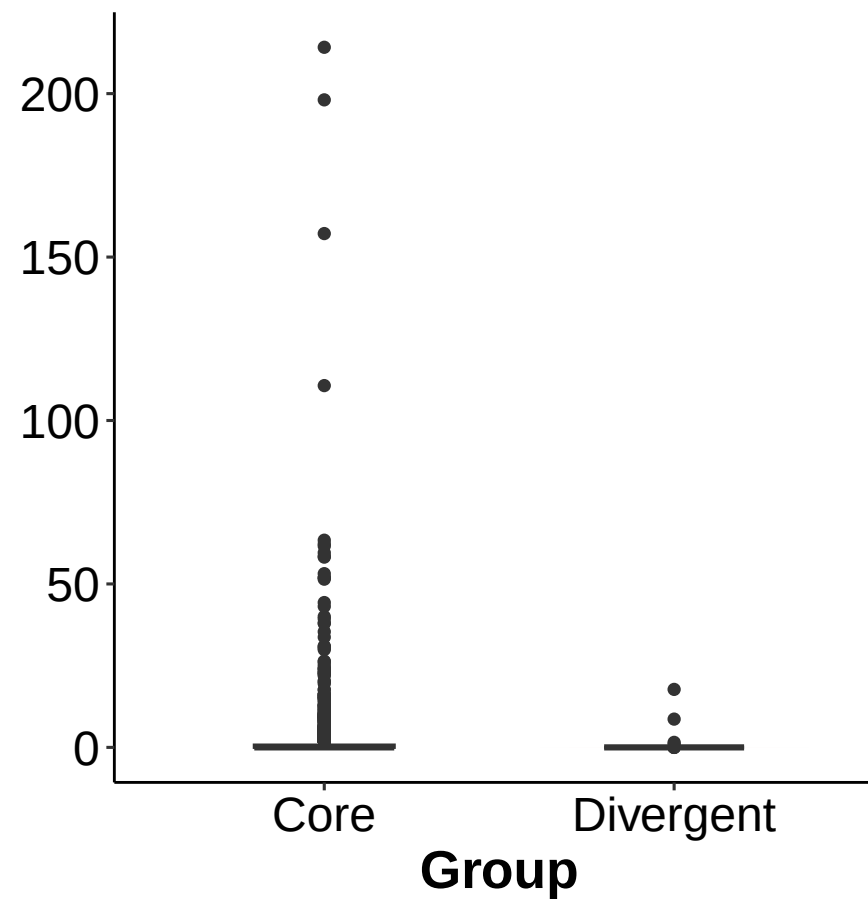
SA_Lysine_metabolism_FBA

p-adj: 0.285 | .=-0.1 (Negligible)



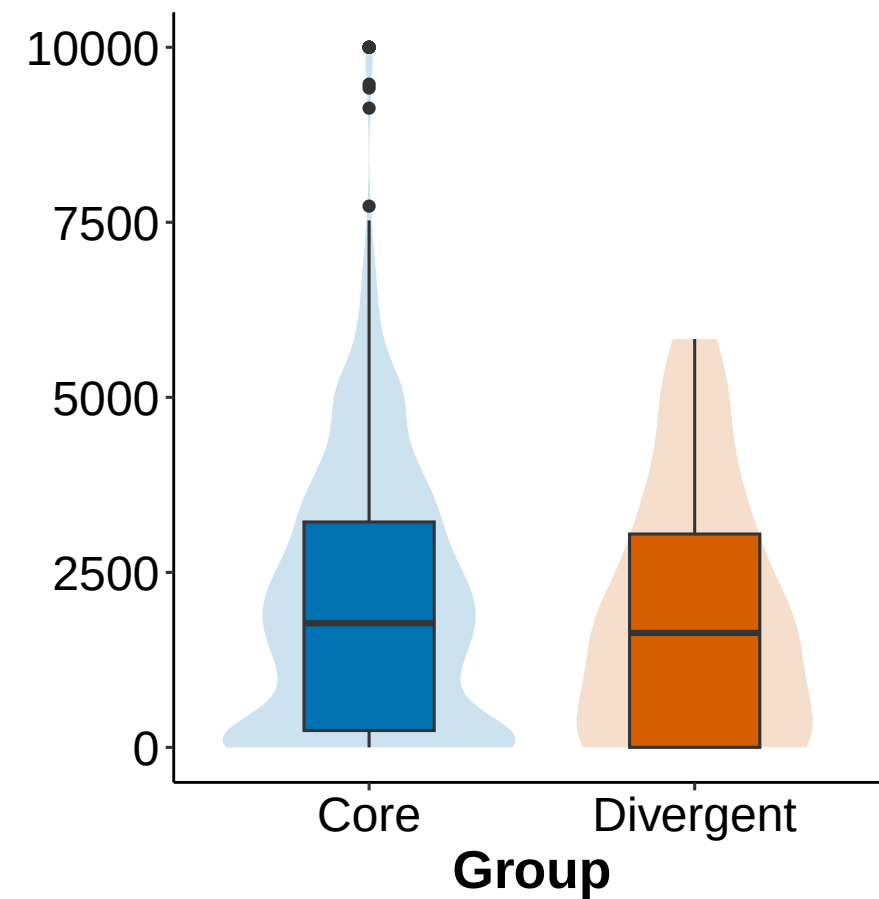
SA_Lysine_metabolism_L1w

p-adj: 0.12 | .=-0.16 (Small)



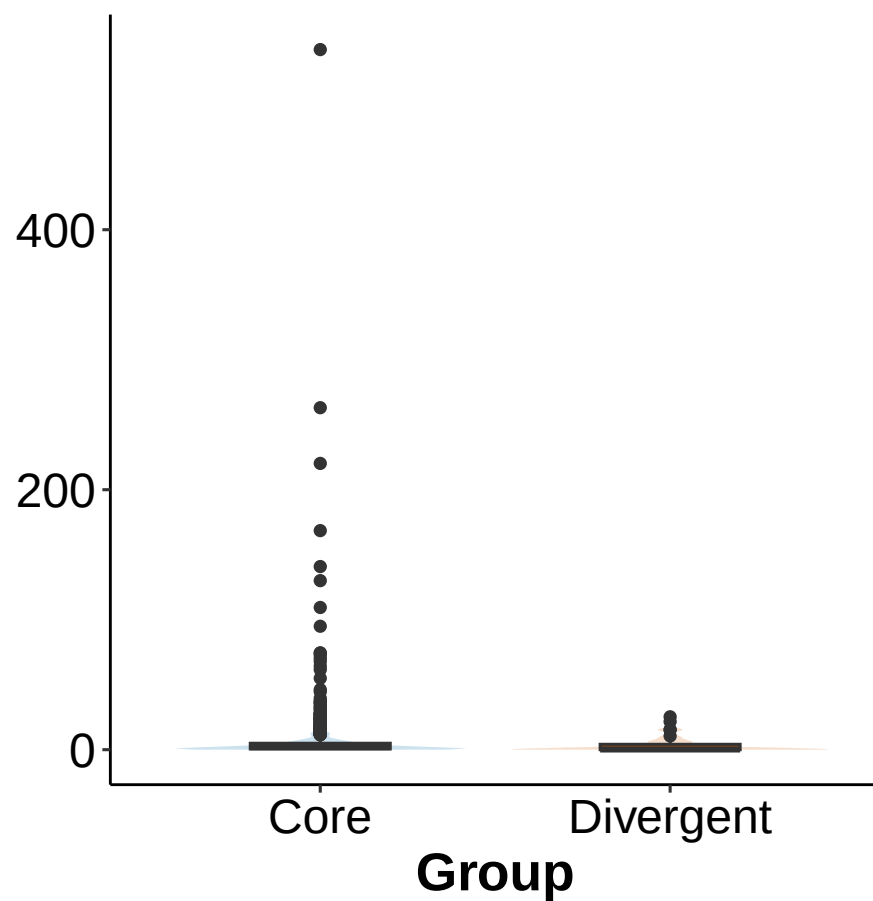
SA_Methionine_and_cysteine_me

p-adj: 0.599 | .=-0.07 (Negligible) | .%: -7.8



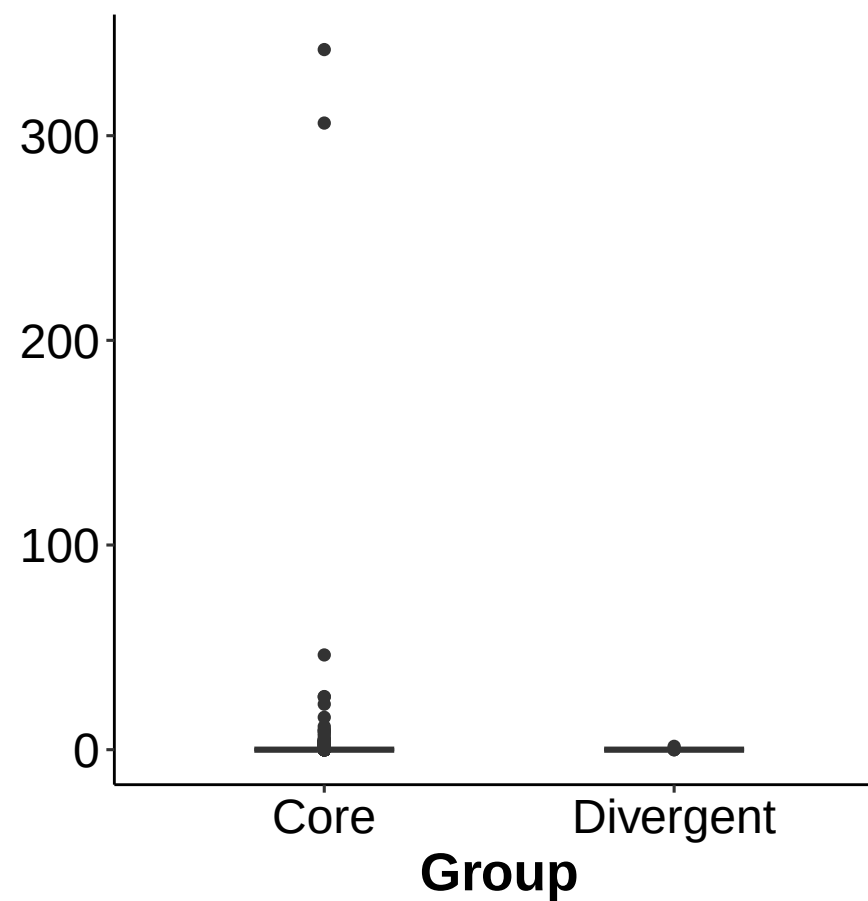
SA_Methionine_and_cysteine_metal

p-adj: 0.00621 | .=-0.29 (Small) | .%: -89.2



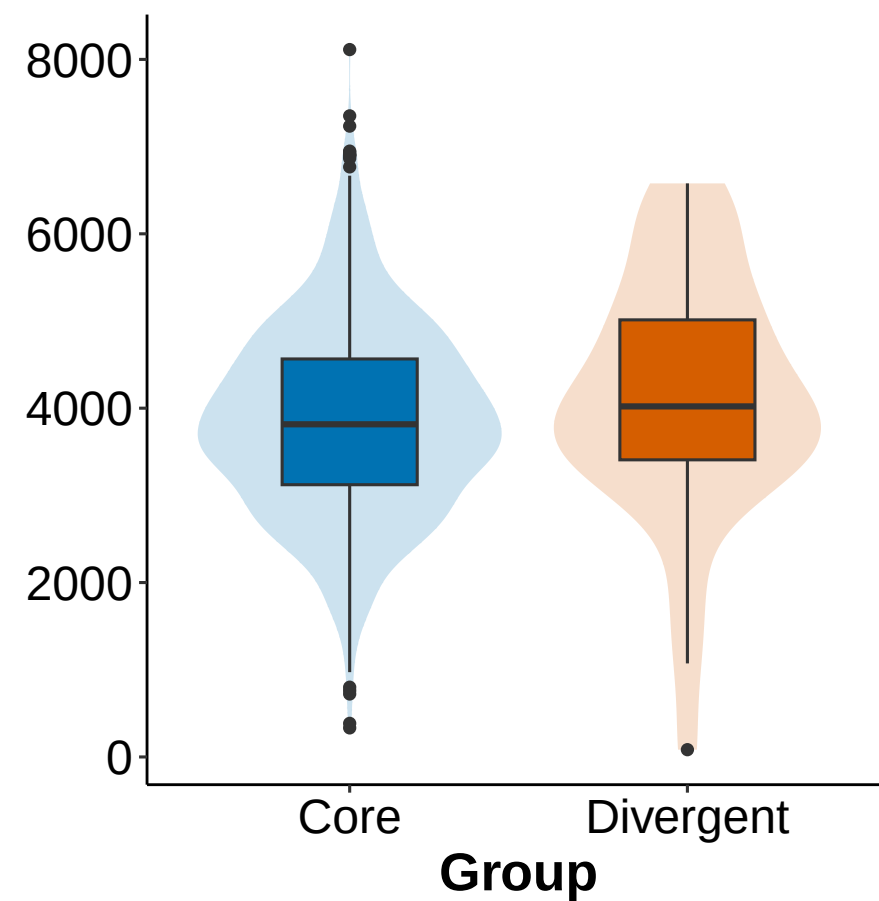
SA_Methionine_and_cysteine_metal

p-adj: 0.58 | .=0.06 (Negligible)



SA_Miscellaneous_FBA

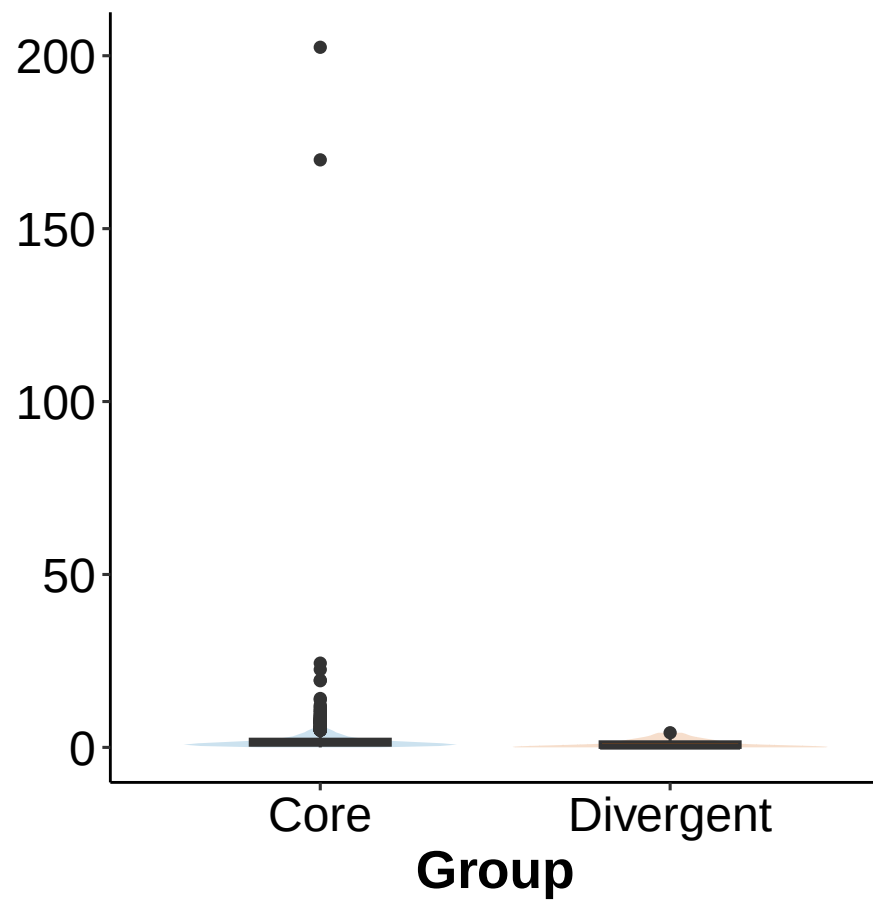
p-adj: 0.253 | .=0.14 (Negligible) | .%: 5.4



Metabolism – . 5. Systemic Reprogramming (SA_*)

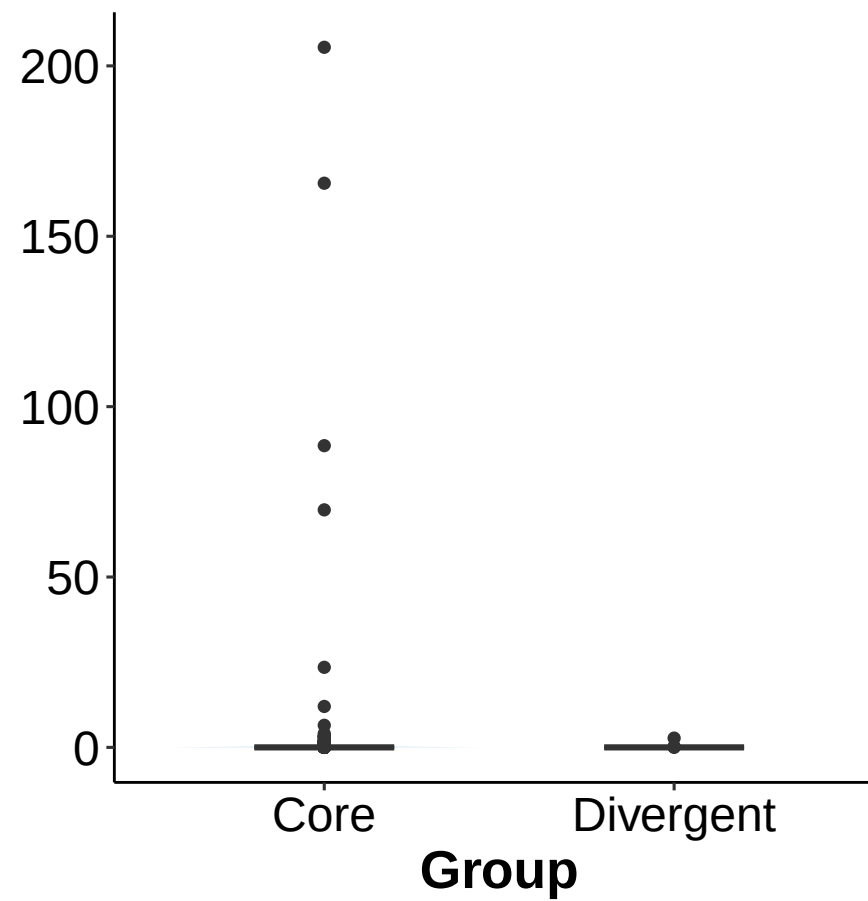
SA_Miscellaneous_L1w

p-adj: 2.83e-05 | .=-0.45 (Medium) | .%: -79.4



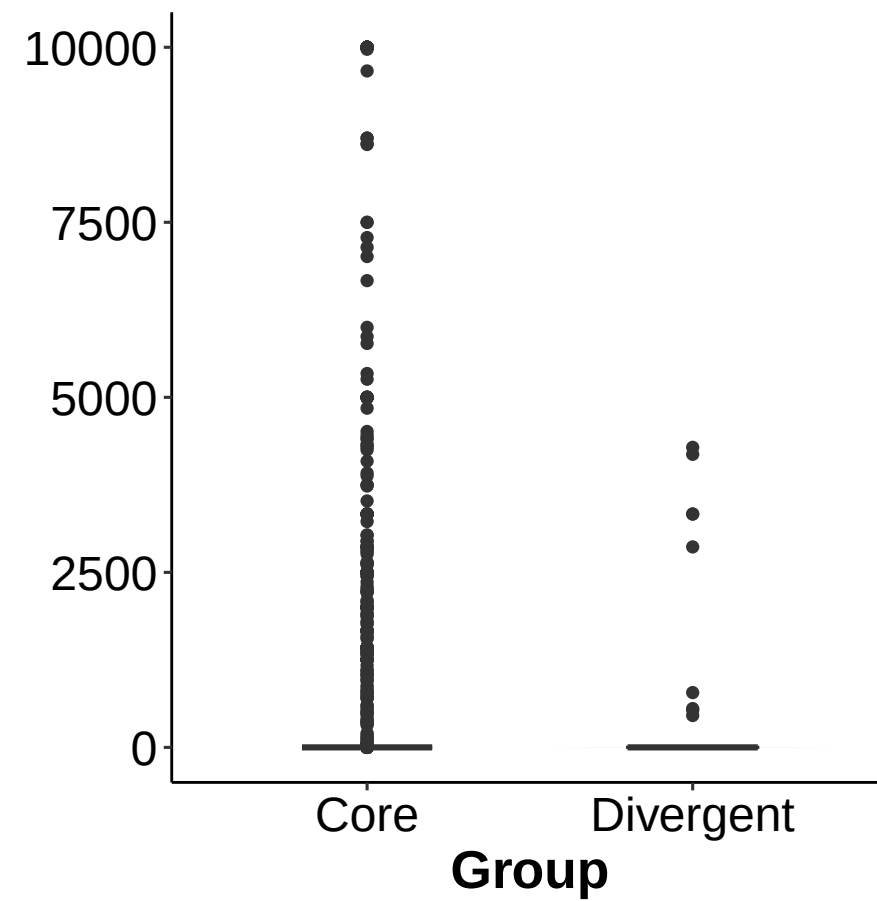
SA_Miscellaneous_pFBA

p-adj: 0.347 | .=-0.12 (Negligible) | .%: -38.4



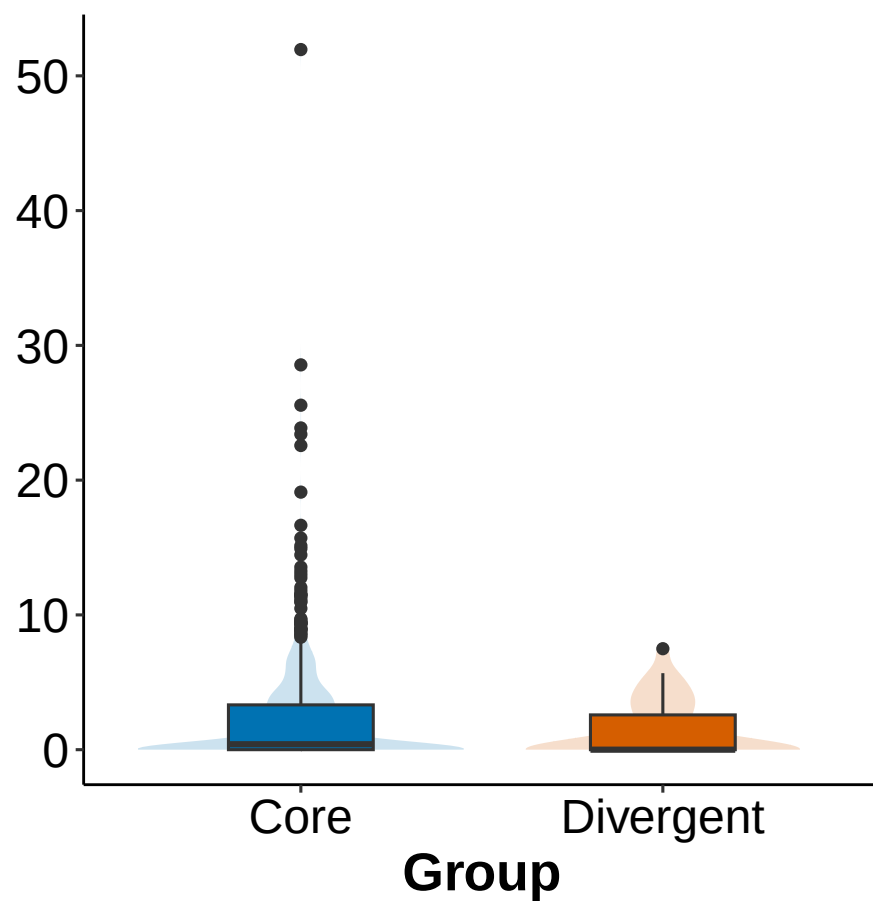
SA_NAD_metabolism_FBA

p-adj: 0.845 | .=0.02 (Negligible)



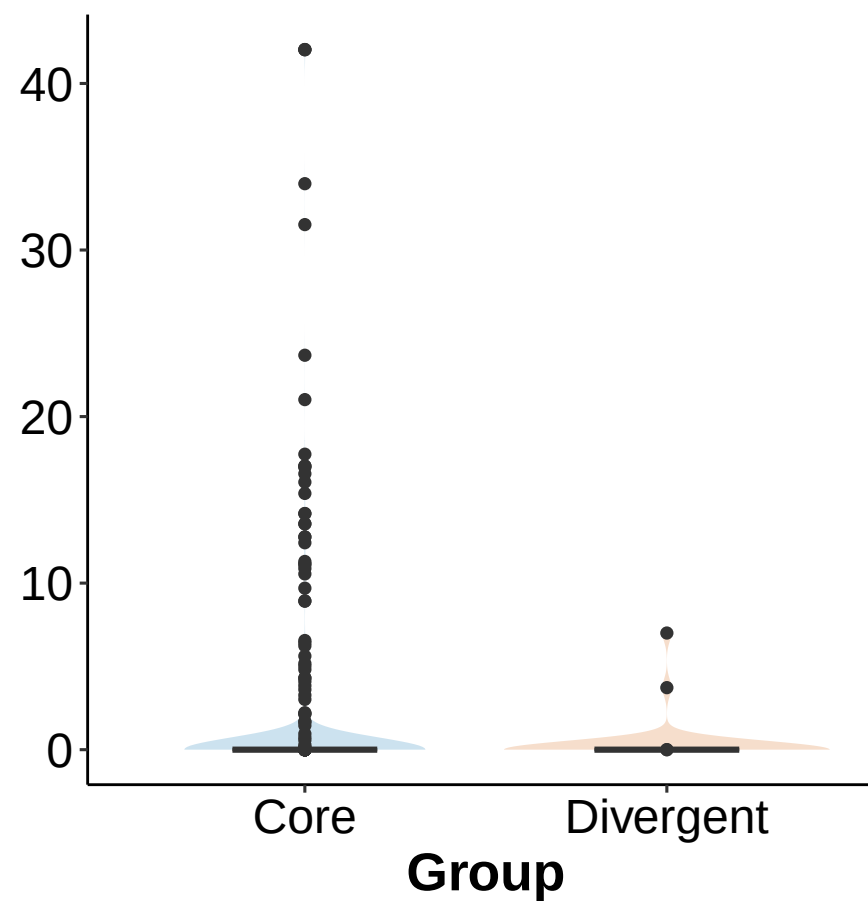
SA_NAD_metabolism_L1w

p-adj: 0.79 | .=-0.04 (Negligible) | .%: -99.9



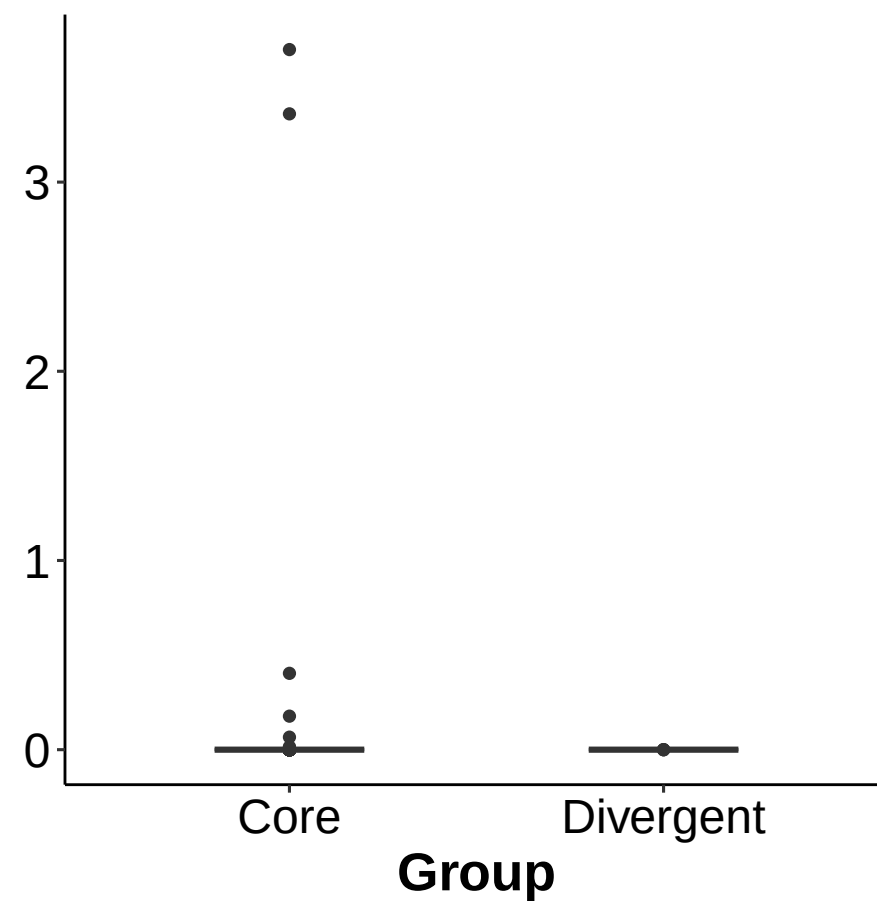
SA_NAD_metabolism_pFBA

p-adj: 0.978 | .=0 (Negligible)



SA_N_glycan_synthesis_L1w

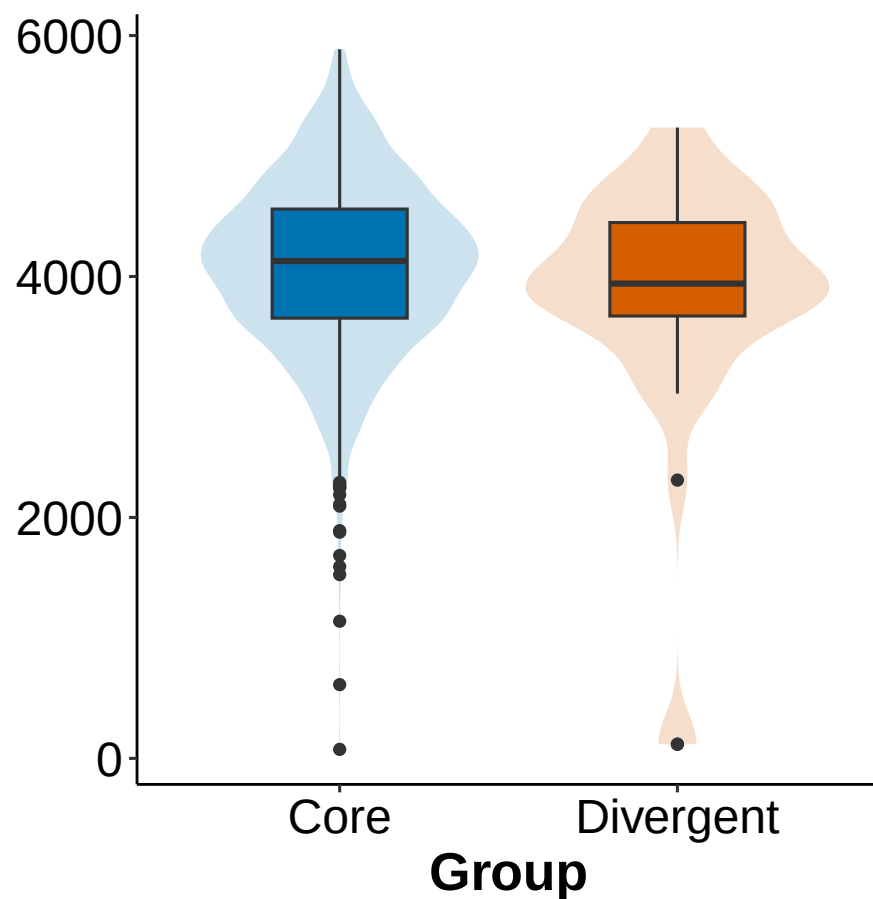
p-adj: 0.949 | .=0 (Negligible)



Metabolism – . 5. Systemic Reprogramming (SA_*)

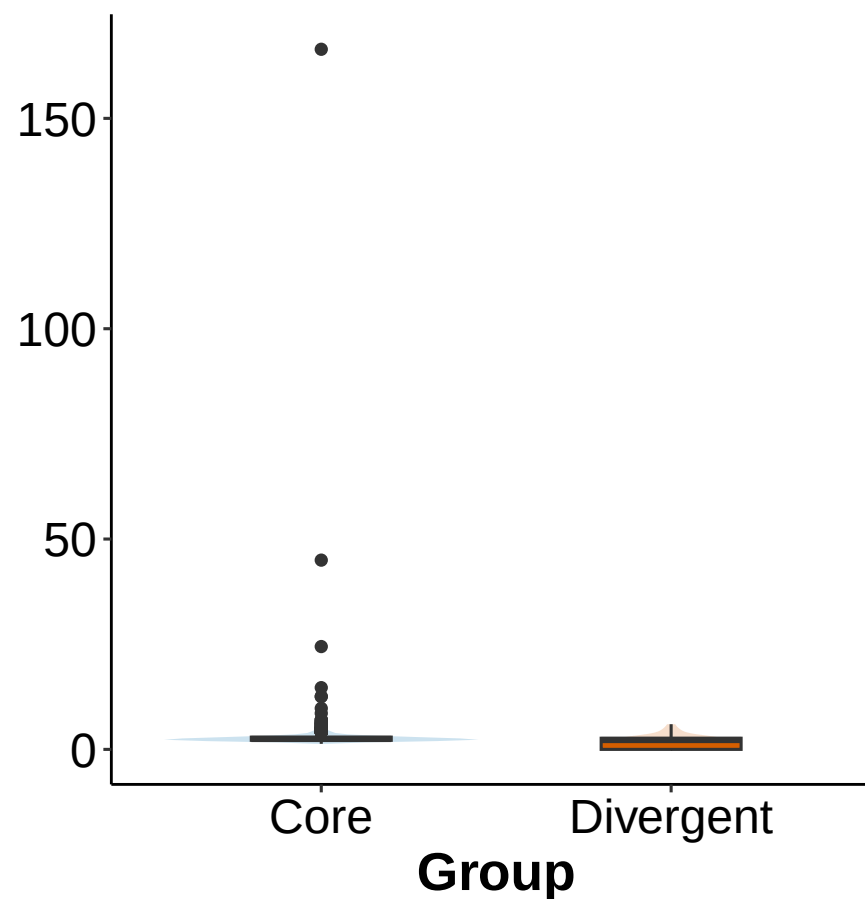
SA_Nucleotide_interconversion_FB

p-adj: 0.38 | .=-0.12 (Negligible) | .%: -4.5



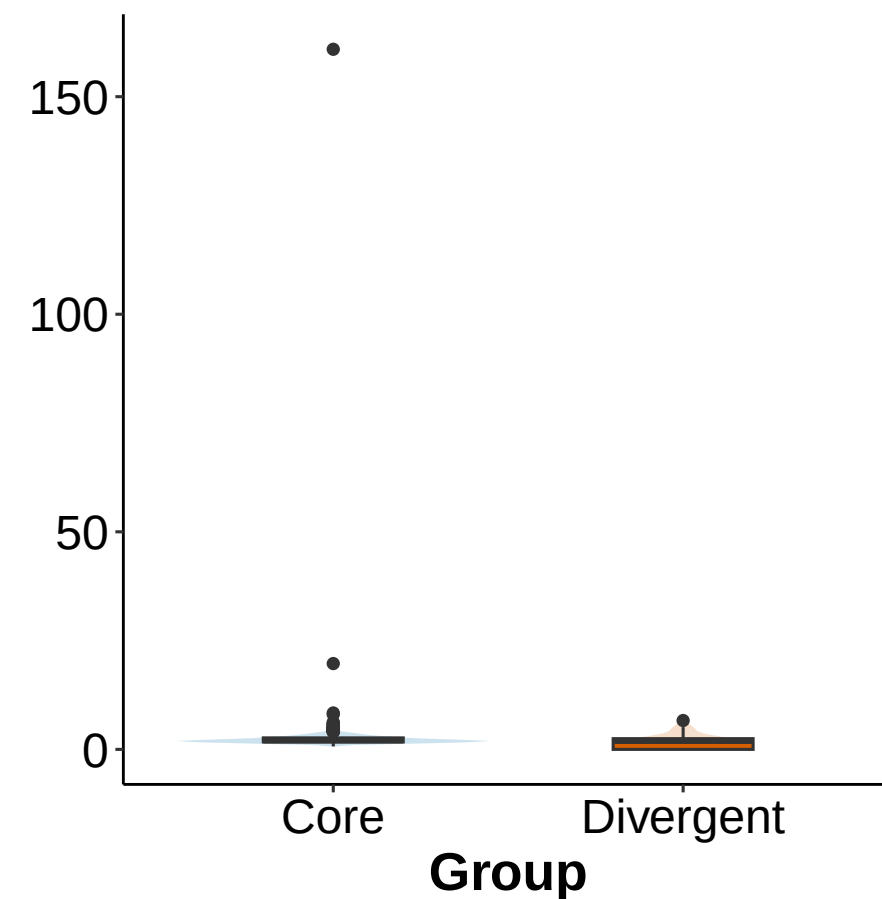
SA_Nucleotide_interconversion_L1w

p-adj: 0.0109 | .=-0.28 (Small) | .%: -10.4



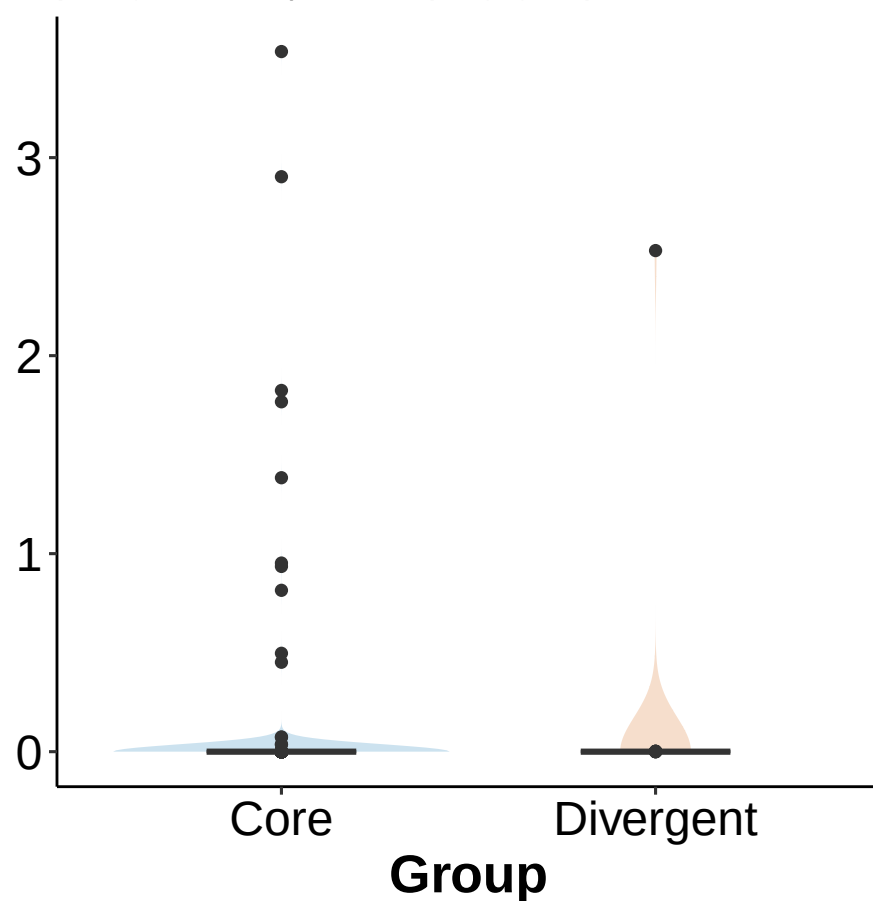
SA_Nucleotide_interconversion_pFB

p-adj: 0.0292 | .=-0.24 (Small) | .%: -5.4



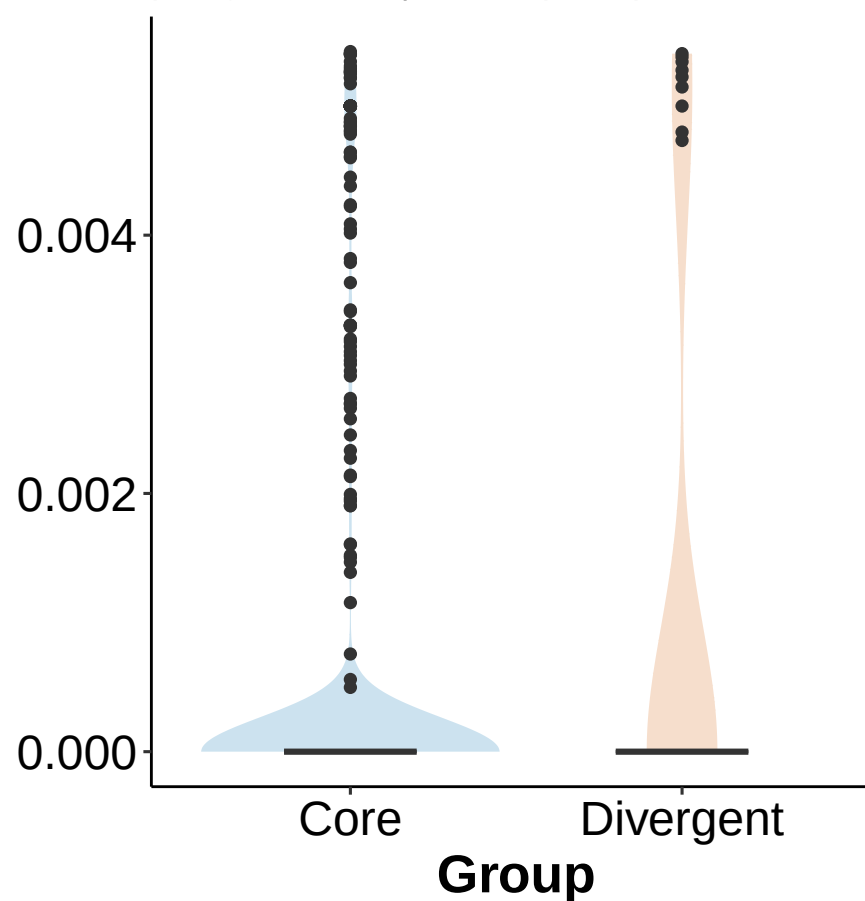
SA_Nucleotide_metabolism_L1w

p-adj: 0.0651 | .=-0.12 (Negligible)



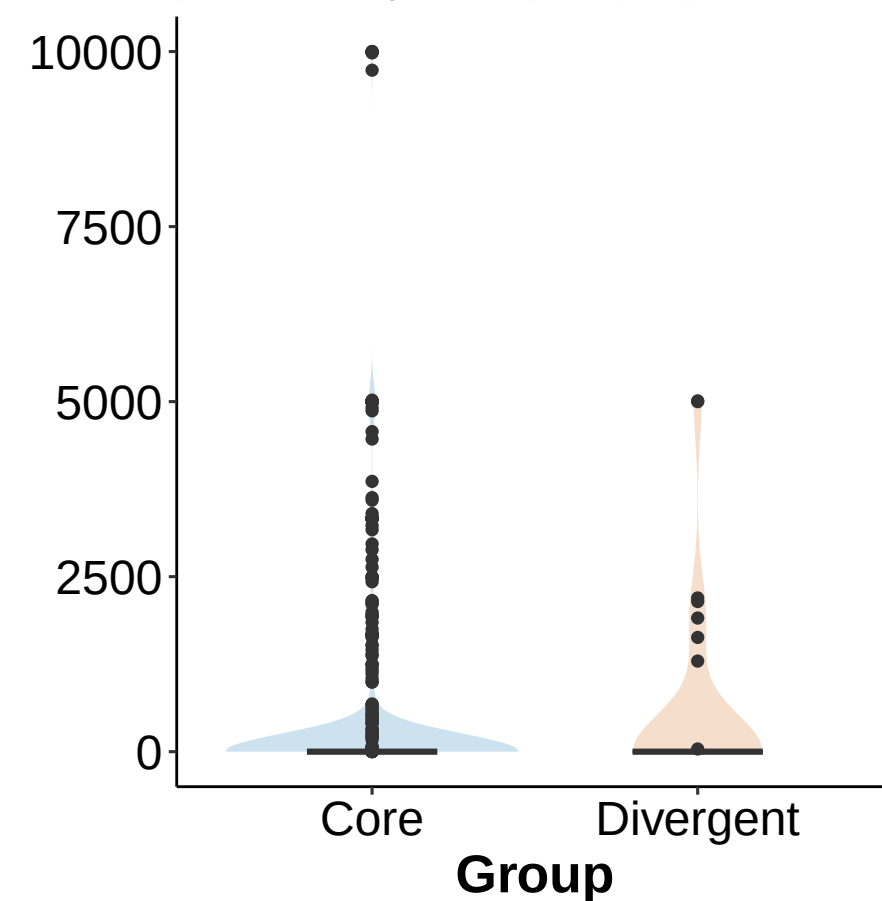
SA_Nucleotide_metabolism_pFBA

p-adj: 0.00621 | .=-0.15 (Small)



SA_Nucleotide_salvage_pathway_pFBA

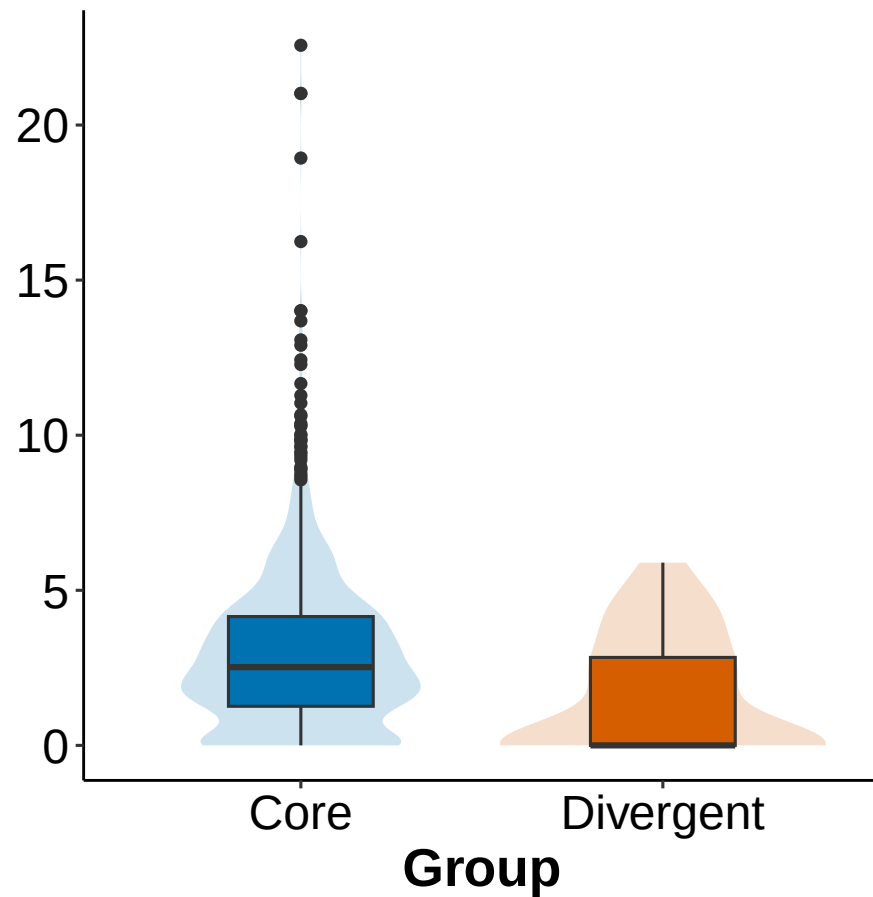
p-adj: 0.823 | .=-0.02 (Negligible)



Metabolism – . 5. Systemic Reprogramming (SA_*)

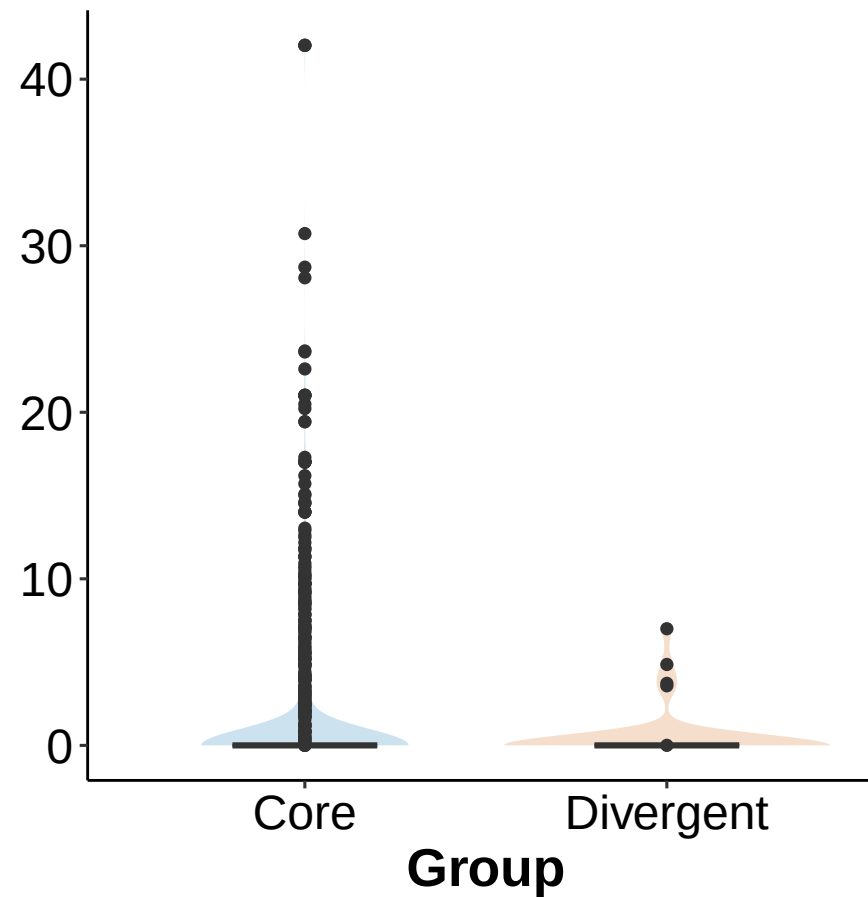
SA_Nucleotide_salvage_pathway_L1v

p-adj: 0.00592 | .=-0.3 (Small) | .%: -100



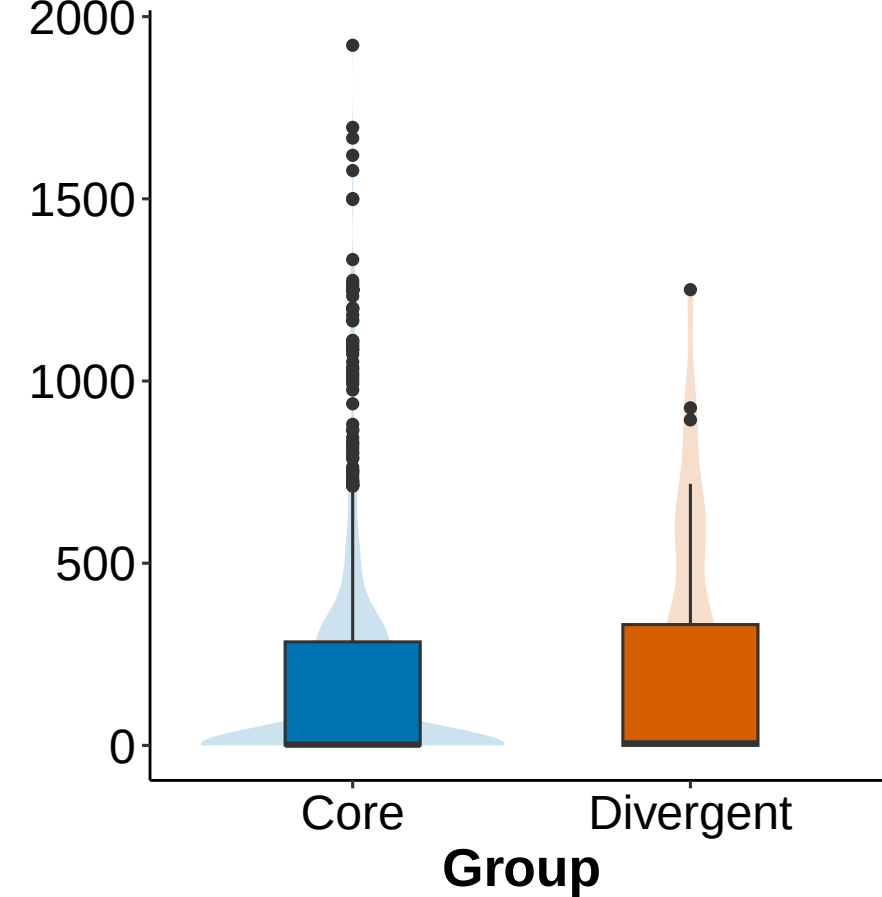
SA_Nucleotide_salvage_pathway_pFE

p-adj: 0.518 | .=-0.06 (Negligible)



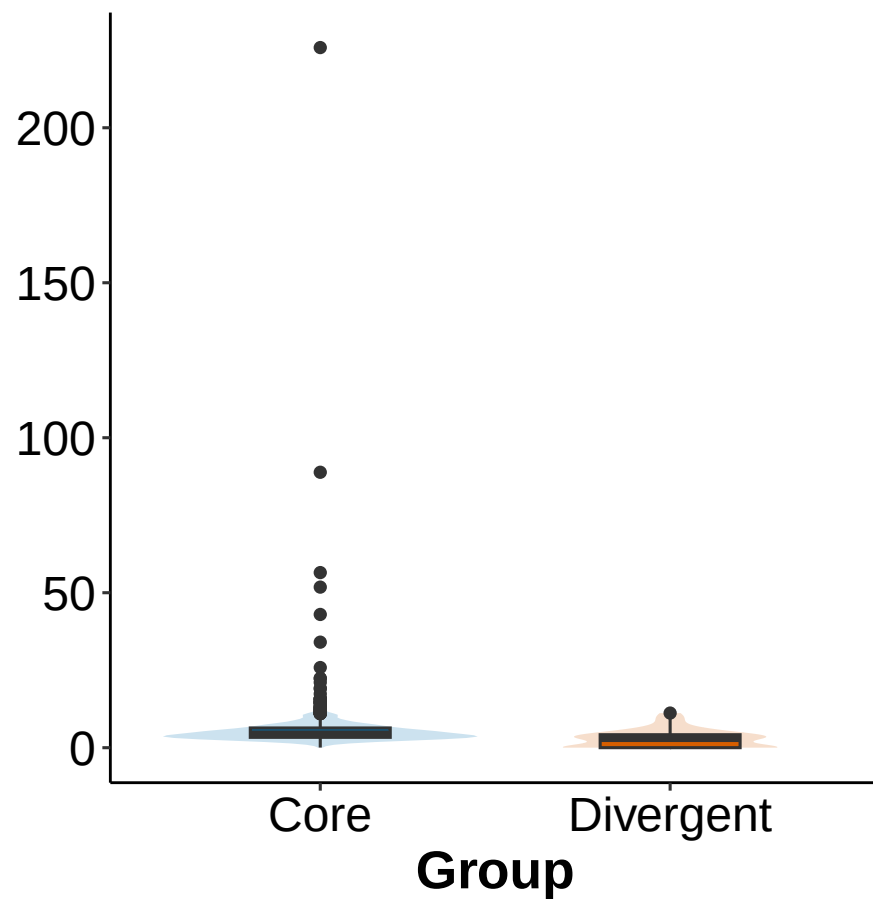
SA_Oxidative_phosphorylation_FB

p-adj: 0.942 | .-=0.01 (Negligible) | .%: 192.6



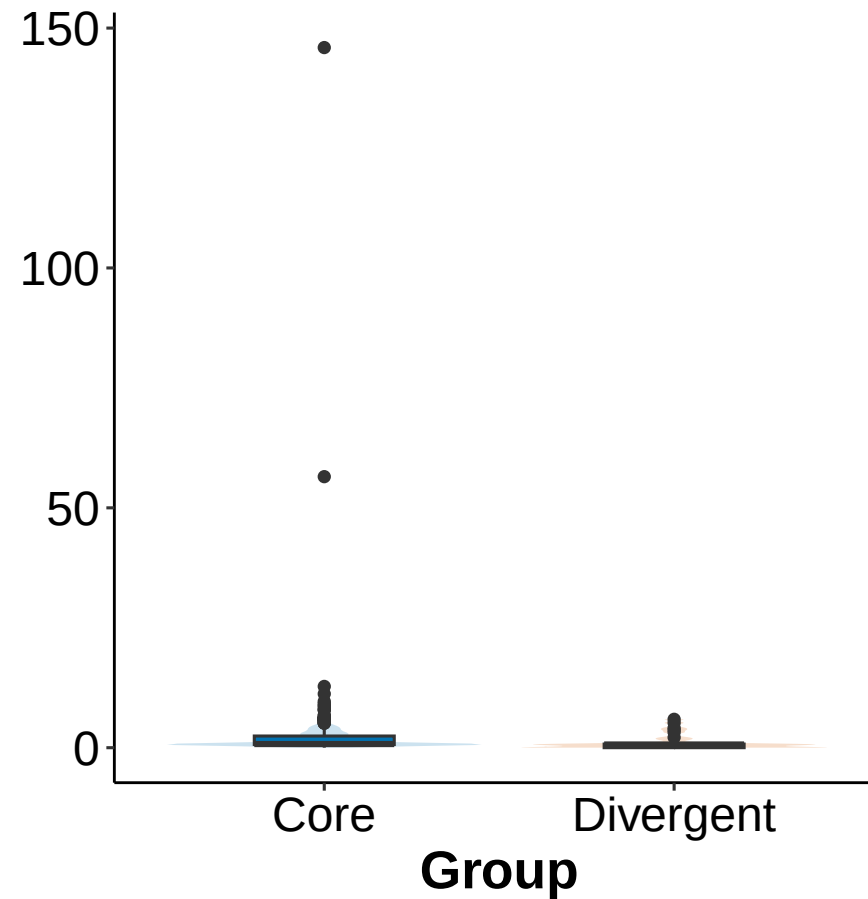
SA_Oxidative_phosphorylation_L1w

p-adj: 1.04e-05 | .=-0.48 (Large) | .%: -38.1



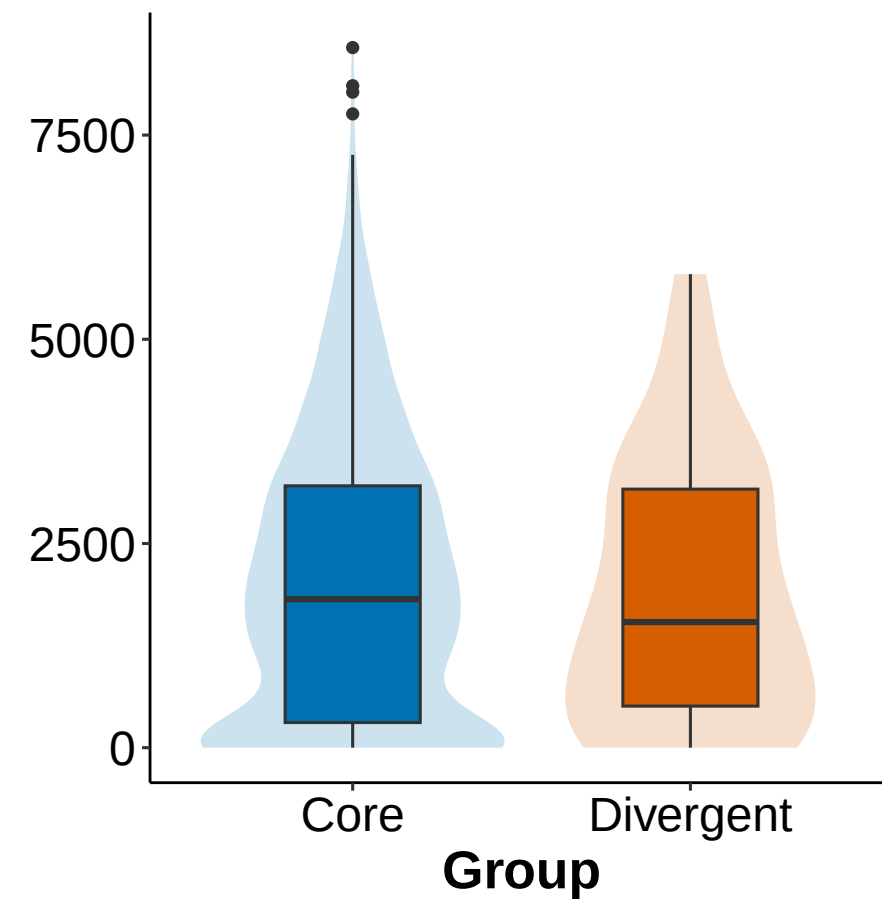
SA_Oxidative_phosphorylation_pFB

p-adj: 0.000148 | .=-0.4 (Medium) | .%: -12.5



SA_Pentose_phosphate_pathway_L1w

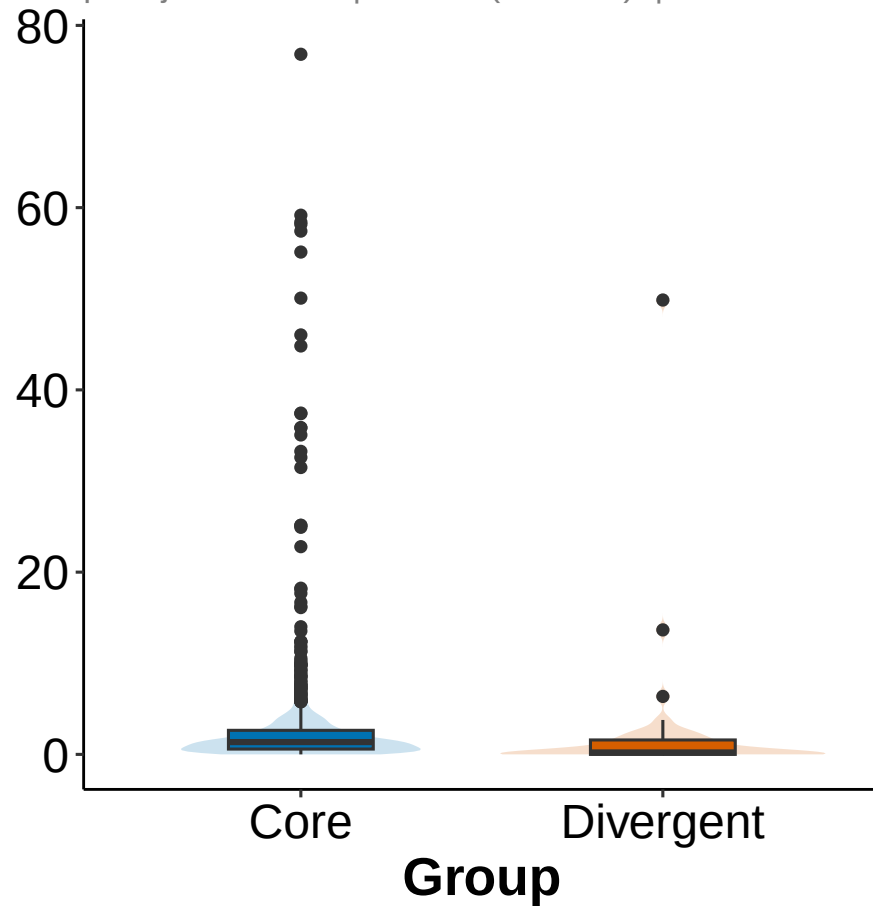
p-adj: 0.759 | .=-0.04 (Negligible) | .%: -15.4



Metabolism – . 5. Systemic Reprogramming (SA_*)

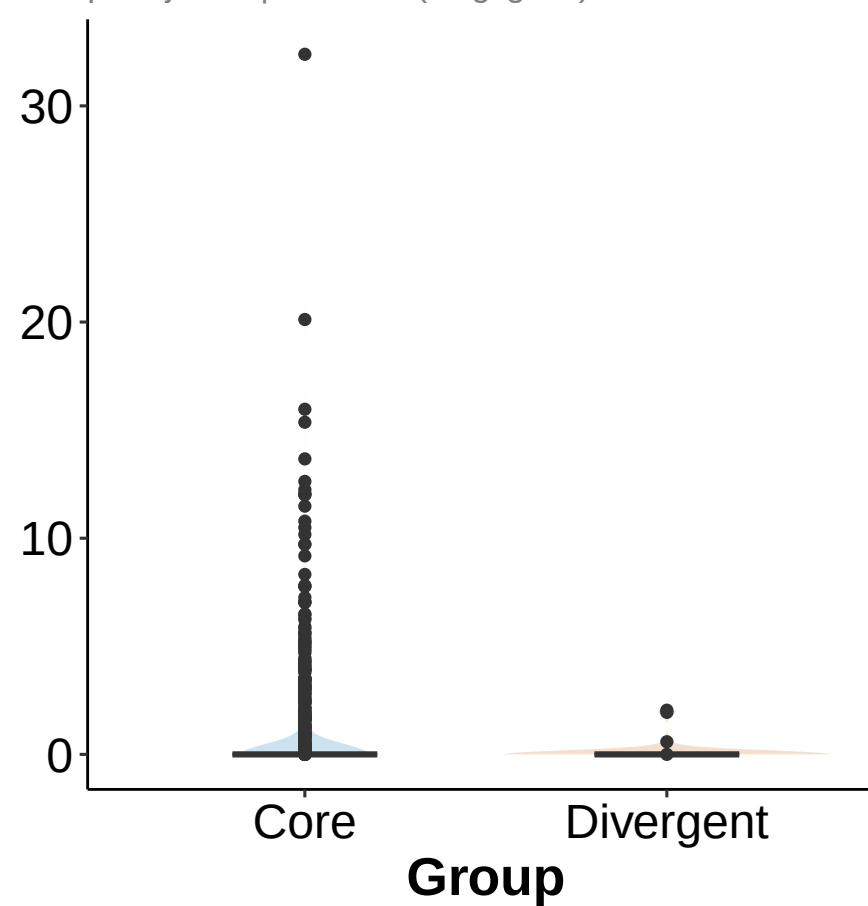
SA_Pentose_phosphate_pathway_L1w

p-adj: 0.000202 | .=-0.4 (Medium) | .%: -84.3



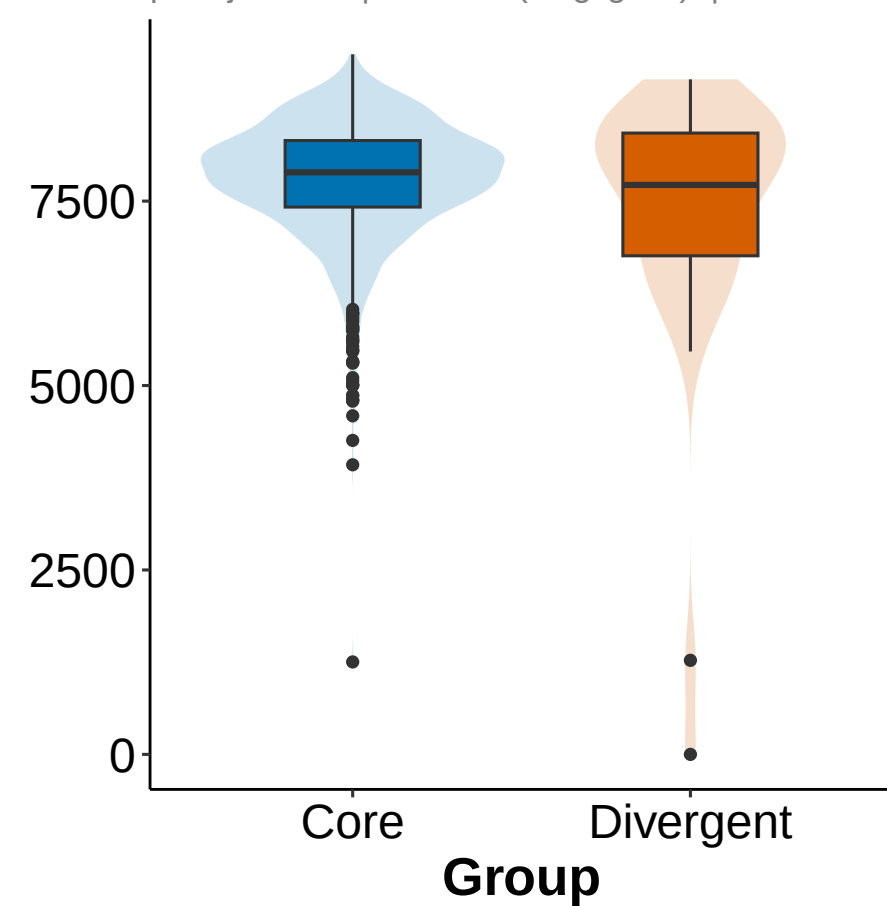
SA_Pentose_phosphate_pathway_pFI

p-adj: 0.3 | .=-0.09 (Negligible)



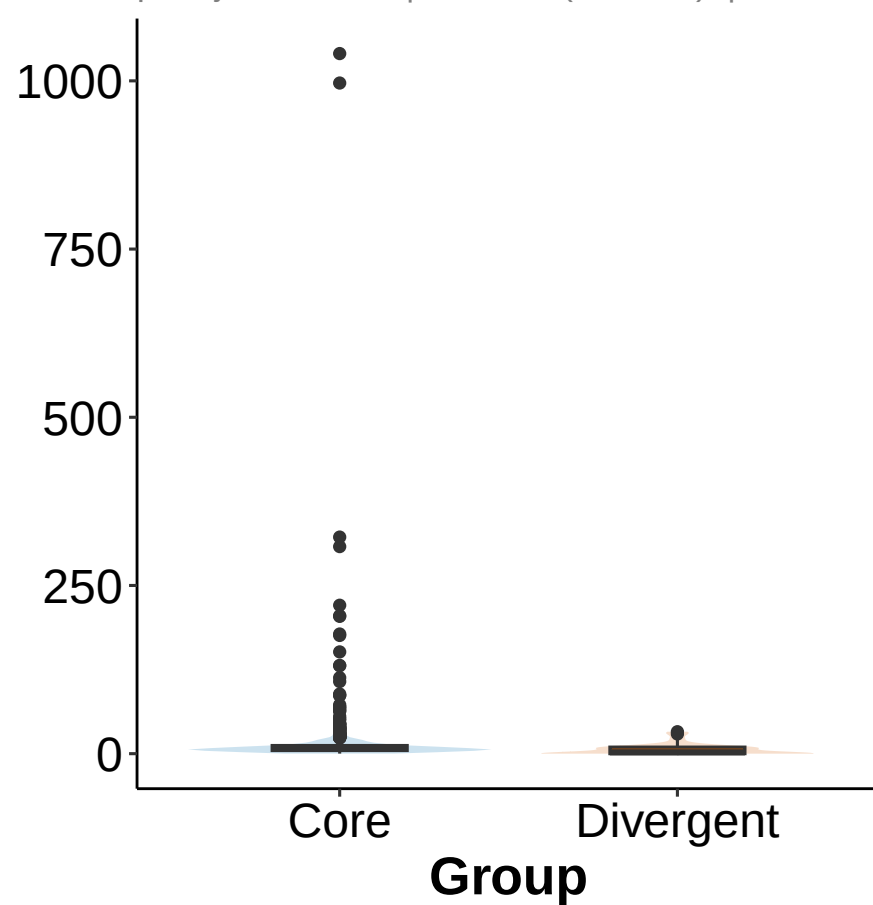
SA_Peptide_metabolism_FBA

p-adj: 0.746 | .=-0.05 (Negligible) | .%: -2.2



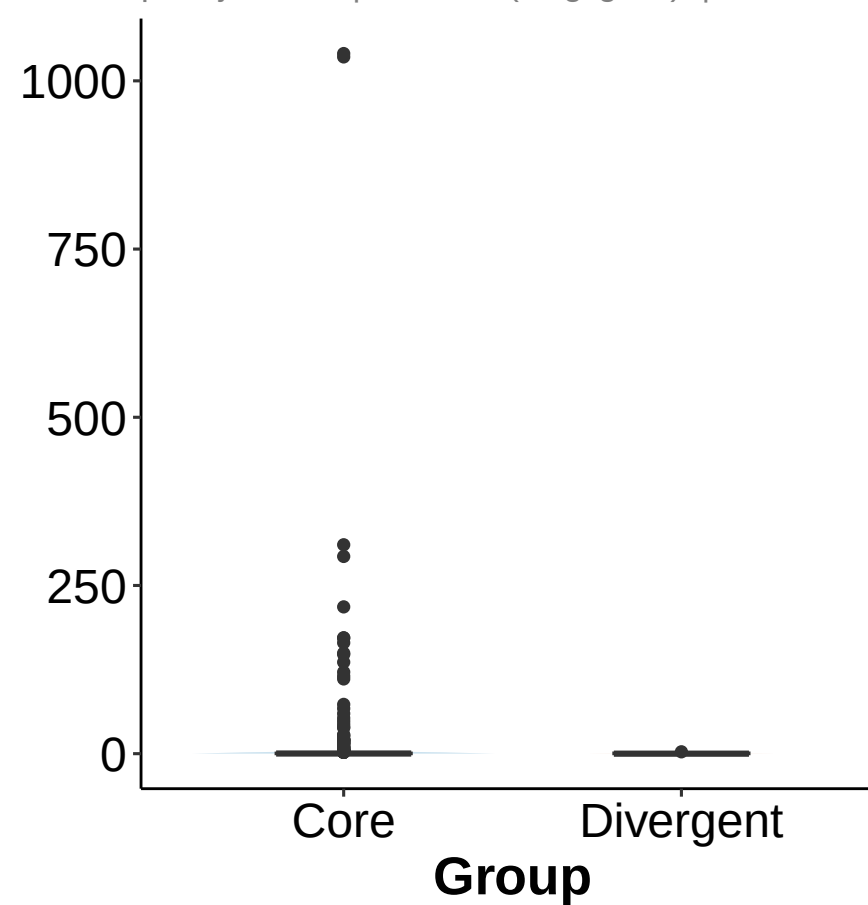
SA_Peptide_metabolism_L1w

p-adj: 0.000518 | .=-0.37 (Medium) | .%: -72.4



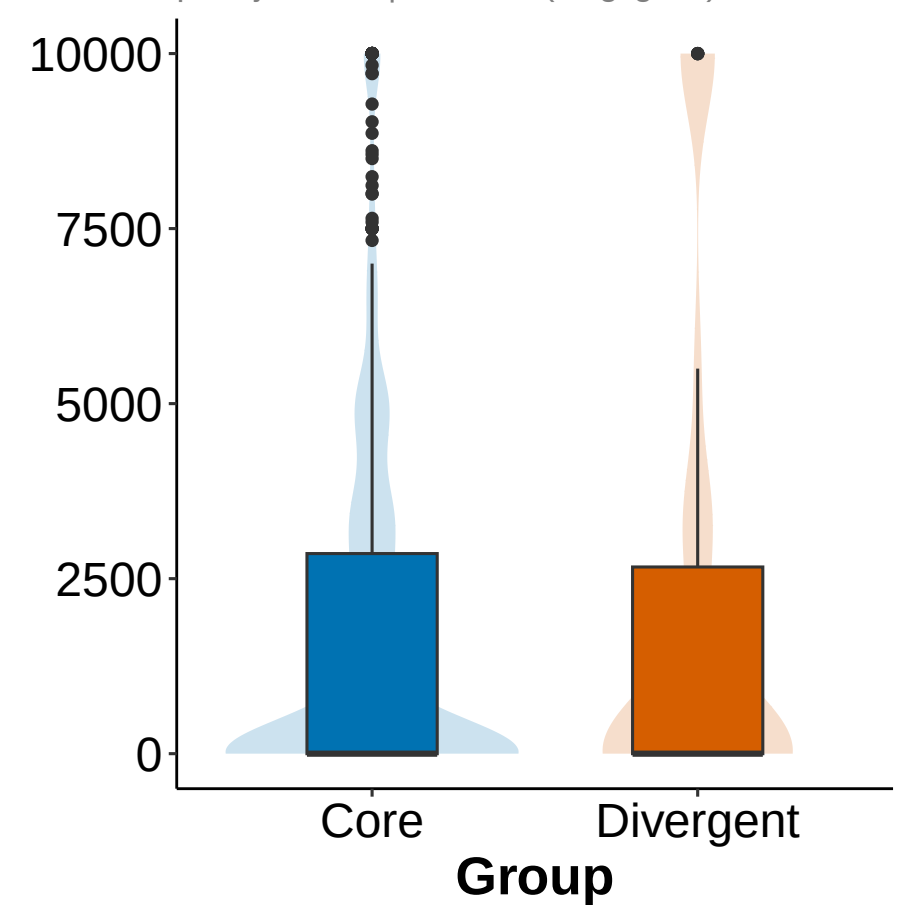
SA_Peptide_metabolism_pFBA

p-adj: 0.566 | .=-0.08 (Negligible) | .%: -100



SA_Phenylalanine_metabolism_FBA

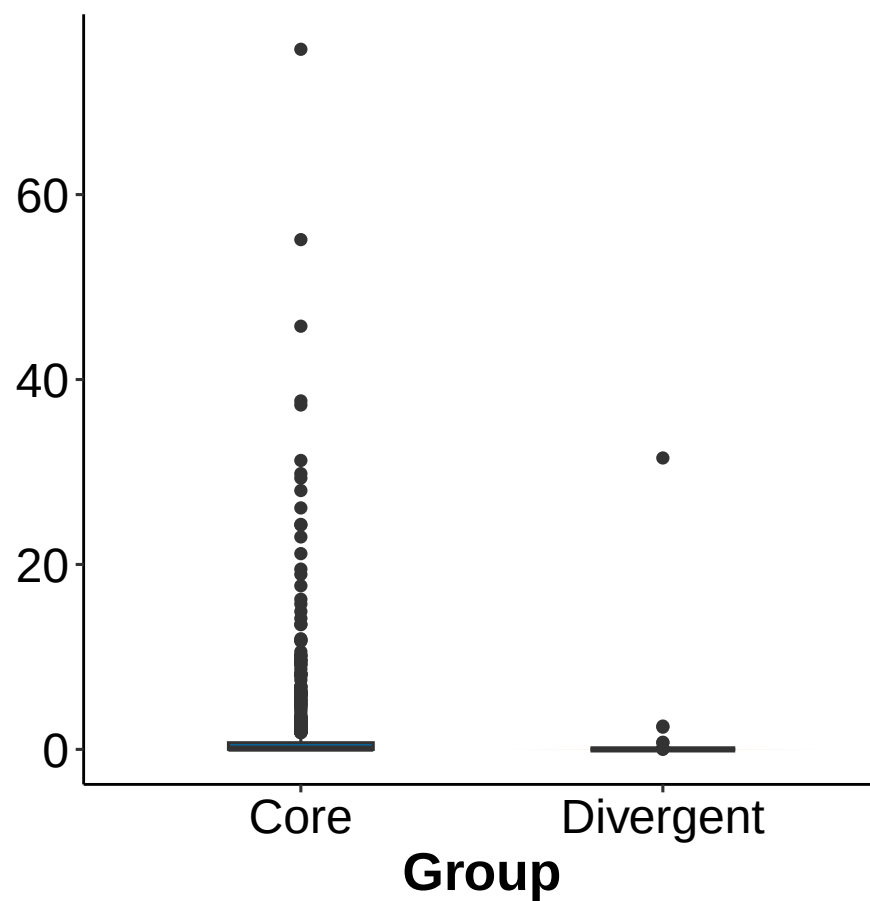
p-adj: 0.644 | .=-0.06 (Negligible)



Metabolism – 5. Systemic Reprogramming (SA_*)

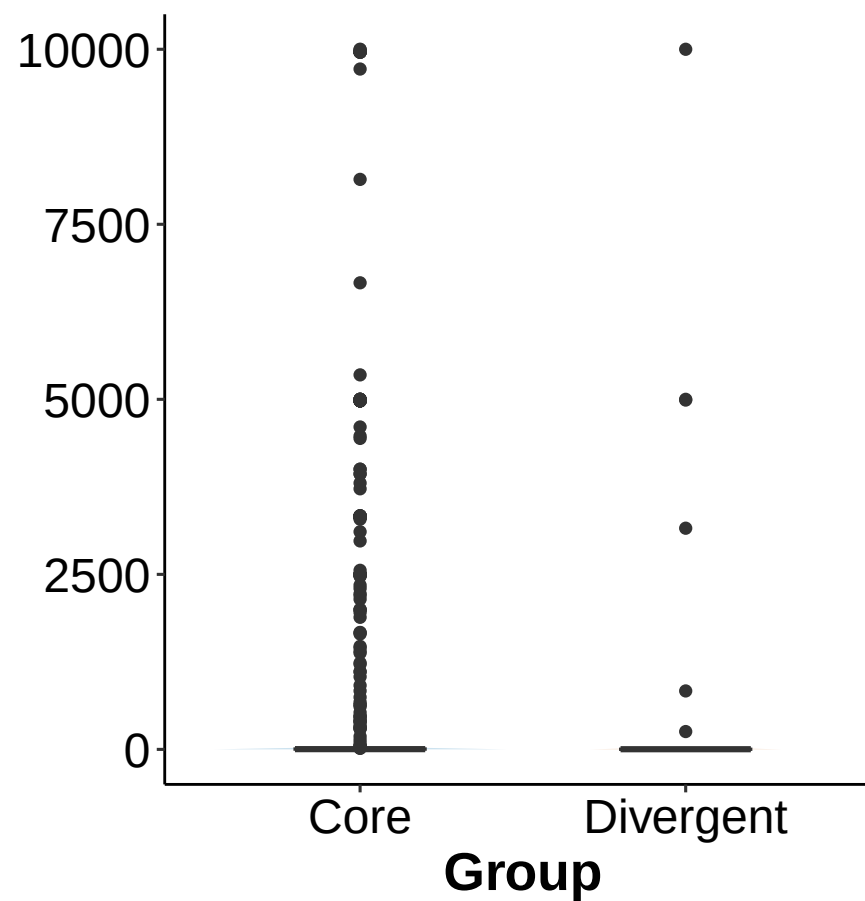
SA_Phenylalanine_metabolism_L1w

p-adj: 0.00108 | .=-0.35 (Medium) | .%: -100



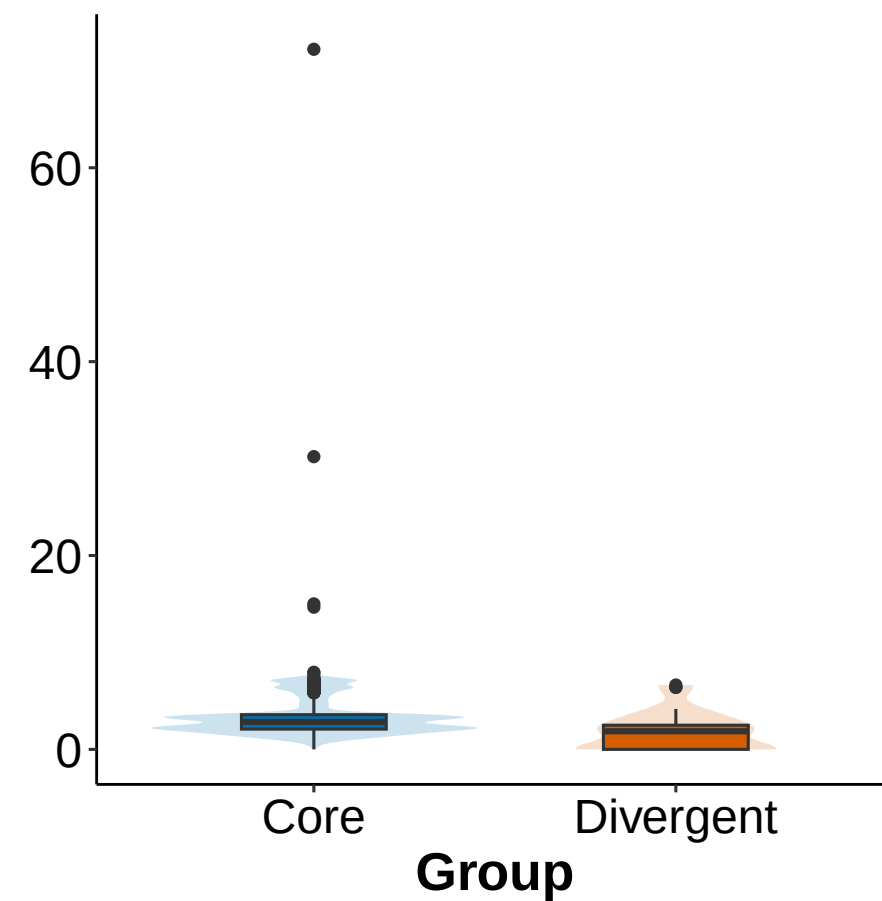
SA_Phosphatidylinositol_phosphatidylcholine

p-adj: 0.000219 | .=-0.38 (Medium) | .%: -33



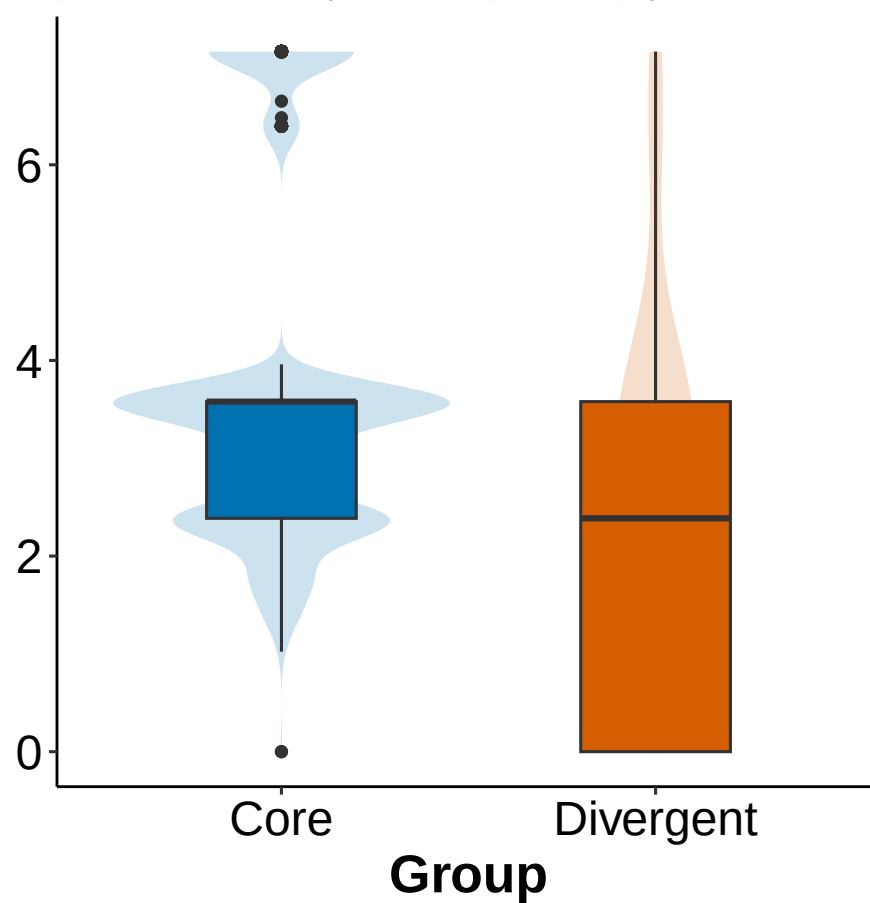
SA_Phosphatidylinositol_phosphate_n

p-adj: 3.91e-05 | .=-0.44 (Medium) | .%: -34.1



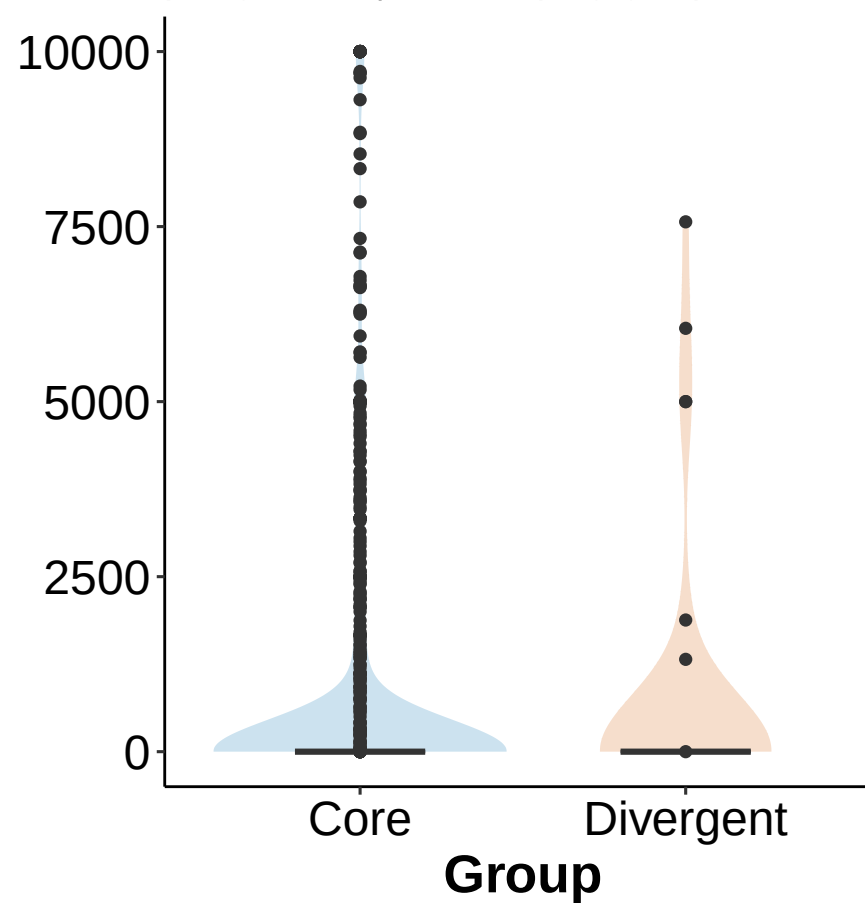
SA_Phosphatidylinositol_phosphate_n

p-adj: 6.64e-05 | .=-0.41 (Medium) | .%: -33.3



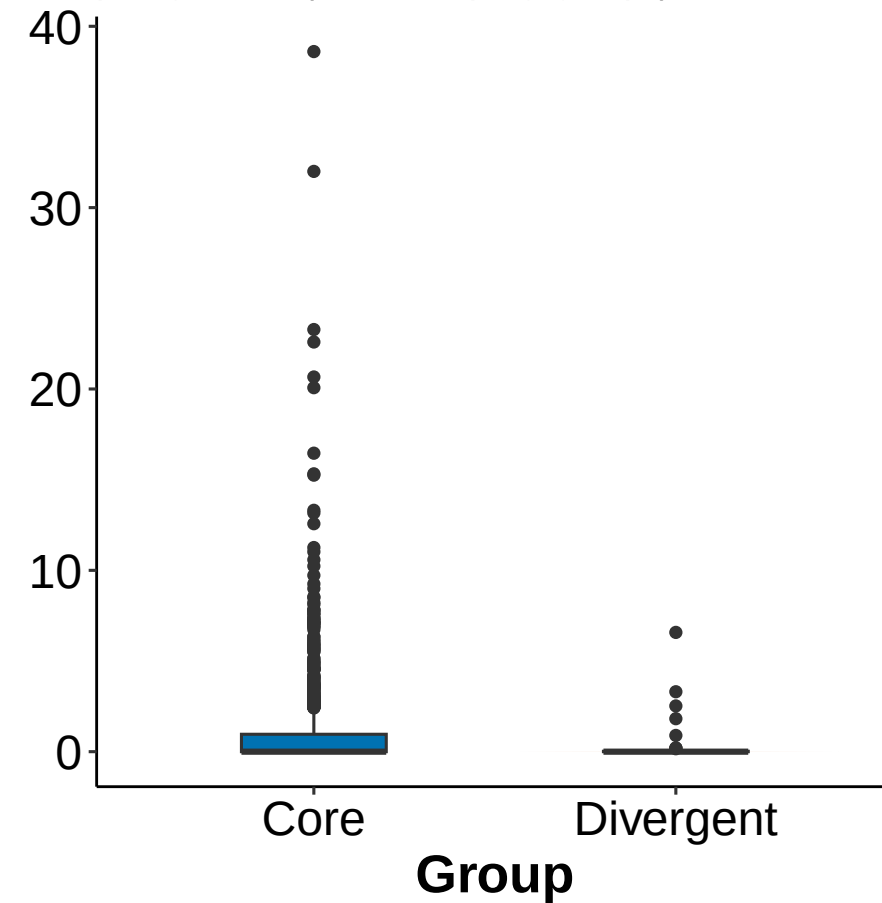
SA_Propanoate_metabolism_FBA

p-adj: 0.727 | .=-0.04 (Negligible)



SA_Propanoate_metabolism_L1w

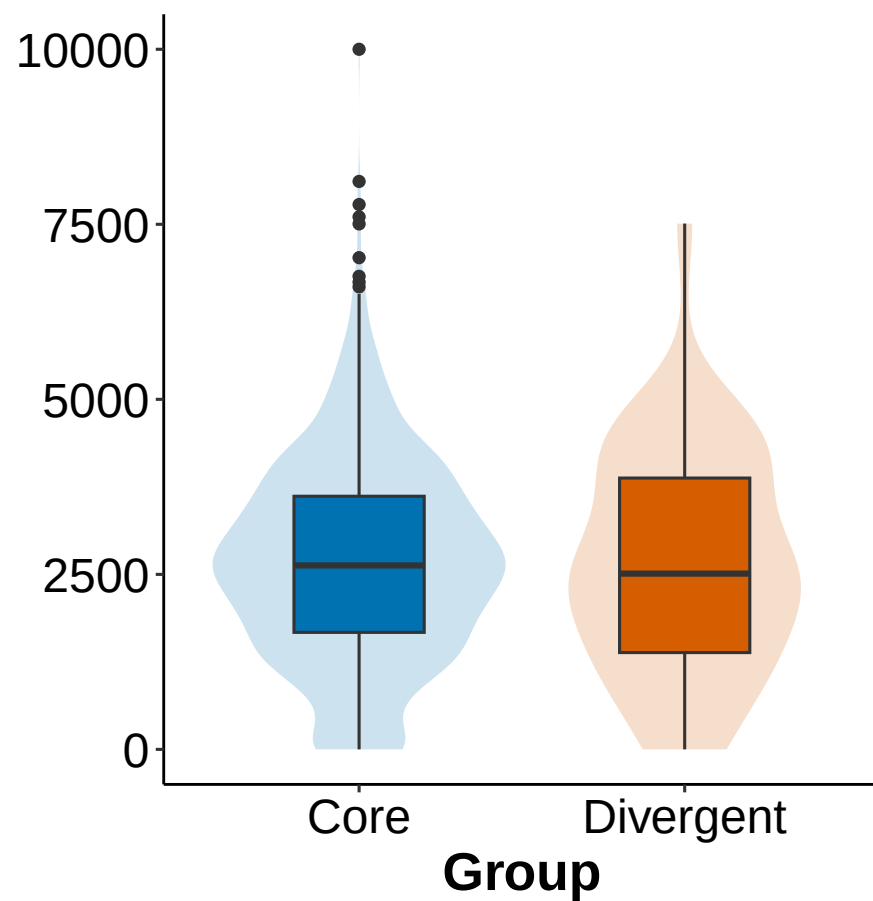
p-adj: 0.424 | .=-0.11 (Negligible) | .%: 129432524



Metabolism – . 5. Systemic Reprogramming (SA_*)

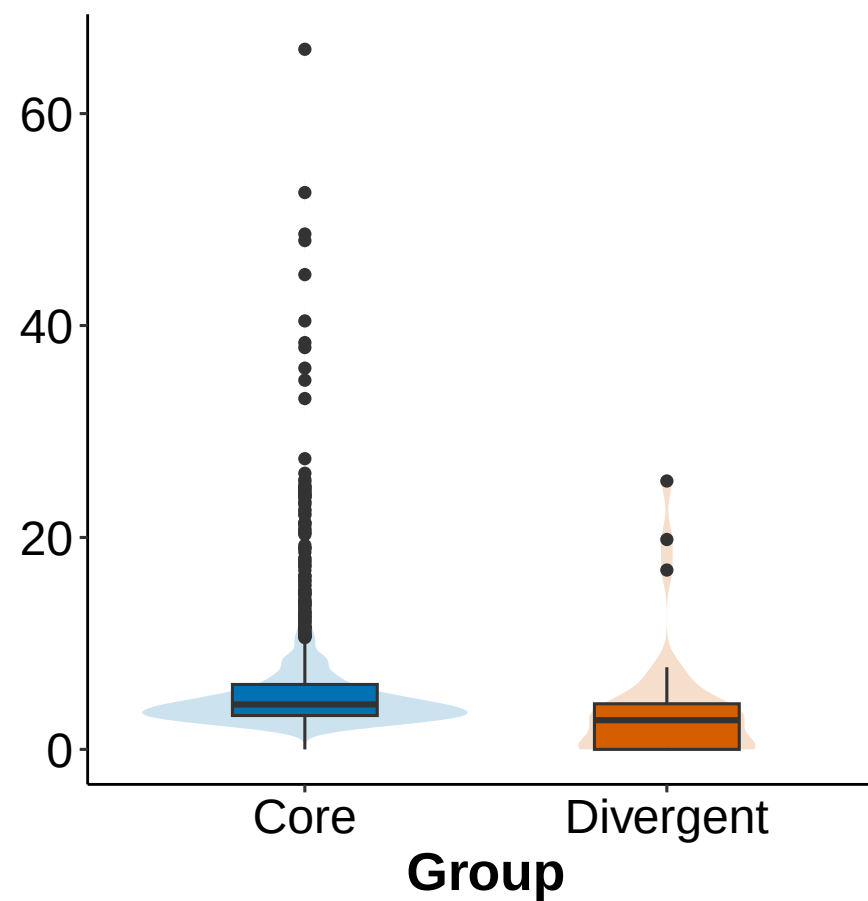
SA_Purine_catabolism_FBA

p-adj: 0.889 | .=-0.02 (Negligible) | .%: -4.4



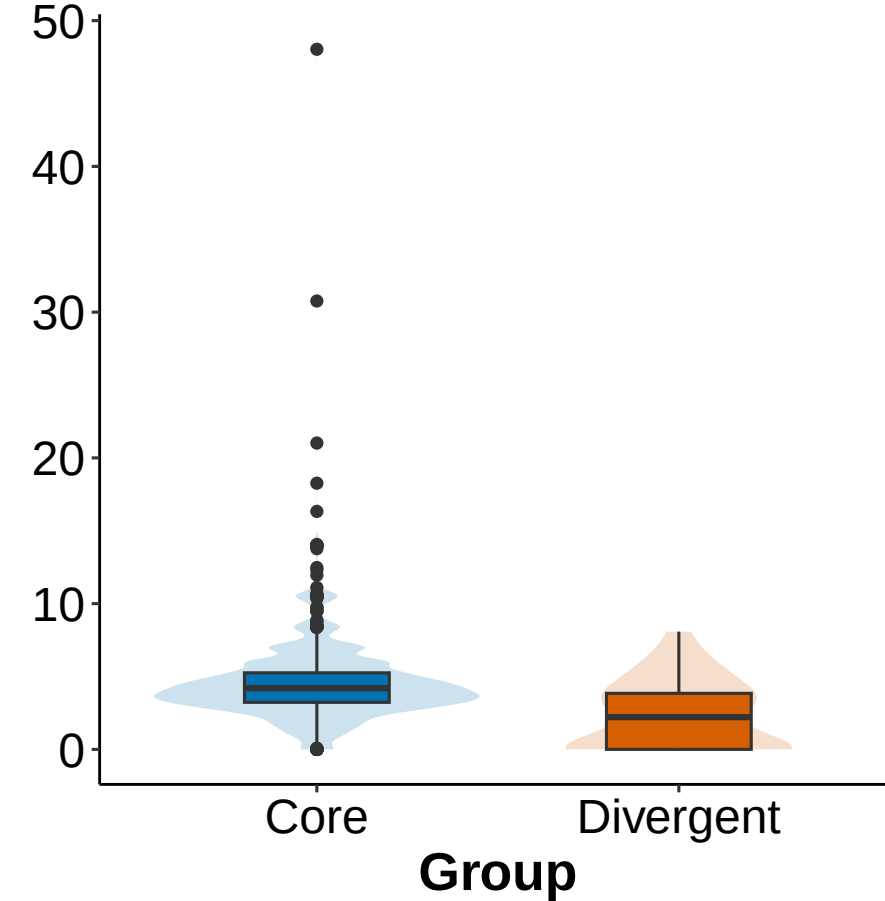
SA_Purine_catabolism_L1w

p-adj: 2.66e-05 | .=-0.45 (Medium) | .%: -35.1



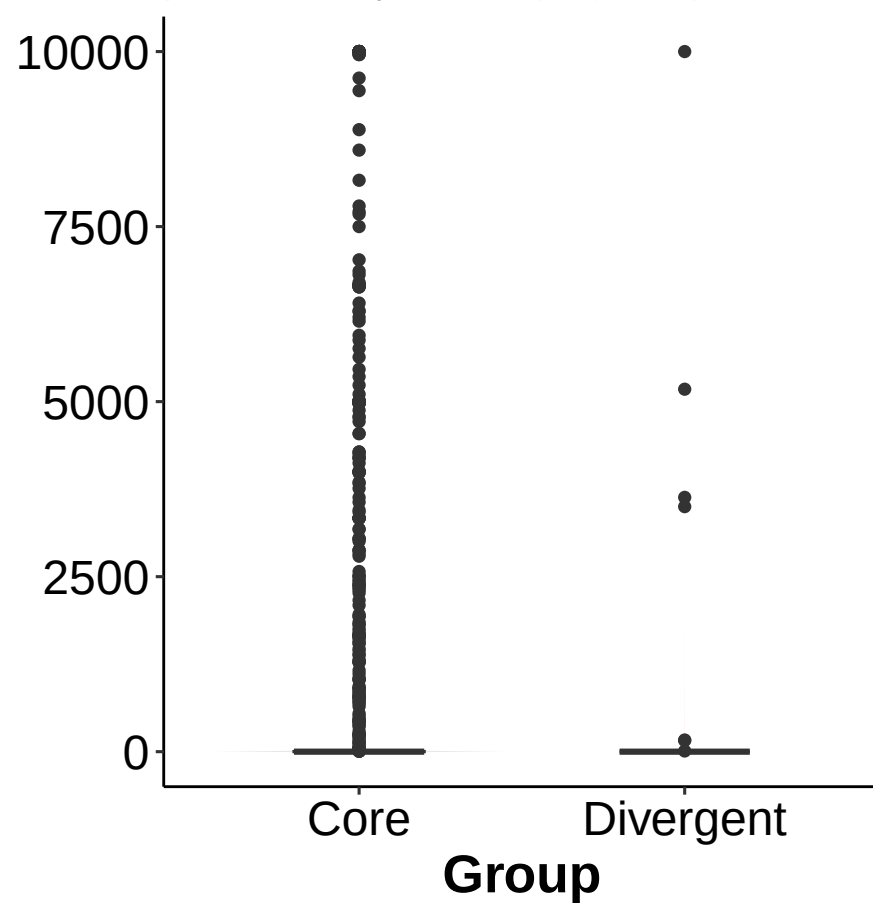
SA_Purine_catabolism_pFBA

p-adj: 1.04e-05 | .=-0.48 (Large) | .%: -47.4



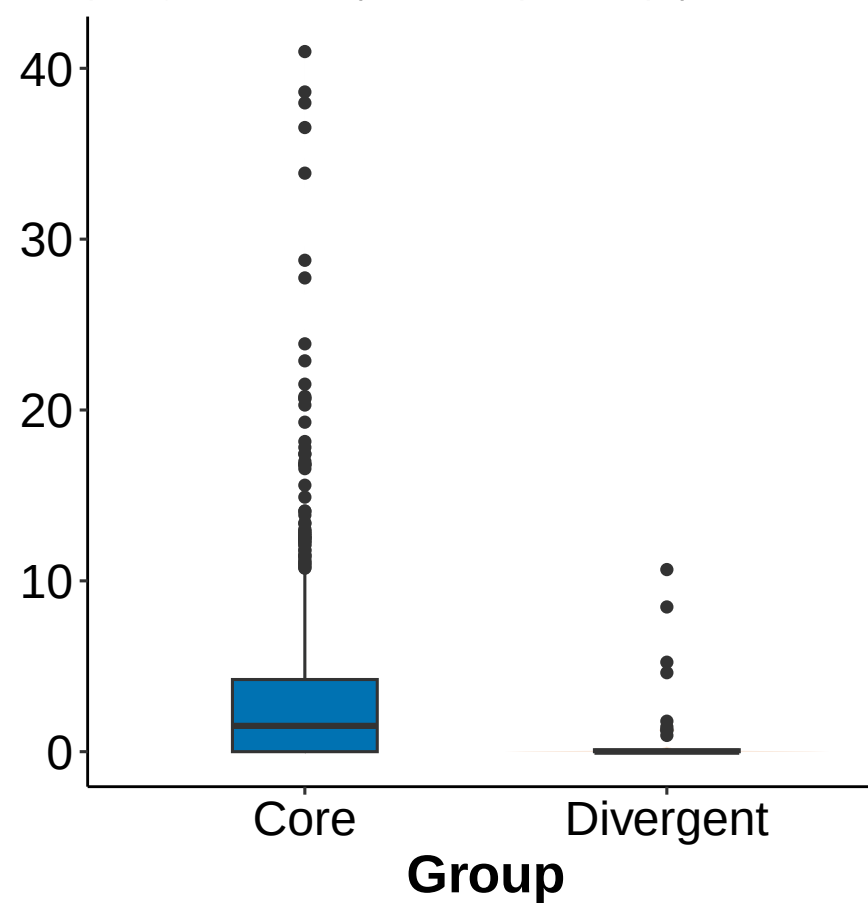
SA_Purine_synthesis_FBA

p-adj: 0.501 | .=-0.07 (Negligible)



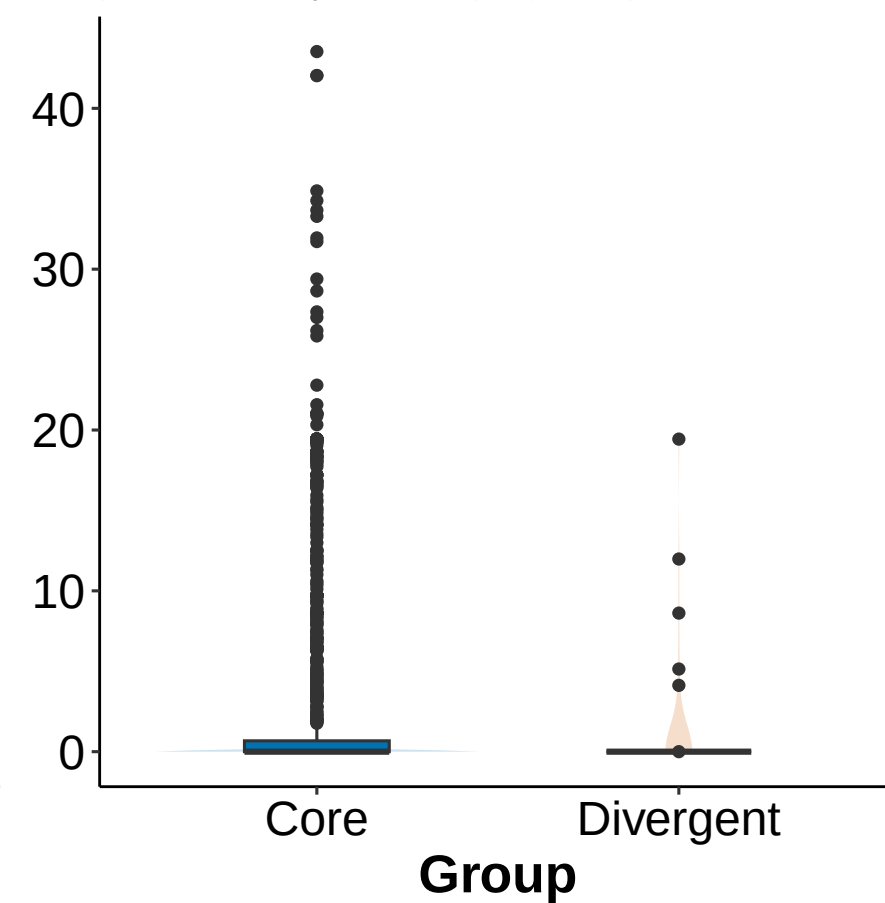
SA_Purine_synthesis_L1w

p-adj: 0.000837 | .=-0.35 (Medium) | .%: -100



SA_Purine_synthesis_pFBA

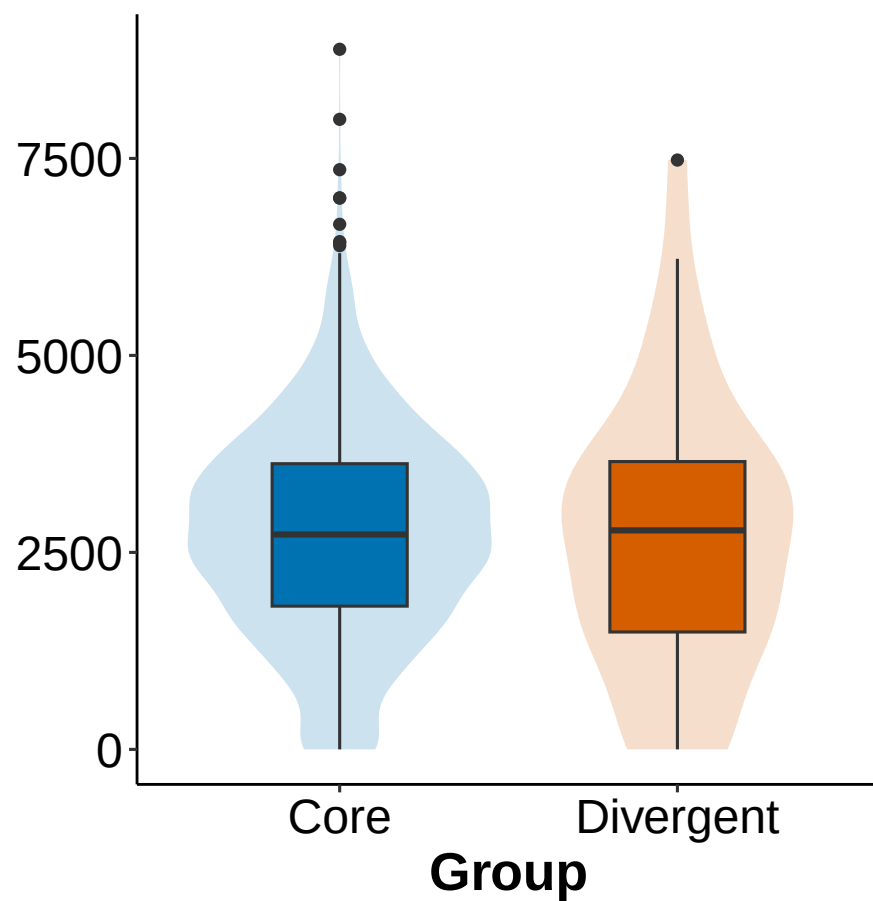
p-adj: 0.184 | .=-0.13 (Negligible)



Metabolism – . 5. Systemic Reprogramming (SA_*)

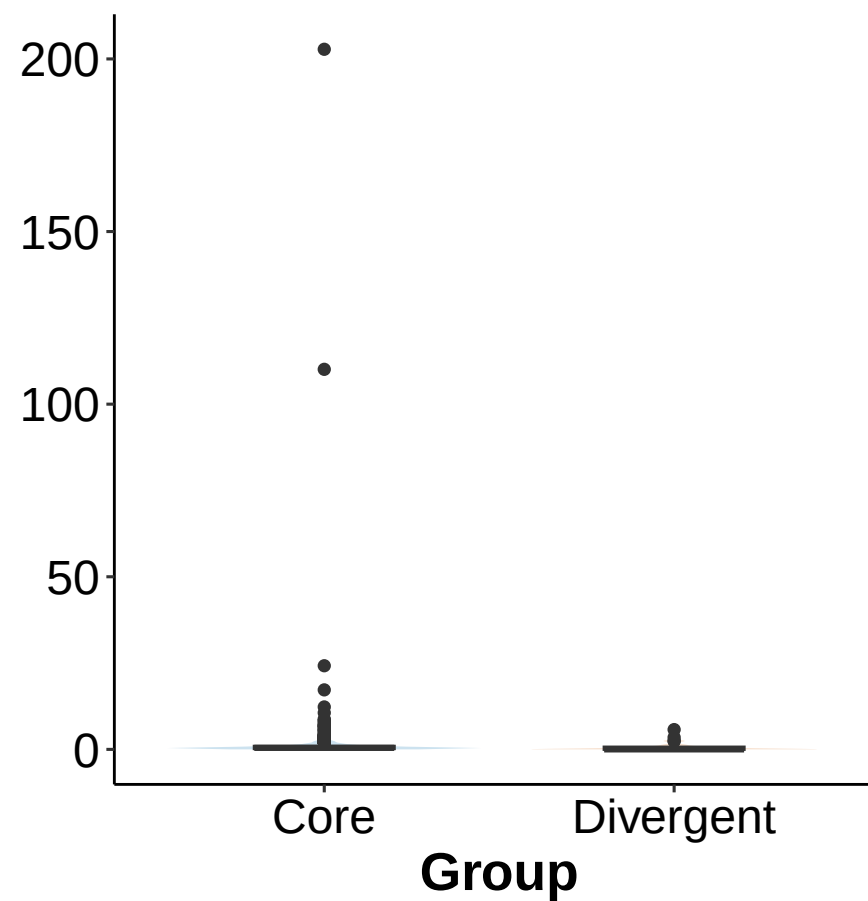
SA_Pyrimidine_catabolism_FBA

p-adj: 0.705 | .=-0.05 (Negligible) | .%: 1.9



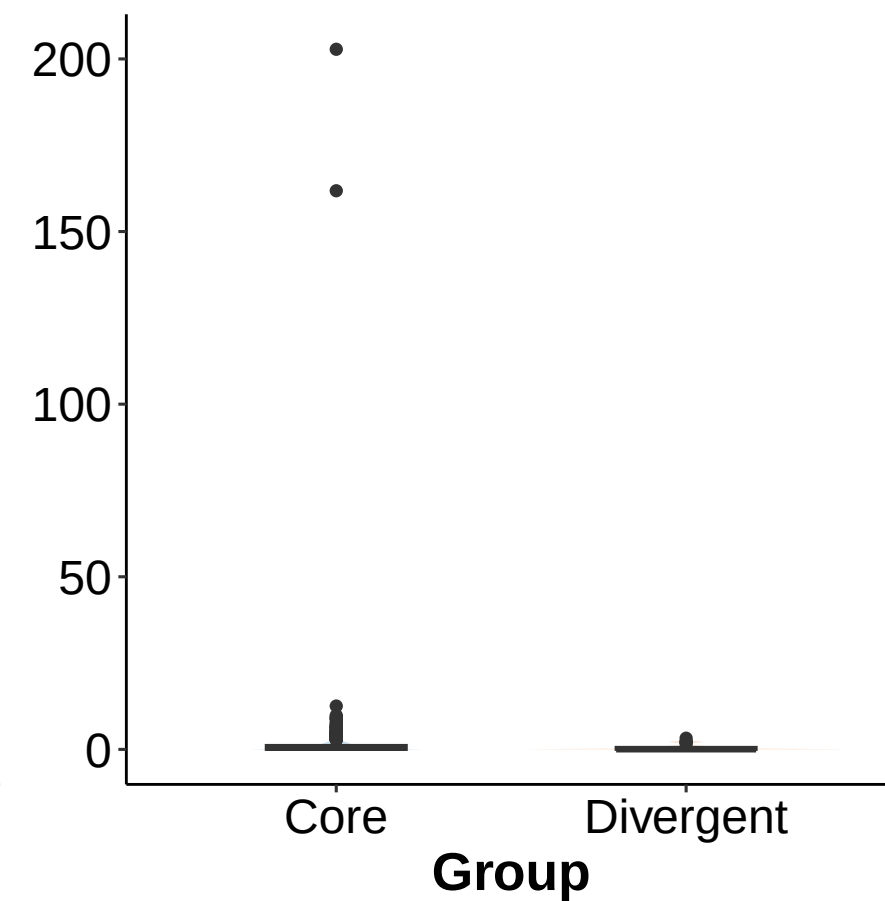
SA_Pyrimidine_catabolism_L1w

p-adj: 0.000837 | .=-0.35 (Medium) | .%: -99.4



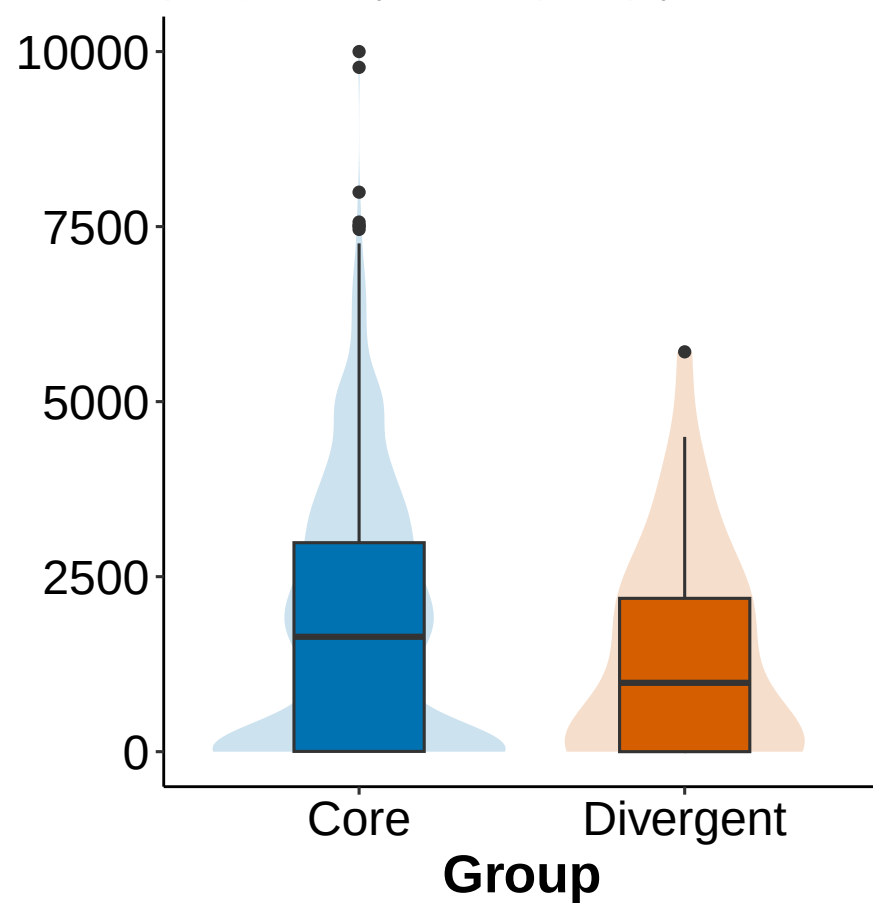
SA_Pyrimidine_catabolism_pFBA

p-adj: 0.0207 | .=-0.24 (Small) | .%: -100



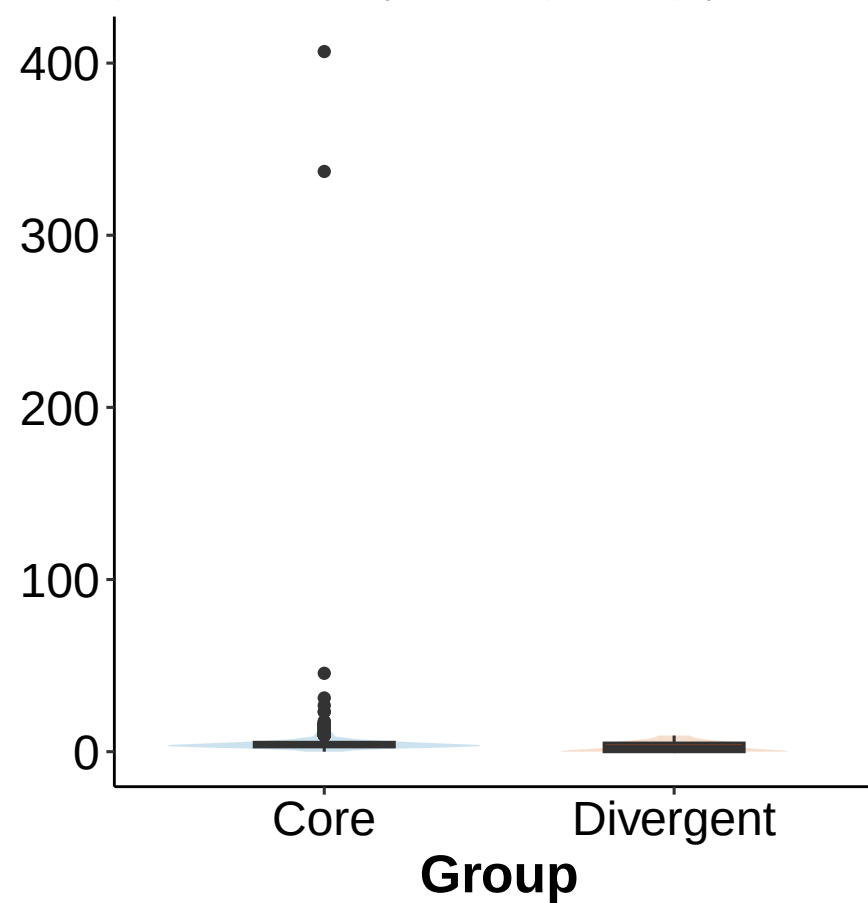
SA_Pyrimidine_synthesis_FBA

p-adj: 0.235 | .=-0.15 (Small) | .%: -40.1



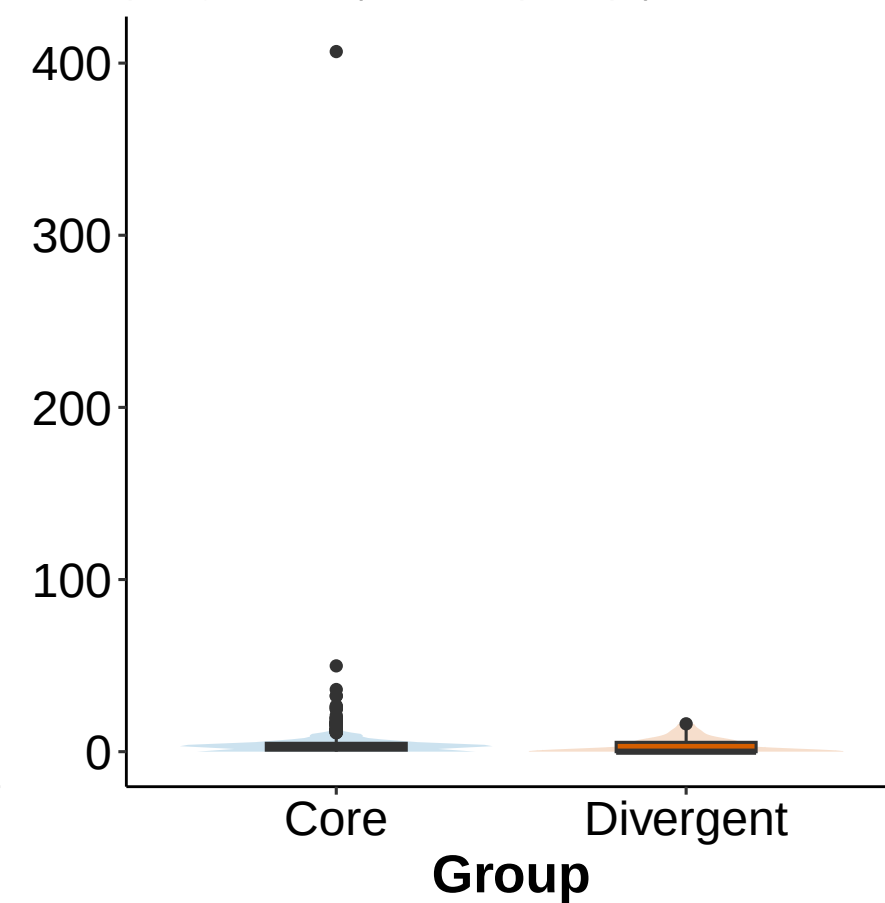
SA_Pyrimidine_synthesis_L1w

p-adj: 0.000687 | .=-0.36 (Medium) | .%: -45.7



SA_Pyrimidine_synthesis_pFBA

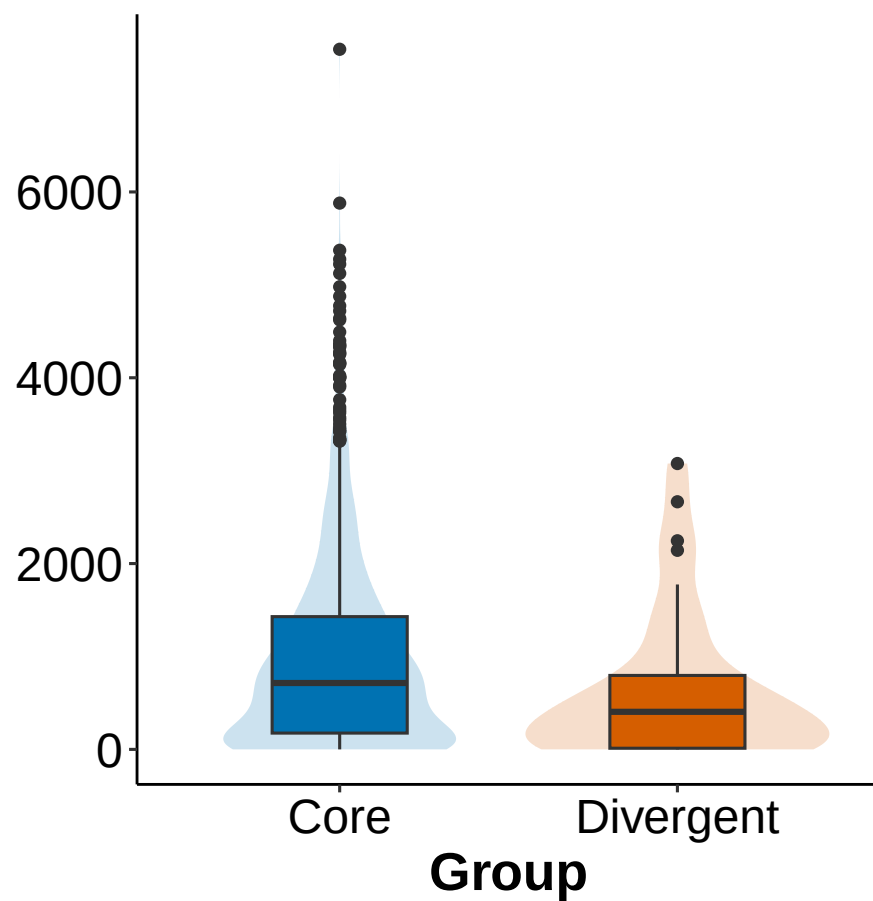
p-adj: 0.0207 | .=-0.25 (Small) | .%: -100



Metabolism – . 5. Systemic Reprogramming (SA_*)

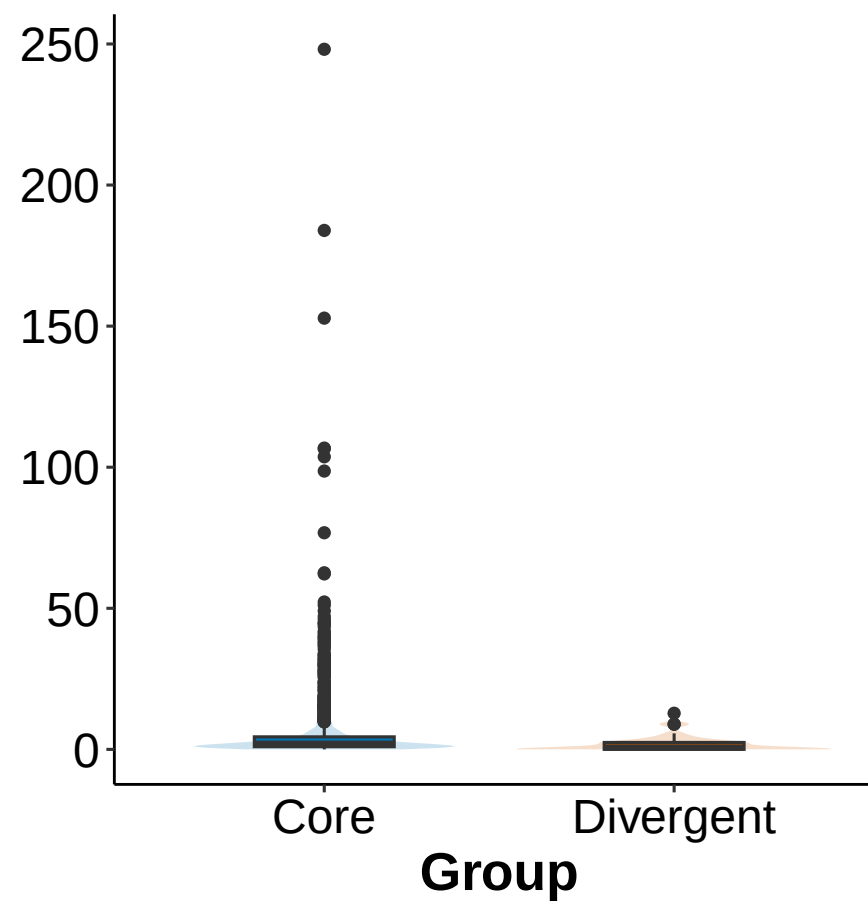
SA_Pyruvate_metabolism_FBA

p-adj: 0.0816 | .=-0.2 (Small) | .%: -43.4



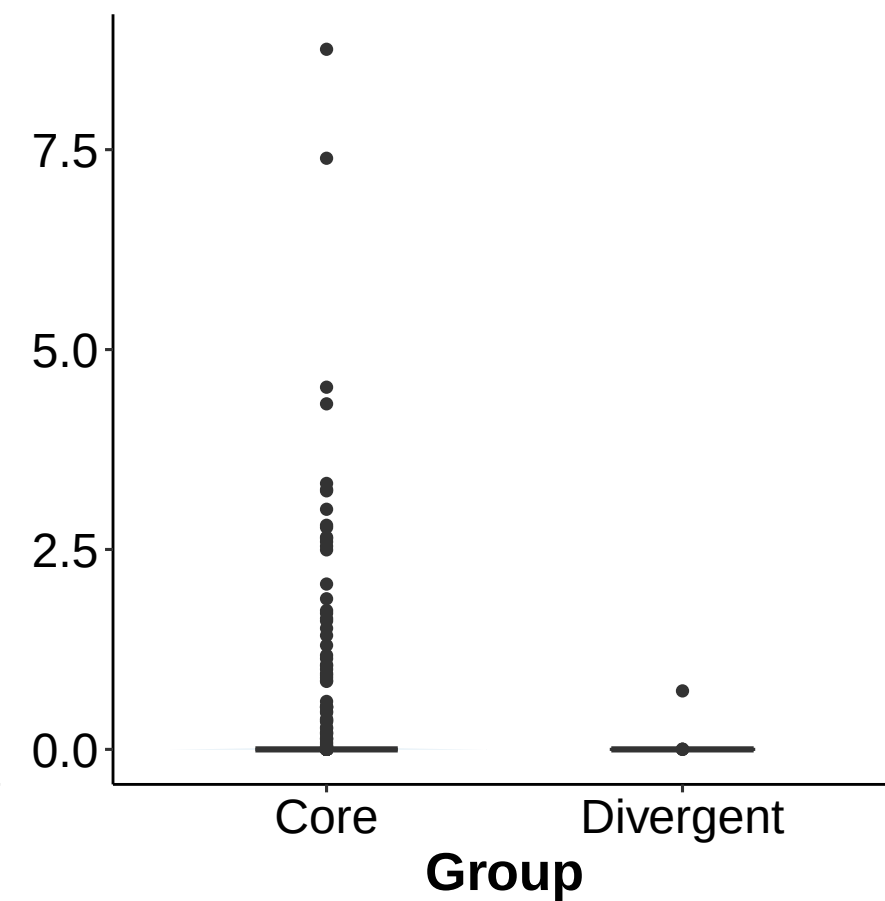
SA_Pyruvate_metabolism_L1w

p-adj: 0.000282 | .=-0.38 (Medium) | .%: -67.4



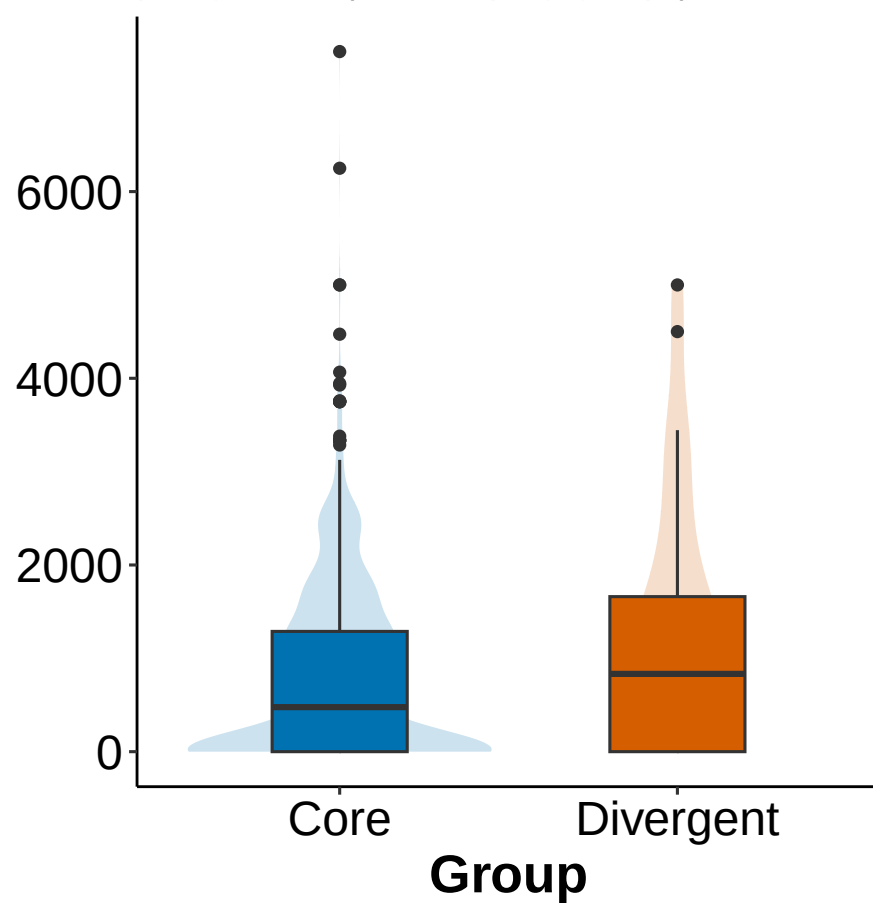
SA_Pyruvate_metabolism_pFBA

p-adj: 0.964 | .=0 (Negligible) | .%: -38.2



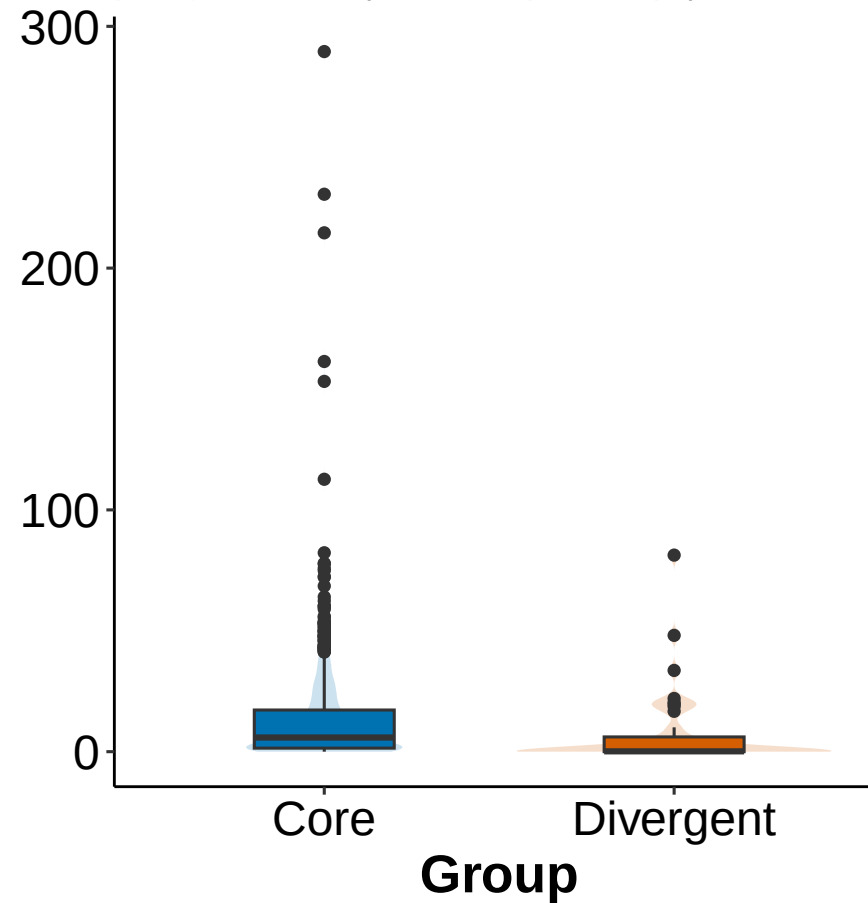
SA_ROS_detoxification_FBA

p-adj: 0.294 | .=0.13 (Negligible) | .%: 74.6



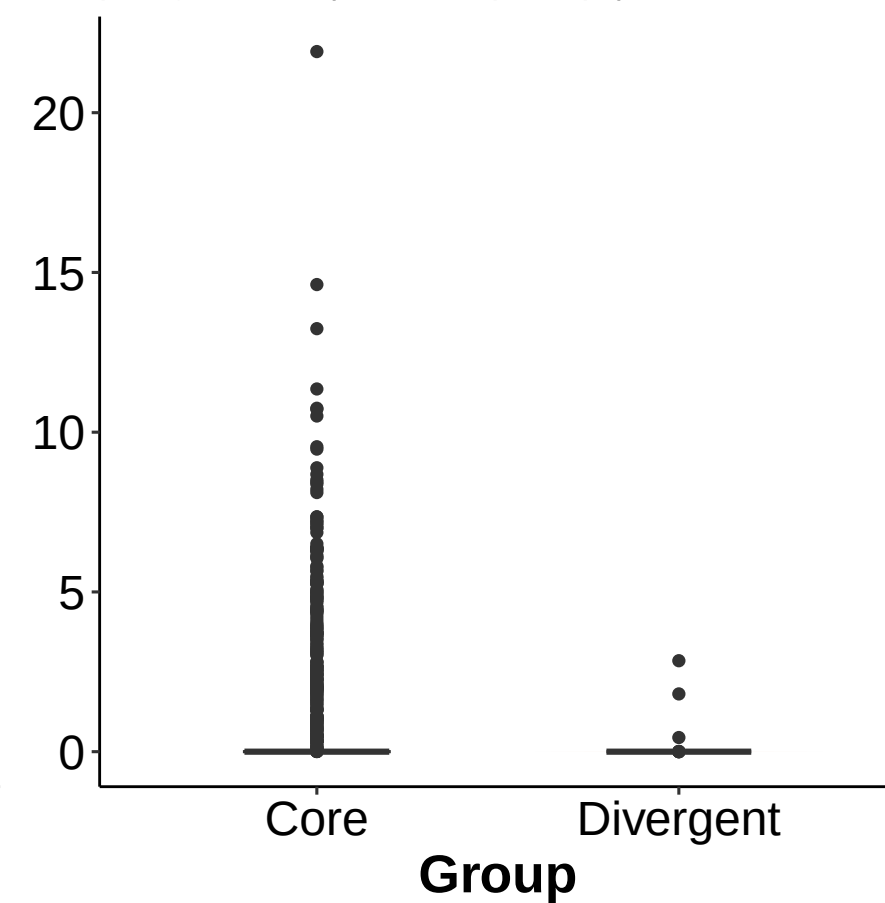
SA_ROS_detoxification_L1w

p-adj: 4.1e-05 | .=-0.44 (Medium) | .%: -97.5



SA_ROS_detoxification_pFBA

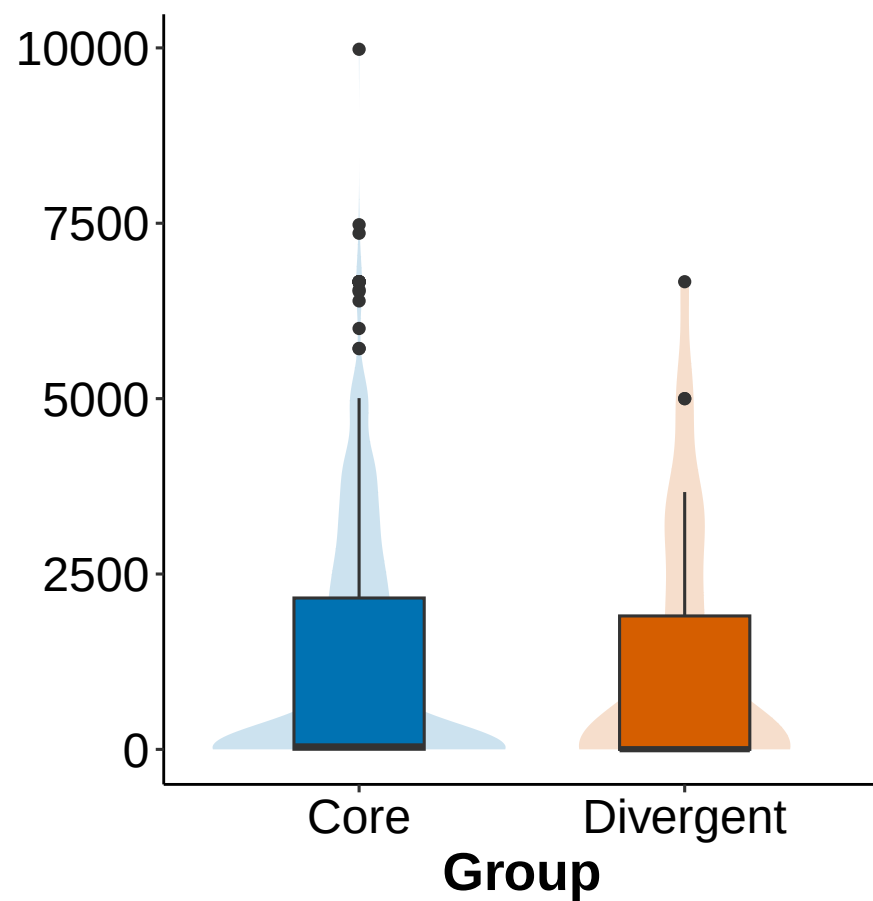
p-adj: 0.0143 | .=-0.26 (Small) | .%: -100



Metabolism – . 5. Systemic Reprogramming (SA_*)

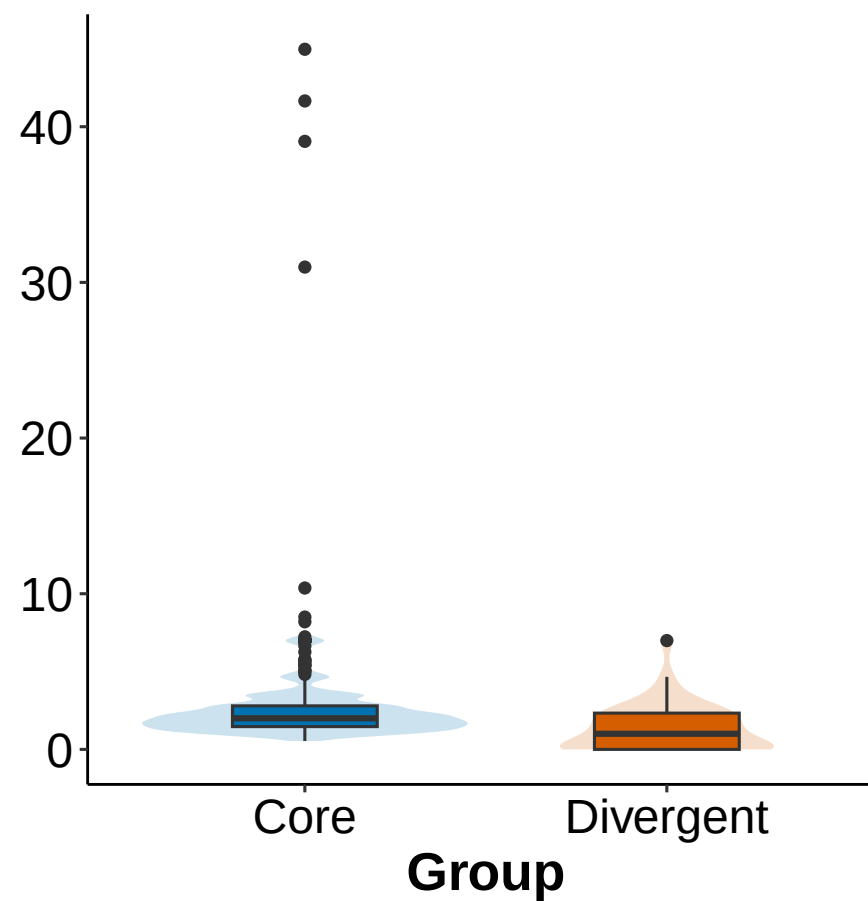
SA_Sphingolipid_metabolism_FB

p-adj: 0.0491 | .=-0.22 (Small) | .%: -93.3



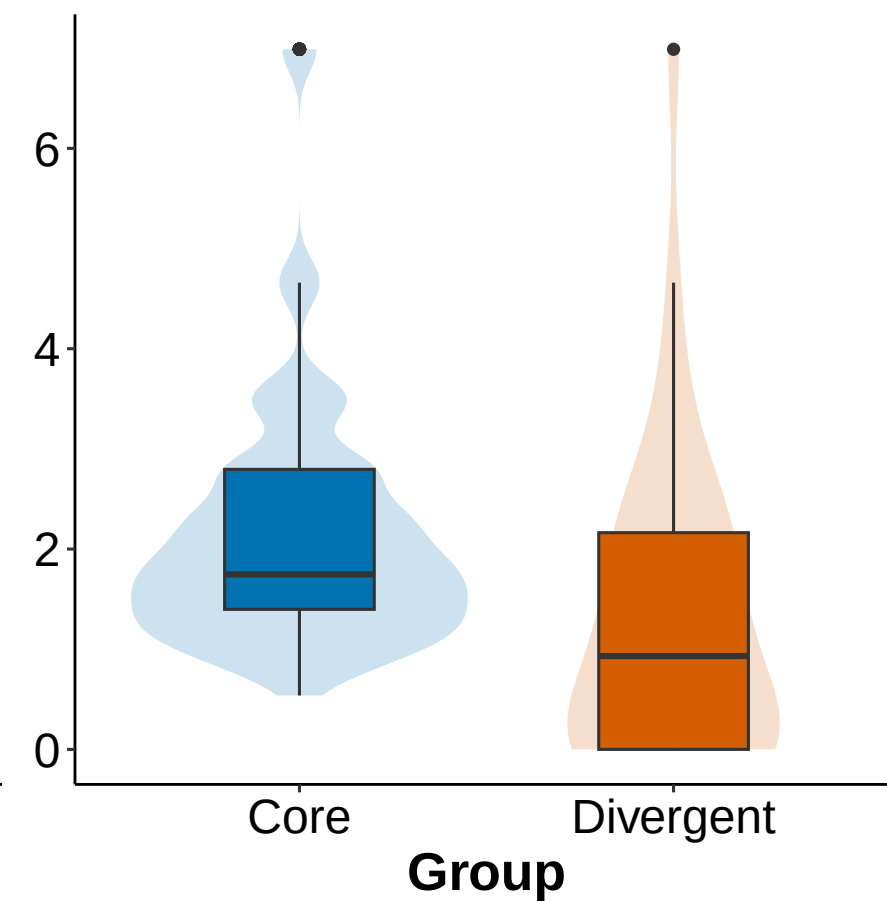
SA_Sphingolipid_metabolism_L1w

p-adj: 2.83e-05 | .=-0.45 (Medium) | .%: -49.8



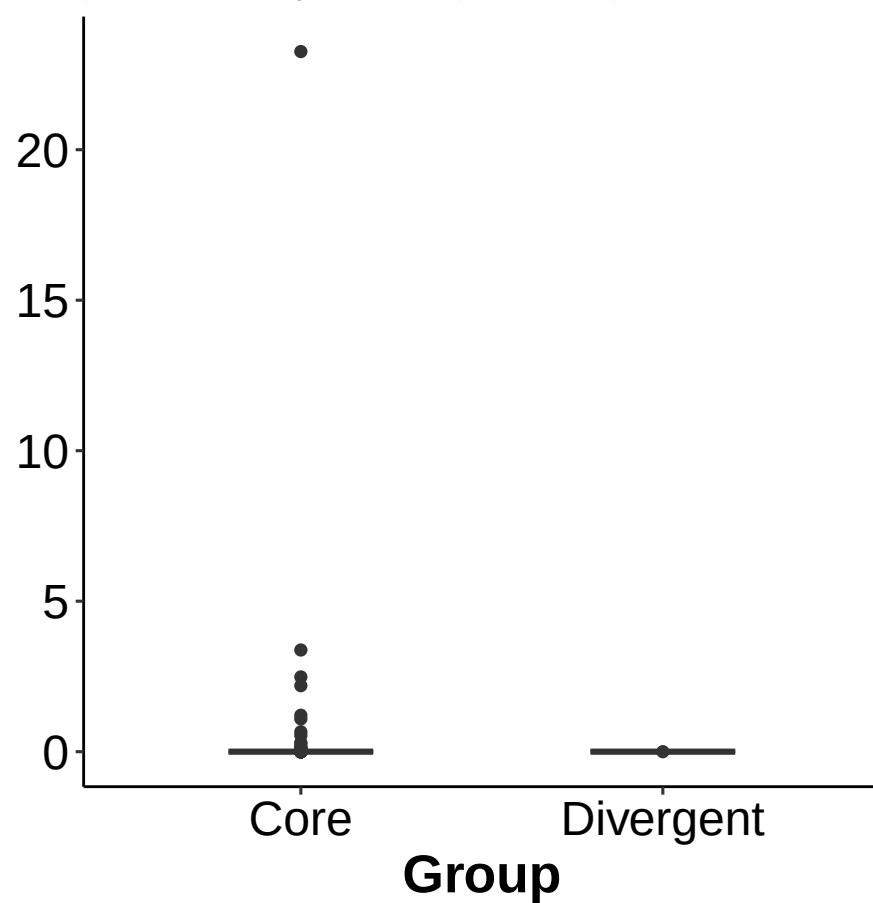
SA_Sphingolipid_metabolism_pFBA

p-adj: 2.83e-05 | .=-0.45 (Medium) | .%: -46.7



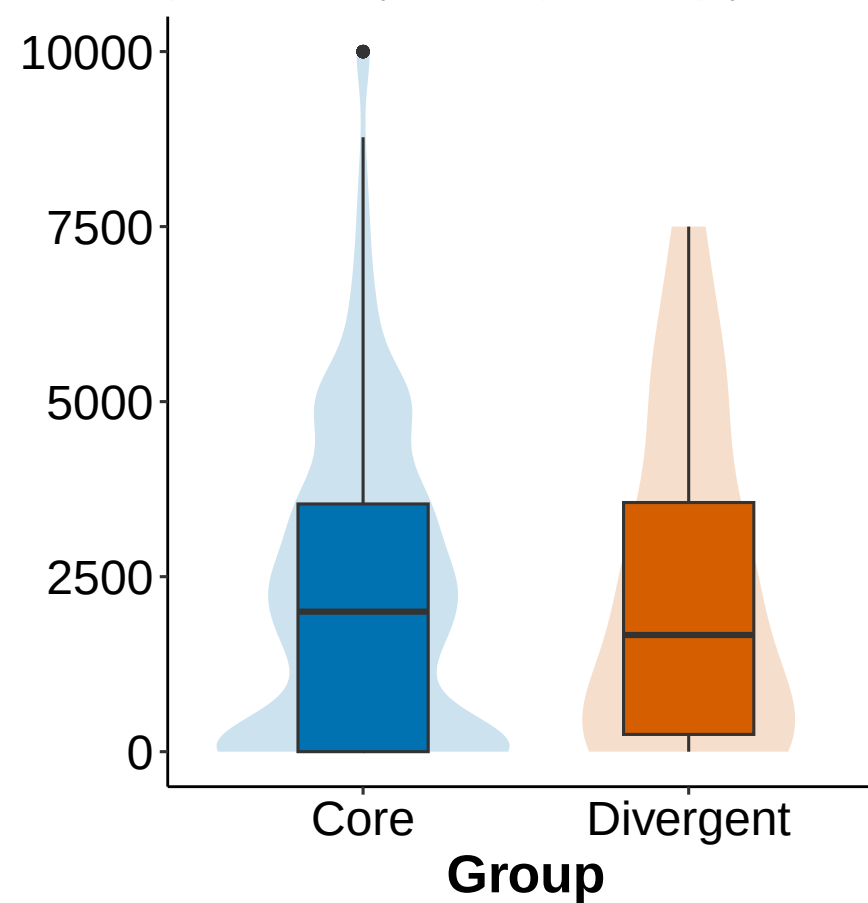
SA_Squalene_and_cholesterol_synth

p-adj: 0.649 | .=-0.02 (Negligible)



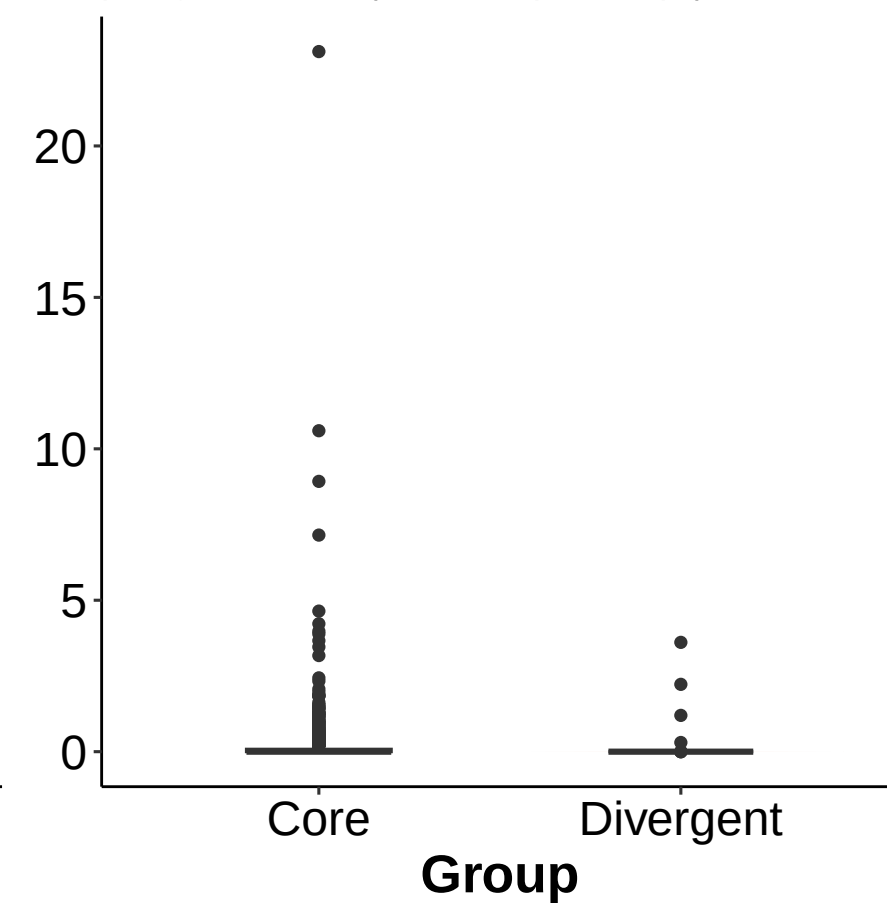
SA_Starch_and_sucrose_metabol

p-adj: 0.949 | .=-0.01 (Negligible) | .%: -16.7

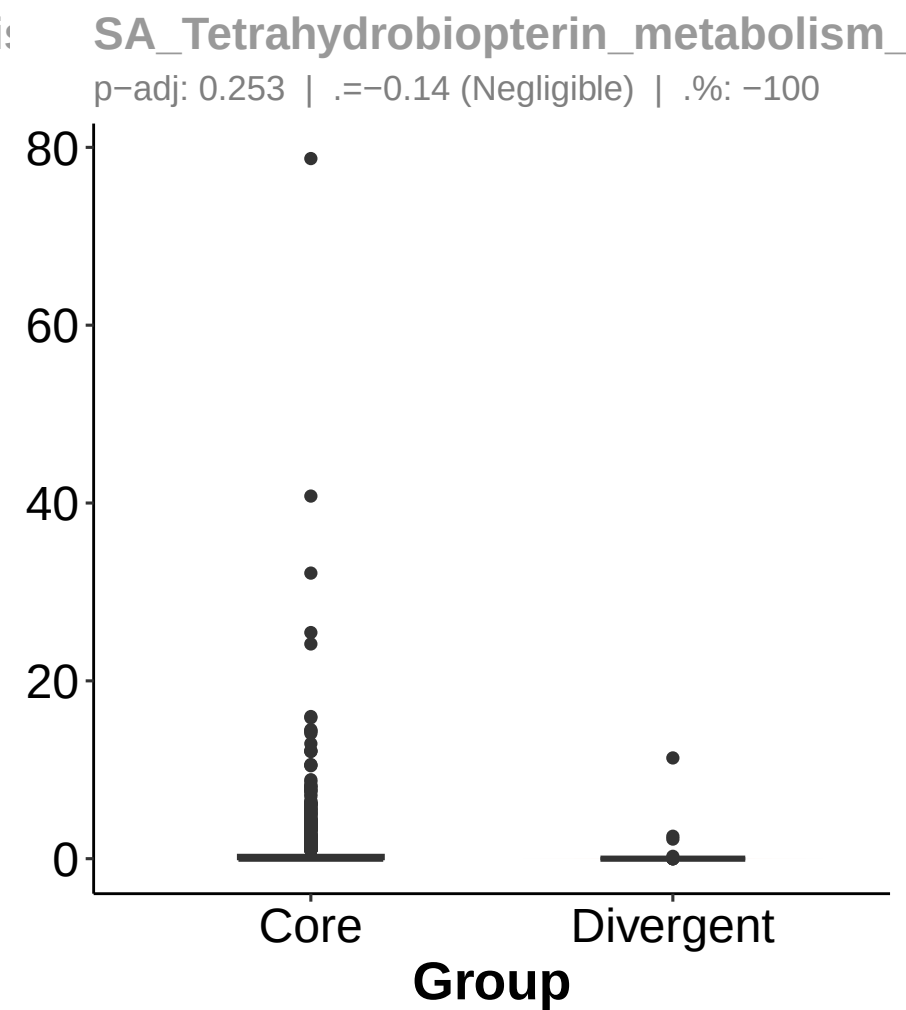
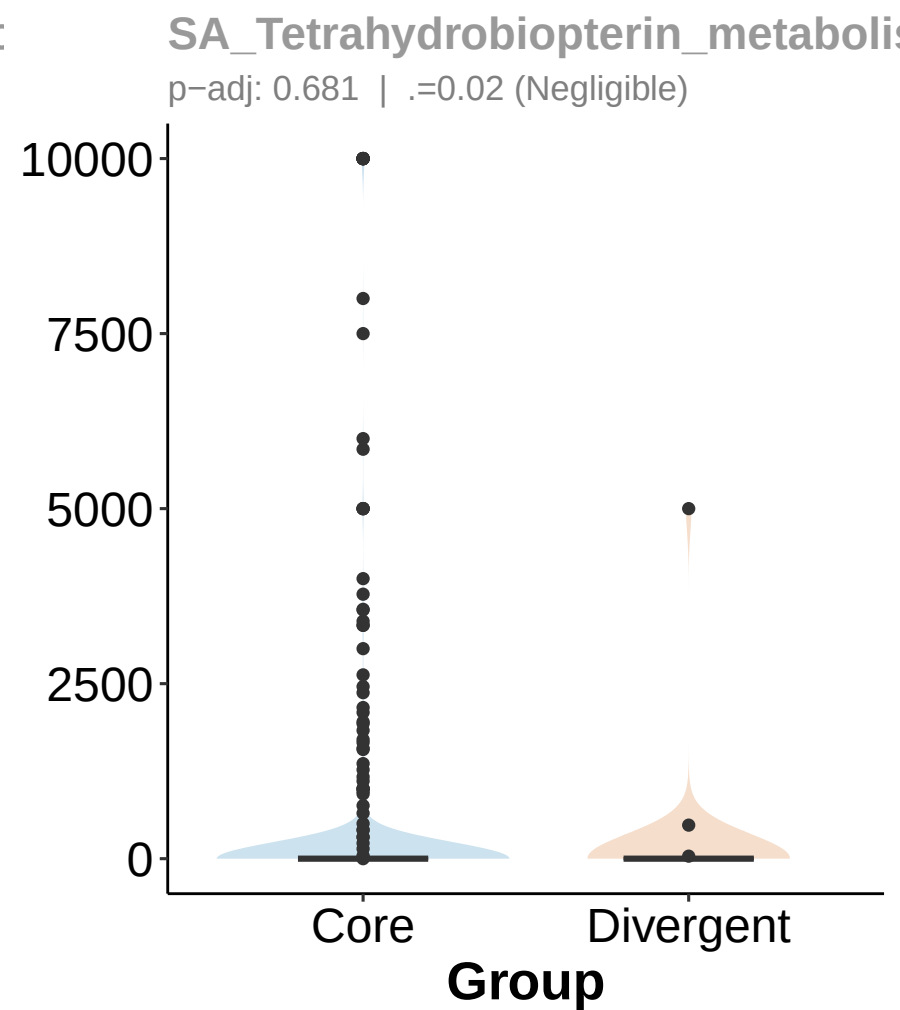
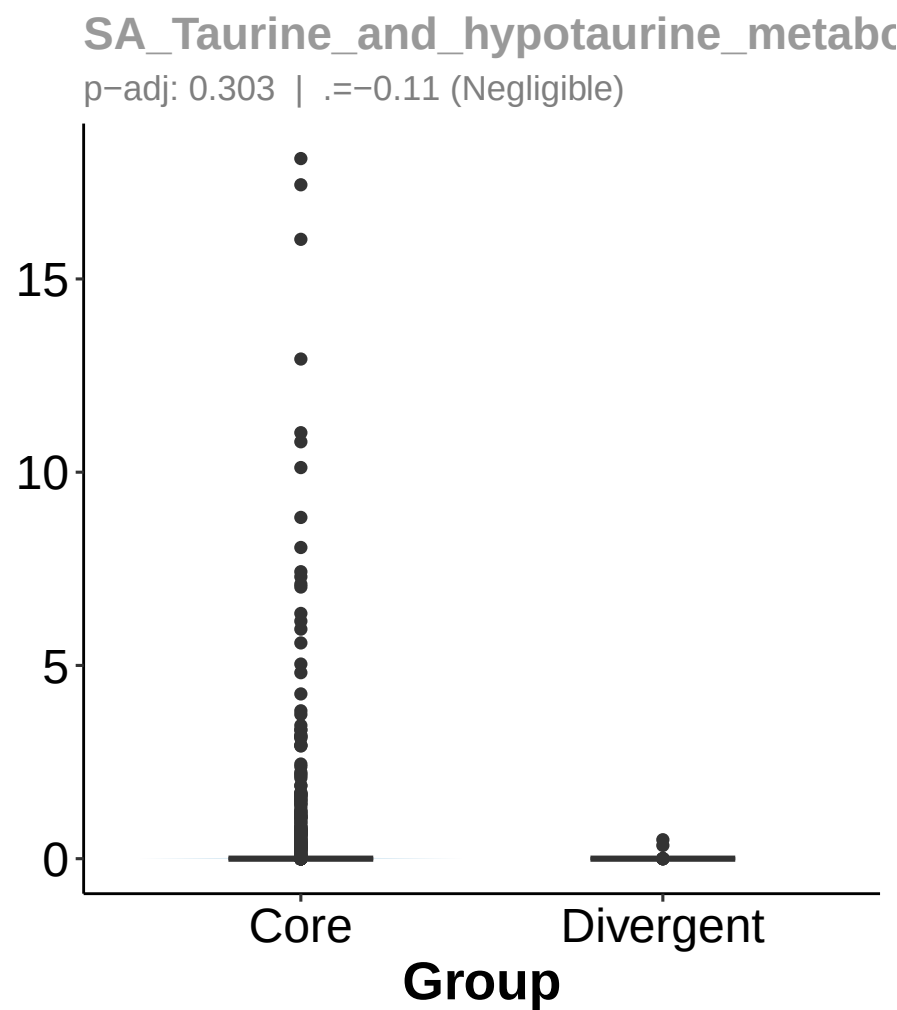
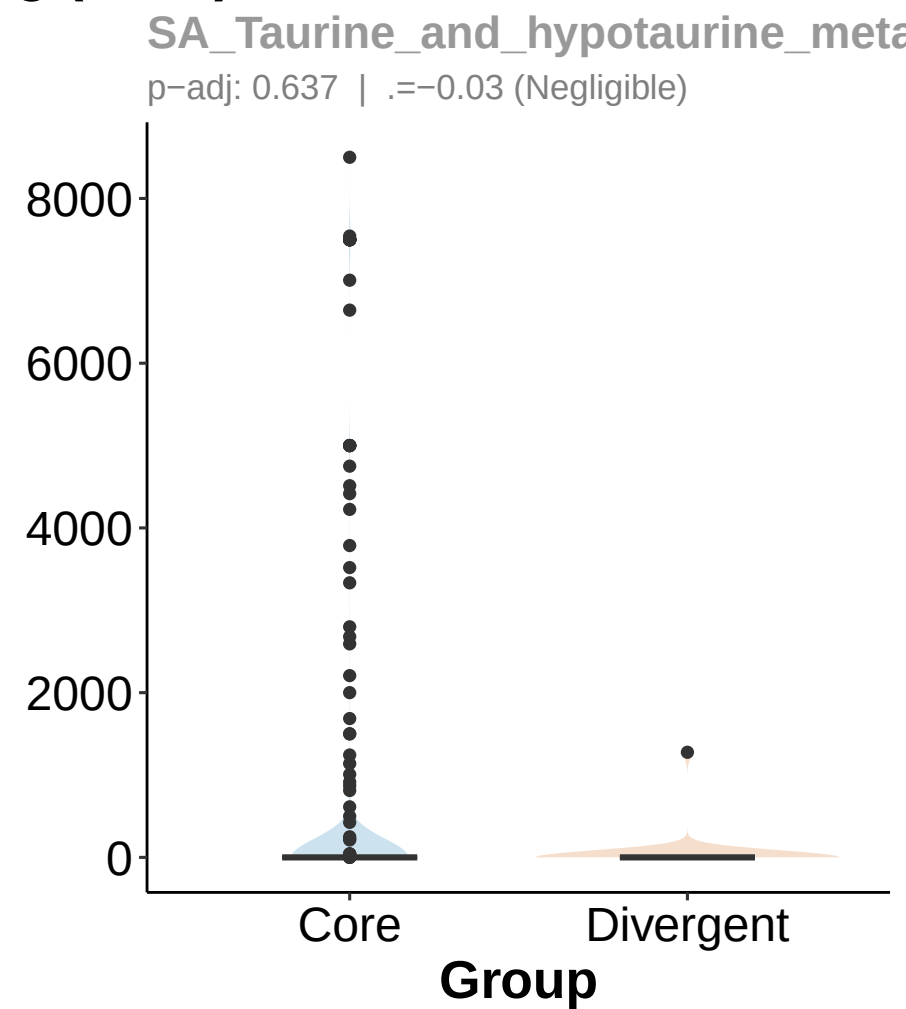
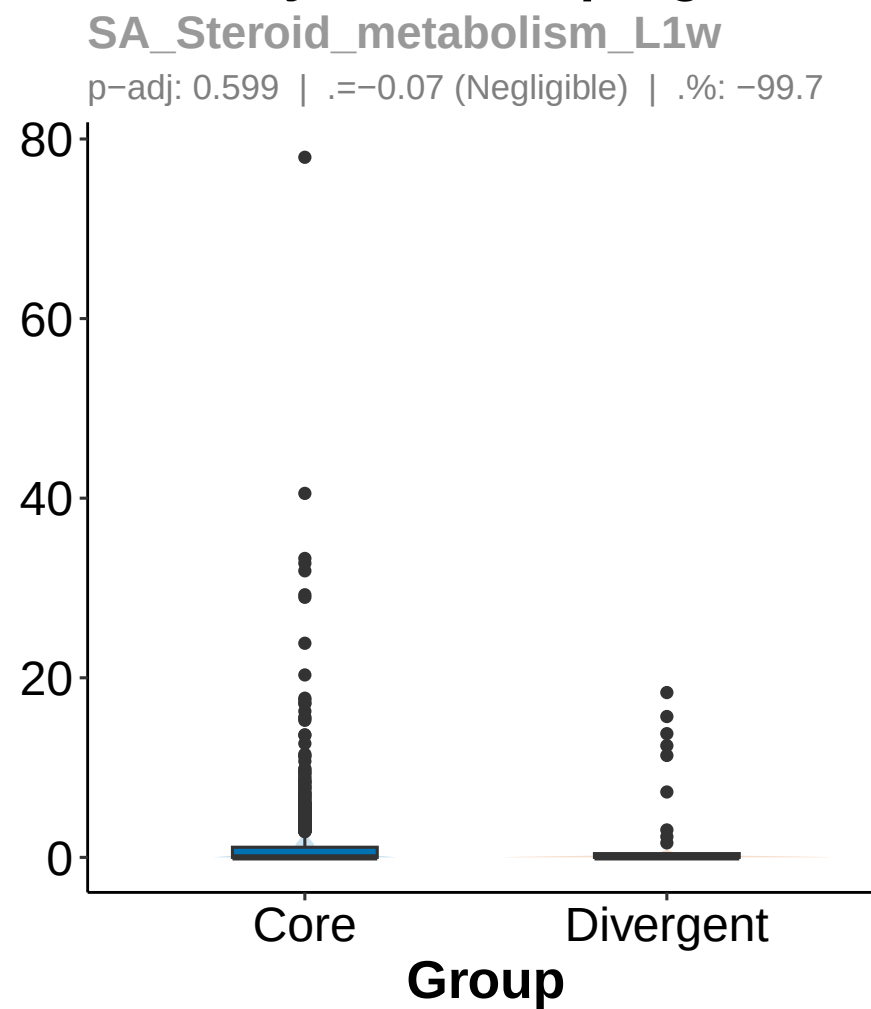
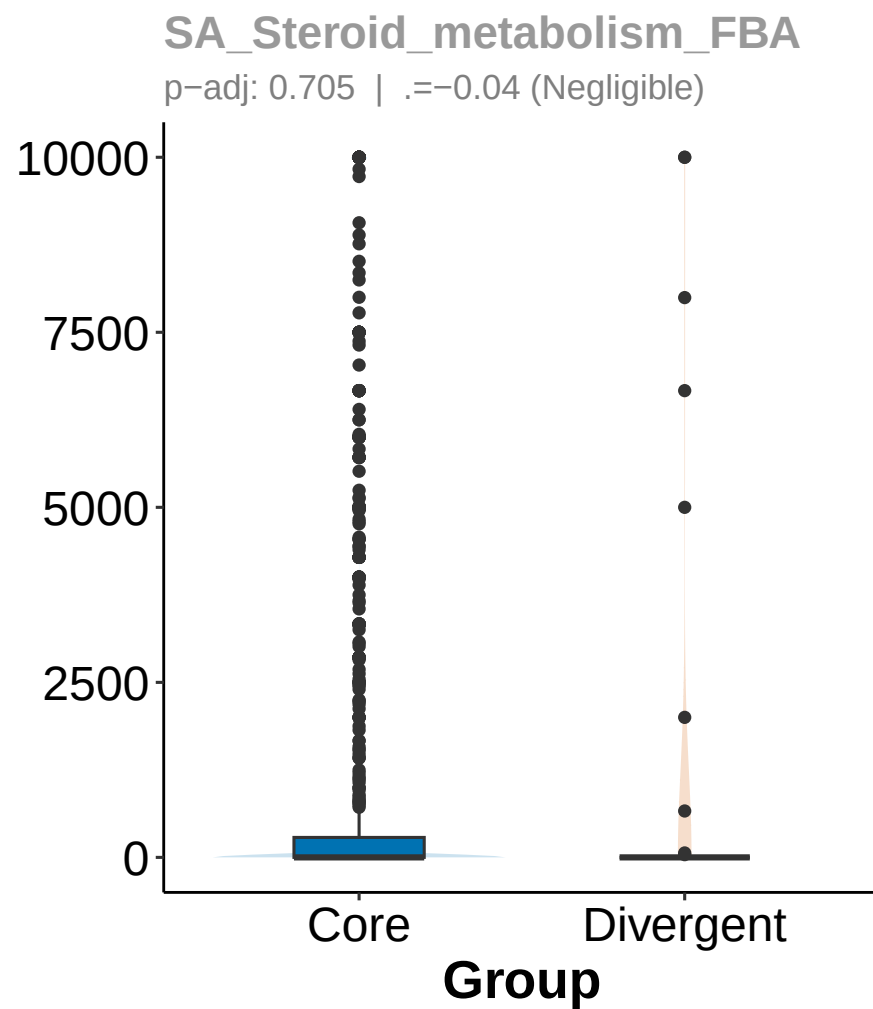


SA_Starch_and_sucrose_metabolism

p-adj: 8.51e-05 | .=-0.42 (Medium) | .%: -86.7



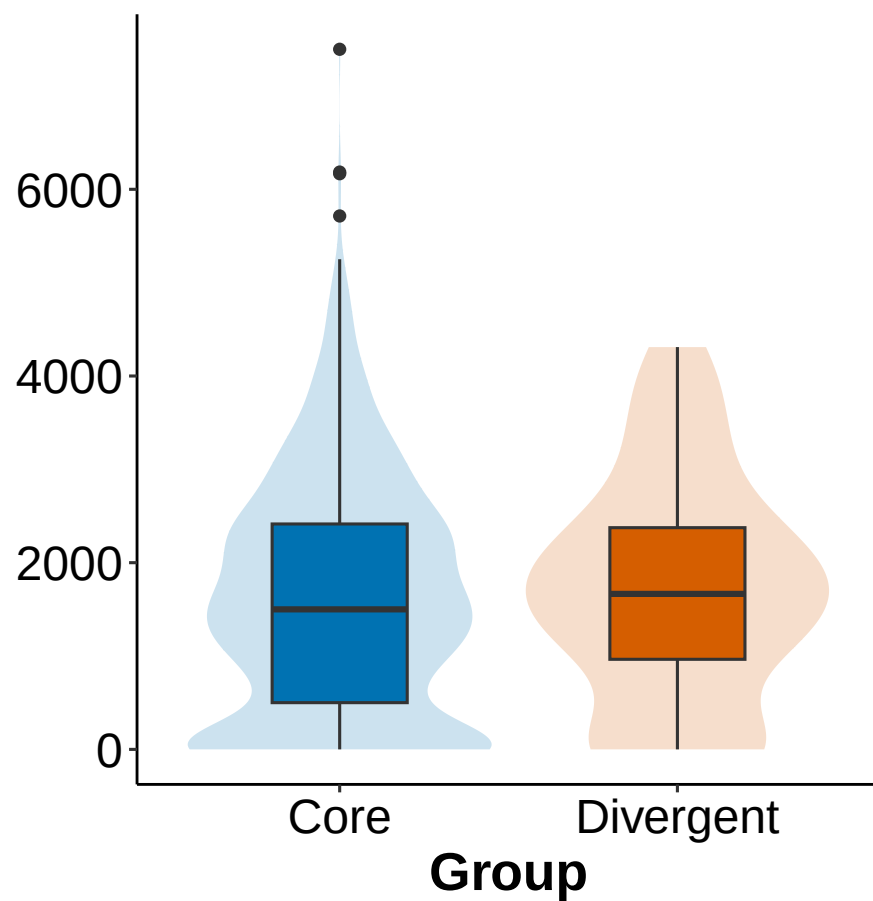
Metabolism – . 5. Systemic Reprogramming (SA_*)



Metabolism – . 5. Systemic Reprogramming (SA_*)

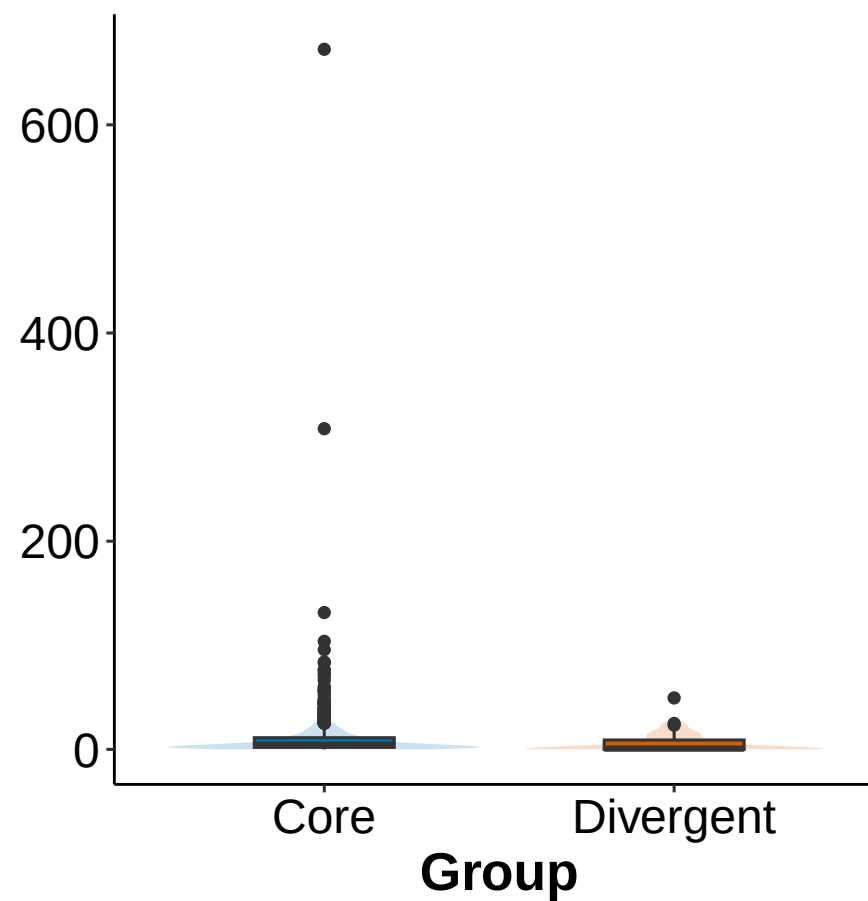
SA_Transport__endoplasmic_reticul

p-adj: 0.705 | .:=0.05 (Negligible) | .%: 11



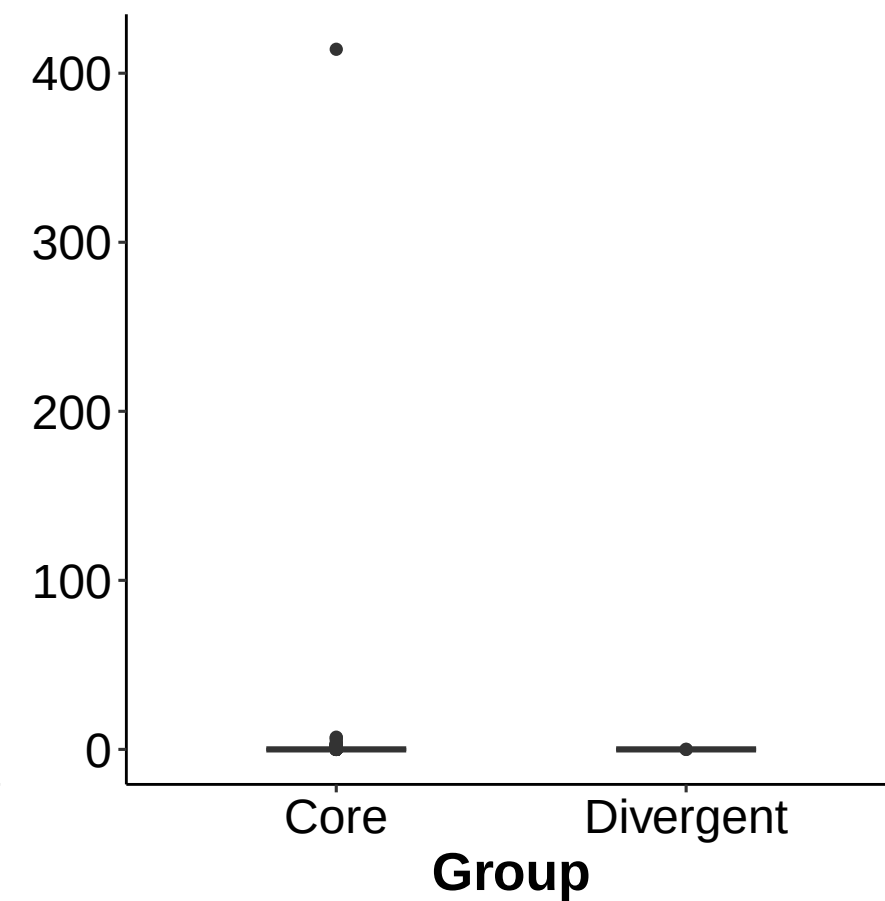
SA_Transport__endoplasmic_reticul

p-adj: 0.000656 | .:=−0.36 (Medium) | .%: −83.9



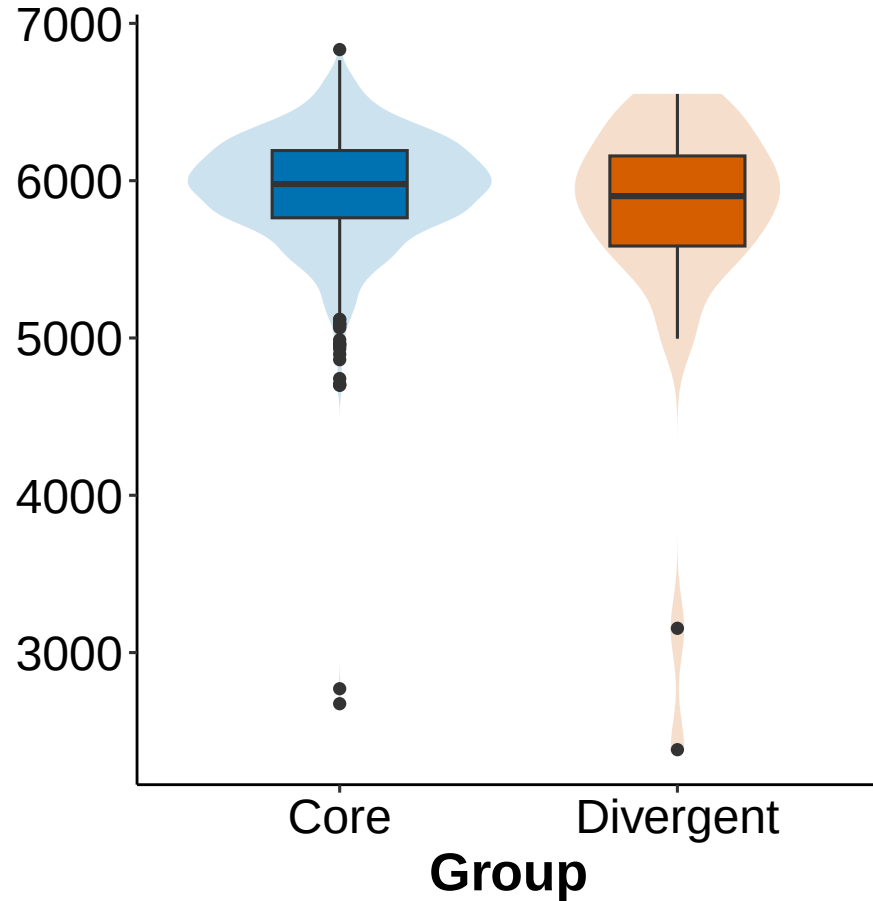
SA_Transport__endoplasmic_reticul

p-adj: 0.563 | .:=−0.04 (Negligible)



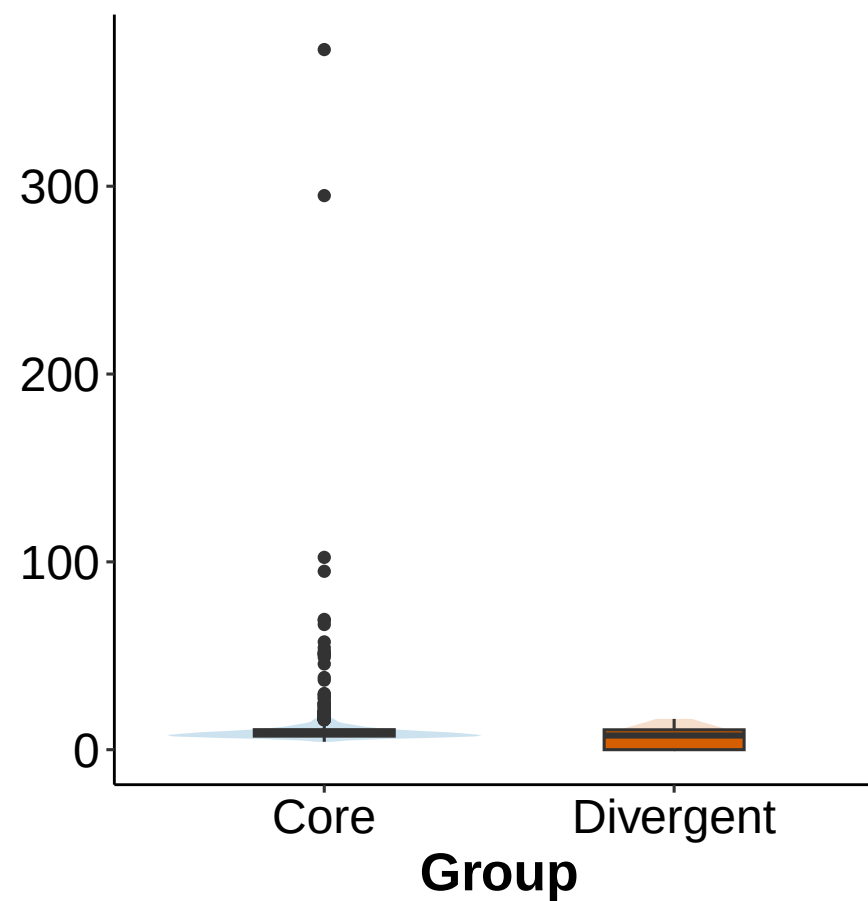
SA_Transport__extracellular_FBA

p-adj: 0.404 | .:=−0.11 (Negligible) | .%: −1.3



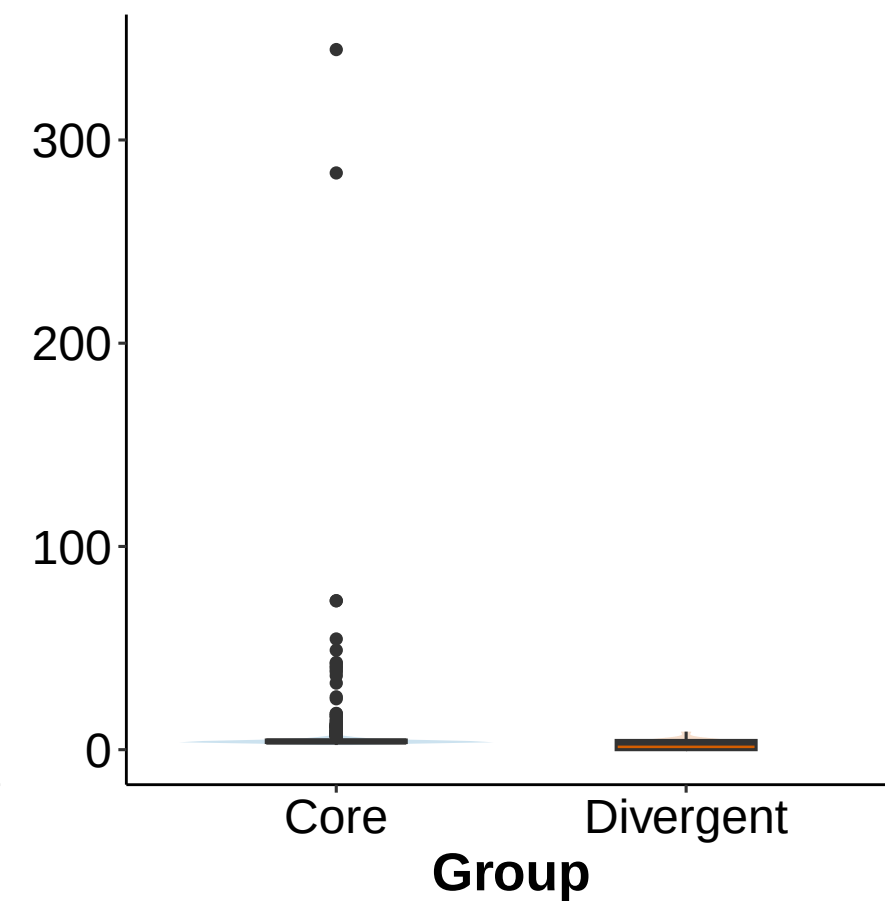
SA_Transport__extracellular_L1w

p-adj: 0.0262 | .:=−0.24 (Small) | .%: −12.1



SA_Transport__extracellular_pFBA

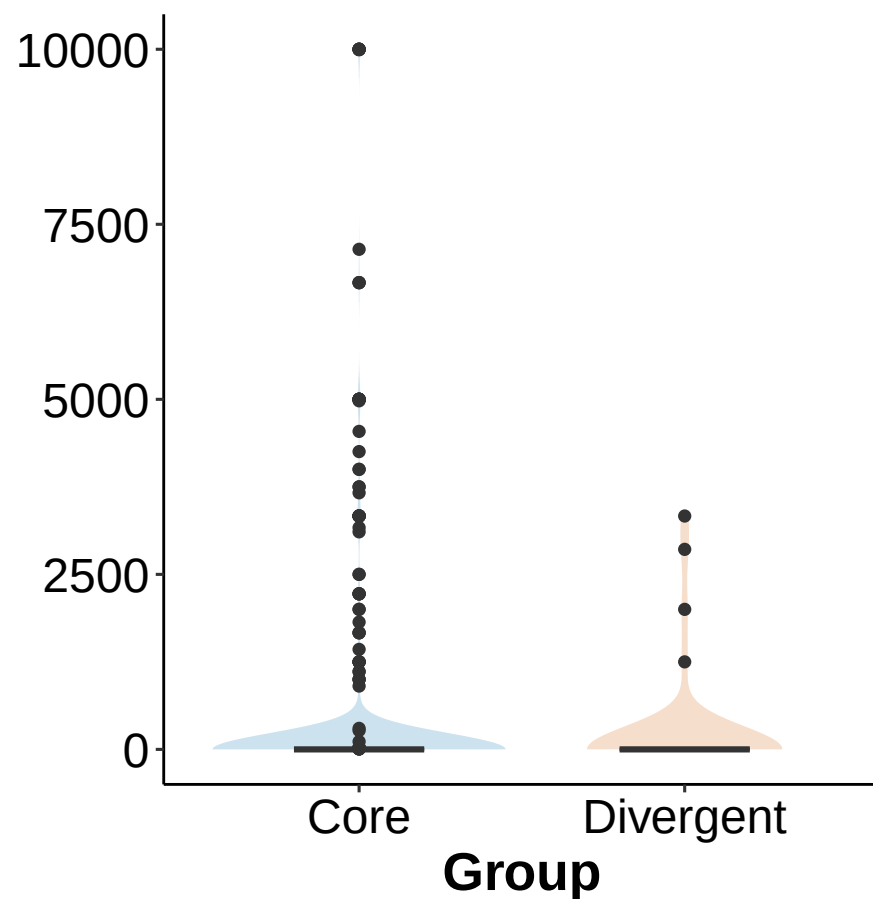
p-adj: 0.00323 | .:=−0.31 (Small) | .%: −11.6



Metabolism – . 5. Systemic Reprogramming (SA_*)

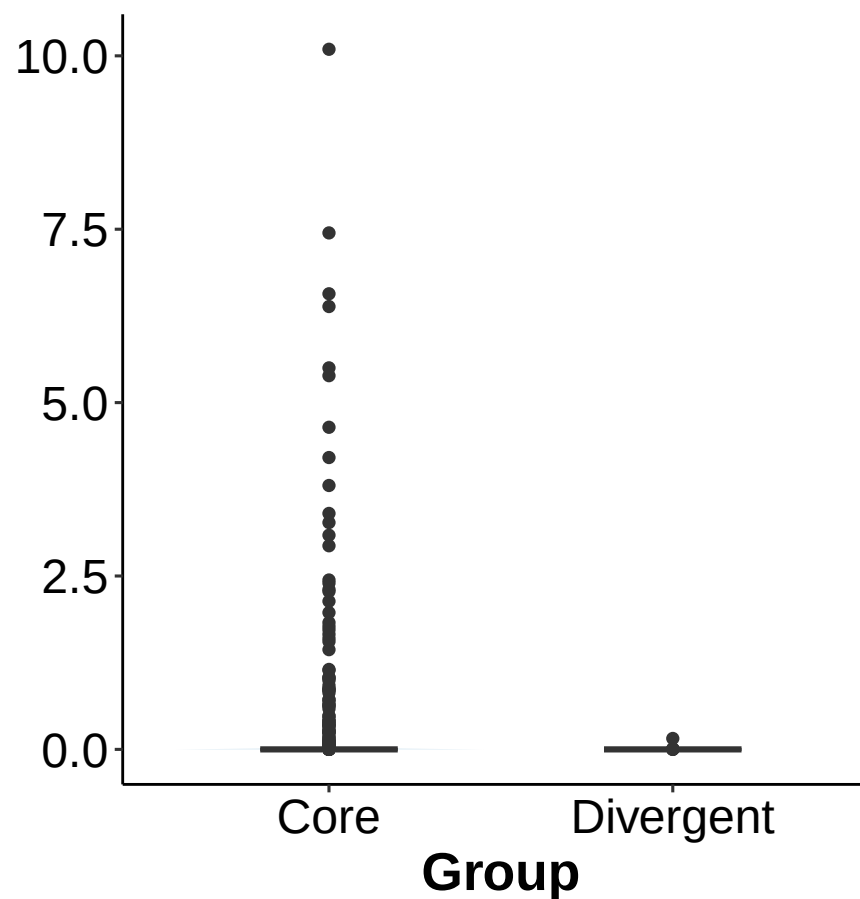
SA_Transport__golgi_apparatus_I

p-adj: 0.574 | .:=0.04 (Negligible)



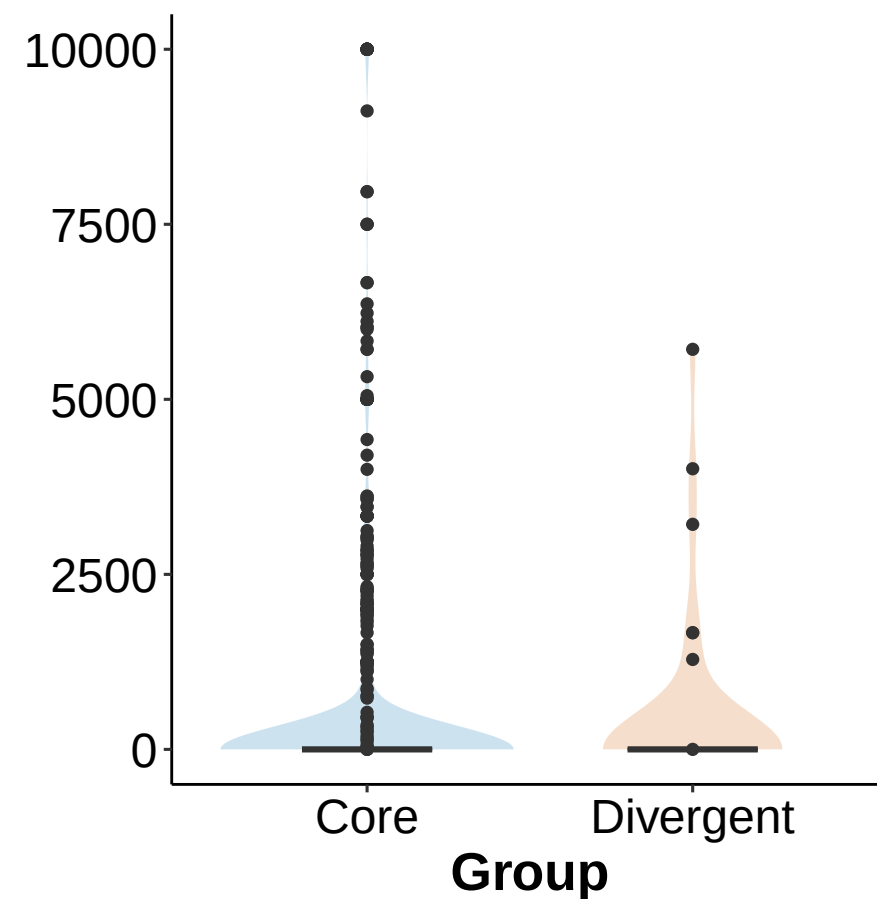
SA_Transport__golgi_apparatus_L1

p-adj: 0.574 | .:=−0.08 (Negligible) | .%: −90.5



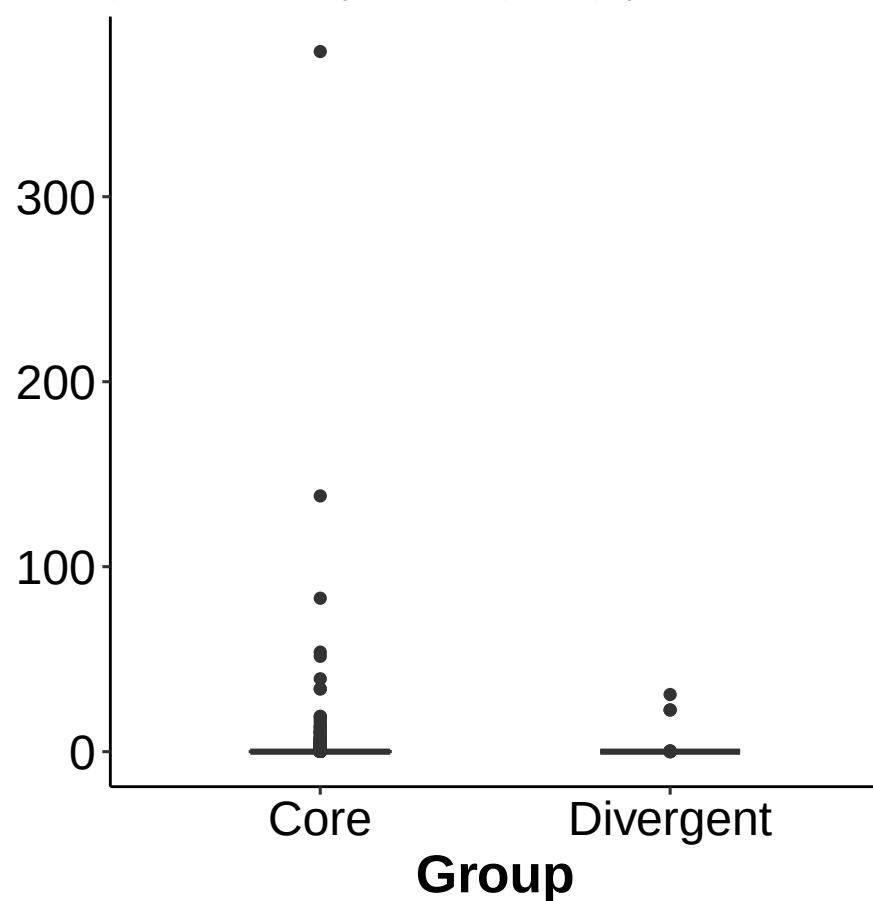
SA_Transport__lysosomal_FBA

p-adj: 0.759 | .:=0.03 (Negligible)



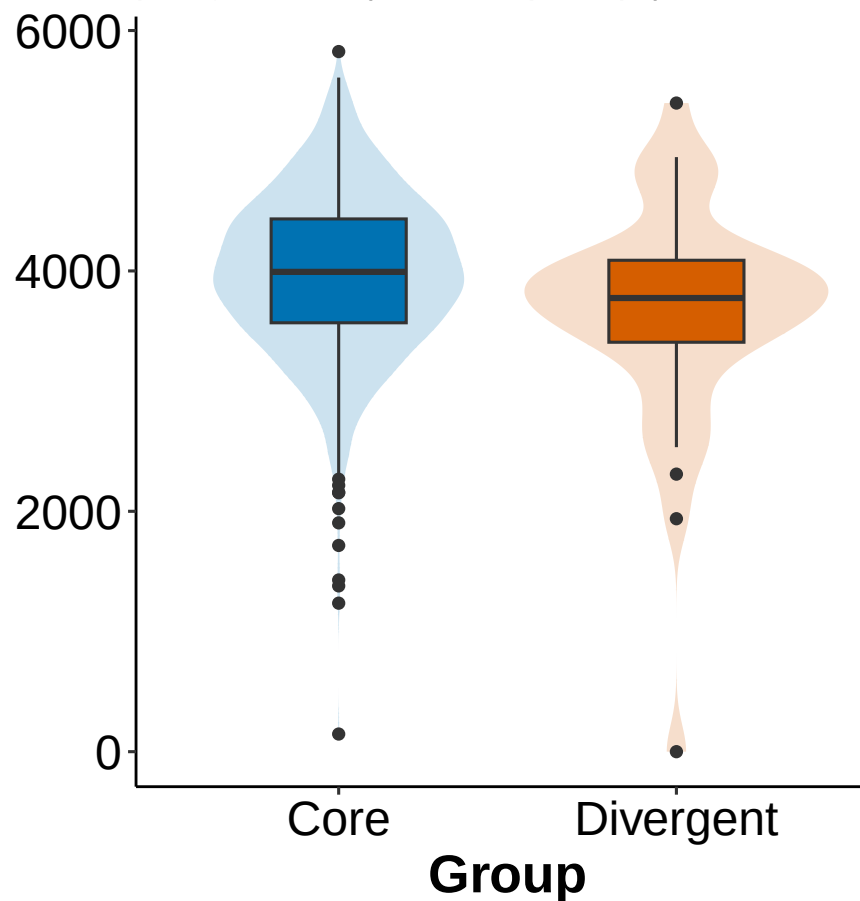
SA_Transport__lysosomal_L1w

p-adj: 0.0651 | .:=−0.21 (Small) | .%: −100



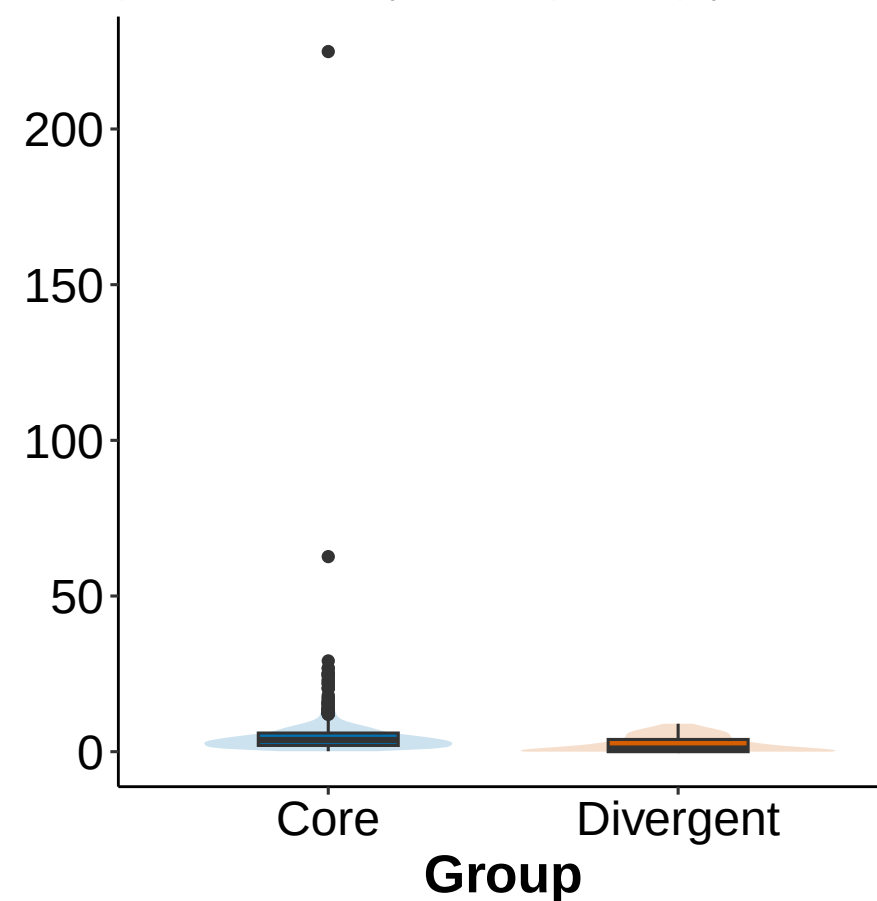
SA_Transport__mitochondrial_FBA

p-adj: 0.0517 | .:=−0.22 (Small) | .%: −5.5



SA_Transport__mitochondrial_L1w

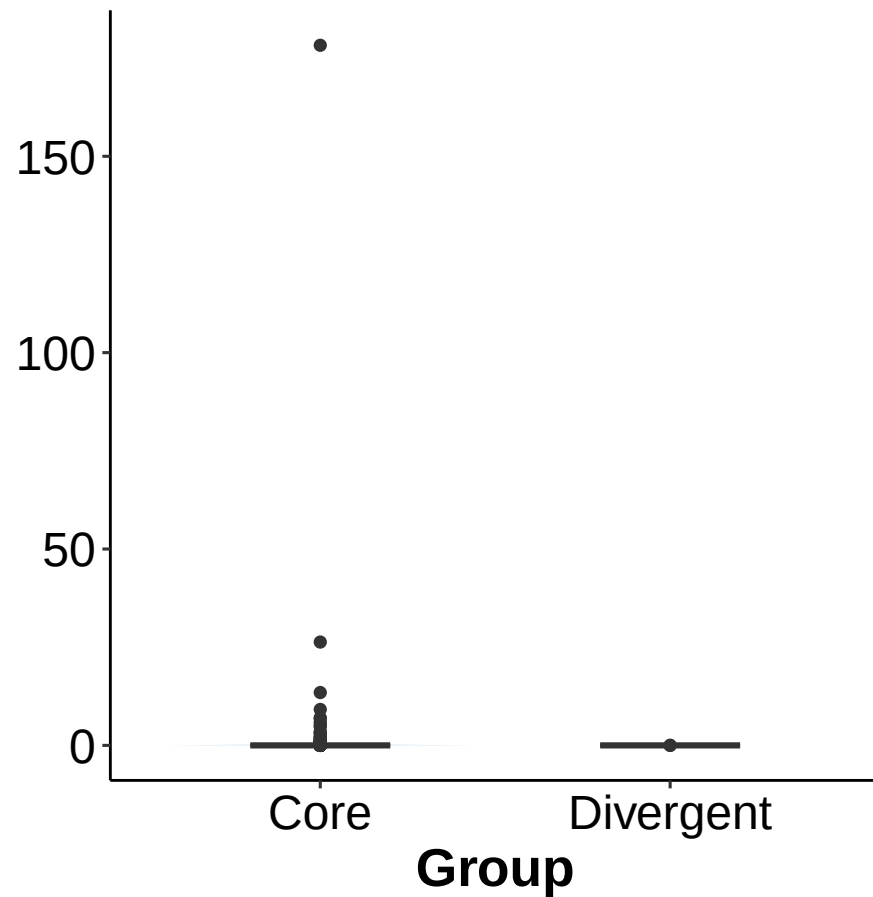
p-adj: 2.75e−05 | .:=−0.45 (Medium) | .%: −73.1



Metabolism – 5. Systemic Reprogramming (SA_*)

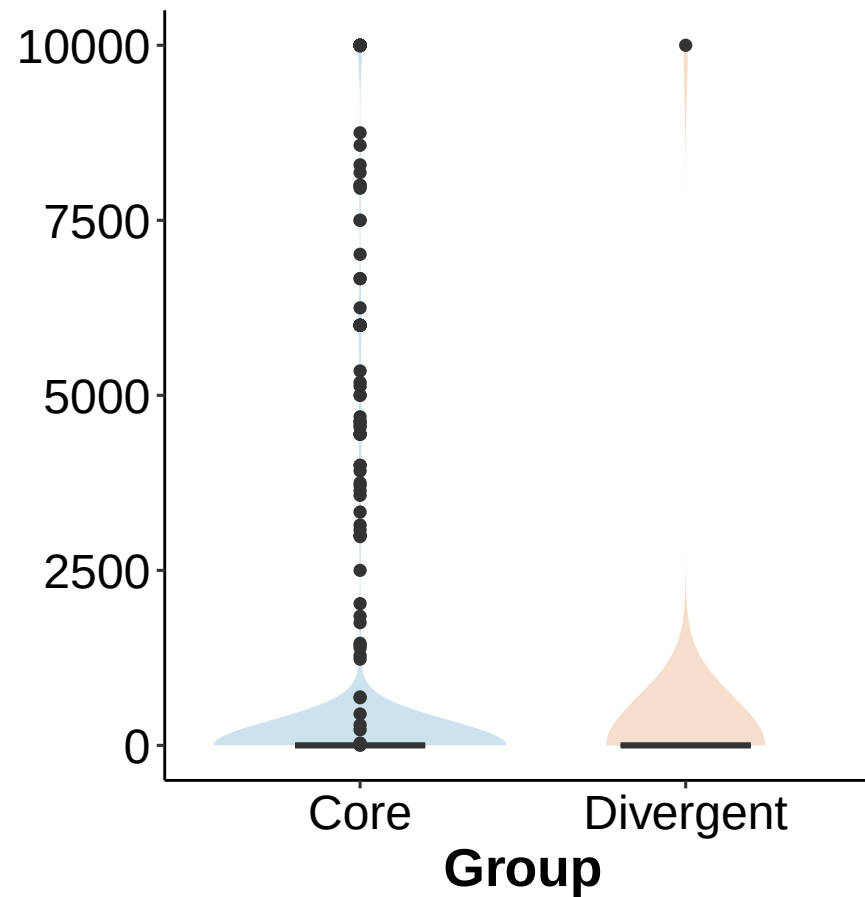
SA_Transport__mitochondrial_pFBA

p-adj: 0.00191 | .=-0.33 (Small) | .%: -41.2



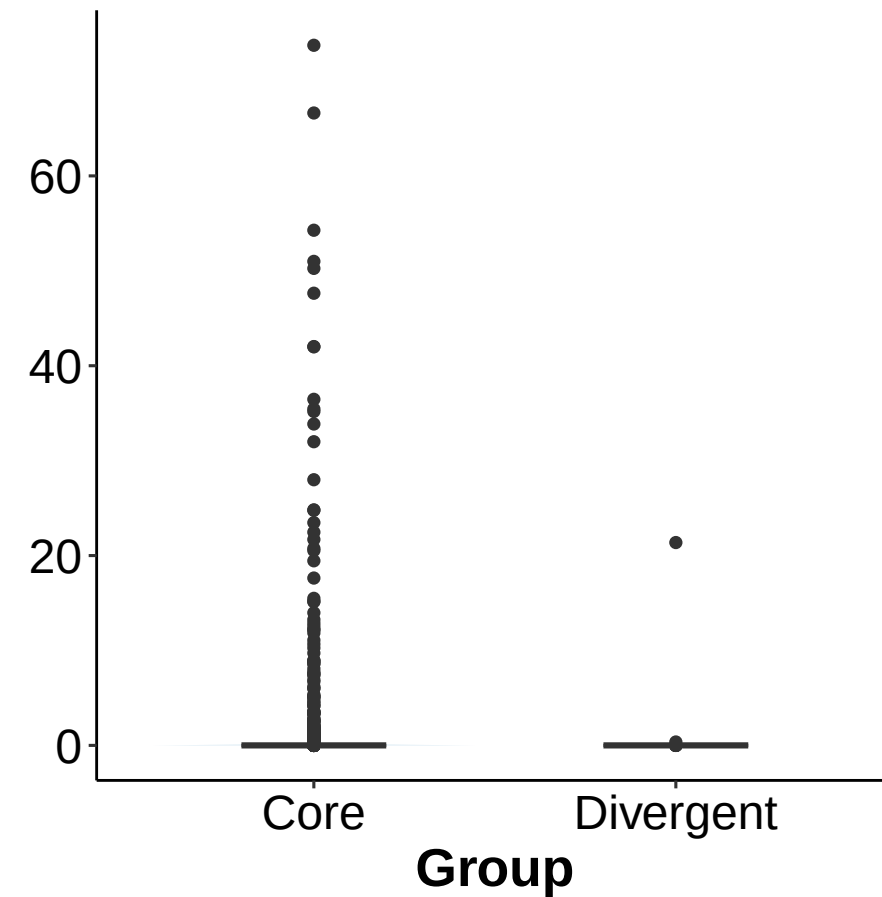
SA_Transport__nuclear_FBA

p-adj: 0.424 | .=-0.05 (Negligible)



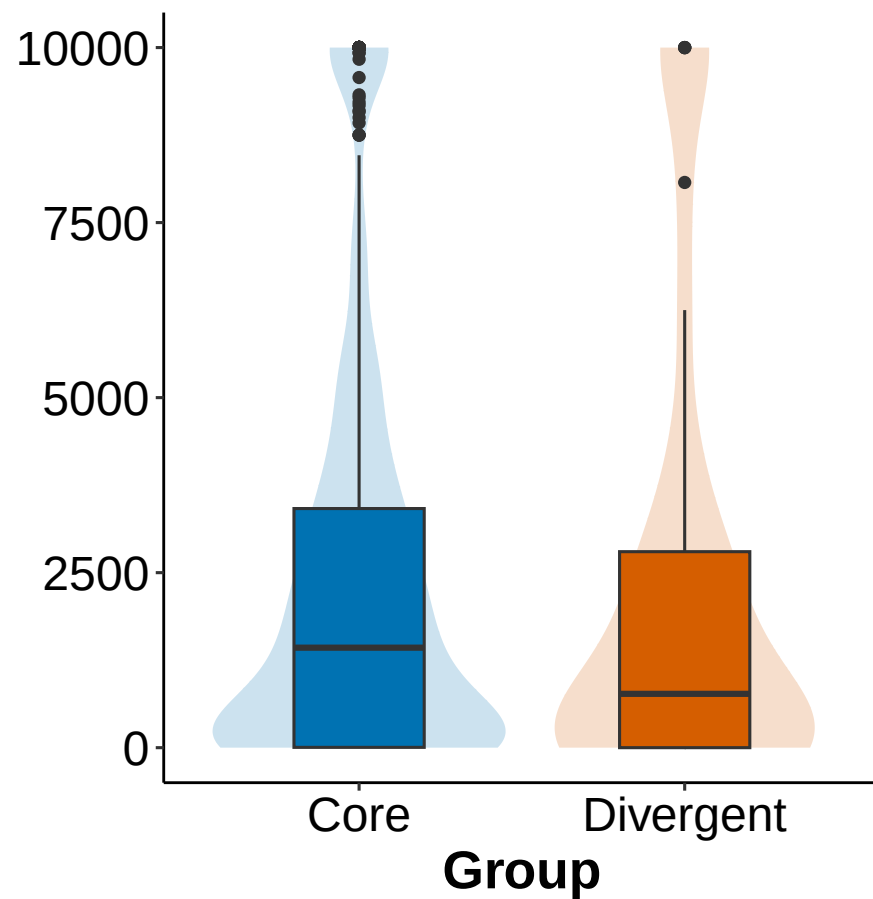
SA_Transport__nuclear_L1w

p-adj: 0.233 | .=-0.13 (Negligible)



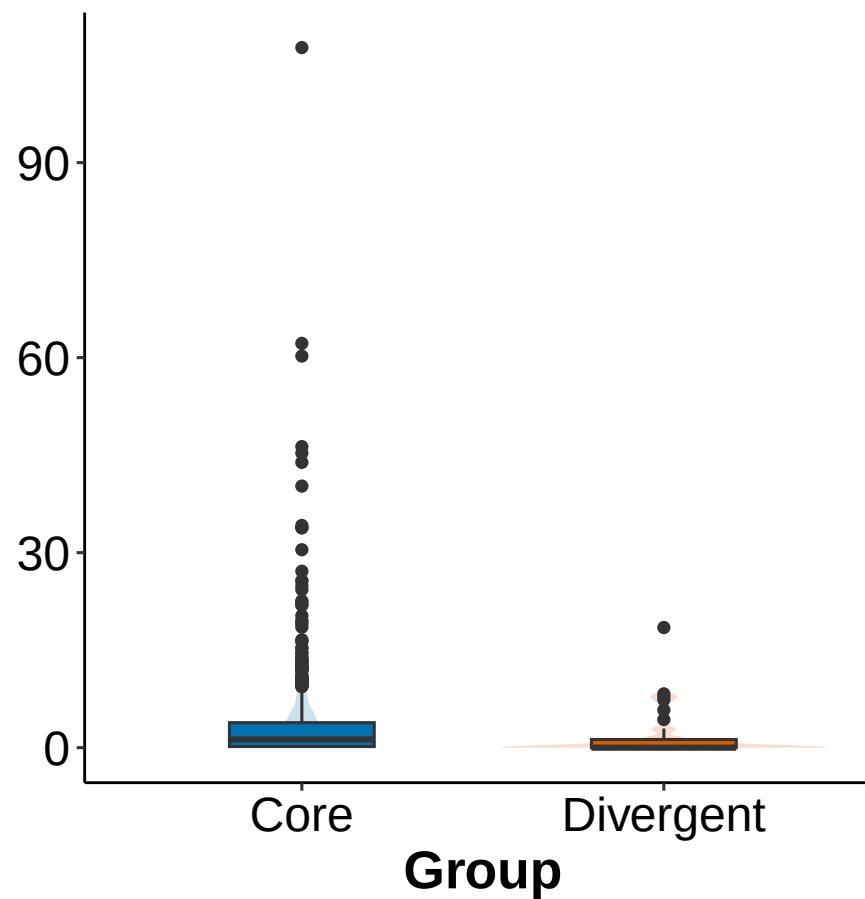
SA_Transport__peroxisomal_FBA

p-adj: 0.333 | .=-0.13 (Negligible) | .%: -46.2



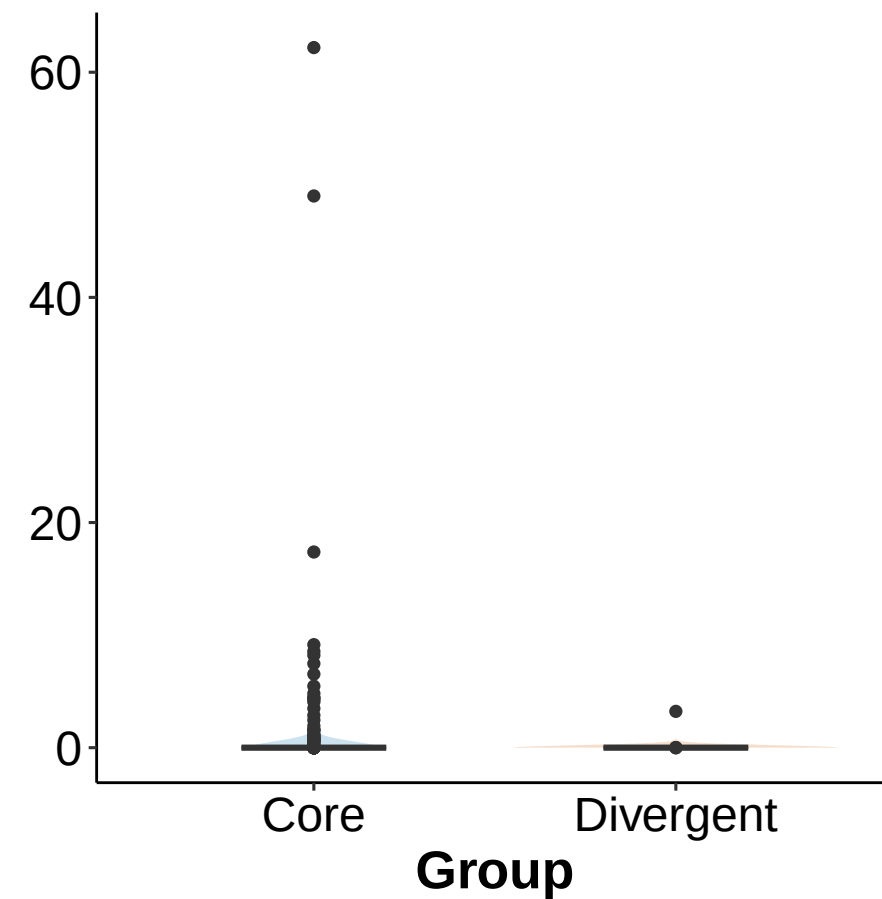
SA_Transport__peroxisomal_L1w

p-adj: 0.00106 | .=-0.35 (Medium) | .%: -100

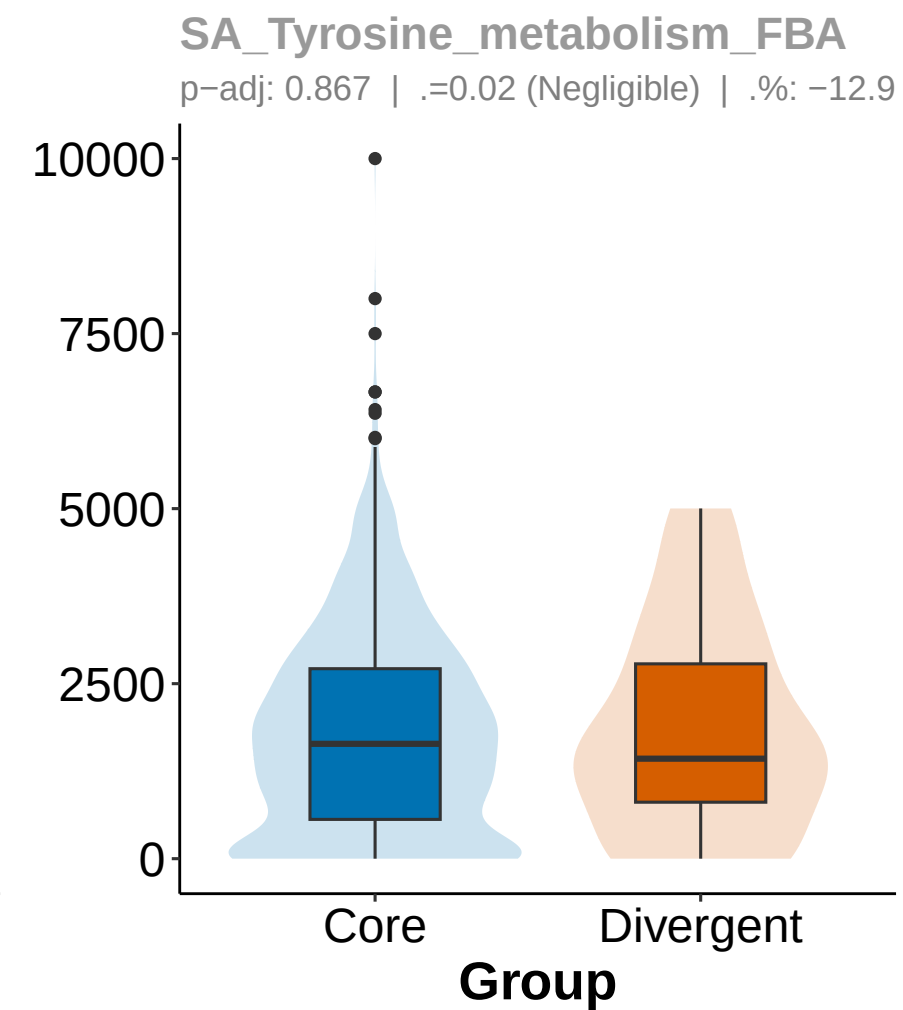
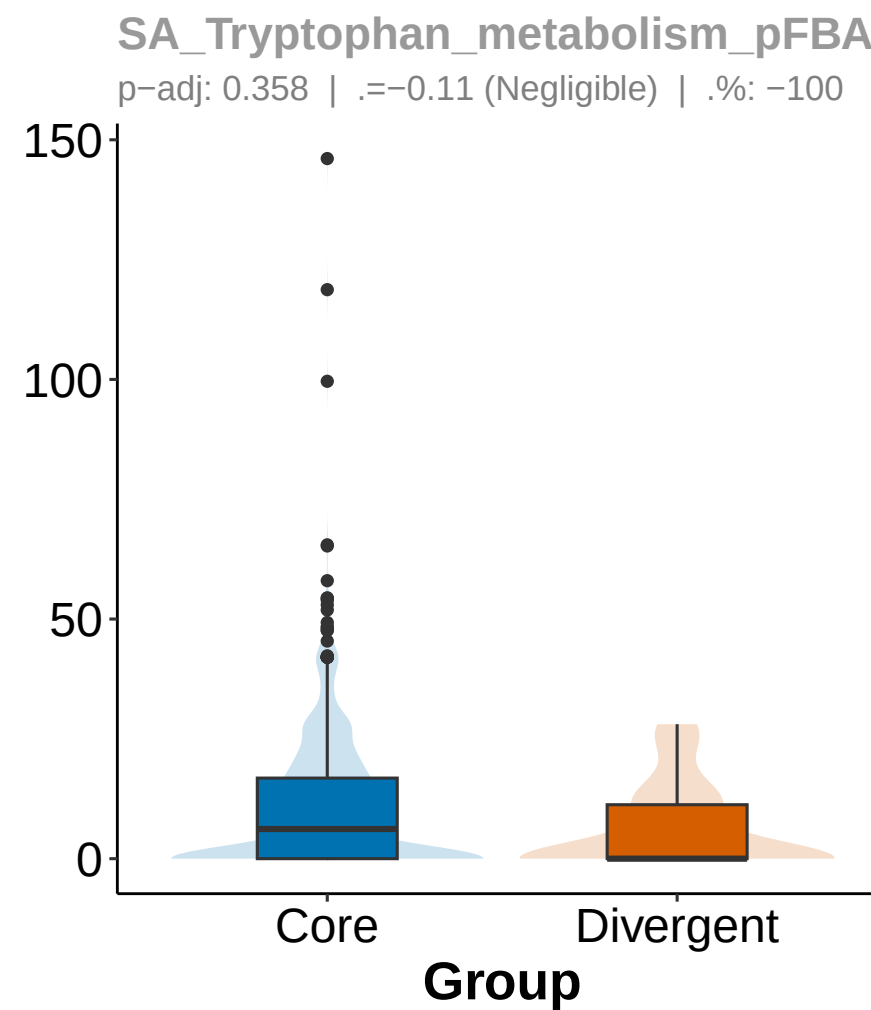
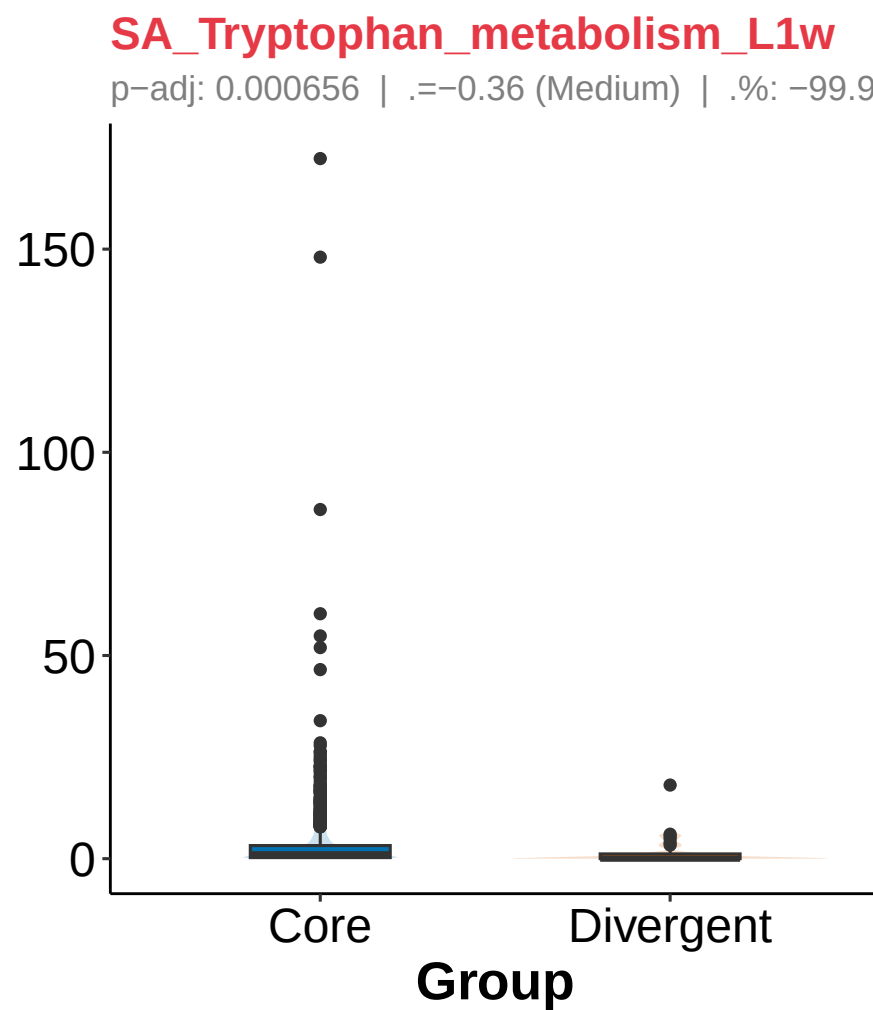
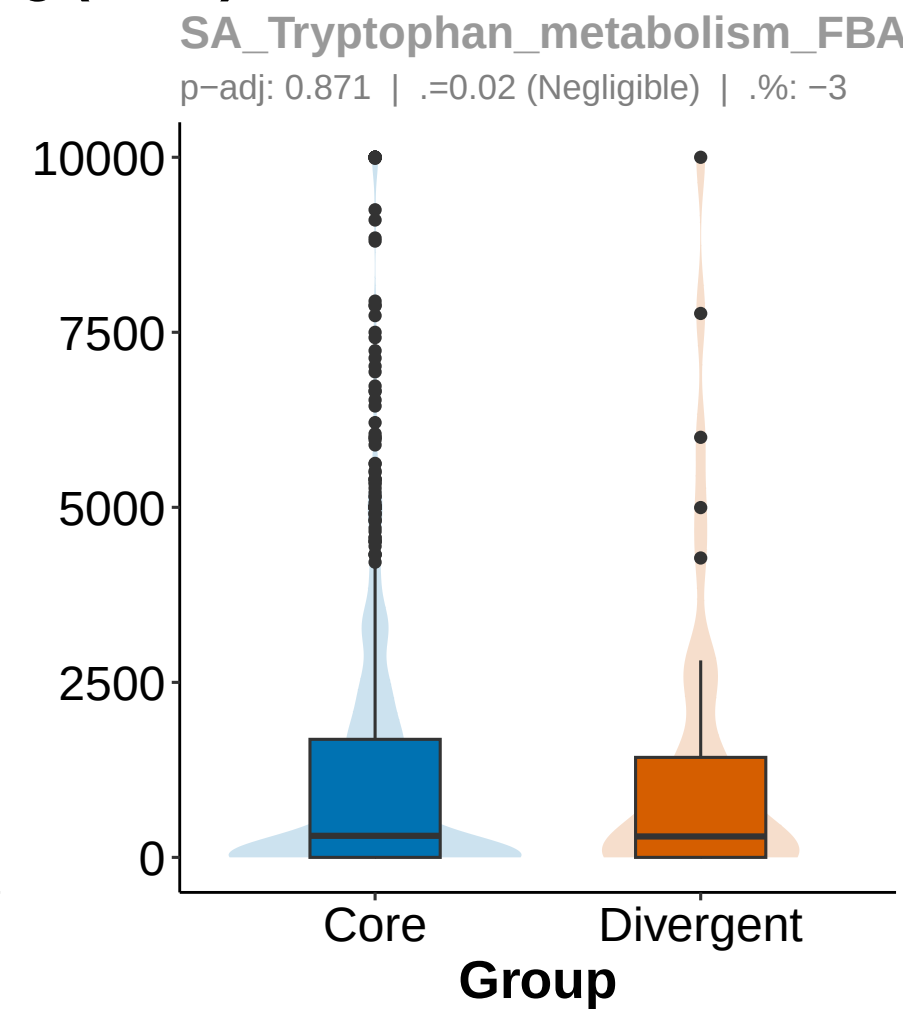
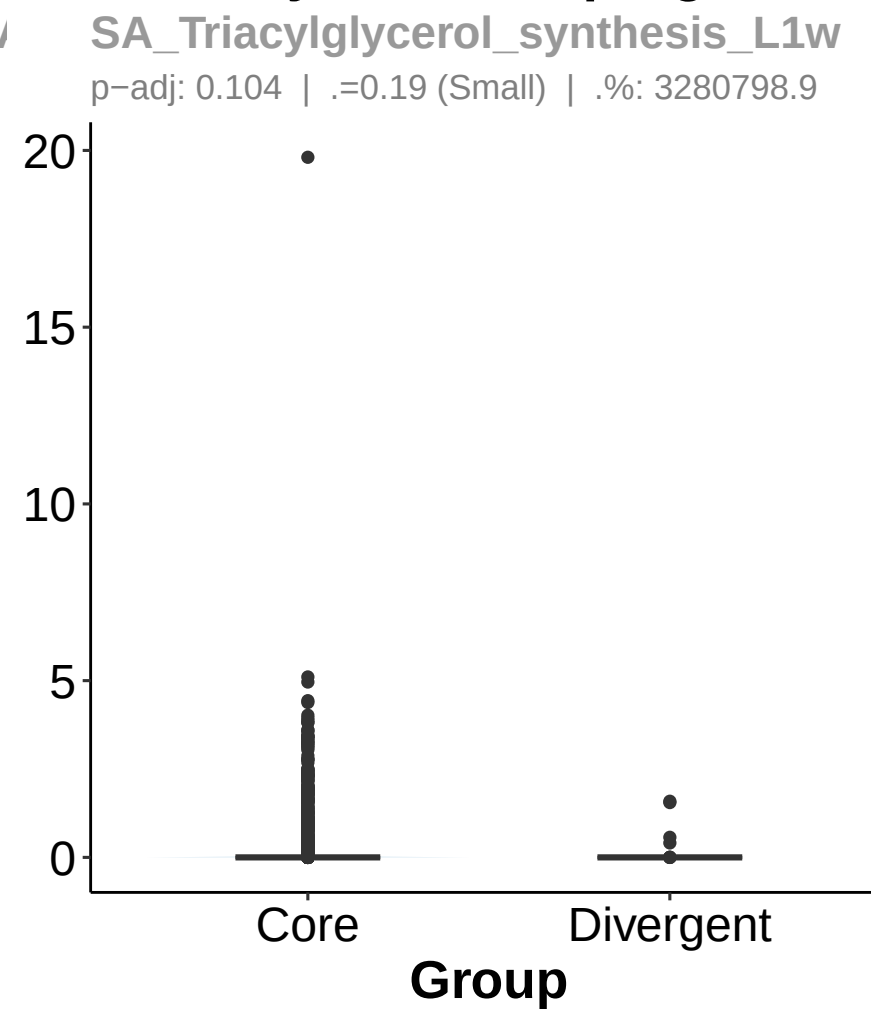
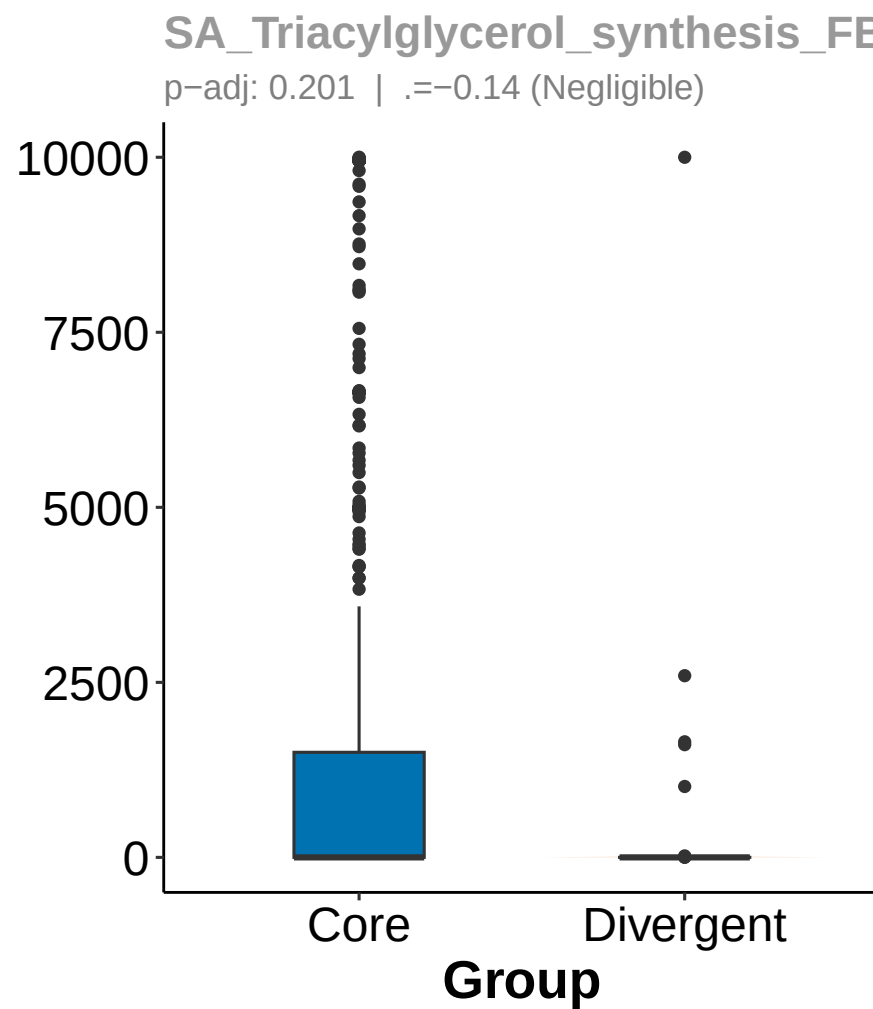


SA_Transport__peroxisomal_pFBA

p-adj: 0.771 | .=0.02 (Negligible)



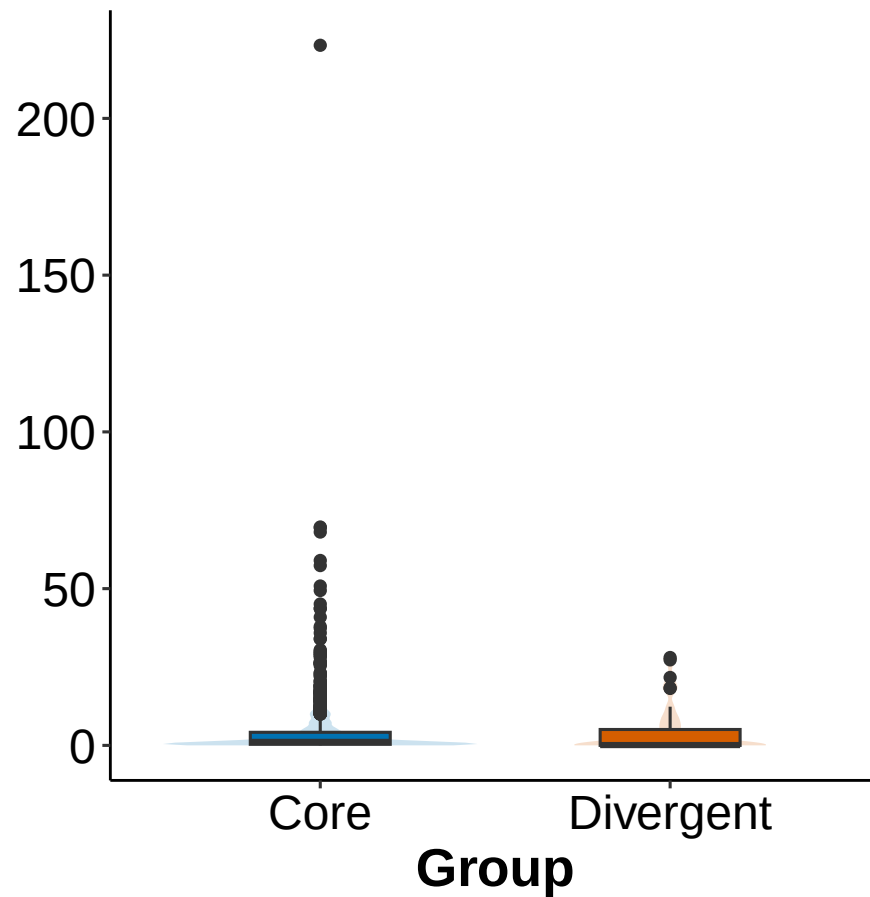
Metabolism – . 5. Systemic Reprogramming (SA_*)



Metabolism – . 5. Systemic Reprogramming (SA_*)

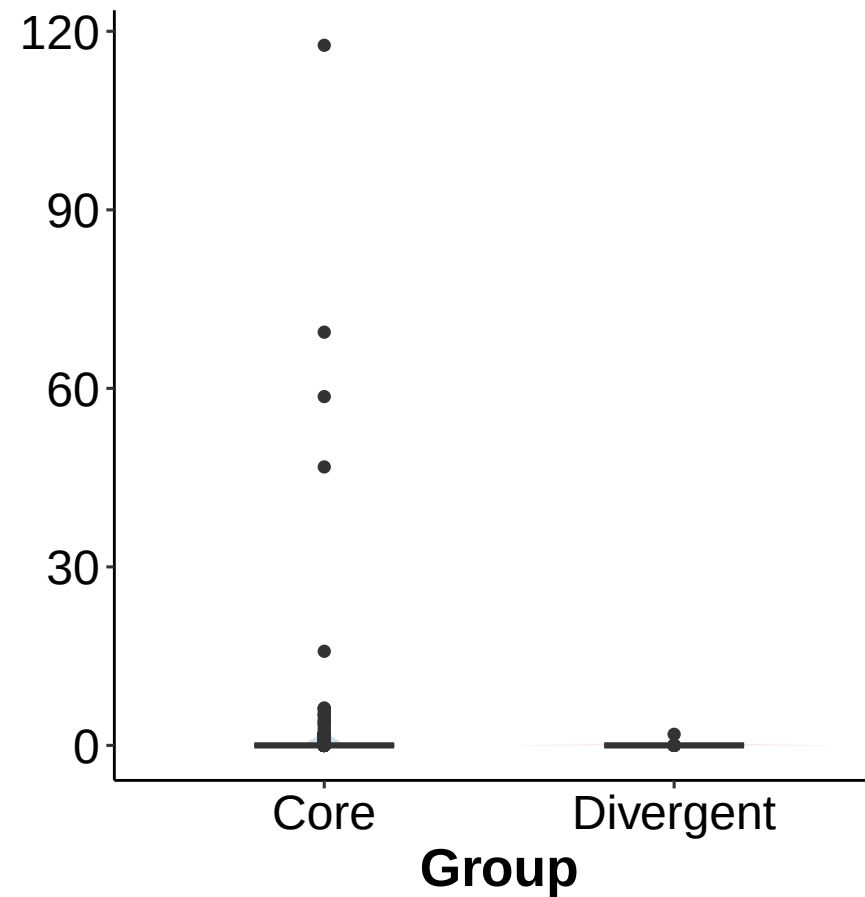
SA_Tyrosine_metabolism_L1w

p-adj: 0.0075 | .=-0.29 (Small) | .%: -90.9



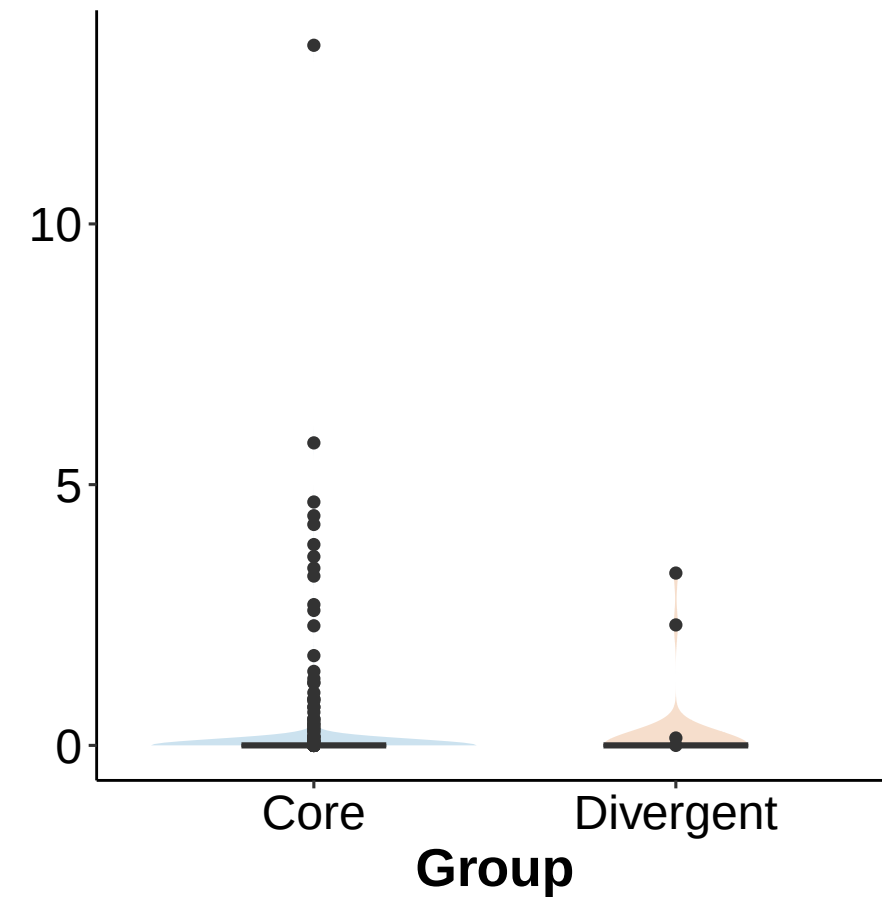
SA_Tyrosine_metabolism_pFBA

p-adj: 0.789 | .=0.02 (Negligible)



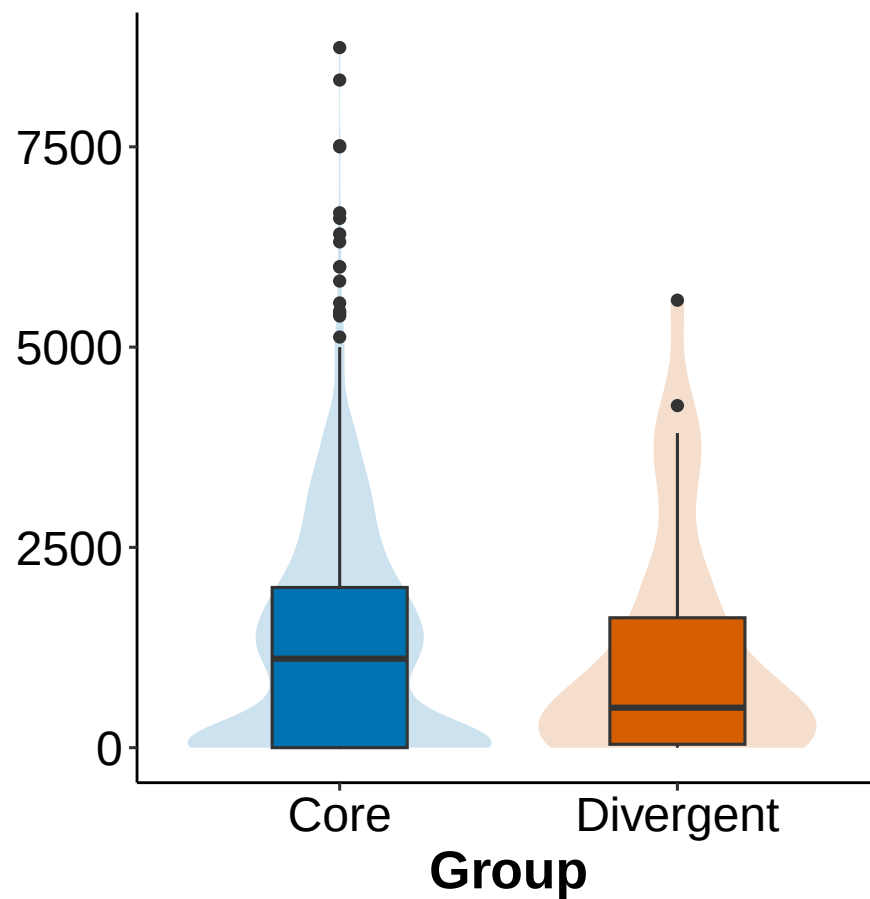
SA_Ubiquinone_synthesis_L1w

p-adj: 0.746 | .=0.02 (Negligible)



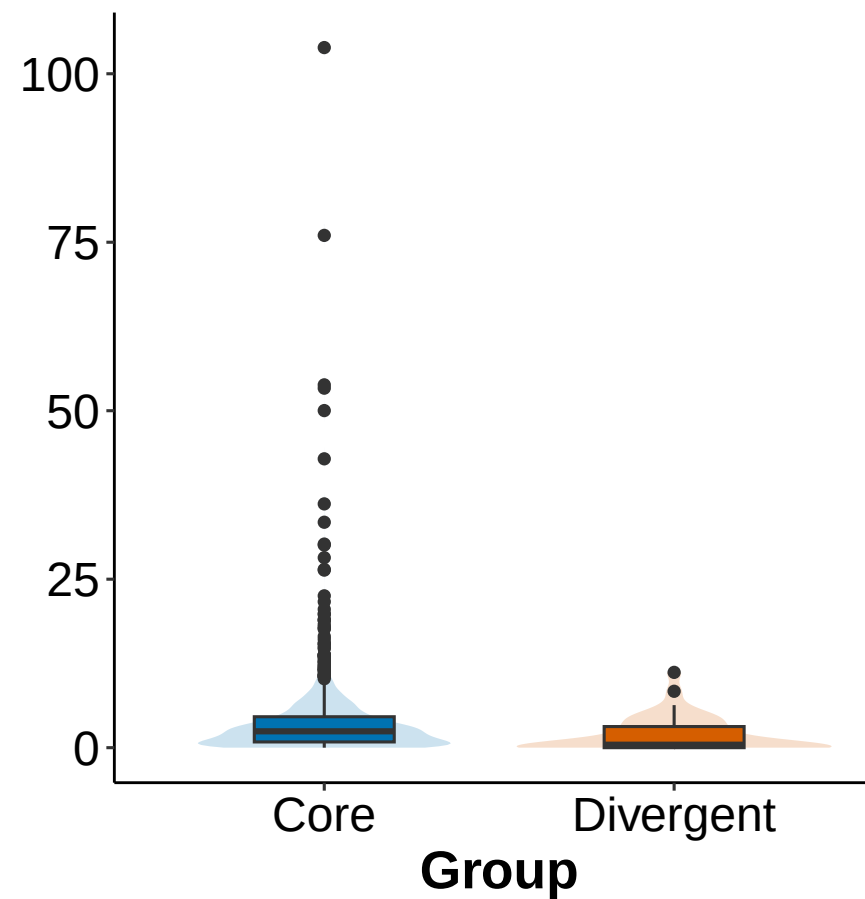
SA_Urea_cycle_FBA

p-adj: 0.566 | .=-0.08 (Negligible) | .%: -55.1



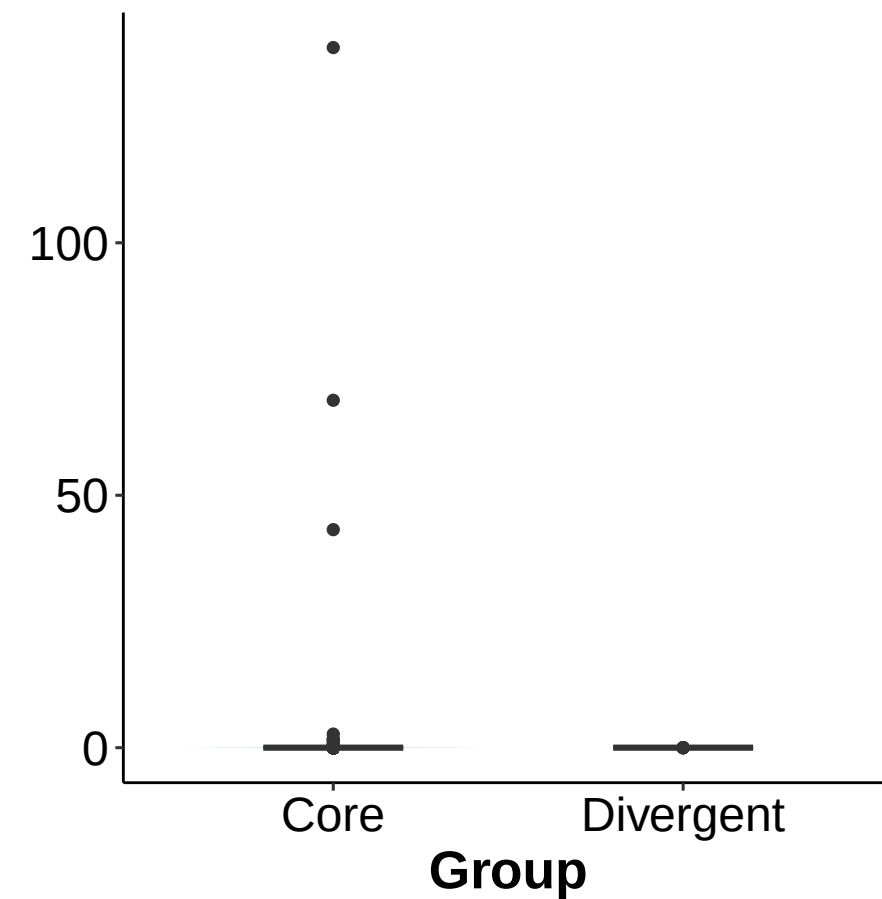
SA_Urea_cycle_L1w

p-adj: 0.000666 | .=-0.36 (Medium) | .%: -81.8

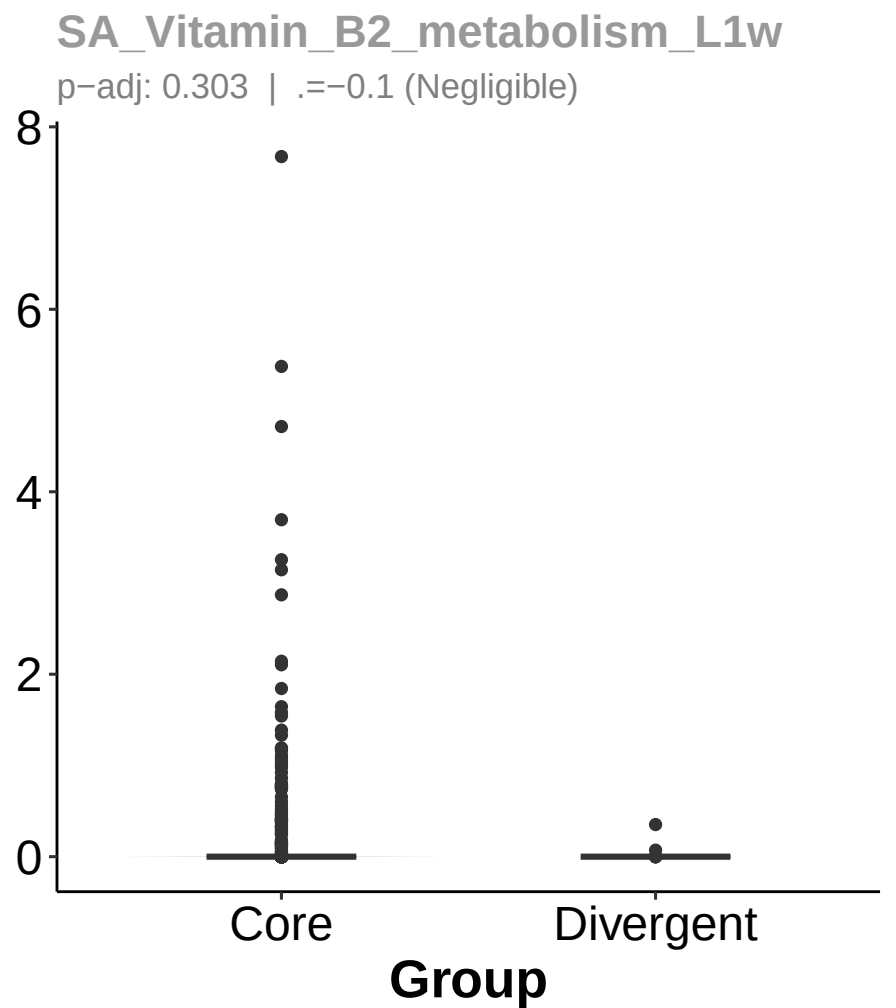
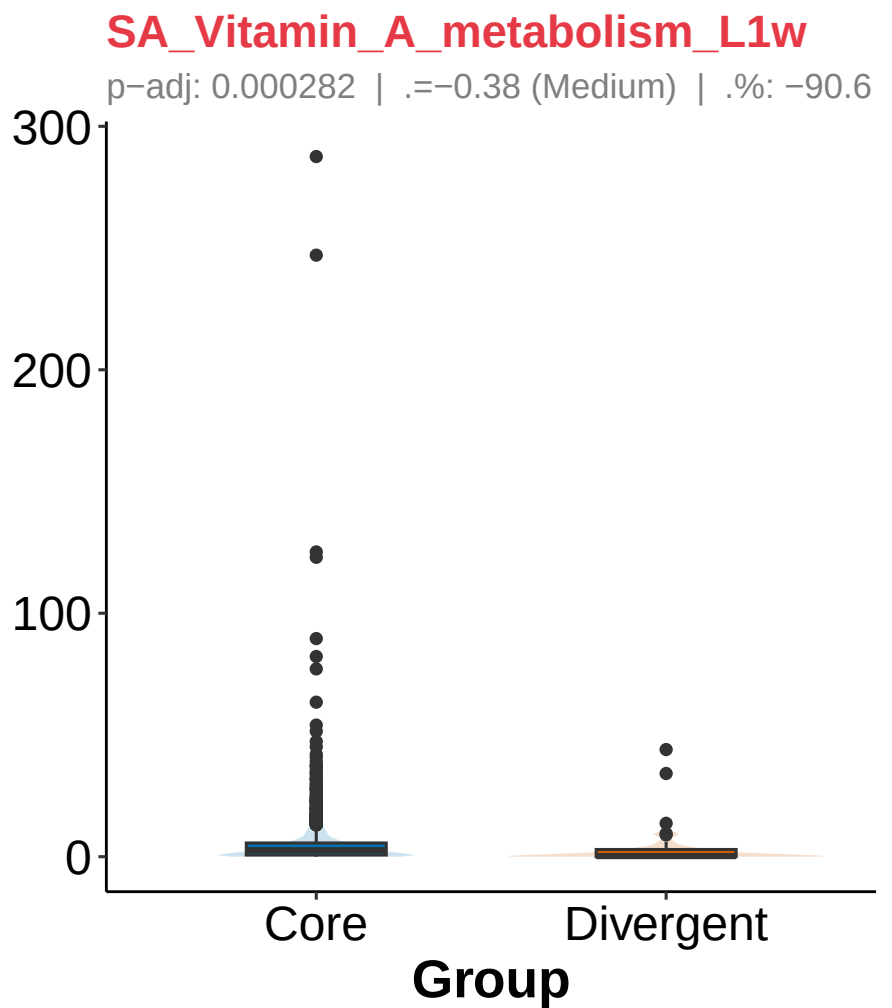
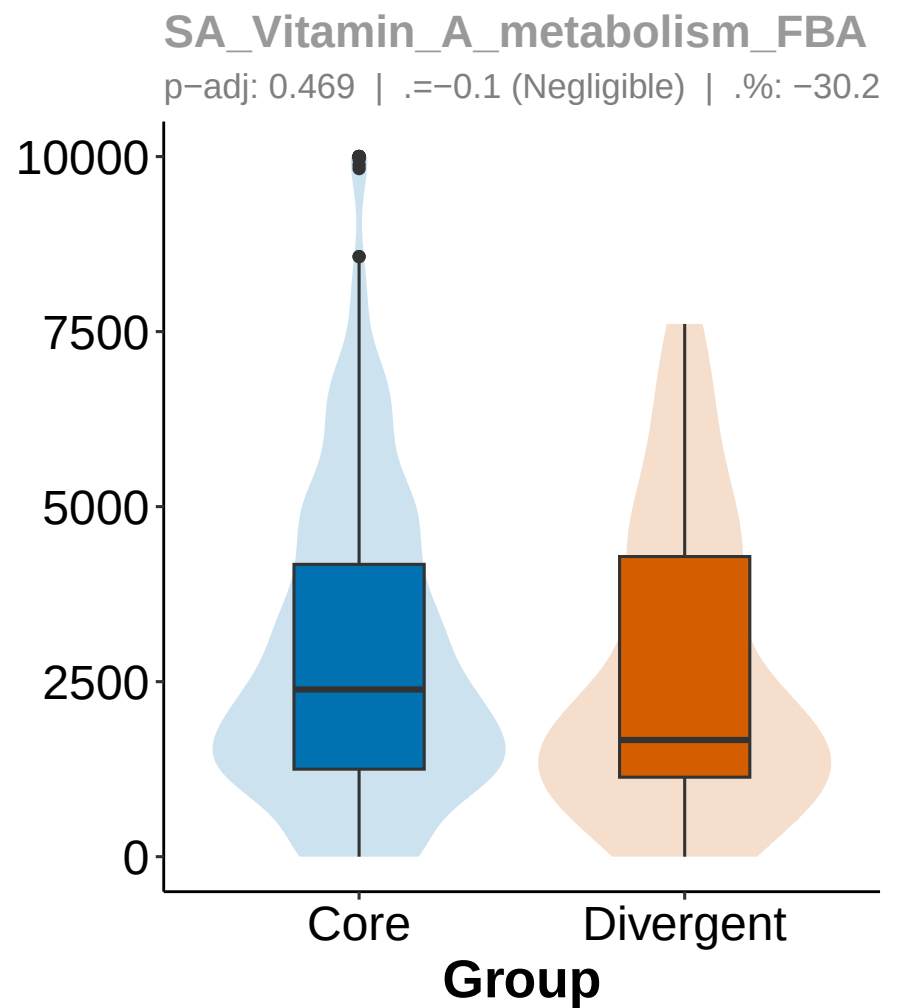
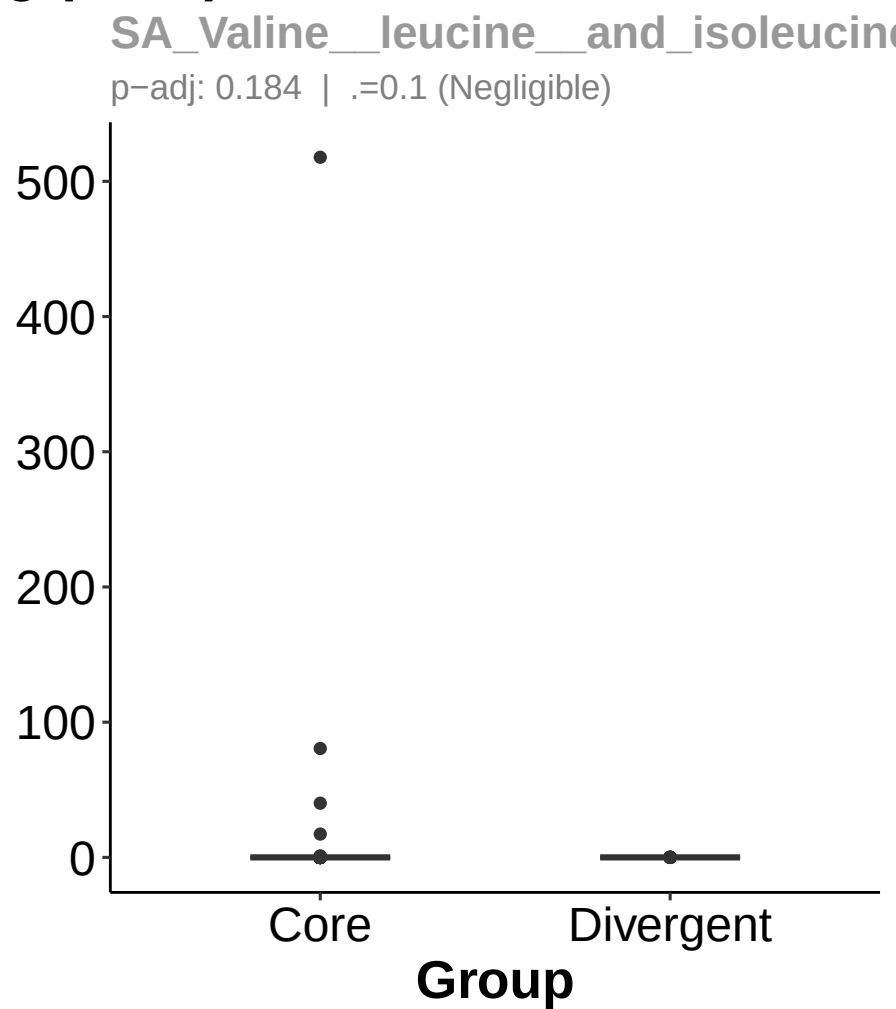
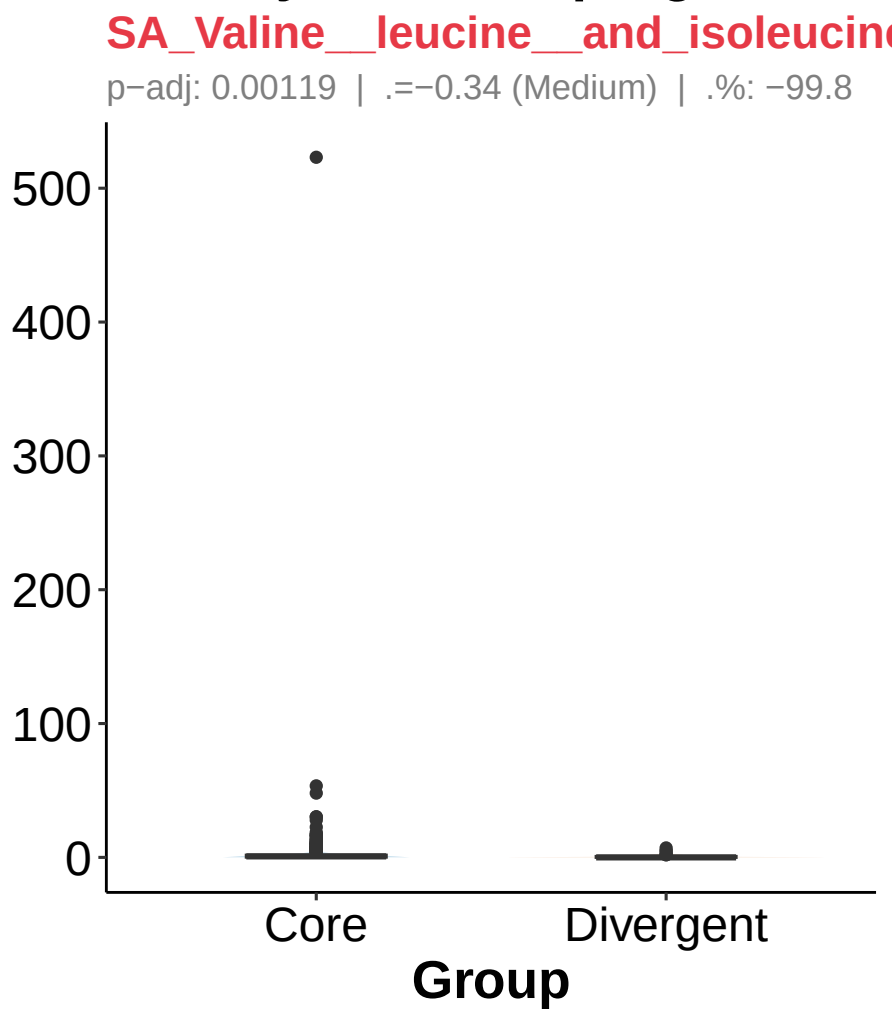
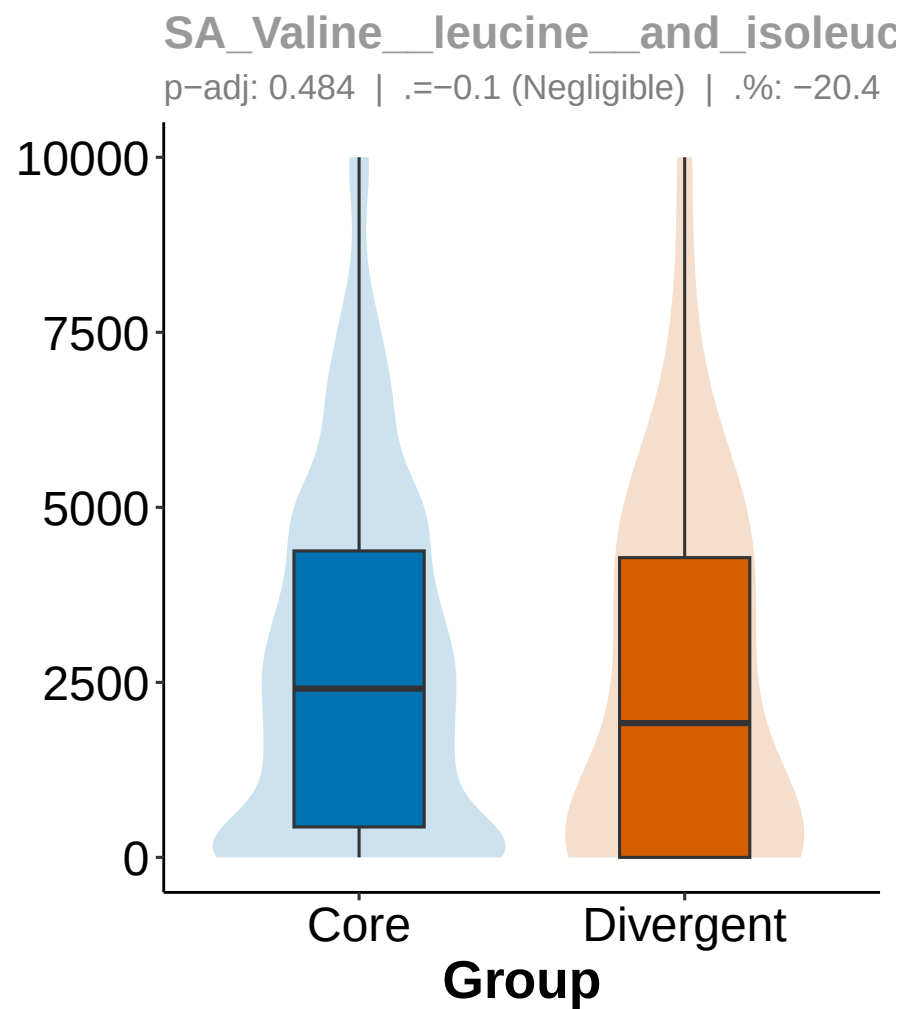


SA_Urea_cycle_pFBA

p-adj: 0.00252 | .=0.25 (Small)



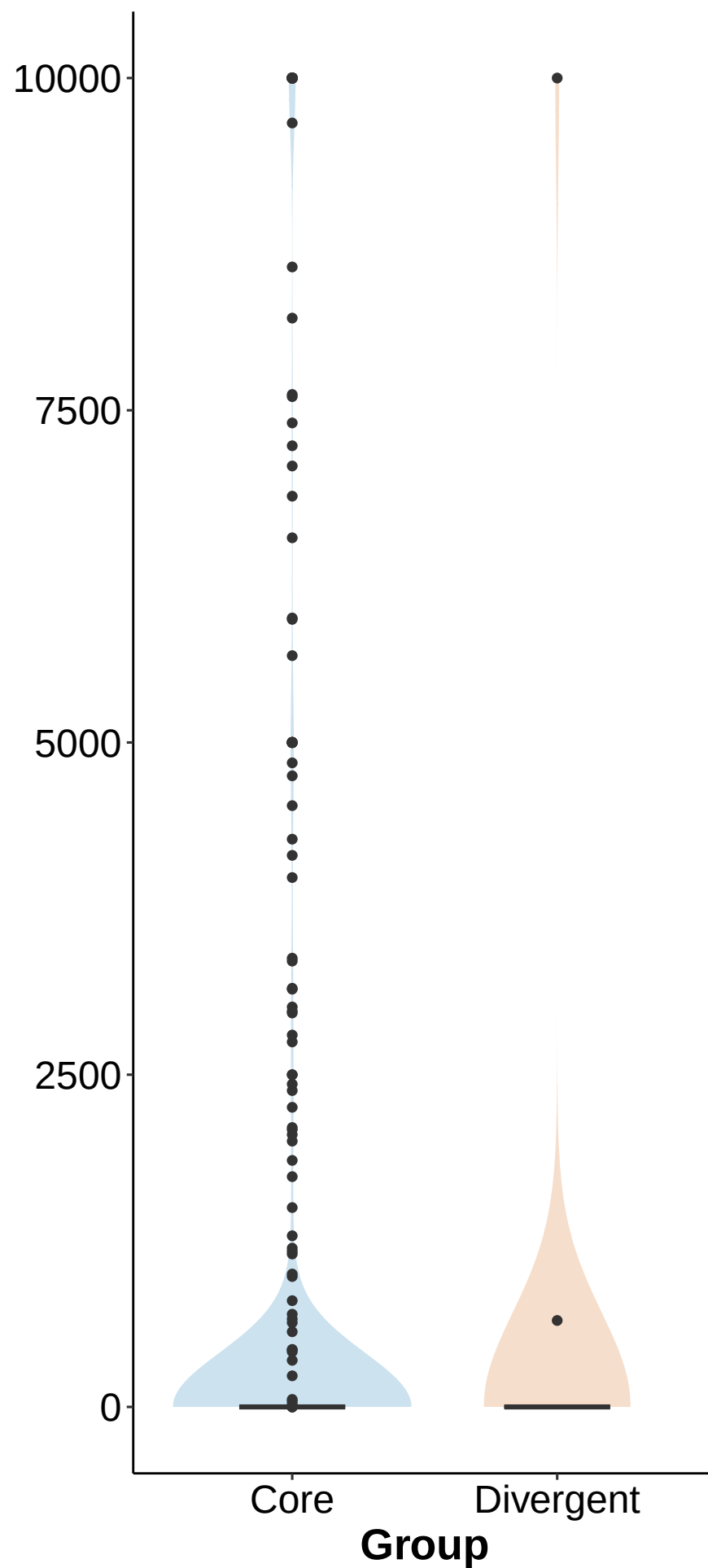
Metabolism – . 5. Systemic Reprogramming (SA_*)



Metabolism – . 5. Systemic Reprogramming (SA_*)

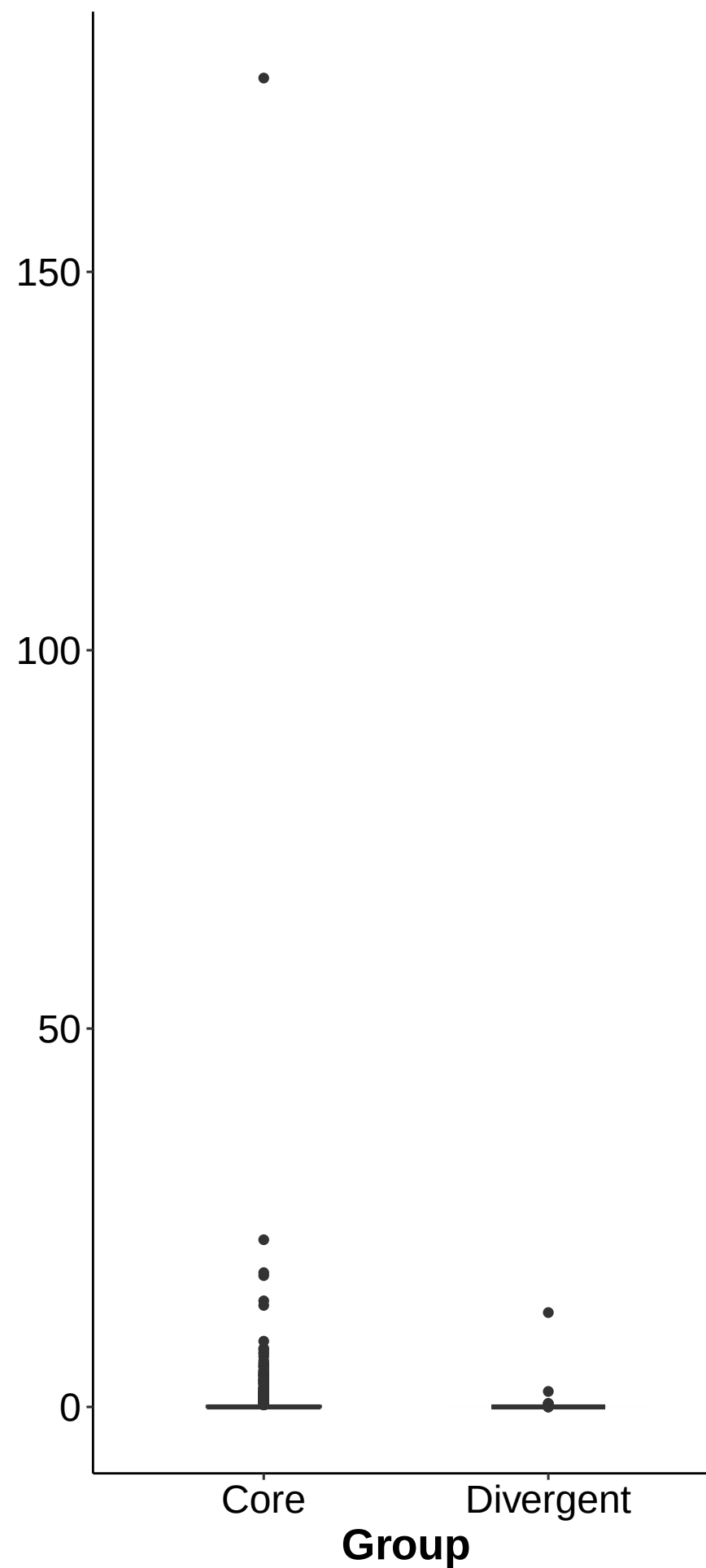
SA_Vitamin_C_metabolism_FBA

p-adj: 0.574 | .=-0.04 (Negligible)



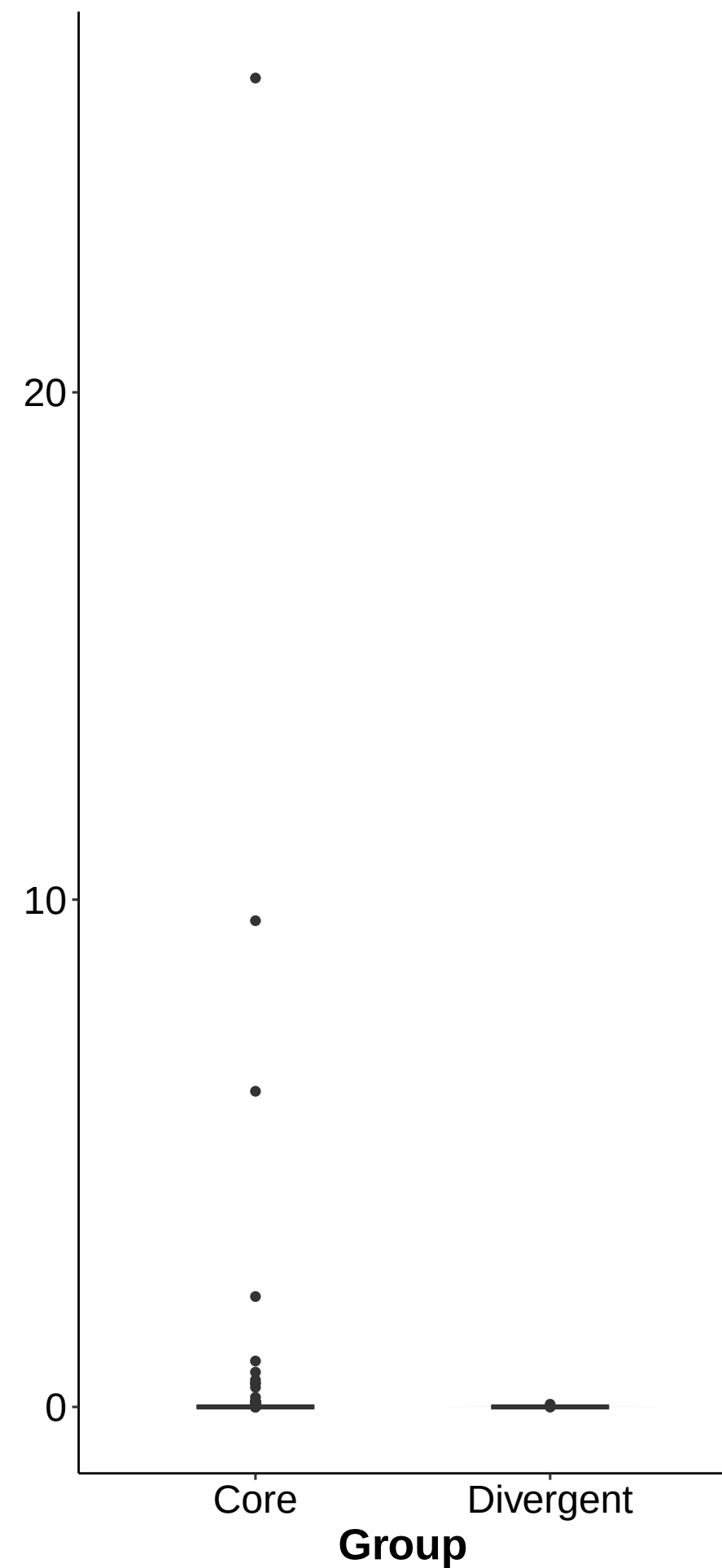
SA_Vitamin_C_metabolism_L1w

p-adj: 0.0894 | .=-0.19 (Small) | .%: -100



SA_Vitamin_D_metabolism_L1w

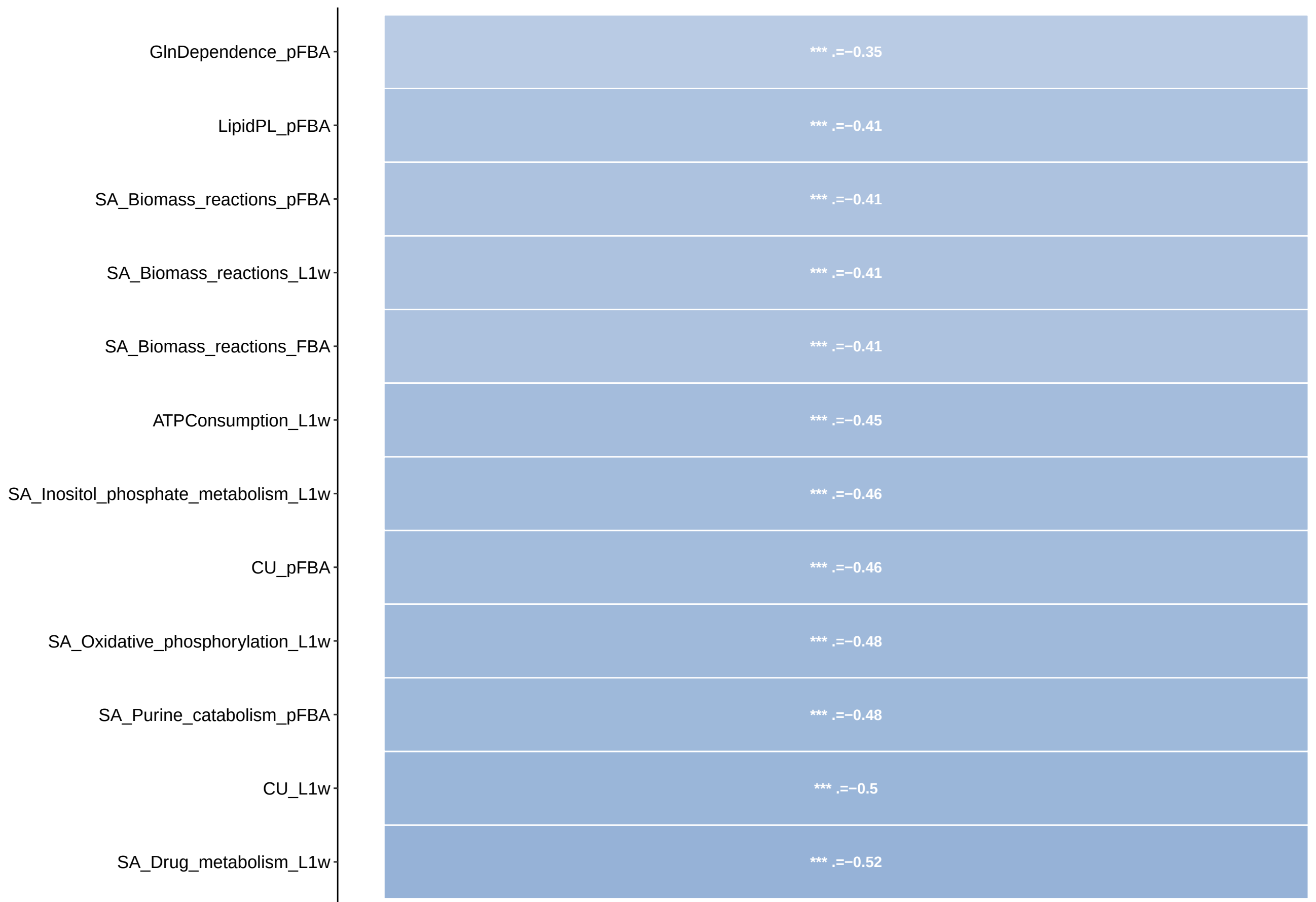
p-adj: 0.924 | .=-0.01 (Negligible)



Cliff's .: Top 12 Variables

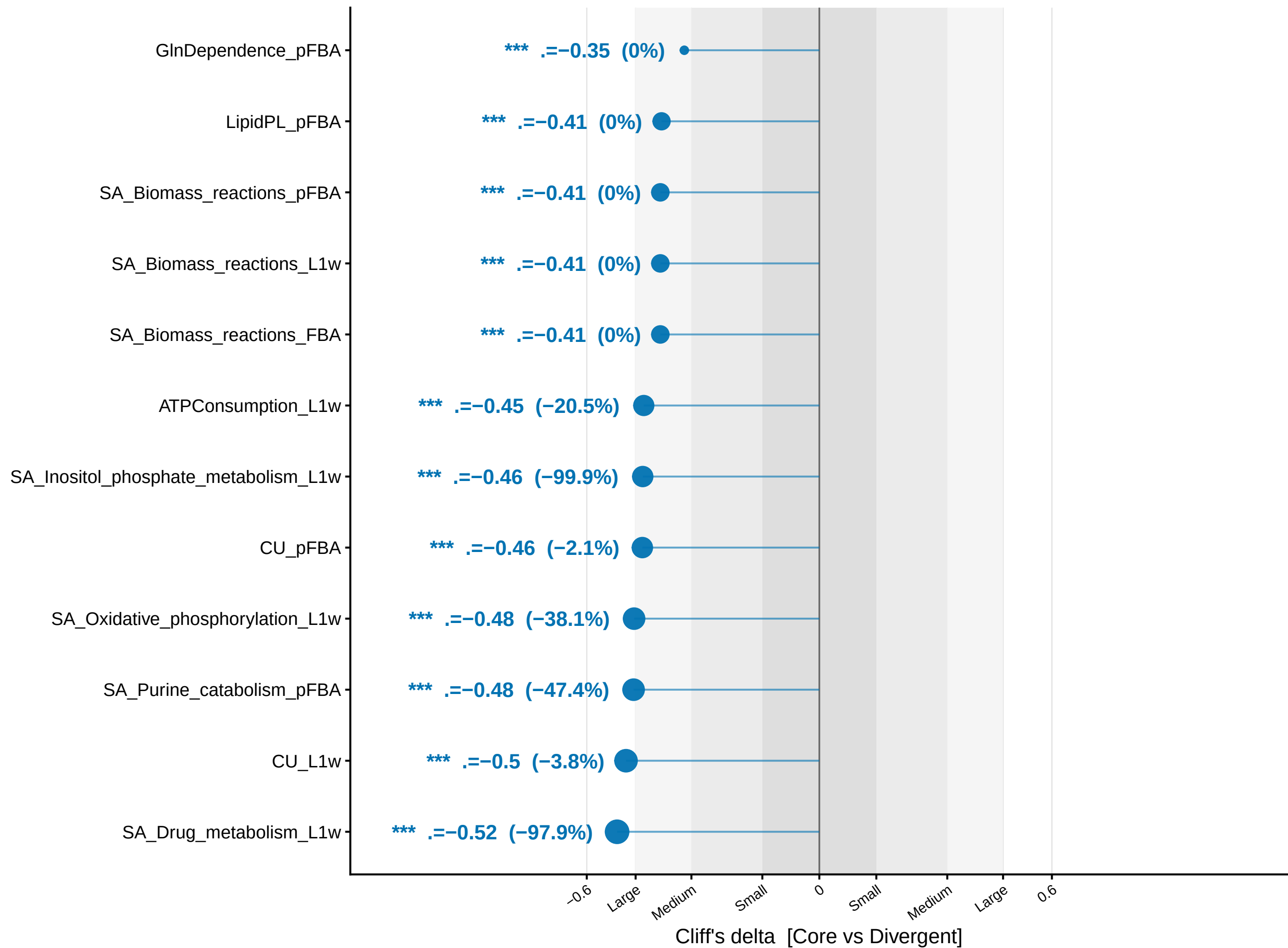
Blue = Core > Divergent | Orange = Divergent > Core | * p_adj<0.05

Cliff's .
Div vs Core



. Top 12 Variables – Cliff's .

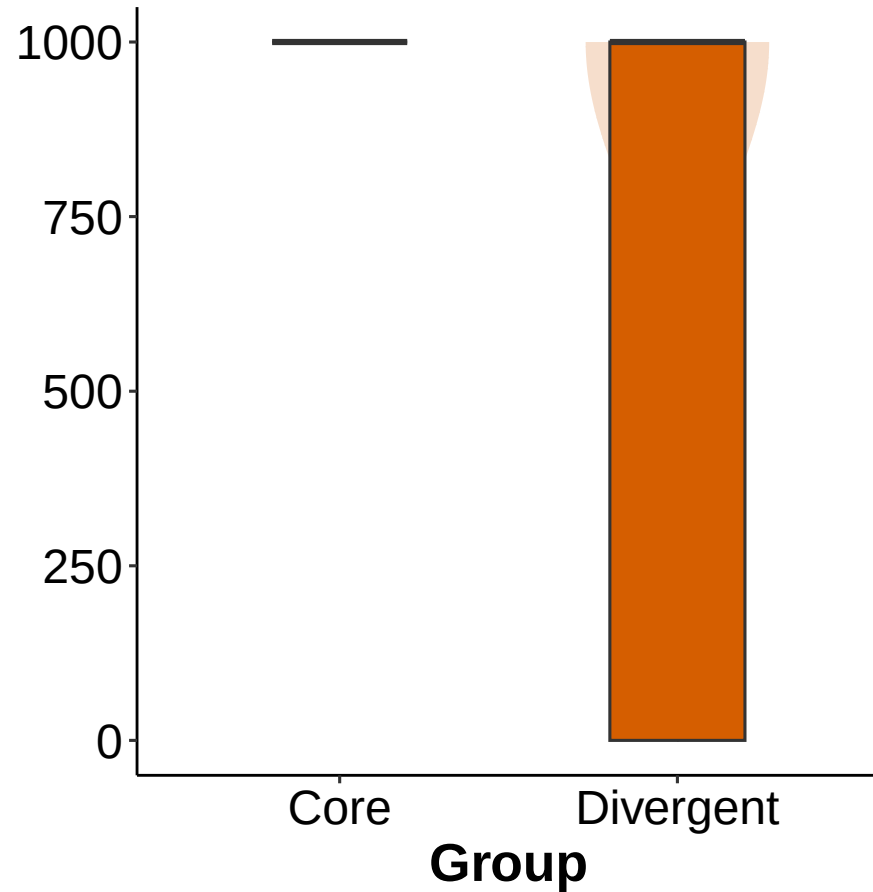
Cliff's . + .% median | . higher in Core · . higher in Divergent



. Top 12 Metabolic Variables (p_adj + Cliff's .)

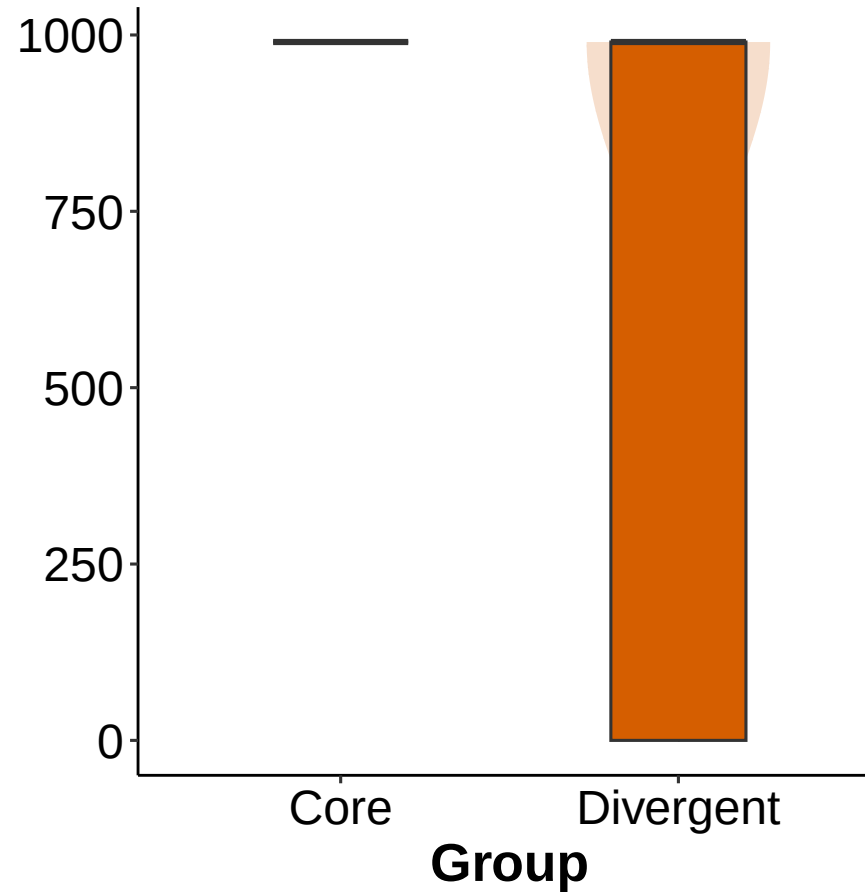
SA_Biomass_reactions_FBA

p-adj: 6.09e-102 | .=-0.41 (Medium) | .%: 0



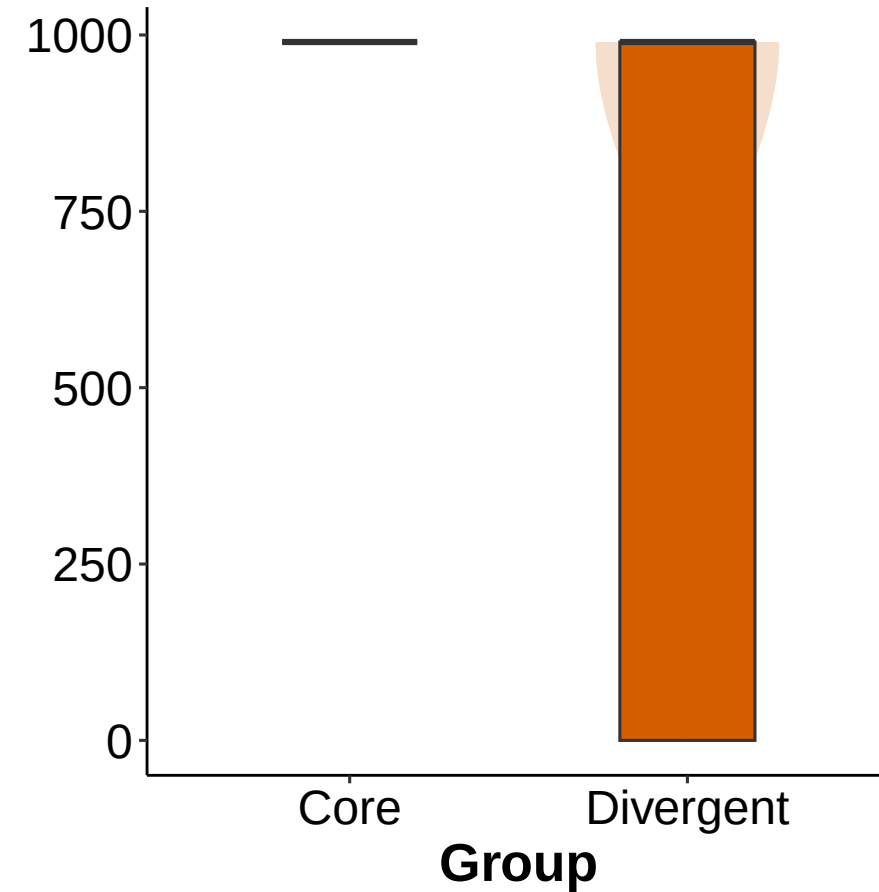
SA_Biomass_reactions_L1w

p-adj: 6.09e-102 | .=-0.41 (Medium) | .%: 0



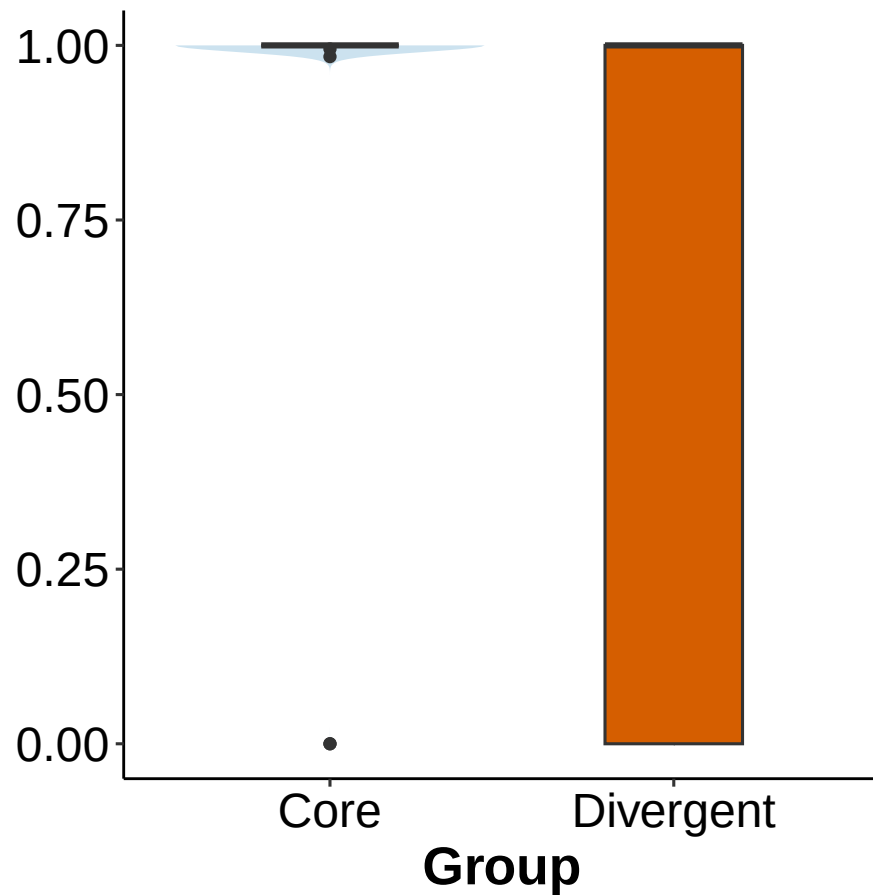
SA_Biomass_reactions_pFBA

p-adj: 6.09e-102 | .=-0.41 (Medium) | .%: 0



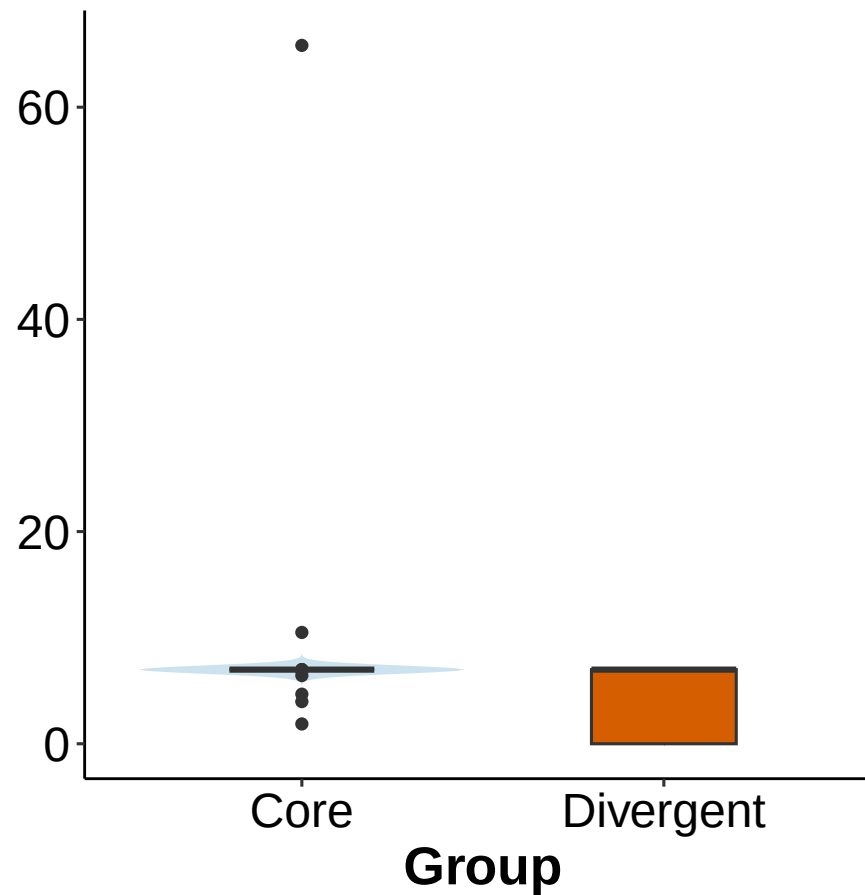
GlnDependence_pFBA

p-adj: 4e-65 | .=-0.35 (Medium) | .%: 0



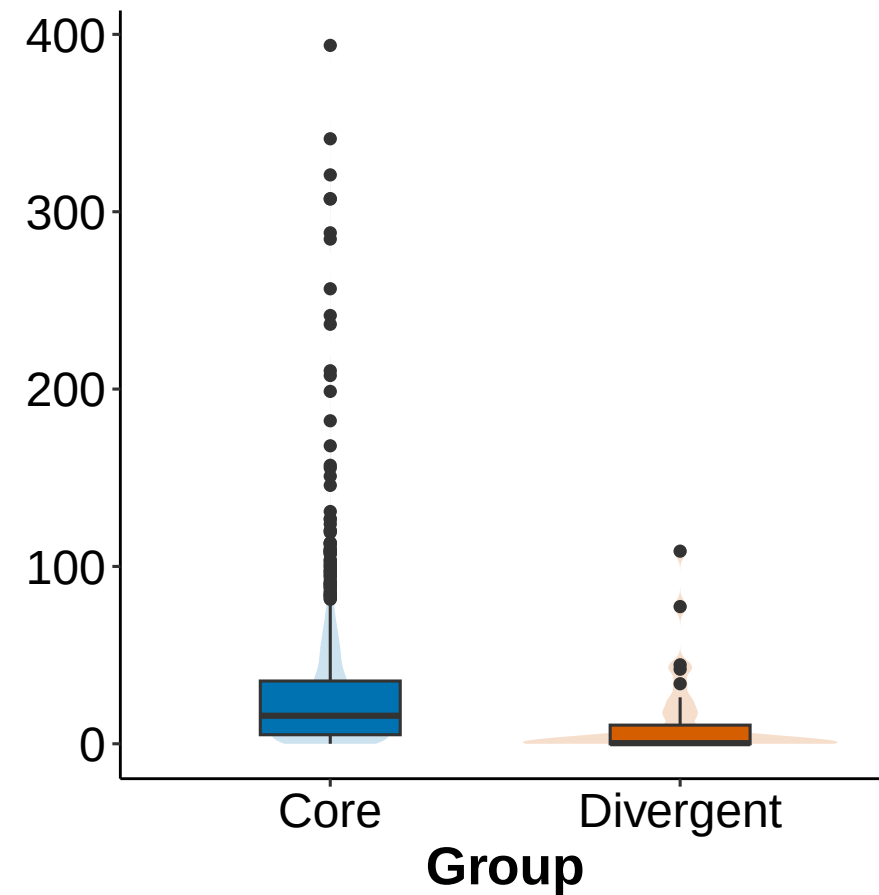
LipidPL_pFBA

p-adj: 1.61e-62 | .=-0.41 (Medium) | .%: 0

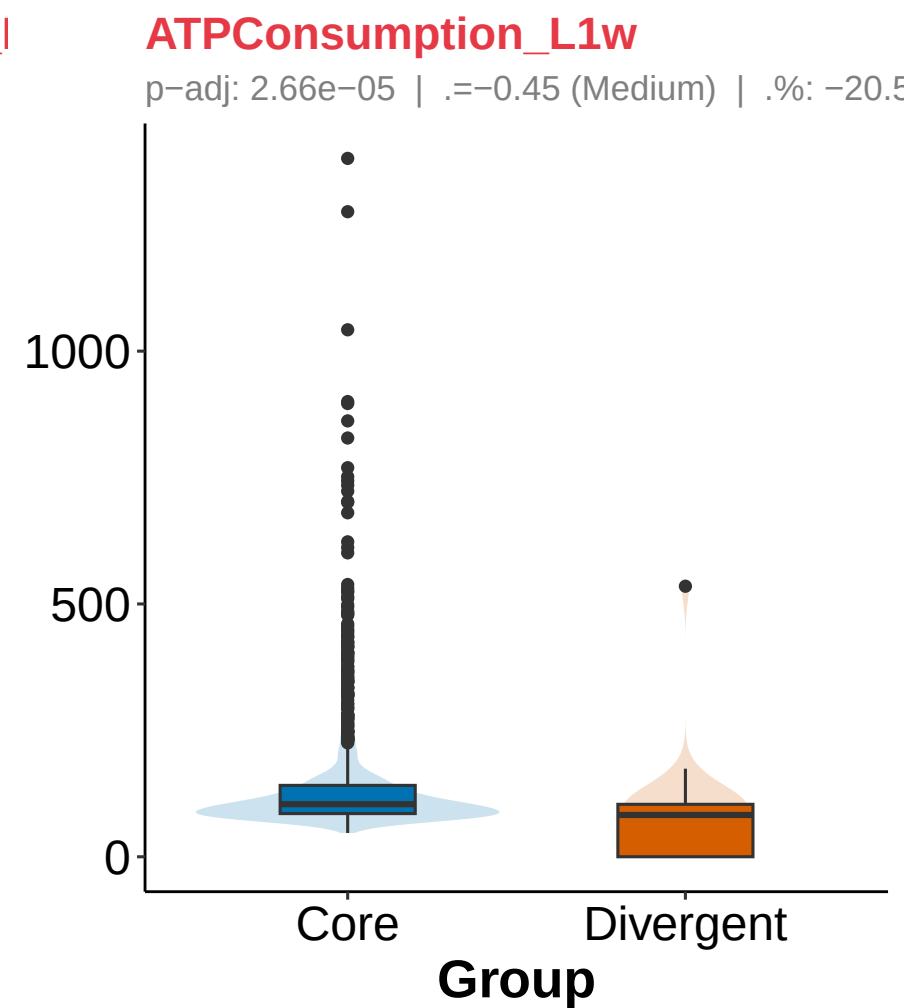
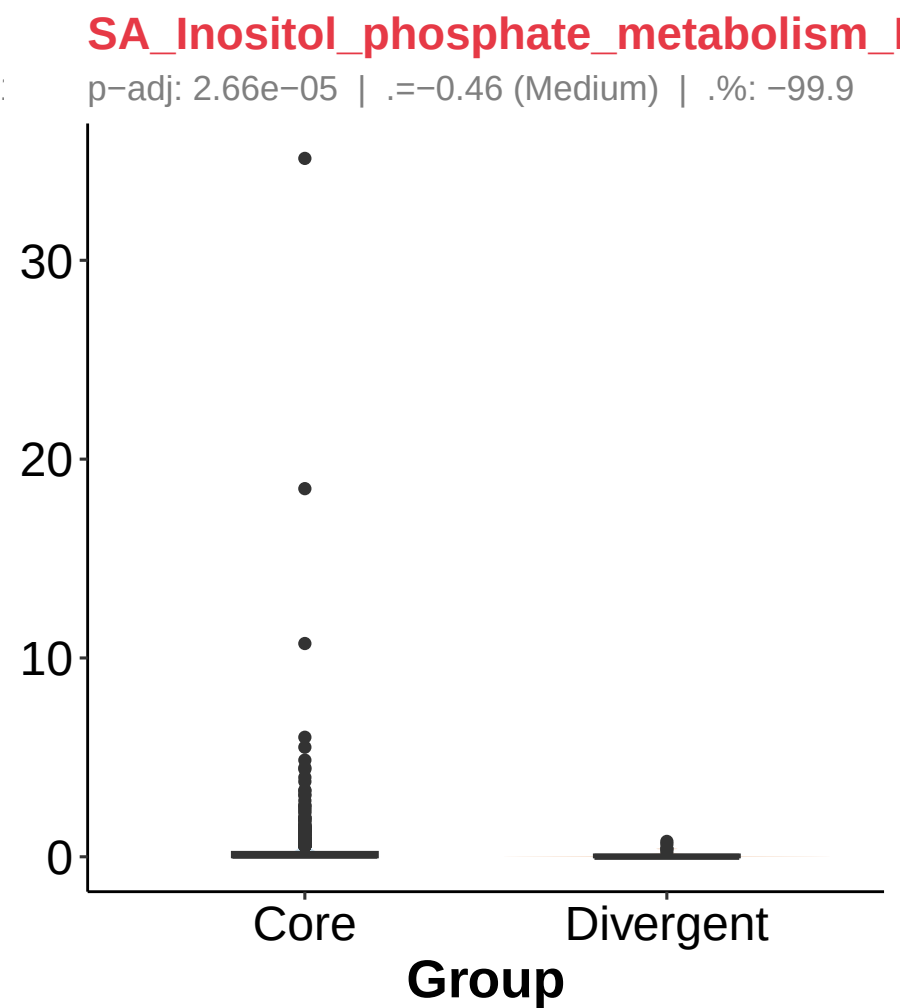
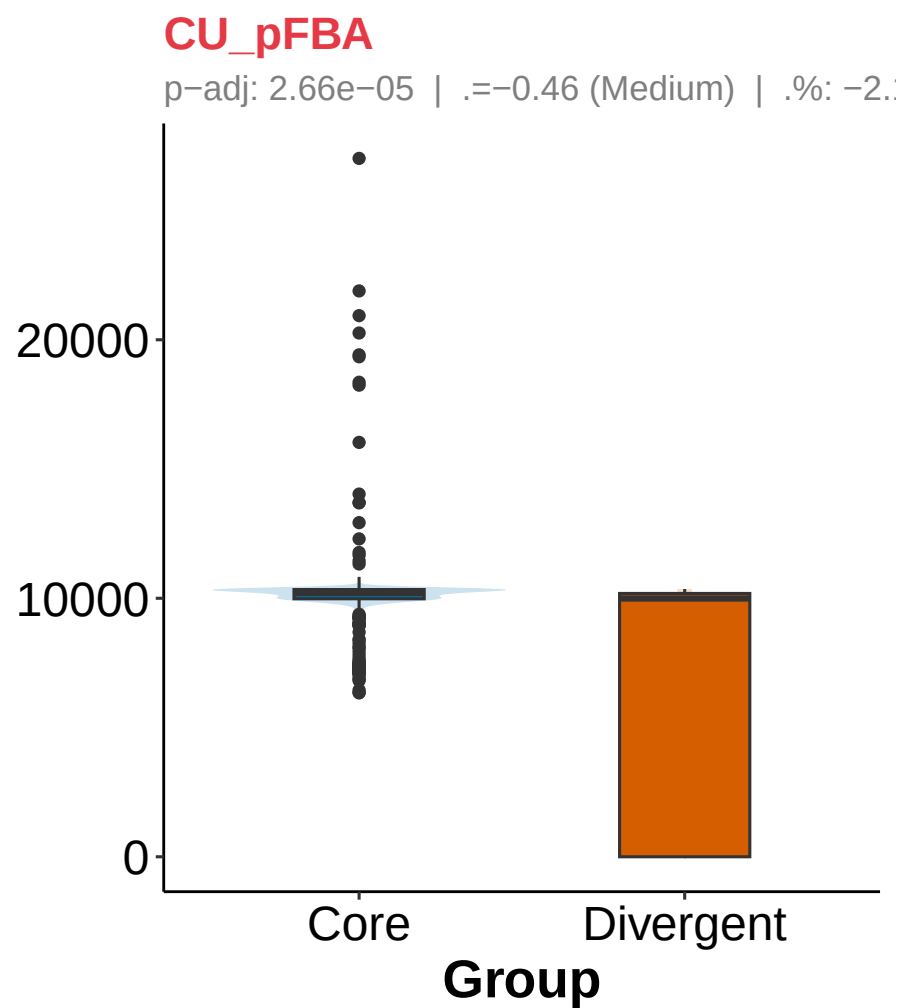
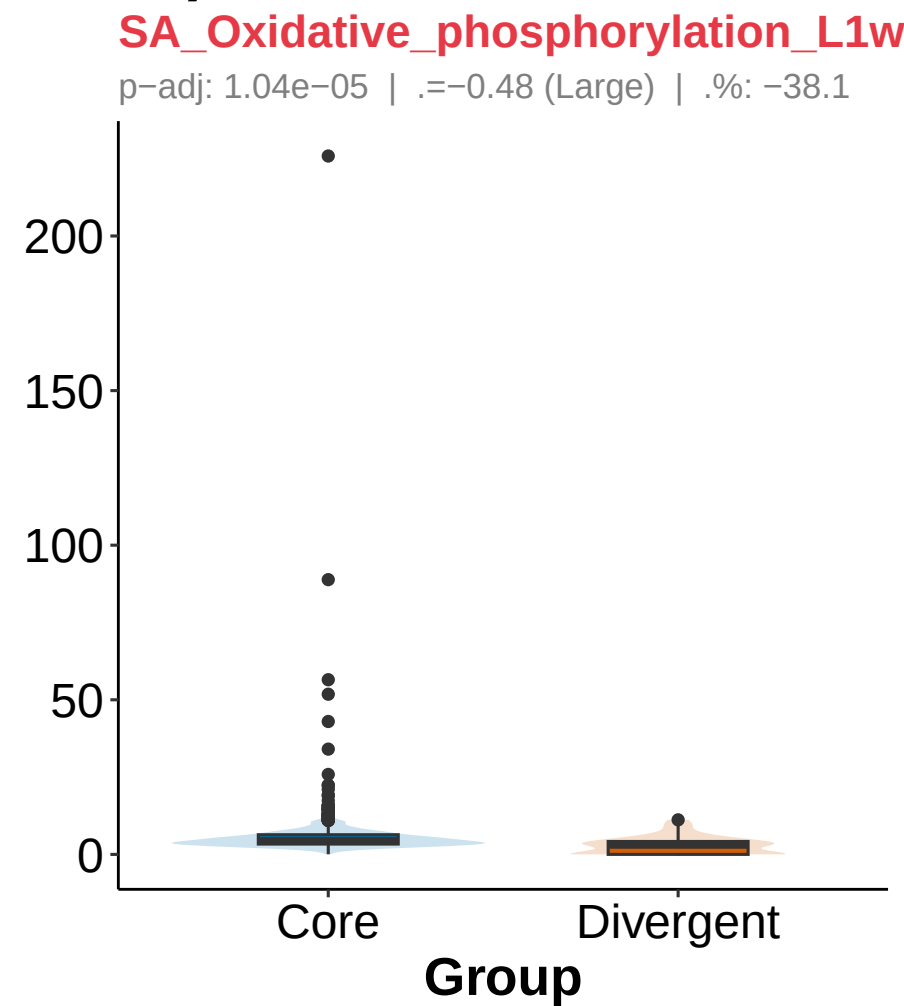
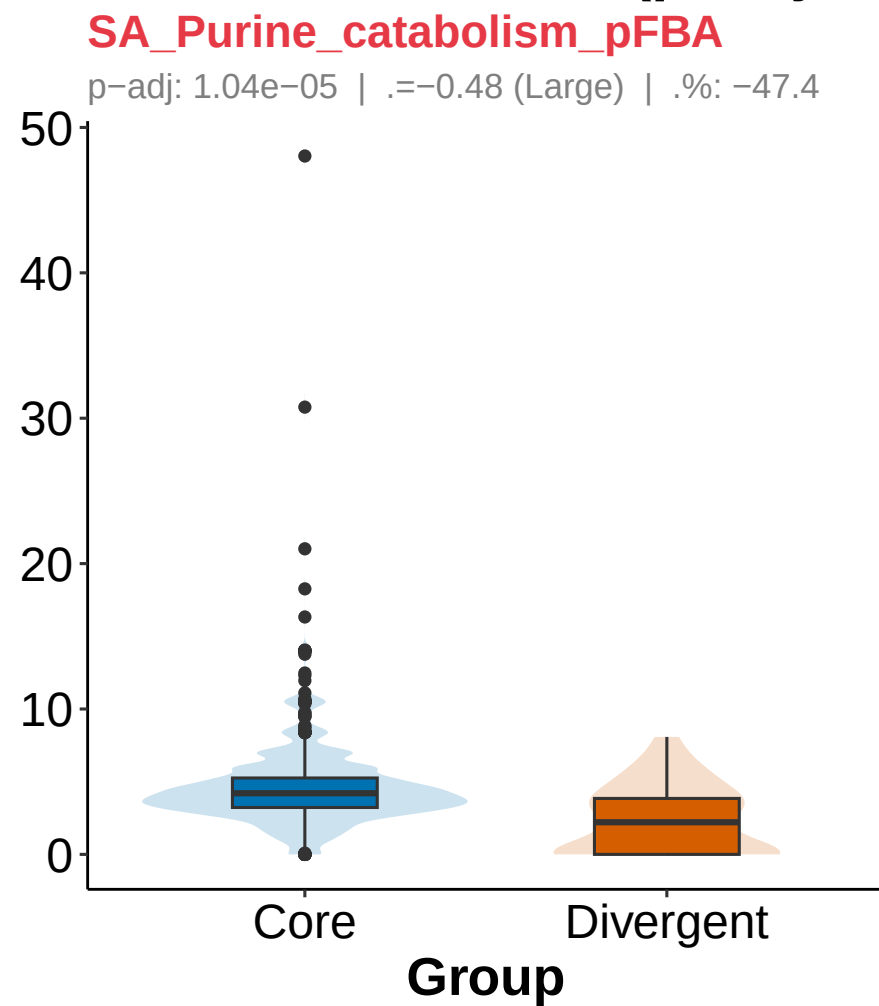
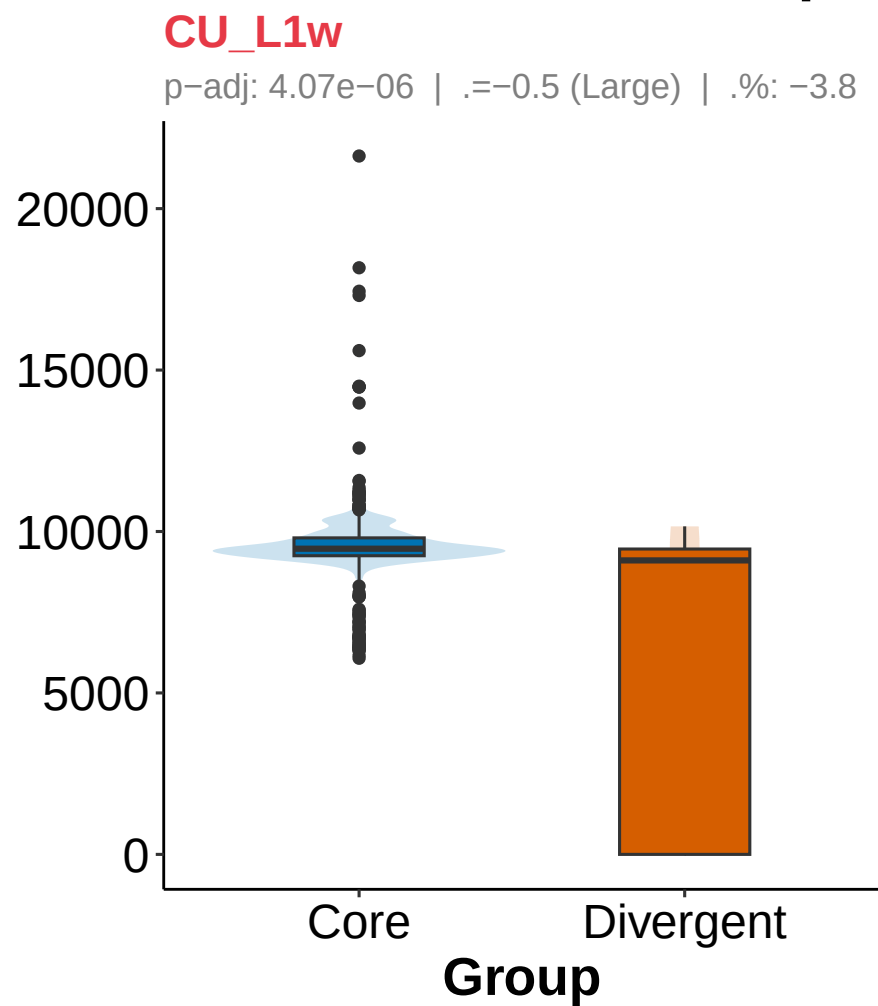


SA_Drug_metabolism_L1w

p-adj: 1.2e-06 | .=-0.52 (Large) | .%: -97.9



. Top 12 Metabolic Variables (p_adj + Cliff's .)



SECTION III

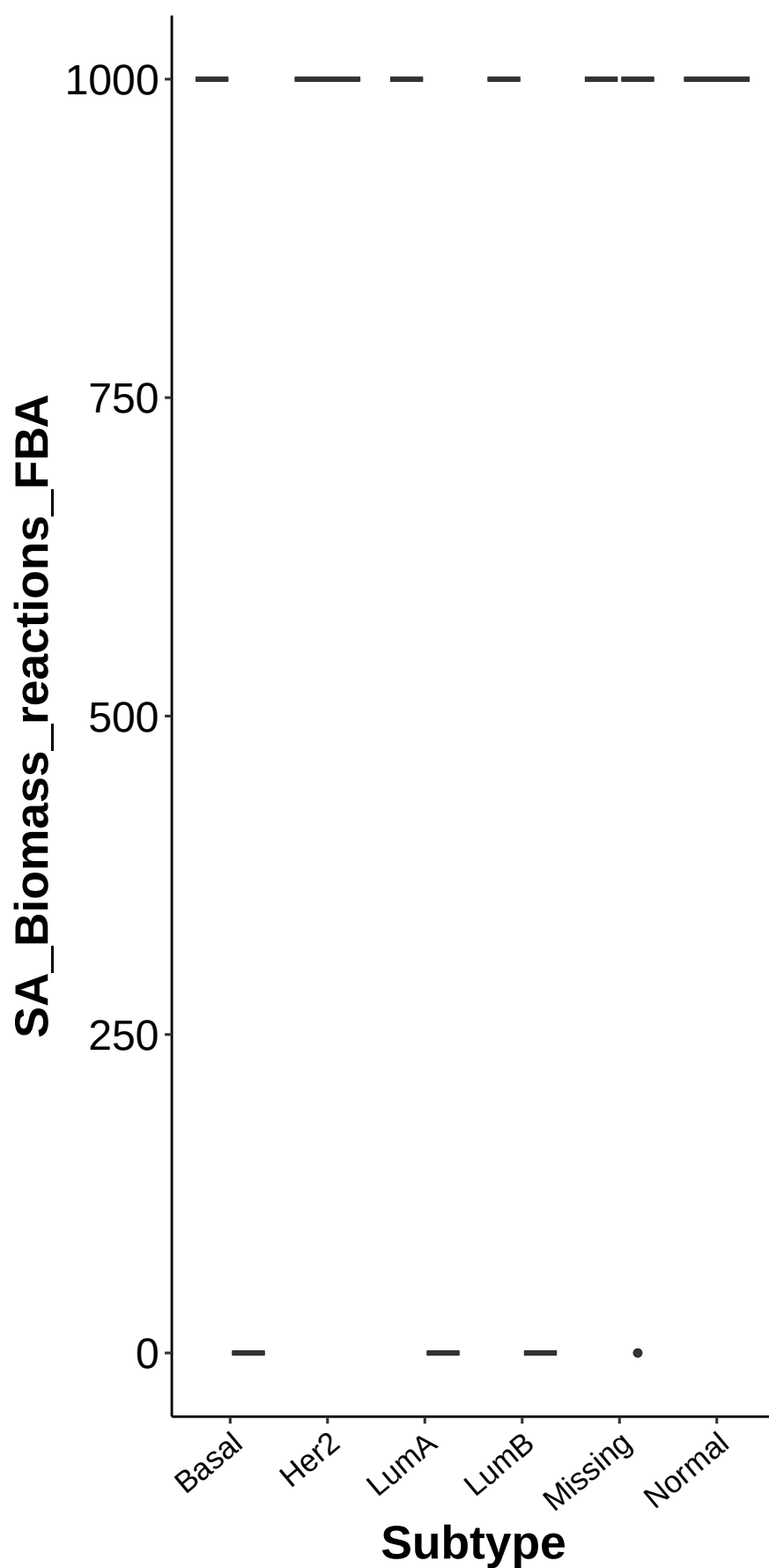
Integrated Analysis: Clinical <—> Metabolism

Is Core vs Divergent a reflection of subtype? Or an independent metabolic reprogramming?

SA_Biomass_reacti

Is the Core/Divergent difference i

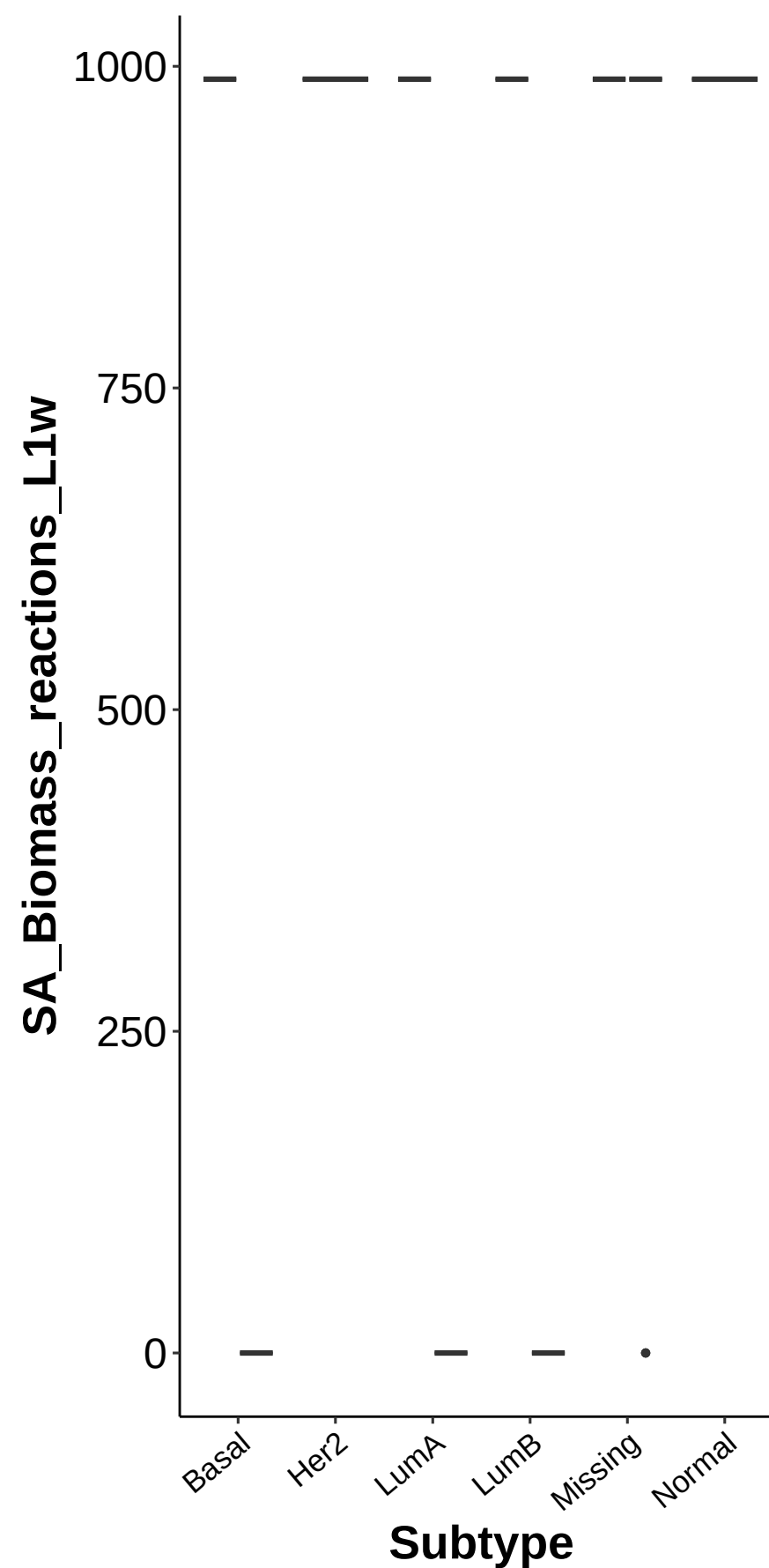
Group Core Divergent



SA_Biomass_reacti

Is the Core/Divergent difference i

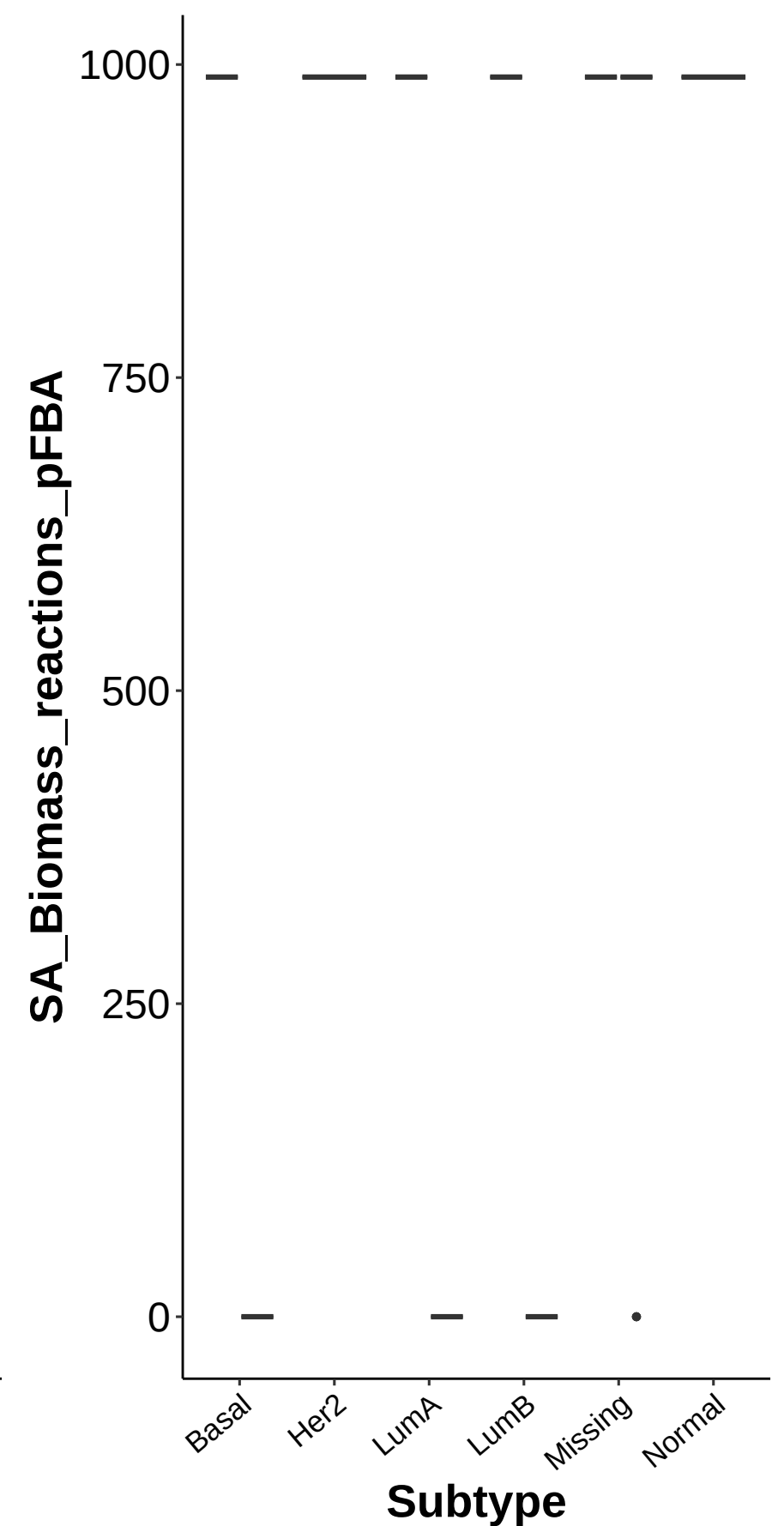
Group Core Divergent



SA_Biomass_reacti

Is the Core/Divergent difference i

Group Core Divergent

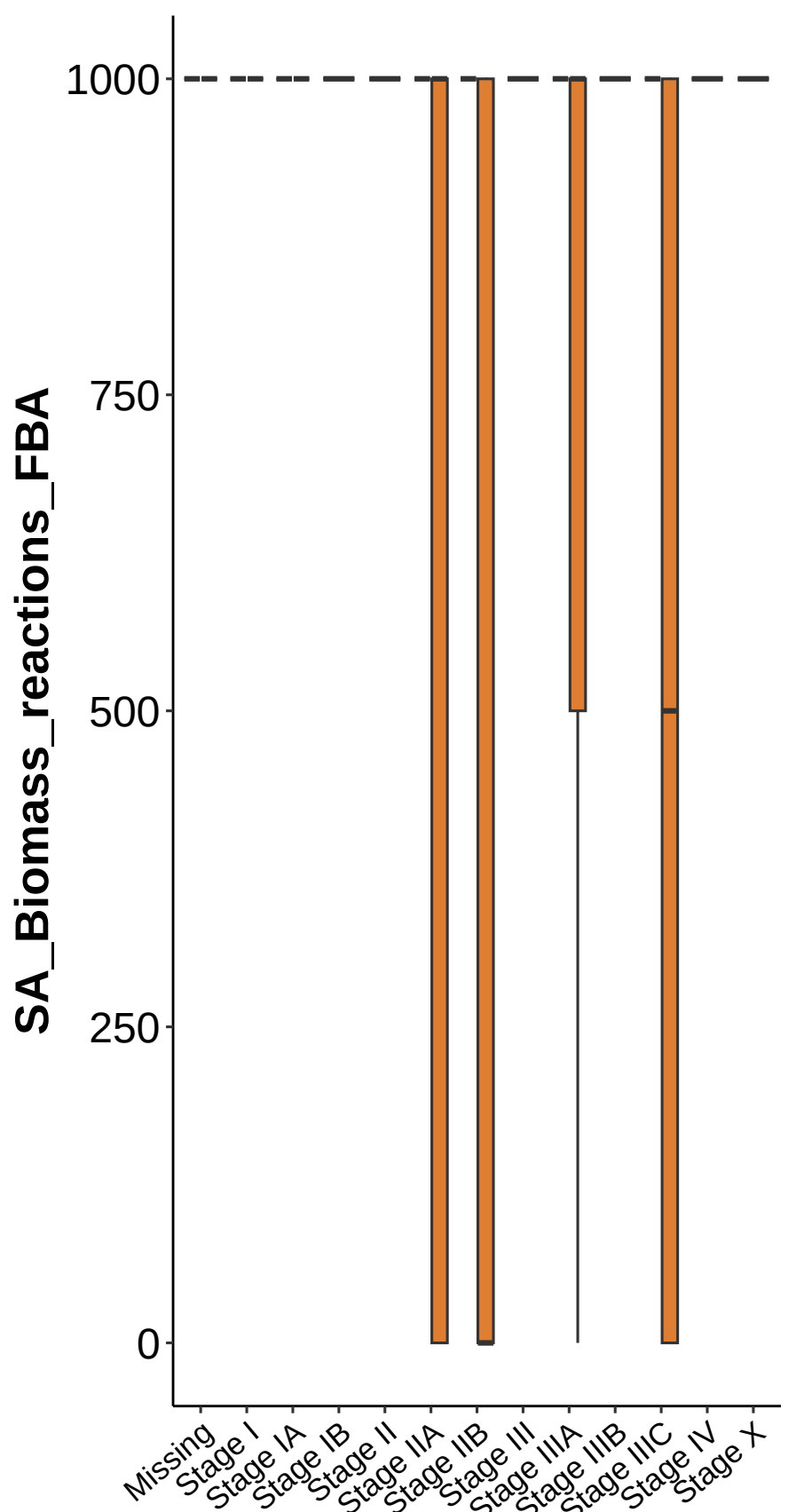


Integration: Top metabolic × ajcc_pathologic_stage.diagnoses

SA_Biomass_reacti

Is the Core/Divergent difference i

Group Core Divergent

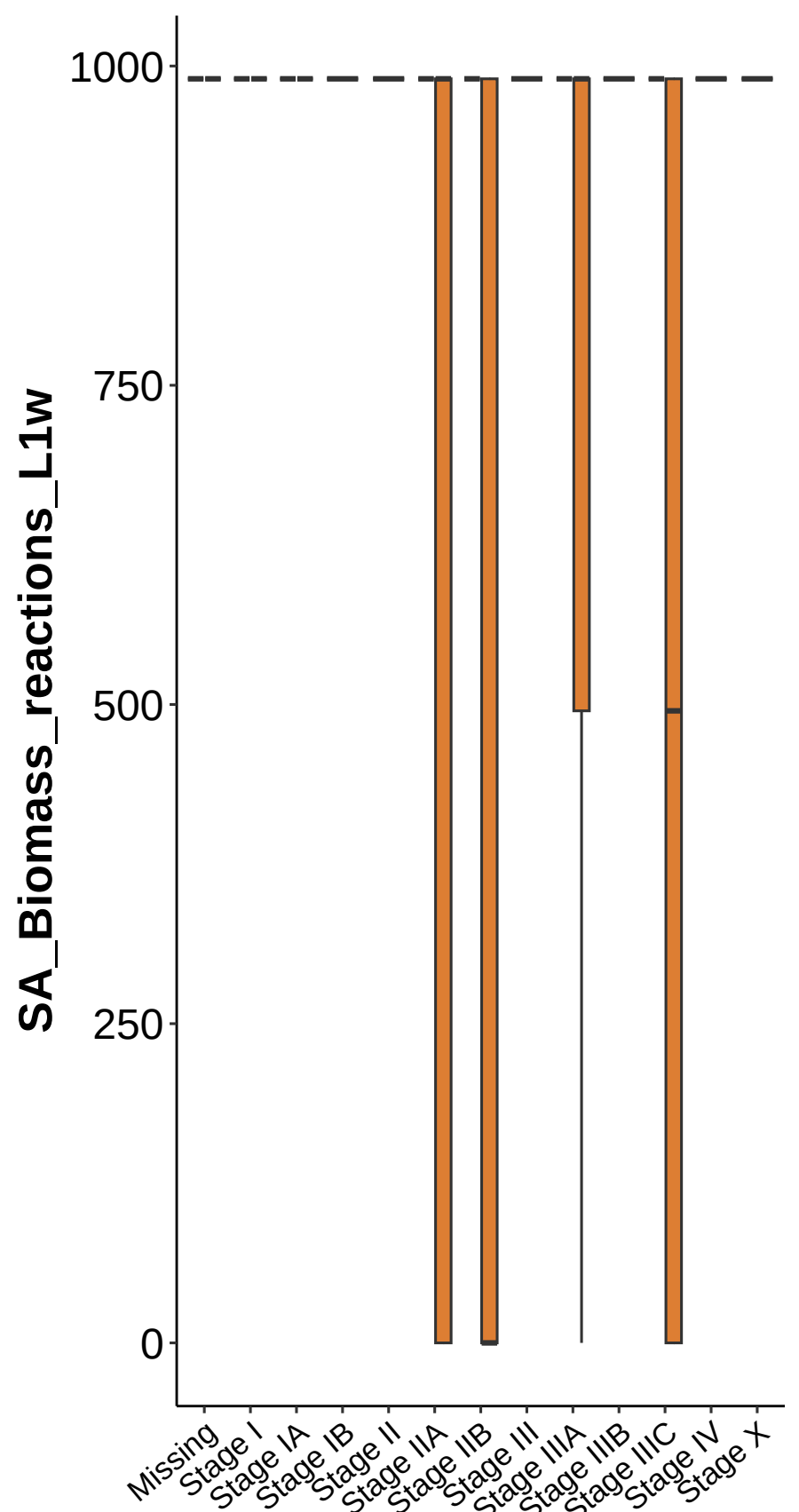


ajcc_pathologic_stage.diagnoses

SA_Biomass_reacti

Is the Core/Divergent difference i

Group Core Divergent

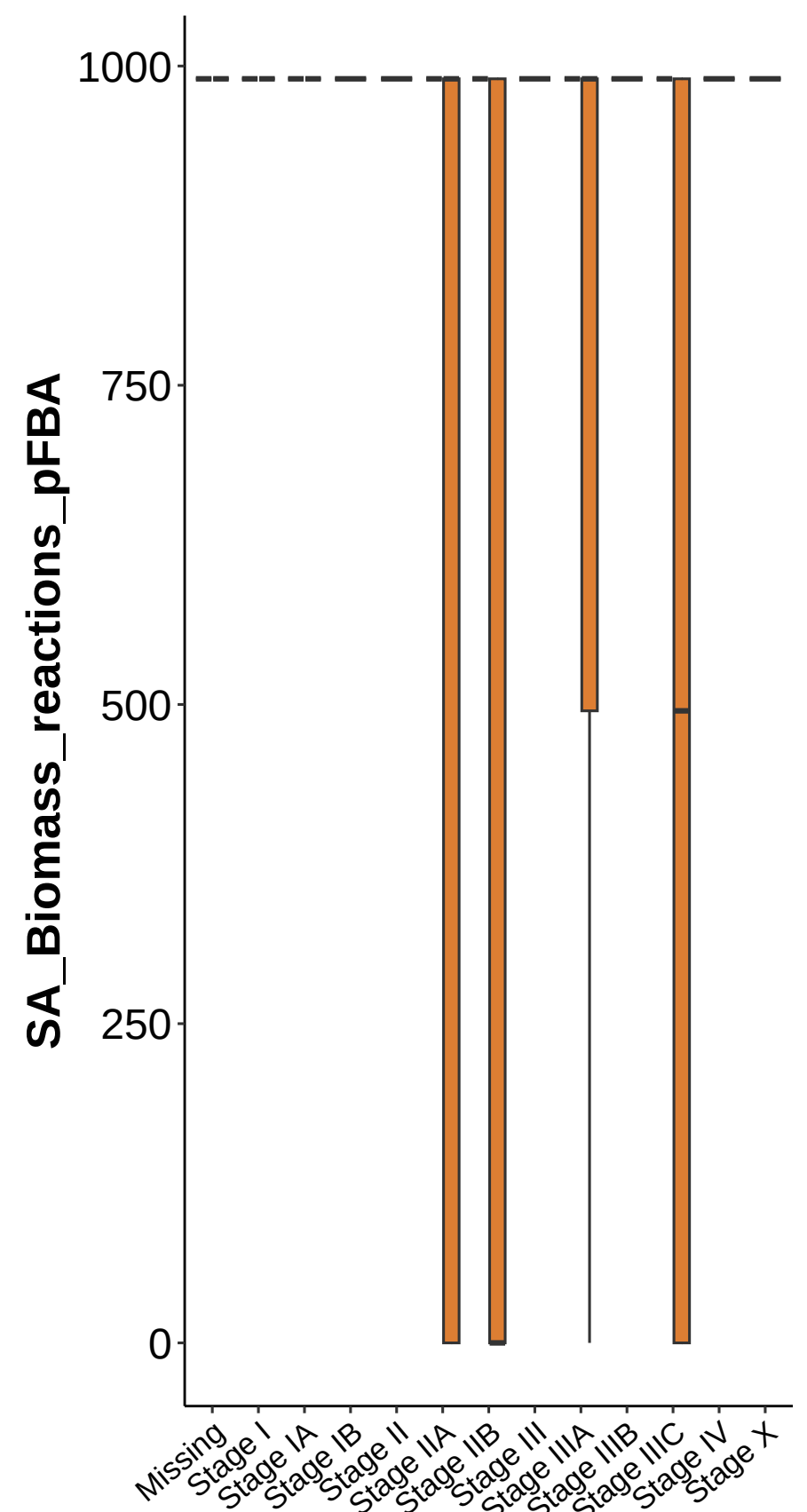


ajcc_pathologic_stage.diagnoses

SA_Biomass_reacti

Is the Core/Divergent difference i

Group Core Divergent



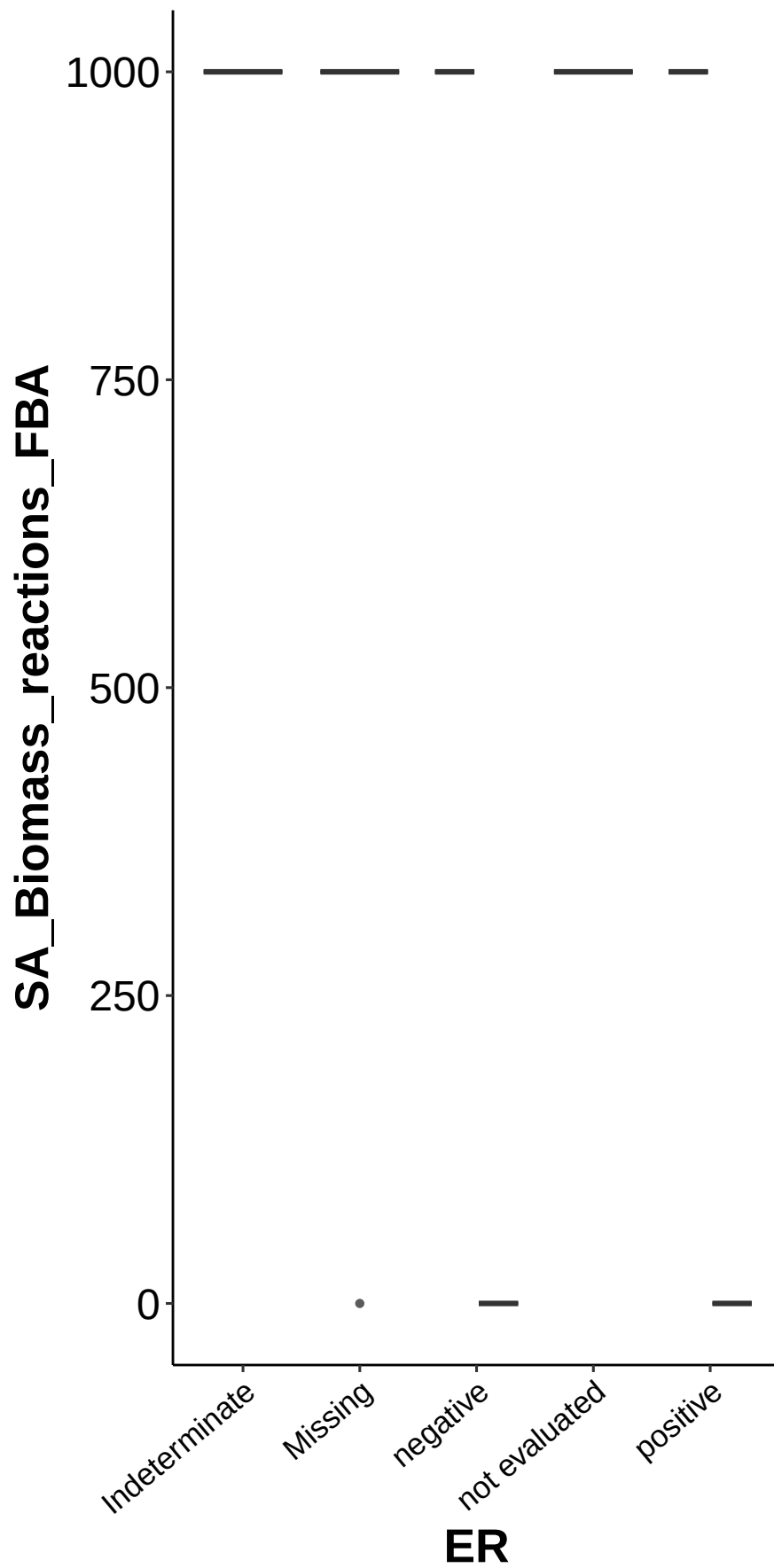
ajcc_pathologic_stage.diagnoses

Integration: Top metabolic × ER

SA_Biomass_reacti

Is the Core/Divergent difference i

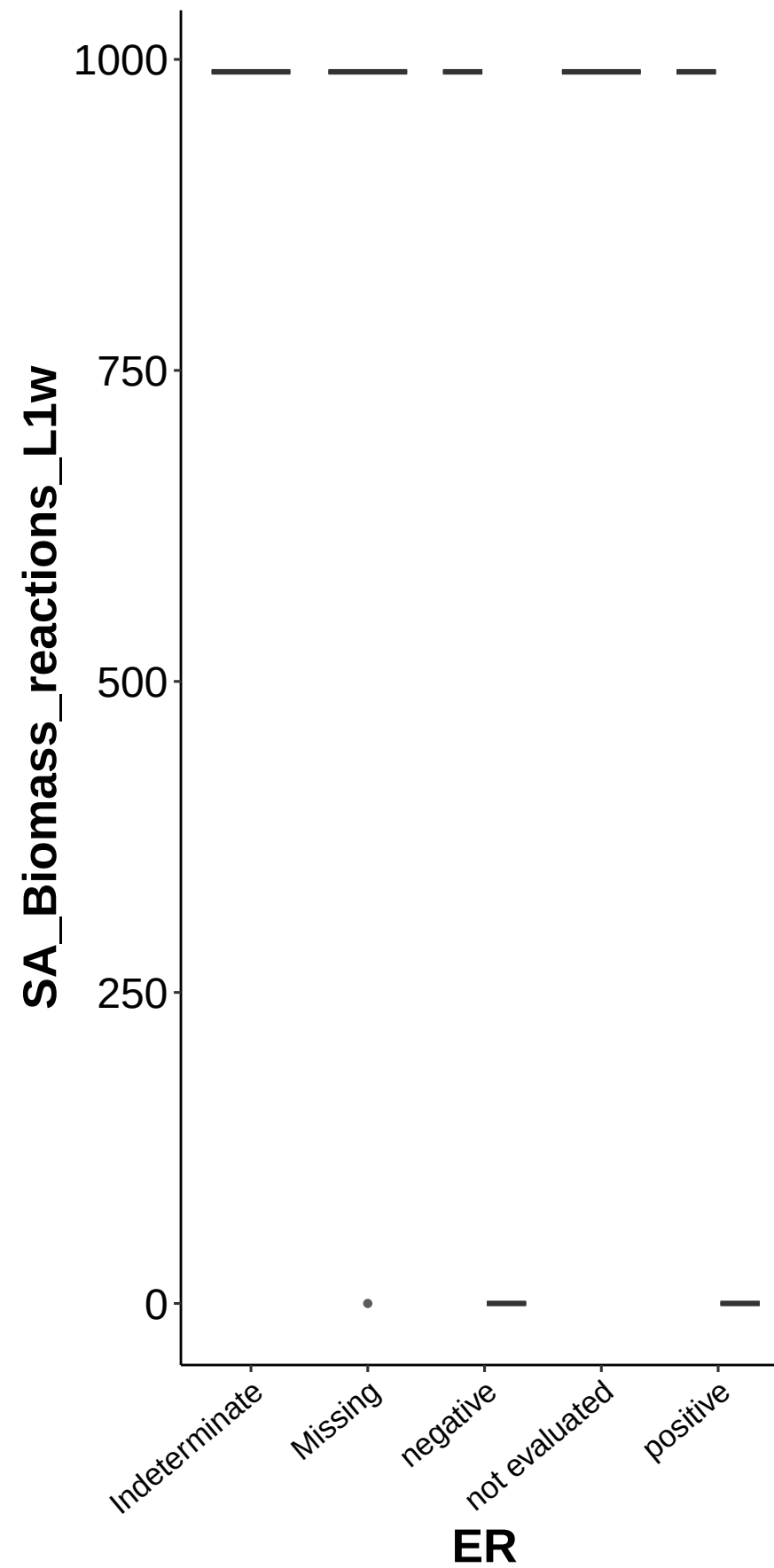
Group  Core  Divergent



SA_Biomass_reacti

Is the Core/Divergent difference i

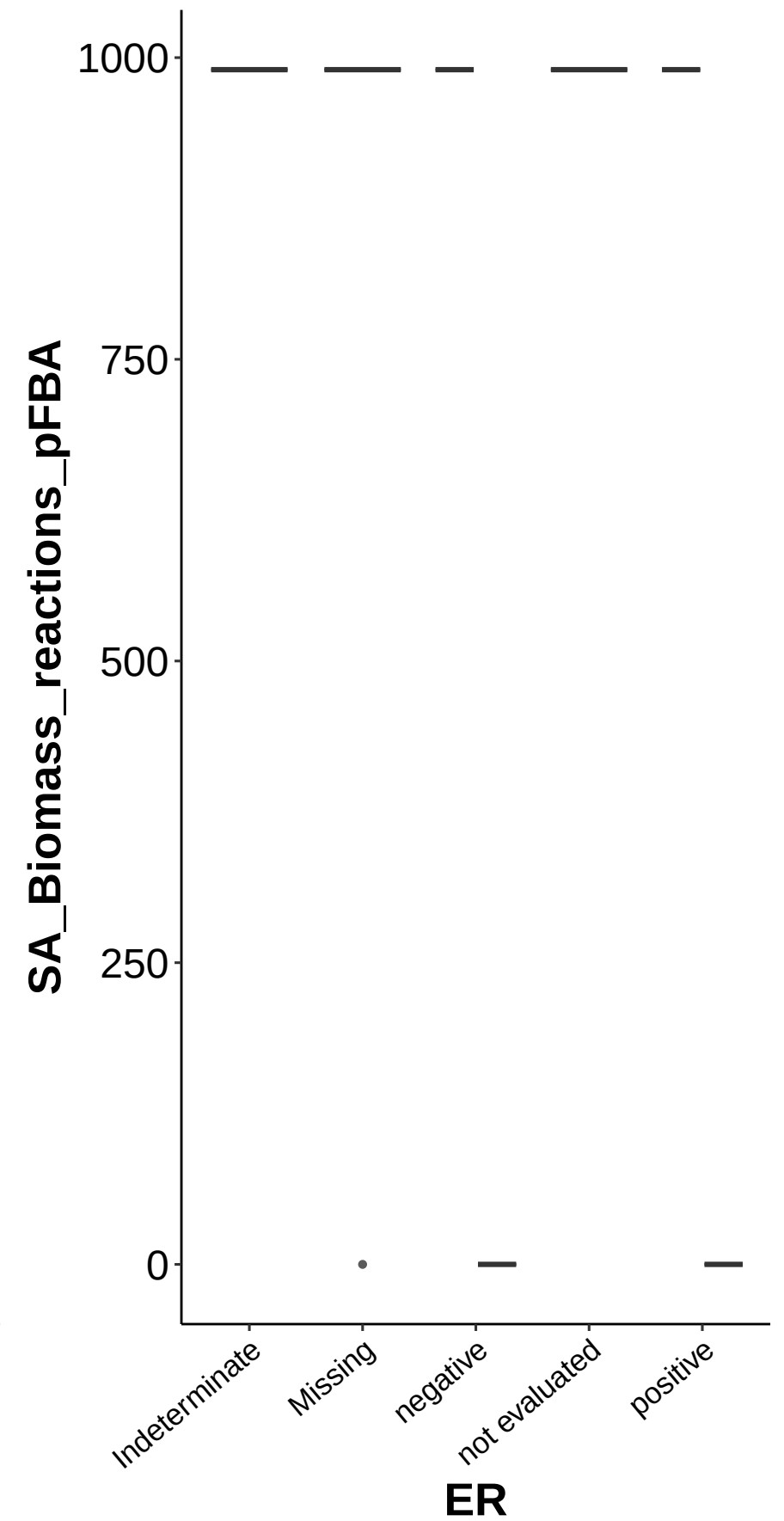
Group  Core  Divergent

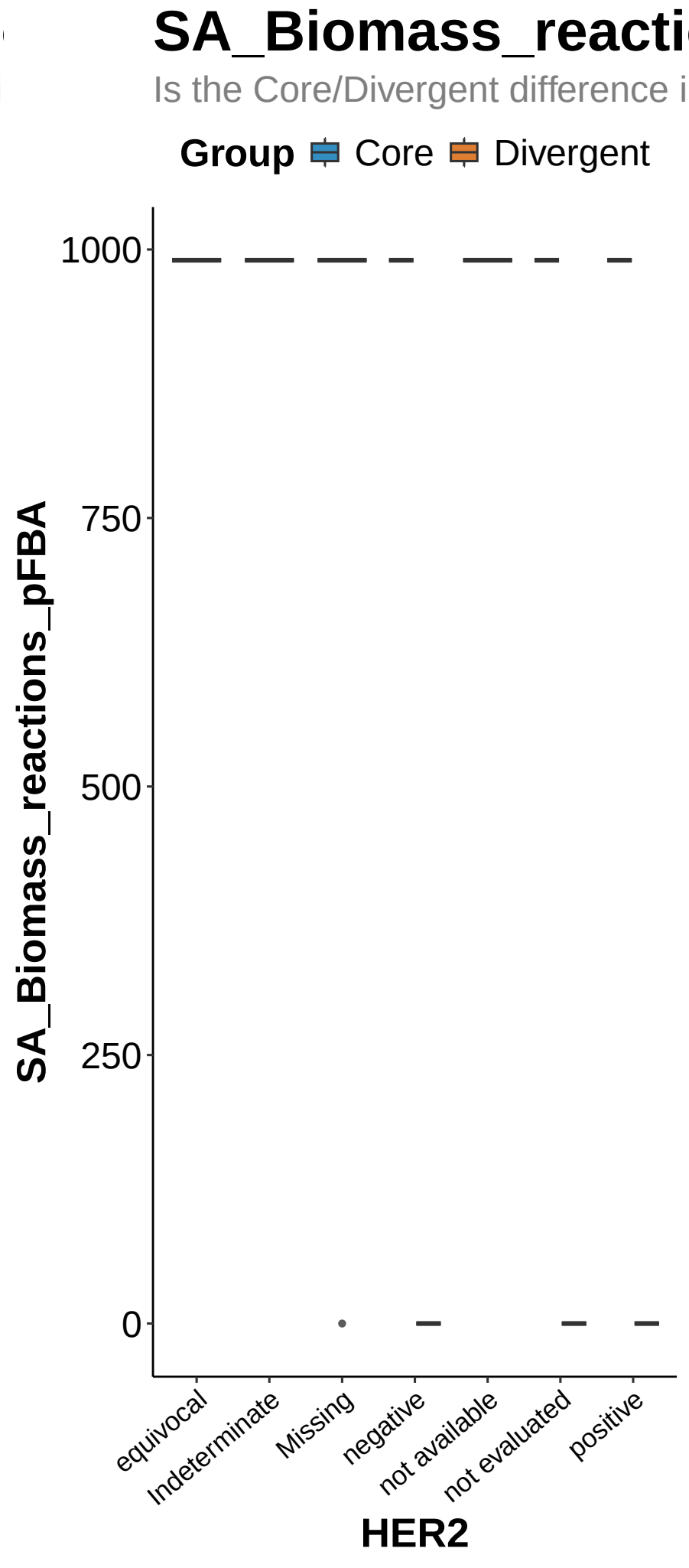
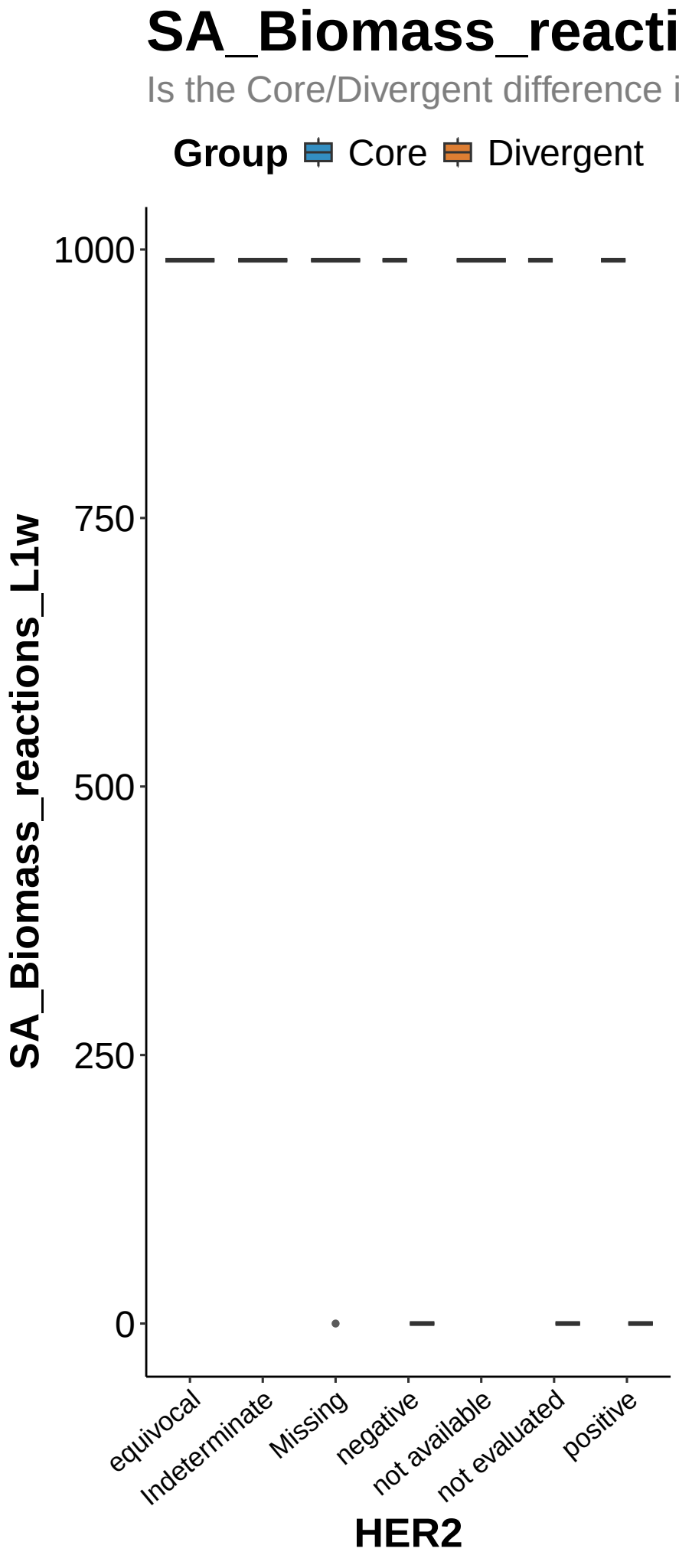
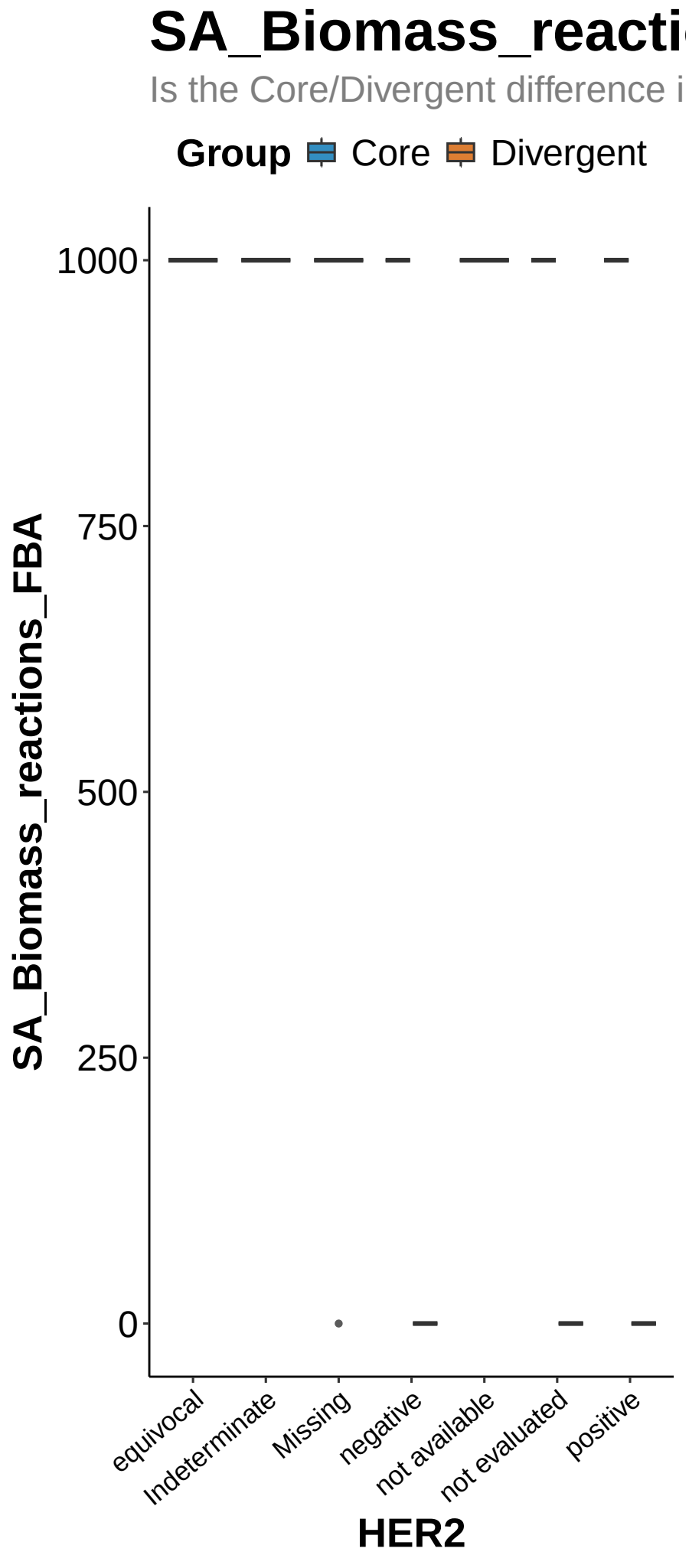


SA_Biomass_reacti

Is the Core/Divergent difference i

Group  Core  Divergent



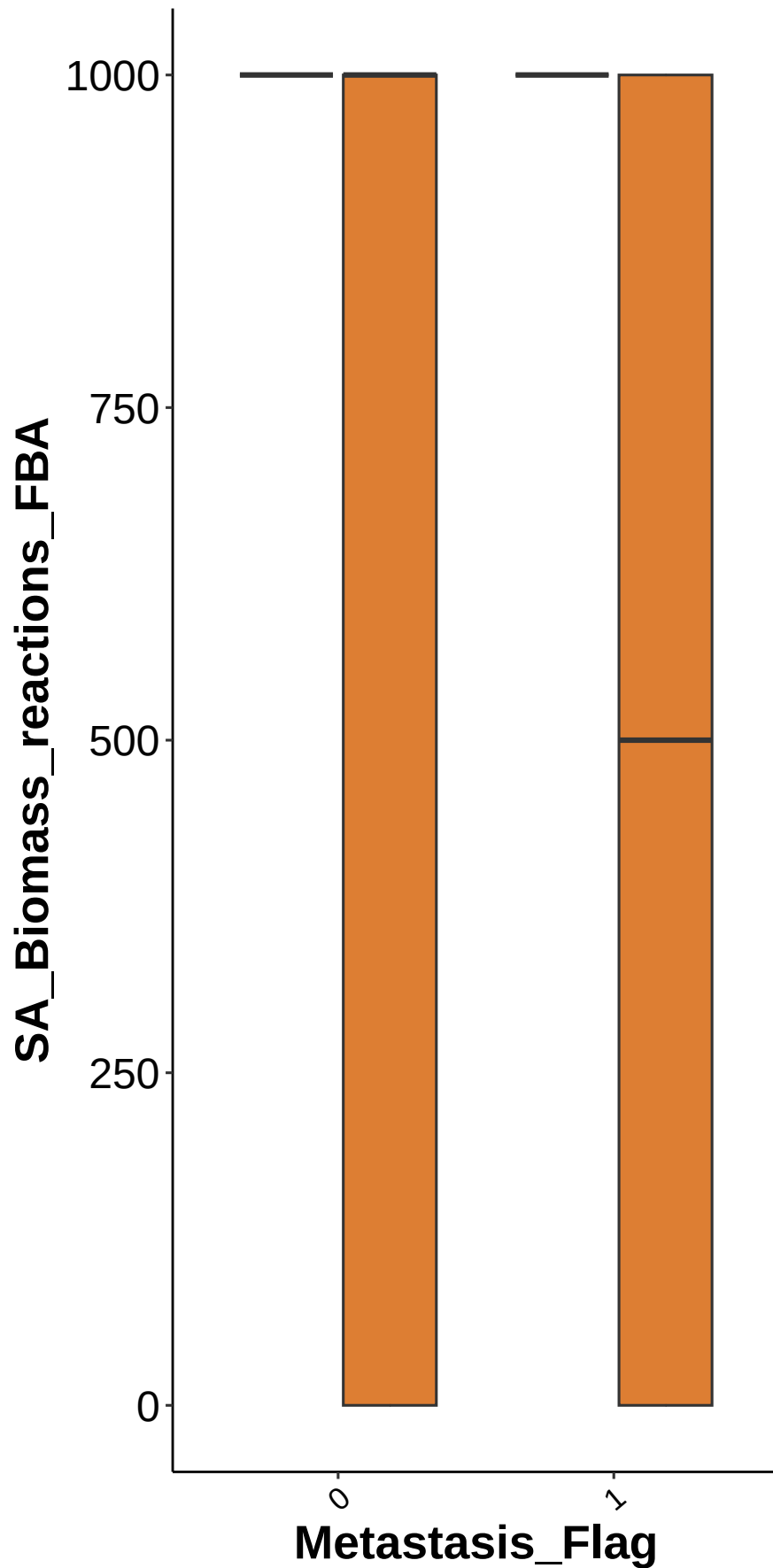


Integration: Top metabolic × Metastasis_Flag

SA_Biomass_reacti

Is the Core/Divergent difference i

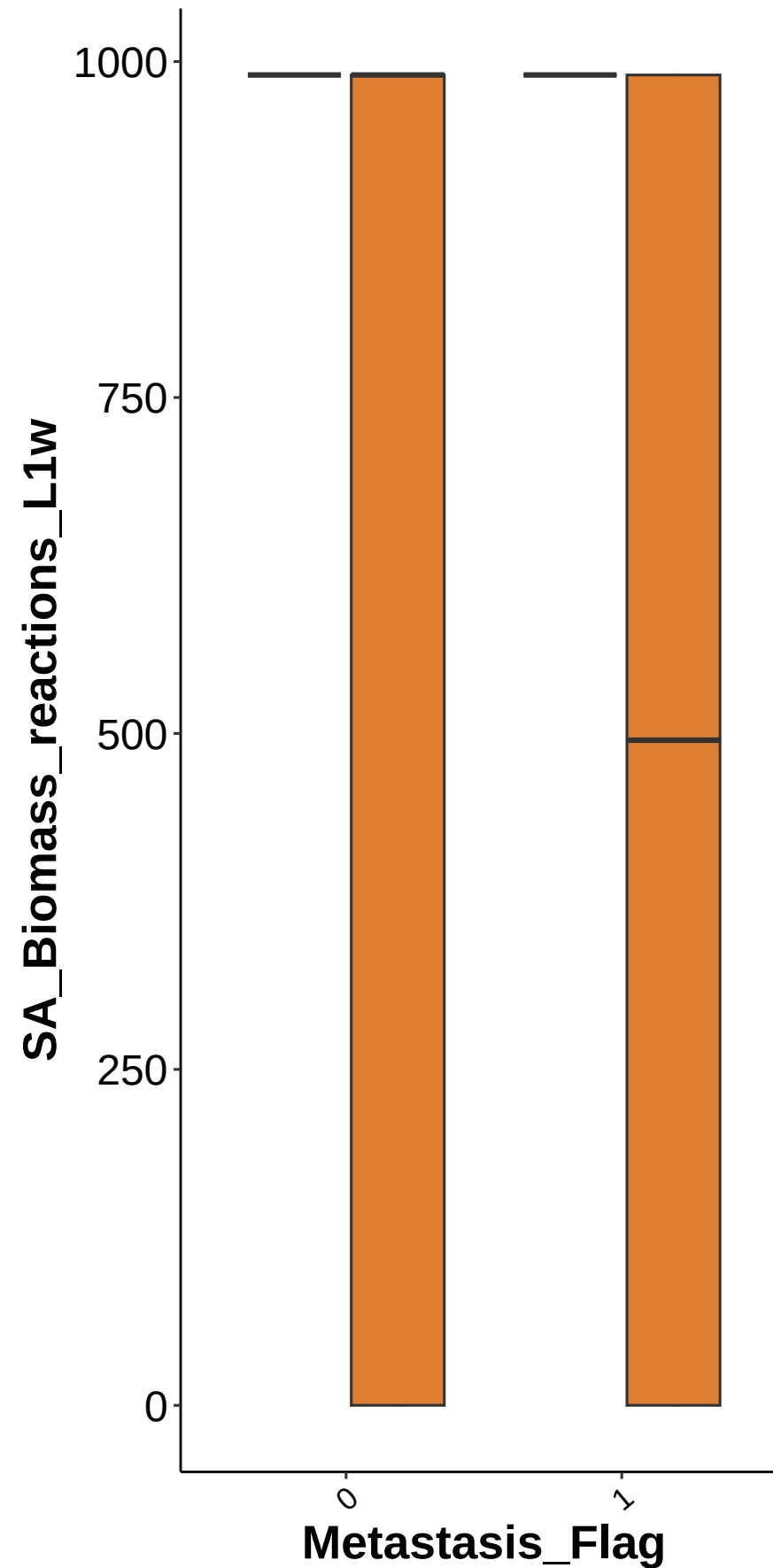
Group Core Divergent



SA_Biomass_reacti

Is the Core/Divergent difference i

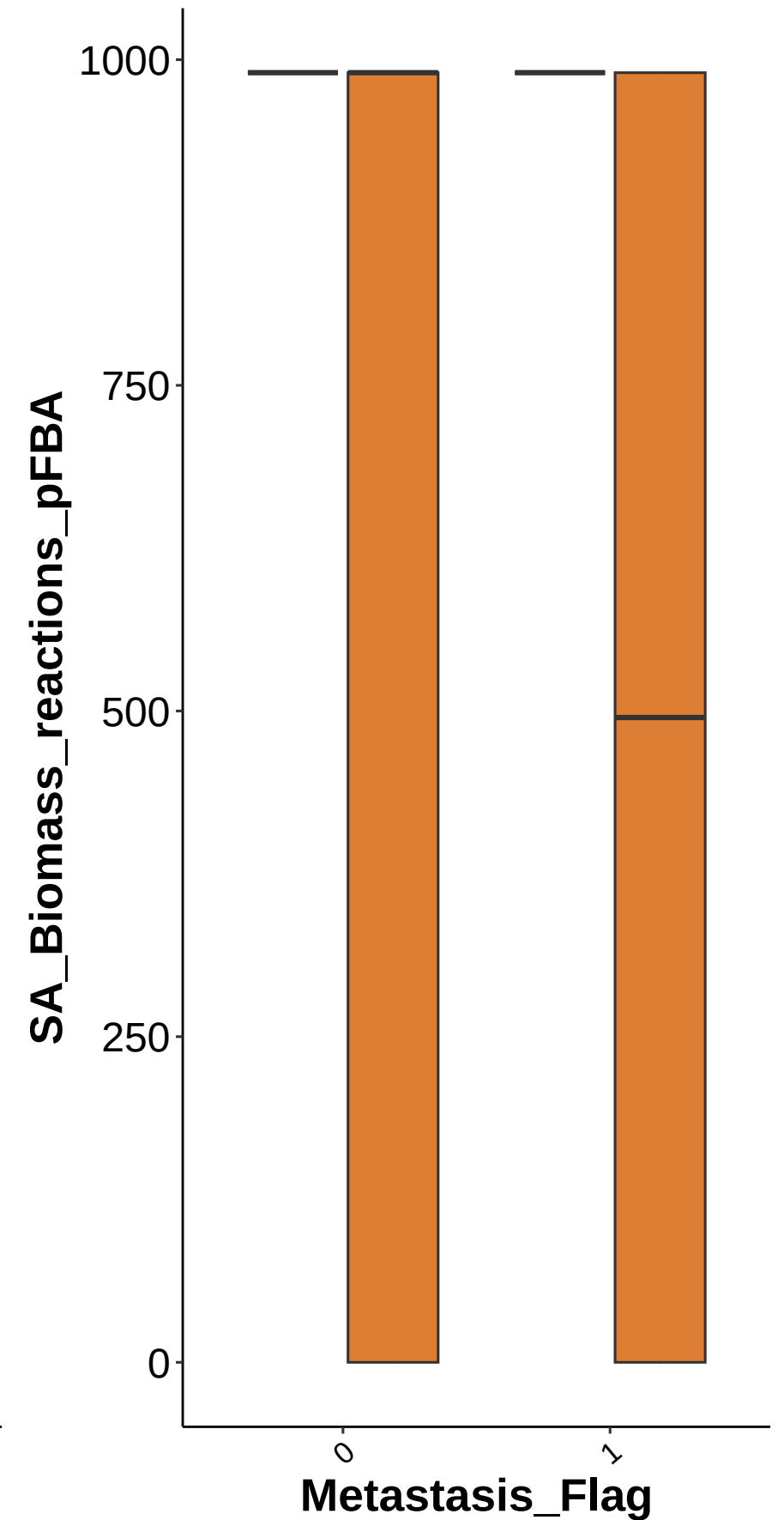
Group Core Divergent



SA_Biomass_reacti

Is the Core/Divergent difference i

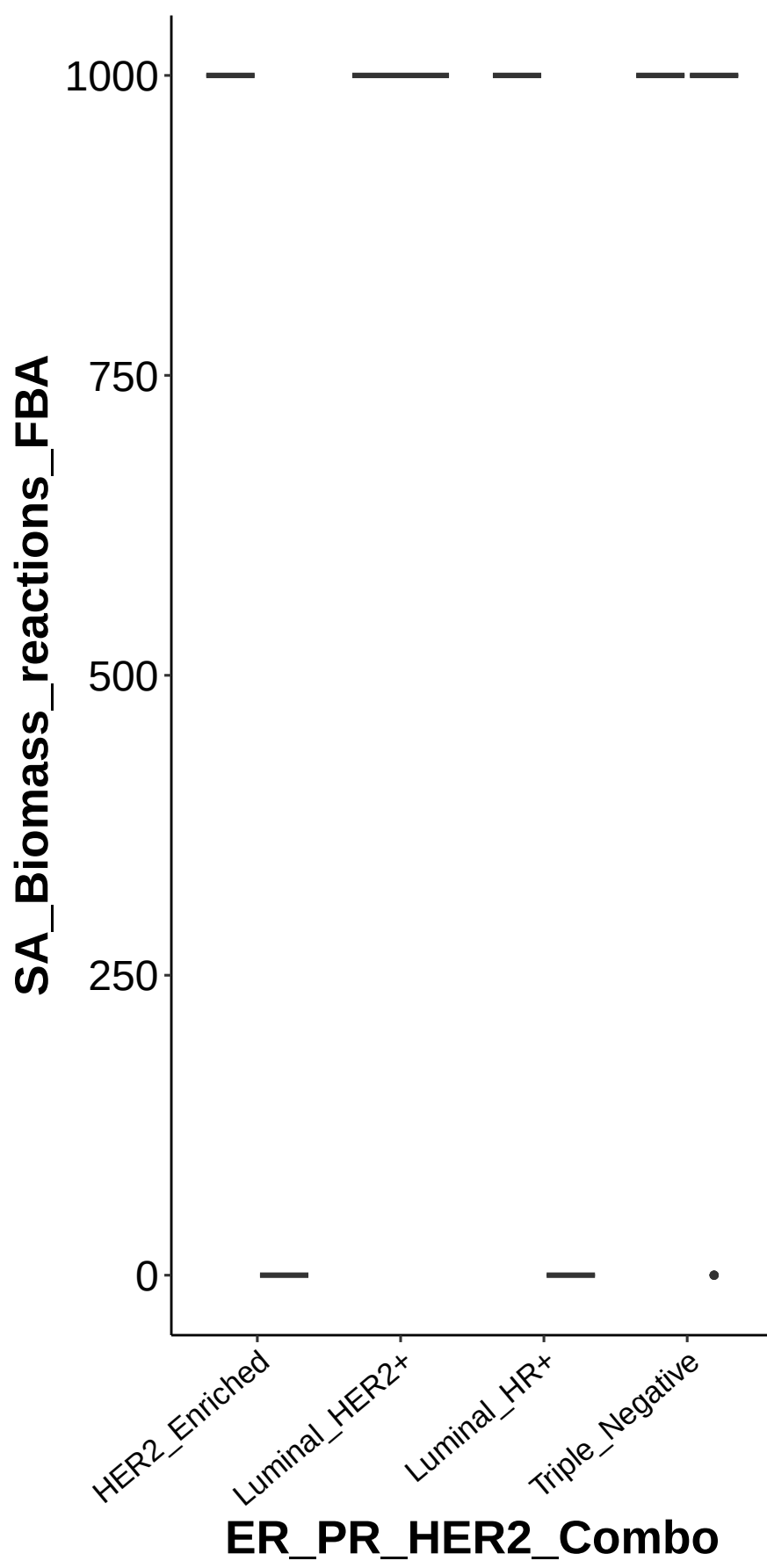
Group Core Divergent



SA_Biomass_reacti

Is the Core/Divergent difference i

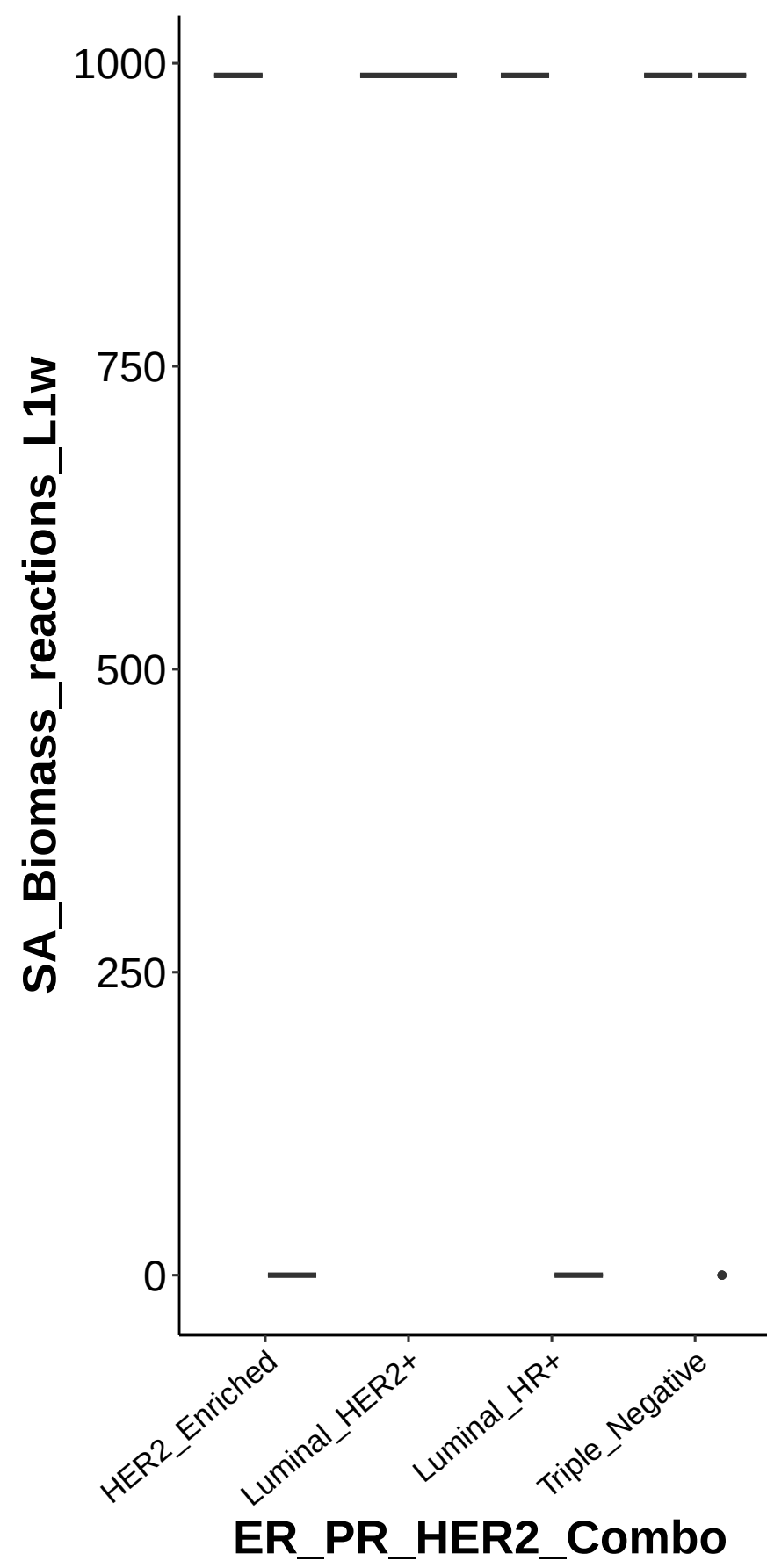
Group Core Divergent



SA_Biomass_reacti

Is the Core/Divergent difference i

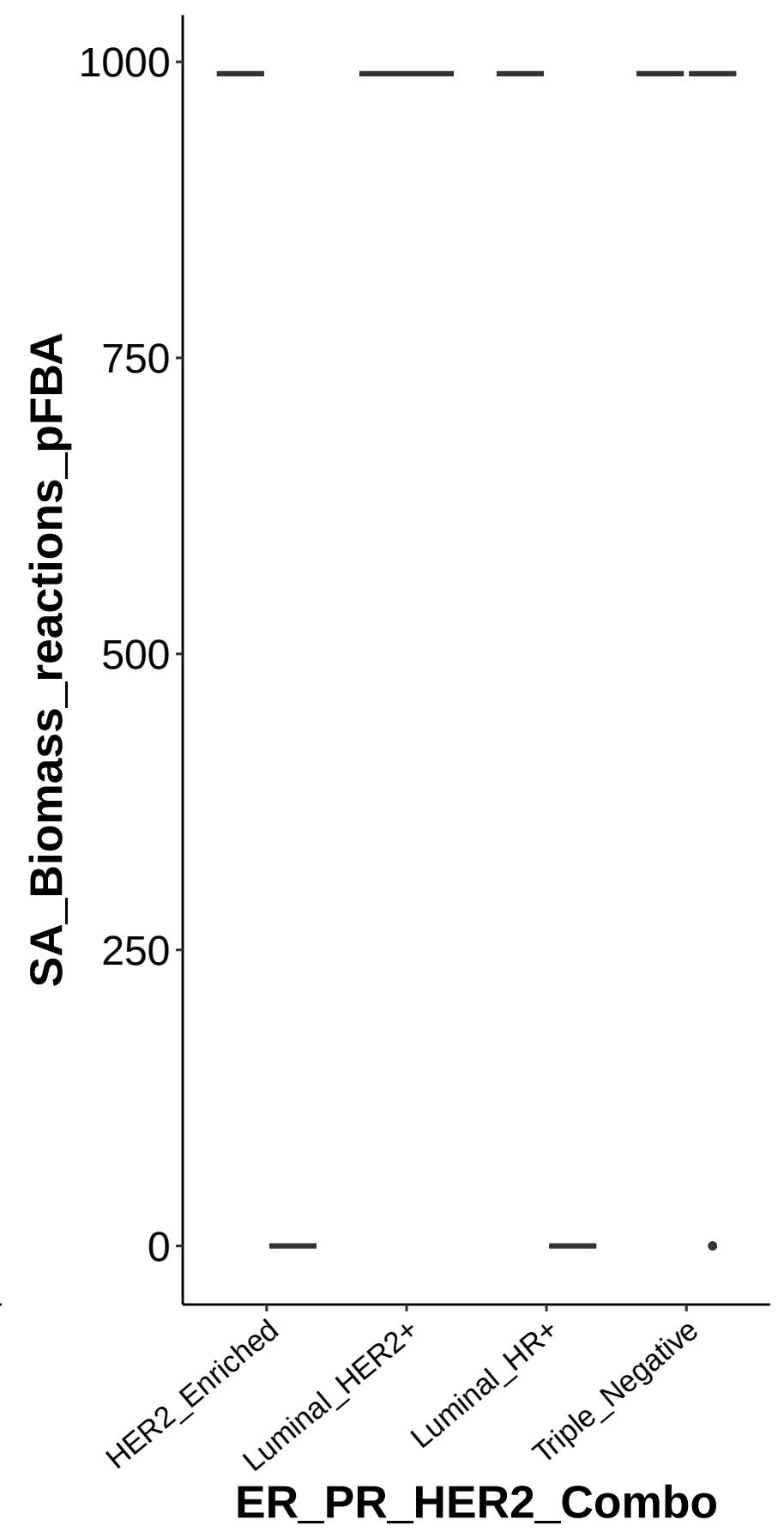
Group Core Divergent



SA_Biomass_reacti

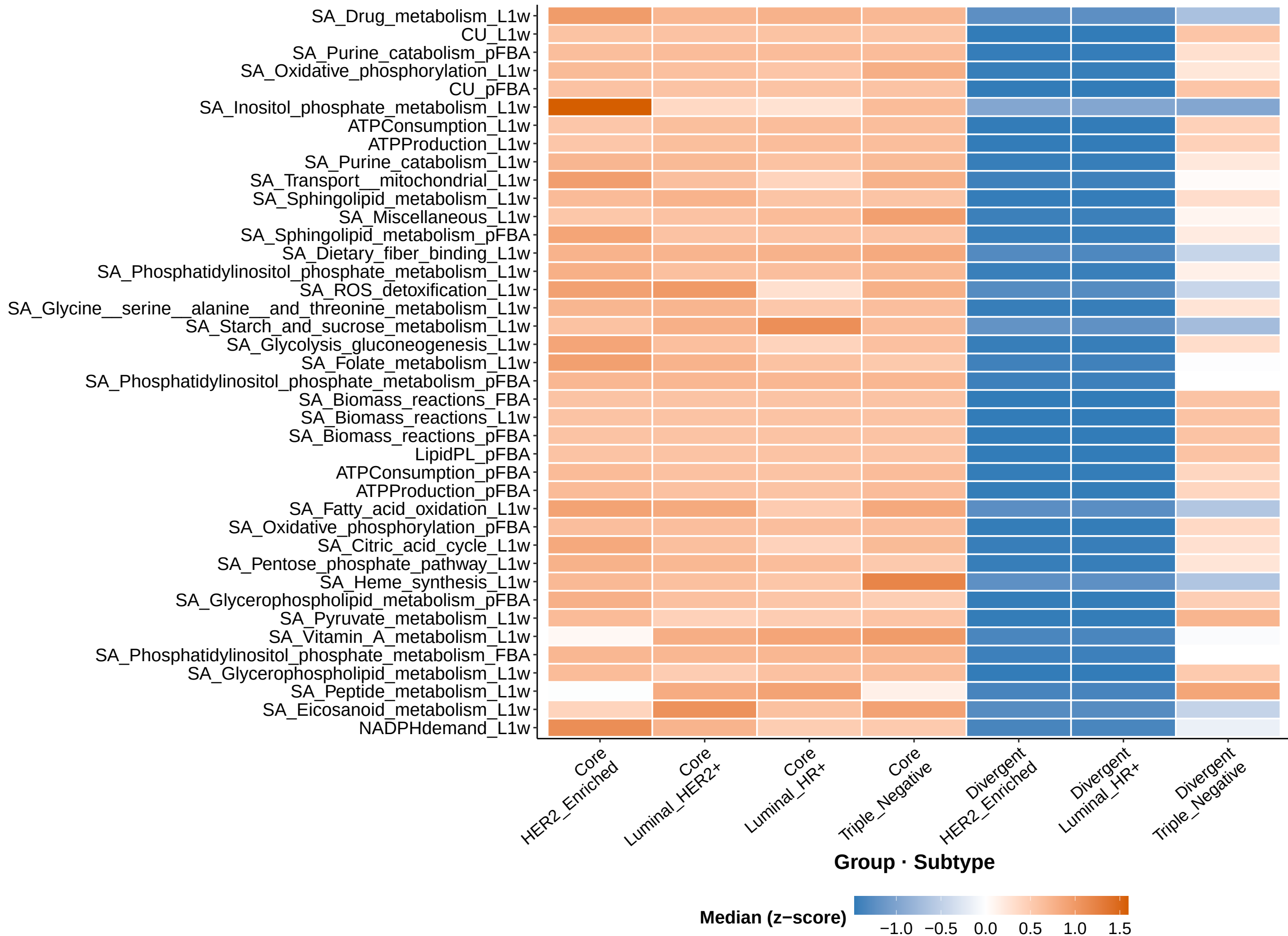
Is the Core/Divergent difference i

Group Core Divergent



Metabolic Signature by Molecular Subtype (ER_PR_HER2_Comb

Variables ordered by Cliff's . | FDR < 0.05



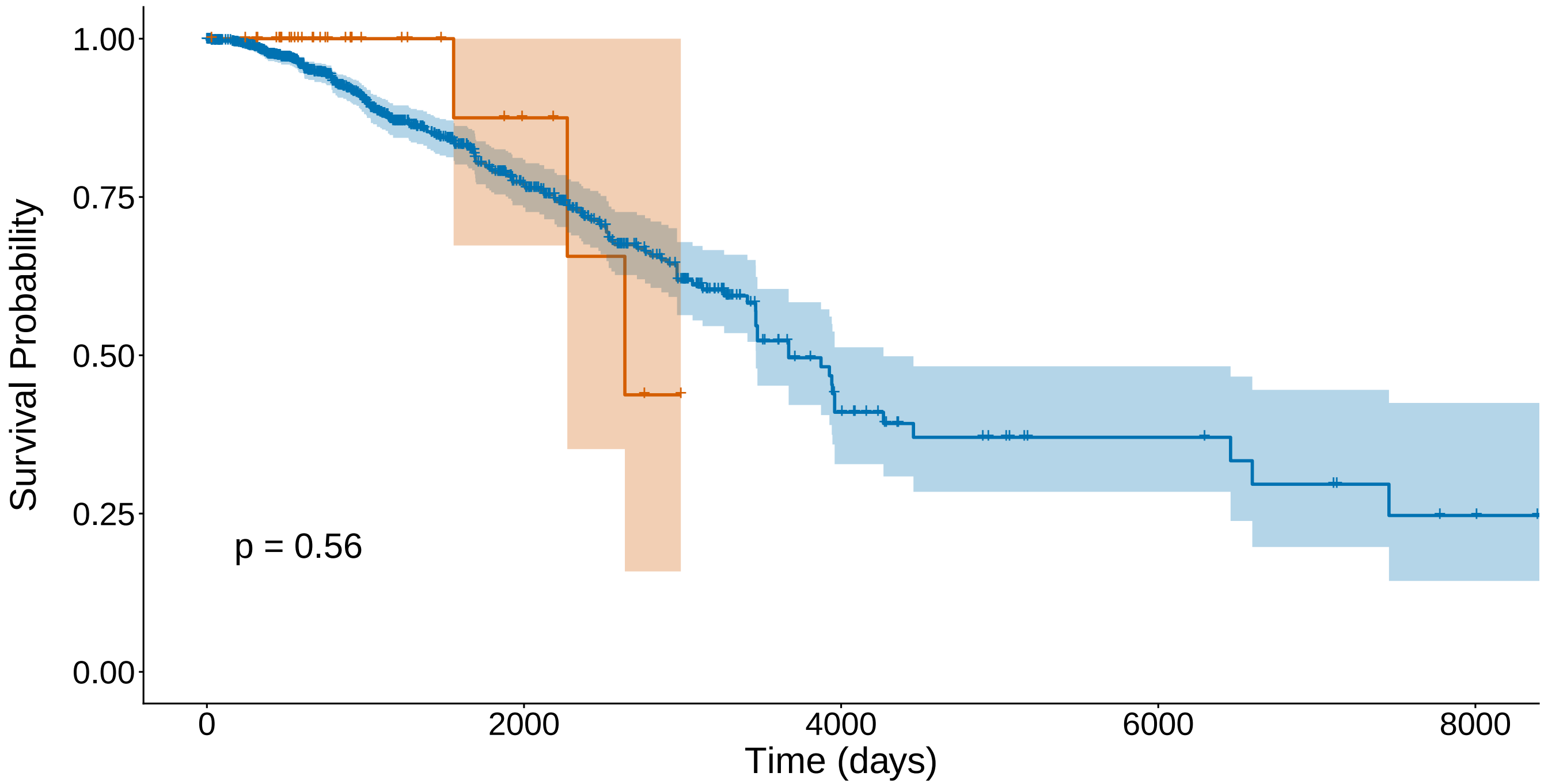
SECTION IV

Survival Analysis

Does the metabolic phenotype Core/Divergent predict prognosis?

Overall Survival Curve: Core vs Divergent

Strata + Core + Divergent



Number at risk

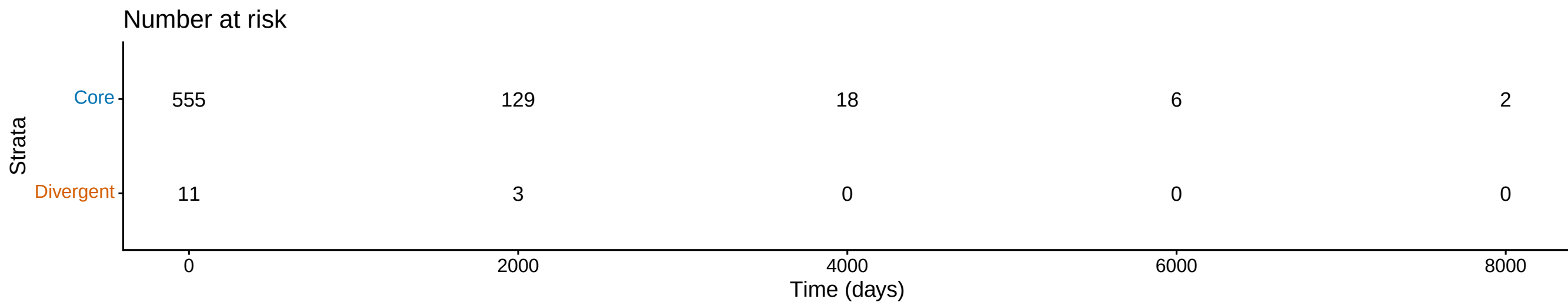
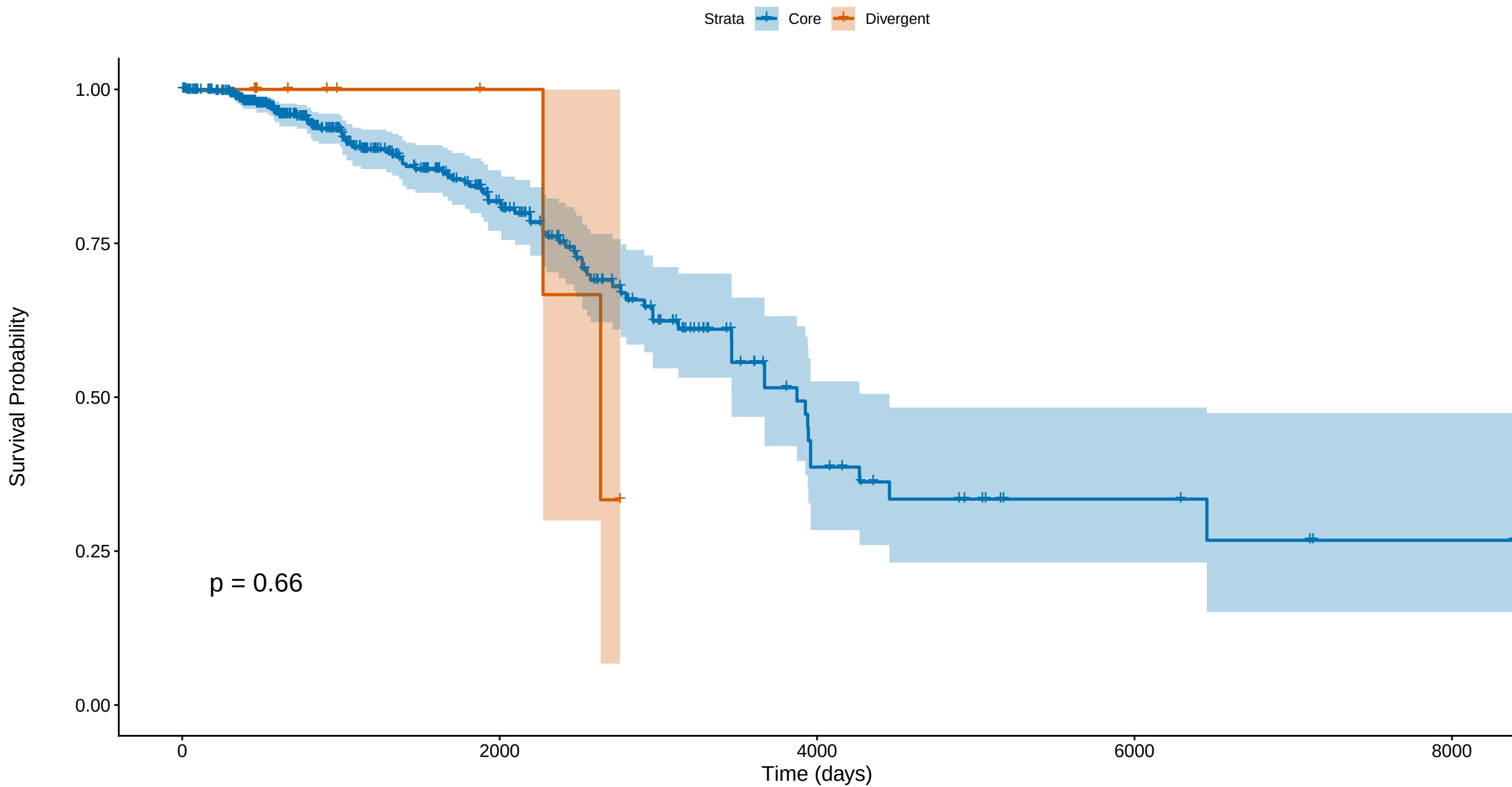
Core	1108	239	28	11	4
Divergent	37	5	0	0	0
	0	2000	4000	6000	8000

SECTION IV–B

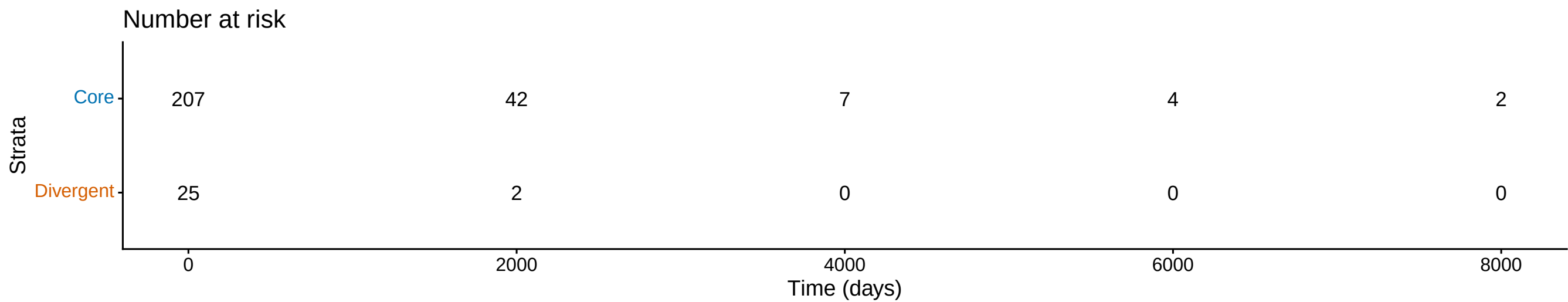
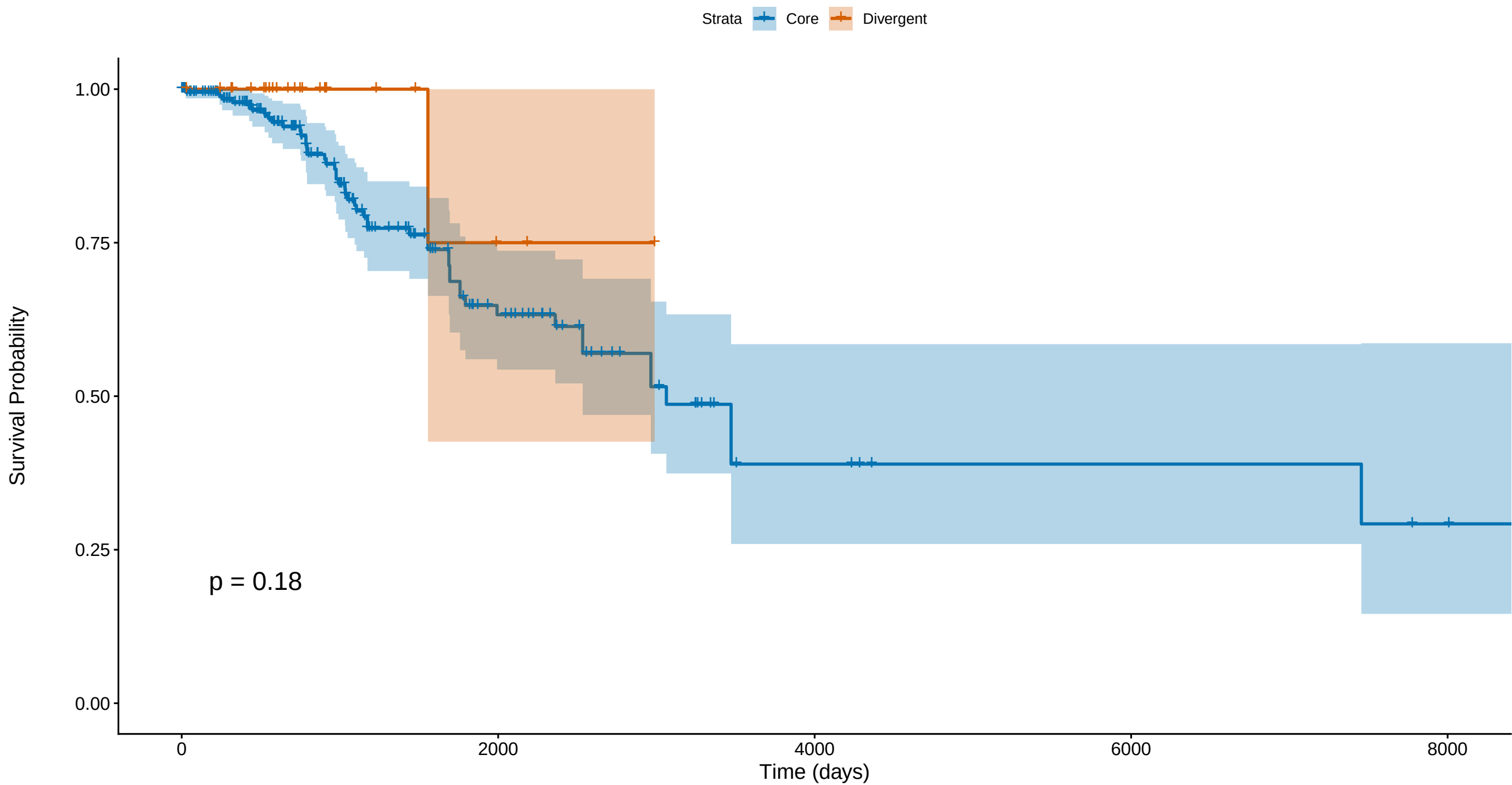
Survival by Molecular Subtype (ER_PR_HER2_Combo)

Each subtype: Core vs Divergent | min. 5 patients per group

Survival: Core vs Divergent
Subtype: Luminal_HR+
Core n=555 | Divergent n=11



Survival: Core vs Divergent
Subtype: Triple_Negative
Core n=207 | Divergent n=25



SECTION IV–C

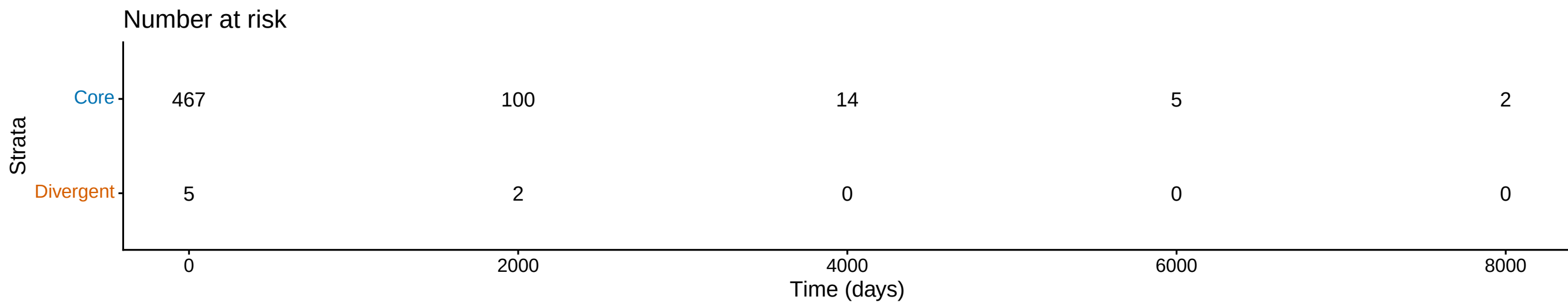
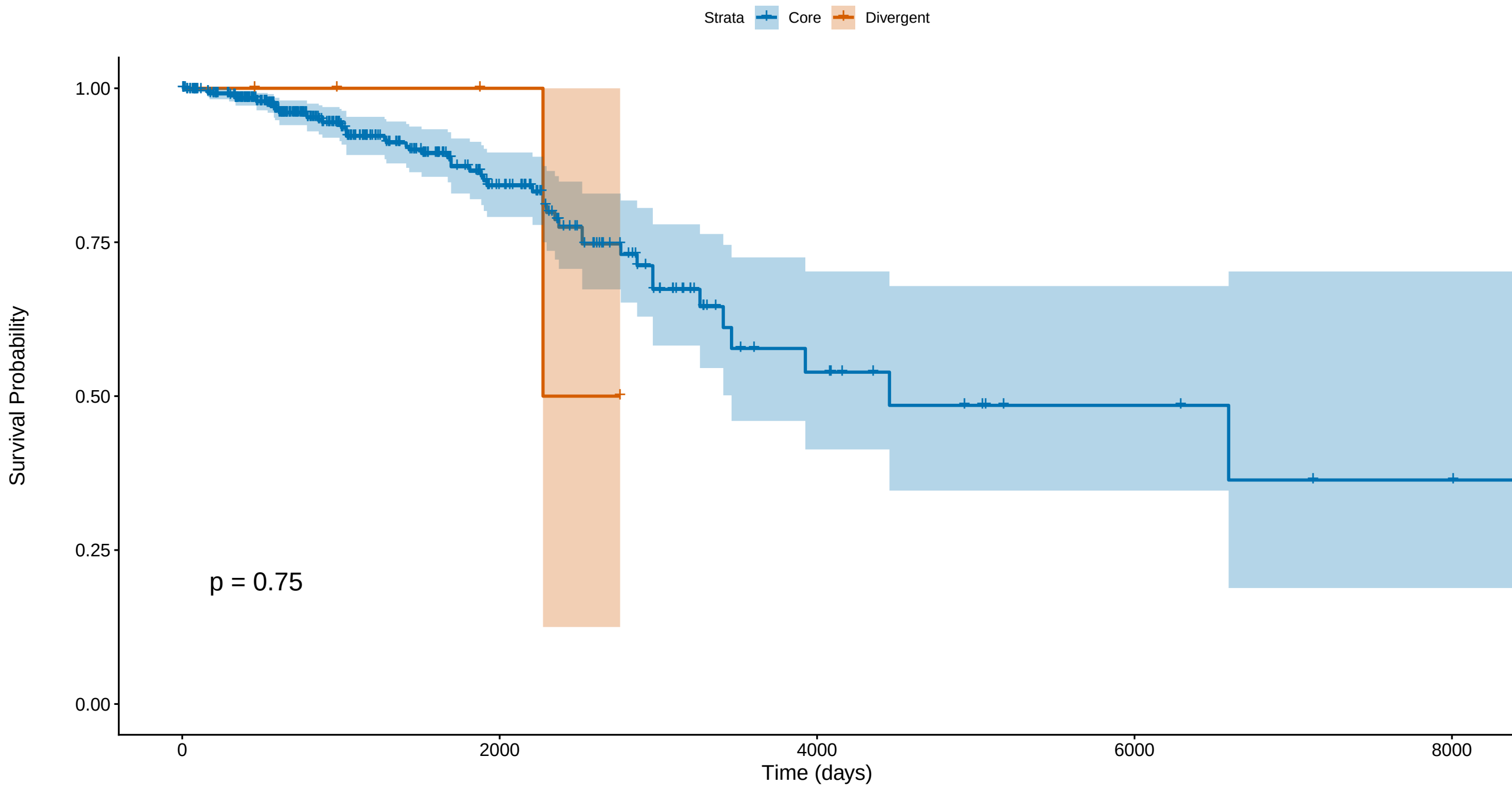
Survival by Molecular Subtype (Subtype / PAM50)

Each subtype: Core vs Divergent | min. 5 patients per group

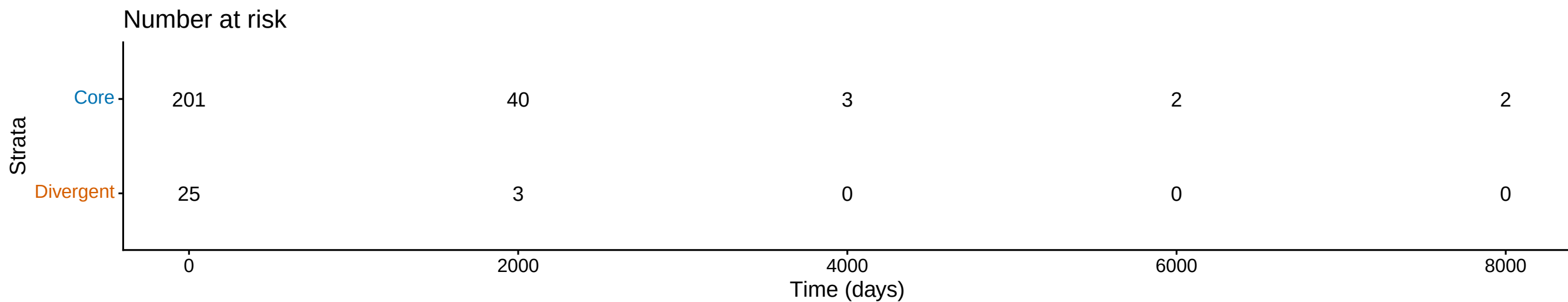
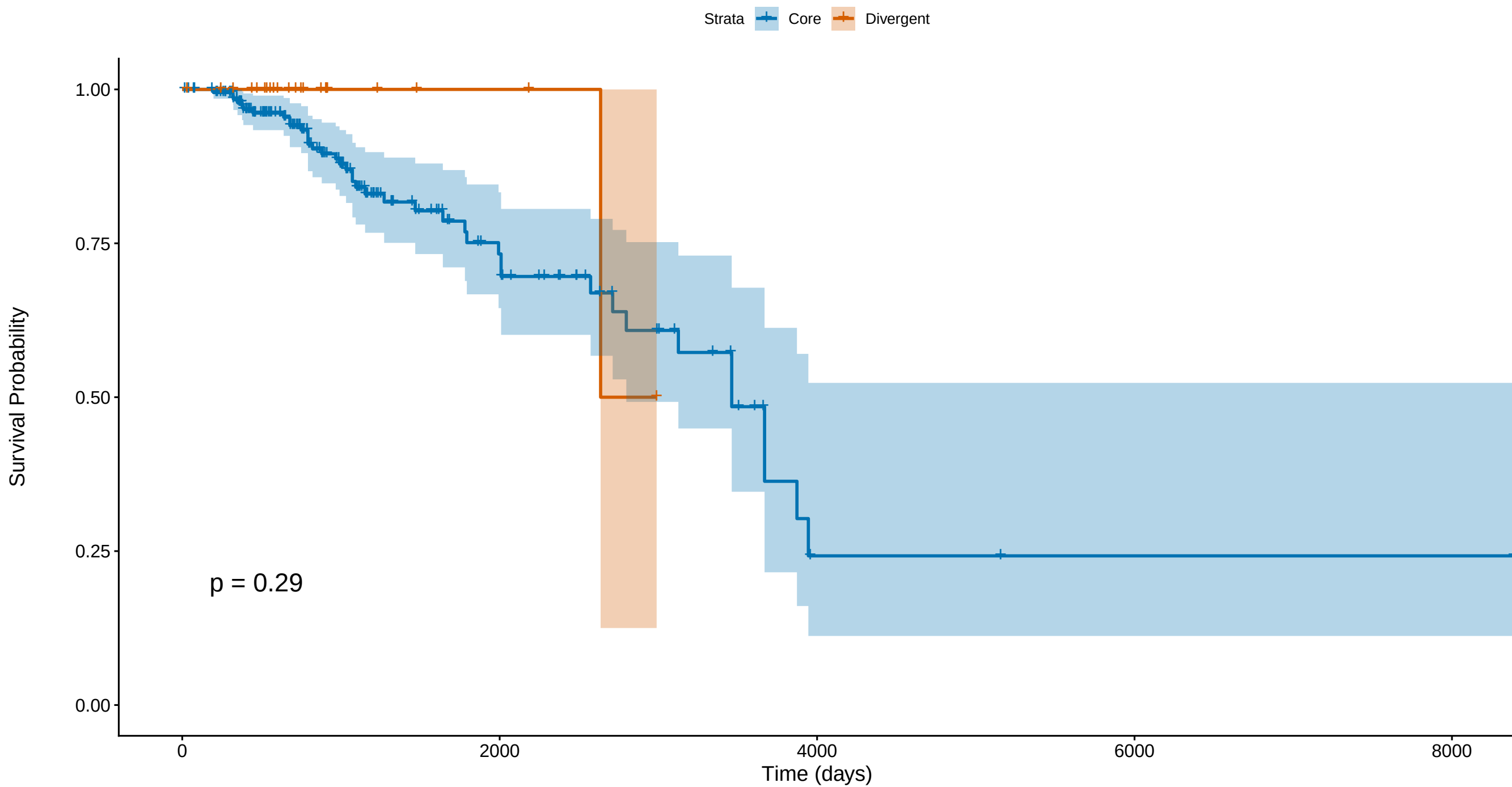
Survival: Core vs Divergent

Subtype: LumA

Core n=467 | Divergent n=5



Survival: Core vs Divergent
Subtype: Missing
Core n=201 | Divergent n=25



Survival Summary

Core: N=1108 | Events=185 | Median OS (events)=1174 days | Median OS (all)=943 days

Divergent: N= 37 | Events= 3 | Median OS (events)=2273 days | Median OS (all)=762 days

N by Molecular Subtype:

HER2_Enriched	Core	N= 72	Events=12
HER2_Enriched	Divergent	N= 1	Events= 0
Luminal_HER2+	Core	N=274	Events=35
Luminal_HR+	Core	N=555	Events=88
Luminal_HR+	Divergent	N= 11	Events= 2
Triple_Negative	Core	N=207	Events=50
Triple_Negative	Divergent	N= 25	Events= 1

Metabolic Signature by Molecular Subtype – Top 40

Variables ordered by Cliff's . | FDR < 0.05

